Engine-to-Driveshaft Coupling Adapter - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Wrench Set	T101588	1	
SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: Wrench Set	*T103063*	*1*	*(412Classic/SP to 412HP Upgrade)*

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-001	AR	
Grease	C-007	AR	

Procedure

1. Position the engine-to-driveshaft coupling adapter (2, Figure 1) on the engine (1). Lubricate the threads and mating face of the retaining nut (4) lightly with grease (C-001) or grease (C-007). Keep the mating faces of the washer (3) and engine-to-driveshaft curvic coupling adapter dry.

2. Install the washer (3) and the retaining nut (4) on the engine and hand-tighten.

3. Position the plate assembly (7 or 8, Figure 2), as applicable, on the adapter (1) and install the nut (4). Position the extension bar (6) in the retaining nut (5).

4. Torque the retaining nut (5) 250 to 275 foot-pounds (339 to 373 Nm) for initial torque to seat the coupling adapter on the engine.

5. Loosen the retaining nut (5) and the washer (2). Remove the extension bar (6) and the plate assembly (7 or 8), as applicable.

6. Remove the retaining nut (5) and the washer (2). Inspect the washer for any deformation or damage on the mating surface and/or the locking tab of the washer. If the washer is damaged, install a new washer and do Step 2. through Step 6. until the inspection of the washer reveals no damage.

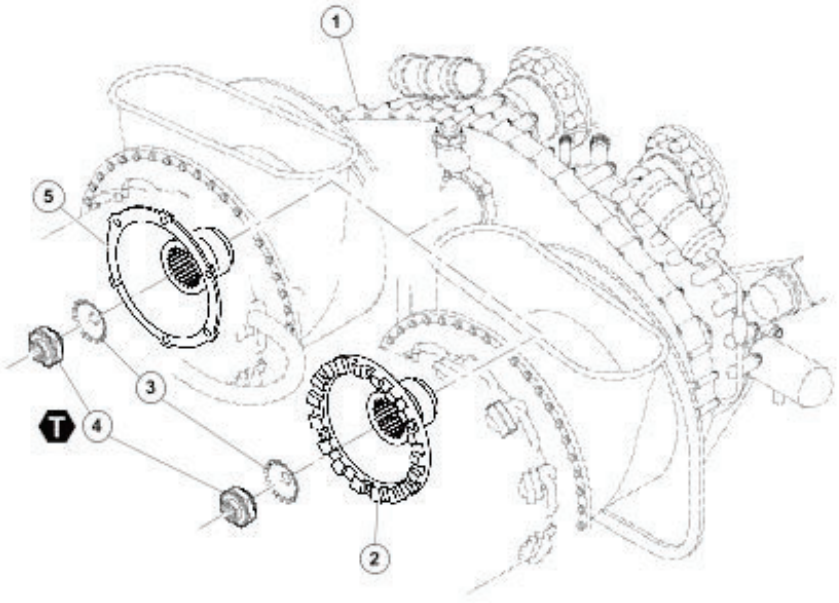
7. Install the washer (2) and the retaining nut (5) on the engine and hand-tighten. Using a suitable marker, mark an index line on the outer lock tang and adapter to enable verification that the lock washer does not rotate during torquing.

8. Position the plate assembly (7 or 8), as applicable, on the adapter (1) and install the nut (4). Position the extension bar (6) in the retaining nut (5).

9. Torque the retaining nut (5) 125 to 200 foot-pounds (170 to 271 Nm).

10. Inspect the washer (3, Figure 1) to ensure the tang indexed in Step 7. has not rotated. If the inspection reveals the washer has rotated, replace the washer and repeat Step 1. through Step 10..

11. Bend the single tab of the washer (3) into the slot of the retaining nut (4).



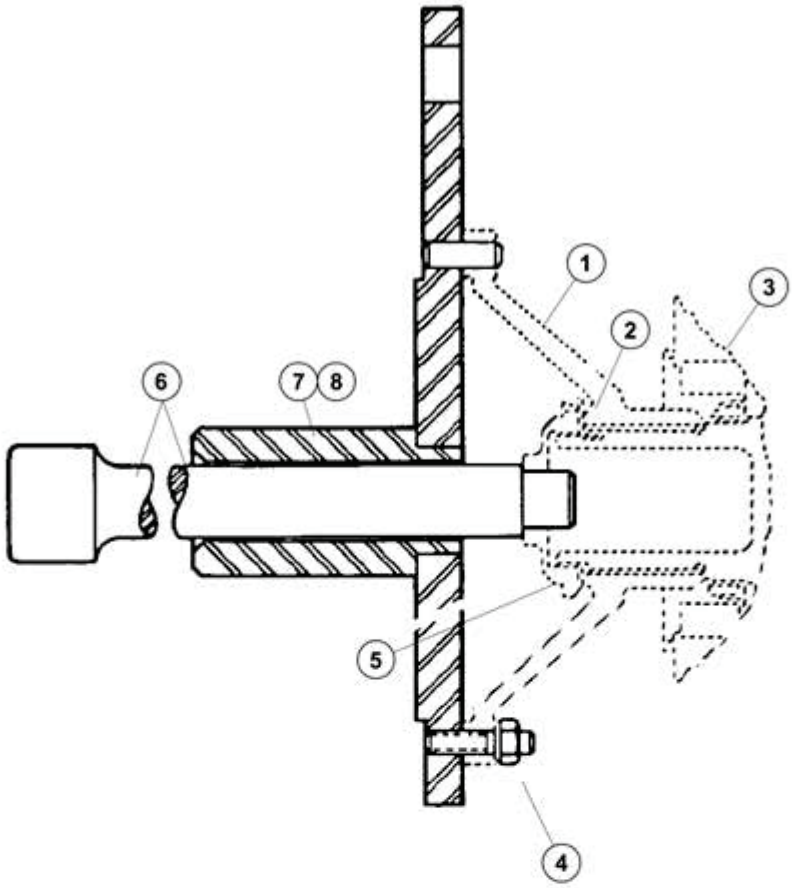
1. Engine
2. Engine-to-driveshaft curvic coupling adapter
3. Washer
4. Retaining nut
5. Engine-to-driveshaft coupling adapter

T 250 TO 275 FT-LBS
(339 TO 373 Nm)
INITIAL TORQUE

125 TO 200 FT-LBS
(170 TO 271 Nm)
FINAL TORQUE

ICN-412-A-631001-B-97499-00005-A-001-01

Figure 1. Engine-to-Driveshaft Coupling Adapter (Sheet 1 of 1)



1. Adapter
2. Washer
3. Engine installation reference
4. Nut
5. Retaining nut
6. Extension bar (L-122)
7. Plate assembly (T101588)
8. Plate assembly (T103063)

NOTES

1. Install T101588 plate assembly (7) on adapter (1). Secure with nut (4).
2. Insert extension bar (6) into plate assembly (7 or 8). Ensure that extension bar is seated all the way into retainer nut (5) before attempting to loosen nut.
3. Install T103063 plate assembly (8) on adapter (1) for model 412 S/N 36020 and subsequent, and helicopters modified by 412-570-001-103 (412SP to 412HP) (BHT-412-SI-74) (412Classic to 412HP) (BHT-412-SI-97).

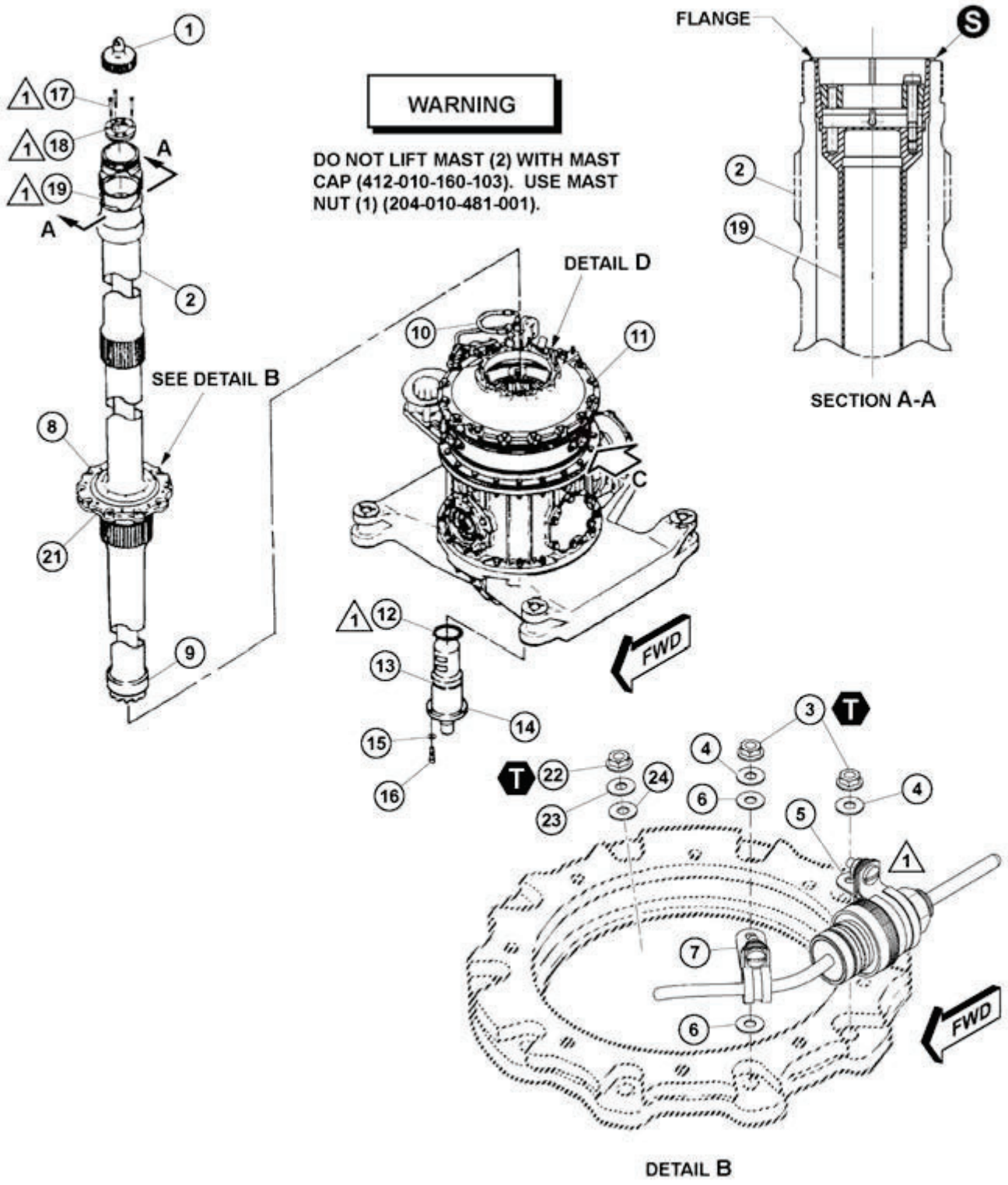
ICN-412-A-631001-B-97499-00006-A-002-01

Figure 2. Engine-to-Driveshaft Coupling Wrench Set - Tool Application (Sheet 1 of 1)

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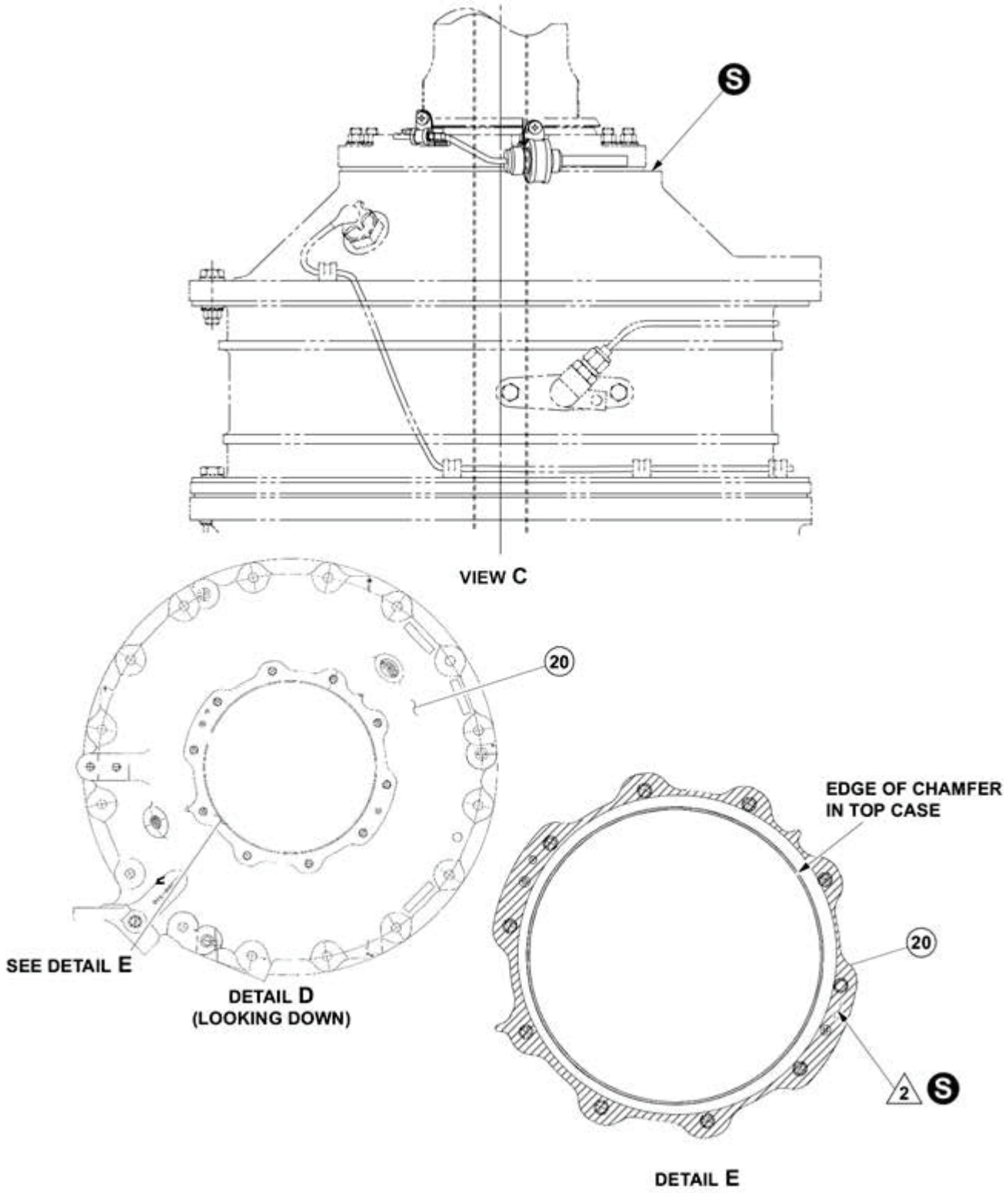
Main Rotor Mast - Description

The main rotor mast (2, Figure 1) is a tubular steel shaft which includes a bearing and plate for attachment to top of transmission. A bearing race is installed on lower end of mast which mates with a roller bearing located in transmission.



ICN-412-A-632101-B-97499-00001-A-002-01

Figure 1. Main Rotor Mast (Sheet 1 of 3)



ICN-412-A-632101-B-97499-00002-A-001-01

Figure 1. Main Rotor Mast (Sheet 2 of 3)

1. Mast nut
2. Mast
3. Nut
4. Steel washer
5. Temperature bulb clamp bracket
6. Aluminum washer
7. Wire clamp bracket
8. Mast bearing retaining plate
9. Lower bearing race
10. Hose
11. Transmission
12. Packing
13. Monopole sensor assembly
14. Spacer
15. Washer
16. Screw
17. Bolt
18. Plug
19. Torque tube
20. Transmission top case
21. Bearing liner
22. Nut
23. Steel washer
24. Aluminum washer

(S) SEALANT (C-308)

(T) 100 TO 140 IN-LBS
(11.3 TO 15.8 Nm)

NOTES

1. S/N 36020 and subsequent and helicopters modified by 412-570-001-103 or post BHT-412-SI-74 (412SP to 412HP upgrade) or post BHT-412-SI-97 (412Classic to 412HP upgrade).

2. Apply a thin uniform coat of sealant (C-308) immediately before the mast installation on the indicated area.

ICN-412-A-632101-B-97499-00940-A-003-01

Figure 1. Main Rotor Mast (Sheet 3 of 3)

Main Rotor Mast - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Hoist	T101581	1	or Equivalent
Cover Lift Plate	T103211-101	1	
Mast Nut	204-010-481-001	1	
Clevis	204-011-178-001	1	
Slide Hammer with 1/4 X 20 UNC Threaded Shaft	Commercial	1	

Procedure

1. Put a sufficient quantity of wooden blocks or equivalent under the skid tubes at the forward crosstube location until transmission top case is reasonably level, so the mast assembly bearing liner (21, Figure 1) will align with the transmission top case (20) opening.

2. Remove main rotor hub (412-A-62-20-00-02A-520A-A / 00948) and blade assembly (412-A-62-10-00-00A-520A-A / 00935).



SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

HELICOPTERS HAVE A TORQUEMETER ASSEMBLY (TORQUE TUBE ASSEMBLY) INSTALLED INSIDE THE MAST. REMOVAL OF PLUG AND TUBE ASSEMBLY REQUIRES REALIGNMENT. HELICOPTERS WITH A 412-375-004-105 MONOPOLE SENSOR INSTALLED REQUIRE THE TORQUE TUBE TO BE REALIGNED AT BELL HELICOPTER. HELICOPTERS WITH A 412-375-004-109 MONOPOLE SENSOR INSTALLED MAY HAVE TORQUE TUBE ALIGNED USING 412-240-028-101 MAST TORQUEMETER ALIGNMENT TOOL. AFTER A MAST IS INSTALLED INTO THE TRANSMISSION, A ZERO TORQUE CALIBRATION IS REQUIRED (412-A-96-12-00-01A-273B-A / 01984).

NOTE

Hub and sleeve assembly may be removed with swashplate and support assembly as a unit. Care must be exercised to prevent damage to mast and hub spline.

3. Remove hub sleeve assembly (412-A-62-30-00-03A-520A-A / 00968).

4. Remove swashplate and support assembly (412-A-62-30-00-01A-520A-A / 00955).



DO NOT LIFT MAST (2, Figure 1) WITH MAST CAP (412-010-160-103). USE mast nut (204-010-481-001).

5. Install mast nut (1). Attach clevis (204-011-178-001) to mast nut. Attach hoist to clevis.

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

6. *Remove monopole sensor assembly (13) from bottom of transmission and mast (2) as follows:*

NOTE

Some oil will be trapped behind monopole sensor assembly (13).

6.1. *Cut and remove lockwire from four screws (16) and remove screws with washers (15).*

6.2. *Remove sealant from around base of monopole sensor assembly (13) using a sharp plastic scraper.*



DO NOT USE JACKSCREW(S) TO PULL SENSOR FROM CASE. DO NOT ALLOW PULLER TO BOTTOM OUT IN THREADED HOLE OF SENSOR.

6.3. *Install* slide hammer with 1/4 x 20 unc threaded shaft *in outboard holes of monopole sensor assembly (13). Pull sensor and spacer (14) from support case. Remove and discard packing (12).*

7. Remove nuts (3 and 22), washers (4,6,23 and 24), temperature bulb clamp bracket (5), and wire clamp bracket (7) from mast bearing retaining plate (8). Remove sealant from around base of mast bearing retaining plate with a sharp plastic scraper.

8. Disconnect mast bearing oil hose (10) from mast bearing retaining plate (8). Cap or plug oil jet and hose.



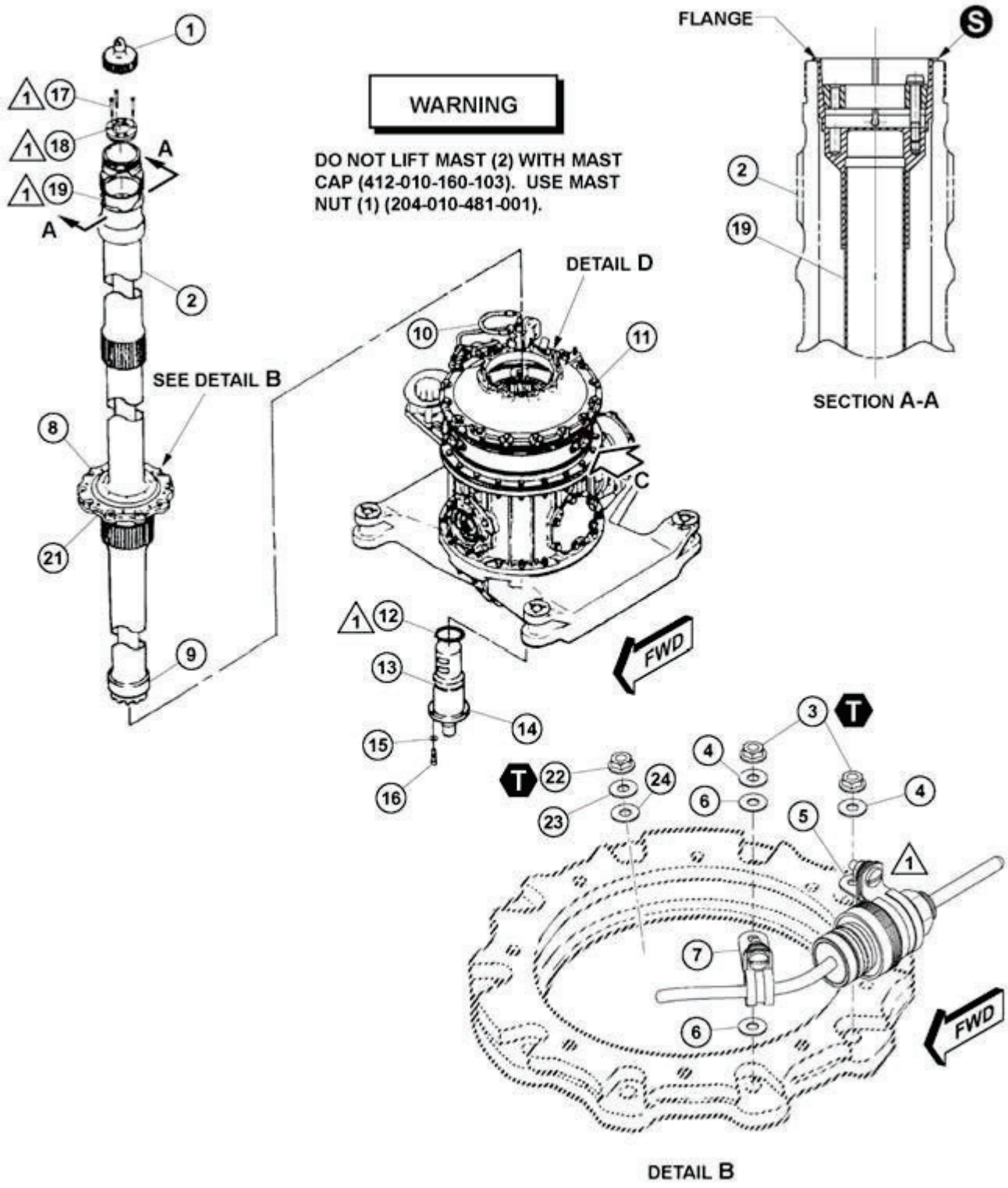
DO NOT TOUCH THE SECTION OF MAST INSTALLED IN TRANSMISSION WITH BARE HANDS.

NOTE

Heating top case of transmission with a heat lamp will aid in mast removal.

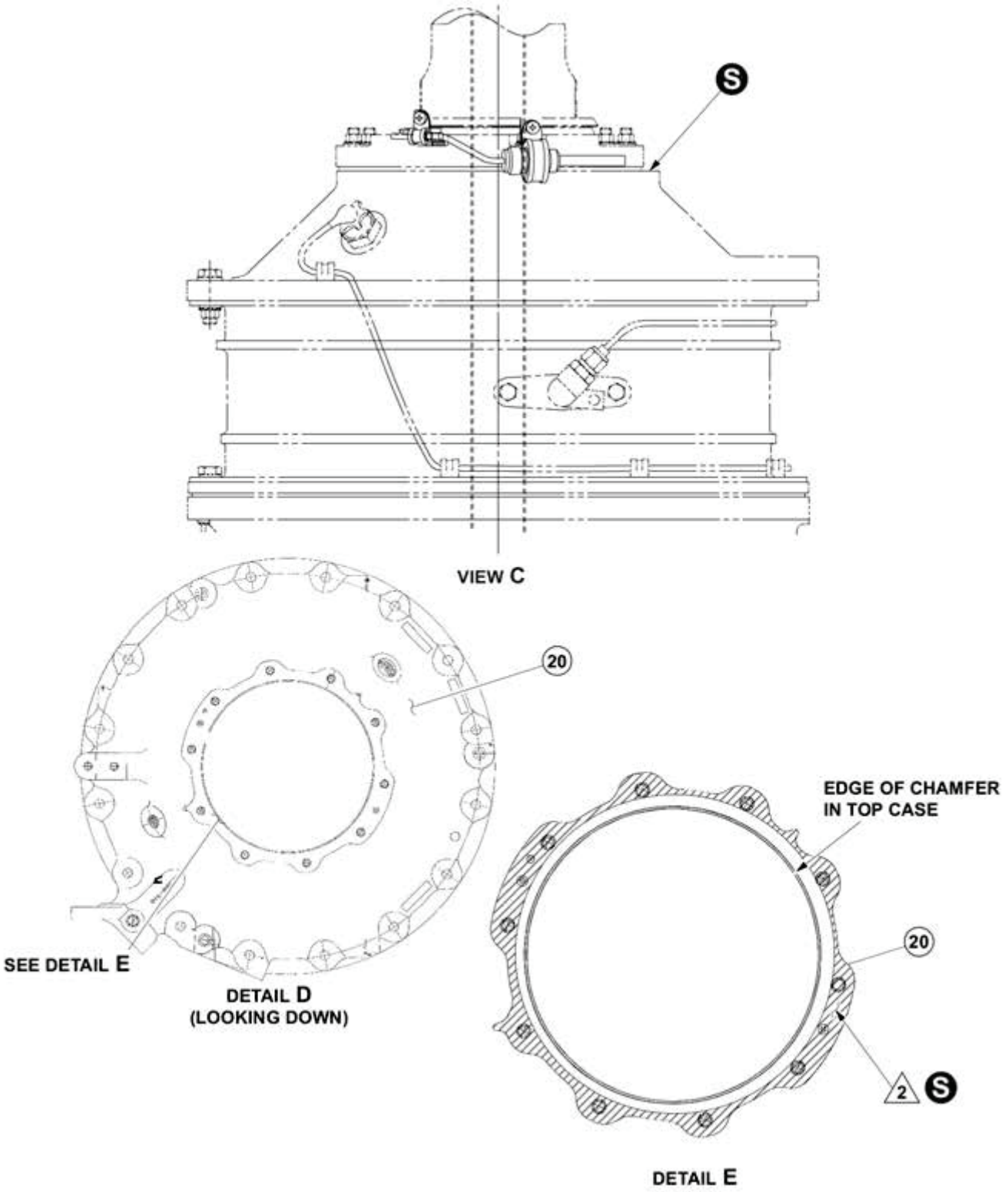
9. Carefully lift mast (2) from transmission. Immediately after removal, install cover lift plate (T103211-101).

10. Position mast (2) on padded stand to prevent damage to mast. Place protective cover on lower bearing race (9). Remove clevis (204-011-178-001). Leave mast nut (1) installed to protect threads.



ICN-412-A-632101-B-97499-00001-A-002-01

Figure 1. Main Rotor Mast - Removal (Sheet 1 of 3)



ICN-412-A-632101-B-97499-00002-A-001-01

Figure 1. Main Rotor Mast - Removal (Sheet 2 of 3)

1. Mast nut
2. Mast
3. Nut
4. Steel washer
5. Temperature bulb clamp bracket
6. Aluminum washer
7. Wire clamp bracket
8. Mast bearing retaining plate
9. Lower bearing race
10. Hose
11. Transmission
12. Packing
13. Monopole sensor assembly
14. Spacer
15. Washer
16. Screw
17. Bolt
18. Plug
19. Torque tube
20. Transmission top case
21. Bearing liner
22. Nut
23. Steel washer
24. Aluminum washer

(S) SEALANT (C-308)

(T) 100 TO 140 IN-LBS
(11.3 TO 15.8 Nm)

NOTES

1. S/N 36020 and subsequent and helicopters modified by 412-570-001-103 or post BHT-412-SI-74 (412SP to 412HP upgrade) or post BHT-412-SI-97 (412Classic to 412HP upgrade).

2. Apply a thin uniform coat of sealant (C-308) immediately before the mast installation on the indicated area.

ICN-412-A-632101-B-97499-00940-A-003-01

Figure 1. Main Rotor Mast - Removal (Sheet 3 of 3)

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Main Rotor Mast - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

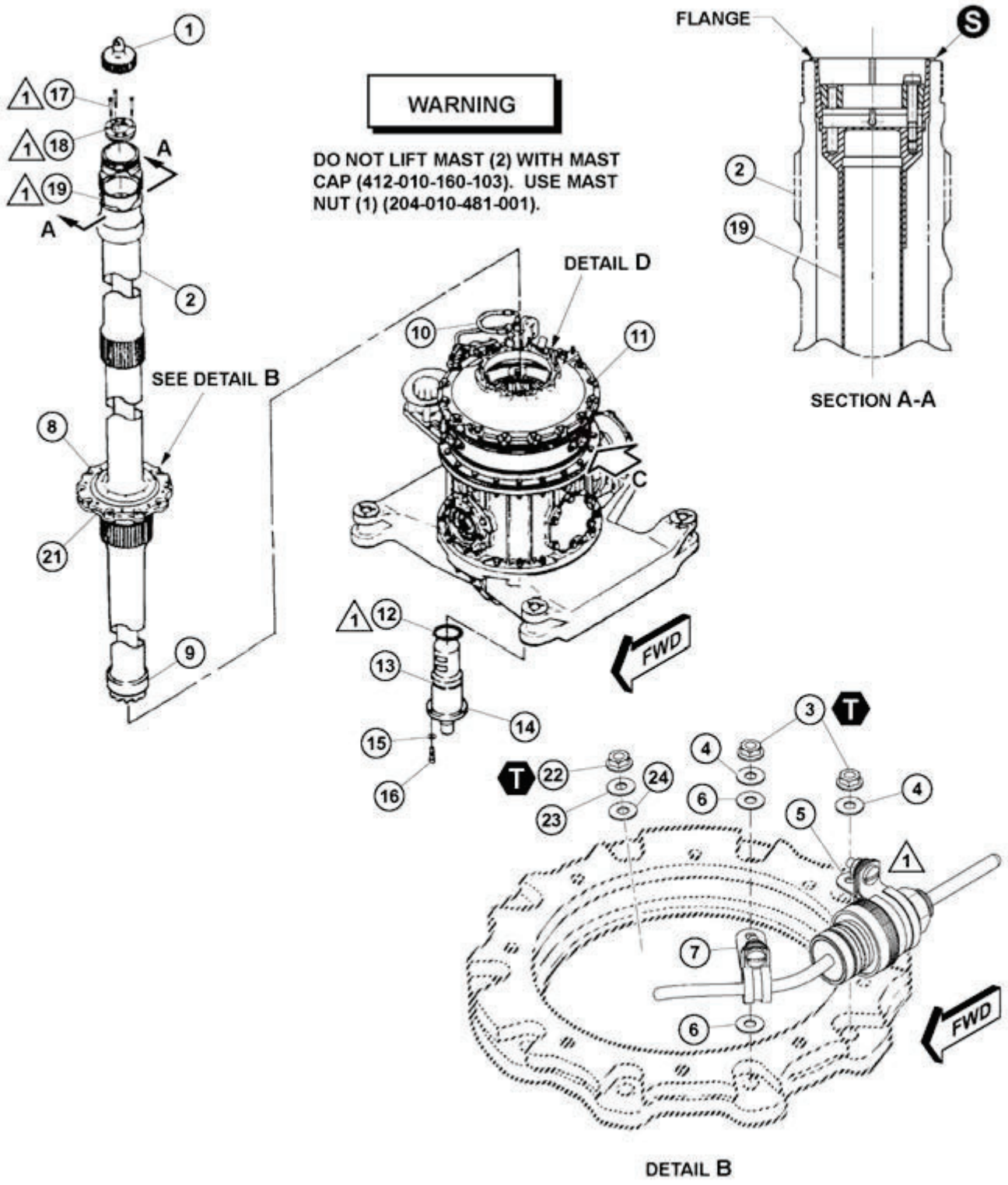
Procedure



DO NOT ALLOW SOLVENT TO ENTER MAST BEARINGS.

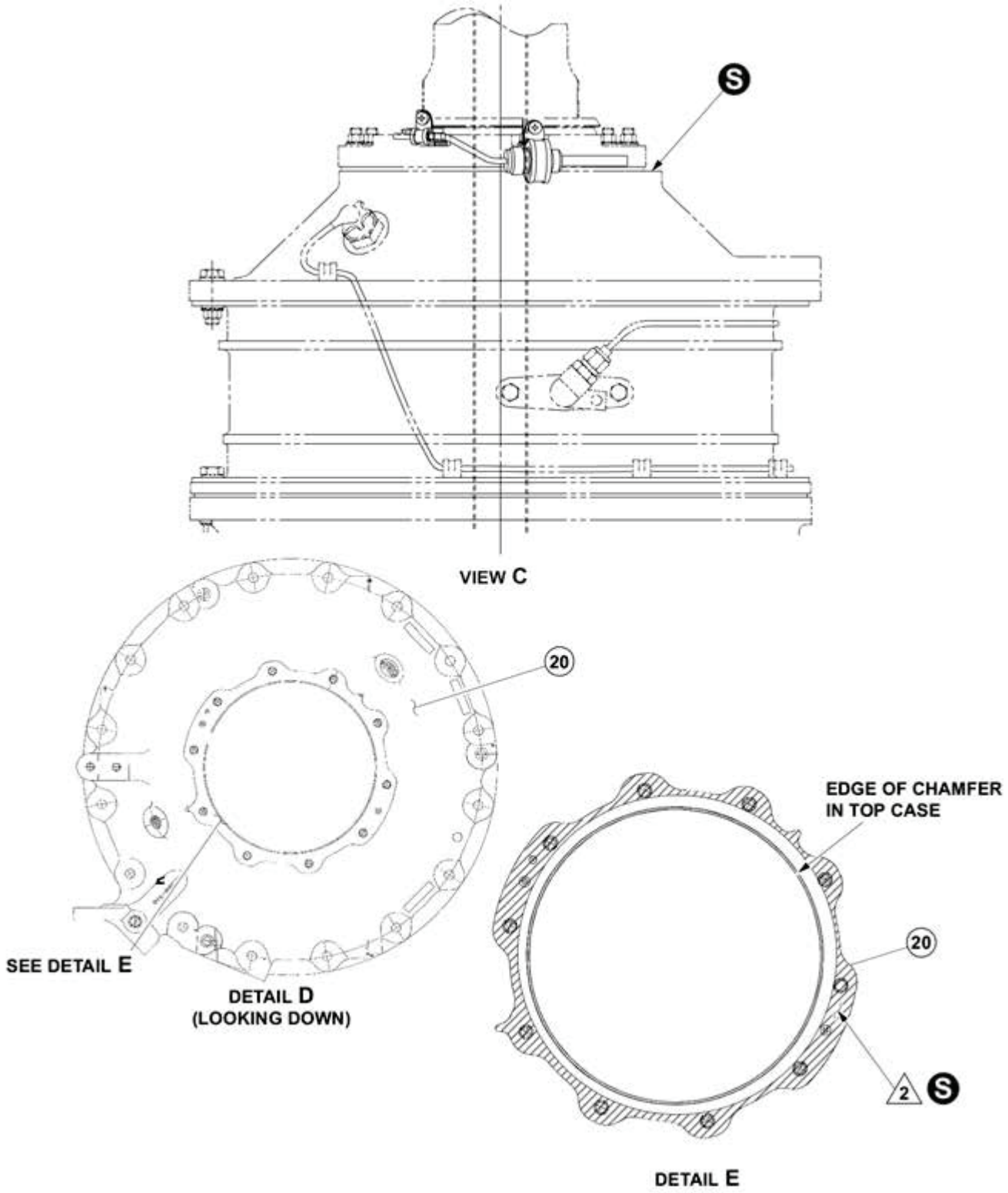
1. Clean main rotor mast (2, Figure 1) with a clean, lint-free cloth damp with drycleaning solvent (C-304).

2. Dry mast with filtered, compressed air.



ICN-412-A-632101-B-97499-00001-A-002-01

Figure 1. Main Rotor Mast (Sheet 1 of 3)



ICN-412-A-632101-B-97499-00002-A-001-01

Figure 1. Main Rotor Mast (Sheet 2 of 3)

1. Mast nut
2. Mast
3. Nut
4. Steel washer
5. Temperature bulb clamp bracket
6. Aluminum washer
7. Wire clamp bracket
8. Mast bearing retaining plate
9. Lower bearing race
10. Hose
11. Transmission
12. Packing
13. Monopole sensor assembly
14. Spacer
15. Washer
16. Screw
17. Bolt
18. Plug
19. Torque tube
20. Transmission top case
21. Bearing liner
22. Nut
23. Steel washer
24. Aluminum washer

(S) SEALANT (C-308)

(T) 100 TO 140 IN-LBS
(11.3 TO 15.8 Nm)

NOTES

(1) S/N 36020 and subsequent and helicopters modified by 412-570-001-103 or post BHT-412-SI-74 (412SP to 412HP upgrade) or post BHT-412-SI-97 (412Classic to 412HP upgrade).

(2) Apply a thin uniform coat of sealant (C-308) immediately before the mast installation on the indicated area.

ICN-412-A-632101-B-97499-00940-A-003-01

Figure 1. Main Rotor Mast (Sheet 3 of 3)

Main Rotor Mast - Inspection and Repair

Procedure

1. Inspect and repair mast in accordance with the BHT-412-CR&O.

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Main Rotor Mast - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Cover Plate	T103211-101	1	
Mast Nut	204-010-481-001	1	
Clevis	204-011-178-001	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	
Aliphatic naphtha	C-305	AR	
Sealant	C-308	AR	
Denatured alcohol	C-326	AR	
Lockwire	C-405	AR	
Clean cloth	C-516	AR	

Procedure

1. Remove cover plate from transmission.

2. Perform dimensional check of transmission to prevent damage to upper planetary assembly as follows:

2.1. Place straight edge (1, Figure 1) on transmission top case (2). Measure from top of mast driving adapter (3) to straight edge and record dimension. If dimension is less than 2.226 inches (56.54 mm), comply with Step 2.2. through Step 2.4..



HANDLE MAST DRIVING ADAPTER (3, Figure 1) CAREFULLY WHEN INDEXING BEARING LINERS. INJURY TO FINGERS MAY RESULT.

2.2. Inspect upper and lower bearing liners (5 and 6) to determine whether tangs are disengaged. If tangs are disengaged, a gap of approximately 0.25 inch (6.0 mm) will be evident.

2.3. Insert hands back-to-back through mast driving adapter (3). Lift mast driving adapter slightly and rotate upper bearing liner (5) until tabs on upper bearing liner align with slots in lower bearing liner (6) and drop into position.

2.4. Repeat measurements described in Step 2.1..



DO NOT LIFT MAST (2, Figure 2) WITH MAST CAP (412-010-160-103). USE ONLY mast nut (204-010-481-001) TO LIFT MAST.



DO NOT TOUCH THE SECTION OF MAST INSTALLED IN TRANSMISSION WITH BARE HANDS.

3. Install mast nut (1, Figure 2). Attach clevis (204-011-178-001) to mast nut. Attach hoist to clevis.

CAUTION

SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

(412CLASSIC/SP TO 412HP UPGRADE) HAVE A TORQUEMETER ASSEMBLY INSTALLED IN MAST (2, Figure 2). FAILURE TO REMOVE MONOPOLE SENSOR ASSEMBLY (13) BEFORE INSTALLING MAST CAN CAUSE DAMAGE TO TORQUEMETER ASSEMBLY.

CAUTION

IF MAST ASSEMBLY IS INSERTED INTO THE TRANSMISSION AT AN ANGLE, DAMAGE TO THE MAST ASSEMBLY BEARING LINER (21) AND TRANSMISSION TOP CASE (20) CAN OCCUR.

4. Put a sufficient quantity of wooden blocks or equivalent under the skid tubes at the forward crosstube location until the transmission top case is reasonably level, so that mast assembly bearing liner (21, Figure 2) will align with the transmission top case (20) opening.

5. Clean the mating surfaces of the mast bearing retaining plate (8) and the transmission top case (20) with a clean cloth (C-516) moistened with aliphatic naphtha (C-305) or denatured alcohol (C-326). Wipe the surfaces dry with a clean cloth before the solvent evaporates.

6. Apply a thin uniform coating of sealant (C-308) to the mating surfaces of the mast bearing retaining plate (8)..

NOTE

Heating transmission top case with a heat lamp will aid in installation.

7. Heat transmission top case (20) opening with a heat lamp or equivalent until hot to touch.

8. Lower the mast (2) until the mast bearing retaining plate (8) is in contact with the transmission top case (20).

9. Install the temperature bulb clamp bracket (5) next to the mast bearing retaining plate (8). Install the steel washer (4) and the nut (3). Install the wire clamp bracket (7) with one aluminum washer (6) under the L bracket and one above the bracket and one steel washer under the nut. Torque the nuts to 100 to 140 inch-pounds (11.3 to 15.8 Nm).

10. Install the aluminum washers (24), a thin steel washers (23), and a nuts (22) on the remaining 8 mounting studs. Torque the nuts (22) 100 to 140 inch-pounds (11.30 to 15.82 Nm).

11. Apply a bead of sealant (C-308) around the mating edges of the mast bearing retaining plate (8) and the transmission top case (20). Make sure to keep the four drain holes open and free of sealant.

12. Connect the oil hose (10) to the mast bearing oil jet.

13. Rotate the transmission input quill by hand to ensure there is no unusual noise or binding.

14. Remove the clevis and the mast nut (1). Use the mast cap assembly for final installation.

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

14.1. *(412Classic/SP to 412HP Upgrade), do a torque tube assembly alignment in accordance with 412-A-63-21-01-00A-272A-A / 01010. Alignment is only required if the torque tube was removed or disturbed during the mast removal or installation.*

NOTE

Replacement monopole sensor assemblies (13) will have the spacer (14) temporarily attached with hardware for shipping. Remove the shipping hardware and retain the spacer for installation with monopole sensor.

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

15. *Install the monopole sensor assembly (13) in the bottom of the transmission (11) as follows:*

15.1. *Lubricate a new packing (12) with transmission oil (412-A-12-10-03-00A-040A-A / 00135) and install on the monopole sensor assembly (13).*

15.2. *Clean the transmission support case free of oil and adhesive residue with a plastic scrapper and a clean cloth dampened with* drycleaning solvent (C-304)*.*

15.3. *Apply a light coat of transmission oil (*412-A-12-10-03-00A-040A-A / 00135*) to the bore in the support case.*

15.4. *Insert the monopole sensor assembly (13), with the spacer (14) installed, into the support case and align the monopole sensor assembly screw holes with the holes in the support case of the transmission (arrow pointing forward).*

15.5. *Install four screws (16) with the washers (15). Secure the screws with* lockwire (C-405)*.*

15.6. *Apply a bead of* sealant (C-308) *around the mating surface of the sensor.*

16. Install the swashplate and the support assembly (412-A-62-30-00-01A-720A-A / 00959).

17. Install the hub and sleeve assembly (412-A-62-30-00-03A-720A-A / 00972).

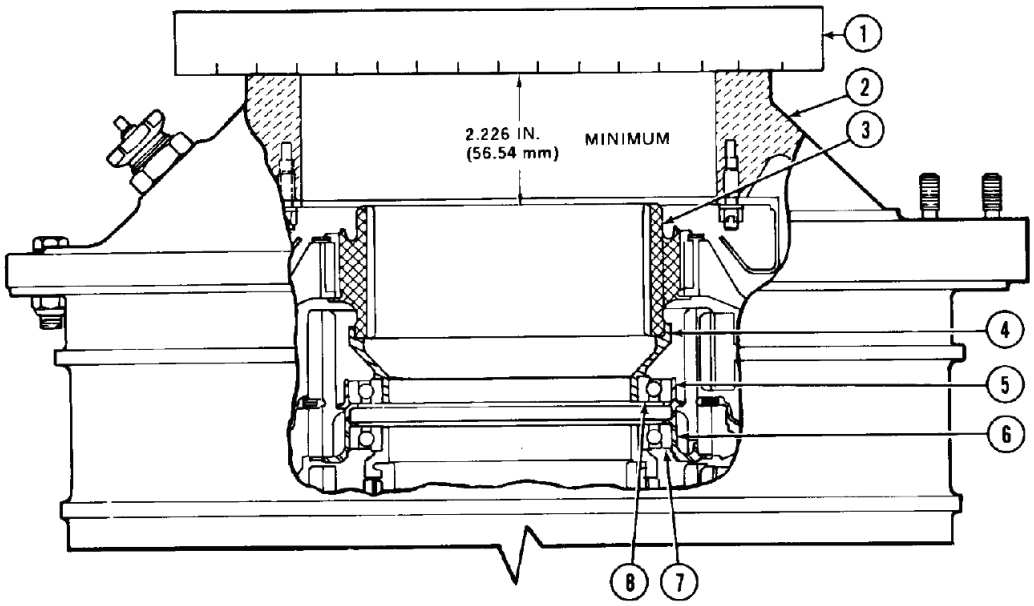
18. Install the main rotor hub (412-A-62-20-00-02A-720A-A / 00952).

19. Remove the wooden blocks or equivalent from under the skid tubes at the forward crosstube location.

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

20. *Calibrate the mast torque system (*412-A-96-12-00-01A-273B-A / 01984*).*

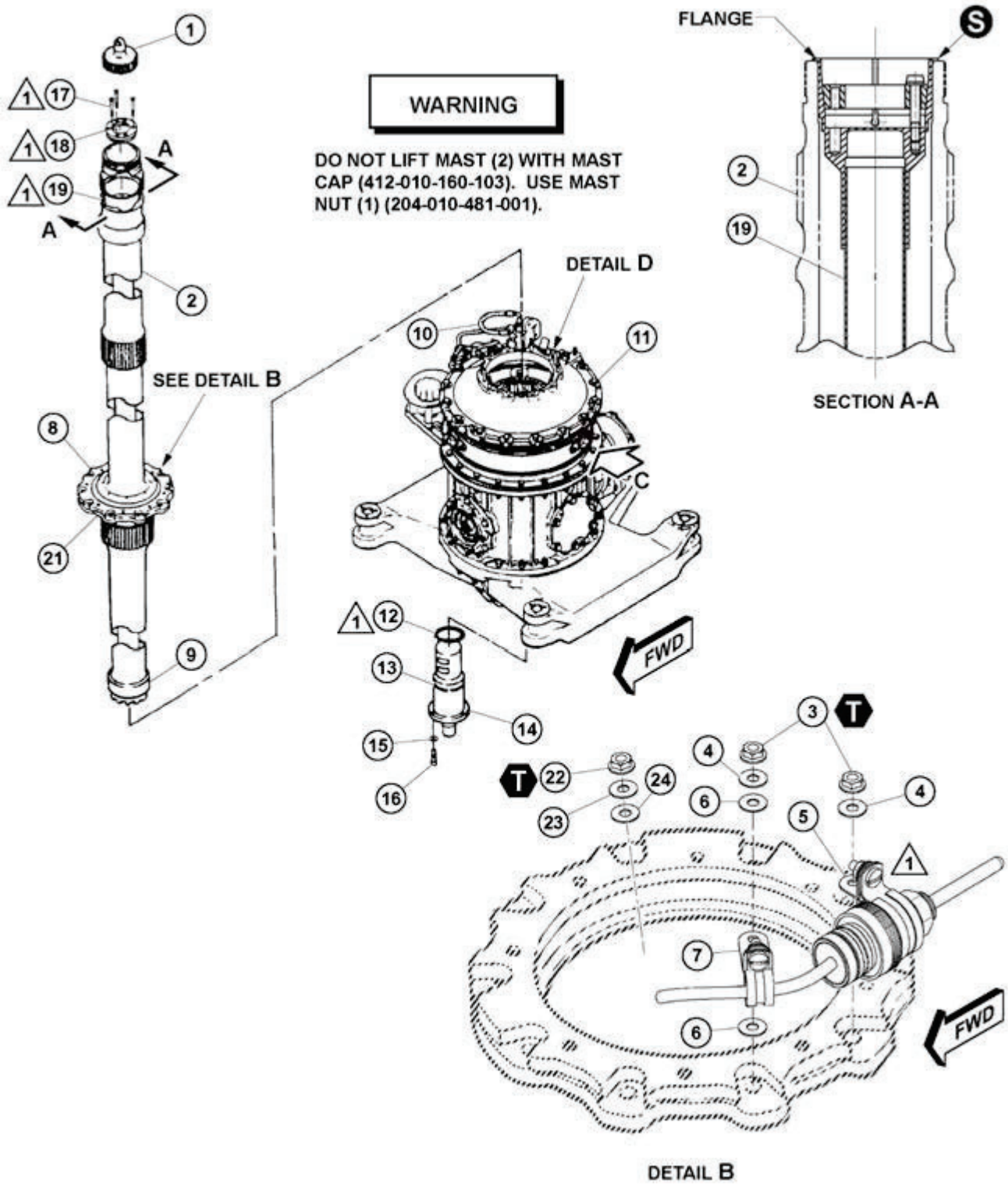
21. Install the main rotor blades (412-A-62-10-00-00A-720A-A / 00946).



1. Straight edge
2. Transmission top case
3. Mast driving adapter
4. Mast driving adapter extension
5. Upper bearing liner
6. Lower bearing liner
7. Bearing
8. Bearing

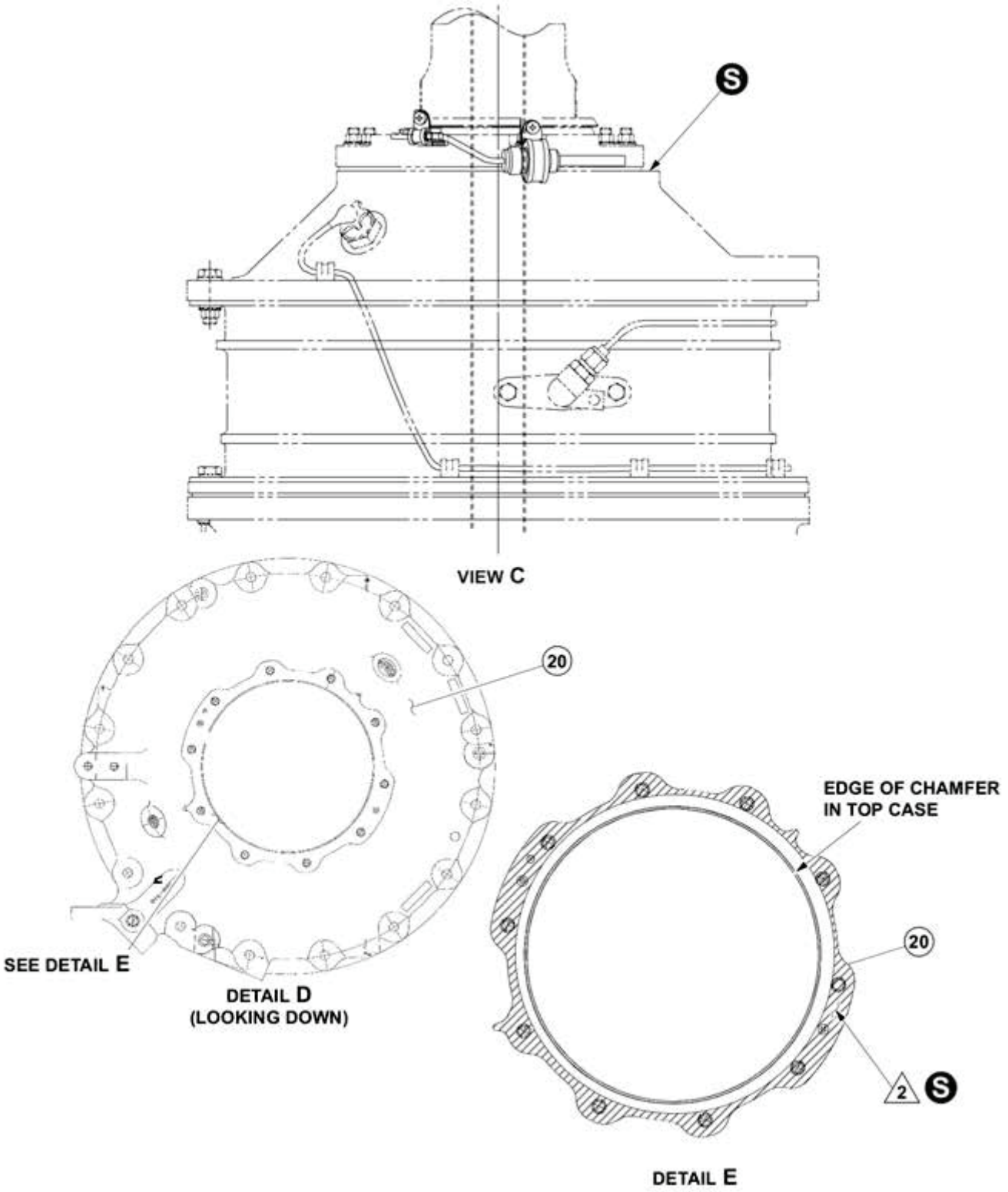
ICN-412-A-632101-B-97499-00003-A-002-01

Figure 1. Mast Driving Adapter to Transmission Top Case Dimension Requirements (Sheet 1 of 1)



ICN-412-A-632101-B-97499-00001-A-002-01

Figure 2. Main Rotor Mast - Installation (Sheet 1 of 3)



ICN-412-A-632101-B-97499-00002-A-001-01

Figure 2. Main Rotor Mast - Installation (Sheet 2 of 3)

1. Mast nut
2. Mast
3. Nut
4. Steel washer
5. Temperature bulb clamp bracket
6. Aluminum washer
7. Wire clamp bracket
8. Mast bearing retaining plate
9. Lower bearing race
10. Hose
11. Transmission
12. Packing
13. Monopole sensor assembly
14. Spacer
15. Washer
16. Screw
17. Bolt
18. Plug
19. Torque tube
20. Transmission top case
21. Bearing liner
22. Nut
23. Steel washer
24. Aluminum washer

(S) SEALANT (C-308)

(T) 100 TO 140 IN-LBS
(11.3 TO 15.8 Nm)

NOTES

1. S/N 36020 and subsequent and helicopters modified by 412-570-001-103 or post BHT-412-SI-74 (412SP to 412HP upgrade) or post BHT-412-SI-97 (412Classic to 412HP upgrade).

2. Apply a thin uniform coat of sealant (C-308) immediately before the mast installation on the indicated area.

ICN-412-A-632101-B-97499-00940-A-003-01

Figure 2. Main Rotor Mast - Installation (Sheet 3 of 3)

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Torque Tube Assembly - Alignment

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Jackscrews	T103222	1	
Mast Torquemeter Alignment Tool	412-240-028-101	1	Only Used with Mast Assemblies 412-040-366-111 and Subsequent

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
Ethyl alcohol	C-339	AR	
Lockwire	C-405	AR	

Procedure

NOTE

SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

The following procedure is applicable to helicopters that have a 412-375-004-109 monopole sensor assembly (13, Figure 1 *and* mast torquemeter alignment tool (412-240-028-101)*).*

NOTE

TB 412-01-179 : Pre:

For helicopters with 412-375-004-105 monopole sensor assembly and 412-040-366-109 mast assembly installed, torque tube alignment for transmissions shall be accomplished at Bell Helicopter only.

1. Remove monopole sensor assembly (13, Figure 1) (412-A-63-21-01-00A-520A-A / 01006).

2. Remove torque tube (19) from mast (2) as follows:

2.1. Remove mast nut (1).

2.2. Using a sharp plastic scraper, remove sealant from torque tube (19) and mast (2).

2.3. Cut and remove lockwire from bolts (17) and remove bolts from plug (18).

2.4. Remove plug (18) using jackscrews (T103222).

2.5. Insert a wooden rod (broom stick) inside torque tube (19) through the opening in the bottom of mast (2).

2.6. Using a hammer, lightly tap the wooden rod against the top of torque tube (19).

2.7. When torque tube (19) breaks loose, remove it from mast (2).

3. Clean surface of rotor in mast assembly which contacts with torque tube (19). Apply light coat of transmission oil to surface. Verify surface is smooth and clean.



DO NOT ROTATE ANY DRIVE SYSTEM COMPONENTS WHILE TOOL IS INSTALLED.

4. Insert mast torquemeter alignment tool (412-240-028-101) in bottom of transmission support case. Work top and bottom splines of tool up through spline of rotor assembly in bottom of mast (2). Rotate tool counterclockwise (looking up at bottom of mast) until splined teeth of tool are bottomed out against teeth of rotor in mast assembly. Hold tool in place with four screws and washers. Leave screws loose enough to allow 1/4 inch movement of tool.

5. Clean outer bearing surface of torque tube (19). Verify surface is smooth. Use care to keep bearing surface clean during installation.

6. Clean the mating surfaces of the torque tube (19) and the mast (2).

7. Apply sealant (C-308) on top of the mast (2).

8. Install torque tube (19) into mast (2) carefully. Rotate torque tube slightly, if required, to engage upper spline of mast torquemeter alignment tool (412-240-028-101). Continue installation until fully seated at top of mast.

9. Ensure slots of torque tube flange are filled with sealant (C-308).

10. Remove any excess sealant from flange and top of mast.

11. Rotate torque tube counterclockwise (looking down from top of mast) as far as alignment tool will allow. Torque tube may be rotated by gently tapping on split flange with soft metal bar and mallet. Hold alignment tool counterclockwise with upward pressure while tube is lowered. If alignment tool is pushed down while lowering torque tube, this is an indication of misalignment. Rotate torque tube gently until alignment tool rises. This indicates the alignment tool has engaged both the signal and reference gears.

12. Install plug (18) into torque tube (19). Install bolts (17) and torque to 50 to 70 inch-pounds (5.65 to 7.90 Nm) in symmetrical manner.

13. Loosen bolts on alignment tool. Verify tool can be rotated slightly and fits loosely in splines of mast (2) and torque tube (19). If tool cannot be rotated, repeat Step 11. and Step 12..

14. Remove alignment tool by pulling tool past torque tube spline, rotating tool a few degrees, and pulling tool out past mast spline.

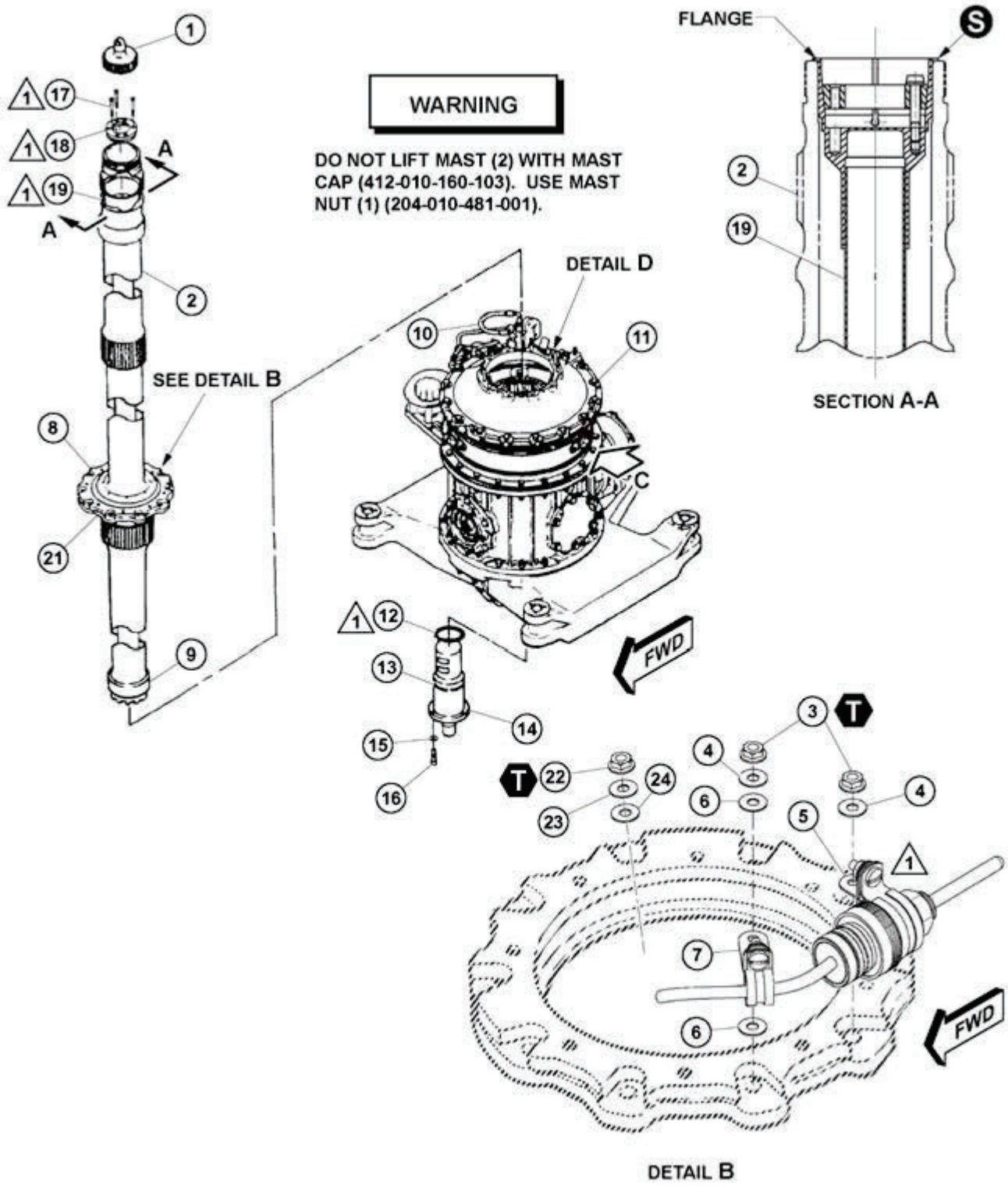
15. Install monopole sensor assembly (13) (412-A-63-21-01-00A-720A-A / 01009).

16. Secure torque tube plug (18) with lockwire (C-405). Apply ethyl alcohol (C-339) to mating surfaces of torque tube (19).

17. Install mast nut (1) (412-A-63-21-01-00A-720A-A / 01009).

18. Perform mast torque system zero torque calibration (412-A-96-12-00-01A-273B-A / 01984).

19. Perform mast torque system flight check (412-A-96-12-00-01A-280B-A / 01985).



ICN-412-A-632101-B-97499-00001-A-002-01

Figure 1. Main Rotor Mast (Sheet 1 of 3)

1. Mast nut
2. Mast
3. Nut
4. Steel washer
5. Temperature bulb clamp bracket
6. Aluminum washer
7. Wire clamp bracket
8. Mast bearing retaining plate
9. Lower bearing race
10. Hose
11. Transmission
12. Packing
13. Monopole sensor assembly
14. Spacer
15. Washer
16. Screw
17. Bolt
18. Plug
19. Torque tube
20. Transmission top case
21. Bearing liner
22. Nut
23. Steel washer
24. Aluminum washer

(S) SEALANT (C-308)

(T) 100 TO 140 IN-LBS
(11.3 TO 15.8 Nm)

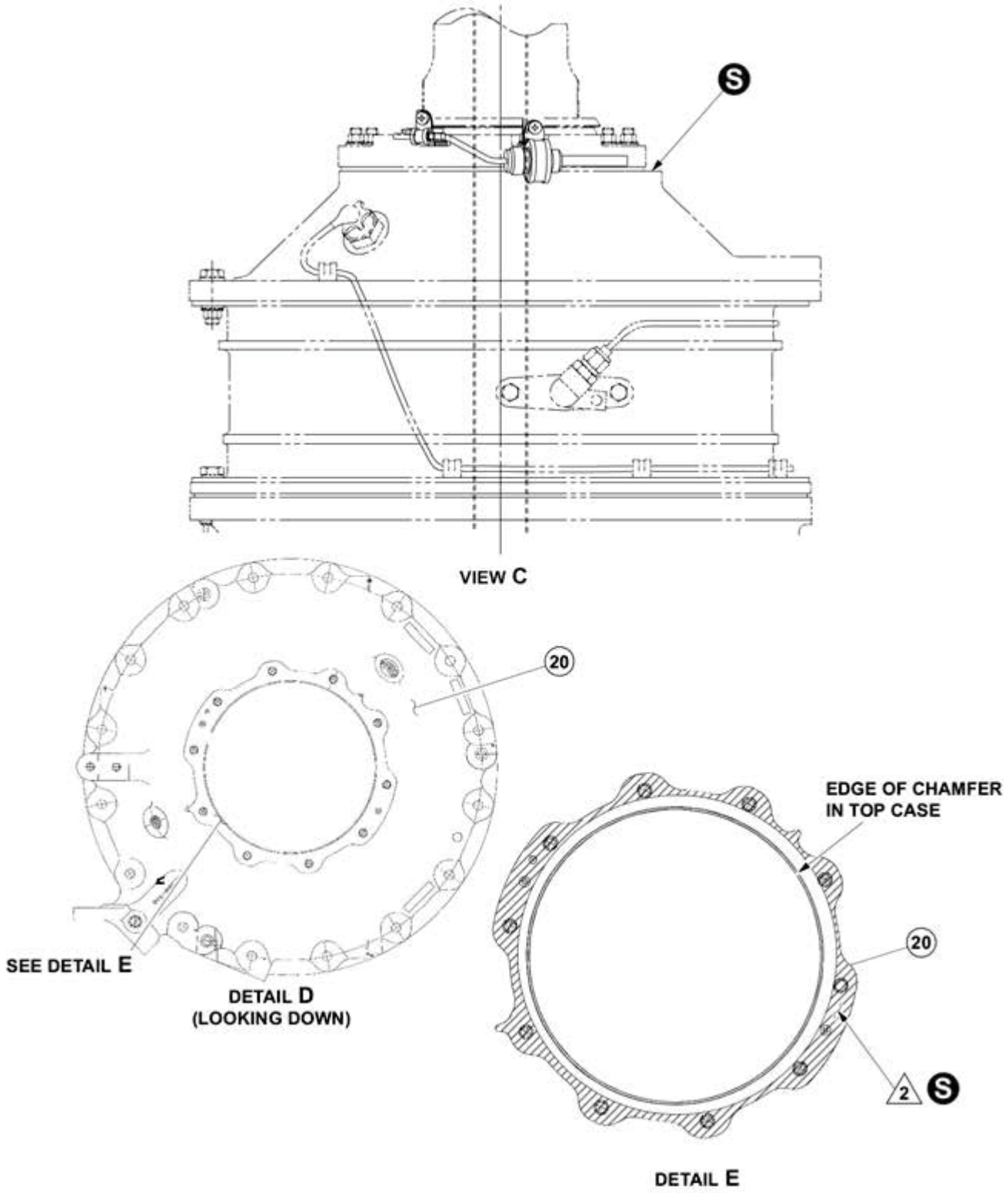
NOTES

1. S/N 36020 and subsequent and helicopters modified by 412-570-001-103 or post BHT-412-SI-74 (412SP to 412HP upgrade) or post BHT-412-SI-97 (412Classic to 412HP upgrade).

2. Apply a thin uniform coat of sealant (C-308) immediately before the mast installation on the indicated area.

ICN-412-A-632101-B-97499-00940-A-003-01

Figure 1. Main Rotor Mast (Sheet 2 of 3)



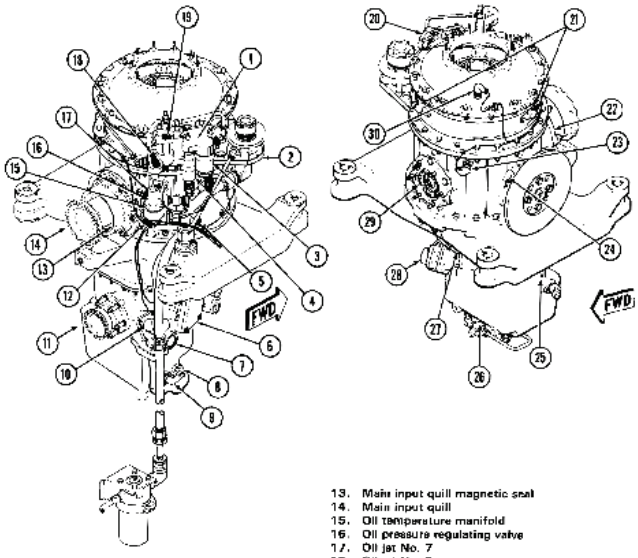
ICN-412-A-632101-B-97499-00002-A-001-01

Figure 1. Main Rotor Mast (Sheet 3 of 3)

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Transmission - Description

The transmission (Figure 1) is a mechanical reduction gearbox attached to the fuselage structure by four isolation mounts at corners of transmission support case and by lift link, which is the main load bearing component. The two aft isolation mounts attach to the helicopter structure and also to the friction dampers. The lift link is attached to the transmission support case and helicopter lift beam.



1. Oil pressure manifold
2. Filler neck, scupper, and mounting bracket
3. Oil pressure transmitter
4. Oil pressure switch
5. Oil jet No. 6
6. No. 1 hydraulic pump drive quill
7. Sight glasses
8. Oil pump inlet screen
9. Sump case electric chip detector
10. Sump internal filter
11. Tail rotor drive quill
12. Oil jet No. 3
13. Main input quill magnetic seal
14. Main input quill
15. Oil temperature manifold
16. Oil pressure regulating valve
17. Oil jet No. 7
18. Oil jet No. 2
19. Oil jet No. 1
20. Cyclic controls spring bracket
21. Planetary oil jets
22. Oil jet No. 5
23. Main input gear quill electric chip detector
24. Quill port cover
25. Oil jet No. 4
26. Oil pump
27. Lift link fitting
28. Rotor tachometer generator
29. No. 2 hydraulic pump drive quill
30. Mast bearing electric chip detector

ICN-412-A-632102-B-97499-00001-A-001-01

Figure 1. Transmission (Sheet 1 of 1)

Transmission - Troubleshooting

Procedure

1. In analysis of transmission oil system troubles, consider the following characteristics of the system:

1.1. Low oil level will not cause low oil pressure indications provided sump contains sufficient oil to cover pump inlet. Oil temperature may rise.

1.2. Effects of an oil leak will depend on location of leak within system and rate of fluid loss. Do not exceed 10 drops per minute from any single source. Do not exceed 15 drops per minute from all sources on the transmission. An external leak can allow sump to be pumped dry, causing internal failure of transmission. While any oil remains in sump, the pressure relief valve will tend to maintain normal system pressure. This applies especially to leaks between pump and relief valve. Leaks occurring beyond the relief valve could cause indications of low oil pressure. Leaks in the interior of transmission will not affect oil level but may starve downstream lubrication areas and may also affect indicated oil pressure and temperature.

1.3. Cumulative clogging of oil filters will not show as a gradual drop in oil pressure because the pressure relief valve maintains normal system pressure. The external filter has a visual indicator which will extend when filter bypass conditions are impending. This indicator should be checked during all normal inspections.

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Transmission - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Cover	AN100036	1	
Adapter	SWE13860	1	
Stand	SWE13875	1	
Hoist	T101581	1	
Dehydrator Assembly	T102118-101	1	
Mast Nut	204-010-481-001	1	
Cover and Lift Plate	204-040-929-101	1	
Clevis	204-011-178-001	1	

Procedure

1. If transmission is not to be reinstalled, accomplish transmission preservation by one of the following methods:

1.1. If helicopter and transmission are operational, perform a 10 minute ground run (BHT-412-FM) prior to removal of transmission.

1.2. If transmission and engine cannot be ground run, accomplish preservation after transmission is removed as follows:

1.2.1. Remove mast assembly (412-A-63-21-01-00A-520A-A / 01006).

1.2.2. Spray inside of transmission through top opening of transmission with 4 quarts of transmission oil (412-A-12-10-03-00A-040A-A / 00135). Rotate gears by hand while spraying oil.

1.2.3. Install cover and lift plate (204-040-929-101) or install mast assembly.

1.2.4. Cut and remove lockwire from plug and remove plug (AN814-6DL) from top case and install dehydrator assembly (T102118-101).

2. Remove the hub and blade assembly (412-A-62-20-00-02A-520A-A / 00948).

NOTE

The hub and sleeve assembly may be removed with the swashplate and support assembly as a unit. Care must be taken to prevent damage to the mast and hub splines.

3. Remove the hub and sleeve assembly (412-A-62-30-00-03A-520A-A / 00968).

4. Remove the swashplate and support assembly (412-A-62-30-00-01A-520A-A / 00955).

5. Remove the main driveshaft (412-A-63-10-01-01A-520A-A / 00994).



SN: 33001-33213 or 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)):

WHEN REMOVING THE TAIL ROTOR DRIVESHAFT (10, Figure 1)), BOTH CLAMP ASSEMBLIES (9 AND 11) MUST BE REMOVED PRIOR TO REMOVING TAIL ROTOR DRIVESHAFT. FAILURE TO COMPLY MAY RESULT IN DAMAGE TO FLEXIBLE COUPLINGS.

NOTE

Tail rotor driveshaft clamp assemblies (9 and 11) are a matched set; keep in sets to maintain balance. To replace one clamp half, or if clamp halves are intermixed, each clamp half shall be re-matched with one of equal weight (within 1 gram).

6. Remove the clamp assemblies (9 and 11) and remove the tail rotor driveshaft (10).

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

7. *Remove the tail rotor driveshaft (10) connected to the transmission (412-A-65-10-00-00A-520A-A / 01259).*

8. Drain oil from the transmission sump (412-A-12-10-03-00A-040A-A / 00135).

9. Disconnect the transmission oil inlet hose (7).

10. Disconnect the oil outlet hose (23).

11. Disconnect the oil sump drain line (22).

12. If installed, disconnect the rotor brake hydraulic hoses and wiring.

13. Disconnect the electrical connector (5).

14. Remove the hydraulic system 1 pump (412-A-29-10-00-01A-520A-A / 00620). Install the cover (AN100036) (6) on the pump drive quill.

15. Remove fluid from the hydraulic system 2 with a suction gun. Disconnect all hoses from the hydraulic system 2 pump. Do not remove the pump at this time. Cap all open ports and hoses.

16. Install the mast nut (204-010-481-001) (3) and clevis (204-011-178-001) (1) with the pin (18). Secure the pin with the quick release pin (2).



BEFORE USING hoist (T101581) TO REMOVE OR INSTALL TRANSMISSION, PLACE A SUPPORT UNDER TAILBOOM TO STEADY FUSELAGE.

NOTE

If the mast is not installed, the cover and lift plate (204-040-929-101) shall be installed on the transmission for lifting.

17. If a suitable stationary or portable hoist is not available, install the hoist (T101581).

18. Attach the hoist to the clevis (204-011-178-001) (1).

19. Remove the bolts (19), washers (20), and shouldered washers (21) from the isolation mounts (24).

20. Tie the forward isolation mounts (24) into position to prevent the mounts from disengaging from the support case.

21. Remove the bolt (26) from the lift link (32).

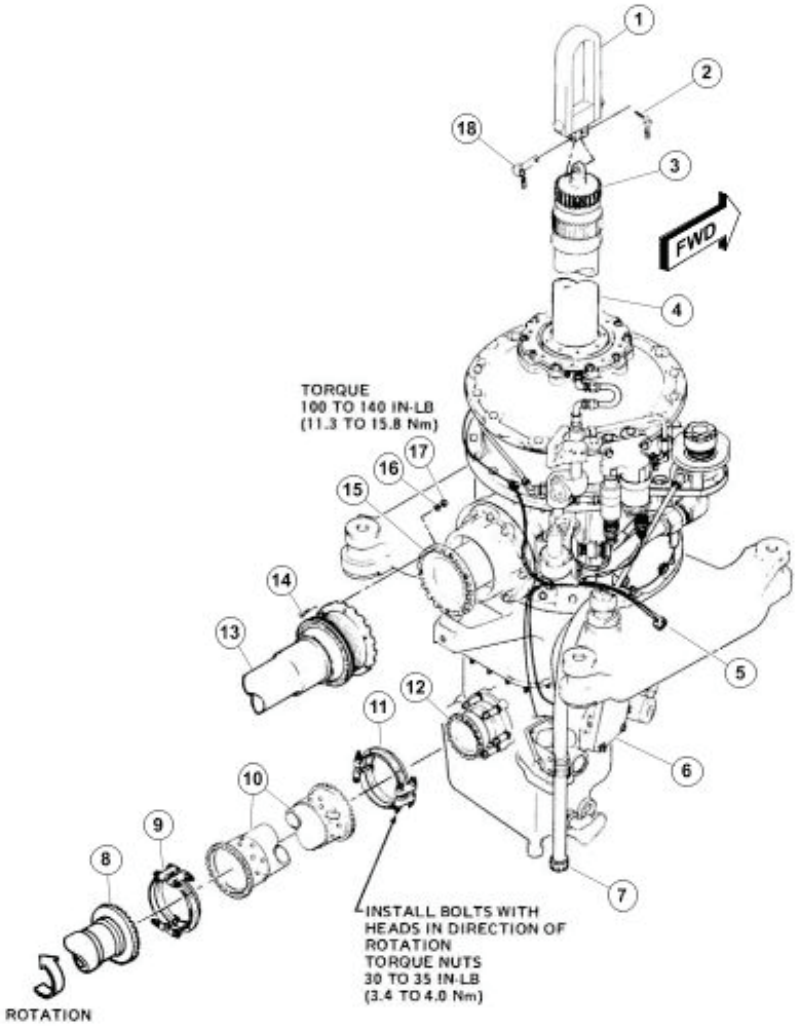
22. Install the adapter (SWE13860) on the stand (SWE13875) if not previously accomplished.



VERIFY PUMP FOR HYDRAULIC SYSTEM 1 AND ALL ELECTRICAL WIRING, HOSES, LINES, AND ATTACHING HARDWARE ARE REMOVED BEFORE ATTEMPTING TO LIFT TRANSMISSION FROM HELICOPTER.

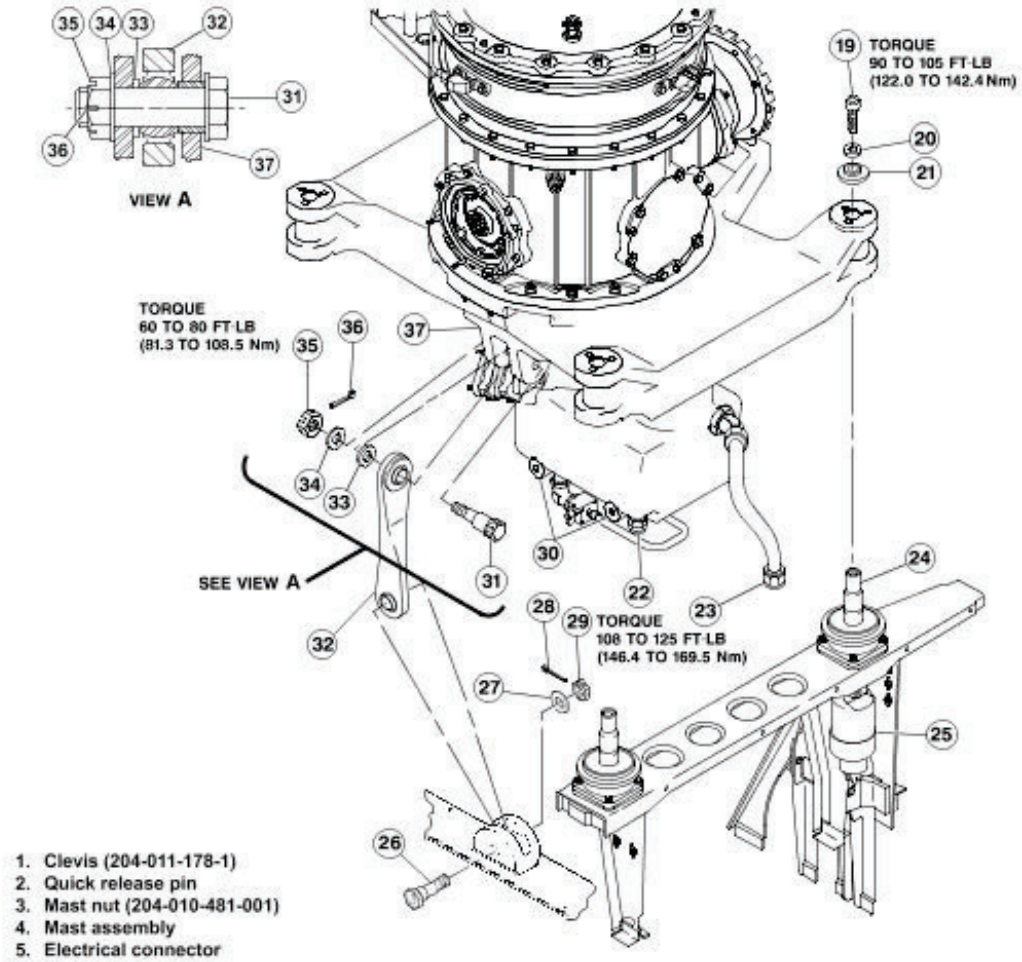
23. Hoist the transmission clear of the airframe. Place the transmission in the support stand and secure to the stand with bolts.

24. If the transmission is not to be reinstalled, remove the bolt assembly (31) and lift link (32).



ICN-412-A-632102-B-97499-00002-A-001-01

Figure 1. Transmission Removal and Installation (Sheet 1 of 2)



1. Clevis (204-011-178-1)
2. Quick release pin
3. Mast nut (204-010-481-001)
4. Mast assembly
5. Electrical connector
6. Cover (AN100036)
7. Transmission oil inlet hose
8. Tail rotor hanger coupling
9. Clamp assembly
10. Tail rotor driveshaft
11. Clamp assembly
12. Tail rotor drive quill
13. Engine-to-transmission driveshaft
14. Bolt
15. Main input quill
16. Thin steel washer
17. Nut
18. Pin
19. Bolt
20. Washer (MS20002C10)
21. Washer (204-030-913-5)
22. Oil sump drain line
23. Oil outlet hose
24. Isolation mount
25. Damper
26. Bolt
27. Thin steel washer
28. Cotter pin
29. Nut
30. Set screw
31. Bolt assembly
32. Lift link
33. Washer (212-040-125-101)
34. Steel washer
35. Nut
36. Cotter pin
37. Transmission lift link fitting

ICN-412-A-632102-B-97499-00003-A-001-01

Figure 1. Transmission Removal and Installation (Sheet 2 of 2)

Transmission - Inspection

Procedure

1. Inspect the three electric chip detectors as follows:

1.1. Remove, inspect, and install the mast bearing electric chip detector (3, Figure 1) (412-A-63-22-00-01A-042A-A / 01018).

1.2. Remove, inspect, and install the main input gear quill electric chip detector (4) (412-A-63-22-00-02A-042A-A / 01023).

1.3. Remove, inspect, and install the sump case electric chip detector (8) (412-A-63-22-00-03A-042A-A / 01028).

2. Inspect the main driveshaft couplings and tail rotor drive quill couplings over-temperature indicator (TEMP-PLATE) for discoloration. A change in color indicates a possible overheat condition and/or component degradation. Determine and correct the cause of discoloration prior to continued operation.

3. Remove, inspect, and install the external oil filter (9) (412-A-63-22-02-01A-042A-A / 01118).

Applicable: TB 412-91-94: Post:

4. Remove, inspect, and install the sump internal oil filter (10) (412-A-63-22-01-17A-042A-A / 01102). Remove, inspect, and install the full flow debris monitor (412-A-63-22-01-18A-042A-A / 01107).

5. Remove, inspect, and install the oil pump inlet screen (7) (412-A-63-22-01-16A-520A-A / 01098, 412-A-63-22-01-16A-280A-A / 01099, 412-A-63-22-01-16A-251A-A / 01100 and 412-A-63-22-01-16A-720A-A / 01101).

6. Inspect the oil level sight glass (6) for crazing, cracks, or staining (412-A-63-22-01-19A-042A-A / 01112).

7. Inspect the transmission for loose, missing, or damaged bolts and studs.

NOTE

Pay special attention to the attaching bolts (26 and 31, Figure 2) for the lift link (32) and the four bolts (19) that secure the transmission support case to the structure.

8. Inspect the isolation mounts (24) and dampers (25) for secure installation and distortion.

9. Inspect the transmission and attached oil lines for leaks. If the main input quill is leaking, replace the magnetic seal (412-A-63-21-02-02A-520A-A / 01140, 412-A-63-21-02-02A-520B-A / 01141, 412-A-63-21-02-02A-251A-A / 01142, 412-A-63-21-02-02A-280A-A / 01143, 412-A-63-21-02-02A-685A-A / 01144, 412-A-63-21-02-02A-720A-A / 01145 and 412-A-63-21-02-02A-720B-A / 01146).

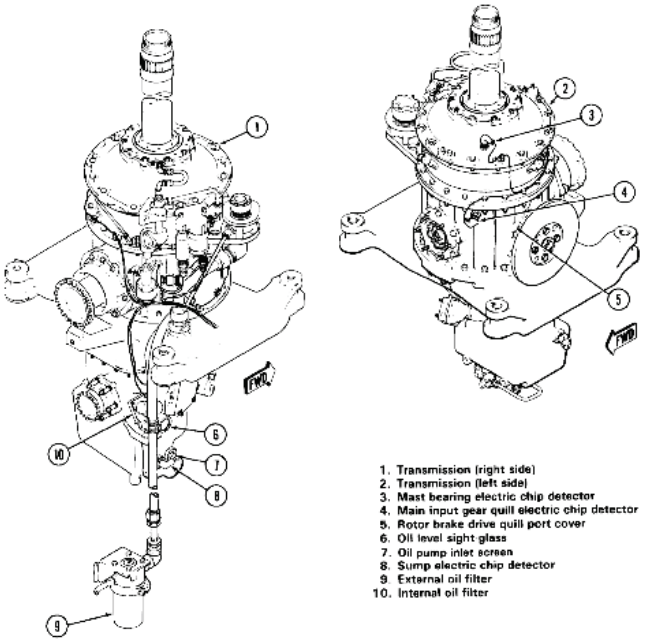
10. Inspect the transmission for mechanical and corrosion damage (BHT-412-CR&O).

11. Inspect the transmission electrical wiring for secure installation and damaged wires.

12. Inspect the main rotor mast (412-A-63-21-01-00A-280A-A / 01008).

NOTE

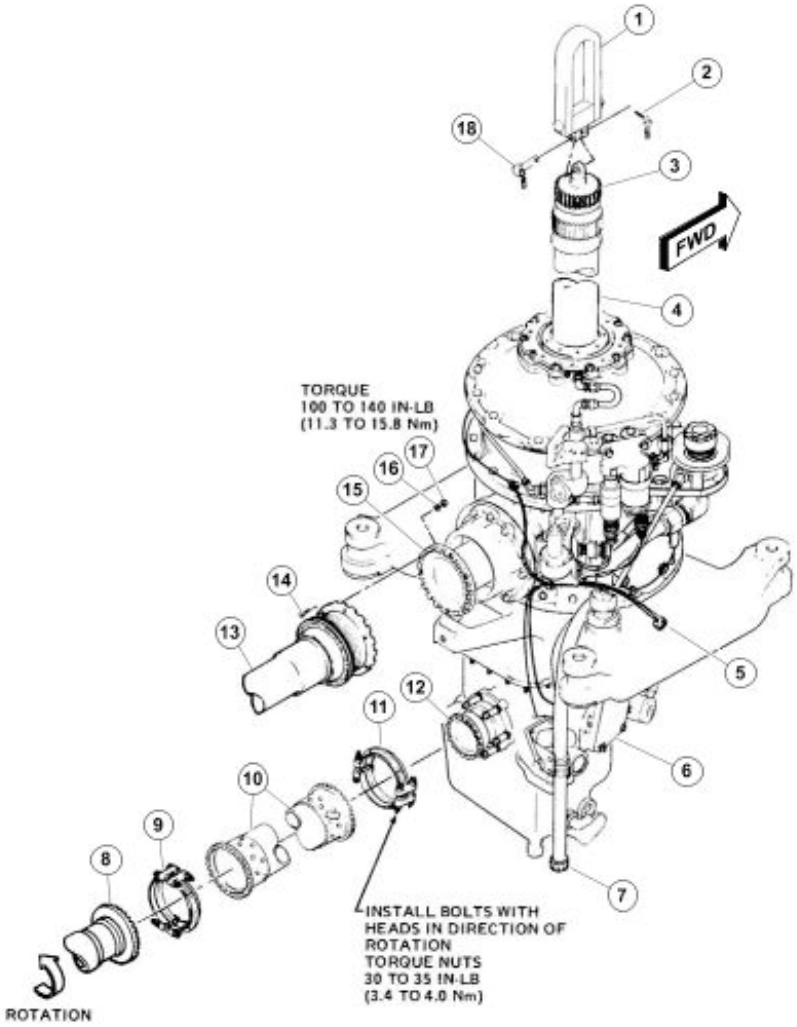
Additional detailed inspections of the transmission and component parts are found in the BHT-412-CR&O.



1. Transmission (right side)
2. Transmission (left side)
3. Mast bearing electric chip detector
4. Main input gear quill electric chip detector
5. Rotor brake drive quill port cover
6. Oil level sight glass
7. Oil pump inlet screen
8. Sump electric chip detector
9. External oil filter
10. Internal oil filter

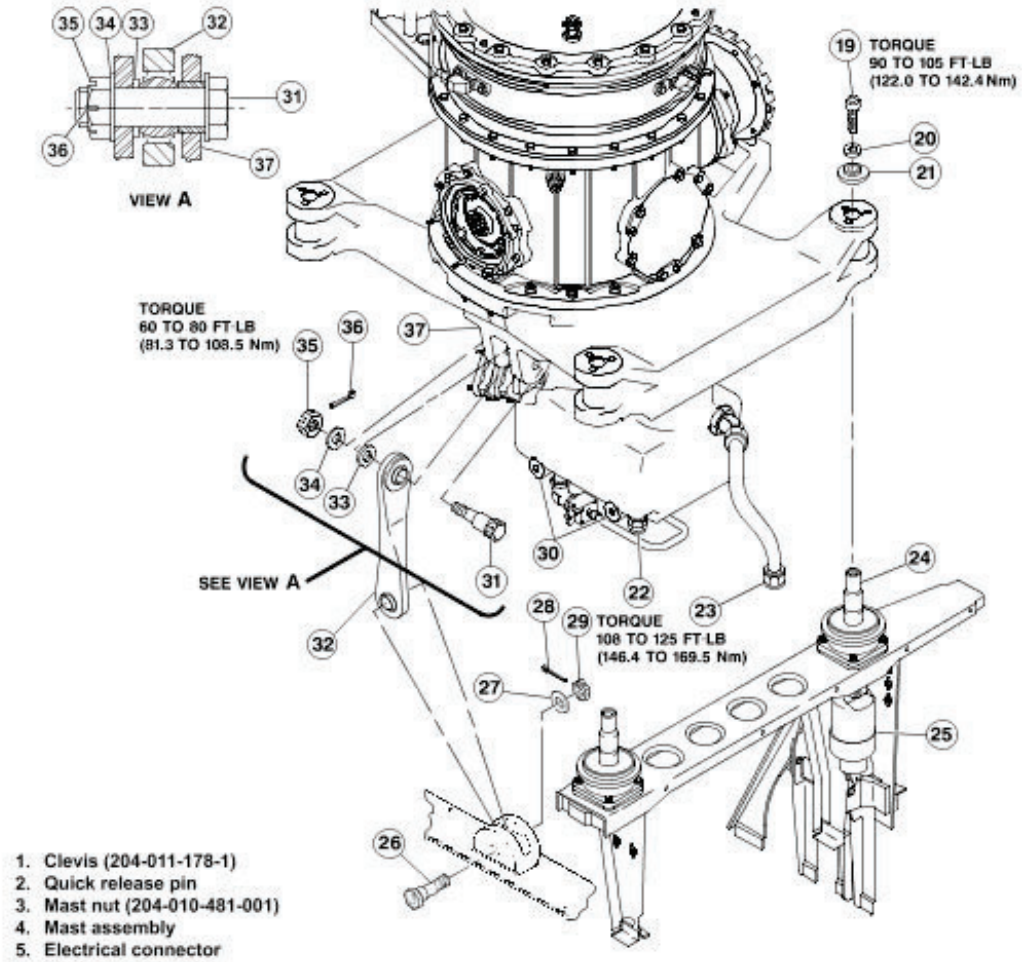
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Figure 1. Transmission Assembly (Sheet 1 of 1)



ICN-412-A-632102-B-97499-00002-A-001-01

Figure 2. Transmission Removal and Installation (Sheet 1 of 2)



1. Clevis (204-011-178-1)
2. Quick release pin
3. Mast nut (204-010-481-001)
4. Mast assembly
5. Electrical connector
6. Cover (AN100036)
7. Transmission oil inlet hose
8. Tail rotor hanger coupling
9. Clamp assembly
10. Tail rotor driveshaft
11. Clamp assembly
12. Tail rotor drive quill
13. Engine-to-transmission driveshaft
14. Bolt
15. Main input quill
16. Thin steel washer
17. Nut
18. Pin
19. Bolt
20. Washer (MS20002C10)
21. Washer (204-030-913-5)
22. Oil sump drain line
23. Oil outlet hose
24. Isolation mount
25. Damper
26. Bolt
27. Thin steel washer
28. Cotter pin
29. Nut
30. Set screw
31. Bolt assembly
32. Lift link
33. Washer (212-040-125-101)
34. Steel washer
35. Nut
36. Cotter pin
37. Transmission lift link fitting

ICN-412-A-632102-B-97499-00003-A-001-01

Figure 2. Transmission Removal and Installation (Sheet 2 of 2)

Transmission - Build-up

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Adapter	SWE13860	1	
Stand	SWE13875	1	
Hoist	T101581	1	
Cover and Lift Plate	204-040-929-101	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Install the adapter (SWE13860) on the stand (SWE13875).

2. Install the cover and lift plate (204-040-929-101) (3, Figure 1) on the transmission with the steel washers (2) and nuts (1).

3. Remove the transmission from the shipping container and position on the stand (SWE13875). Secure the transmission to the stand with bolts.

4. If the transmission is to be installed in the helicopter immediately, open the drain valve (9) and allow all preservative oil to drain. Close the valve.



IF ANY OIL LINES FROM A TRANSMISSION THAT WAS REMOVED DUE TO INTERNAL FAILURE OR METAL CONTAMINATION ARE TO BE INSTALLED ON REPLACEMENT TRANSMISSION, MAKE SURE ALL FOREIGN PARTICLES ARE CLEANED FROM HOSES. MAST ASSEMBLIES FROM TRANSMISSIONS PREVIOUSLY DESCRIBED SHALL BE OVERHAULED PRIOR TO FURTHER USE

NOTE

If components from the transmission removed from the helicopter are to be used for build-up of the replacement transmission, position the transmissions near each other to facilitate transfer of components.

5. Install the mast in the transmission (412-A-63-21-01-00A-720A-A / 01009).

6. Remove the cover (15) and install the hydraulic system 2 pump (412-A-29-10-00-02A-720A-A / 00628). Cap or plug open ports on the hydraulic line.

NOTE

The hydraulic system 1 pump will be installed after the transmission is installed in the helicopter. Leave the hydraulic system 1 pump drive quill cover (27) installed at this time.

7. Remove the cover (11) and install the rotor tachometer generator (20) (412-A-63-21-02-06A-720A-A / 01169).

8. Remove the plug (25) and packing (24). Install the oil pressure transmitter (44) and packing (43) (412-A-63-22-01-11A-720A-A / 01083).

9. Remove the cap (23). Install the oil pressure switch (40) and packing (42) (412-A-63-22-01-12A-720A-A / 01086).

10. Remove the set screws (8).

11. Remove the plug (29). Install the sump electric chip detector (28) (412-A-63-22-00-03A-720A-A / 01032).

12. Remove the cover (31). Install the full flow debris monitor (31) (412-A-63-22-01-18A-720A-A / 01111)

NOTE

Refer to 412-A-98-00-00-39A-051A-A / 02499 for electrical wiring diagrams to confirm wire numbers.

13. Install the electrical harness (36) as follows:

13.1. Connect the electrical connector (46) to the oil pressure switch (40).

13.2. Connect the electrical connector (45) to the oil pressure transmitter (44).

13.3. Connect the ground wire to the ground connection (37).

13.4. Install six upper clamps (34) (remaining four are installed in Step 13.8.).

13.5. Remove the cover (22). Install the electrical terminal and nipple (39) on the thermoswitch (26).

13.6. Connect the electrical terminal and nipple (41) to the main input quill electric chip detector (7) (412-A-63-22-00-02A-720A-A / 01027).

13.7. Connect the electrical terminal and nipple (38) to the mast bearing electric chip detector (4) (412-A-63-22-00-01A-720A-A / 01022).

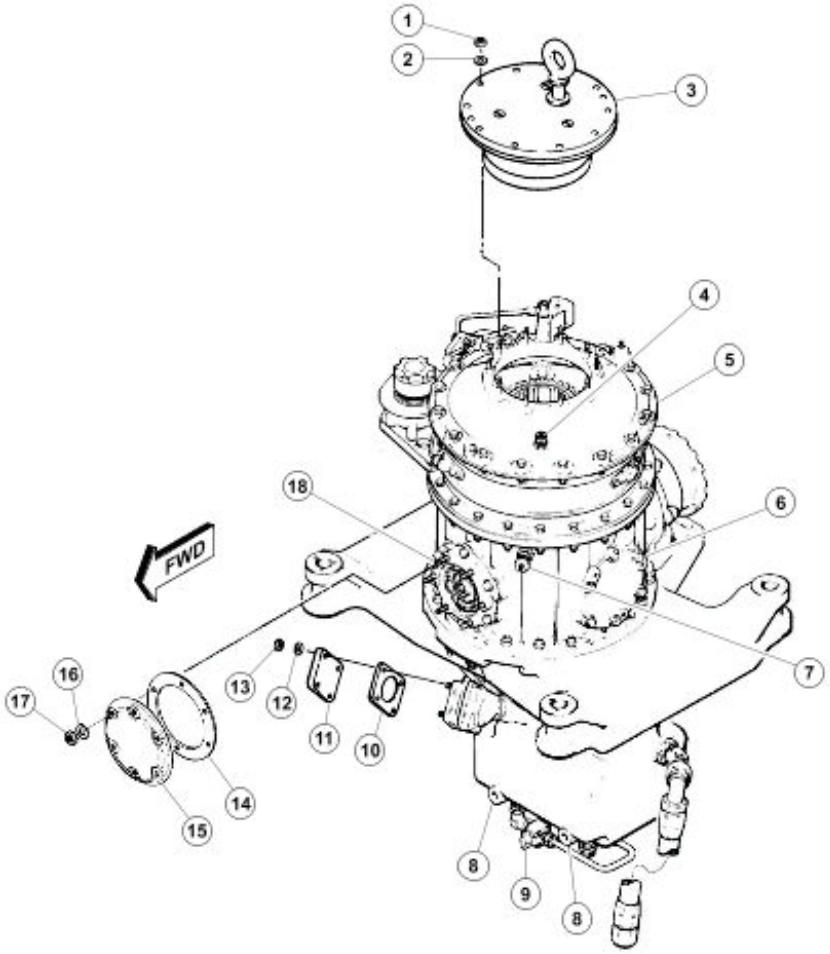
13.8. Install four lower clamps (34).

13.9. *Install electrical terminal and nipple (29) on sump electric chip detector (27) and connect electrical harness to full flow debris monitor for (412Classic/SP to 412HP Upgrade).*

13.10. Install electrical connector (28) on rotor tachometer generator (19) and secure with lockwire (C-405).

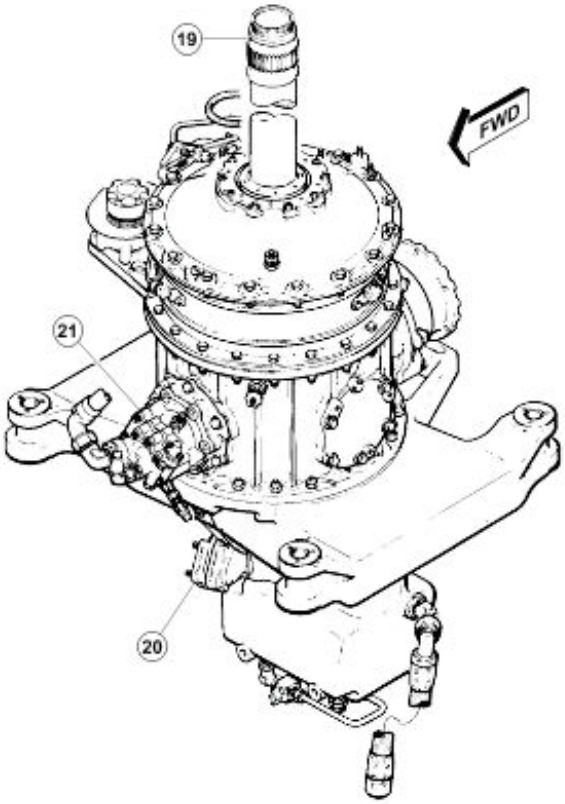
13.11. If transmission is to be installed in a helicopter equipped with a rotor brake, remove rotor brake quill port cover (6) and install rotor brake quill (412-A-63-21-02-08A-720A-A / 01181).

13.12. Install caps, covers, and plugs removed in preceding steps.



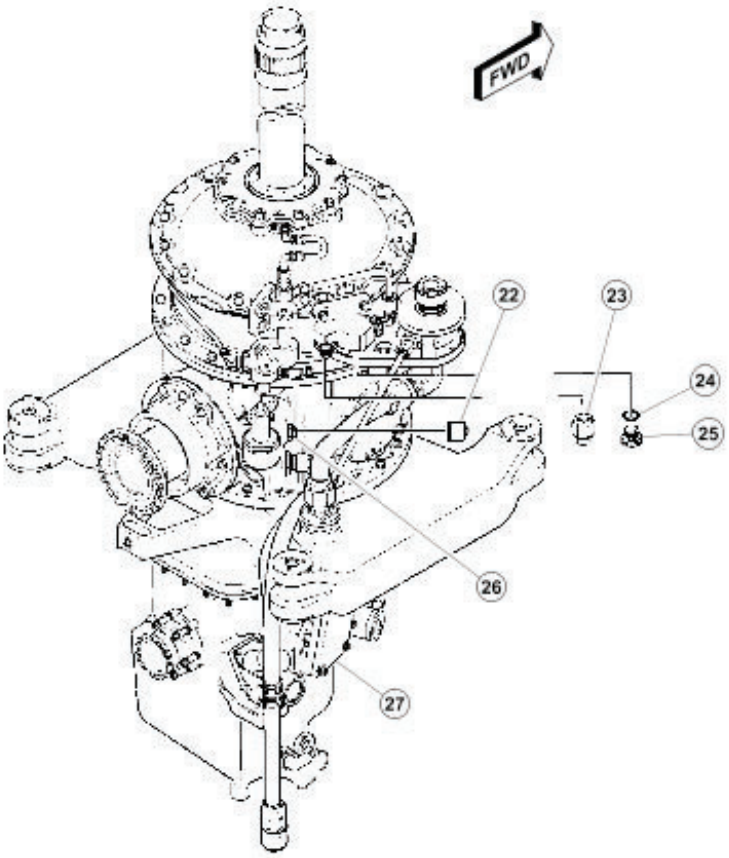
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Figure 1. Transmission Build-up (Sheet 1 of 5)



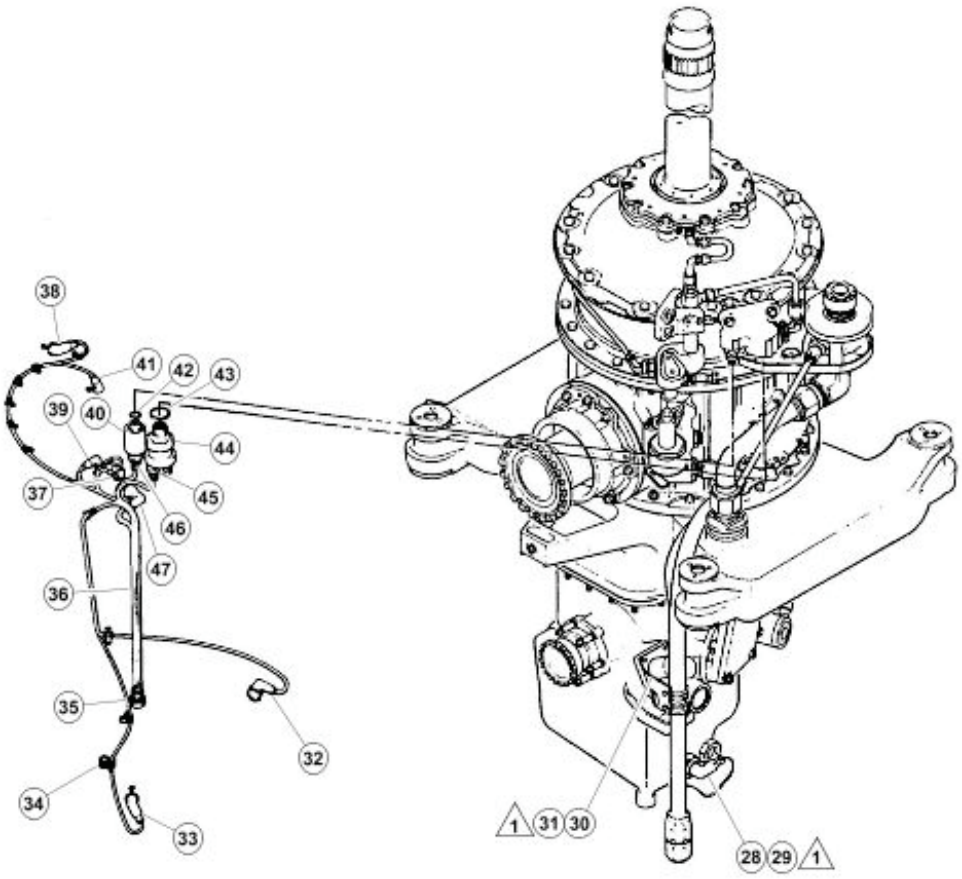
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Figure 1. Transmission Build-up (Sheet 2 of 5)



ICN-412-A-632102-B-97499-00007-A-001-01

Figure 1. Transmission Build-up (Sheet 3 of 5)



ICN-412-A-632102-B-97499-00008-A-001-01

Figure 1. Transmission Build-up (Sheet 4 of 5)

1. Nut
2. Steel washer
3. Cover and lift plate
4. Mast bearing electric chip detector
5. Transmission
6. Rotor brake quill port cover
7. Main input quill electric chip detector
8. Set screw
9. Drain valve
10. Gasket
11. Cover
12. Washer (AN960C416)
13. Nut
14. Gasket
15. Cover
16. Steel washer
17. Nut
18. Hydraulic pump (system II) drive quill
19. Main rotor mast assembly
20. Rotor tachometer generator
21. Hydraulic pump (system II)
22. Cover (thermoswitch)
23. Cap
24. Packing
25. Plug
26. Thermoswitch
27. Hydraulic pump (system I) drive quill cover
28. Sump electric chip detector
29. Plug
30. Internal oil filter
31. Cover assembly
32. Electrical connector
33. Electrical terminal and nipple
34. Clamps (10 Reqd)
35. Electrical connector
36. Electrical harness
37. Ground connection
38. Electrical terminal and nipple
39. Electrical terminal and nipple
40. Oil pressure switch
41. Electrical terminal and nipple
42. Packing
43. Packing
44. Oil pressure transmitter
45. Electrical connector
46. Electrical connector
47. Electrical connector

NOTE

1 For helicopters S/N 36694 and subsequent

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Figure 1. Transmission Build-up (Sheet 5 of 5)

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Transmission - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Hoist Assembly	T101581	1	
Mast Nut	204-010-481-001	1	
Clevis	204-011-178-001	1	
Lift Plate Cover	204-040-929-101	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure



IF REPLACING TRANSMISSION FOR INTERNAL FAILURE OR METAL PARTICLES IN OIL, CLEAN ALL OIL LINES AND REPLACE OIL COOLER, EXTERNAL OIL FILTER, AND MAST ASSEMBLY.

1. Inspect condition of pylon structure and repair or replace any unserviceable isolation mounts (24, Figure 1) and/or dampers (25) (412-A-63-21-04-00A-028A-A / 01185).

Applicable: TB 412-98-150: Post:

2. Install mast assembly (412-A-63-21-01-00A-720A-A / 01009) and replace dehydrator assembly (T102118-101) in top case with plug (AN814-6DL). Replace dehydrator assembly with breather (A800V). Torque plug or breather to 80 to 120 inch-pounds (9.04 to 13.56 Nm). Secure plug with lockwire (C-405).

3. If not installed before, position the lift link (32) and the washer (33) between the lugs of the transmission lift-link fitting (37). Install the bolt assembly (31), steel washer (34) and nut (35). Torque the nut (35) to 60 to 80 ft-lb (81.3 to 108.5 Nm). Install the cotter pin (36).

4. Erect hoist assembly (T101581) (or equivalent). Install clevis (204-011-178-001) (1) on mast nut (204-010-481-001) (3). Attach hoist to clevis.

NOTE

If mast is not installed, lift plate cover (204-040-929-101) must be installed on transmission for lifting.

5. Lift transmission from stand and lower into position on the isolation mounts (24) while you guide the lift link (32) into position on the lift beam.

6. Align the lift link (32) with the lift beam and install the bolt (26), thin steel washer (27) and nut (29). Torque the nut (29) to 108 to 125 ft-lb (146.4 to 169.5 Nm). Install the cotter pin (28).

NOTE

Ensure countersunk side of washer (20) is toward head of bolt (19). Washer (21) is installed with beveled side down.

7. Install bolts (19), washers (20), and washers (21) on isolation mounts (24). Torque bolts (19) to 90 to 105 foot-pounds (122 to 142 Nm). Remove ties holding forward isolation mounts in position.

Applicable: SN: 33001-33213 or 36001-36019 or (412-570-001-103: Not installed BHT-412-SI-74: Not installed BHT-412-SI-97: Not installed):

8. *Install tail rotor driveshaft (10) as follows:*

NOTE

Clamp assemblies (9 and 11) are a matched set. To replace one clamp half, or if clamp halves are inadvertently mixed, each clamp half shall be rematched with one of equal weight (within 1 gram) to maintain balance.

8.1. *Position tail rotor driveshaft (10) between tail rotor drive quill (12) and tail rotor hanger coupling (8).*

NOTE

All nuts used on a clamp shall be the same type and style. Nuts are self-locking and shall not be used unless nut drag torque is 3.5 inch-pounds (0.40 Nm) minimum and 30 inch-pounds (3.39 Nm) maximum.

8.2. *Install clamp assembly (9) with bolt heads in direction of rotation. Tighten clamp assembly bolts evenly. Torque nuts to 30 to 35 inch-pounds (3.39 to 3.95 Nm) above nut friction. Tap lightly around outer surface of clamp assembly with plastic mallet to seat clamp. Repeat torquing procedure.*

8.3. *Install clamp assembly (11) with bolt heads in direction of rotation and with clamp joints 90° to clamp assembly (9). Torque clamp assembly bolts in same manner as described in* Step 8.2..

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

9. *Install forward tail rotor driveshaft (*412-A-65-10-00-00A-720A-A / 01263*).*

10. Install engine-to-transmission driveshaft (13) (412-A-63-10-01-01A-720A-A / 00999).

11. Install hydraulic system 1 pump (412-A-29-10-00-01A-720A-A / 00623).

12. Connect electrical connector (5) on right side of transmission.

13. Connect transmission oil inlet hose (7) to oil filter.

14. Connect transmission oil outlet hose (23) to manifold line.

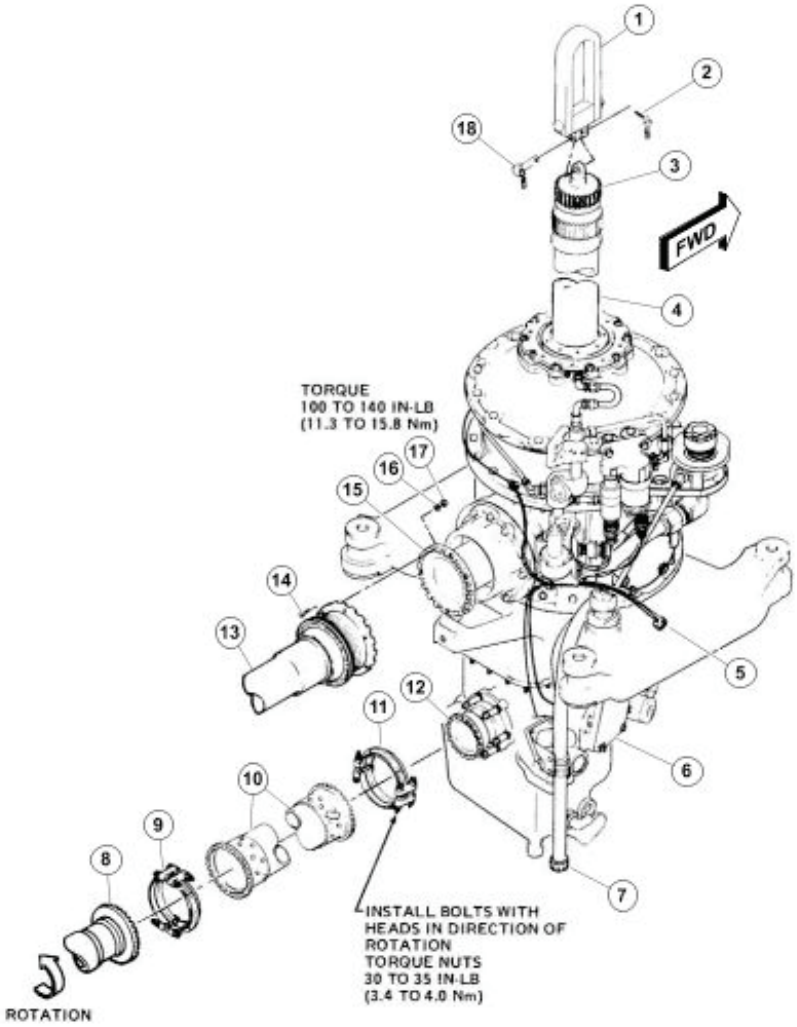
15. Connect oil sump drain line (22).

16. If equipped, connect rotor brake hydraulic hose and electrical wiring.

17. Connect hydraulic system 2 inlet and pressure hoses, case drain hose, and seal drain hose to pump fittings.

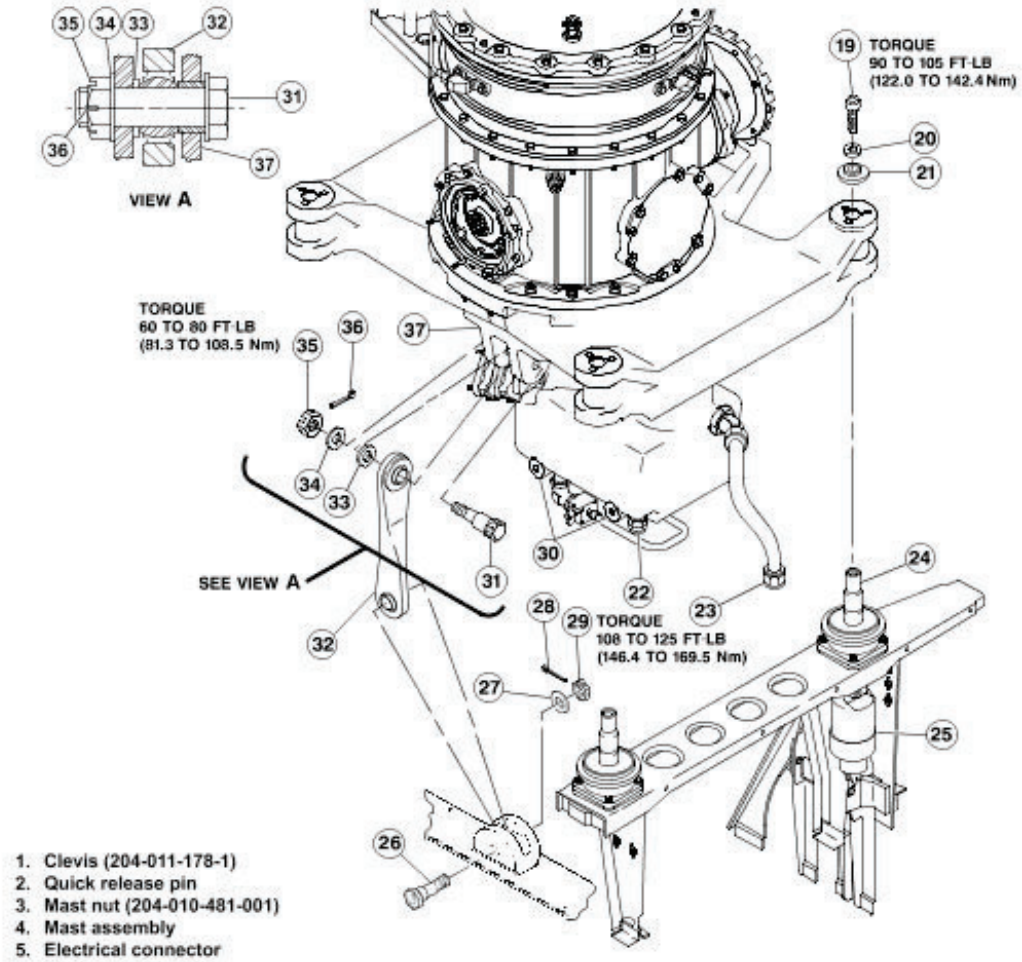
18. Service transmission (412-A-12-10-03-00A-040A-A / 00135). Verify actual presence of oil in sight gauge.

19. If transmission has not been previously qualified on run-in test stand, accomplish alternate run-in procedures in 412-A-63-21-02-00A-322A-A / 01017.



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Figure 1. Transmission Removal and Installation (Sheet 1 of 2)



1. Clevis (204-011-178-1)
2. Quick release pin
3. Mast nut (204-010-481-001)
4. Mast assembly
5. Electrical connector
6. Cover (AN100036)
7. Transmission oil inlet hose
8. Tail rotor hanger coupling
9. Clamp assembly
10. Tail rotor driveshaft
11. Clamp assembly
12. Tail rotor drive quill
13. Engine-to-transmission driveshaft
14. Bolt
15. Main input quill
16. Thin steel washer
17. Nut
18. Pin
19. Bolt
20. Washer (MS20002C10)
21. Washer (204-030-913-5)
22. Oil sump drain line
23. Oil outlet hose
24. Isolation mount
25. Damper
26. Bolt
27. Thin steel washer
28. Cotter pin
29. Nut
30. Set screw
31. Bolt assembly
32. Lift link
33. Washer (212-040-125-101)
34. Steel washer
35. Nut
36. Cotter pin
37. Transmission lift link fitting

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Figure 1. Transmission Removal and Installation (Sheet 2 of 2)

Transmission - Alternate Run-in Procedures

Procedure

NOTE

After overhaul, it is desirable that transmission be test run and inspected in accordance with publication furnished with run-in test stand. In the event a test stand is not available, the following functional test may be accomplished as the minimum acceptable alternate test.

1. Install sufficient ballast in cabin to obtain maximum gross weight (BHT-412-FM).

2. Verify helicopter is properly serviced and lubricated (412-A-12-00-00-00A-040A-A / 00126). Visually inspect the following:

2.1. Oil system for security and leakage.

2.2. Fuel system for security and leakage.

2.3. Hydraulic system for security and leakage.

2.4. Flight controls for security.

3. Inspect engine accessories and controls for security, loose connections, and proper operation.

NOTE

Throughout the entire procedure which follows, carefully check for evidence of leakage and abnormal noises.

4. Start engines (BHT-412-FM) and run at 60% RPM for 5 minutes. Verify temperatures and pressures are within limits.

5. Gradually increase RPM from 60 to 88% at the rate of 7% each 6 minutes of operation.

6. Hold 88% for 2 minutes with collective applied to the maximum which can be attained without becoming airborne or exceeding 100% torque.

7. Shut down engines. Remove, inspect, clean, and install the following components:

7.1. Three electric chip detectors (412-A-63-22-00-01A-042A-A / 01018, 412-A-63-22-00-01A-520A-A / 01019, 412-A-63-22-00-01A-280A-A / 01020, 412-A-63-22-00-01A-251A-A / 01021, 412-A-63-22-00-01A-720A-A / 01022, 412-A-63-22-00-02A-042A-A / 01023, 412-A-63-22-00-02A-520A-A / 01024, 412-A-63-22-00-02A-280A-A / 01025, 412-A-63-22-00-02A-251A-A / 01026, 412-A-63-22-00-02A-720A-A / 01027, 412-A-63-22-00-03A-042A-A / 01028, 412-A-63-22-00-03A-520A-A / 01029, 412-A-63-22-00-03A-280A-A / 01030, 412-A-63-22-00-03A-251A-A / 01031 and 412-A-63-22-00-03A-720A-A / 01032).

7.2. Oil pump inlet screen (412-A-63-22-01-16A-520A-A / 01098, 412-A-63-22-01-16A-280A-A / 01099, 412-A-63-22-01-16A-251A-A / 01100 and 412-A-63-22-01-16A-720A-A / 01101).

7.3. Sump case internal filter (412-A-63-22-01-17A-042A-A / 01102).

7.4. External oil filter (412-A-63-22-02-01A-042A-A / 01118).

7.5. Planetary oil jets (two) (412-A-63-22-01-01A-042A-A / 01033).

7.6. Oil jet No. 1 (412-A-63-22-01-02A-520A-A / 01039, 412-A-63-22-01-02A-280A-A / 01040, 412-A-63-22-01-02A-251A-A / 01041, 412-A-63-22-01-02A-685A-A / 01042 and 412-A-63-22-01-02A-720A-A / 01043)

7.7. Oil jet No. 2 (412-A-63-22-01-03A-520A-A / 01044, 412-A-63-22-01-03A-280A-A / 01045, 412-A-63-22-01-03A-251A-A / 01046, 412-A-63-22-01-03A-685A-A / 01047 and 412-A-63-22-01-03A-720A-A / 01048).

7.8. Oil jet No. 3 (412-A-63-22-01-04A-520A-A / 01049, 412-A-63-22-01-04A-280A-A / 01050, 412-A-63-22-01-04A-251A-A / 01051, 412-A-63-22-01-04A-685A-A / 01052 and 412-A-63-22-01-04A-720A-A / 01053).

7.9. Oil jet No. 4 (412-A-63-22-01-05A-520A-A / 01054, 412-A-63-22-01-05A-280A-A / 01055, 412-A-63-22-01-05A-251A-A / 01056, 412-A-63-22-01-05A-685A-A / 01057 and 412-A-63-22-01-05A-720A-A / 01058).

7.10. Oil jet No. 5 (412-A-63-22-01-06A-520A-A / 01059, 412-A-63-22-01-06A-280A-A / 01060, 412-A-63-22-01-06A-251A-A / 01061, 412-A-63-22-01-06A-685A-A / 01062 and 412-A-63-22-01-06A-720A-A / 01063).

7.11. Oil jet No. 6 (412-A-63-22-01-07A-520A-A / 01064, 412-A-63-22-01-07A-280A-A / 01065, 412-A-63-22-01-07A-251A-A / 01066, 412-A-63-22-01-07A-685A-A / 01067 and 412-A-63-22-01-07A-720A-A / 01068).

7.12. Oil jet No. 7 (412-A-63-22-01-08A-520A-A / 01069, 412-A-63-22-01-08A-280A-A / 01070, 412-A-63-22-01-08A-251A-A / 01071, 412-A-63-22-01-08A-685A-A / 01072 and 412-A-63-22-01-08A-720A-A / 01073).

8. Restart engines (BHT-412-FM) and gradually increase to 88% RPM. Operate at 88% for 6 minutes. Move collective to the maximum position which can be attained without becoming airborne. At the rate of 3% each 6 minutes, gradually increase RPM to 100% without exceeding 100% torque.

9. *At 100% RPM, check transmission oil pressure. The pressure should be between 52 and 58 PSI when oil temperature is 90° to 100°C. If pressure is not within these limits, adjust pressure relief valve as required (412-A-63-22-01-09A-271A-A / 01078.1).*

9.1. *DELETED*

9.2. *DELETED*

9.3. *DELETED*

9.4. *DELETED*

9.5. *DELETED*

9.6. *DELETED*

10. Operate at 100% RPM for 30 minutes with collective applied to maximum attainable without becoming airborne or exceeding 100% torque.

11. Shut down helicopter and repeat inspections listed in Step 7..

12. *Inspect main driveshaft couplings, tail rotor drive quill coupling, and surrounding area for evidence of grease leakage. If there is evidence of leakage, replace or repair defective components. Repeat leak test run in* Step 10..

13. Inspect main and tail rotor driveshaft couplings over-temperature indicator (TEMP-PLATE) for possible overheat condition. A change in color (from white to black) indicates possible overheating and/or component degradation. If overheating is indicated, cause shall be determined and corrected prior to helicopter being released for flight. Refer to the BHT-412-CR&O for inspection and lubrication procedures for main driveshaft couplings and for tail rotor driveshaft couplings.

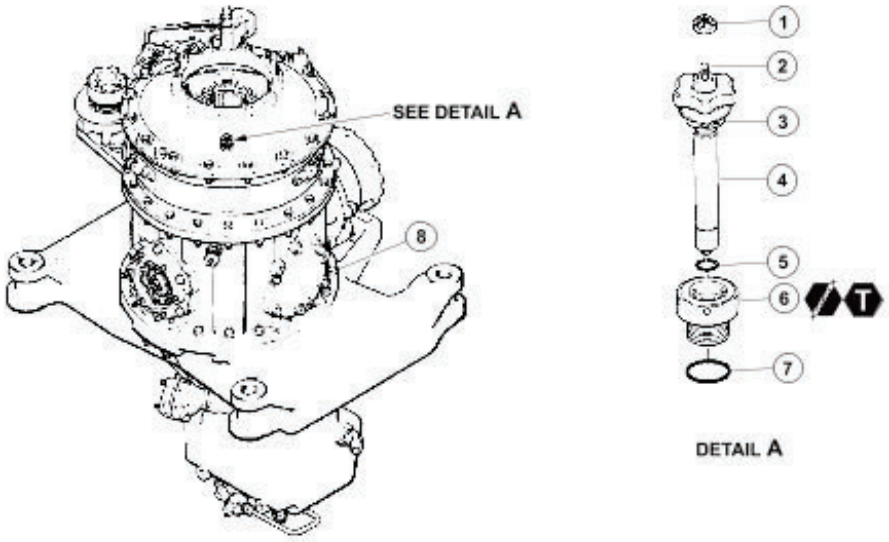
14. Check transmission oil level. Service as required (412-A-12-10-03-00A-040A-A / 00135).

15. Install engine and transmission cowling.

16. Check for leaks at next helicopter ground run.

Mast Bearing Electric Chip Detector - Description

The mast bearing electric chip detector is installed in the transmission top case (Figure 1) to detect foreign particles in the transmission oil originating from the main rotor mast upper bearing.



1. Nut
2. Stud
3. Locking lugs
4. Chip detector
5. Packing
6. Chip detector housing
7. Packing
8. Transmission

LOCKWIRE (C-405)

30 TO 108 IN-LBS
(3.4 TO 12.2 Nm)

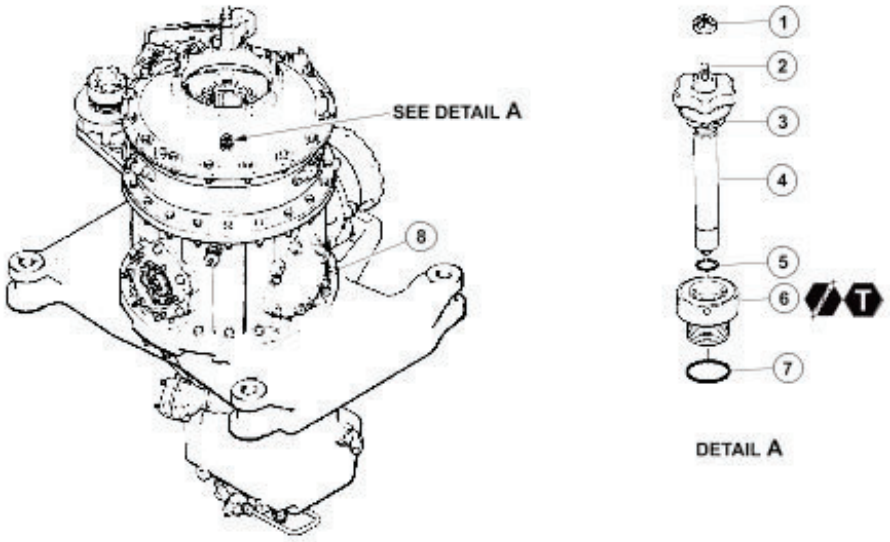
ICN-412-A-632200-B-97499-00001-A-001-01

Figure 1. Mast Bearing Electric Chip Detector (Sheet 1 of 1)

Mast Bearing Electric Chip Detector - Removal

Procedure

1. Remove nut (1, Figure 1) and disconnect wire from stud (2).
2. Push and twist chip detector (4) counterclockwise to disengage locking lugs (3). Remove chip detector from chip detector housing (6).
3. Cut and remove lockwire from chip detector housing (6), and remove chip detector housing from transmission.
4. Remove and discard packing (5) from chip detector (4) and, if removed, discard packing (7) from chip detector housing (6).



1. Nut
2. Stud
3. Locking lugs
4. Chip detector
5. Packing
6. Chip detector housing
7. Packing
8. Transmission

LOCKWIRE (C-405)

30 TO 108 IN-LBS
(3.4 TO 12.2 Nm)

ICN-412-A-632200-B-97499-00001-A-001-01

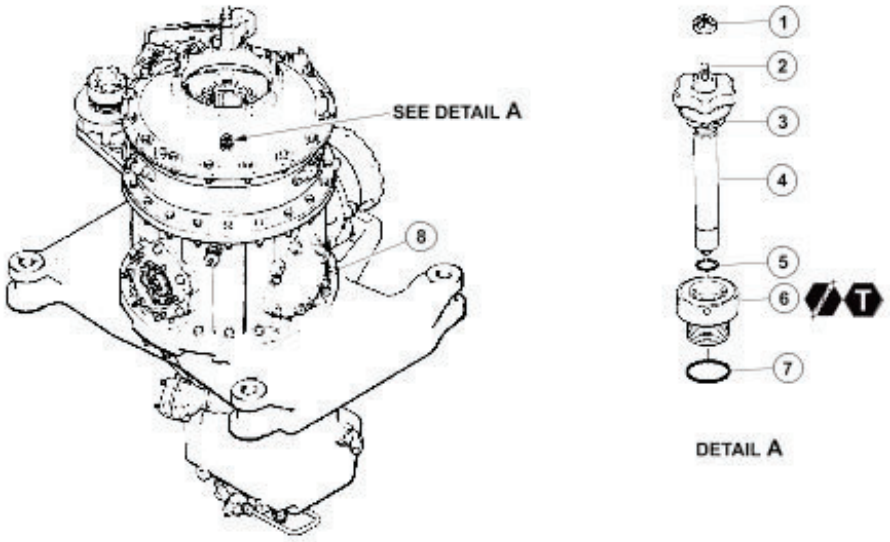
Figure 1. Mast Bearing Electric Chip Detector (Sheet 1 of 1)

Mast Bearing Electric Chip Detector - Inspection

Procedure

1. Inspect chip detector (4, Figure 1) for presence of metal particles. If any particles are found, refer to 412-A-63-00-00-01A-280A-A / 00989 and 412-A-63-00-00-01A-310A-A / 00990 for required action.

2. Inspect chip detector (4) components in accordance with Figure 2.



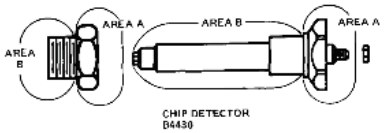
1. Nut
2. Stud
3. Locking lugs
4. Chip detector
5. Packing
6. Chip detector housing
7. Packing
8. Transmission

LOCKWIRE (C-405)

30 TO 108 IN-LBS
(3.4 TO 12.2 Nm)

ICN-412-A-632200-B-97499-00001-A-001-01

Figure 1. Mast Bearing Electric Chip Detector (Sheet 1 of 1)



AREA	LIMITS
All	No cracks allowed.
A	Maximum depth of pitting is 0.030 inch (0.762 millimeter) with no more than 40 percent of any 1.0 inch square (645.16 sq. millimeters) or 20 percent of any total area of any pitted surface.
B	Maximum depth of pitting is 0.020 inch (0.508 millimeter) with no more than 40 percent of any 1.0 inch square (645.16 sq. millimeters) or 20 percent of total area of any pitted surface. Thread damage is not permitted.

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Figure 2. Electric Chip Detector (Typical) - Damage Limits (Sheet 1 of 1)

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Mast Bearing Electric Chip Detector - Cleaning

Preliminary requirements

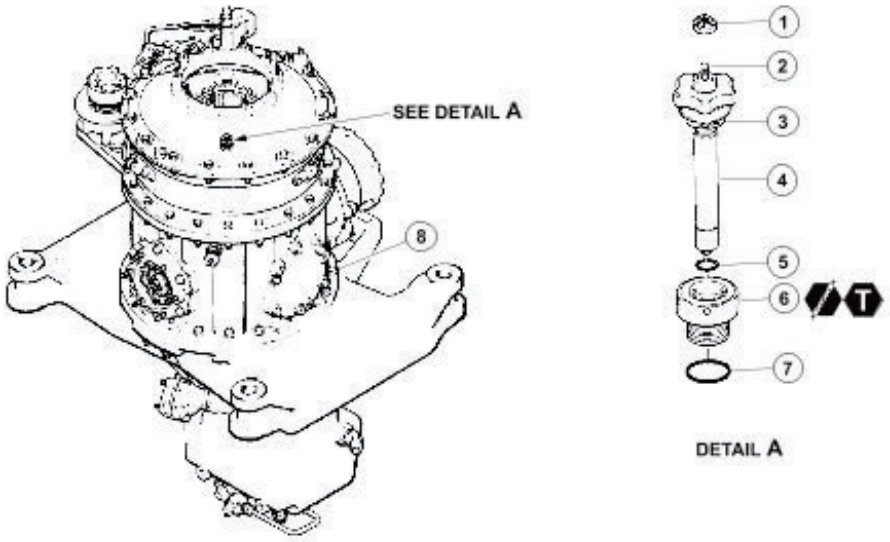
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean electric chip detector (4, Figure 1) with drycleaning solvent (C-304).

2. Dry with filtered, compressed air.



1. Nut
2. Stud
3. Locking lugs
4. Chip detector
5. Packing
6. Chip detector housing
7. Packing
8. Transmission

LOCKWIRE (C-405)

30 TO 108 IN-LBS
(3.4 TO 12.2 Nm)

ICN-412-A-632200-B-97499-00001-A-001-01

Figure 1. Mast Bearing Electric Chip Detector (Sheet 1 of 1)

Mast Bearing Electric Chip Detector - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Lubricate new packing (7, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packing on chip detector housing (6).

NOTE

If a plug has been installed where the chip detector housing (6) is normally installed, perform Step 2..

2. Remove plug from top of transmission case were chip detector housing (6) is normally located.

3. Install chip detector housing (6) in top of transmission (8).

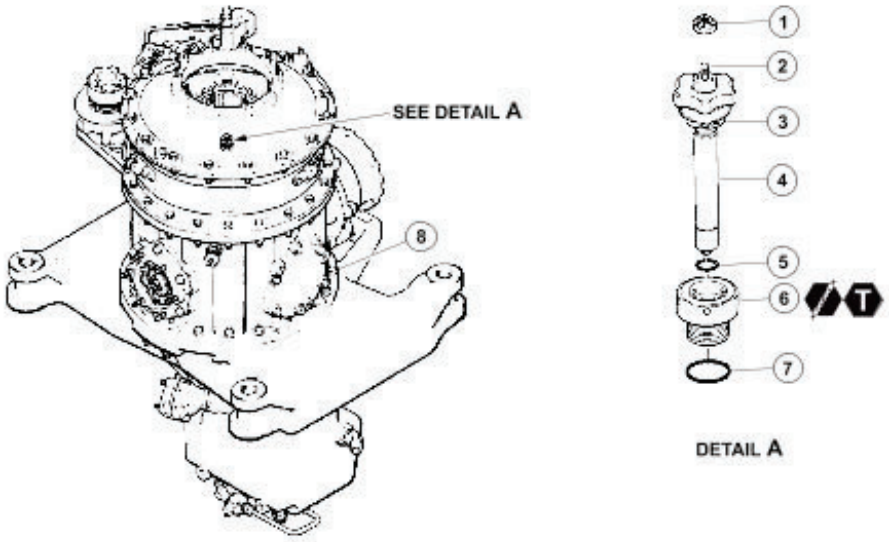
4. Torque chip detector housing (6) to 30 to 108 inch-pounds (3.4 to 12.2 Nm). Secure housing to transmission with lockwire (C-405).

5. Lubricate new packing (5) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install new packing on chip detector (4).

6. Insert chip detector (4) into chip detector housing (6). Push in on electric detector and twist clockwise to engage locking lugs (3).

7. Install electrical wire on stud (2).

8. Install nut (1) on stud (2). Tighten nut to a maximum of 4 inch-pounds (0.45 Nm) torque. Position nipple over nut.



1. Nut
2. Stud
3. Locking lugs
4. Chip detector
5. Packing
6. Chip detector housing
7. Packing
8. Transmission

LOCKWIRE (C-405)

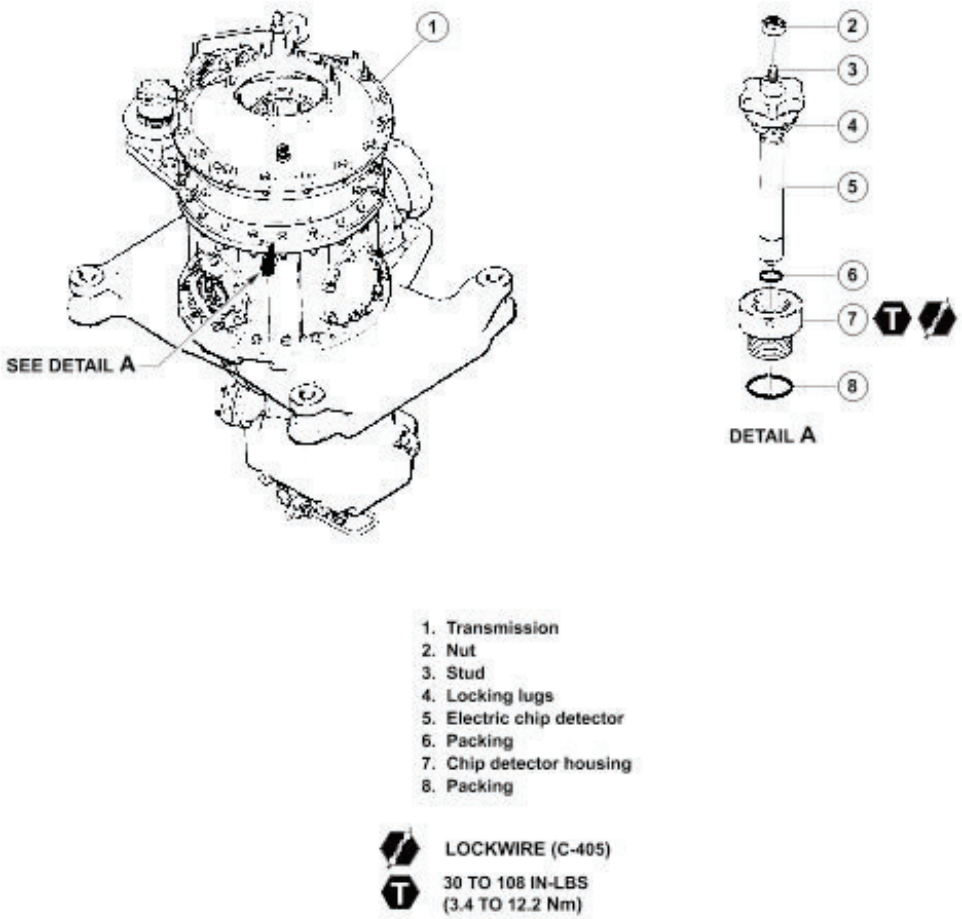
30 TO 108 IN-LBS
(3.4 TO 12.2 Nm)

ICN-412-A-632200-B-97499-00001-A-001-01

Figure 1. Mast Bearing Electric Chip Detector (Sheet 1 of 1)

Main Input Gear Quill Electric Chip Detector - Description

The main input gear quill electric chip detector (Figure 1) is installed to detect metal particles in the transmission oil originating from the main input gear quill.



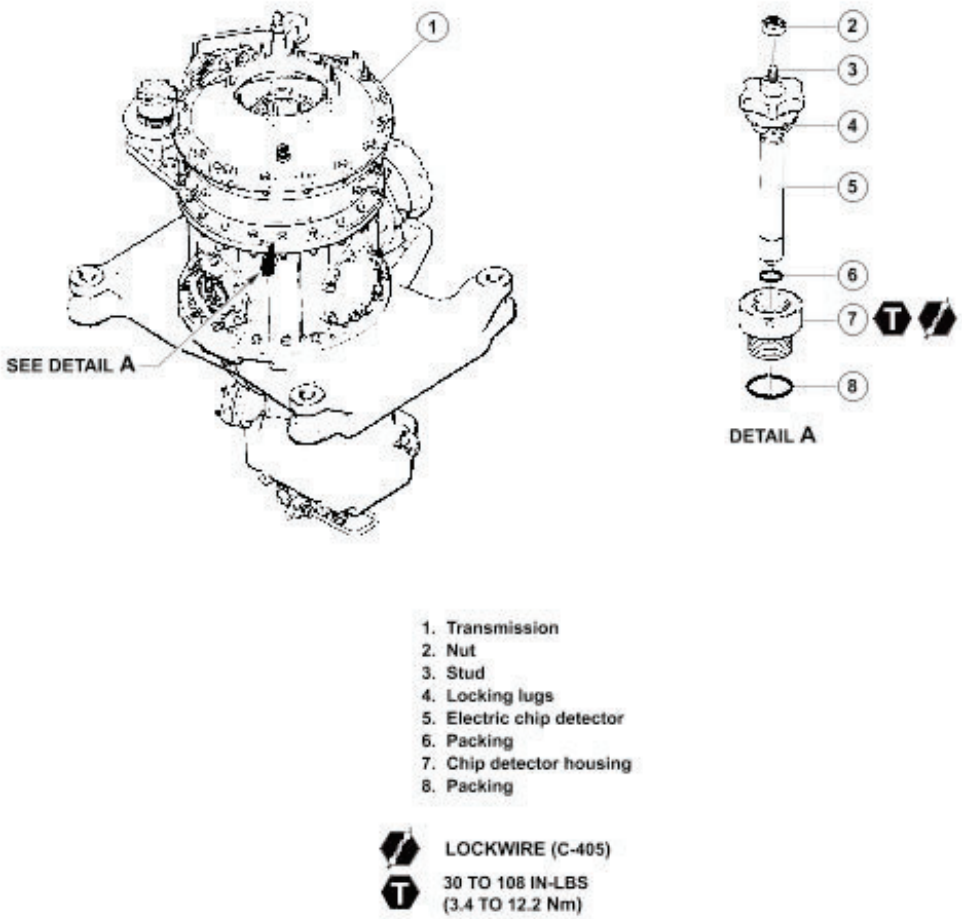
ICN-412-A-632200-B-97499-00003-A-001-01

Figure 1. Main Input Gear Quill Electric Chip Detector (Sheet 1 of 1)

Main Input Gear Quill Electric Chip Detector - Removal

Procedure

1. Remove nut (2, Figure 1) and remove wire from stud (3).
2. Push in and twist electric chip detector (5) counterclockwise to disengage locking lugs (4).
3. Remove electric chip detector (5) from chip detector housing (7).
4. Cut and remove lockwire from chip detector housing (7), and remove chip detector housing from transmission (1).
5. Remove and discard packing (6) from electric chip detector (5) and, if removed, discard packing (8) from chip detector housing (7).



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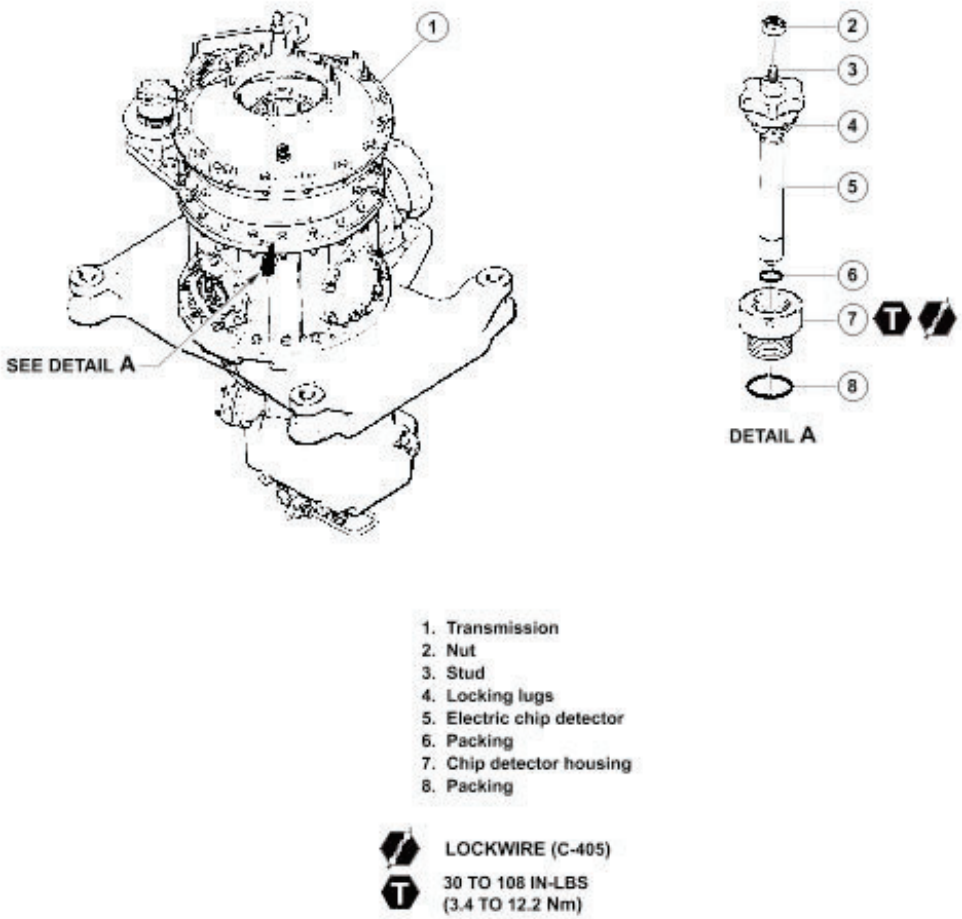
Figure 1. Main Input Gear Quill Electric Chip Detector (Sheet 1 of 1)

Main Input Gear Quill Electric Chip Detector - Inspection

Procedure

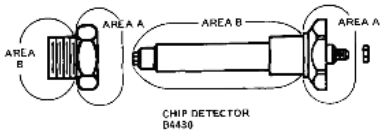
1. Inspect electric chip detector (5, Figure 1) for presence of metal particles. If any particles are found, refer to 412-A-63-00-00-01A-280A-A / 00989 and 412-A-63-00-00-01A-310A-A / 00990 for required action.

2. Inspect electric chip detector (5) components in accordance with Figure 2.



ICN-412-A-632200-B-97499-00003-A-001-01

Figure 1. Main Input Gear Quill Electric Chip Detector (Sheet 1 of 1)



AREA	LIMITS
All	No cracks allowed.
A	Maximum depth of pitting is 0.030 inch (0.762 millimeter) with no more than 40 percent of any 1.0 inch square (645.16 sq. millimeters) or 20 percent of any total area of any pitted surface.
B	Maximum depth of pitting is 0.020 inch (0.508 millimeter) with no more than 40 percent of any 1.0 inch square (645.16 sq. millimeters) or 20 percent of total area of any pitted surface. Thread damage is not permitted.

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Figure 2. Electric Chip Detector (Typical) - Damage Limits (Sheet 1 of 1)

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Main Input Gear Quill Electric Chip Detector - Cleaning

Preliminary requirements

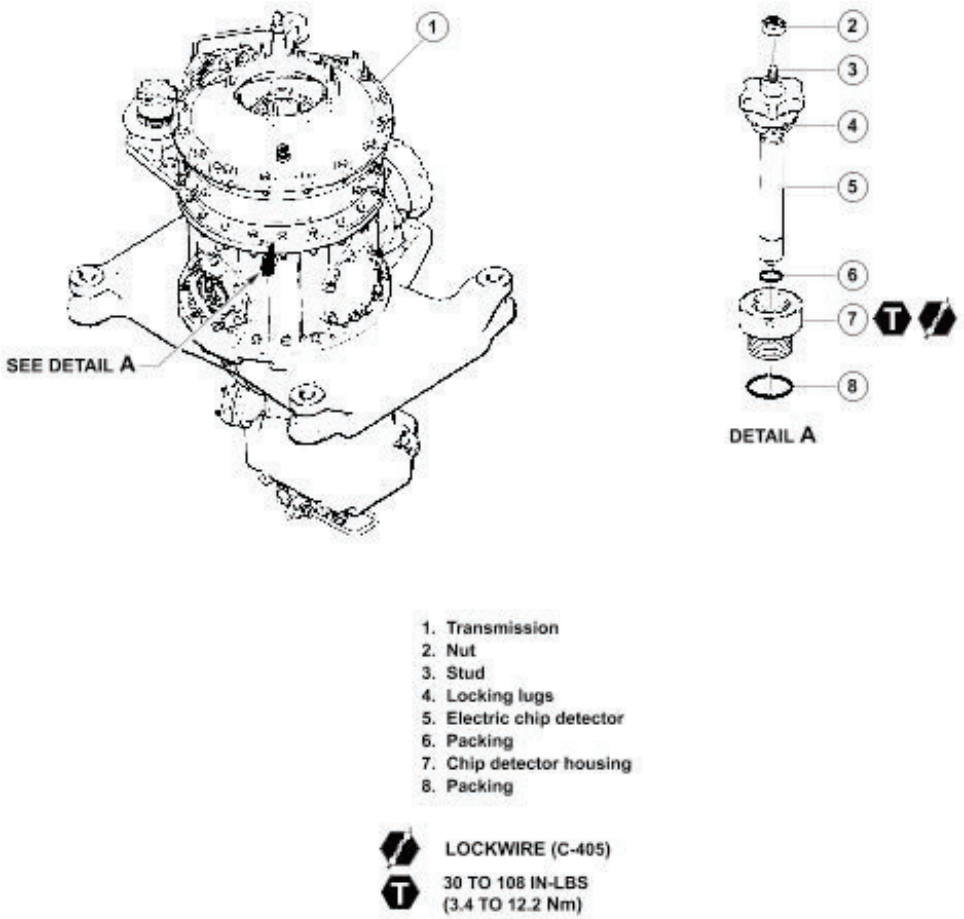
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean electric chip detector (5, Figure 1) with drycleaning solvent (C-304).

2. Dry with filtered, compressed air.



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Figure 1. Main Input Gear Quill Electric Chip Detector (Sheet 1 of 1)

Main Input Gear Quill Electric Chip Detector - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

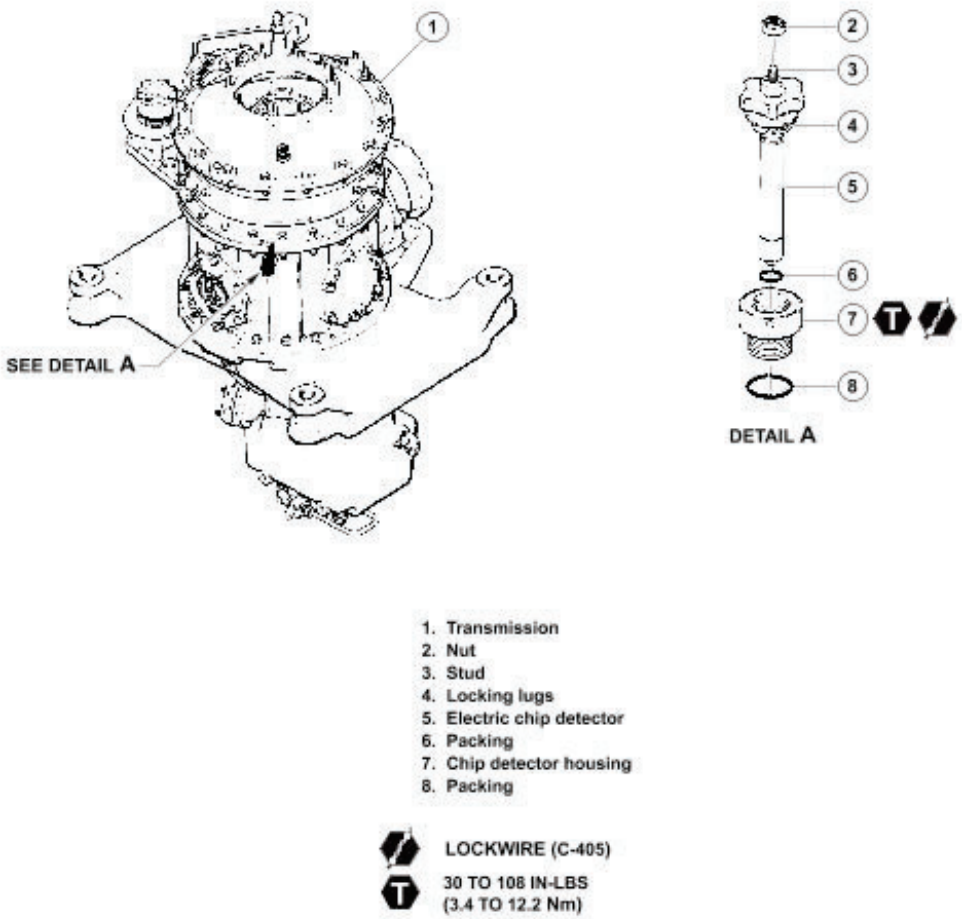
1. Lubricate new packings (6 and 8, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141).Install packing (8) on chip detector housing (7) and packing (6) on electric chip detector (5).

2. Install chip detector housing (7) on transmission (1). Torque housing to 30 to 108 inch-pounds (3.4 to 12 Nm). Secure housing to transmission (1) with lockwire (C-405).

3. Insert electric chip detector (5) into chip detector housing (7). Push in on electric chip detector and twist clockwise to engage locking lugs (4).

4. Install electrical wire on stud (3).

5. Install nut (2) on stud (3). Tighten nut to a maximum of 4 inch-pounds (0.45 Nm) torque. Position nipple over nut.

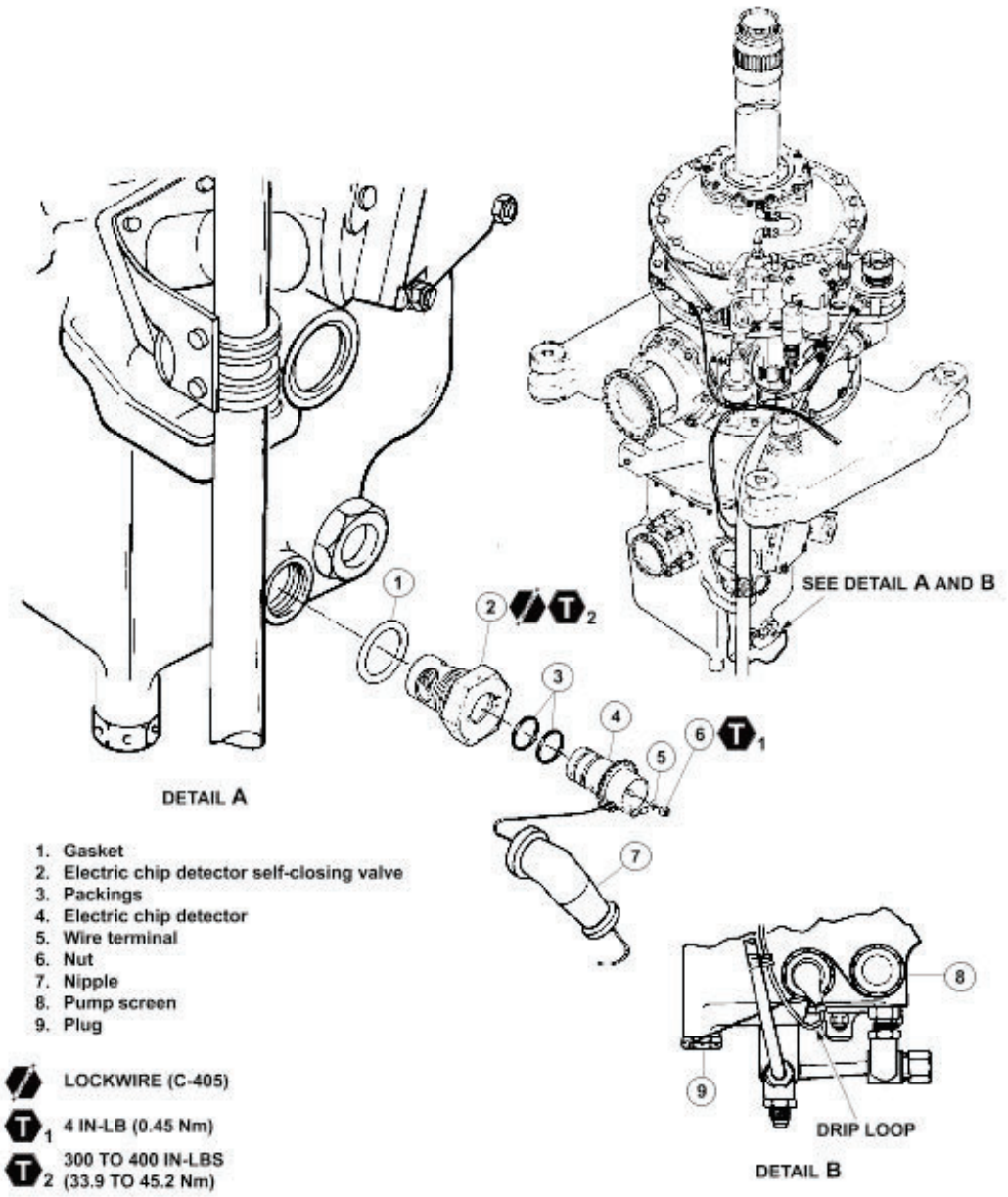


ICN-412-A-632200-B-97499-00003-A-001-01

Figure 1. Main Input Gear Quill Electric Chip Detector (Sheet 1 of 1)

Sump Case Electric Chip Detector - Description

The sump case electric chip detector (4, Figure 1) is installed to detect metal particles which collect in sump.



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Figure 1. Sump Case Electric Chip Detector (Sheet 1 of 1)

Sump Case Electric Chip Detector - Removal

Procedure

NOTE

The sump case electric chip detector (4, Figure 1) may be removed for inspection without draining sump. However, if the electric chip detector self-closing valve (2) is to be removed, the sump must be drained (412-A-12-10-03-00A-040A-A / 00135).

1. Remove oil level access door on right side of pylon.

2. Move nipple (7, Figure 1) back on wire terminal (5) to expose nut (6). Remove nut and wire terminal from electric chip detector (4).

3. Push in and twist electric chip detector (4) counterclockwise to disengage from locking detents.

4. Remove electric chip detector (4) from electric chip detector self-closing valve (2).

5. Remove and discard packings (3) from electric chip detector (4).

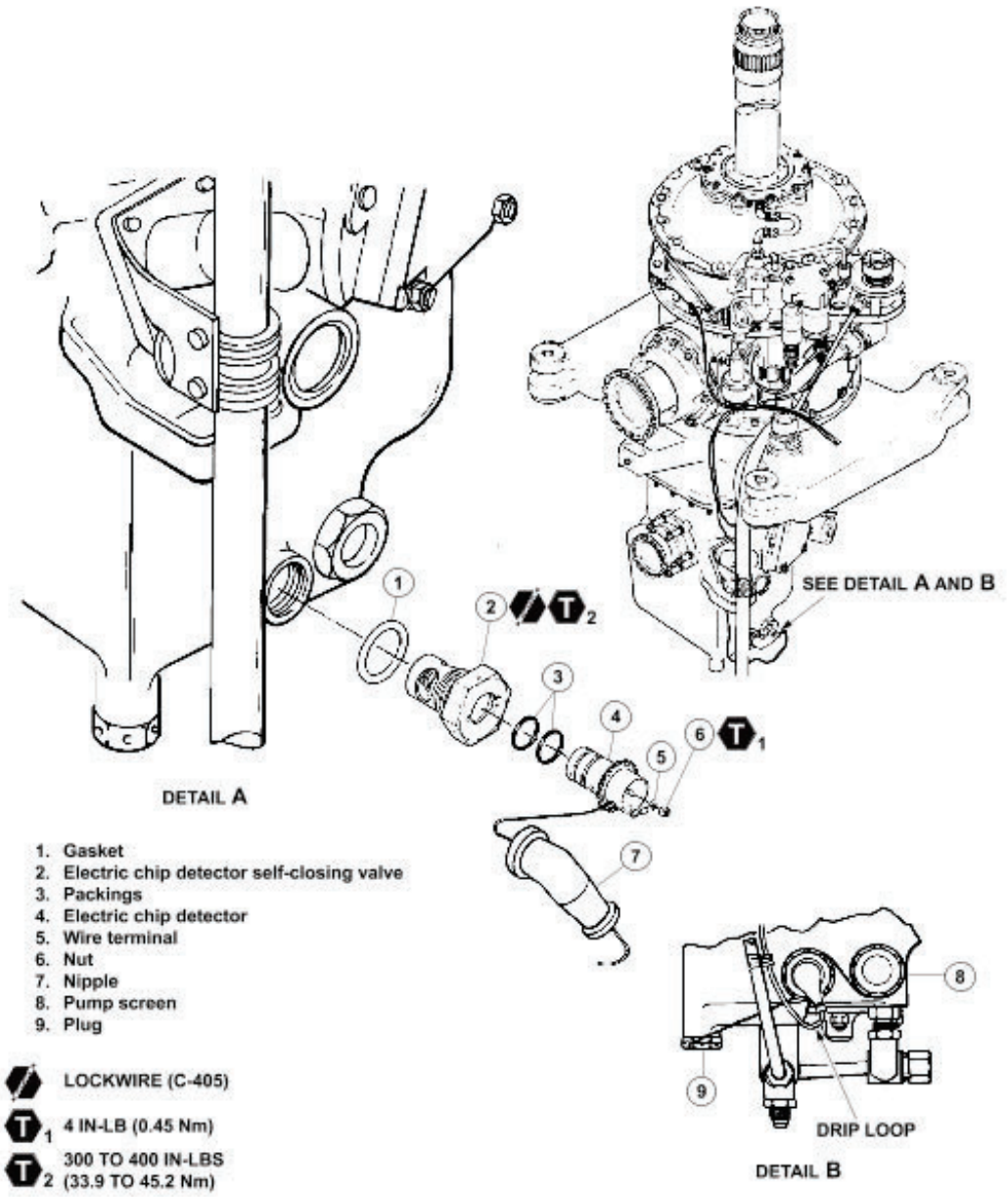
6. If the electric chip detector self-closing valve (2) is being removed, perform the following steps:

6.1. Drain transmission sump case (412-A-12-10-03-00A-040A-A / 00135).

6.2. Cut and remove lockwire from electric chip detector self-closing valve (2), pump screen (8), and plug (9).

6.3. Remove electric chip detector self-closing valve (2) from transmission sump case.

6.4. Remove gasket (1) from electric chip detector self-closing valve (2).



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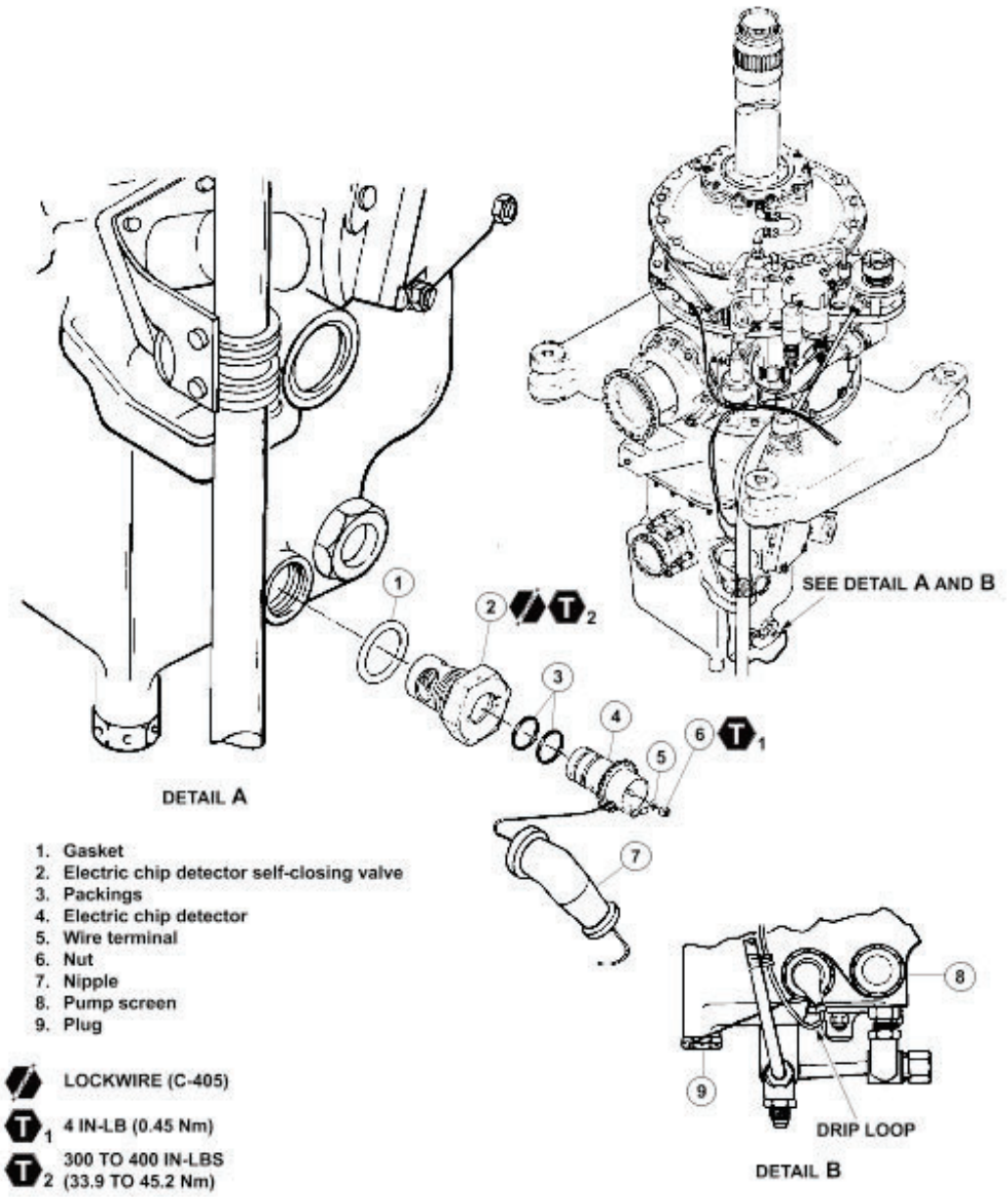
Figure 1. Sump Case Electric Chip Detector (Sheet 1 of 1)

Sump Case Electric Chip Detector - Inspection

Procedure

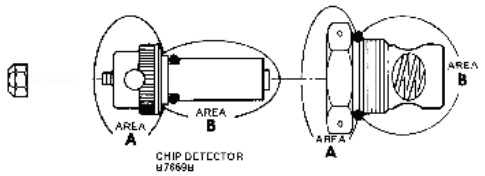
1. Inspect electric chip detector (4, Figure 1) for presence of metal particles. If any particles are found, refer to 412-A-63-00-00-01A-280A-A / 00989 and 412-A-63-00-00-01A-310A-A / 00990 for required action.

2. Inspect electric chip detector (4) components in accordance with Figure 2.



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Figure 1. Sump Case Electric Chip Detector (Sheet 1 of 1)



AREA	LIMITS
All	No cracks allowed.
A	Maximum depth of pitting is 0.030 inch (0.762 millimeter) with no more than 40 percent of any 1.0 square inch (645.16 mm^2) or 20 percent of total area of any pitted surface.
B	Maximum depth of pitting is 0.020 inch (0.508 millimeter) with no more than 40 percent of any 1.0 square inch (645.16 mm^2) or 20 percent of total area of any pitted surface. Thread damage is not permitted.

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Figure 2. Sump Case Electric Chip Detector - Damage Limits (Sheet 1 of 1)

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Sump Case Electric Chip Detector - Cleaning

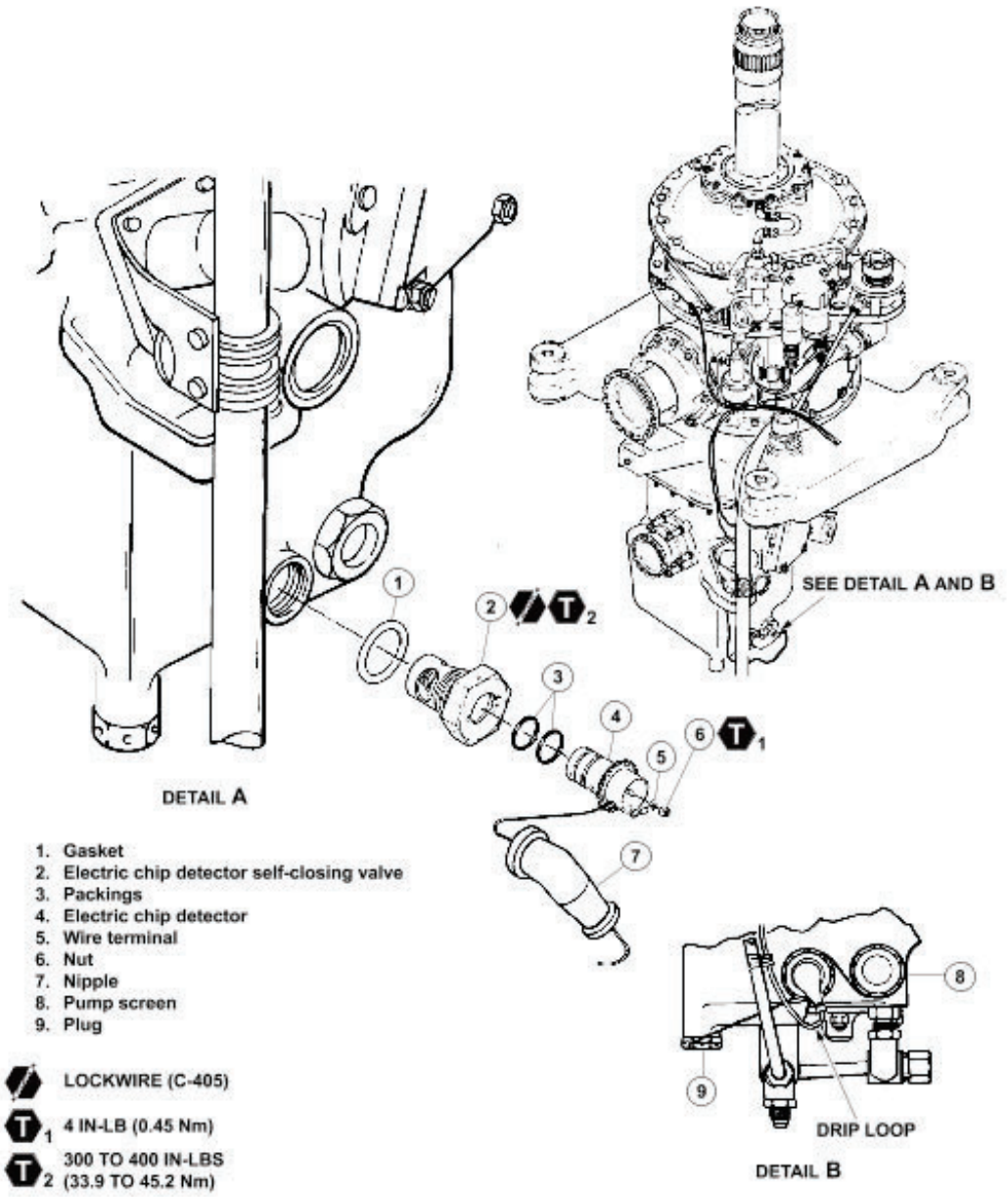
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean electric chip detector (4, Figure 1) with drycleaning solvent (C-304).
2. Dry with filtered, compressed air.



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Figure 1. Sump Case Electric Chip Detector (Sheet 1 of 1)

Sump Case Electric Chip Detector - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. If the electric chip detector self-closing valve (2, Figure 1) was removed, perform the following steps:

1.1. Lubricate new gasket (1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install gasket on electric chip detector self-closing valve (2).

1.2. Install electric chip detector self-closing valve (2) on transmission sump case and torque to 300 to 400 inch-pounds (33.9 to 45.2 Nm).

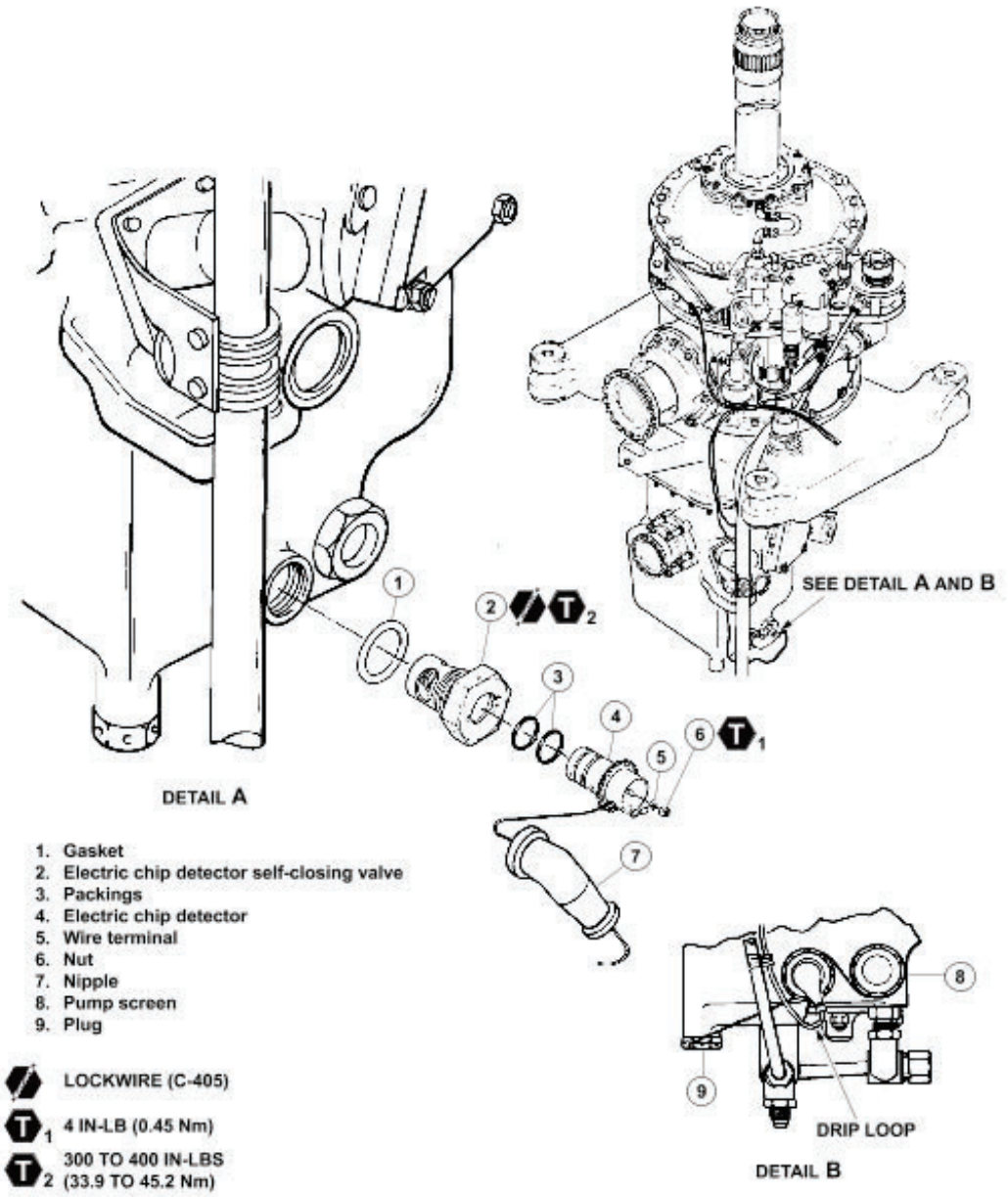
1.3. Secure electric chip detector self-closing valve (2) to pump screen (8) and plug (9) with lockwire (C-405).

2. Lubricate new packings (3) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packings (3) on electric chip detector (4).

3. Insert electric chip detector (4) into electric chip detector self-closing valve (2). Push in on electric chip detector and turn clockwise to engage the locking detents.

4. Install wire terminal (5) on electric chip detector (4).

5. Install nut (6) on electric chip detector (4). Tighten nut to a maximum of 4 inch-pounds (0.45 Nm) torque. Position nipple (7) over nut.

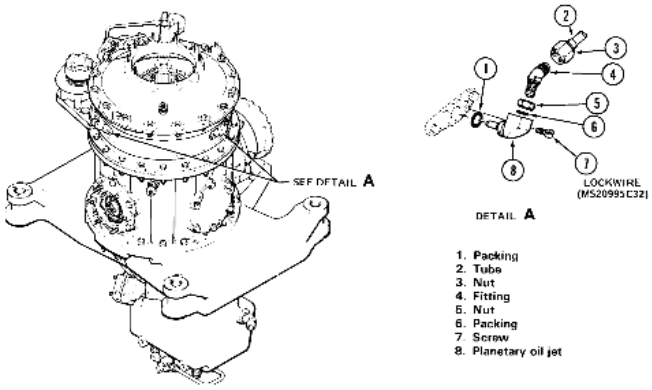


ICN-412-A-632200-B-97499-00004-A-001-01

Figure 1. Sump Case Electric Chip Detector (Sheet 1 of 1)

Planetary Oil Jets - Description

Two planetary oil jets (8, Figure 1) are installed in the planetary gear case of the transmission to provide positive lubrication to planetary gearing.



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Figure 1. Planetary Oil Jet (Sheet 1 of 1)

Planetary Oil Jets - Removal

Procedure

NOTE

The planetary oil jet on the forward side of transmission is shown in Figure 1. Maintenance procedures for the planetary oil jet on the left side of transmission are similar.

1. Disconnect tube (2, Figure 1) from fitting (4).

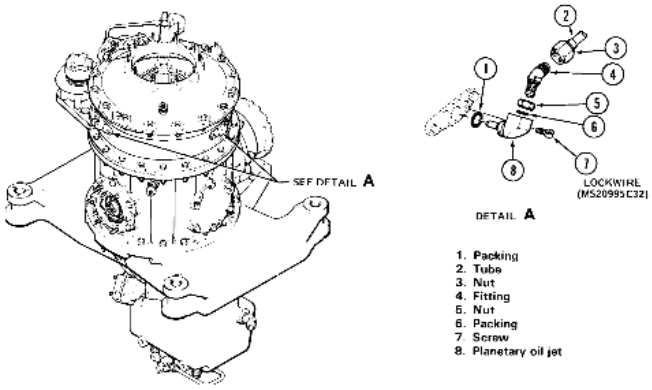
2. Cut and remove lockwire from screw (7) and remove screw.

3. Remove planetary oil jet (8).

4. Disassemble planetary oil jet (8) as follows:

4.1. Loosen nut (5) and remove fitting (4) with packings (6).

4.2. Remove and discard packings (1 and 6).



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Figure 1. Planetary Oil Jet (Sheet 1 of 1)

Planetary Oil Jets - Inspection

Procedure

1. Inspect planetary oil jet (8) for clogging. If passages are clogged, determine source(s) of material (412-A-63-00-00-01A-280A-A / 00989).
2. Inspect for corrosion. Superficial corrosion damage on plate on exterior of transmission is acceptable if removed.
3. Inspect for mechanical damage. Cracks and distortion are not acceptable.

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Planetary Oil Jets - Cleaning

Preliminary requirements

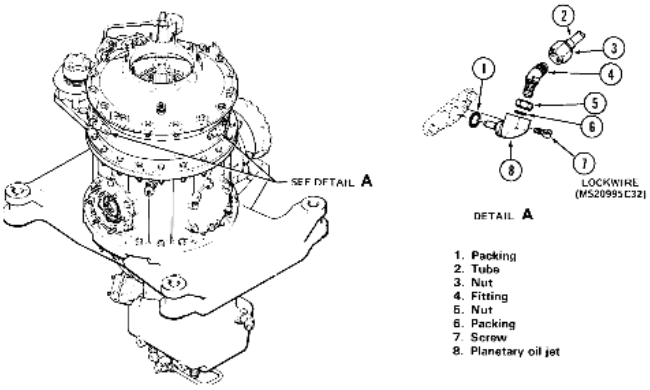
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean planetary oil jets (8, Figure 1) using drycleaning solvent (C-304).

2. Dry with filtered, compressed air.



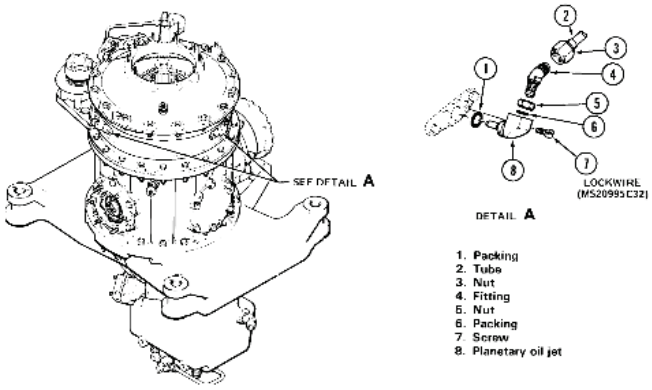
ICN-412-A-632201-B-97499-00001-A-001-01

Figure 1. Planetary Oil Jet (Sheet 1 of 1)

Planetary Oil Jets - Repair

Procedure

1. Repair of planetary jets (8, Figure 1) is limited to removal of corrosion products from plate at base of jet on exterior of transmission.



ICN-412-A-632201-B-97499-00001-A-001-01

Figure 1. Planetary Oil Jet (Sheet 1 of 1)

Planetary Oil Jets - Installation

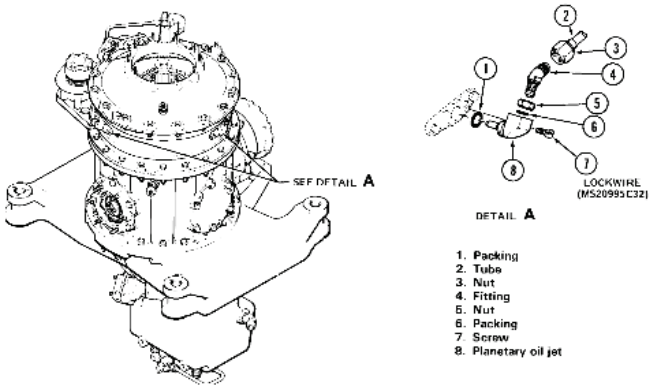
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Lubricate new packing (1, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packing on planetary oil jet (8).
2. Install nut (5) on fitting (4), and install fitting on planetary oil jet (8).
3. Lubricate new packing (6) with transmission oil and install on fitting (4).
4. Insert planetary oil jet (8) into transmission housing and install screw (7). Secure screw (7) on planetary oil jet with lockwire (C-405).
5. Connect tube (2) and tighten nut (3).
6. Check for oil leaks at next helicopter ground run.



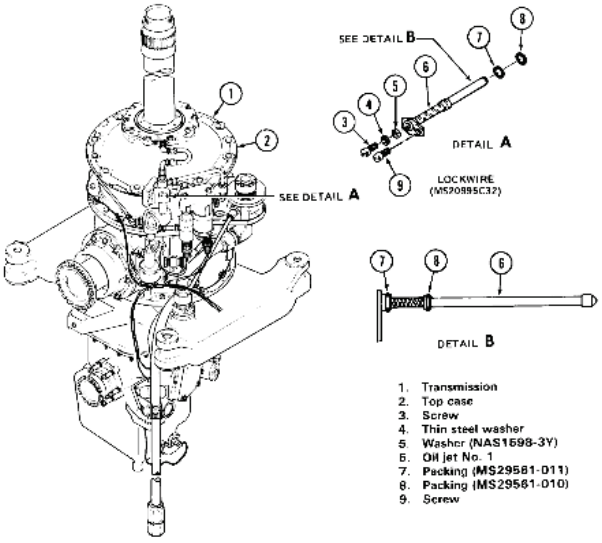
ICN-412-A-632201-B-97499-00001-A-001-01

Figure 1. Planetary Oil Jet (Sheet 1 of 1)

Oil Jet No. 1 - Removal

Procedure

1. Cut and remove lockwire from screw (9, Figure 1) and screw (3), and remove screw (9).
2. Remove oil jet No. 1 (6) from housing in top case (2) on transmission (1).
3. Remove screw (3) and washers (4 and 5).
4. Remove and discard packings (7 and 8).



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Figure 1. Oil Jet No. 1 (Sheet 1 of 1)

Oil Jet No. 1 - Inspection

Procedure

1. Refer to 412-A-63-22-01-01A-280A-A / 01035.

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Oil Jet No. 1 - Cleaning

Procedure

1. Refer to 412-A-63-22-01-01A-251A-A / 01036.

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Oil Jet No. 1 - Repair

Procedure

1. Refer to 412-A-63-22-01-01A-685A-A / 01037.

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Oil Jet No. 1 - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

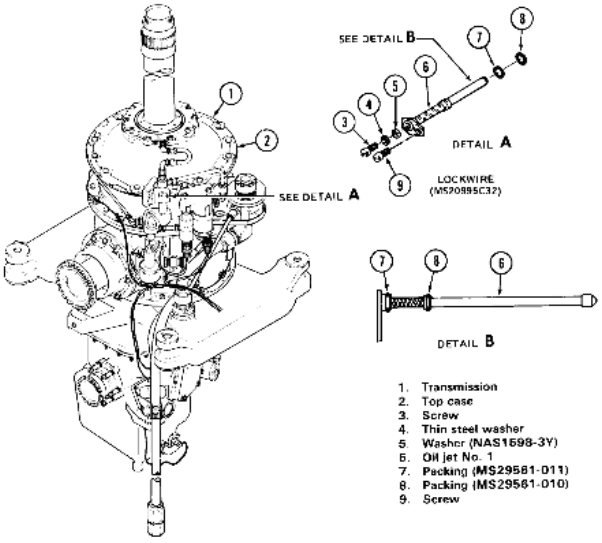
1. Install screw (3, Figure 1) with washers (4 and 5) on oil jet No. 1 (6).

2. Lubricate new packings (7 and 8) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install in grooves of oil jet No. 1 (6).

3. Carefully insert oil jet No. 1 (6) into housing in top case (2).

4. Install screw (9). Secure screw (9) to screw (3) using lockwire (C-405).

5. Check for oil leaks at next helicopter ground run.



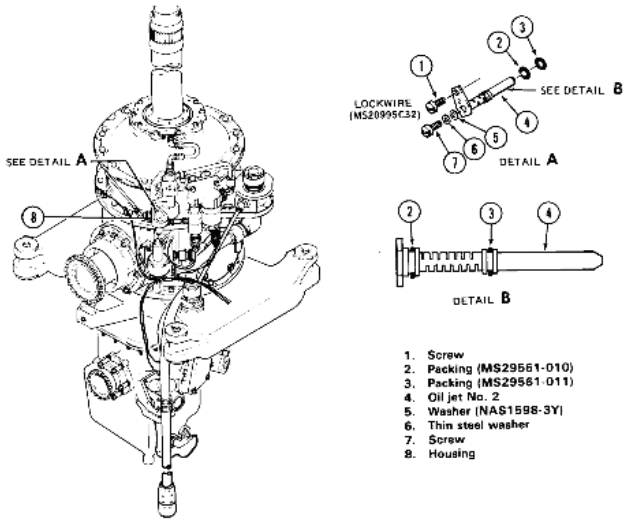
ICN-412-A-632201-B-97499-00002-A-001-01

Figure 1. Oil Jet No. 1 (Sheet 1 of 1)

Oil Jet No. 2 - Removal

Procedure

1. Cut and remove lockwire from screw (1, Figure 1), and remove screw from oil jet No. 2 (4).
2. Remove oil jet No. 2 (4) from housing (8) on transmission.
3. Remove screw (7) and washers (5 and 6).
4. Remove and discard packings (2 and 3).



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Figure 1. Oil Jet No. 2 (Sheet 1 of 1)

Oil Jet No. 2 - Inspection

Procedure

1. Refer to 412-A-63-22-01-01A-280A-A / 01035.

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Oil Jet No. 2 - Cleaning

Procedure

1. Refer to 412-A-63-22-01-01A-251A-A / 01036.

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Oil Jet No. 2 - Repair

Procedure

1. Refer to 412-A-63-22-01-01A-685A-A / 01037.

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Oil Jet No. 2 - Installation

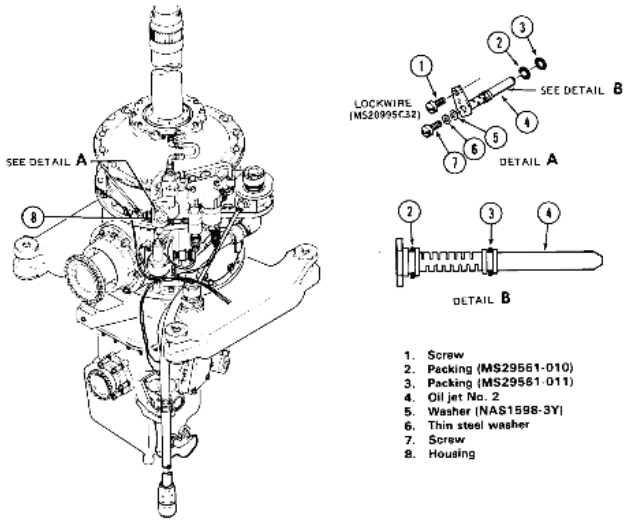
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Install screw (7, Figure 1) with washers (5 and 6) on oil jet No. 2 (4).
2. Lubricate new packings (2 and 3) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install in grooves of oil jet No. 2.
3. Carefully insert oil jet No. 2 (4) into housing (8) on transmission.
4. Install screw (1). Secure screw (1) to screw (7) using lockwire (C-405).
5. Check for oil leaks at next helicopter ground run.



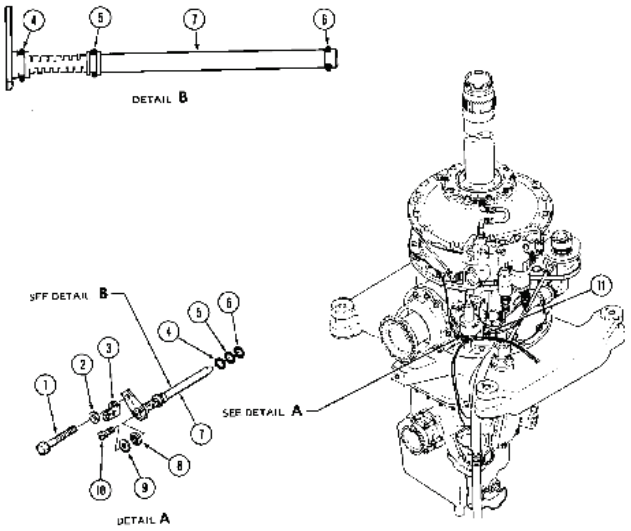
ICN-412-A-632201-B-97499-00003-A-001-01

Figure 1. Oil Jet No. 2 (Sheet 1 of 1)

Oil Jet No. 3 - Removal

Procedure

1. Cut and remove lockwire from bolt (1, Figure 1) and screw (10).
2. Remove bolt (1), thin steel washer (2), and bracket (3) from oil jet No. 3 (7).
3. Remove oil jet No. 3 (7) with packing (4, 5, and 6).
4. Remove screw (10) and washers (8 and 9).
5. Remove and discard packings (4, 5, and 6).



1. Bolt
2. Thin steel washer
3. Bracket
4. Packing (MS29561-011)
5. Packing (MS29561-010)
6. Packing (MS29561-008)
7. Oil jet No. 3
8. Washer (NAS1598 3Y)
9. Thin steel washer
10. Screw
11. Manifold

ICN-412-A-632201-B-97499-00004-A-001-01

Figure 1. Oil Jet No. 3 (Sheet 1 of 1)

Oil Jet No. 3 - Inspection

Procedure

1. Refer to 412-A-63-22-01-01A-280A-A / 01035.

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Oil Jet No. 3 - Cleaning

Procedure

1. Refer to 412-A-63-22-01-01A-251A-A / 01036.

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Oil Jet No. 3 - Repair

Procedure

1. Refer to 412-A-63-22-01-01A-685A-A / 01037.

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Oil Jet No. 3 - Installation

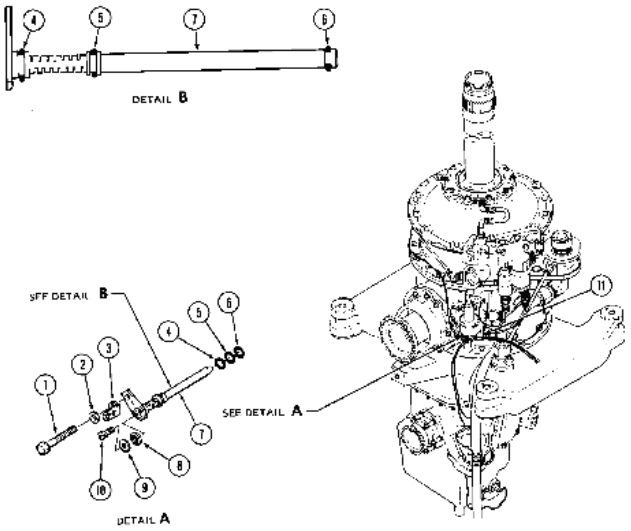
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Carefully install screw (10, Figure 1) with washers (8 and 9) on oil jet No. 3 (7).
2. Lubricate new packings (4, 5, and 6) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install in grooves of oil jet No. 3 (7).
3. Carefully insert oil jet No. 3 (7) into manifold (11).
4. Install bolt (1), thin steel washer (2), and bracket (3). Torque bolt (1) to 50 to 70 inch-pounds (5.7 to 7.9 Nm).
5. Secure bolt (1) to screw (10) using lockwire (C-405).
6. Check for oil leaks at next helicopter ground run.



1. Bolt
2. Thin steel washer
3. Bracket
4. Packing (MS29561-011)
5. Packing (MS29561-010)
6. Packing (MS29561-008)
7. Oil jet No. 3
8. Washer (NAS1598 3Y)
9. Thin steel washer
10. Screw
11. Manifold

ICN-412-A-632201-B-97499-00004-A-001-01

Figure 1. Oil Jet No. 3 (Sheet 1 of 1)

Oil Jet No. 4 - Removal

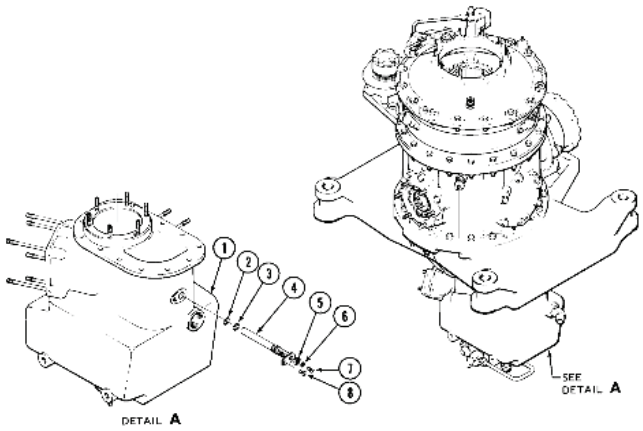
Procedure

1. Remove access door on left side of pylon support in cabin.
2. Cut and remove lockwire from screw (8, Figure 1) and screw (7). Remove screw (8).

NOTE

When removing oil jet No. 4 (4) from sump case (1), make sure packings (2 and 3) are extracted with the oil jet.

3. Remove oil jet No. 4 (4) with packings (2 and 3) from sump case (1).
4. Remove screw (7) and washers (5 and 6) from oil jet No. 4 (4).
5. Remove and discard packings (2 and 3).



1. Sump case
2. Packing (MS29561-010)
3. Packing (MS29561-011)
4. Oil Jet No. 4
5. Washer (NAS1598-3Y)
6. Thin steel washer
7. Screw
8. Screw

ICN-412-A-632201-B-97499-00005-A-001-01

Figure 1. Oil Jet No. 4 (Sheet 1 of 1)

Oil Jet No. 4 - Inspection

Procedure

1. Refer to 412-A-63-22-01-01A-280A-A / 01035.

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Oil Jet No. 4 - Cleaning

Procedure

1. Refer to 412-A-63-22-01-01A-251A-A / 01036.

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Oil Jet No. 4 - Repair

Procedure

1. Refer to 412-A-63-22-01-01A-685A-A / 01037.

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Oil Jet No. 4 - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

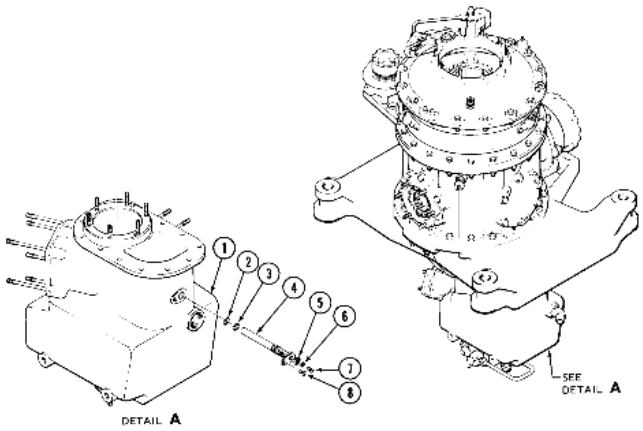
1. Install screw (7, Figure 1) with washers (5 and 6) on oil jet No. 4 (4).

2. Lubricate new packings (2 and 3) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install in grooves of oil jet No. 4 (4).

3. Carefully insert oil jet No. 4 (4) into sump case (1).

4. Install screw (8) to secure oil jet No. 4 (4) into sump case (1). Secure screw (8) to screw (7) using lockwire (C-405).

5. Check for oil leaks at next helicopter ground run.



1. Sump case
2. Packing (MS29561-010)
3. Packing (MS29561-011)
4. Oil Jet No. 4
5. Washer (NAS1598-3Y)
6. Thin steel washer
7. Screw
8. Screw

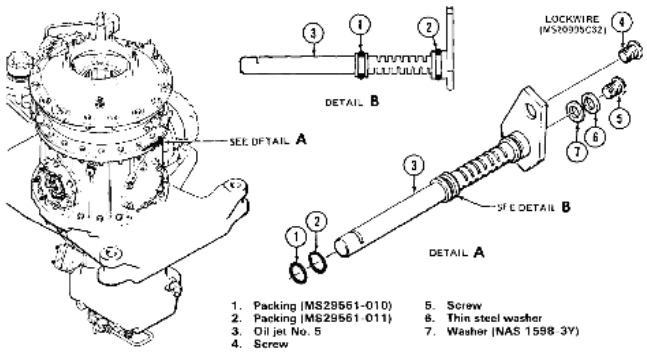
ICN-412-A-632201-B-97499-00005-A-001-01

Figure 1. Oil Jet No. 4 (Sheet 1 of 1)

Oil Jet No. 5 - Removal

Procedure

1. Cut and remove lockwire from screw (4, Figure 1) and screw (5). Remove screw (4).
2. Carefully remove oil jet No. 5 (3) with packing (1 and 2).
3. Remove screw (5) and washers (6 and 7) from oil jet No. 5 (3).
4. Remove and discard packings (1 and 2).



ICN-412-A-632201-B-97499-00006-A-001-01

Figure 1. Oil Jet No. 5 (Sheet 1 of 1)

Oil Jet No. 5 - Inspection

Procedure

1. Refer to 412-A-63-22-01-01A-280A-A / 01035.

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Oil Jet No. 5 - Cleaning

Procedure

1. Refer to 412-A-63-22-01-01A-251A-A / 01036.

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Oil Jet No. 5 - Repair

Procedure

1. Refer to 412-A-63-22-01-01A-685A-A / 01037.

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Oil Jet No. 5 - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

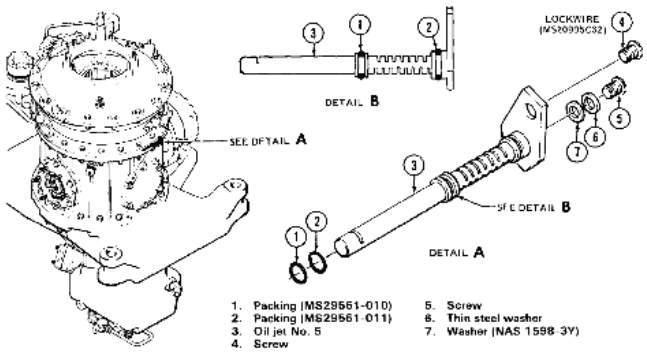
1. Install screw (5, Figure 1) with washers (6 and 7) on oil jet No. 5 (3).

2. Lubricate new packings (1 and 2) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install in grooves of oil jet No. 5 (3).

3. Carefully insert oil jet No. 5 (3) into transmission case.

4. Install screw (4), and secure screw (5) to screw (4) using lockwire (C-405).

5. Check for oil leaks at next helicopter ground run.



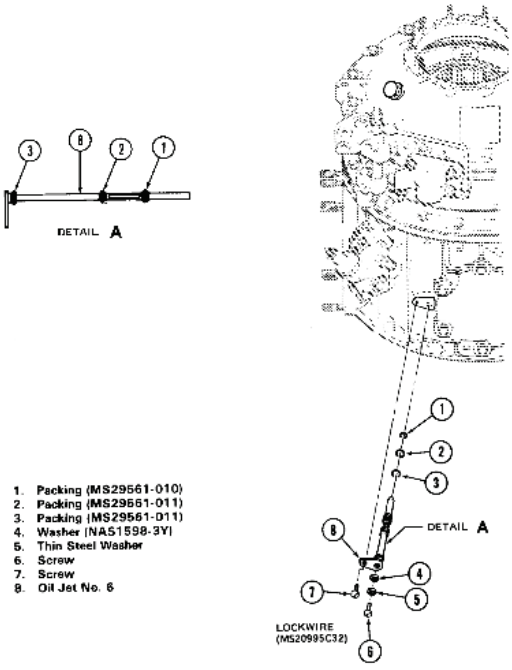
ICN-412-A-632201-B-97499-00006-A-001-01

Figure 1. Oil Jet No. 5 (Sheet 1 of 1)

Oil Jet No. 6 - Removal

Procedure

1. Cut and remove lockwire from screw (7, Figure 1) and screw (6), and remove screw (7).
2. Remove oil jet No. 6 (8) with packings (1, 2, and 3).
3. Remove screw (6) and washers (4 and 5).
4. Remove and discard packings (1, 2, and 3).



ICN-412-A-632201-B-97499-00007-A-001-01

Figure 1. Oil Jet No. 6 (Sheet 1 of 1)

Oil Jet No. 6 - Inspection

Procedure

1. Refer to 412-A-63-22-01-01A-280A-A / 01035.

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Oil Jet No. 6 - Cleaning

Procedure

1. Refer to 412-A-63-22-01-01A-251A-A / 01036.

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Oil Jet No. 6 - Repair

Procedure

1. Refer to 412-A-63-22-01-01A-685A-A / 01037.

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Oil Jet No. 6 - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

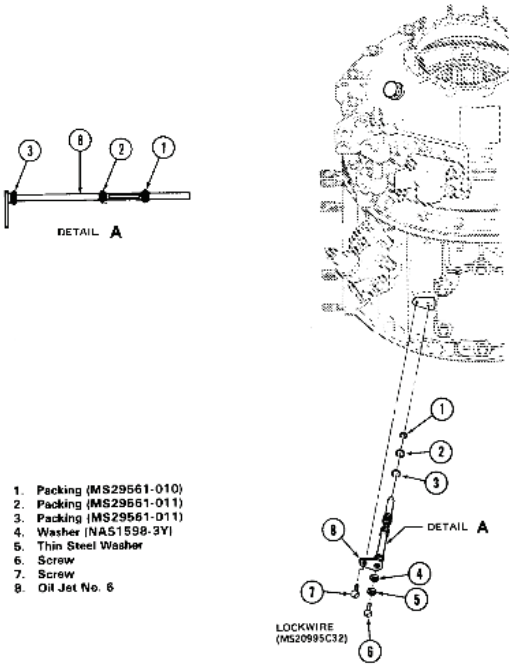
1. Install screw (6, Figure 1) with washers (4 and 5) on oil jet No. 6 (8).

2. Lubricate new packings (1, 2, and 3) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install in grooves of oil jet No. 6 (8).

3. Carefully insert oil jet No. 6 (8) into transmission case.

4. Install screw (7). Secure screw (7) to screw (6) with lockwire (C-405).

5. Check for oil leaks at next helicopter ground run.



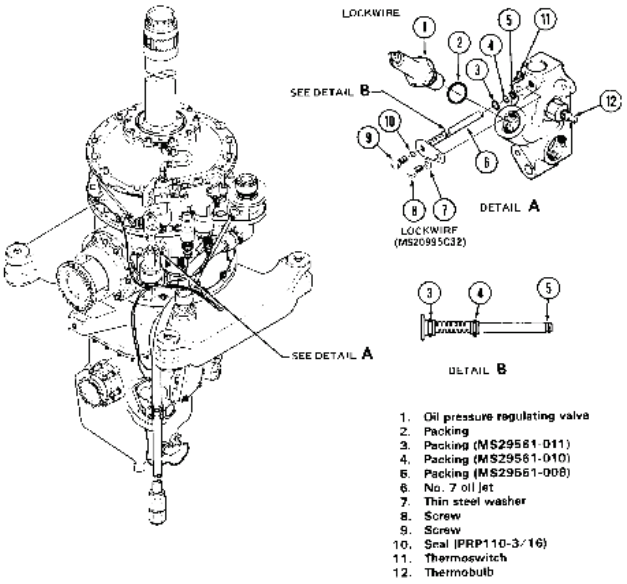
ICN-412-A-632201-B-97499-00007-A-001-01

Figure 1. Oil Jet No. 6 (Sheet 1 of 1)

Oil Jet No. 7 - Removal

Procedure

1. Cut and remove lockwire from oil pressure regulating valve (1, Figure 1), thermoswitch (11), and thermobulb (12).
2. Remove oil pressure regulating valve (1, Figure 1) with packing (2). Discard packing.
3. Cut and remove lockwire from screw (8) and screw (9), and remove screw (8) and thin steel washer (7).
4. Remove oil jet No. 7 (6) with packings (3, 4, and 5).
5. Remove screw (9) and seal (10). Discard seal.
6. Discard packings (3, 4, and 5).



ICN-412-A-632201-B-97499-00008-A-001-01

Figure 1. Oil Jet No. 7 (Sheet 1 of 1)

Oil Jet No. 7 - Inspection

Procedure

1. Refer to 412-A-63-22-01-01A-280A-A / 01035.

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Oil Jet No. 7 - Cleaning

Procedure

1. Refer to 412-A-63-22-01-01A-251A-A / 01036.

Intentionally blank page

Oil Jet No. 7 - Repair

Procedure

1. Refer to 412-A-63-22-01-01A-685A-A / 01037.

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Oil Jet No. 7 - Installation

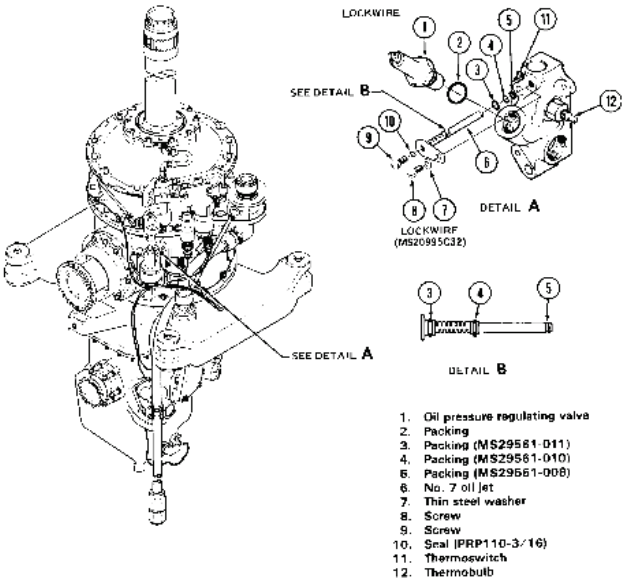
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Install screw (9, Figure 1) with new seal (10).
2. Lubricate new packings (3, 4, and 5) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install in grooves of oil jet No. 7 (6).
3. Carefully insert oil jet No. 7 (6) into transmission case.
4. Install screw (8) and thin steel washer (7). Secure screw (8) to screw (9) with lockwire (C-405).
5. Lubricate new packing (2) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install on oil pressure regulating valve (1).
6. Install oil pressure regulating valve (1) in manifold.
7. Secure oil pressure regulating valve (1) to thermoswitch (11) and thermobulb (12) with lockwire (C-405).
8. Check for oil leaks at next helicopter ground run.



ICN-412-A-632201-B-97499-00008-A-001-01

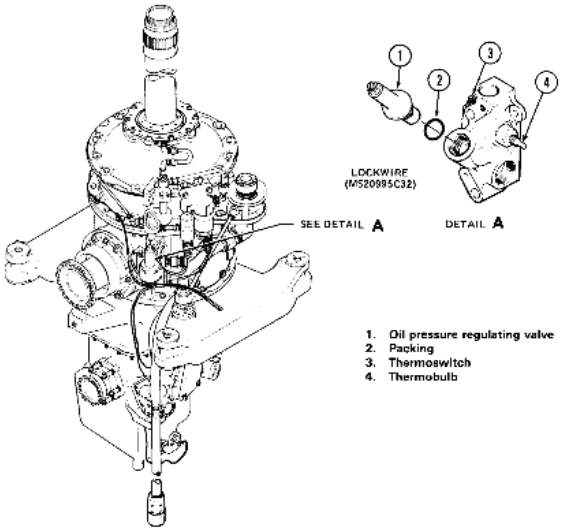
Figure 1. Oil Jet No. 7 (Sheet 1 of 1)

Oil Pressure Regulating Valve - Removal

Procedure

1. Cut and remove lockwire from oil pressure regulating valve (1, Figure 1), thermoswitch (3), and thermobulb (4).

2. Remove oil pressure regulating valve (1) and packing (2). Discard packing.



ICN-412-A-632201-B-97499-00009-A-001-01

Figure 1. Oil Pressure Regulator Valve (Sheet 1 of 1)

Oil Pressure Regulating Valve - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

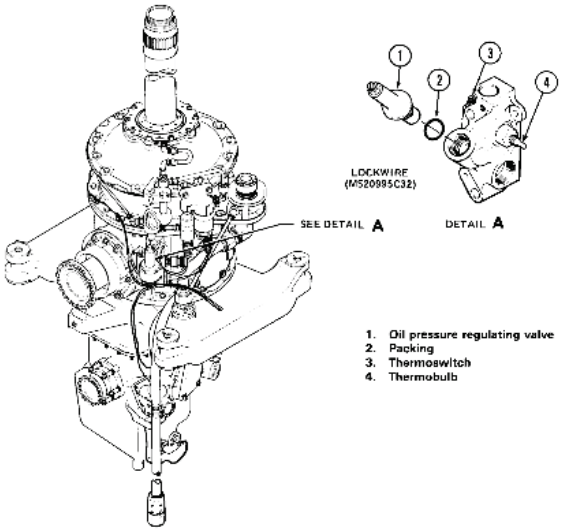
Procedure



DO NOT IMMERSE OIL PRESSURE REGULATING VALVE IN SOLVENT.

1. Wipe oil pressure regulating valve (1, Figure 1) with clean cloth dampened with drycleaning solvent (C-304).

2. Dry using filtered, compressed air.



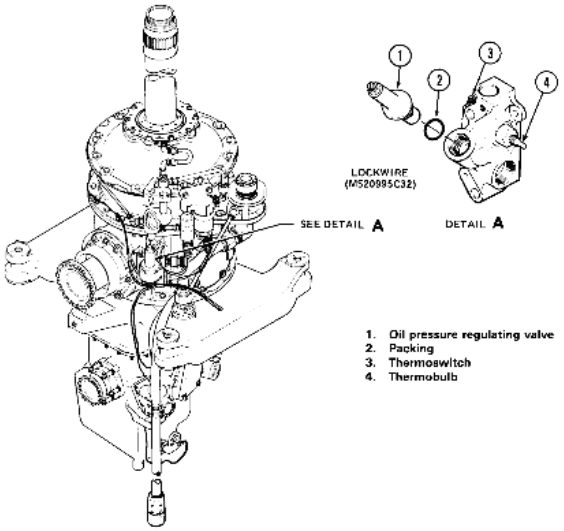
ICN-412-A-632201-B-97499-00009-A-001-01

Figure 1. Oil Pressure Regulator Valve (Sheet 1 of 1)

Oil Pressure Regulating Valve - Inspection

Procedure

1. Inspect oil pressure regulating valve (1, Figure 1) for damaged threads and for mechanical or corrosion damage. Damage in excess of superficial is unacceptable. Corrosion damage shall be removed.



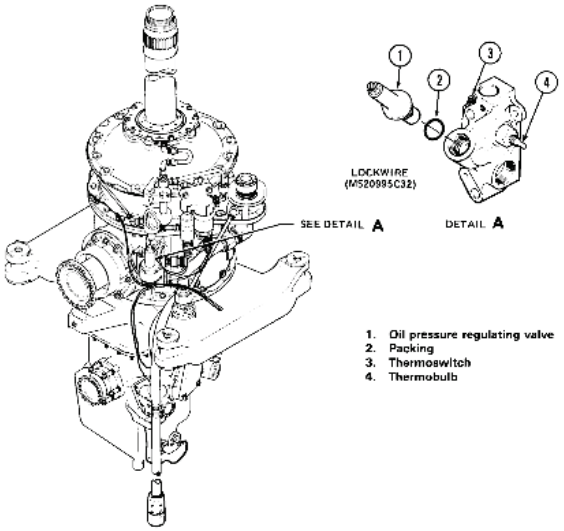
ICN-412-A-632201-B-97499-00009-A-001-01

Figure 1. Oil Pressure Regulator Valve (Sheet 1 of 1)

Oil Pressure Regulating Valve - Repair

Procedure

1. Repair of oil pressure regulating valve (1, Figure 1) is limited to removal of superficial corrosion from areas exposed on exterior of transmission.



ICN-412-A-632201-B-97499-00009-A-001-01

Figure 1. Oil Pressure Regulator Valve (Sheet 1 of 1)

Oil Pressure Regulating Valve - Installation

Preliminary requirements

Consumables, materials and expendables

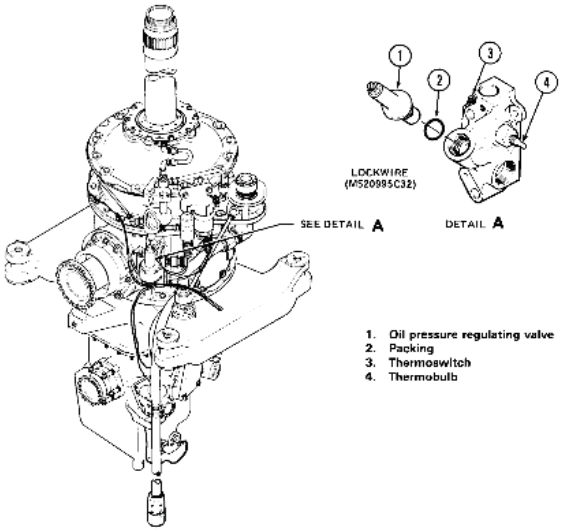
Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Lubricate new packing (2, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141), and install packing on oil pressure regulating valve (1).

2. Install oil pressure regulating valve (1) in manifold (Detail A).

3. Secure oil pressure regulating valve (1) to thermoswitch (3) and thermobulb (4) with lockwire (C-405).



ICN-412-A-632201-B-97499-00009-A-001-01

Figure 1. Oil Pressure Regulator Valve (Sheet 1 of 1)

Oil Pressure Regulating Valve - Adjustment

Procedure

1. Adjust the oil pressure regulating valve (1, Figure 1) as follows:

1.1. Shut down the helicopter (BHT-412-FM).

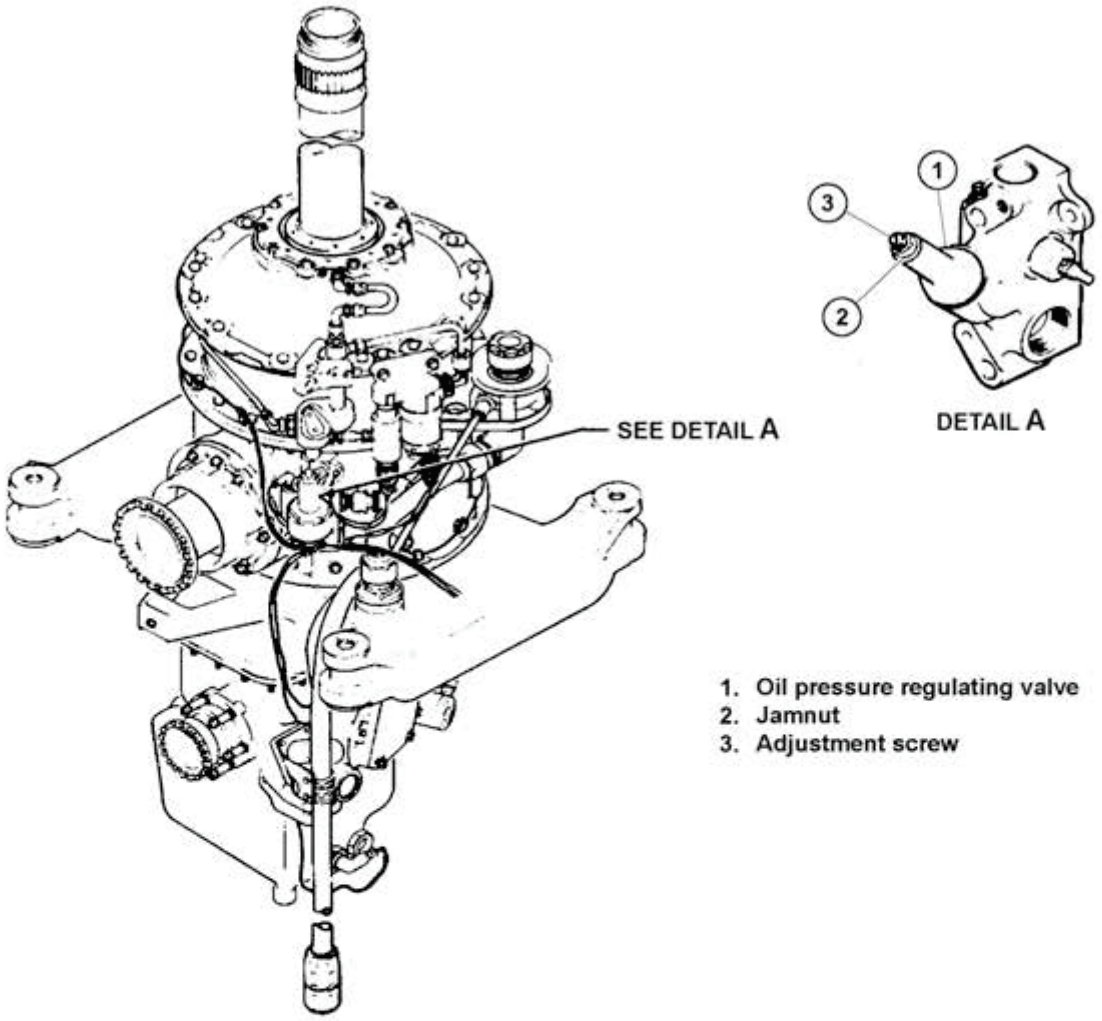
1.2. Loosen the jamnut (2) on adjustment screw (3).

1.3. Turn the adjustment screw (3) clockwise to increase pressure, or counterclockwise to decrease pressure, as required.

1.4. Tighten the jam nut (2).

1.5. Start the engines (BHT-412-FM) and gradually increase RPM to 100% and recheck the pressure.

1.6. Repeat Step 1.1. through Step 1.5., as required to obtain proper oil pressure.



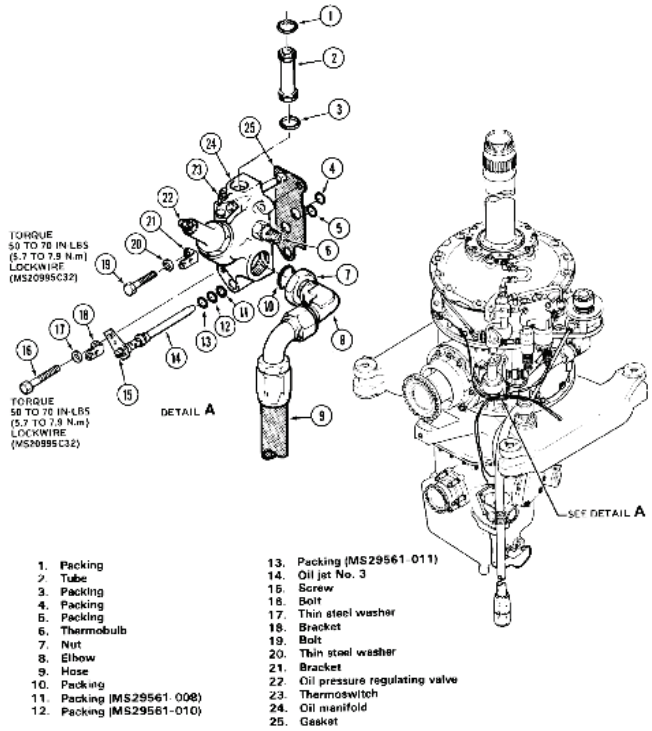
ICN-412-A-632201-B-97499-03021-A-001-01

Figure 1. Oil Pressure Regulating Valve - Adjustment (Sheet 1 of 1)

Oil Temperature Manifold - Removal

Procedure

1. Cut and remove lockwire from bolt (16, Figure 1) and screw (15).
2. Remove bolt (16), thin steel washer (17), and bracket (18).
3. Remove oil jet No. 3 (14) and packings (11, 12, and 13). Discard packings.
4. Cut and remove lockwire from electrical connector and disconnect electrical connector from thermobulb (6).
5. Disconnect hose (9) from elbow (8).
6. Identify and tag electrical wire installed on thermoswitch (23), and disconnect electrical wire from thermoswitch.
7. Identify and tag electrical wires installed on bracket (21). Remove electrical wires from bracket.
8. Remove bolt (19), thin steel washer (20), and bracket (21).
9. Cut sealant from around tube (2) using a sharp plastic scraper.
10. Carefully work oil manifold (24) loose and remove manifold with tube (2), packings (1, 3, 4, and 5), and gasket (25). Discard packings and gasket.



ICN-412-A-632201-B-97499-00010-A-001-01

Figure 1. Oil Temperature Manifold (Sheet 1 of 1)

Oil Temperature Manifold - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
Lockwire	C-405	AR	
Lockwire	C-508	AR	

Procedure

1. If removed, install nut (7, Figure 1) on elbow (8), coat new packing (10) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install on elbow. Install elbow, nut and packing on manifold. Do not tighten nut at this time.

2. Lubricate new packings (1 and 3) with transmission oil and install on tube (2). Install tube in oil manifold (24).

3. Lubricate new packings (4 and 5) with transmission oil and install in oil manifold (24). Lubricate new gasket (25) with transmission oil and install on oil manifold.

4. Carefully work oil manifold (24) into position on transmission case. Ensure tube (2) is seated in oil manifold. Ensure bushings on oil manifold are seated on transmission case.

5. Install bolt (19) with thin steel washer (20) and bracket (21). Do not torque bolt at this time.

6. Lubricated new packings (11, 12, and 13) and install packings on oil jet No. 3 (14).

7. Carefully install oil jet No. 3 (14) in oil manifold (24) and install bolt (16) with thin steel washer (17) and bracket (18).

8. Torque bolt (16) to 50 to 70 inch-pounds (5.6 to 7.9 Nm). Secure bolt to screw (15) with lockwire (C-405).

9. Torque bolt (19) to 50 to 70 inch-pounds (5.6 to 7.9 Nm). Secure bolt to thermoswitch (23) and thermobulb (6) with lockwire (C-405).

10. Install hose (9) on elbow (8). Tighten nut (7).

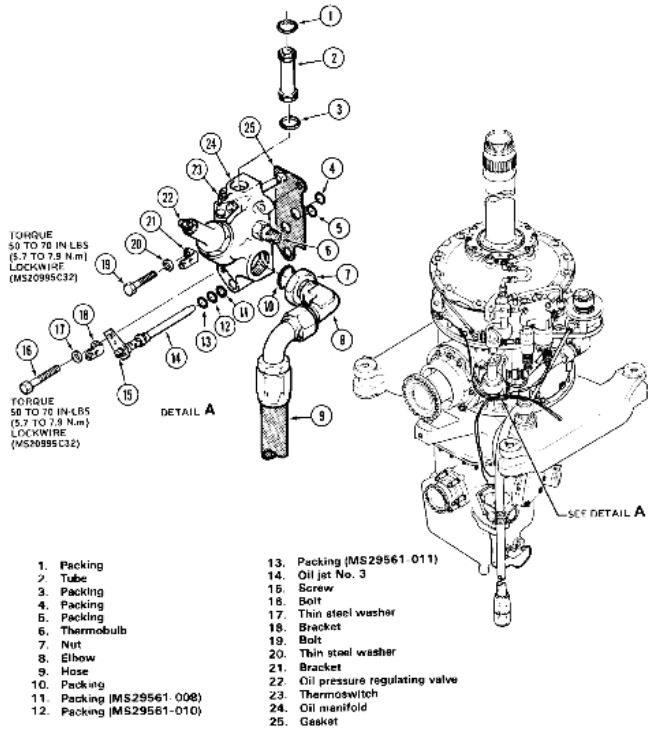
11. Connect electrical plug to thermobulb (6) and secure with lockwire (C-508).

12. Connect electrical wire to thermoswitch (23).

13. Install electrical wires on bracket (21). Remove identification tag.

14. Apply a bead of sealant (C-308) around each end of tube (2) at mating joint.

15. Check for oil leaks at next helicopter ground run.



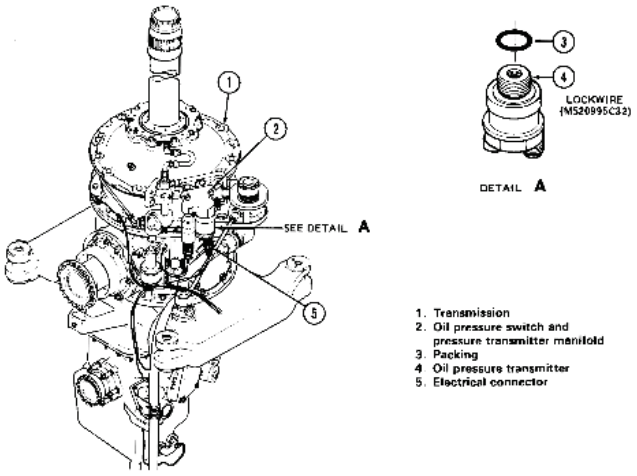
ICN-412-A-632201-B-97499-00010-A-001-01

Figure 1. Oil Temperature Manifold (Sheet 1 of 1)

Oil Pressure Transmitter - Removal

Procedure

1. Cut and remove lockwire from electrical connector (5, Figure 1), and disconnect electrical connector.
2. Cut and remove lockwire securing oil pressure transmitter (4).
3. Remove oil pressure transmitter (4) with packing (3). Discard packing.
4. Cap or plug exposed cavity in oil pressure switch and pressure transmitter manifold (2).



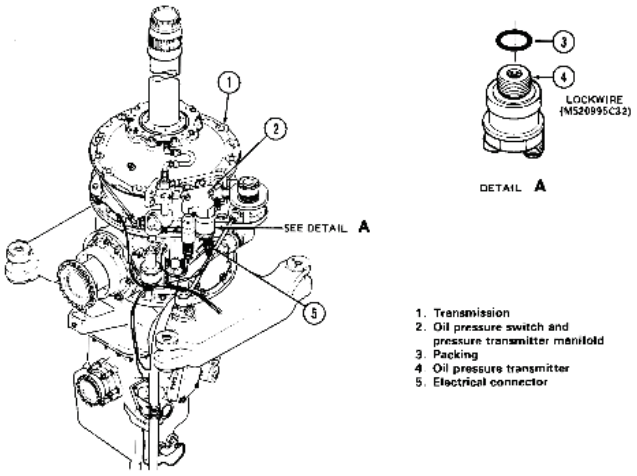
ICN-412-A-632201-B-97499-00011-A-001-01

Figure 1. Oil Pressure Transmitter (Sheet 1 of 1)

Oil Pressure Transmitter - Inspection

Procedure

1. Inspect oil pressure transmitter (4, Figure 1) and electrical connector for bent pins, damaged threads, dents, or other mechanical damage rendering transmitter unserviceable.



ICN-412-A-632201-B-97499-00011-A-001-01

Figure 1. Oil Pressure Transmitter (Sheet 1 of 1)

Oil Pressure Transmitter - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	
Lockwire	C-508	AR	

Procedure

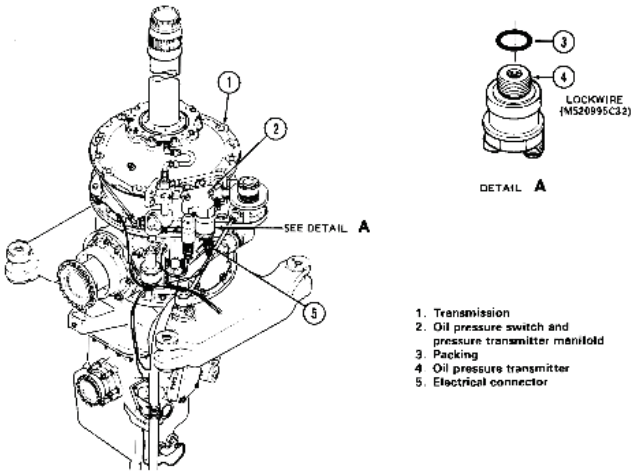
1. Lubricate packing (3, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141).

2. Install packing (3) on oil pressure transmitter (4).

3. Install oil pressure transmitter (4) in oil pressure switch and pressure transmitter manifold (2). Secure oil pressure transmitter with lockwire (C-405).

4. Install electrical connector (5) and secure electrical connector with lockwire (C-508).

5. Perform operational check of oil pressure transmitter (4) (412-A-96-11-00-03A-320A-A / 01951) and check for oil leaks at next helicopter ground run.



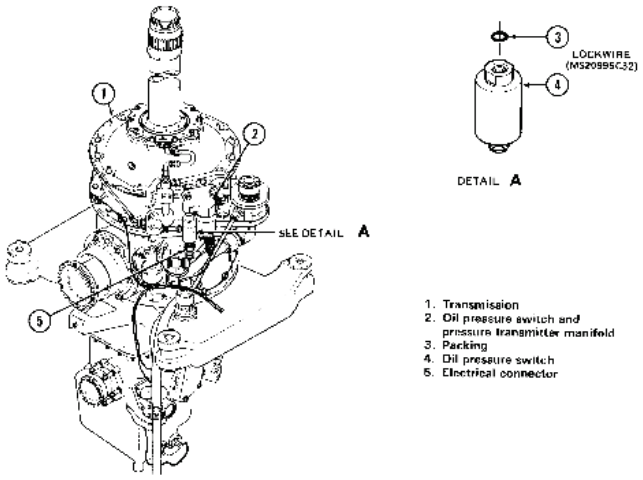
ICN-412-A-632201-B-97499-00011-A-001-01

Figure 1. Oil Pressure Transmitter (Sheet 1 of 1)

Oil Pressure Switch - Removal

Procedure

1. Cut and remove lockwire from electrical connector (5, Figure 1), and disconnect electrical connector.
2. Cut and remove lockwire from oil pressure switch (4), and remove oil pressure switch with packing (3). Discard packing.
3. Cap or plug cavity in oil pressure switch and pressure transmitter manifold (2).



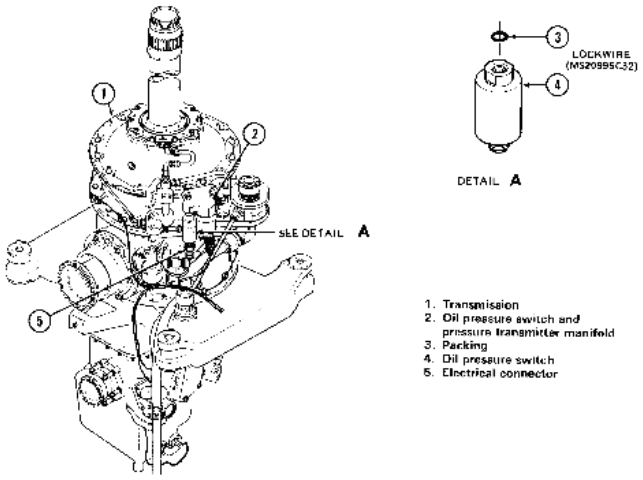
ICN-412-A-632201-B-97499-00012-A-001-01

Figure 1. Oil Pressure Switch (Sheet 1 of 1)

Oil Pressure Switch - Inspection

Procedure

1. Inspect oil pressure switch (4, Figure 1) and electrical connector for bent pins, damaged threads, dents, or other mechanical damage which would render switch unserviceable.



ICN-412-A-632201-B-97499-00012-A-001-01

Figure 1. Oil Pressure Switch (Sheet 1 of 1)

Oil Pressure Switch - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	
Lockwire	C-508	AR	

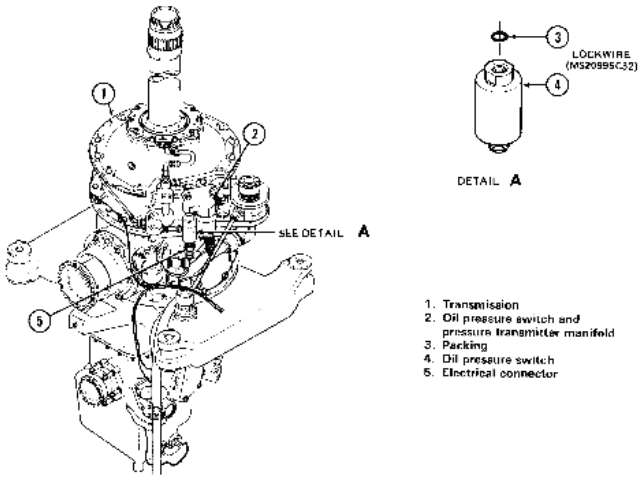
Procedure

1. Lubricate packing (3, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141), and install packing on oil pressure switch (4).

2. Install oil pressure switch (4) in oil pressure switch and pressure transmitter manifold (2), and secure oil pressure switch with lockwire (C-405).

3. Install electrical connector (5). Secure connector with lockwire (C-508).

4. Perform operational check of oil pressure switch (4) (412-A-96-03-01-04A-320A-A / 02049) and check for oil leaks at next helicopter ground run.



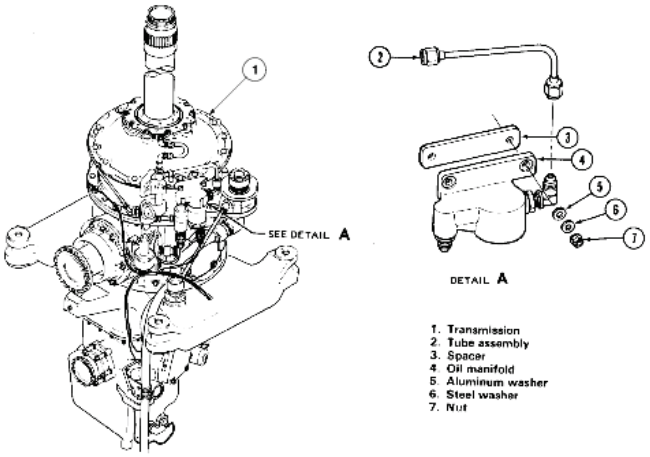
ICN-412-A-632201-B-97499-00012-A-001-01

Figure 1. Oil Pressure Switch (Sheet 1 of 1)

Oil Pressure Manifold - Removal

Procedure

1. Remove oil pressure switch (412-A-63-22-01-12A-520A-A / 01084).
2. Remove oil pressure transmitter (412-A-63-22-01-11A-520A-A / 01081).
3. Remove tube assembly (2, Figure 1).
4. Remove nuts (7) and washers (5 and 6), securing oil manifold (4). Remove oil manifold and spacer (3) from transmission (1).



ICN-412-A-632201-B-97499-00013-A-001-01

Figure 1. Oil Pressure Manifold (Sheet 1 of 1)

Oil Pressure Manifold - Cleaning

Preliminary requirements

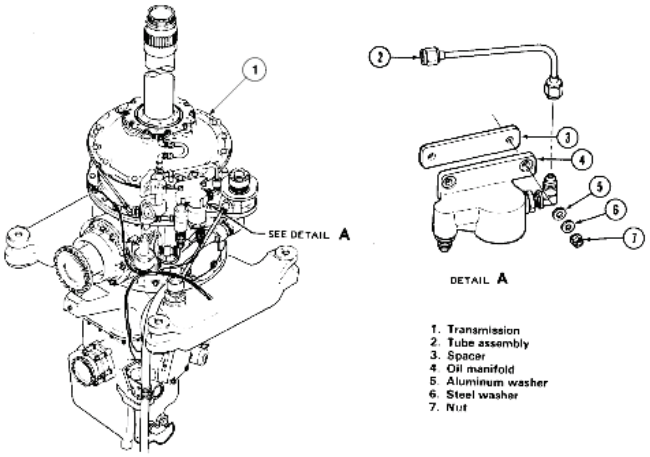
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Wipe oil manifold (4, Figure 1) with clean cloth dampened with drycleaning solvent (C-304).

2. Dry using filtered, compressed air.



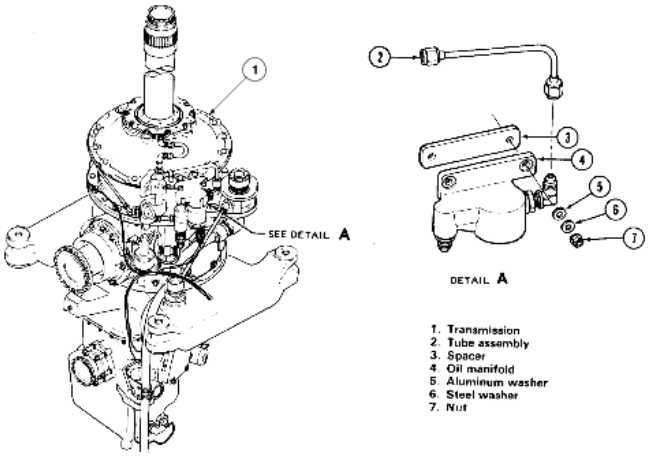
ICN-412-A-632201-B-97499-00013-A-001-01

Figure 1. Oil Pressure Manifold (Sheet 1 of 1)

Oil Pressure Manifold - Inspection

Procedure

1. Inspect oil manifold (4, Figure 1) for damaged threads and for mechanical or corrosion damage. Damage in excess of superficial is unacceptable. Corrosion damage shall be removed.



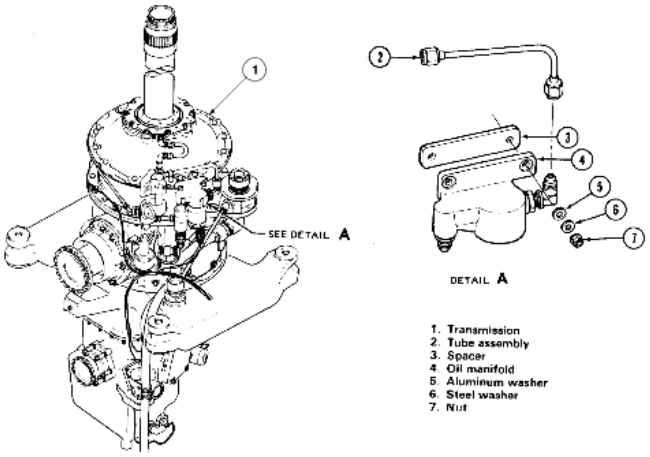
ICN-412-A-632201-B-97499-00013-A-001-01

Figure 1. Oil Pressure Manifold (Sheet 1 of 1)

Oil Pressure Manifold - Repair

Procedure

1. Repair of oil manifold (4, Figure 1) is limited to removal of superficial corrosion from areas exposed on exterior of transmission.



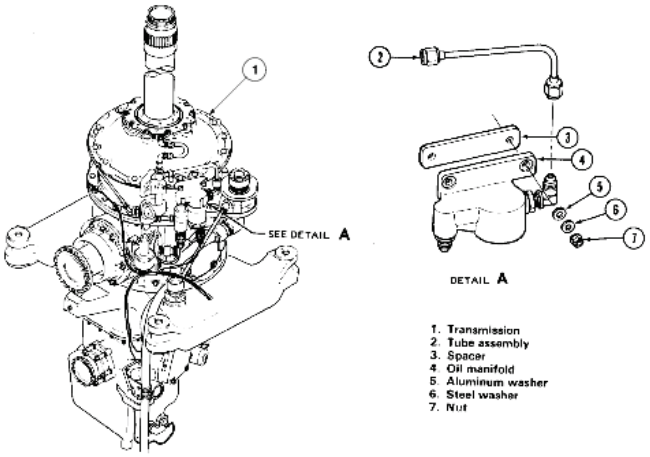
ICN-412-A-632201-B-97499-00013-A-001-01

Figure 1. Oil Pressure Manifold (Sheet 1 of 1)

Oil Pressure Manifold - Installation

Procedure

1. Position spacer (3, Figure 1) and oil manifold (4) on transmission (1).
2. Install aluminum washers (5) next to oil manifold (4), then install steel washers (6) and nuts (7).
3. Install tube assembly (2).
4. Install oil pressure transmitter (412-A-63-22-01-11A-720A-A / 01083) and oil pressure switch (412-A-63-22-01-12A-720A-A / 01086).
5. Perform operational check of oil pressure switch and transmitter (412-A-96-11-00-03A-320A-A / 01951) and check for oil leaks at next helicopter ground run.



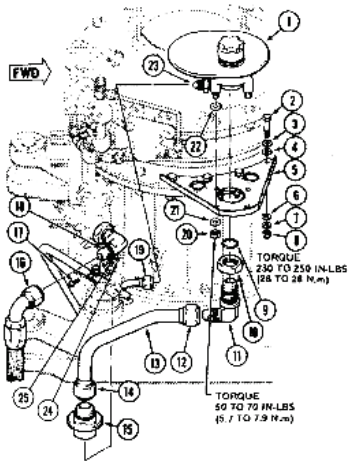
ICN-412-A-632201-B-97499-00013-A-001-01

Figure 1. Oil Pressure Manifold (Sheet 1 of 1)

Filler Neck, Scupper, and Mounting Bracket - Removal

Procedure

1. Disconnect hose assembly (16, Figure 1) from elbow (18).
2. Remove nut (24), washer (25), and screw (17).
3. Disconnect tube assembly (19).
4. Loosen and remove nut (14) from extension cap (15), and loosen nut (12) from elbow (11) and nut (14) from extension cap (15). Remove tube (13).
5. Loosen nut (10) and remove elbow (11) with packing (9). Discard packing.
6. Remove nuts (20) and aluminum washers (21). Remove scupper assembly (1) from bracket (5) and remove aluminum washers (22) from bracket.
7. Remove nuts (8) and washers (6 and 7) from bolts (2), and remove bracket (5).
8. Remove bolts (2) and washers (3 and 4) from bracket (5).



1. Scupper assembly
2. Bolt
3. Thin steel washer
4. Thin aluminum washer
5. Bracket
6. Thin aluminum washer
7. Thin steel washer
8. Nut
9. Packing
10. Nut
11. Elbow
12. Nut
13. Tube
14. Nut
15. Extension cap
16. Hose assembly
17. Screw
18. Elbow
19. Tube assembly
20. Nut
21. Aluminum washer
22. Aluminum washer
23. Fitting
24. Nut
25. Washer

ICN-412-A-632201-B-97499-00014-A-001-01

Figure 1. Filler Neck, Scupper, and Mounting Bracket (Sheet 1 of 1)

Filler Neck, Scupper, and Mounting Bracket - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Wipe all parts with clean cloth dampened with drycleaning solvent (C-304).
2. Dry using filtered, compressed air.

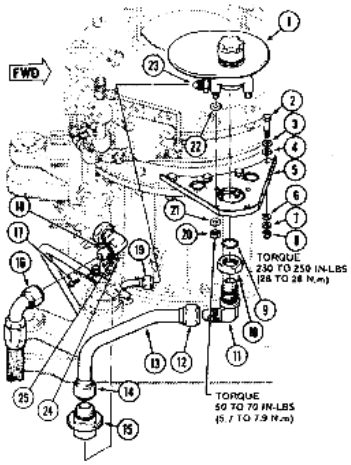
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Filler Neck, Scupper, and Mounting Bracket - Inspection

Procedure

1. Inspect scupper (1, Figure 1) and oil filler cap for cracks and distortion or other damage rendering either component unserviceable. A loose fitting cap and/or cracked or distorted scupper is not acceptable. Unserviceable components shall be replaced. Superficial damage such as dents, nicks, or scratches not affecting serviceability are acceptable without repair. Light surface corrosion is acceptable provided it is removed and original surface finish restored.

2. Inspect bracket (5) and tube (13) for damage such as distortion and cracks. Loose fitting, cracked, and/or severely distorted parts are unacceptable and shall be replaced. Superficial dents, nicks, and scratches not affecting serviceability are acceptable without repair.



1. Scupper assembly
2. Bolt
3. Thin steel washer
4. Thin aluminum washer
5. Bracket
6. Thin aluminum washer
7. Thin steel washer
8. Nut
9. Packing
10. Nut
11. Elbow
12. Nut
13. Tube
14. Nut
15. Extension cap
16. Hose assembly
17. Screw
18. Elbow
19. Tube assembly
20. Nut
21. Aluminum washer
22. Aluminum washer
23. Fitting
24. Nut
25. Washer

ICN-412-A-632201-B-97499-00014-A-001-01

Figure 1. Filler Neck, Scupper, and Mounting Bracket (Sheet 1 of 1)

Filler Neck, Scupper, and Mounting Bracket - Installation

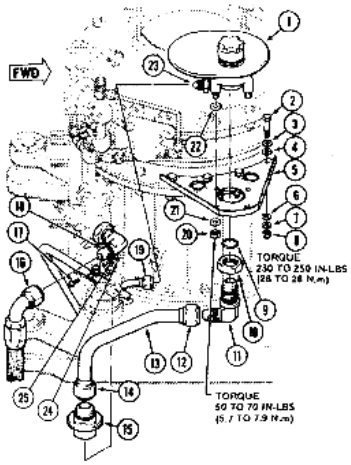
Procedure

1. Install aluminum washers (22, Figure 1) on scupper assembly (1). Position scupper assembly on bracket (5).
2. Install aluminum washers (21) and nuts (20). Torque nuts (20) to 50 to 70 inch-pounds (5.6 to 7.9 Nm).
3. Install nut (10) on elbow (11).
4. Lubricate packing (9) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packing on elbow (11).
5. Install elbow (11) on scupper assembly (1). Do not tighten nut (10) at this time.
6. Position bracket (5) on transmission and install bolts (2) with thin steel washers (3) under bolt heads and thin aluminum washers (4) next to bracket (5).
7. Install aluminum washers (6) next to transmission case with thin steel washer (7) next to nuts (8). Torque nuts (8) to 230 to 250 inch-pounds (25.9 to 28.2 Nm).
8. Install tube (13) on elbow (11). Tighten nuts (12 and 14) and tighten nut (10).
9. Connect tube assembly (19) to fitting (23).
10. Install screw (17), washer (25), and nut (24).
11. Connect hose assembly (16) to elbow (18).

NOTE

Extension cap (15) may be installed by inserting cap on transmission, pressing down on cap, and twisting clockwise to engage lock.

12. Check for leaks at next helicopter ground run.



1. Scupper assembly
2. Bolt
3. Thin steel washer
4. Thin aluminum washer
5. Bracket
6. Thin aluminum washer
7. Thin steel washer
8. Nut
9. Packing
10. Nut
11. Elbow
12. Nut
13. Tube
14. Nut
15. Extension cap
16. Hose assembly
17. Screw
18. Elbow
19. Tube assembly
20. Nut
21. Aluminum washer
22. Aluminum washer
23. Fitting
24. Nut
25. Washer

ICN-412-A-632201-B-97499-00014-A-001-01

Figure 1. Filler Neck, Scupper, and Mounting Bracket (Sheet 1 of 1)

Oil Pump - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Slide Hammer with 1/4 X 28 UNF Threaded Shaft	Commercial	1	

Procedure

1. Remove access door in front of pylon support in cabin area.

2. Drain transmission oil (412-A-12-10-03-00A-040A-A / 00135) from transmission sump assembly (1, Figure 1).

3. Cut and remove lockwire securing oil pump screen (18) and remove oil pump screen with packing (17). Discard packing.

4. Remove tubes (11 and 16).

5. Remove nut (9) and steel washer (8). Remove bracket (10), tee connection (15), nut (12), and washer (13) together as an assembly.

6. Clean sealant from puller hole in bottom center of pump (4) using a sharp plastic scraper.



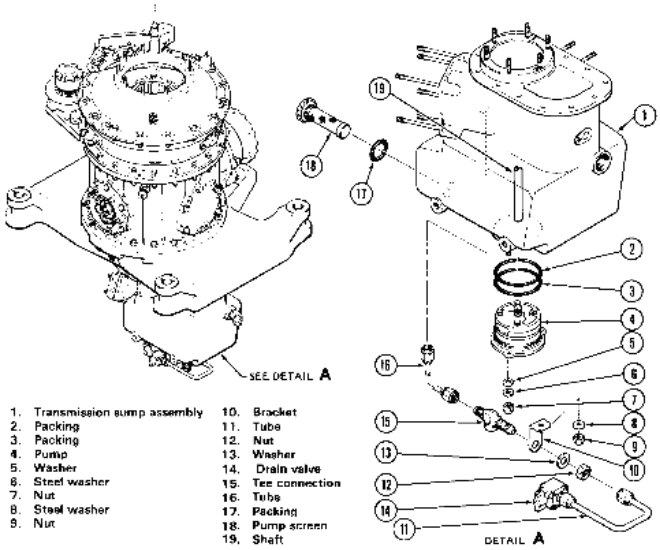
DO NOT USE JACKSCREWS TO REMOVE PUMP (4). DO NOT ALLOW PULLER BOLT TO BOTTOM OUT IN THREADED HOLE OF PUMP.

7. Install slide hammer with 1/4 x 28 unf threaded shaft into center of pump base.

8. Loosen jamnut on drain valve (14) and rotate counterclockwise.

9. Remove remaining nuts (7) with washers (5 and 6) and remove pump (4) with packings (2 and 3).

10. Remove and discard packings (2 and 3).



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Figure 1. Transmission Oil Pump (Sheet 1 of 1)

Oil Pump - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

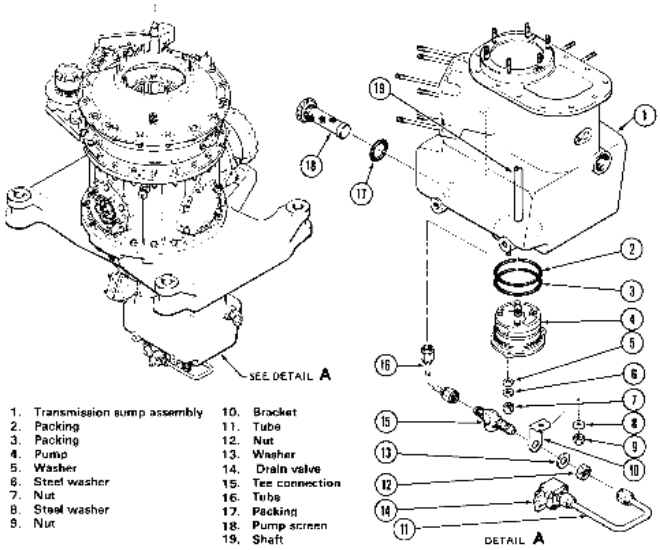
Procedure

1. Lubricate packings (2 and 3, Figure 1) and mating surfaces of pump (4) and transmission sump assembly (1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packings on pump.

NOTE

The main rotor can be manually rotated, as necessary, to obtain alignment of oil pump shaft.

2. Warm sump case port with a heat lamp and insert pump into mounting port. Align pump shaft with internal splines of transmission shaft (19). Use a flashlight to verify engagement of pump and shaft (19) through open port of oil pump screen (18).
3. Install bracket (10), tee connection (15), washer (13), and nut (12) on pump mounting stud.
4. Install steel washer (8) and nut (9). Position drain valve (14), as required, and tighten jamnut.
5. Install washers (5), steel washers (6), and nuts (7). Torque nuts (7) and nut (9) to 50 to 70 inch-pounds (5.6 to 7.0 Nm).
6. Connect tubes (11 and 16).
7. Lubricate packing (17) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packing on oil pump screen (18).
8. Install oil pump screen (18) and torque oil pump screen to 300 to 400 inch-pounds (33.9 to 45.1 Nm).
9. Secure oil pump screen (18) to electric chip detector base with lockwire (C-405).
10. Service transmission (412-A-12-10-03-00A-040A-A / 00135).
11. Install access door.
12. Check for oil leaks at next helicopter ground run.



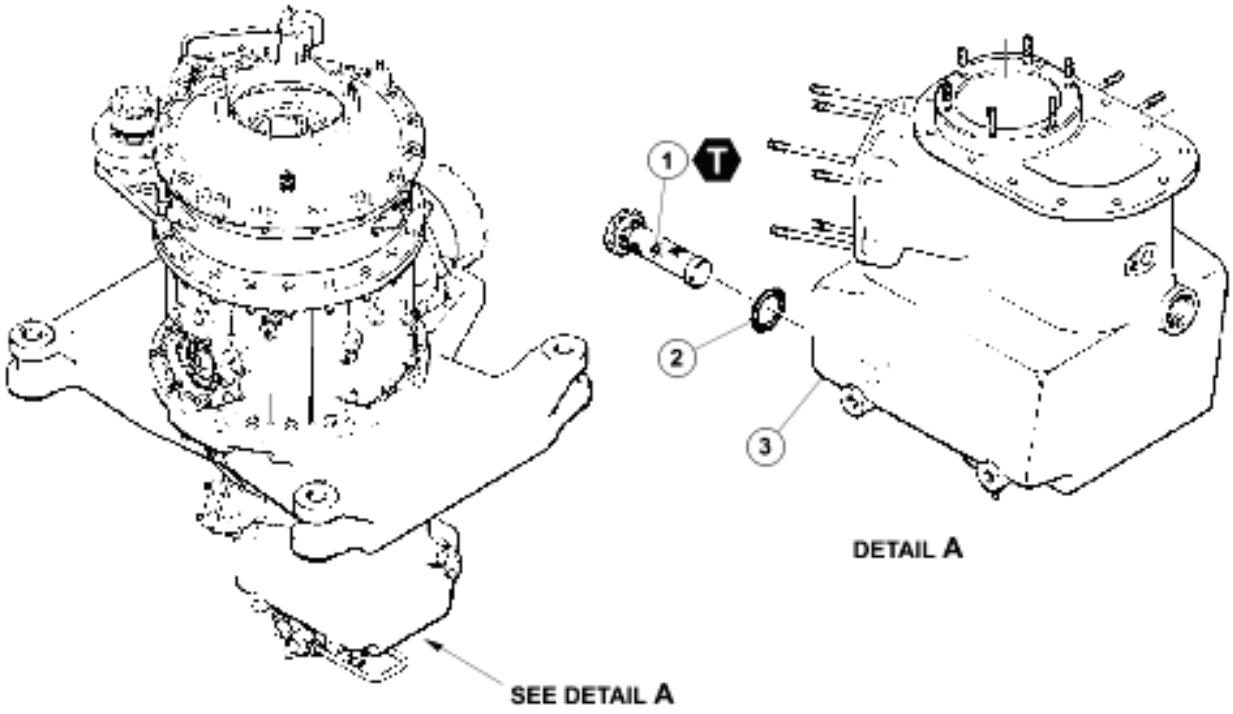
ICN-412-A-632201-B-97499-00015-A-001-01

Figure 1. Transmission Oil Pump (Sheet 1 of 1)

Oil Pump Inlet Screen - Removal

Procedure

1. Drain transmission oil sump (412-A-12-10-03-00A-040A-A / 00135).
2. Cut and remove lockwire from oil pump inlet screen (1, Figure 1).
3. Remove oil pump inlet screen (1) from transmission sump assembly (3).
4. Remove and discard gasket (2).



1. Screen
2. Gasket
3. Transmission sump assembly

T 300 TO 400 IN-LBS
(33.9 TO 45.1 Nm)

ICN-412-A-632201-B-97499-00016-A-001-01

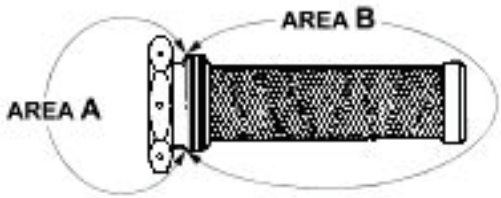
Figure 1. Oil Pump Inlet Screen (Sheet 1 of 1)

Oil Pump Inlet Screen - Inspection

Procedure

1. Inspect oil pump inlet screen (Figure 1) for metal particles. If metal particles are present, refer to 412-A-63-00-00-01A-280A-A / 00989.

2. Inspect inlet oil pump inlet screen (1, Figure 2) for damage in accordance with Figure 1.

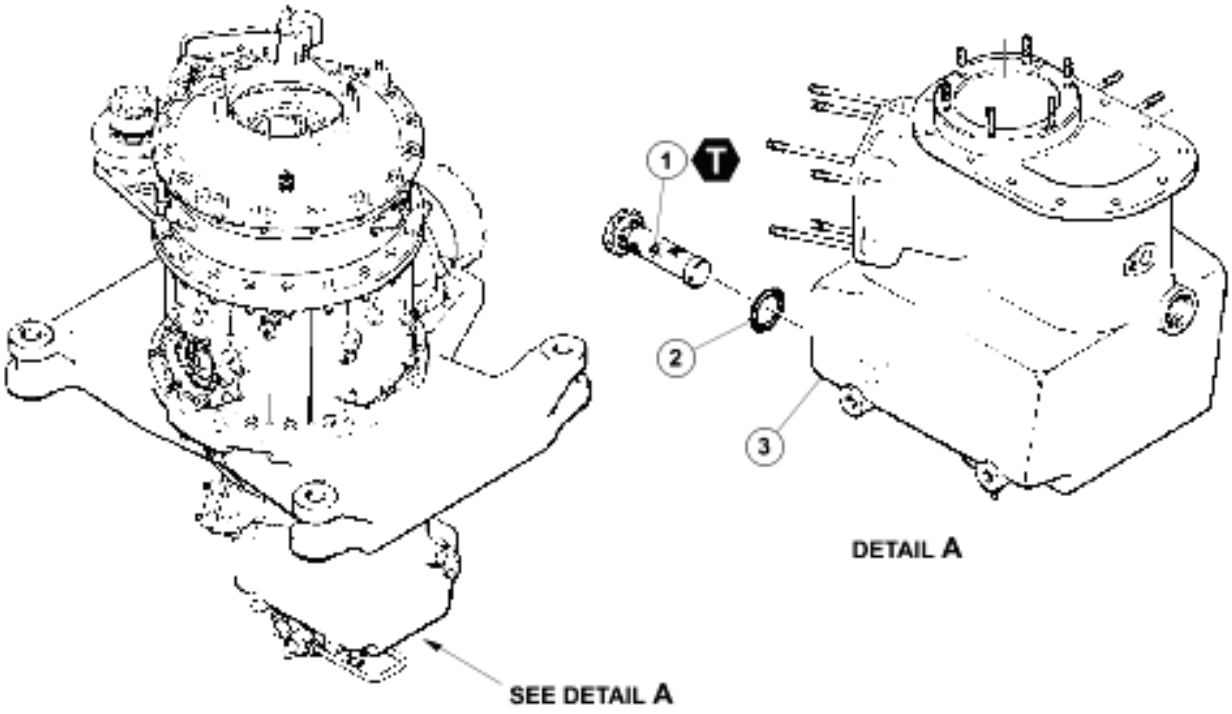


SCREEN
204-040-237-001 AND -009

AREA	LIMITS
All	No cracks allowed.
A	Maximum depth of pitting is 0.030 inch (0.762 mm) with no more than 40% of any 1.0 square inch (645.16 mm^2) or 20% of any total area of any pitted surface.
B	Maximum depth of pitting is 0.020 inch (0.508 mm) with no more than 40% of any 1.0 square inch (645.16 mm^2) or 20% of any total area of any pitted surface. Corrosion is not permitted on screen. Thread damage is not permitted.

ICN-412-A-632201-B-97499-00017-A-001-01

Figure 1. Oil Pump Inlet Screen - Damage Limits (Sheet 1 of 1)



1. Screen
2. Gasket
3. Transmission sump assembly

T 300 TO 400 IN-LBS
(33.9 TO 45.1 Nm)

ICN-412-A-632201-B-97499-00016-A-001-01

Figure 2. Oil Pump Inlet Screen (Sheet 1 of 1)

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Oil Pump Inlet Screen - Cleaning

Preliminary requirements

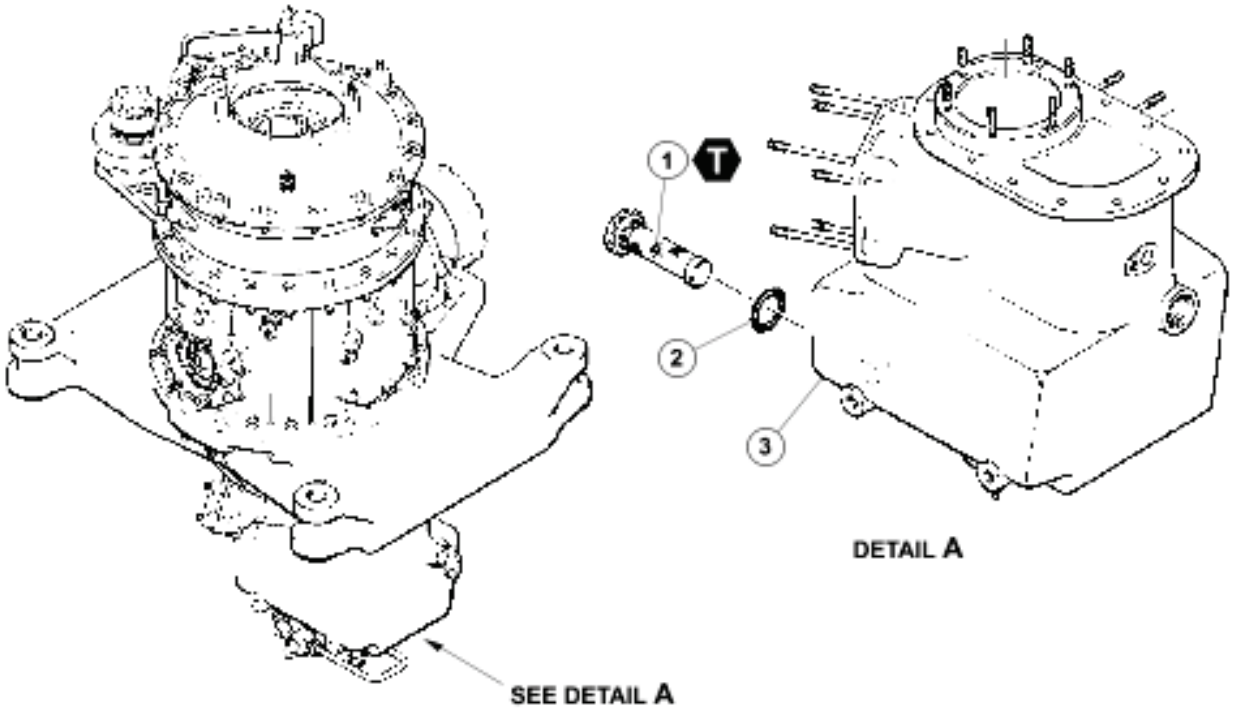
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Rinse inlet oil pump screen (1, Figure 1) with drycleaning solvent (C-304).

2. Dry using filtered, compressed air.



1. Screen
2. Gasket
3. Transmission sump assembly

T 300 TO 400 IN-LBS
(33.9 TO 45.1 Nm)

ICN-412-A-632201-B-97499-00016-A-001-01

Figure 1. Oil Pump Inlet Screen (Sheet 1 of 1)

Oil Pump Inlet Screen - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

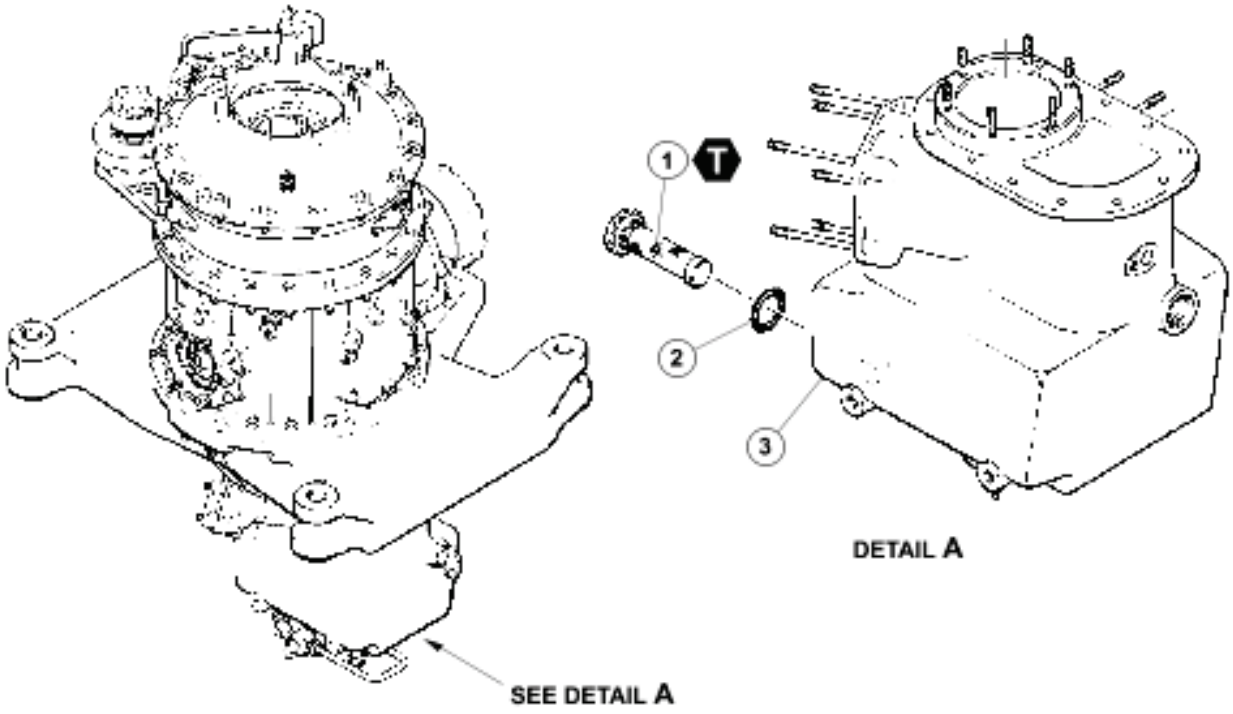
1. *Lubricate new gasket (2, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install gasket (2) on oil pump inlet screen (1).*

2. Install oil pump inlet screen (1) on transmission sump assembly (3).

3. Torque oil pump inlet screen (1) to 300 to 400 inch-pounds (33.9 to 45.1 Nm).

4. Secure oil pump inlet screen (1) to transmission sump assembly (3) with lockwire (C-405).

5. Check for oil leaks at next helicopter ground run.



1. Screen
2. Gasket
3. Transmission sump assembly

T 300 TO 400 IN-LBS
(33.9 TO 45.1 Nm)

ICN-412-A-632201-B-97499-00016-A-001-01

Figure 1. Oil Pump Inlet Screen (Sheet 1 of 1)

Sump Case Internal Filter - Description

Applicable: SN: 33001-33214 TB 412-91-94: Pre:

The maintenance procedures in 412-A-63-22-01-17A-520A-A / 01103, 412-A-63-22-01-17A-280A-A / 01104, 412-A-63-22-01-17A-251A-A / 01105 and 412-A-63-22-01-17A-720A-A / 01106 apply. For maintenance procedures applicable to helicopters modified by subject TB (installation of full flow debris monitor), refer to 412-A-63-22-01-18A-042A-A / 01107.

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Sump Case Internal Filter - Removal

Procedure

NOTE

For maintenance procedures applicable to helicopters modified by subject TB (installation of full flow debris monitor), refer to 412-A-63-22-01-18A-042A-A / 01107.

1. Remove access door on right side of pylon support in cabin area.

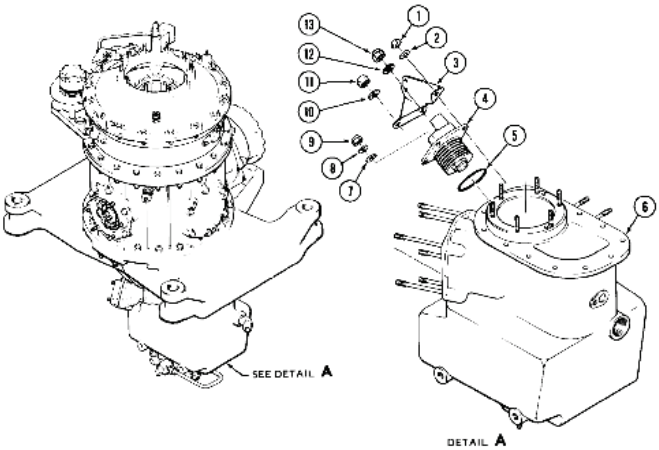
2. Remove nuts (1, 11, and 13, Figure 1) with washers (2, 10, and 12). Remove bracket (3).

3. Remove nut (9) with washers (7 and 8).

4. Remove filter assembly (4).

5. Remove and discard packing (5) from filter assembly (4).

6. Cap or cover filter mounting port in sump case (6).



1. Nut
2. Thin steel washer
3. Bracket
4. Filter assembly
5. Packing
6. Sump case
7. Thin aluminium washer
8. Thin steel washer
9. Nut
10. Thin steel washer
11. Nut
12. Thin steel washer
13. Nut

ICN-412-A-632201-B-97499-00018-A-001-01

Figure 1. Sump Case Internal Filter (Pre TB 412-91-94) (Sheet 1 of 1)

Sump Case Internal Filter - Inspection

Procedure

NOTE

For maintenance procedures applicable to helicopters modified by subject TB (installation of full flow debris monitor), refer to 412-A-63-22-01-18A-042A-A / 01107.

NOTE

If the sump case internal filter must be disassembled for inspection, refer to 412-A-63-22-01-17A-251A-A / 01105 and perform step 1.1 and step 1.2.

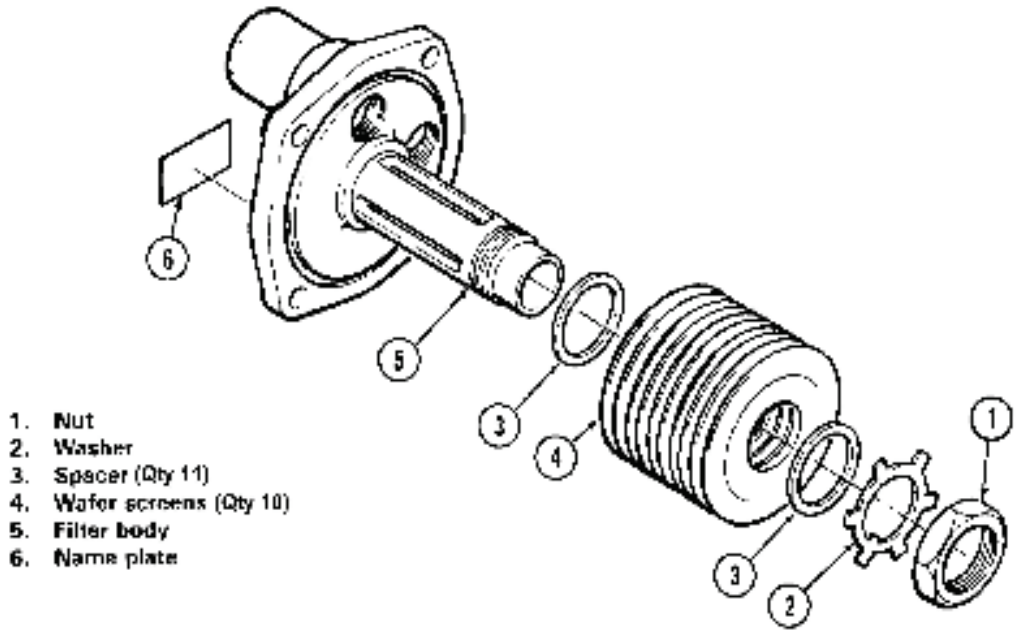
1. Inspect filter for metal particles and/or other material. If metal particles are found, refer to 412-A-63-00-00-01A-280A-A / 00989.

2. Inspect wafer screens (4, Figure 1) for distortion or tears. Damage that can be detected visually is not acceptable.

3. Inspect spacers (3) for corrosion and deformation. Damage in excess of superficial is not acceptable.

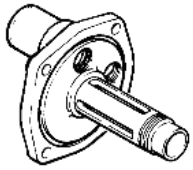
4. Inspect nut (1) for corrosion and thread damage. Damage in excess of superficial is not acceptable.

5. Inspect filter body (5) for mechanical and corrosion damage (Figure 2).

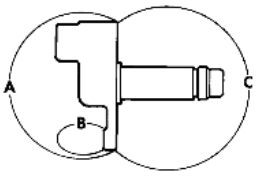


ICN-412-A-632201-B-97499-00019-A-001-01

Figure 1. Sump Case Internal Filter (Sheet 1 of 1)



FILTER BODY
(02W10116)



AREA	LIMITS
All	No cracks allowed.
A	Maximum depth of pitting is 0.030 inch (0.762 millimeter) with no more than 80% of any 1.0 inch square (645.16 square millimeter) or 30% of any pitted surface.
B	Maximum depth of pitting (including inside of bolt holes) is 0.020 inch (0.508 millimeter) with no more than 20% of area or wall of any hole to be pitted.
C	Maximum depth of pitting is 0.020 inch (0.508 millimeter) with no more than 30% of any 1.0 inch square (645.16 square millimsters) or 30% of total area of any pitted surface. Thread damage is not permitted.

ICN-412-A-632201-B-97499-00020-A-001-01

Figure 2. Sump Case Internal Filter - Damage Limits (Sheet 1 of 1)

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Sump Case Internal Filter - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

NOTE

For maintenance procedures applicable to helicopters modified by subject TB (installation of full flow debris monitor), refer to 412-A-63-22-01-18A-042A-A / 01107.

NOTE

Perform Step 1. and Step 2. if it is not possible to sufficiently clean the filters while assembled.

Perform Step 3. and Step 4. if it is possible to sufficiently clean the filters while assembled.

1. To disassemble filter, proceed as follows:

1.1. Straighten tangs on washer (2, Figure 1) and remove nut (1).

1.2. Remove washer (2), spacers (3), and wafer screens (4).

1.3. Wash wafer screens (4) with drycleaning solvent (C-304). Dry wafer screens with filtered, compressed air.

2. To assemble filter, proceed as follows:

2.1. Install 10 spacers (3) and 10 wafer screens (4). Starting with spacer, alternately install spacers and wafer screens.

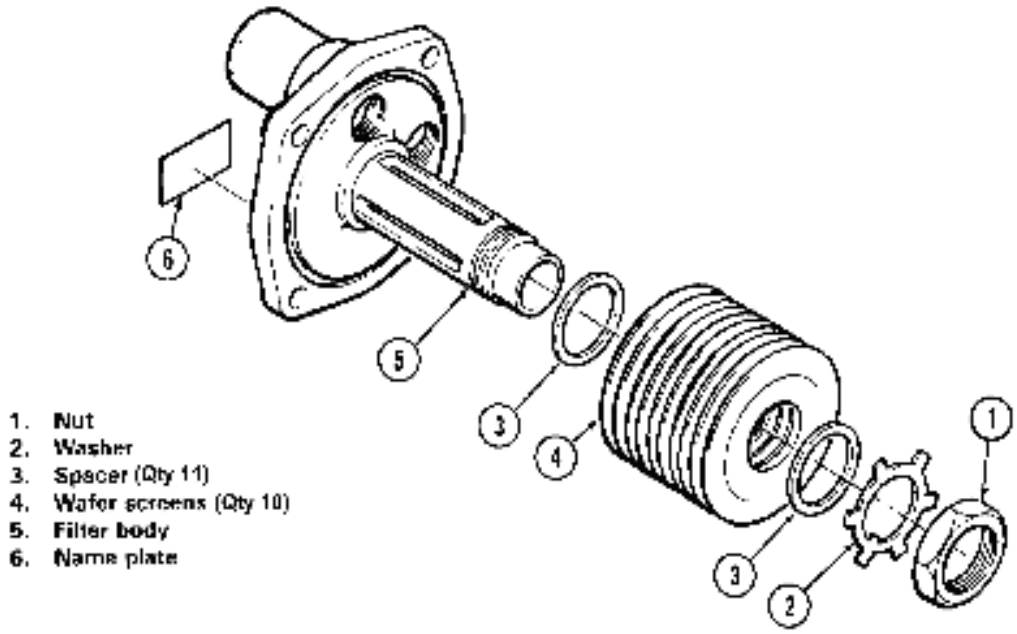
2.2. Install eleventh spacer (3), washer (2), and nut (1).

2.3. Tighten nut (1) until wafer screens (4) will not rotate when turned by hand.

2.4. Bend one tang of thin steel washer (2) over a flat side of nut (1).

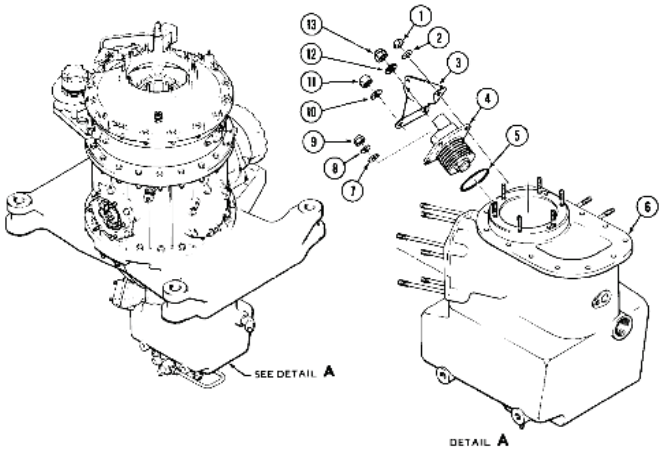
3. Wash filter assembly (4, Figure 2) with drycleaning solvent (C-304).

4. Dry with filtered, compressed air.



ICN-412-A-632201-B-97499-00019-A-001-01

Figure 1. Sump Case Internal Filter (Sheet 1 of 1)



1. Nut
2. Thin steel washer
3. Bracket
4. Filter assembly
5. Packing
6. Sump case
7. Thin aluminium washer
8. Thin steel washer
9. Nut
10. Thin steel washer
11. Nut
12. Thin steel washer
13. Nut

ICN-412-A-632201-B-97499-00018-A-001-01

Figure 2. Sump Case Internal Filter (Pre TB 412-91-94) (Sheet 1 of 1)

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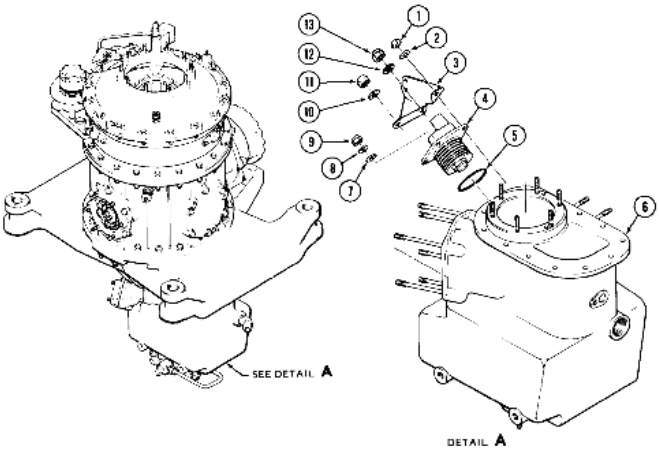
Sump Case Internal Filter - Installation

Procedure

NOTE

For maintenance procedures applicable to helicopters modified by subject TB (installation of full flow debris monitor), refer to 412-A-63-22-01-18A-042A-A / 01107.

1. Lubricate the new packing (5, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install the packing on the filter assembly (4).
2. Insert the filter assembly (4) in the mounting port of the sump case (6).
3. Position the bracket (3) over the filter assembly (4) and the studs on the sump case (6).
4. Install the washers (2, 10, and 12) and the nuts (1, 11, and 13).
5. Install the thin aluminum washer (7) next to the filter housing and the thin steel washer (8) adjacent to the nut (9).
6. Install the nut (9). Torque the nuts (1, 9, 11, and 13) 50 to 70 inch-pounds (5.6 to 7.9 Nm).
7. Check for oil leaks at the next helicopter ground run.



1. Nut
2. Thin steel washer
3. Bracket
4. Filter assembly
5. Packing
6. Sump case
7. Thin aluminium washer
8. Thin steel washer
9. Nut
10. Thin steel washer
11. Nut
12. Thin steel washer
13. Nut

ICN-412-A-632201-B-97499-00018-A-001-01

Figure 1. Sump Case Internal Filter (Pre TB 412-91-94) (Sheet 1 of 1)

Full Flow Debris Monitor - Description

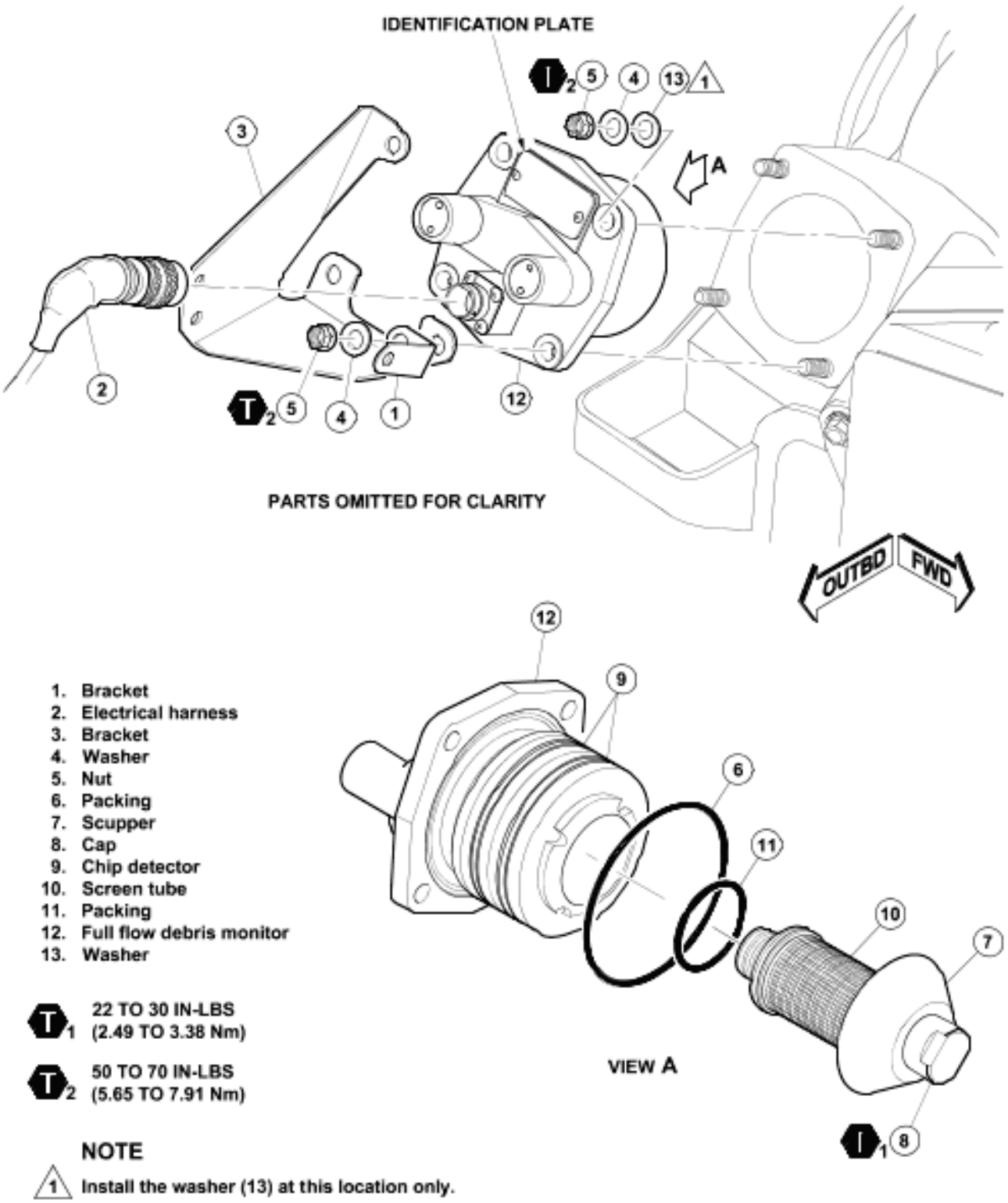
The maintenance procedures in 412-A-63-22-01-18A-520A-A / 01108, 412-A-63-22-01-18A-280A-A / 01109, 412-A-63-22-01-18A-251A-A / 01110 and 412-A-63-22-01-18A-720A-A / 01111 are applicable to the helicopters modified by 412 Classic to 412HP Upgrade kit, 412SP to 412HP Upgrade kit and TB 412-91-94.

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Full Flow Debris Monitor - Removal

Procedure

1. Remove the access door on the right side of the pylon support in the cabin area (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect the electrical harness (2, Figure 1).
3. Remove the nuts (5), washers (4 and 13), and brackets (1 and 3).
4. Remove the full flow debris monitor (12) from the sump case.
5. Remove the cap (8), scupper (7), and screen tube (10).
6. Remove and discard the packings (6 and 11).



ICN-412-A-632201-B-97499-00021-A-001-01

Figure 1. Full Flow Debris Monitor (Sheet 1 of 1)

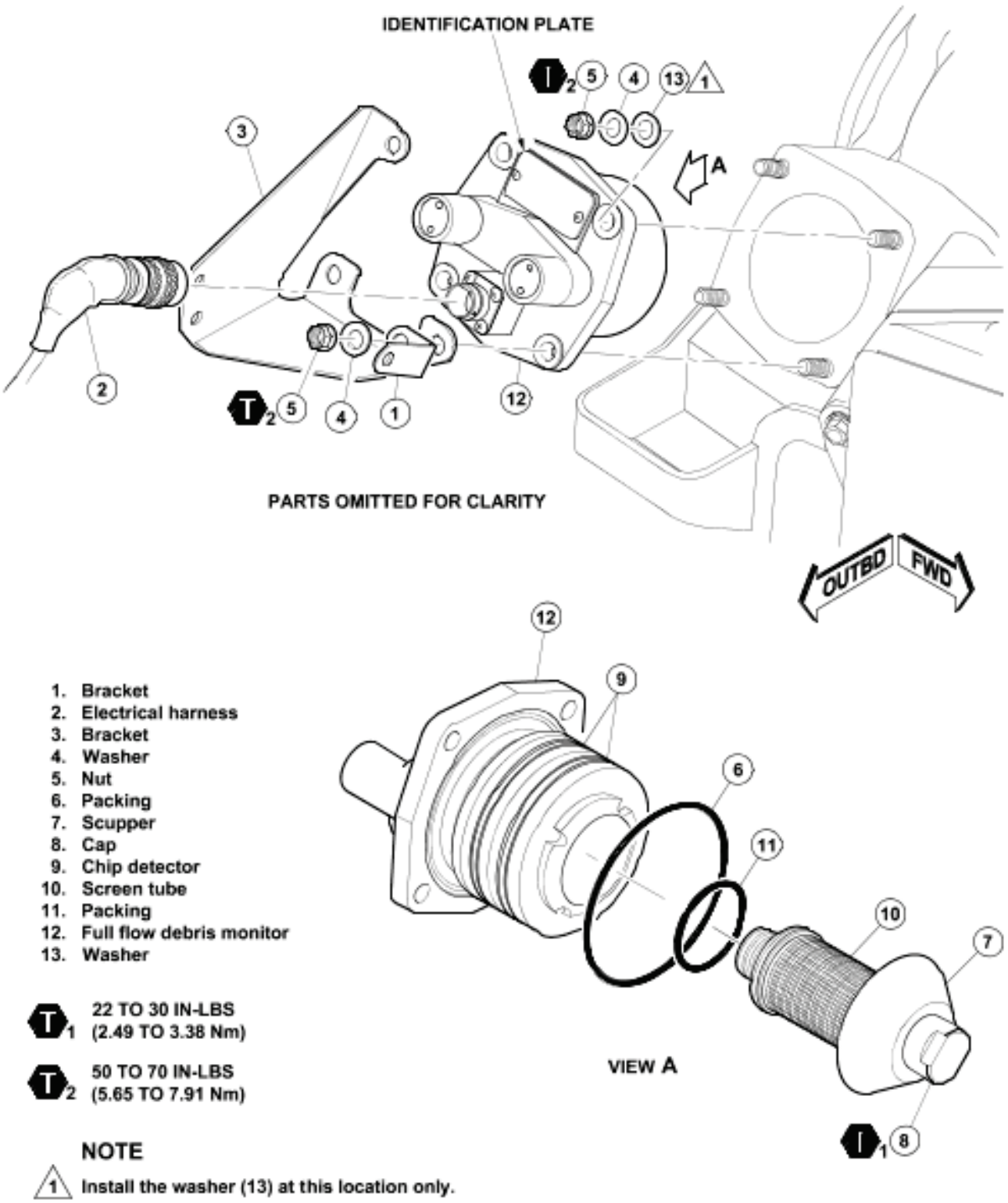
Full Flow Debris Monitor - Inspection

Procedure

1. Inspect the chip detectors (9, Figure 1) and the screen tube (10) for metal particles and/or other material. If metal particles are found, refer to 412-A-63-00-00-01A-280A-A / 00989 and 412-A-63-00-00-01A-310A-A / 00990.

2. Inspect the chip detectors (9) and the screen tube (10) for distortion or tears. Damage that can be detected visually is not acceptable.

3. Inspect the nuts (5) for corrosion and thread damage. In excess of superficial is not acceptable.



ICN-412-A-632201-B-97499-00021-A-001-01

Figure 1. Full Flow Debris Monitor (Sheet 1 of 1)

Full Flow Debris Monitor - Cleaning

Preliminary requirements

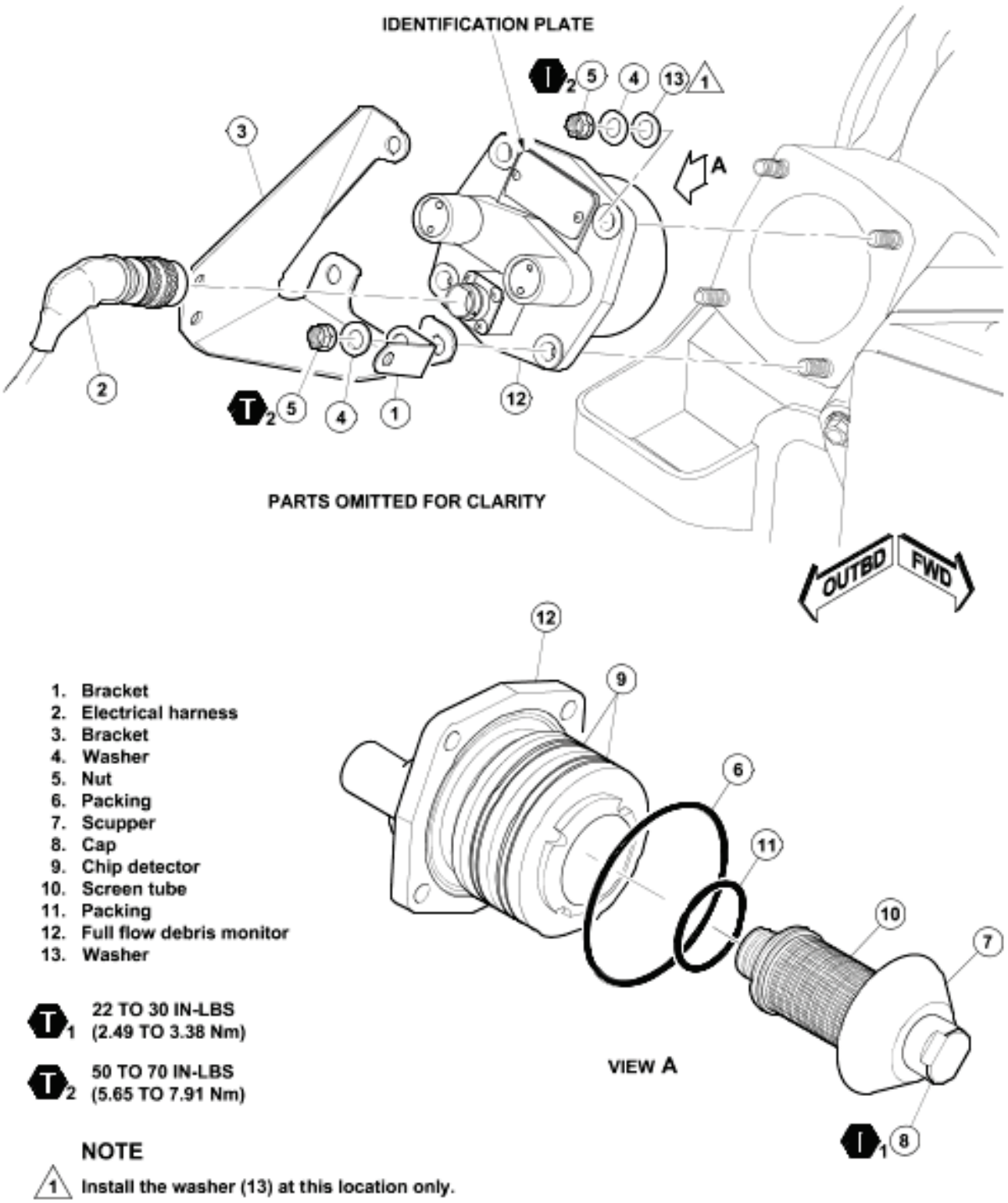
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Wash the chip detectors (9, Figure 1) and the screen tube (10) with drycleaning solvent (C-304).

2. Dry with filtered, compressed air.



ICN-412-A-632201-B-97499-00021-A-001-01

Figure 1. Full Flow Debris Monitor (Sheet 1 of 1)

Full Flow Debris Monitor - Installation

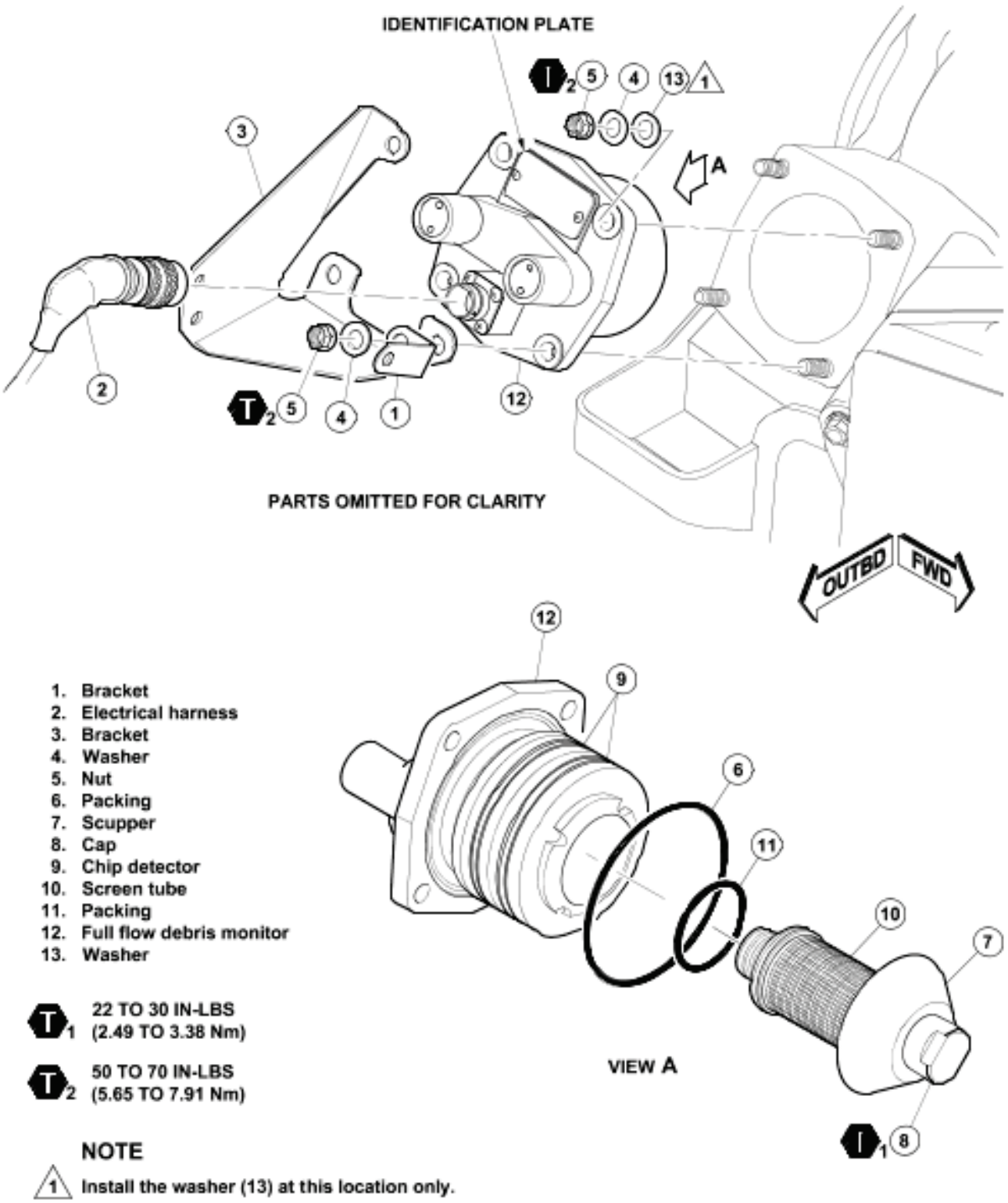
Procedure

1. Lubricate the new packings (6 and 11, Figure 1) with transmission oil (412-A-12-10-03-00A-040A-A / 00135).
2. Install the packings (6 and 11) on the full flow debris monitor (12).
3. Install the screen tube (10), scupper (7), and cap (8) on the full flow debris monitor (12).
4. Torque the cap (8) 22 to 30 inch-pounds (2.5 to 3.4 Nm).
5. Perform an operational check (412-A-96-03-01-12A-320A-A / 02058).



FAILURE TO CORRECTLY INSTALL THE FULL FLOW DEBRIS MONITOR COULD CAUSE LOSS OF TRANSMISSION OIL PRESSURE.

6. Install the full flow debris monitor (12) in the transmission sump case with the connector at the bottom and the data plate at the top.
7. Install the brackets (1 and 3) with the washers (4 and 13) and the nuts (5).
8. Torque the nuts (5) 50 to 70 inch-pounds (5.6 to 7.0 Nm).
9. Connect the electrical harness (2).
10. Install the access door on the right side of the pylon support in the cabin area.
11. Check for oil leaks at the next helicopter ground run.



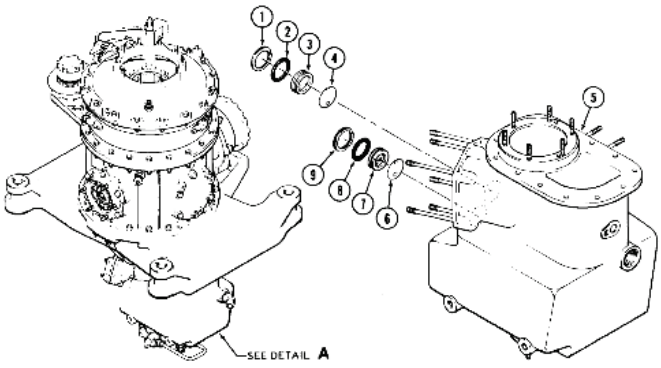
ICN-412-A-632201-B-97499-00021-A-001-01

Figure 1. Full Flow Debris Monitor (Sheet 1 of 1)

Sight Glasses - Description

NOTE

The maintenance procedures in 412-A-63-22-01-19A-520A-A / 01113, 412-A-63-22-01-19A-251A-A / 01114, 412-A-63-22-01-19A-280A-A / 01115 and 412-A-63-22-01-19A-720A-A / 01116 are detailed for the indicator (4, Figure 1). Maintenance procedures for the indicator (6) are similar.



1. Retainer ring
2. Packing
3. Glass
4. Indicator (204-040-508-5)
5. Transmission sump case
6. Indicator (204-040-508-7)
7. Glass
8. Packing
9. Retainer ring

ICN-412-A-632201-B-97499-00022-A-001-01

Figure 1. Sight Glass (Sheet 1 of 1)

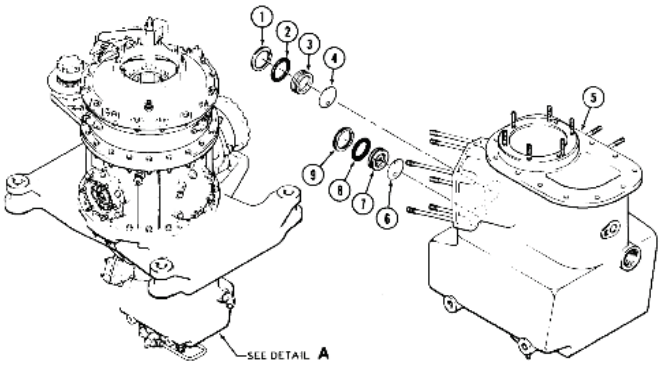
Sight Glasses - Removal

Procedure

NOTE

The following removal procedures are detailed for the indicator (4, Figure 1). The removal procedures for the indicator (6) are similar.

1. Drain the transmission oil as necessary (412-A-12-10-03-00A-040A-A / 00135) to a level below the sight gauge.
2. Remove the retainer ring (1, Figure 1).
3. Carefully remove the glass (3) with the packing (2).
4. Remove the indicator (4).
5. Remove and discard the packing (2).
6. Cover the opening in the transmission sump case (5).



1. Retainer ring
2. Packing
3. Glass
4. Indicator (204-040-508-5)
5. Transmission sump case
6. Indicator (204-040-508-7)
7. Glass
8. Packing
9. Retainer ring

ICN-412-A-632201-B-97499-00022-A-001-01

Figure 1. Sight Glass (Sheet 1 of 1)

Sight Glasses - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

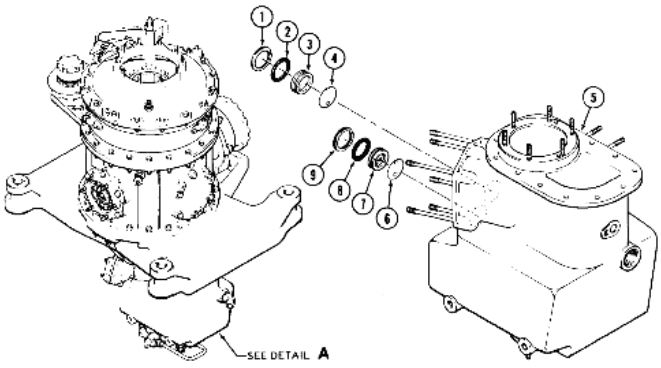
Procedure

NOTE

The following cleaning procedures are detailed for the indicator (4, Figure 1). The cleaning procedures for the indicator (6) are similar.

1. Wipe the components with a clean cloth dampened with drycleaning solvent (C-304).

2. Dry with filtered, compressed air.



1. Retainer ring
2. Packing
3. Glass
4. Indicator (204-040-508-5)
5. Transmission sump case
6. Indicator (204-040-508-7)
7. Glass
8. Packing
9. Retainer ring

ICN-412-A-632201-B-97499-00022-A-001-01

Figure 1. Sight Glass (Sheet 1 of 1)

Sight Glasses - Inspection

Procedure

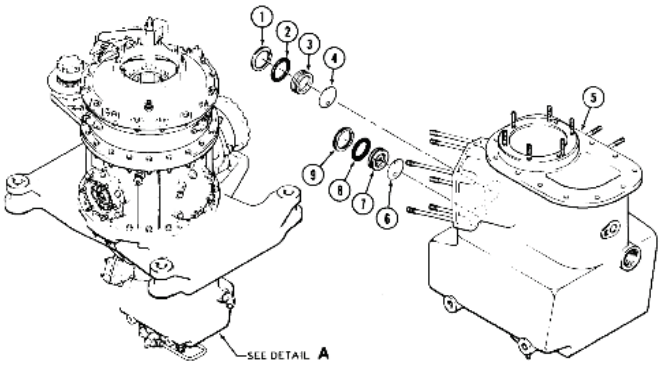
NOTE

The following inspection procedures are detailed for the indicator (4, Figure 1). The inspection procedures for the indicator (6) are similar.

1. Inspect the glass (3, Figure 1) for cracks, crazing, and stains, which would prevent an accurate determination of oil level. If any of these conditions are present, replace the glass.

2. Inspect the indicator (4) for stains that might present a false oil level reading. If such a condition exists, replace the indicator.

3. Inspect the retainer ring (1) for distortion, cracks, tool marks, and insufficient tension. If any of these conditions exist, replace the retainer.



1. Retainer ring
2. Packing
3. Glass
4. Indicator (204-040-508-5)
5. Transmission sump case
6. Indicator (204-040-508-7)
7. Glass
8. Packing
9. Retainer ring

ICN-412-A-632201-B-97499-00022-A-001-01

Figure 1. Sight Glass (Sheet 1 of 1)

Sight Glasses - Installation

Procedure

NOTE

The following installation procedures are detailed for the indicator (4, Figure 1). The installation procedures for the indicator (6) are similar.

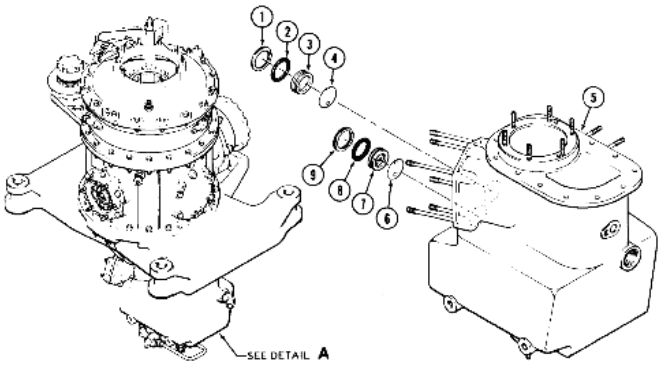
1. Lubricate the new packing (2, Figure 1) with transmission oil (412-A-12-10-03-00A-040A-A / 00135). Install the indicator (4) in the sump.

2. Install the new packing (2) on the sight glass (3) and insert the glass in the sump over the indicator (4).

3. Install the retainer ring (1).

4. Service the transmission (412-A-12-10-03-00A-040A-A / 00135).

5. Check for oil leaks at the next helicopter ground run.



1. Retainer ring
2. Packing
3. Glass
4. Indicator (204-040-508-5)
5. Transmission sump case
6. Indicator (204-040-508-7)
7. Glass
8. Packing
9. Retainer ring

ICN-412-A-632201-B-97499-00022-A-001-01

Figure 1. Sight Glass (Sheet 1 of 1)

Transmission Oil Cooling System - Description

The transmission oil cooling system detailed in this chapter consists of an external oil filter, thermal oil bypass valve, two drain valves, and connecting tubes and hoses. Maintenance procedures applicable to the oil cooler and blower is described in 412-A-79-20-00-04A-280A-A / 01778.

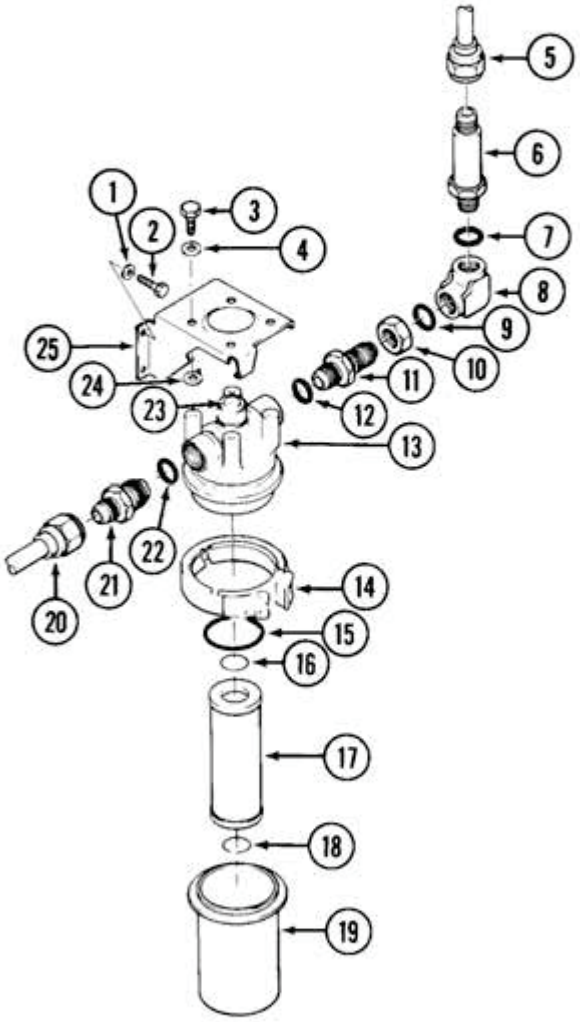
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External Oil Filter - Description

The external oil filter (Figure 1) is located inside the right pylon support below the sump case and is connected to an oil cooler outlet line. The filter element is replaceable and is made of pleated paper. The filter head includes a bypass valve to assure oil flow in the event of filter clogging. If a bypass condition is impending, a red indicator button extends to provide a positive visual indication.

If external oil filter is found with an impending bypass indication, some components may require replacement, refer to 412-A-63-00-00-01A-280A-A / 00989.

1. Thin aluminum washers (4 places)
2. Bolt (4 places)
3. Bolt (4 places)
4. Thin aluminum washers (4 places)
5. Outlet hose
6. Fitting
7. Packing
8. Elbow
9. Packing
10. Checknut
11. Union
12. Packing
13. Filter head
14. V-band clamp
15. Packing
16. Packing
17. Filter element
18. Packing
19. External oil filter body
20. Inlet tube
21. Reducer
22. Packing
23. Bypass indicator
24. Spacer (4 places)
25. Filter bracket



ICN-412-A-632202-B-97499-00001-A-001-01

Figure 1. Transmission External Oil Filter (Sheet 1 of 1)

External Oil Filter - Removal

Procedure

1. *Remove access door on front of pylon support in cabin area. Put a container under external oil filter body (19, Figure 1) to catch oil spillage.*

2. *Open v-band clamp (14). Remove external oil filter body (19).*

3. Remove filter element (17) and make a visual check for evidence of contamination of metal particles.

4. Remove packing (16) from center boss under filter head (13). Remove packing (18) from boss inside bottom of external oil filter body (19). Discard packings (16) and (18).

5. Remove and discard packing (15).

6. If it is necessary to remove the filter head (13), perform the following:

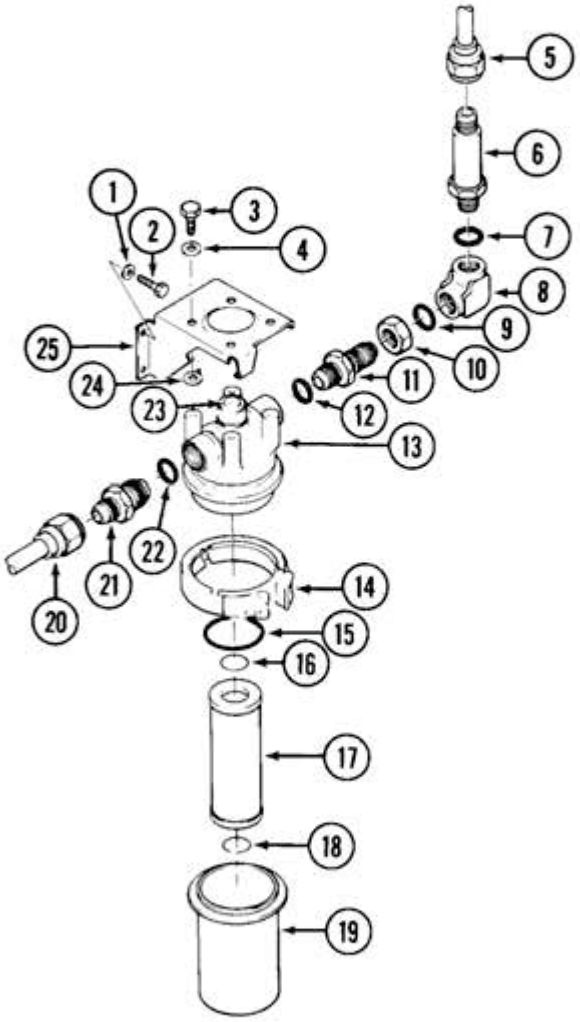
6.1. Disconnect inlet tube (20) and outlet hose (5).

6.2. Cut and remove lockwire from bolts (3).

6.3. Remove bolts (3), thin aluminum washers (4), and spacers (24) from filter bracket (25). Remove filter head (13).

7. Cap or plug all open lines and hoses to prevent contamination.

1. Thin aluminum washers (4 places)
2. Bolt (4 places)
3. Bolt (4 places)
4. Thin aluminum washers (4 places)
5. Outlet hose
6. Fitting
7. Packing
8. Elbow
9. Packing
10. Checknut
11. Union
12. Packing
13. Filter head
14. V-band clamp
15. Packing
16. Packing
17. Filter element
18. Packing
19. External oil filter body
20. Inlet tube
21. Reducer
22. Packing
23. Bypass indicator
24. Spacer (4 places)
25. Filter bracket



ICN-412-A-632202-B-97499-00001-A-001-01

Figure 1. Transmission External Oil Filter (Sheet 1 of 1)

External Oil Filter - Inspection

Procedure

NOTE

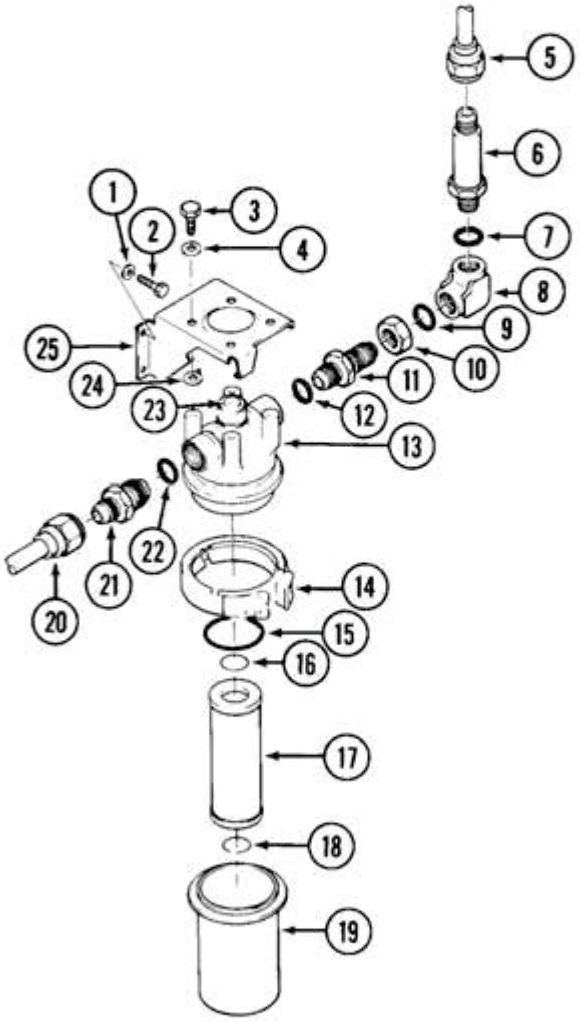
Inspect oil filter components both before and after cleaning.

1. Inspect oil filter element (17, Figure 1) for contamination. If metal particles are present, refer to 412-A-63-00-00-01A-280A-A / 00989. Do not reuse element.

2. Inspect flange areas of external oil filter body (19) and filter head (13) for cracks. If cracks are found, replace part.

3. Inspect area contacted by packing (15) for damage such as distortion and sharp edges. Minor surface irregularities may be removed. Distortion or sharp edges necessitates replacement.

1. Thin aluminum washers (4 places)
2. Bolt (4 places)
3. Bolt (4 places)
4. Thin aluminum washers (4 places)
5. Outlet hose
6. Fitting
7. Packing
8. Elbow
9. Packing
10. Checknut
11. Union
12. Packing
13. Filter head
14. V-band clamp
15. Packing
16. Packing
17. Filter element
18. Packing
19. External oil filter body
20. Inlet tube
21. Reducer
22. Packing
23. Bypass indicator
24. Spacer (4 places)
25. Filter bracket



ICN-412-A-632202-B-97499-00001-A-001-01

Figure 1. Transmission External Oil Filter (Sheet 1 of 1)

External Oil Filter - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

NOTE

Inspect oil filter components both before and after cleaning.

1. Wipe components with a clean cloth damp with drycleaning solvent (C-304).

2. Dry with filtered, compressed air.

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External Oil Filter - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. If removed, install filter head (13, Figure 1) with outlet port aft. Install spacers (24) between filter bracket (25) and filter head (13). Install bolts (3), thin aluminum washers (4), and secure with lockwire (C-405).

2. Remove plugs or caps and install inlet tube (20) from thermal bypass valve to reducer (21) in filter inlet.

3. Install outlet hose (5) to fitting (6) in filter outlet.

4. Lubricate packings (15, 16, and 18) with transmission oil (412-A-12-01-00-00A-040A-A / 00141).

5. Place packing (18) on boss inside bottom of external oil filter body (19). Insert new filter element (17) and seat firmly on boss. Place large packing (15) around upper lip of external oil filter body (19) next to flange.

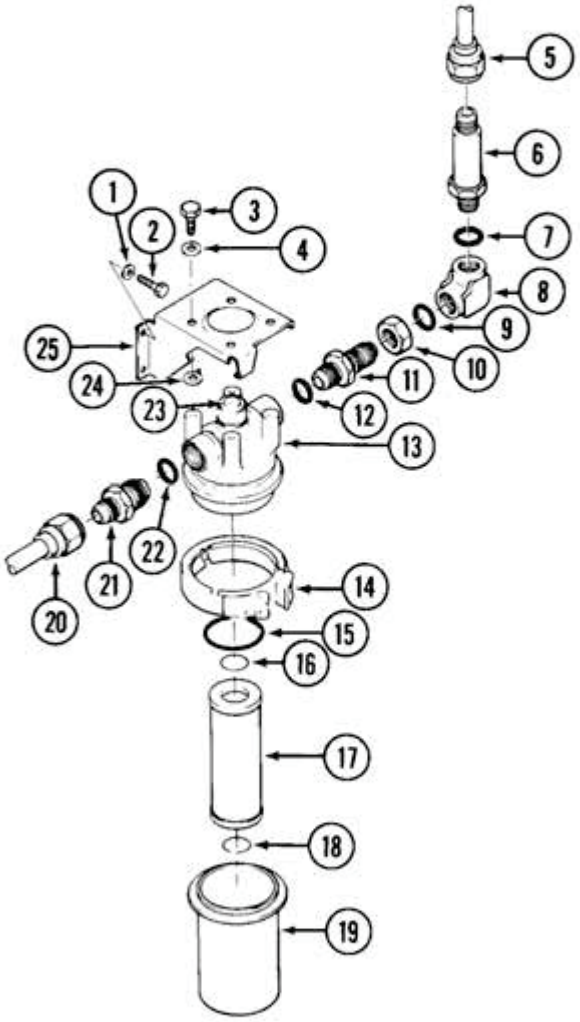
6. Install packing (16) around center boss under filter head (13). Insert external oil filter body (19) into filter head while pressing upward until seated.

7. *Install v-band clamp (14) around mating flanges of head and body. While tapping v-band clamp lightly to aid in seating, torque nut on clamp to 50 inch-pounds (5.7 Nm).*

8. Service transmission (412-A-12-10-03-00A-040A-A / 00135).

9. Check for oil leaks at next helicopter ground run.

1. Thin aluminum washers (4 places)
2. Bolt (4 places)
3. Bolt (4 places)
4. Thin aluminum washers (4 places)
5. Outlet hose
6. Fitting
7. Packing
8. Elbow
9. Packing
10. Checknut
11. Union
12. Packing
13. Filter head
14. V-band clamp
15. Packing
16. Packing
17. Filter element
18. Packing
19. External oil filter body
20. Inlet tube
21. Reducer
22. Packing
23. Bypass indicator
24. Spacer (4 places)
25. Filter bracket

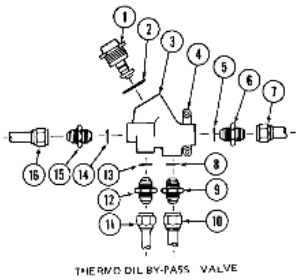


ICN-412-A-632202-B-97499-00001-A-001-01

Figure 1. Transmission External Oil Filter (Sheet 1 of 1)

Thermal Bypass Valve - Description

The thermal oil bypass valve (Figure 1) is installed on the rear bulkhead of cargo hook compartment. The valve is located in oil lines between transmission and oil cooler. The assembly consists of a valve body and a thermostat. The thermostat will be completely open when transmission oil temperature is 130°F (55°C) or below, causing oil to bypass cooler and flow directly to transmission. The thermostat partially closes at temperatures between 130 and 175°F (55 and 79°C), passing some oil through oil cooler while allowing remainder to bypass directly to transmission. At higher temperatures, all oil flow is directed through oil cooler before being routed to transmission.



1. Thermostat
2. Aluminum gasket
3. Valve body
4. Screw and washer (2 places)
5. Packing
6. Union
7. Oil inlet tube
8. Packing
9. Union
10. Oil drain tube
11. Oil drain tube
12. Union
13. Packing
14. Packing
15. Union
16. Oil outlet tube

ICN-412-A-632202-B-97499-00002-A-001-01

Figure 1. Thermal Oil Bypass Valve (Sheet 1 of 1)

Thermal Bypass Valve - Removal

Procedure

NOTE

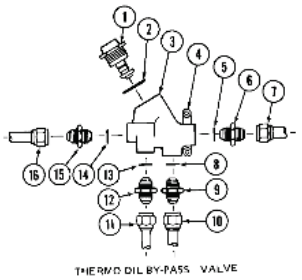
Thermostat (1, Figure 1) may be removed without removing valve body (3).

1. Disconnect tubes (7, 10, 11, and 16) from valve body (3).

2. Remove screws and washers (4) and remove valve body (3) from bulkhead.

3. Remove unions (6, 9, 12, and 15) with packings (5, 8, 13, and 14) and discard packings.

4. Cut and remove lockwire from thermostat (1). Remove thermostat and aluminum gasket (2) from valve body (3) and discard gasket.



1. Thermostat
2. Aluminum gasket
3. Valve body
4. Screw and washer (2 places)
5. Packing
6. Union
7. Oil inlet tube
8. Packing
9. Union
10. Oil drain tube
11. Oil drain tube
12. Union
13. Packing
14. Packing
15. Union
16. Oil outlet tube

ICN-412-A-632202-B-97499-00002-A-001-01

Figure 1. Thermal Oil Bypass Valve (Sheet 1 of 1)

Thermal Bypass Valve - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Wipe components with a clean cloth damp with drycleaning solvent (C-304).
2. Dry with filtered, compressed air.

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Thermal Bypass Valve - Installation

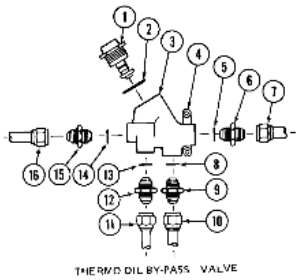
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Lubricate new aluminum gasket (2, Figure 1) and new packings (5, 8, 13, 14) with transmission oil (412-A-12-01-00-00A-040A-A / 00141).
2. Install thermostat (1) with aluminum gasket (2) in valve body (3). Tighten thermostat sufficiently to prevent leakage. Secure thermostat with lockwire (C-405).
3. Install unions (6, 9, 12, and 15) with new packings (5, 8, 13, and 14).
4. Install valve body (3) on bulkhead bracket with screws and washers (4).
5. Service transmission (412-A-12-10-03-00A-040A-A / 00135).
6. Check for oil leaks and proper operation at next helicopter ground run.



1. Thermostat
2. Aluminum gasket
3. Valve body
4. Screw and washer (2 places)
5. Packing
6. Union
7. Oil inlet tube
8. Packing
9. Union
10. Oil drain tube
11. Oil drain tube
12. Union
13. Packing
14. Packing
15. Union
16. Oil outlet tube

ICN-412-A-632202-B-97499-00002-A-001-01

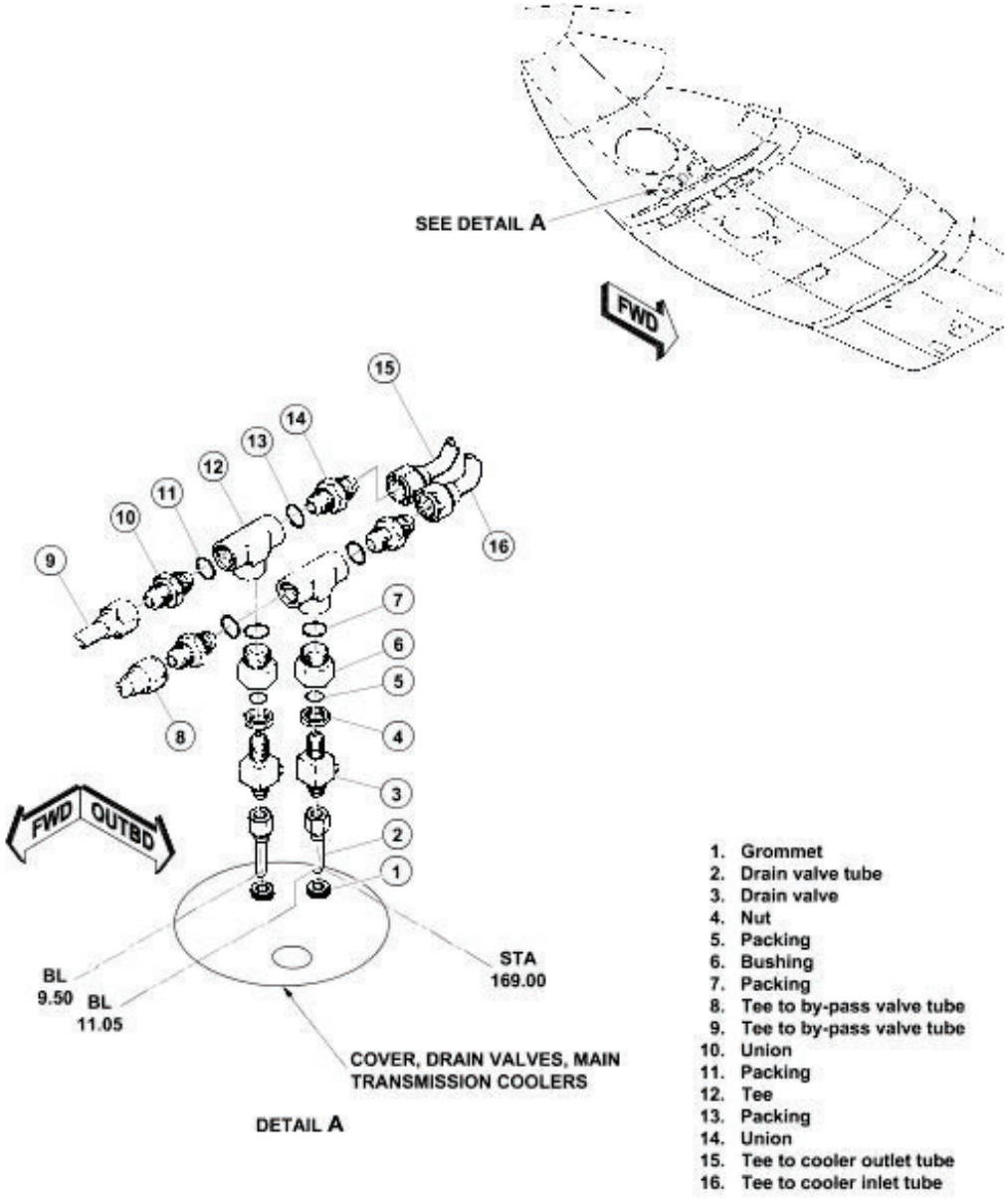
Figure 1. Thermal Oil Bypass Valve (Sheet 1 of 1)

Oil Drain Valves - Description

Two ball type drain valves (one for inlet and one for outlet oil) are installed in lowest part of oil system (Figure 1) between thermal oil bypass valve and transmission oil cooler. The valves are provided to drain oil trapped in lines of oil cooler to transmission system.

NOTE

Maintenance procedures for drain valves are identical. The procedures for inlet drain valve are provided.



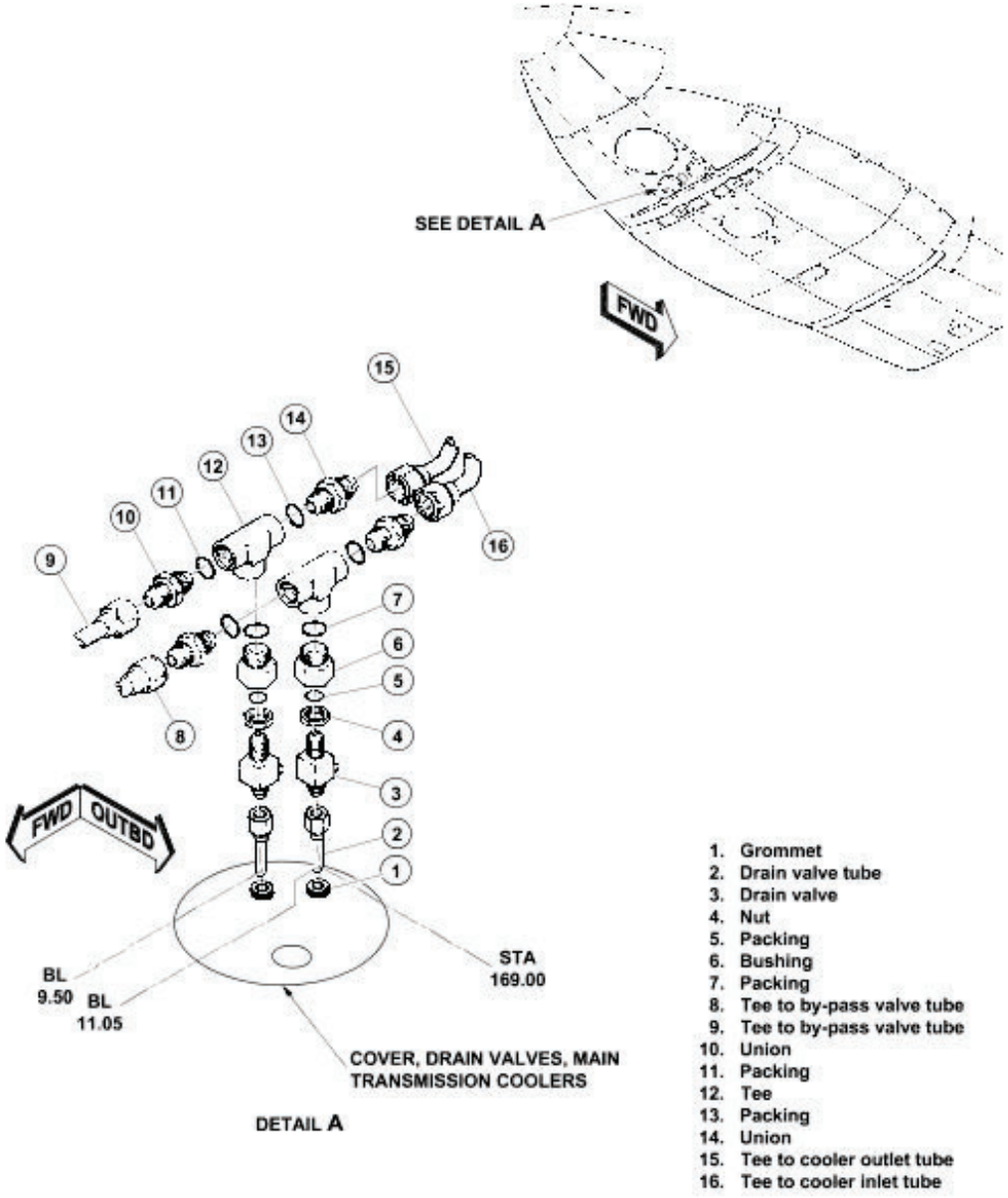
ICN-412-A-632202-B-97499-00003-A-001-01

Figure 1. Oil Drain Valve (Sheet 1 of 1)

Oil Drain Valves - Removal

Procedure

1. Place a container beneath drain valve (3, Figure 1) to catch oil spillage.
2. Open drain valve (3) and drain oil.
3. Disconnect tubes (2, 9, and 15). Cap or plug tubes to prevent contamination.
4. Remove tee (12) with drain valve (3) as an assembly.
5. Complete disassembly by loosening nut (4) and removing drain valve (3) and packing (5). If required, remove bushing (6) with packing (7), union (10) with packing (11), and union (14) with packing (13).
6. Discard packings (5, 7, 11, and 13).



ICN-412-A-632202-B-97499-00003-A-001-01

Figure 1. Oil Drain Valve (Sheet 1 of 1)

Oil Drain Valves - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

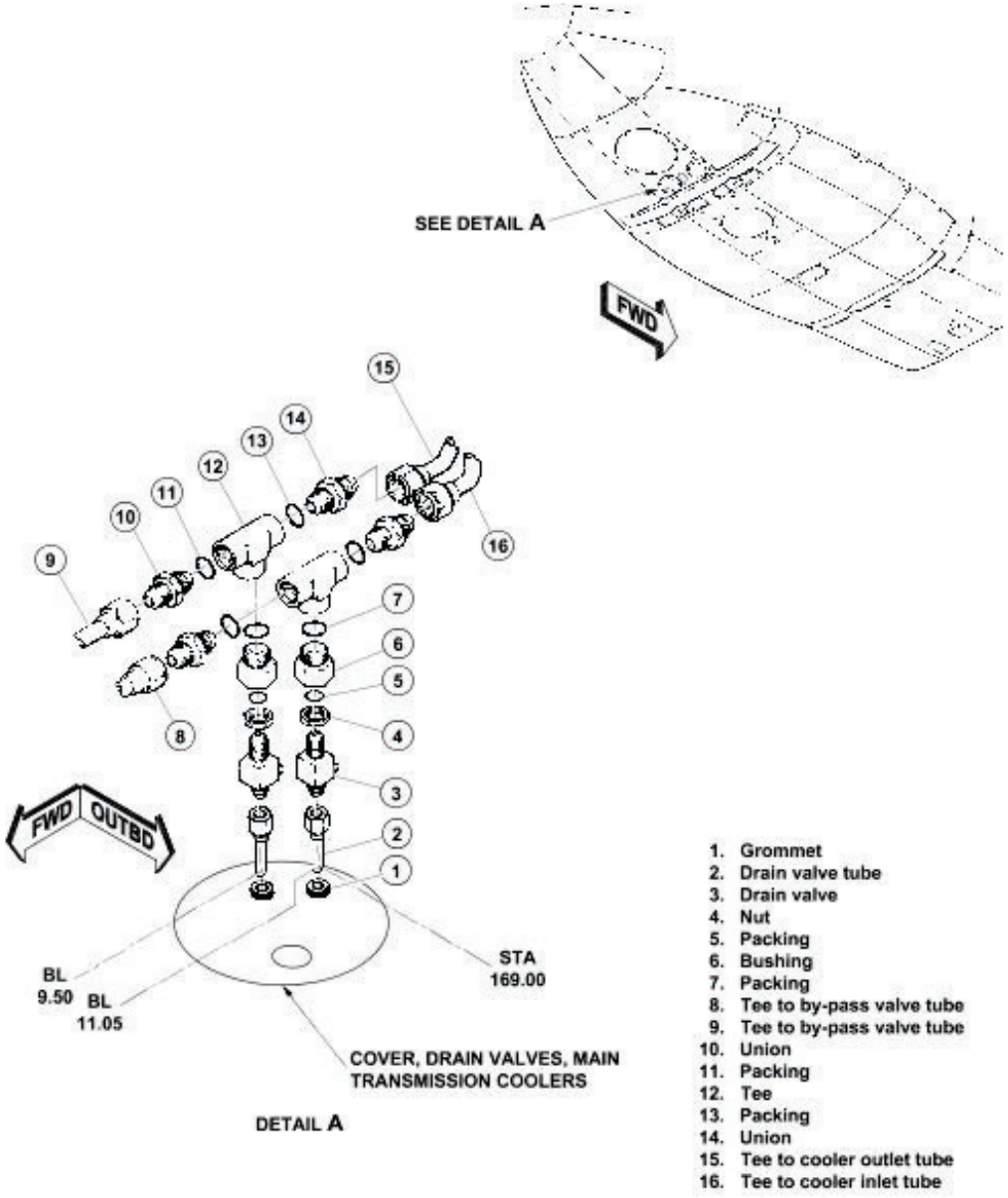
1. Wipe components with a clean cloth dampened with drycleaning solvent (C-304).
2. Dry with filtered, compressed air.

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Oil Drain Valves - Inspection

Procedure

1. Inspect components for cracks and thread damage. No cracks or thread damage is permissible.
2. Inspect drain valve (3, Figure 1) for proper operation.
3. Discard components that are not serviceable.



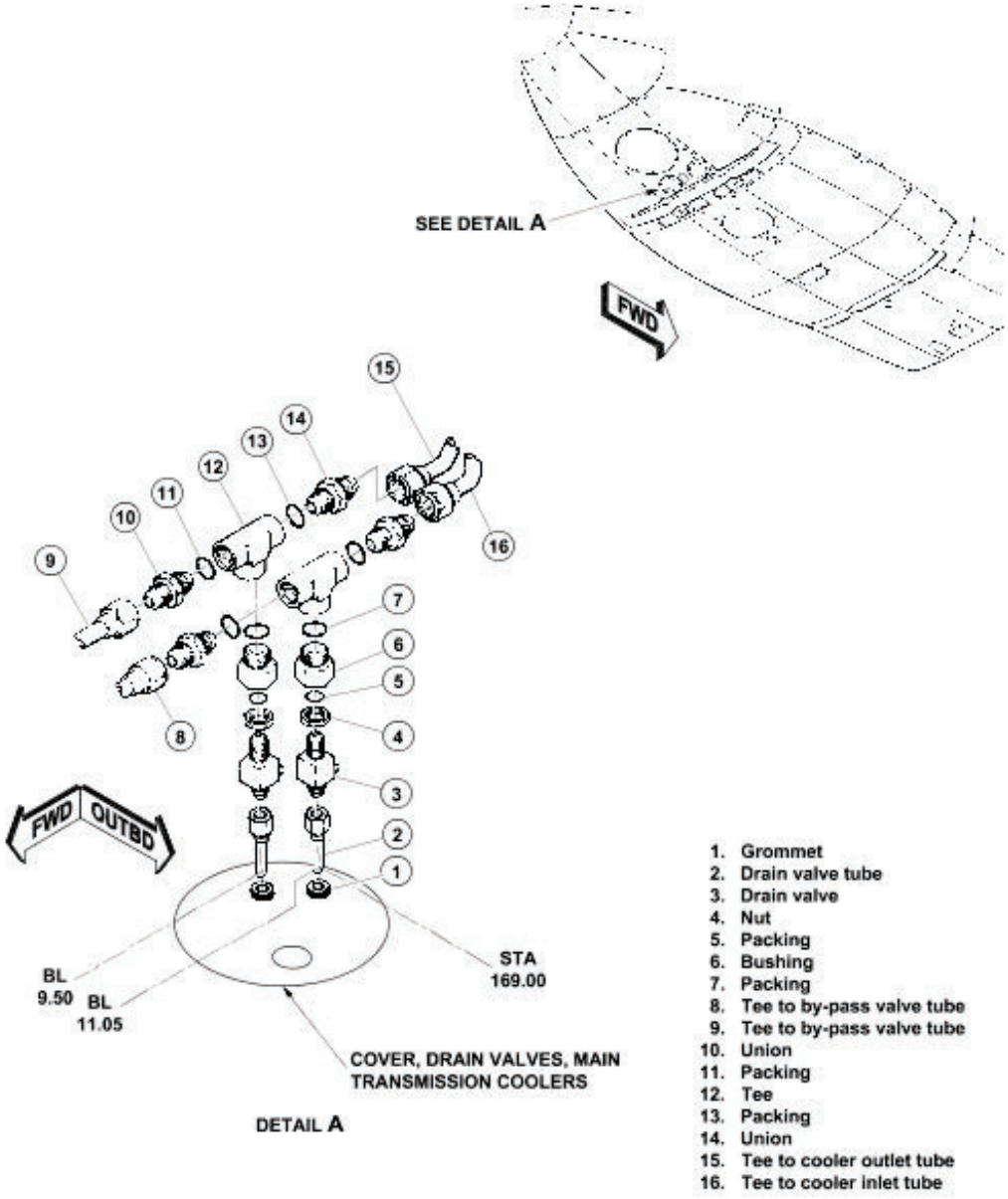
ICN-412-A-632202-B-97499-00003-A-001-01

Figure 1. Oil Drain Valve (Sheet 1 of 1)

Oil Drain Valves - Installation

Procedure

1. Lubricate new packings (5, 7, 11, and 13, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141).
2. Install unions (10 and 14) with new packings (11 and 13) into side ports of tee (12).
3. Install packing (7) on bushing (6) and install bushing in lower port of tee (12).
4. Install drain valve (3) with packing (5) and nut (4) in bushing (6). Tighten nut sufficiently to prevent leakage.
5. Place tee and drain valve assembly in position in helicopter after removing protective cover/caps from lines, hoses, and ports.
6. Install tubes (9 and 15) and install drain valve tube (2) with new grommet (1) (if required) on lower portion of valve.
7. Service transmission (412-A-12-10-03-00A-040A-A / 00135).
8. Check for oil leaks and proper operation at next helicopter ground run.



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Figure 1. Oil Drain Valve (Sheet 1 of 1)

Transmission Oil Cooler and Blower - Maintenance

Procedure

1. Refer to 412-A-79-20-00-06A-280A-A / 01785 for maintenance procedures applicable to the combination reduction gearbox oil and transmission oil cooler and combination engine oil and transmission oil cooler blower assemblies.

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Transmission Quills - Description

The transmission quills consist of main input quill, hydraulic system 2 drive quill, hydraulic system 1 and tachometer drive quill, rotor brake quill, and tail rotor drive quill.

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Main Input Quill - Description

The main input quill provides the means for the engine to provide power to the transmission through the main driveshaft.

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Main Input Quill - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Jackscrew Set	T101308	1	

Procedure

1. Open engine and transmission cowling.

2. Remove main driveshaft (412-A-63-10-01-01A-520A-A / 00994).

3. Disconnect drain tube (24, Figure 1).

4. Remove nut (11) with clip (10) from lower right stud on transmission (1).

5. Remove nuts (12).

6. Using a sharp plastic scraper, cut sealant around edge of quill.

7. Cut and remove lockwire from screws (8) and remove screws.

8. Using a heat lamp or a 250 watt light bulb, heat case of the transmission (1) in area of quill until it is hot to the touch with bare hands. Install three jackscrew set (T101308) in holes where screws (8) were removed. Tighten jackscrews evenly and carefully to separate main input quill (6) from transmission (1).

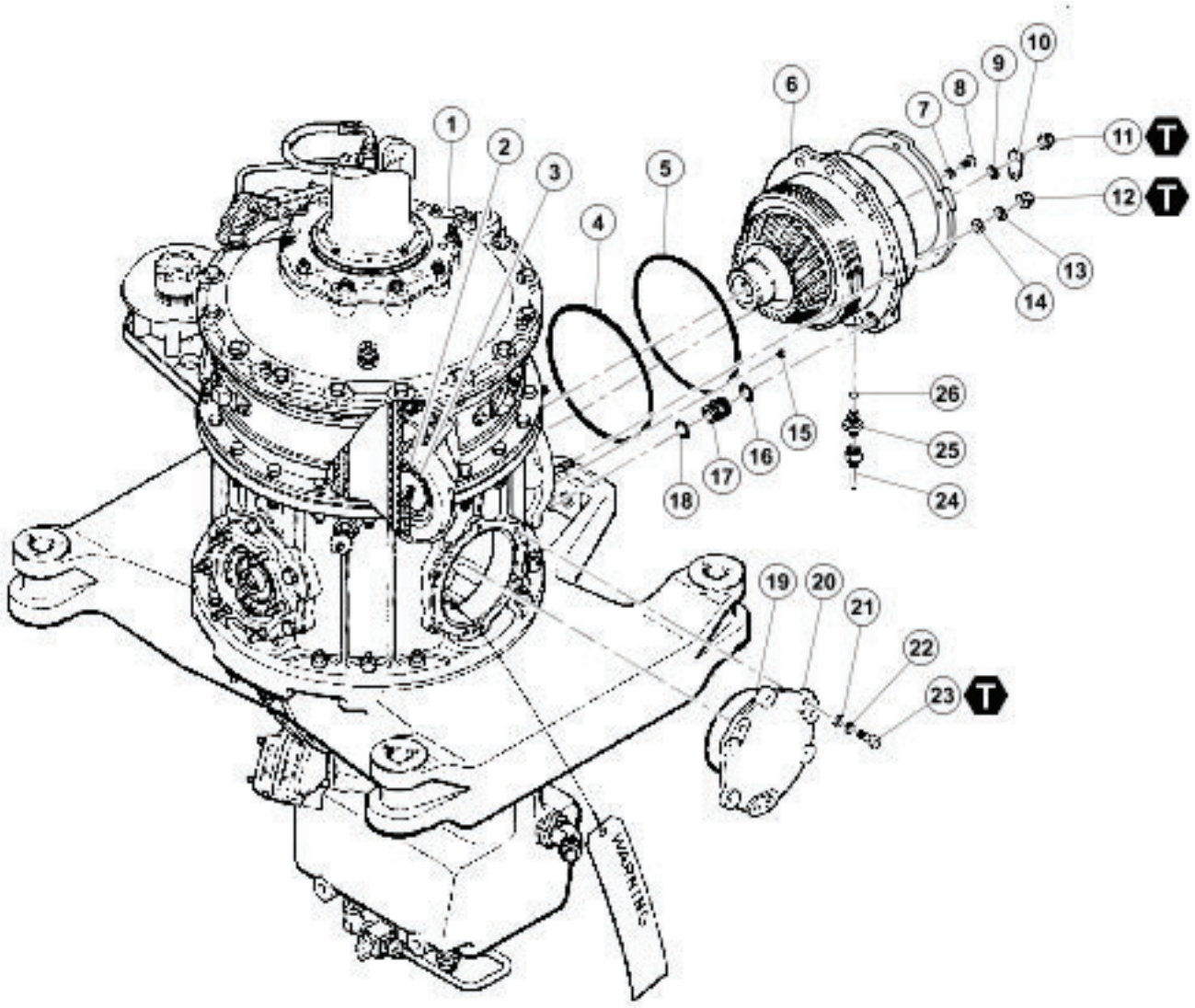
9. After main input quill (6) has completely separated from case, remove jackscrews and install three screws (8) with aluminum washers (7) to secure cap to quill sleeve.

10. Position main input quill (6) at right angle for clearance, and remove from transmission pylon cavity.

11. Remove and discard packings (4 and 5).

12. Remove oil transfer tube (17) with packings (16 and 18). Discard packings.

13. Install cover (412-A-63-21-03-00A-720A-A / 01184) over main input quill port of transmission (1).



1. Transmission
2. Bearing
3. Workaid (rubber plug)
4. Packing
5. Packing
6. Main input quill
7. Aluminum washer
8. Screw
9. Aluminum washer
10. Clip
11. Nut
12. Nut
13. Thin steel washer
14. Aluminum washer
15. Setscrew
16. Packing
17. Oil transfer tube
18. Packing
19. Packing
20. Cover
21. Aluminum washer
22. Steel washer
23. Bolt
24. Tube (seal drain)
25. Union
26. Packing

T 160 TO 190 IN-LBS (18.1 TO 21.4 Nm)

ICN-412-A-632102-B-97499-00010-A-001-01

Figure 1. Main Input Quill (Sheet 1 of 1)

Main Input Quill - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

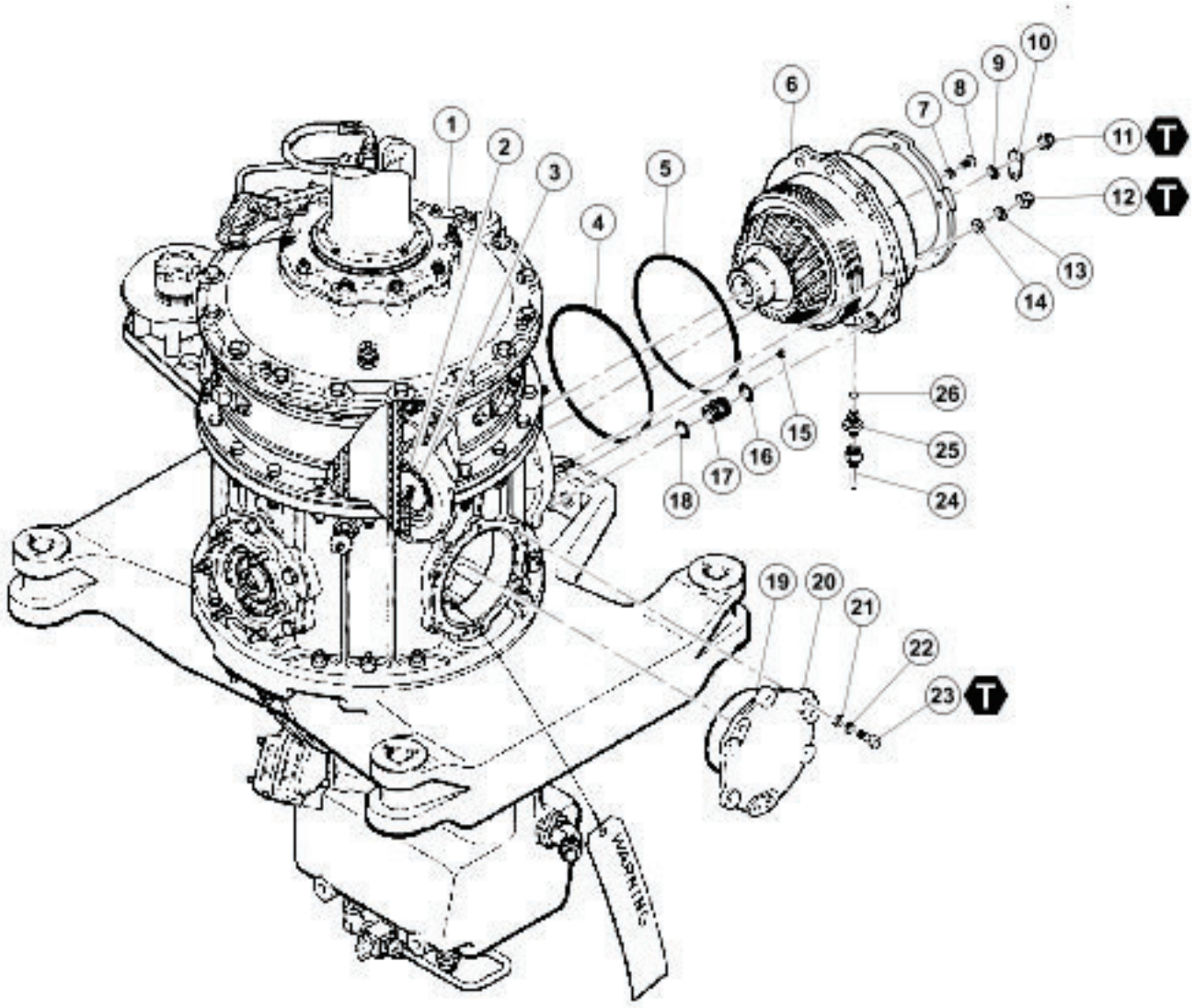
Procedure

NOTE

Inspect quill for evidence of leakage prior to cleaning. The cause of any leakage shall be corrected prior to further use. For a definition of acceptable leakage, refer to 412-A-63-21-02-01A-280A-A / 01137.

1. Wipe external surfaces of main input quill (6, Figure 1) with a clean cloth dampened with drycleaning solvent (C-304).

2. Dry with filtered, compressed air.



1. Transmission
2. Bearing
3. Workaid (rubber plug)
4. Packing
5. Packing
6. Main input quill
7. Aluminum washer
8. Screw
9. Aluminum washer
10. Clip
11. Nut
12. Nut
13. Thin steel washer
14. Aluminum washer
15. Setscrew
16. Packing
17. Oil transfer tube
18. Packing
19. Packing
20. Cover
21. Aluminum washer
22. Steel washer
23. Bolt
24. Tube (seal drain)
25. Union
26. Packing

T 160 TO 190 IN-LBS (18.1 TO 21.4 Nm)

ICN-412-A-632102-B-97499-00010-A-001-01

Figure 1. Main Input Quill (Sheet 1 of 1)

Main Input Quill - Inspection

Procedure

1. Inspect main input quill (6, Figure 1) for evidence of leakage. External leakage around seals is not acceptable. A small amount of seepage, however, is acceptable. A continuous flow of droplets is considered leakage.



THIS COMPONENT IS DYNAMICALLY BALANCED. SIGNIFICANTLY REWORKED DAMAGED AREAS MAY CAUSE HIGH FREQUENCY VIBRATIONS AFTER INSTALLATION.

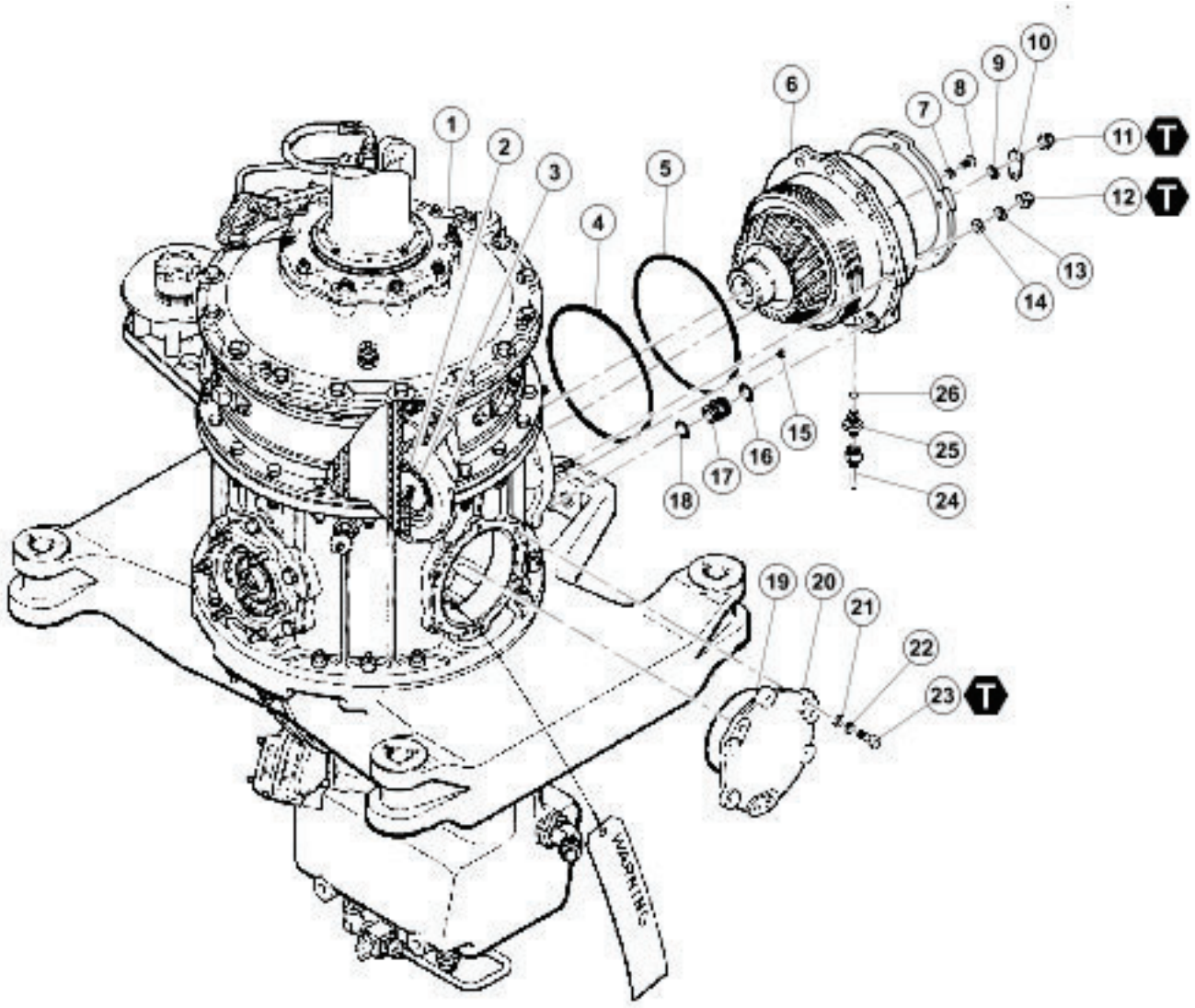
2. Inspect visible portion of main driveshaft curvic coupling adapter (4, Figure 2). Superficial damage is permissible without rework.

3. Inspect visible portions of input quill cap assembly (3). Maximum depth of pitting is 0.030 inch (0.76 mm), provided no more than 40% of any 1.0 square inch (645.15 mm^2) or 20% of total area of any surface is affected. Repair acceptable damage in accordance with 412-A-63-21-02-01A-685A-A / 01138.

4. Inspect visible portion of input sleeve (2). Light oxidation is permitted provided no more than 40% of any 1.0 square inch (645.15 mm^2) or 20% of total area of any surface is affected. Remove acceptable oxidation as described in 412-A-63-21-02-01A-685A-A / 01138.

5. Inspect teeth on quill pinion (1) for abnormal gear wear pattern (BHT-412-CR&O). Abnormal gear wear pattern is not acceptable and is cause to inspect mating gear in transmission.

6. Inspect bearings inside quill by rotating main driveshaft curvic coupling adapter (4) while holding input sleeve (2). Bearing roughness which can be detected by feel is not acceptable. If bearings exhibit roughness, refer to the BHT-412-CR&O for overhaul procedures.

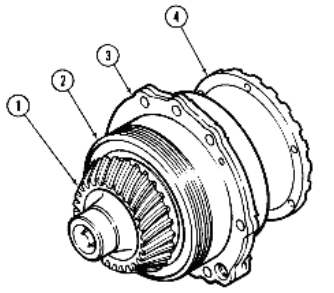


1. Transmission
2. Bearing
3. Workaid (rubber plug)
4. Packing
5. Packing
6. Main input quill
7. Aluminum washer
8. Screw
9. Aluminum washer
10. Clip
11. Nut
12. Nut
13. Thin steel washer
14. Aluminum washer
15. Setscrew
16. Packing
17. Oil transfer tube
18. Packing
19. Packing
20. Cover
21. Aluminum washer
22. Steel washer
23. Bolt
24. Tube (seal drain)
25. Union
26. Packing

T 160 TO 190 IN-LBS (18.1 TO 21.4 Nm)

ICN-412-A-632102-B-97499-00010-A-001-01

Figure 1. Main Input Quill (Sheet 1 of 1)



1. Pinion
2. Input sleeve
3. Input quill cap assembly
4. Main driveshaft curvic coupling adapter

ICN-412-A-632102-B-97499-00011-A-001-01

Figure 2. Main Input Quill Assembly (Sheet 1 of 1)

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Main Input Quill - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	
Abrasive cloth or paper	C-423	AR	
India stone	C-464	AR	

Procedure

1. Repair quills that have leakage by installing a new seal (412-A-63-21-02-02A-720A-A / 01145 or 412-A-63-21-02-02A-720B-A / 01146, as applicable).

2. Polish out acceptable corrosion and mechanical damage (as defined in 412-A-63-21-02-01A-280A-A / 01137) using a fine india stone (C-464) and 600 grit abrasive cloth or paper (C-423).

3. Touch-up repaired areas that were previously painted using epoxy polyamide primer (C-204).

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Main Input Quill - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Pusher Set	T101586-5	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
Lockwire	C-405	AR	
Anti-seize thread compound	C-452	AR	

Procedure

1. Remove setscrews (15, Figure 1) from studs in transmission (1).

NOTE

If rotor brake quill is installed, remove cover from port on opposite side of transmission (412-A-63-21-03-00A-520A-A / 01183). If this port is removed, it will be necessary to remove oil jet No. 6 (412-A-63-22-01-07A-520A-A / 01064) to allow clearance for work and plug.

2. Remove cover (20) (412-A-63-21-03-00A-520A-A / 01183).

3. Reach through port where cover was removed and position work aid (3, Figure 1 and 2, Figure 2) to hold rollers of bearing (2, Figure 1) against outer race of bearing.

4. Using a heat lamp or a 250 watt light bulb, heat transmission (1) case in area of quill port until it is hot to the touch with bare hands.

5. Lubricate new packings (16 and 18) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install on oil transfer tube (17). Position tube in transmission case.

6. Lubricate new packings (4 and 5) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packing (5) in groove of main input quill (6) nearest flange and packing (4) in groove nearest pinion. Leave center groove open.



THE CENTER GROOVE IS AN OIL PASSAGE. IT MUST STAY OPEN.

7. Position main input quill (6) into port with drain tube union (15, Figure 2) for drain tube at bottom.

8. Install the three pusher set studs (13) in holes where setscrews (15, Figure 1) were removed in Step 1..

9. Carefully slide input quill pinion (6, Figure 2) into transmission case. Align main input quill pinion teeth with input gear quill (5) teeth. Align front of input quill pinion with bearing rollers (3).

10. Slide spacers (11) over studs and install nuts (12) on studs.

11. Tighten nuts (12) evenly to keep quill aligned. When quill is started into bearing, remove work aid plug (2) to prevent pushing work aid into oil jet No. 6 (if installed).

12. Continue tightening nuts (12) to fully seat quill. When quill is fully seated, loosen nuts (12) and remove studs (13) and spacers (11).

NOTE

In the following steps, for aluminum washers (7 and 14, Figure 1), use only AN960JD616 (aluminum) washer next to transmission case. Do not substitute any other type (including NAS1197).

13. Install aluminum washer (9), clip (10), and nut (11) on lower right stud. Install six aluminum washers (14) (next to transmission case), six thin steel washers (13) (under nut), and six nuts (12) on remaining studs. Torque nuts (11 and 12) to 160 to 190 inch-pounds (18.1 to 21.4 Nm).

14. Rotate main driveshaft curvic coupling adapter (4, Figure 3) on outboard end of main input quill (6, Figure 1) by hand. Check for binding and verify presence of backlash. Any binding or roughness is not acceptable.

15. Place a small amount of anti-seize thread compound (C-452) on threads of union (25).

16. If removed, lubricate a new packing (26) with transmission oil (412-A-12-01-00-00A-040A-A / 00141), and install on union (25). Install union in main input quill (6) and install tube (24) in union.

17. Install three aluminum washers (7) (AN960JD616 only) on screws (8) and install screws. Secure screws to main input quill (6) with lockwire (C-405).

18. Install three setscrews (15) into studs on transmission case.

NOTE

If port cover opposite cover (20) was removed, install oil jet No. 6 (412-A-63-22-01-07A-720A-A / 01068) and cover (412-A-63-21-03-00A-720A-A / 01184).

19. Lubricate new packing (19) with transmission oil (412-A-12-01-00-00A-040A-A / 00141), and install packing on cover (20).

NOTE

Aluminum washers (21) are installed next to transmission case and steel washers (22) are installed directly under heads of bolt (23).

20. Install steel washers (22) and then aluminum washers (21) under heads of bolt (23).

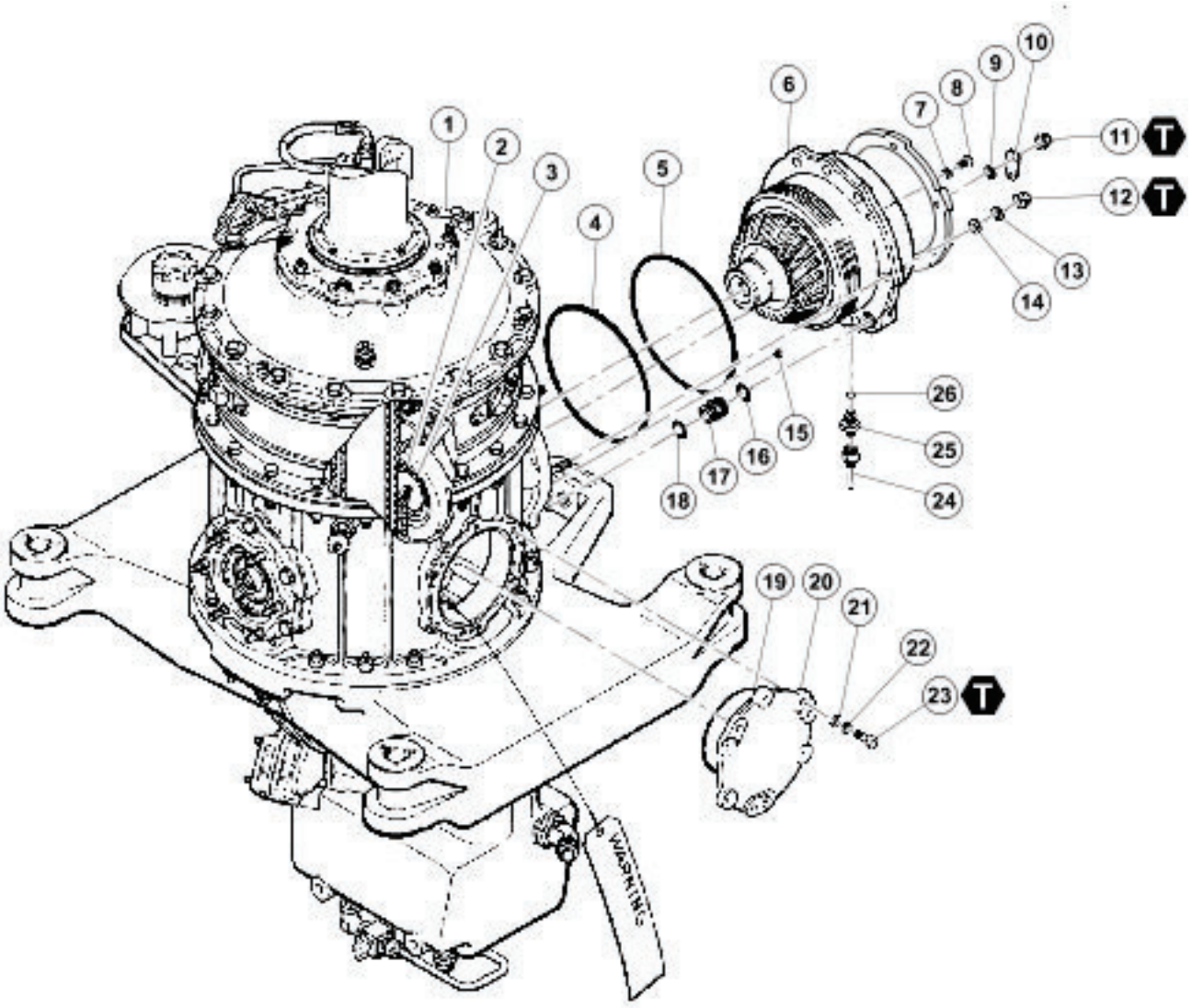
21. Install bolts (23) on cover (20). Torque bolts to 160 to 190 inch-pounds (18.1 to 21.4 Nm).

22. Apply a bead of sealant (C-308) around flange of quill and cover which was removed where there is contact with case.

23. Install main driveshaft (412-A-63-10-01-01A-720A-A / 00999).

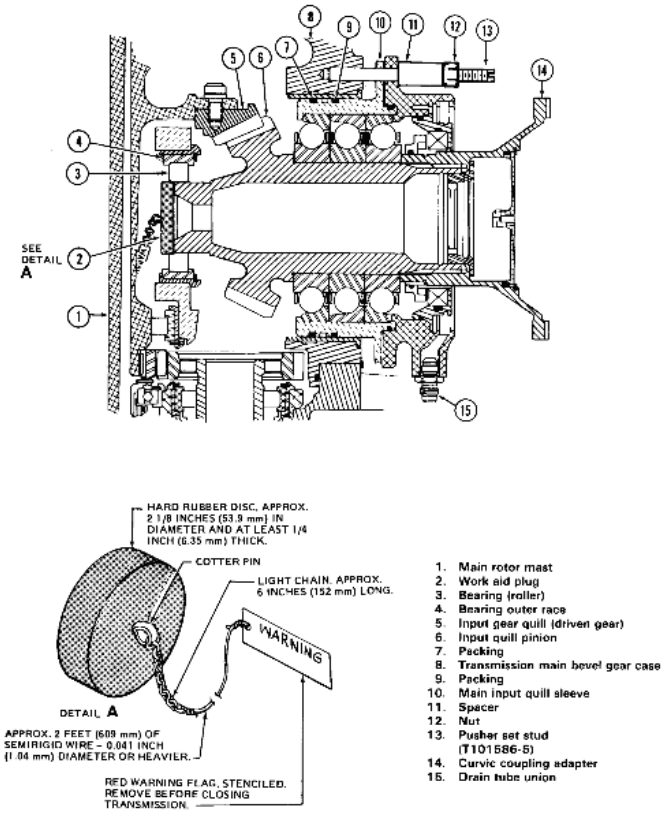
24. Close engine and transmission cowling.

25. Perform engine ground run (BHT-412-FM) and check for proper operation and leakage.



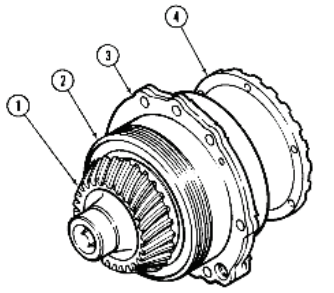
ICN-412-A-632102-B-97499-00010-A-001-01

Figure 1. Main Input Quill (Sheet 1 of 1)



ICN-412-A-632102-B-97499-00012-A-001-01

Figure 2. Pusher Set (T101586-5) Application (Sheet 1 of 1)



1. Pinion
2. Input sleeve
3. Input quill cap assembly
4. Main driveshaft curvic coupling adapter

ICN-412-A-632102-B-97499-00011-A-001-01

Figure 3. Main Input Quill Assembly (Sheet 1 of 1)

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Main Input Quill Magnetic Seal - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Wrench	T101306	1	
Wrench Set	T101588	1	

Procedure

1. Open engine and transmission cowling.

2. Remove main driveshaft (412-A-63-10-01-01A-520A-A / 00994).

3. Remove retainer ring (1, Figure 1), plate (2), and packing (3). Discard packing.

4. Remove lock-spring (4).

5. Insert wrench (T101306) (9, Figure 2) in nut (5). Install wrench set (T101588) (3) on main driveshaft curvic coupling adapter (1) so splines on wrench (3) engage splines of wrench (9). Tighten wing nuts (2) to lock wrench (3) to main driveshaft curvic coupling adapter.

6. Insert extension bar (4) through wrench (3) and engage wrench (9).

7. Hold wrench (3) and loosen nut (5) so it can be removed by hand. Leave nut installed. Remove extension bar (4), wrench (3), and wrench (9).

8. Remove retainer ring (7, Figure 1).

9. Remove nut (5, Figure 2).

10. Use two screwdrivers to remove adapter and magnetic seal assembly (12, Figure 1) as an assembly. Position two screwdrivers 180° apart and engage groove in seal retainer (8). Pry against input quill cap assembly (9) to remove adapter and magnetic seal assembly (12). Carefully place assembly on bench with main driveshaft curvic coupling adapter (8, Figure 3) facing down.



MAGNETIC RING (3) AND CARBON SEAL (2) MUST BE KEPT CLEAN IF THEY ARE TO BE REINSTALLED. STORE THESE PARTS IN A CLEAN PLASTIC BAG IMMEDIATELY AFTER REMOVAL TO PREVENT METAL PARTICLES OR OTHER MATERIAL FROM COLLECTING.

11. Remove carbon seal (2) and seal retainer (6) and magnetic ring (3) from main driveshaft curvic coupling adapter (8) as an assembly (Detail B).



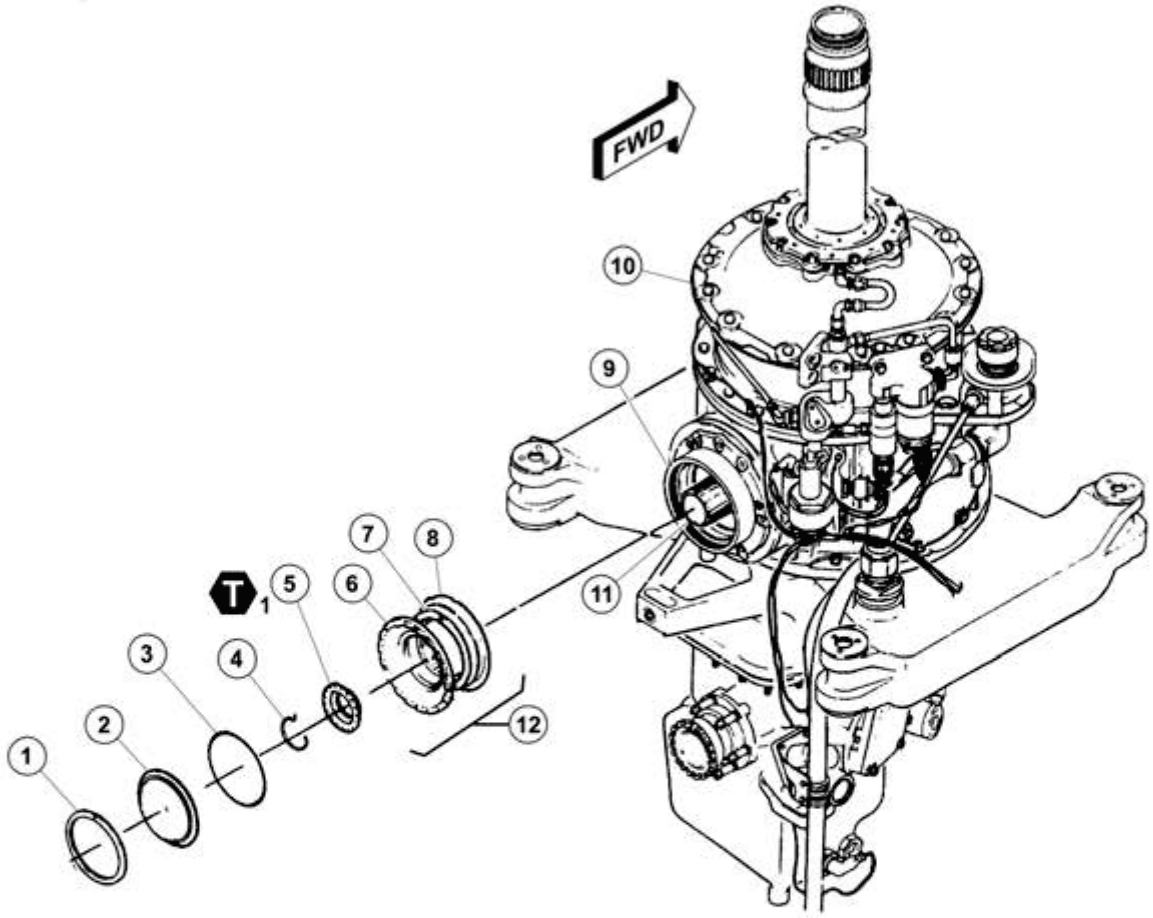
MAGNETIC RING (3) AND CARBON SEAL (2) SHALL BE SEPARATED BY PULLING HALVES STRAIGHT AWAY FROM EACH OTHER. ALLOWING HALVES TO SLIDE AGAINST EACH OTHER MAY DAMAGE MATING SURFACES.

12. Remove carbon seal (2) from seal retainer (6).

13. Remove packing (1) from carbon seal (2). Discard packing.

14. Remove magnetic ring (3) from seal retainer (6) (Detail C).

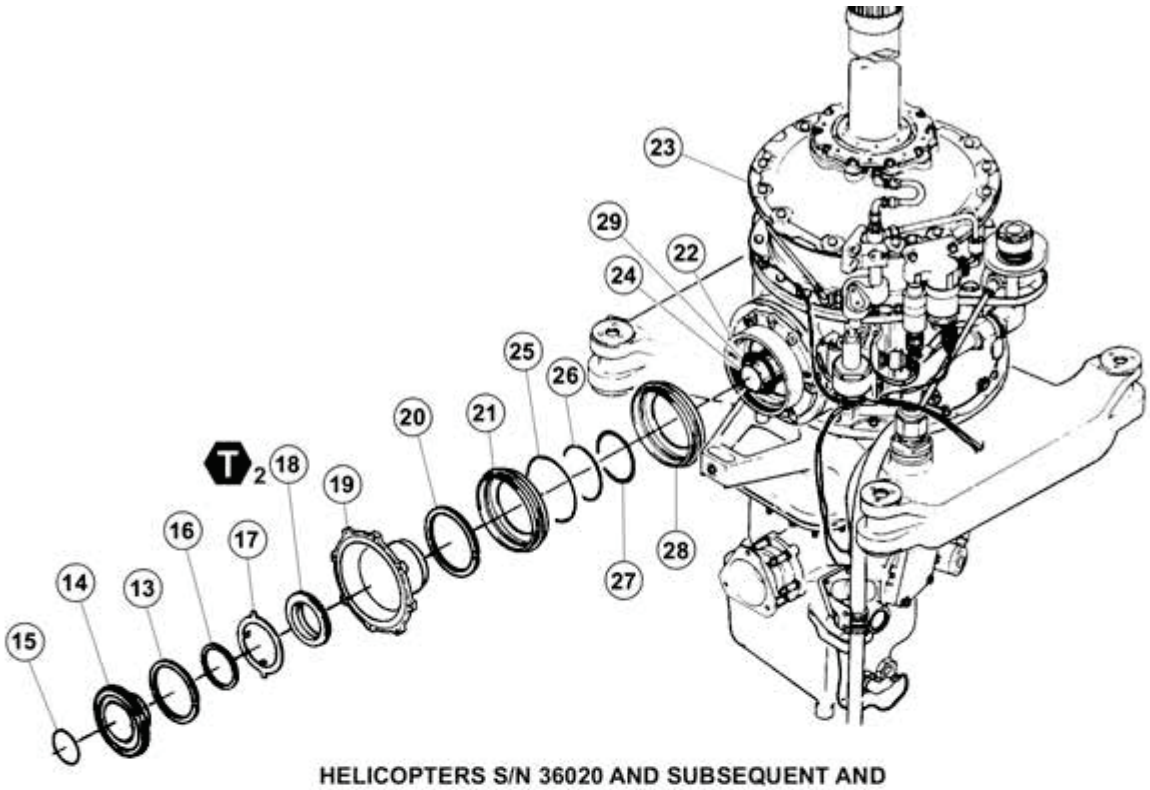
15. Remove packings (4 and 5) from seal retainer (6). Discard packings.



HELICOPTERES PRIOR TO S/N 36020 AND NOT MODIFIED
BY 412-570-001-103 OR PRE BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR PRE BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

ICN-412-A-632102-B-97499-00013-A-002-01

Figure 1. Main Input Quill Magnetic Seal (Sheet 1 of 2)



HELICOPTERS S/N 36020 AND SUBSEQUENT AND
HELICOPTERS MODIFIED BY 412-570-001-103 OR
POST BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR POST BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

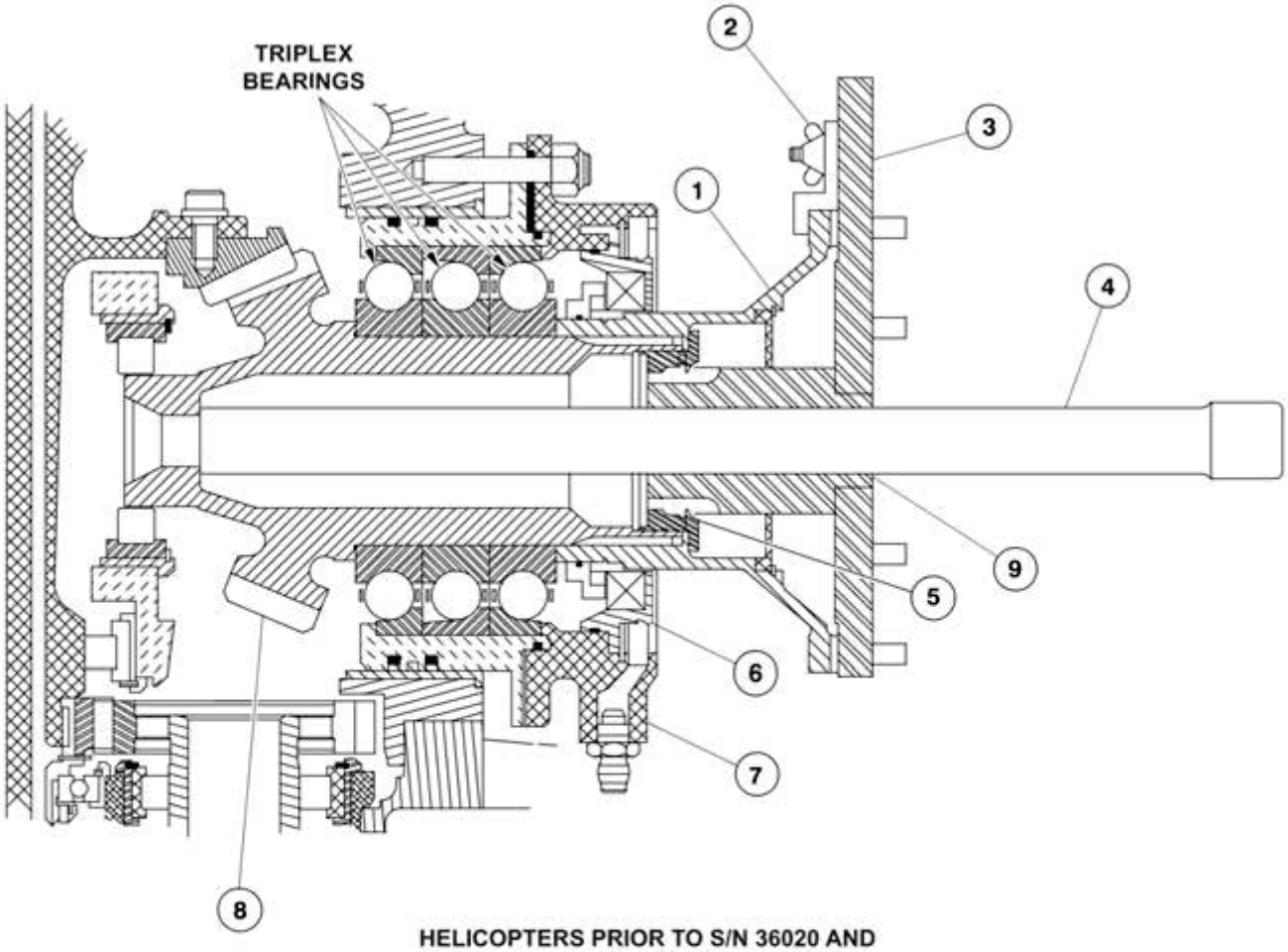
1. Retainer ring
2. Plate
3. Packing
4. Lock-spring
5. Nut
6. Main driveshaft curvic coupling
7. Retainer ring
8. Seal retainer
9. Input quill cap assembly
10. Transmission
11. Input quill pinion
12. Adapter and magnetic seal assembly
13. Retainer ring
14. Plate
15. Packing
16. Retainer ring
17. Lockwasher
18. Nut
19. Main driveshaft adapter
20. Retainer ring
21. Seal retainer
22. Input quill cap assembly
23. Transmission
24. Input quill pinion
25. Packing
26. Packing
27. Packing
28. Magnetic seal
29. Packing

T1 350 TO 400 FT-LBS (476 TO 542 Nm)

T2 450 TO 500 FT-LBS (610.12 TO 677.91 Nm)

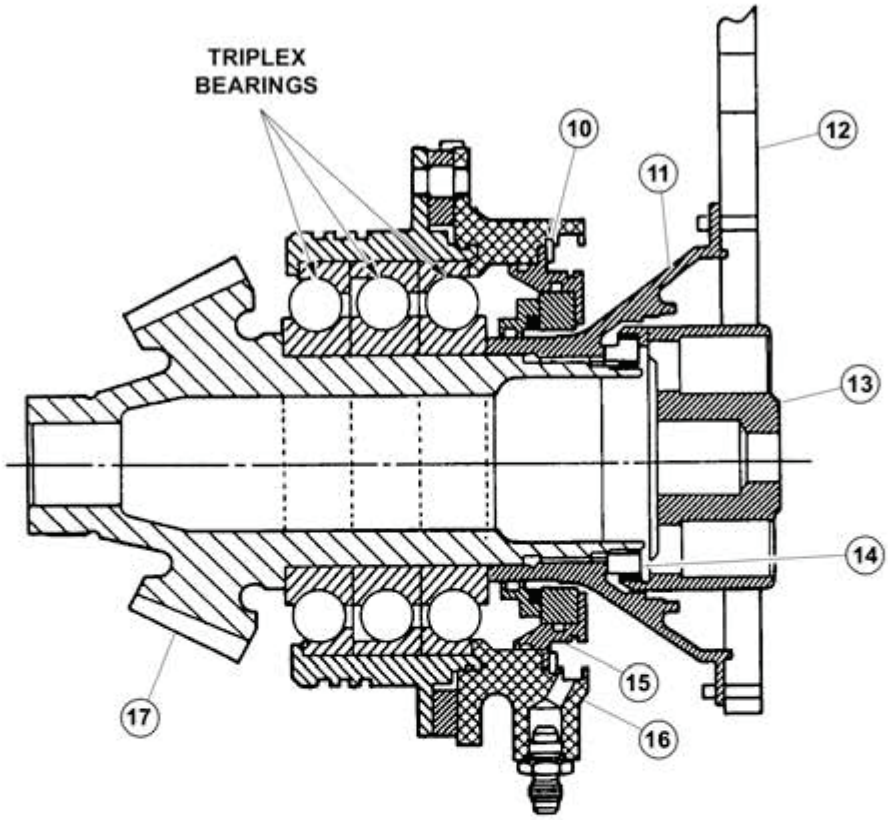
ICN-412-A-632102-B-97499-00014-A-002-01

Figure 1. Main Input Quill Magnetic Seal (Sheet 2 of 2)



ICN-412-A-632102-B-97499-00015-A-002-01

Figure 2. Tool Application (Sheet 1 of 2)

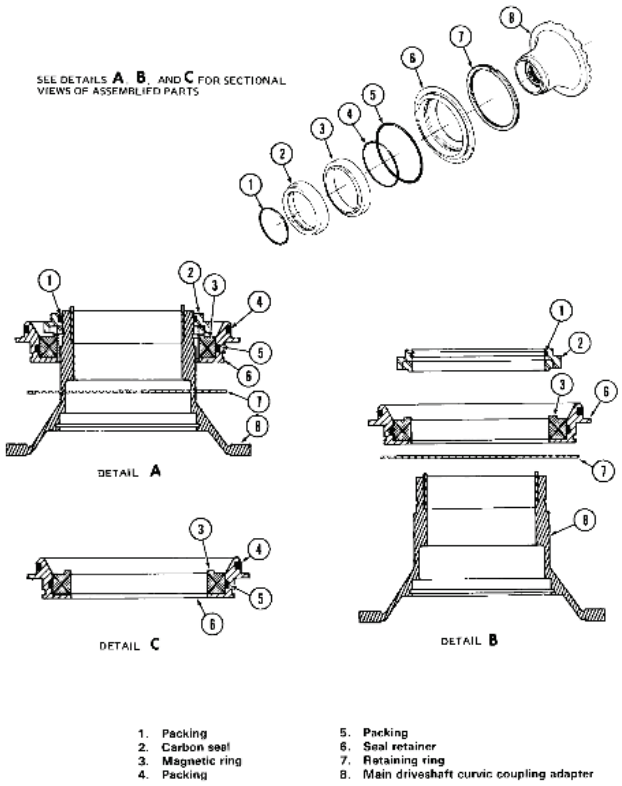


HELICOPTERS S/N 36020 AND SUBSEQUENT AND
HELICOPTERS MODIFIED BY 412-570-001-103 OR POST BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR POST BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

1. Main driveshaft curvic coupling adapter
2. Wing nut (MS35426-14)
3. Wrench set (T101588)
4. Extension bar (L-122)
5. Nut
6. Seal retainer
7. Input quill cap assembly
8. Input quill pinion
9. Tool (T101306)
10. Retainer ring
11. Main driveshaft adapter
12. Wrench set 412-240-002-101
13. Wrench 412-240-001-101
14. Nut
15. Seal retainer
16. Input quill cap assembly
17. Input quill pinion

ICN-412-A-632102-B-97499-00016-A-002-01

Figure 2. Tool Application (Sheet 2 of 2)



ICN-412-A-632102-B-97499-00017-A-001-01

Figure 3. Main Input Quill Curvic Coupling Adapter and Magnetic Seal Assembly (Sheet 1 of 1)

Main Input Quill Magnetic Seal - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Wrench	412-240-001-101	1	
Wrench Set	412-240-002-101	1	

Procedure

1. Open the engine and transmission cowling.

2. Remove the main driveshaft (412-A-63-10-01-01A-520A-A / 00994).

3. Remove the retainer ring (13, Figure 1) and the plate (14).

4. Remove and discard the packing (15).

5. Remove the retainer ring (16) and the lockwasher (17).

6. Install the wrench set (412-240-002-101) (Figure 2).

7. Install the wrench (412-240-001-101) on the nut (18, Figure 1).

8. Loosen the nut but leave it installed.

9. Remove the wrench (412-240-001-101) and the wrench set (412-240-002-101) (Figure 2).

10. Remove the retainer ring (20, Figure 1).

11. Remove the nut (18).

NOTE

Adapter and magnetic seal assembly consists of all parts shown in Figure 1. These parts may be removed as an assembly.

12. Use two screwdrivers to remove the main driveshaft adapter (19) and the magnetic seal (28) as an assembly.

13. Position two screwdrivers 180° apart and engage them in the grooves of the seal retainer (21).

14. Apply a force against the input quill cap assembly (22) to remove the main driveshaft adapter (19) and the magnetic seal (28).

15. Put the assembly on a bench with the adapter (19) face down.

16. Remove and discard the packing (29) from the input quill pinion (24).



MAGNETIC RING (3, Figure 3) AND CARBON SEAL (2) MUST BE KEPT CLEAN IF THEY ARE TO BE REINSTALLED. STORE THESE PARTS IN A CLEAN PLASTIC BAG IMMEDIATELY AFTER REMOVAL TO PREVENT METAL PARTICLES OR OTHER MATTER FROM ADHERING TO MAGNETIC AND/OR CARBON SEAL.

17. Remove carbon seal (2, Figure 3), seal retainer (6), and magnetic ring (3) from the main driveshaft curvic coupling adapter (8) as an assembly. Refer to Detail B.



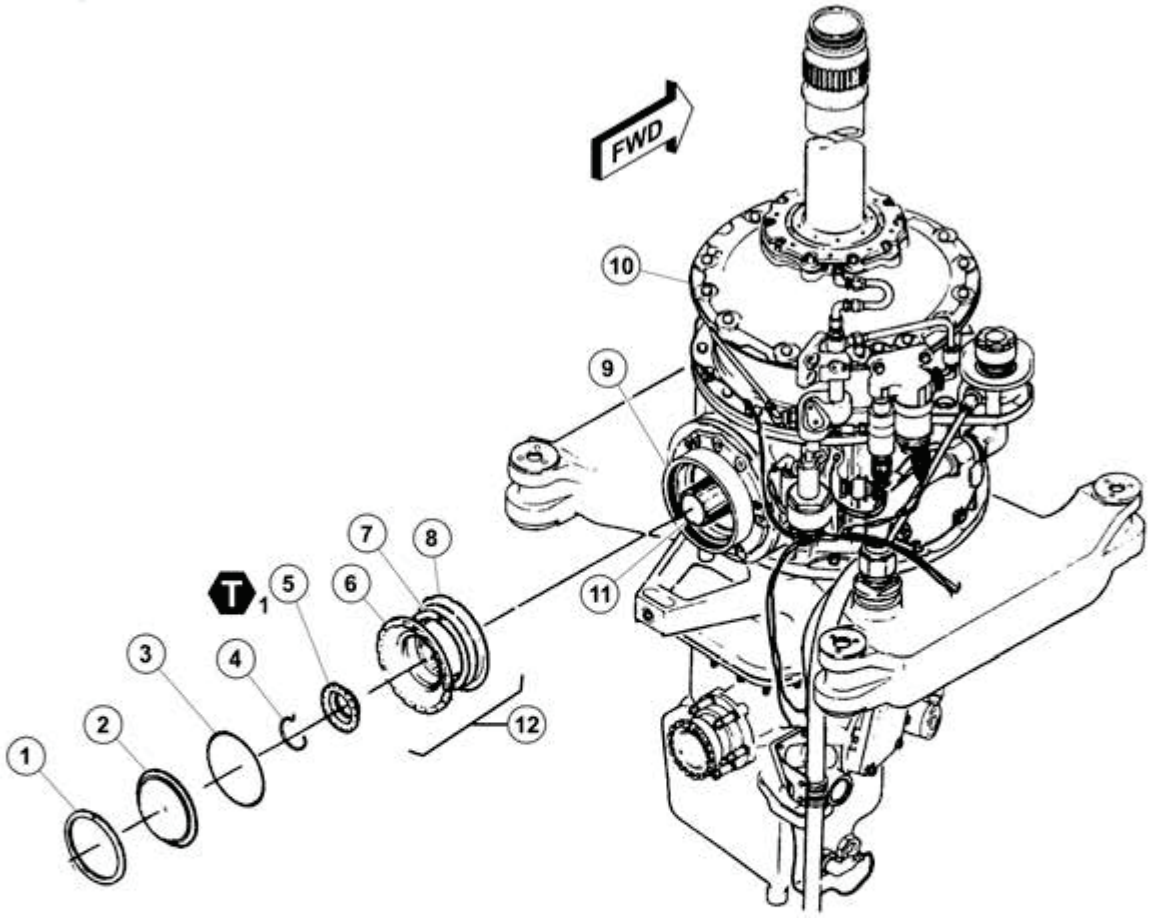
MAGNETIC RING (3) AND CARBON SEAL (2) MUST BE SEPARATED BY PULLING HALVES AWAY FROM EACH OTHER. SLIDING THE SEAL HALVES CAN DAMAGE THE MATING SURFACES.

18. Remove the carbon seal (2) from the seal retainer (6).

19. Remove the packing (1) from the carbon seal (2).

20. Remove the magnetic ring (3) from the seal retainer (6).

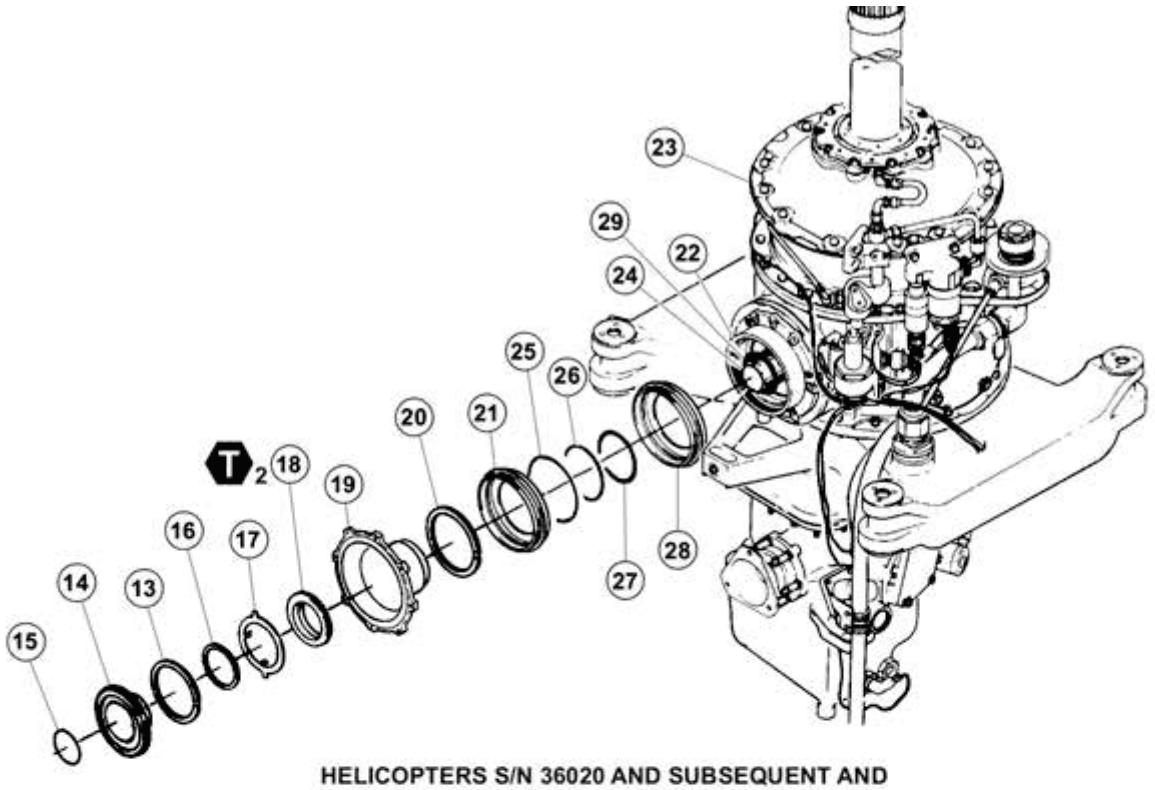
21. Remove the packing (4 and 5) from the seal retainer (6).



HELICOPTERES PRIOR TO S/N 36020 AND NOT MODIFIED
BY 412-570-001-103 OR PRE BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR PRE BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

ICN-412-A-632102-B-97499-00013-A-002-01

Figure 1. Main Input Quill Magnetic Seal (Sheet 1 of 2)



HELICOPTERS S/N 36020 AND SUBSEQUENT AND
HELICOPTERS MODIFIED BY 412-570-001-103 OR
POST BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR POST BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

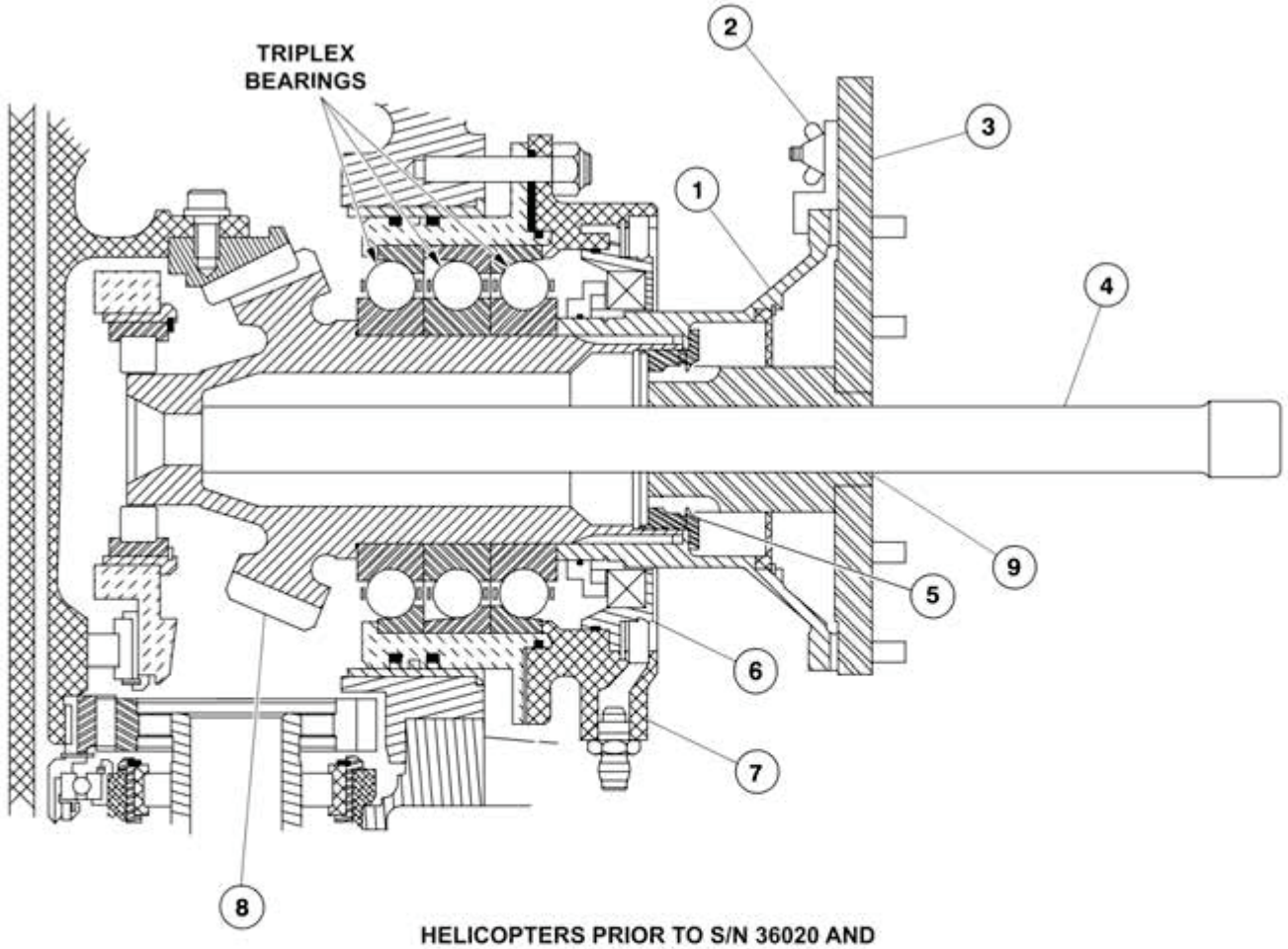
1. Retainer ring
2. Plate
3. Packing
4. Lock-spring
5. Nut
6. Main driveshaft curvic coupling
7. Retainer ring
8. Seal retainer
9. Input quill cap assembly
10. Transmission
11. Input quill pinion
12. Adapter and magnetic seal assembly
13. Retainer ring
14. Plate
15. Packing
16. Retainer ring
17. Lockwasher
18. Nut
19. Main driveshaft adapter
20. Retainer ring
21. Seal retainer
22. Input quill cap assembly
23. Transmission
24. Input quill pinion
25. Packing
26. Packing
27. Packing
28. Magnetic seal
29. Packing

T1 350 TO 400 FT-LBS (476 TO 542 Nm)

T2 450 TO 500 FT-LBS (610.12 TO 677.91 Nm)

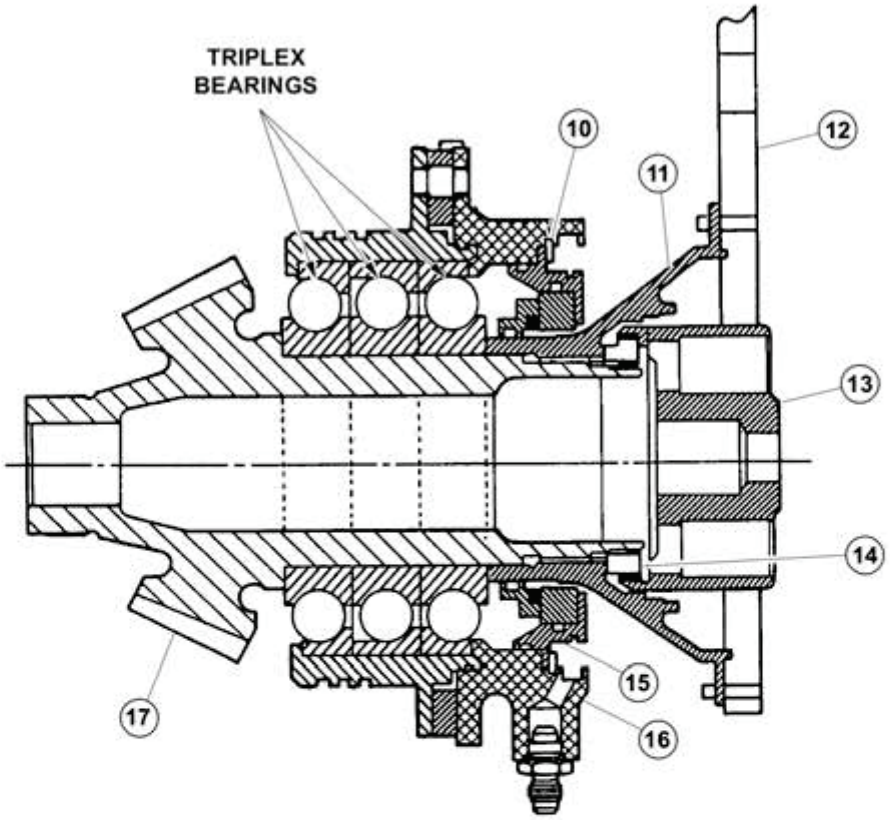
ICN-412-A-632102-B-97499-00014-A-002-01

Figure 1. Main Input Quill Magnetic Seal (Sheet 2 of 2)



ICN-412-A-632102-B-97499-00015-A-002-01

Figure 2. Tool Application (Sheet 1 of 2)

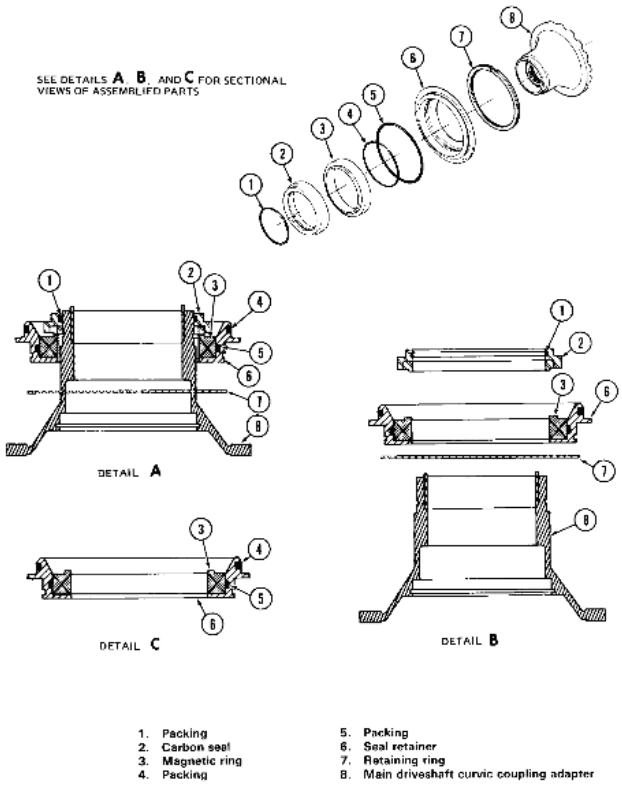


HELICOPTERS S/N 36020 AND SUBSEQUENT AND
HELICOPTERS MODIFIED BY 412-570-001-103 OR POST BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR POST BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

1. Main driveshaft curvic coupling adapter
2. Wing nut (MS35426-14)
3. Wrench set (T101588)
4. Extension bar (L-122)
5. Nut
6. Seal retainer
7. Input quill cap assembly
8. Input quill pinion
9. Tool (T101306)
10. Retainer ring
11. Main driveshaft adapter
12. Wrench set 412-240-002-101
13. Wrench 412-240-001-101
14. Nut
15. Seal retainer
16. Input quill cap assembly
17. Input quill pinion

ICN-412-A-632102-B-97499-00016-A-002-01

Figure 2. Tool Application (Sheet 2 of 2)



ICN-412-A-632102-B-97499-00017-A-001-01

Figure 3. Main Input Quill Curvic Coupling Adapter and Magnetic Seal Assembly (Sheet 1 of 1)

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Main Input Quill Magnetic Seal - Cleaning

Preliminary requirements

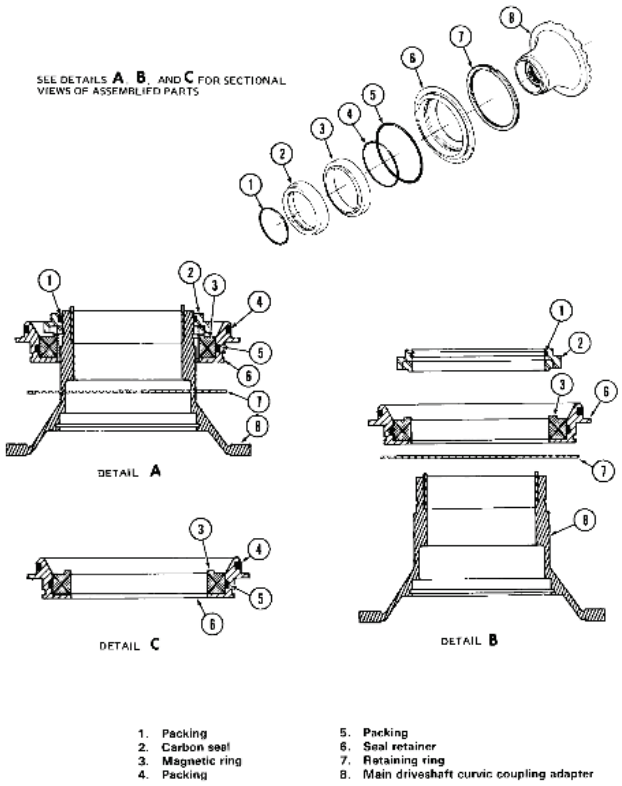
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean seal retainer (6, Figure 1) and main driveshaft curvic coupling adapter (8) with drycleaning solvent (C-304).

2. Dry with filtered, compressed air.



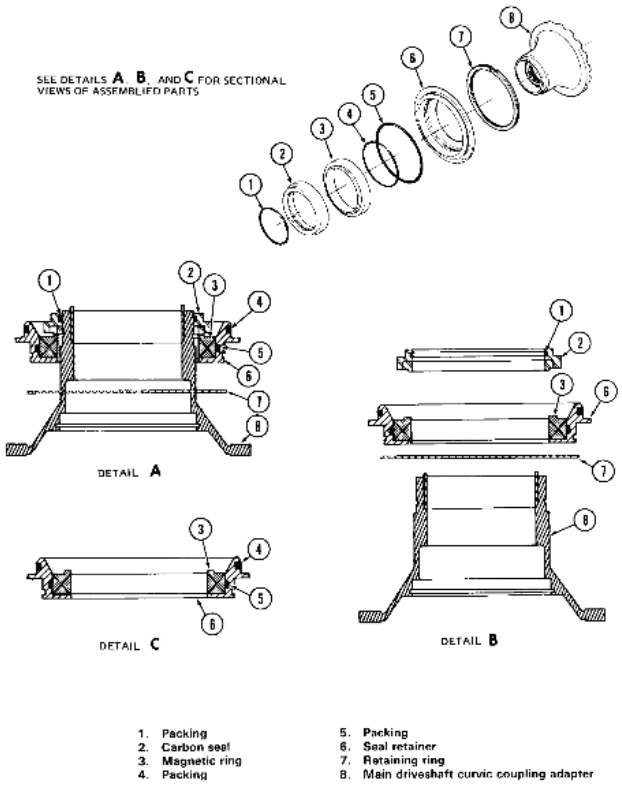
ICN-412-A-632102-B-97499-00017-A-001-01

Figure 1. Main Input Quill Curvic Coupling Adapter and Magnetic Seal Assembly (Sheet 1 of 1)

Main Input Quill Magnetic Seal - Inspection

Procedure

1. Inspect main driveshaft curvic coupling adapter (8, Figure 1) (412-A-63-10-01-02A-280A-A / 01002).
2. Inspect seal retainer (6) (BHT-412-CR&O).



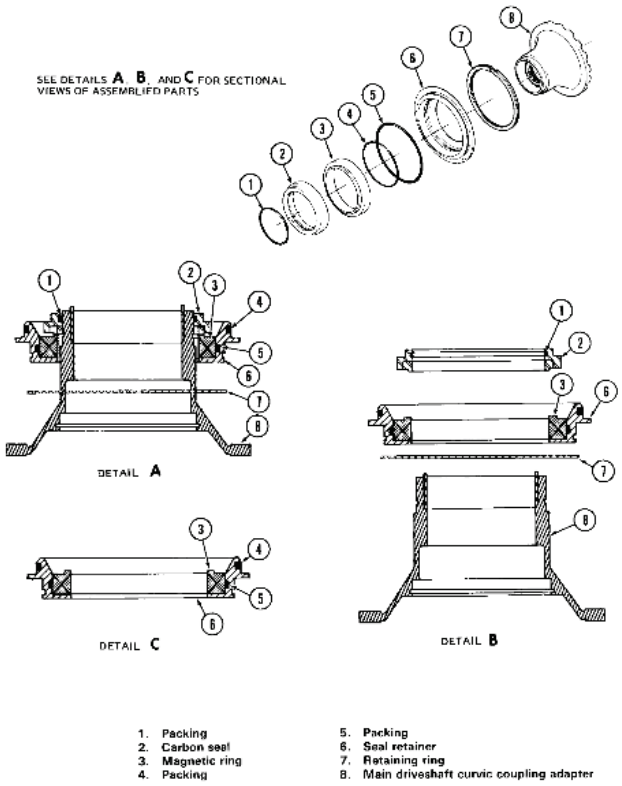
ICN-412-A-632102-B-97499-00017-A-001-01

Figure 1. Main Input Quill Curvic Coupling Adapter and Magnetic Seal Assembly (Sheet 1 of 1)

Main Input Quill Magnetic Seal - Repair

Procedure

1. Polish out nicks and burrs on seal retainer (6, Figure 1) which could damage packings.



ICN-412-A-632102-B-97499-00017-A-001-01

Figure 1. Main Input Quill Curvic Coupling Adapter and Magnetic Seal Assembly (Sheet 1 of 1)

Main Input Quill Magnetic Seal - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Wrench	T101306	1	
Wrench Set	T101588	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-001	AR	
Assembly fluid	C-024	AR	

Procedure

1. Assemble adapter and magnetic seal assembly (12, Figure 1) as follows:



LEAVE SEAL IN PACKAGING UNTIL READY FOR INSTALLATION. SEAL CAVITY SHALL BE CLEAN AND FREE OF BURRS ON PACKING CONTACT DIAMETERS. WHEN SEAL PACKAGE IS OPENED, SEAL SHALL BE SEPARATED BY PULLING HALVES DIRECTLY AWAY FROM EACH OTHER TO PREVENT DAMAGE TO MATING SURFACES. INSTALL SEAL BY HAND PRESSURE ONLY.

1.1. Coat packings (4 and 5, Figure 2) and seal retainer (6) contact diameters with assembly fluid (C-024).

1.2. Install packing (5) in inner groove of seal retainer (6). Install magnetic ring (3) in seal retainer.

1.3. Install packing (4) on outer groove of seal retainer (6).

1.4. Coat packing (1) sparingly with assembly fluid (C-024) and install in inner groove of carbon seal (2) as shown in Detail B.



DO NOT SLIDE SEAL HALVES ON EACH OTHER DURING ASSEMBLY.

1.5. Place carbon seal (2) on magnetic ring (3) very carefully. Align so magnetic seal assembly may be installed on main driveshaft curvic coupling adapter (8) (Detail A).

2. Position retaining ring (7) on main driveshaft curvic coupling adapter (8) (Detail A).

3. Position seal retainer (6), and parts assembled in Step 1., on main driveshaft curvic coupling adapter (8) (Detail A).

4. Coat outer diameter of packing (4) and contact area of seal retainer (6) with assembly fluid (C-024).



DO NOT PUSH ON SEAL RETAINER (6) WHILE INSTALLING MAGNETIC SEAL AND ADAPTER ASSEMBLY. CARBON SEAL (2) MAY BE PUSHED OFF EDGE OF ADAPTER.

5. Align splines of main driveshaft curvic coupling (6, Figure 1) with splines of input quill pinion (11). Using hand pressure only, push main driveshaft curvic coupling inward to obtain contact with splines of pinion.

6. Locate groove in edge of seal retainer (8). Rotate retainer to align groove with pin at upper side of input quill cap assembly (9), then push main driveshaft curvic coupling (6) in against triplex bearings. Refer to Figure 3 for a sectional view of installed adapter and magnetic seal assembly.



DO NOT SEAT SEAL RETAINER (6, Figure 3) INTO INPUT QUILL CAP ASSEMBLY (7) AT THIS TIME.

7. Install nut (5, Figure 1) finger tight to hold adapter and magnetic seal assembly (12) in place.

8. Install wrench (T101306) and wrench set (T101588) as described in 412-A-63-21-02-02A-520A-A / 01140. Torque nut (5) to 350 to 400 foot-pounds (475 to 542 Nm).

9. Install lock-spring (4). Ensure tang on lock-spring fully engages nut (5) and input quill pinion (11).

10. Using hand pressure only, seat seal retainer (6, Figure 3) in groove on input quill cap assembly (7). Ensure seal retainer is fully seated.

11. Install retainer ring (7, Figure 1) in groove on input quill cap assembly (9). Ensure seal retainer (6, Figure 3) is fully seated.

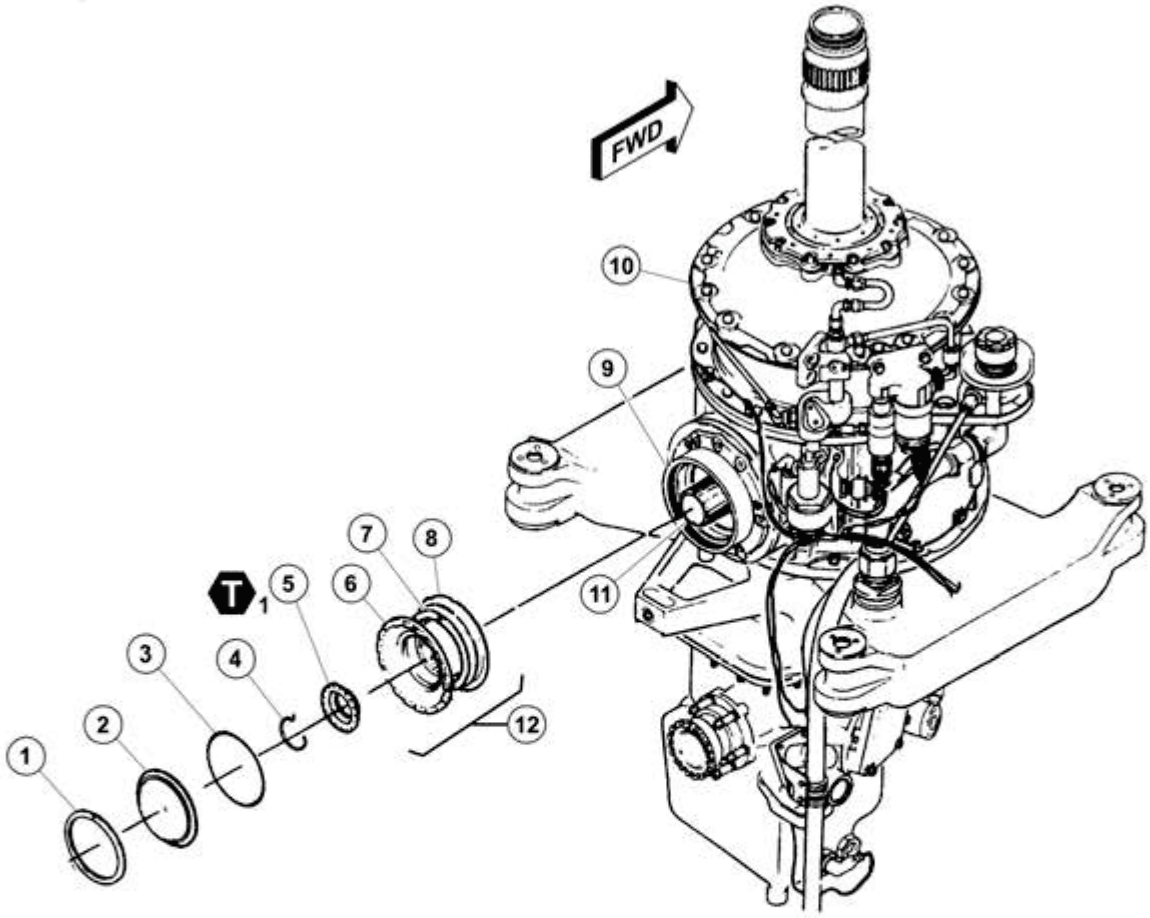
12. Install packing (3, Figure 1) on plate (2). Install plate on main driveshaft curvic coupling (6).

13. Install retainer ring (1). Ensure retainer ring is fully seated.

14. Install main driveshaft (412-A-63-10-01-01A-720A-A / 00999).

15. Close and secure engine and transmission cowling.

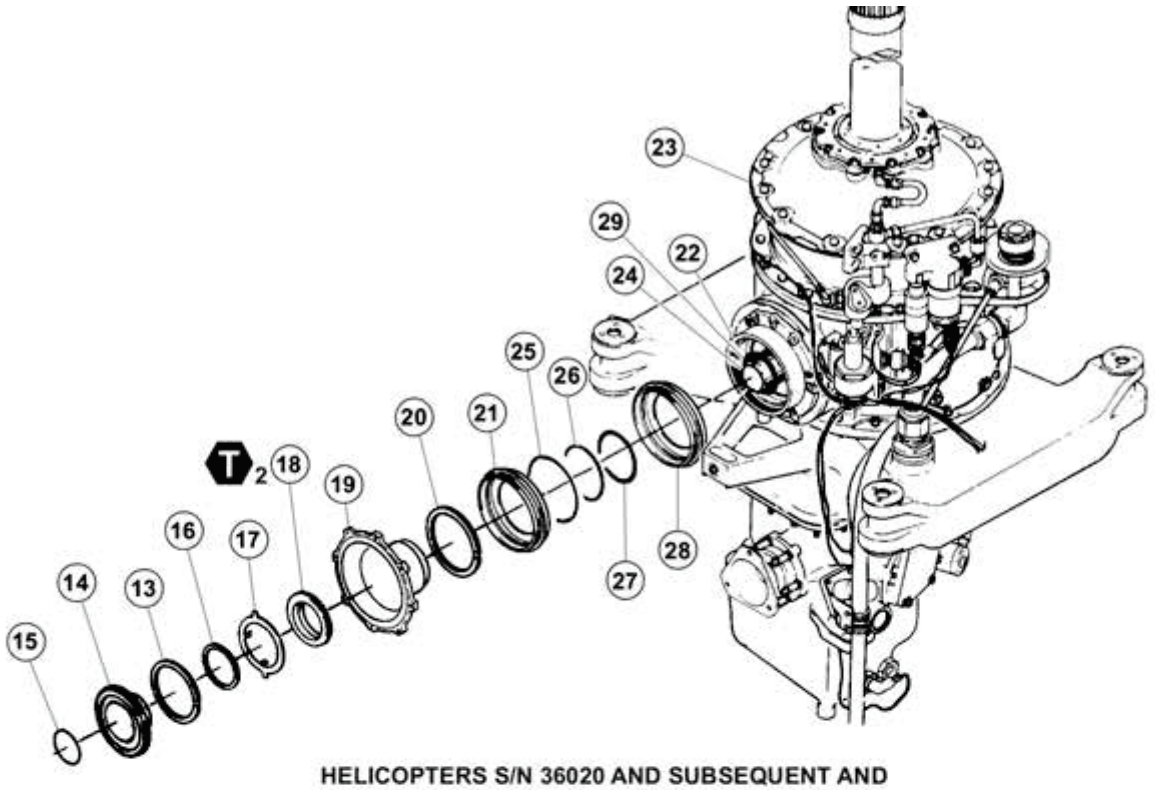
16. Perform engine ground run (BHT-412-FM) to check for operation and leaks.



HELICOPTERES PRIOR TO S/N 36020 AND NOT MODIFIED
BY 412-570-001-103 OR PRE BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR PRE BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

ICN-412-A-632102-B-97499-00013-A-002-01

Figure 1. Main Input Quill Magnetic Seal (Sheet 1 of 2)



HELICOPTERS S/N 36020 AND SUBSEQUENT AND
HELICOPTERS MODIFIED BY 412-570-001-103 OR
POST BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR POST BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

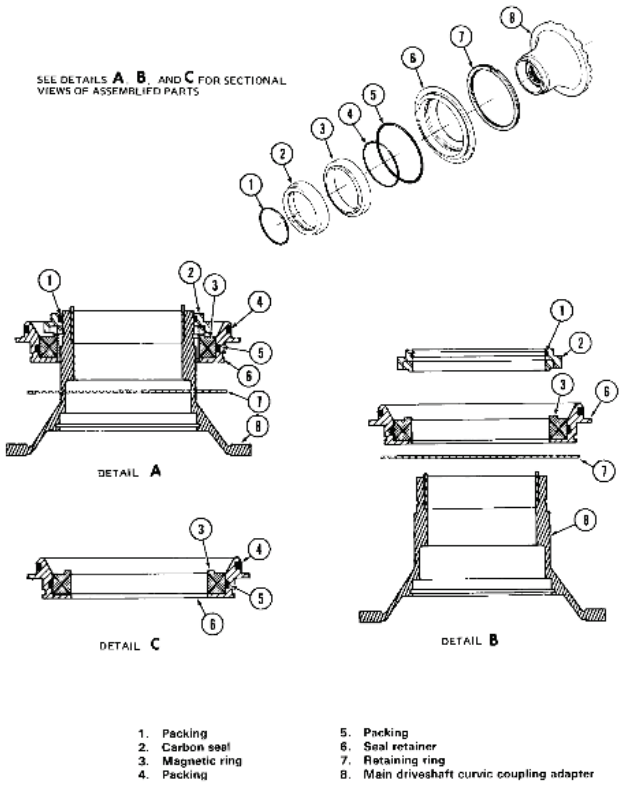
1. Retainer ring
2. Plate
3. Packing
4. Lock-spring
5. Nut
6. Main driveshaft curvic coupling
7. Retainer ring
8. Seal retainer
9. Input quill cap assembly
10. Transmission
11. Input quill pinion
12. Adapter and magnetic seal assembly
13. Retainer ring
14. Plate
15. Packing
16. Retainer ring
17. Lockwasher
18. Nut
19. Main driveshaft adapter
20. Retainer ring
21. Seal retainer
22. Input quill cap assembly
23. Transmission
24. Input quill pinion
25. Packing
26. Packing
27. Packing
28. Magnetic seal
29. Packing

T 1 350 TO 400 FT-LBS (476 TO 542 Nm)

T 2 450 TO 500 FT-LBS (610.12 TO 677.91 Nm)

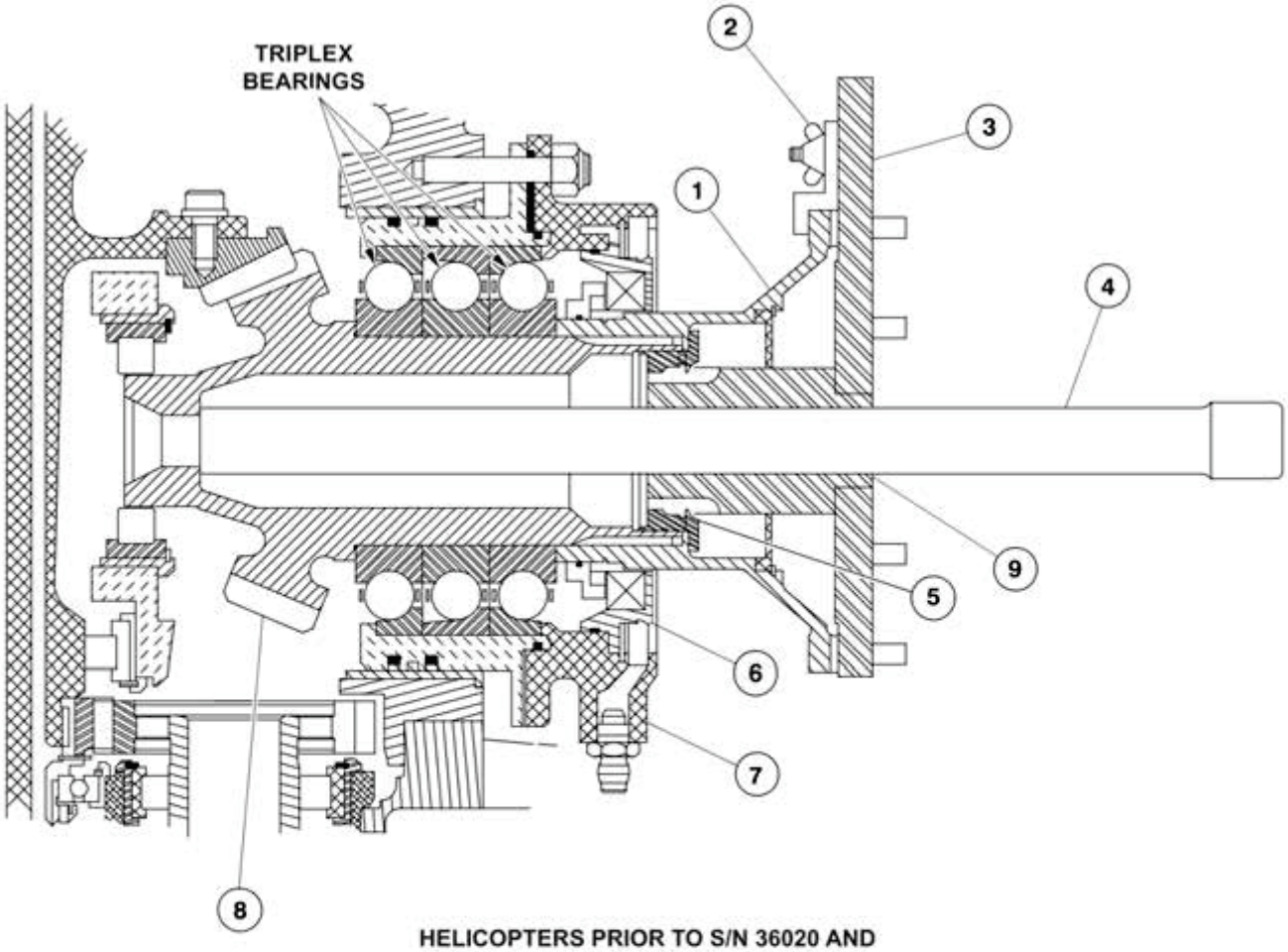
ICN-412-A-632102-B-97499-00014-A-002-01

Figure 1. Main Input Quill Magnetic Seal (Sheet 2 of 2)



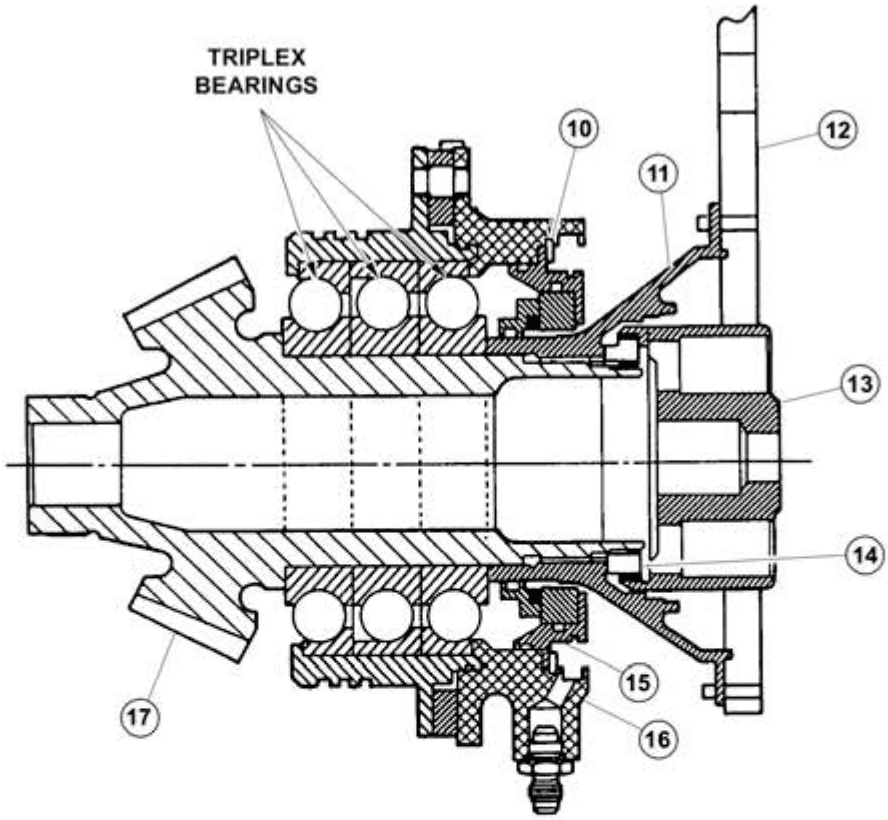
ICN-412-A-632102-B-97499-00017-A-001-01

Figure 2. Main Input Quill Curvic Coupling Adapter and Magnetic Seal Assembly (Sheet 1 of 1)



ICN-412-A-632102-B-97499-00015-A-002-01

Figure 3. Tool Application (Sheet 1 of 2)



HELICOPTERS S/N 36020 AND SUBSEQUENT AND
HELICOPTERS MODIFIED BY 412-570-001-103 OR POST BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR POST BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

1. Main driveshaft curvic coupling adapter
2. Wing nut (MS35426-14)
3. Wrench set (T101588)
4. Extension bar (L-122)
5. Nut
6. Seal retainer
7. Input quill cap assembly
8. Input quill pinion
9. Tool (T101306)
10. Retainer ring
11. Main driveshaft adapter
12. Wrench set 412-240-002-101
13. Wrench 412-240-001-101
14. Nut
15. Seal retainer
16. Input quill cap assembly
17. Input quill pinion

ICN-412-A-632102-B-97499-00016-A-002-01

Figure 3. Tool Application (Sheet 2 of 2)

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Main Input Quill Magnetic Seal - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Wrench	412-240-001-101	1	
Wrench Set	412-240-002-101	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-001	AR	
Assembly fluid	C-024	AR	

Procedure

1. Assemble the magnetic seal (28, Figure 1) and the main driveshaft adapter (19) as follows:



LEAVE THE SEAL IN PACKAGING UNTIL READY FOR INSTALLATION. THE SEAL CAVITY MUST BE CLEAN AND FREE OF BURRS ON PACKING CONTACT DIAMETER. WHEN THE SEAL PACKAGE IS OPEN, THE SEAL MUST BE SEPARATED BY PULLING HALVES AWAY FROM EACH OTHER TO PREVENT DAMAGE TO MATING SURFACES. INSTALL THE SEAL BY HAND PRESSURE ONLY.

1.1. Apply assembly fluid (C-024) to the packings (4 and 5, Figure 2) and the seal retainer (6) contact diameter.

1.2. Install the packing (5) in the inner groove of the seal retainer (6).

1.3. Install the magnetic ring (3) in the seal retainer (6).

1.4. Install the packing (4) on the outer groove of the seal retainer (6).

1.5. Lightly apply assembly fluid (C-024) to the packing (1).

1.6. Install the packing (1) in the inner groove of the carbon seal (2). Refer to Detail B.



DO NOT SLIDE SEAL HALVES ON EACH OTHER DURING ASSEMBLY.

1.7. Very carefully, put the carbon seal (2) in position on the magnetic ring (3).

1.8. Align to make sure that the magnetic seal assembly can be installed on the main driveshaft curvic coupling adapter (8) (Detail A).

2. Put the retainer ring (7) in position on the main driveshaft curvic coupling adapter (8) (Detail A).

3. Put the seal retainer (6) and the assembly from Step 1. in position on the main driveshaft curvic coupling adapter (8) (Detail A).

4. Apply assembly fluid (C-024) to the outside diameter of the packing (4) and to the contact area of the seal retainer (6).

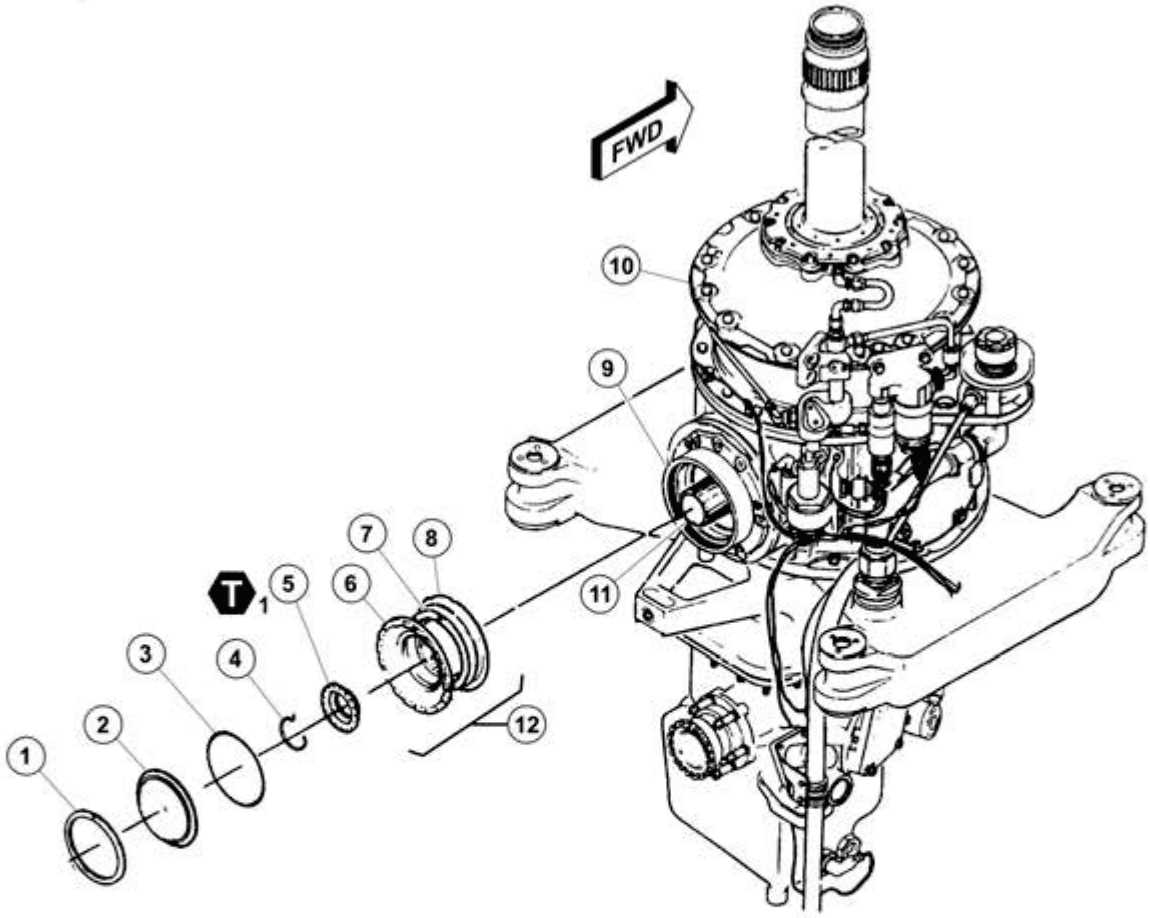
5. Install a new packing (29, Figure 1) on the input quill pinion (24).



DO NOT PUSH THE SEAL RETAINER (6, Figure 2) WHILE INSTALLING THE ADAPTER AND MAGNETIC SEAL. THE CARBON SEAL (2) CAN BE PUSHED OFF THE EDGE OF THE ADAPTER.

6. Align the splines of the main driveshaft adapter (19, Figure 1) with the splines of the input quill pinion (24).

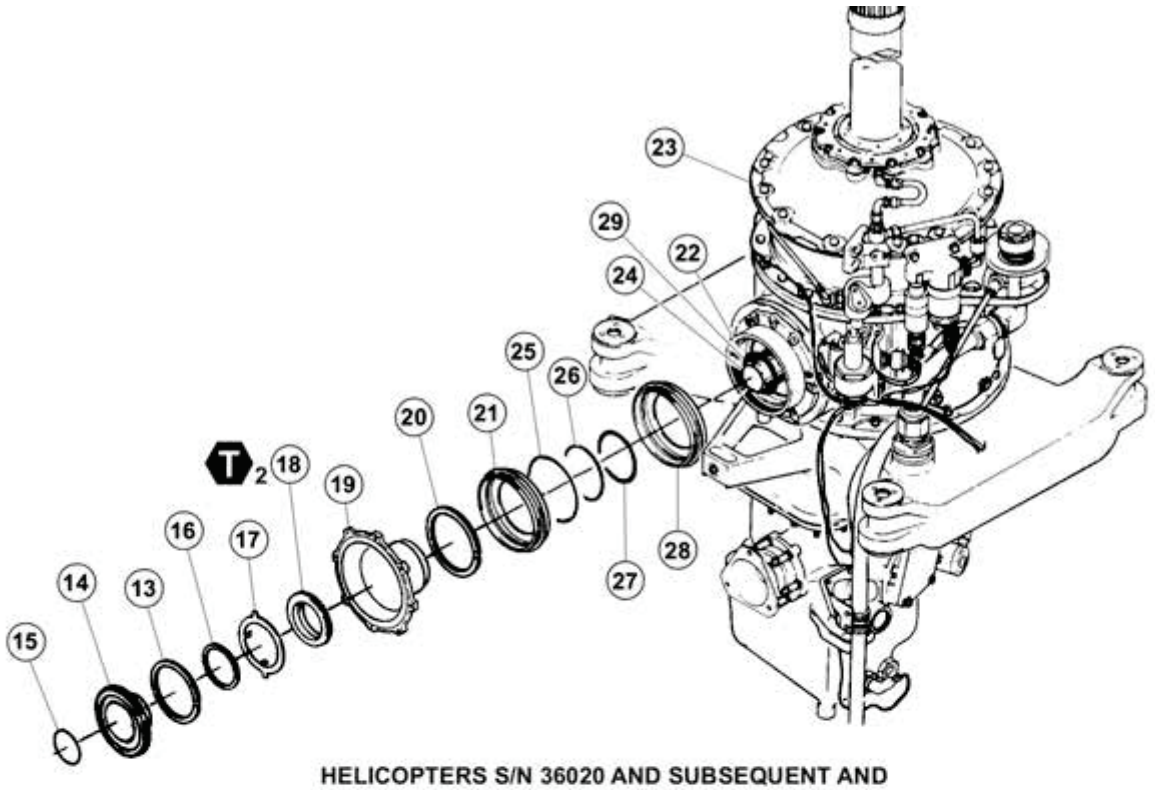
7. With your hands, push on the main driveshaft adapter (19) to get in contact with the splines of the input quill pinion (24).
8. Find the groove in the edge of the seal retainer (21).
9. Rotate the seal retainer (21) to align the groove with the pin at the upper side of the input quill cap assembly (22).
10. Push the main driveshaft adapter (19) against the triplex bearings (refer to Figure 3 for a sectional view of the assembly).
11. Apply a thin layer of grease (C-001) to the threads of the nut (18) and transmission oil to the flange face (412-A-12-10-03-00A-040A-A / 00135).
12. Install the wrench set (412-240-002-101) (Figure 3).
13. With the wrench (412-240-001-101), install the nut (18, Figure 1).
14. Torque the nut (18) .
15. Remove and clean the nut (18).
16. Repeat Step 11., Step 13., and Step 14..
17. Clean all excess grease from the threads and around the flange of the nut (18).
18. Remove the wrench (412-240-001-101) (Figure 3) and the wrench set (412-240-002-101).
19. Install the lockwasher (17, Figure 3) and the retainer ring (16).
20. With hand pressure only, push the seal retainer (17, Figure 3) in the groove of the input quill cap assembly (7).
21. Make sure that the seal retainer (6) is installed against the input quill cap assembly (7).
22. Install the retainer ring (20, Figure 1) in the groove of the input quill cap assembly (22).
23. Make sure that the seal retainer (6, Figure 3) continues to be in the correct position.
24. Install the packing (15, Figure 1) on the plate (14).
25. Install the plate (14) on the main driveshaft adapter (19).
26. Install the retainer ring (13).
27. Make sure that the retainer ring (13) is in the correct position.
28. Install the main driveshaft (412-A-63-10-01-01A-720A-A / 00999).
29. Close and safety the engine and transmission cowling.
30. Do an engine ground run (BHT-412-FM) to examine for leaks and operation.



HELICOPTERES PRIOR TO S/N 36020 AND NOT MODIFIED
BY 412-570-001-103 OR PRE BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR PRE BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

ICN-412-A-632102-B-97499-00013-A-002-01

Figure 1. Main Input Quill Magnetic Seal (Sheet 1 of 2)



HELICOPTERS S/N 36020 AND SUBSEQUENT AND
HELICOPTERS MODIFIED BY 412-570-001-103 OR
POST BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR POST BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

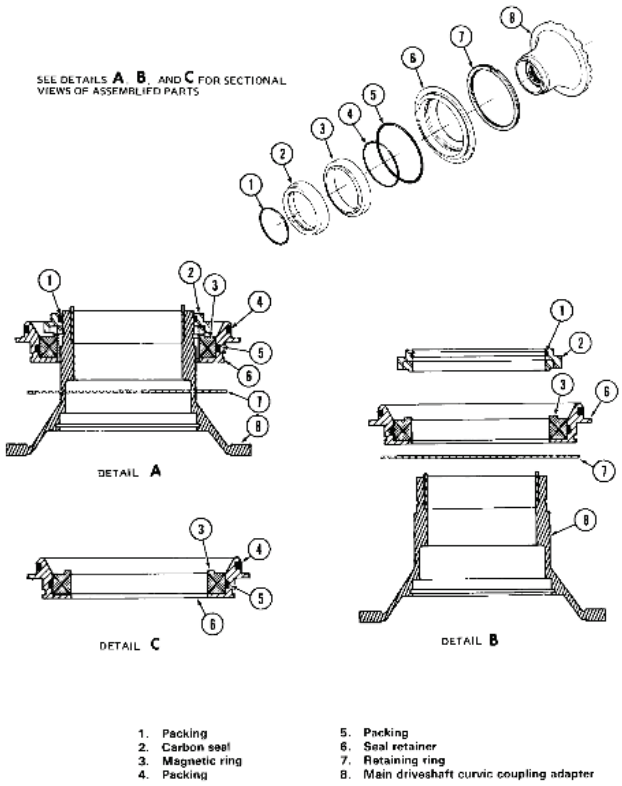
1. Retainer ring
2. Plate
3. Packing
4. Lock-spring
5. Nut
6. Main driveshaft curvic coupling
7. Retainer ring
8. Seal retainer
9. Input quill cap assembly
10. Transmission
11. Input quill pinion
12. Adapter and magnetic seal assembly
13. Retainer ring
14. Plate
15. Packing
16. Retainer ring
17. Lockwasher
18. Nut
19. Main driveshaft adapter
20. Retainer ring
21. Seal retainer
22. Input quill cap assembly
23. Transmission
24. Input quill pinion
25. Packing
26. Packing
27. Packing
28. Magnetic seal
29. Packing

T1 350 TO 400 FT-LBS (476 TO 542 Nm)

T2 450 TO 500 FT-LBS (610.12 TO 677.91 Nm)

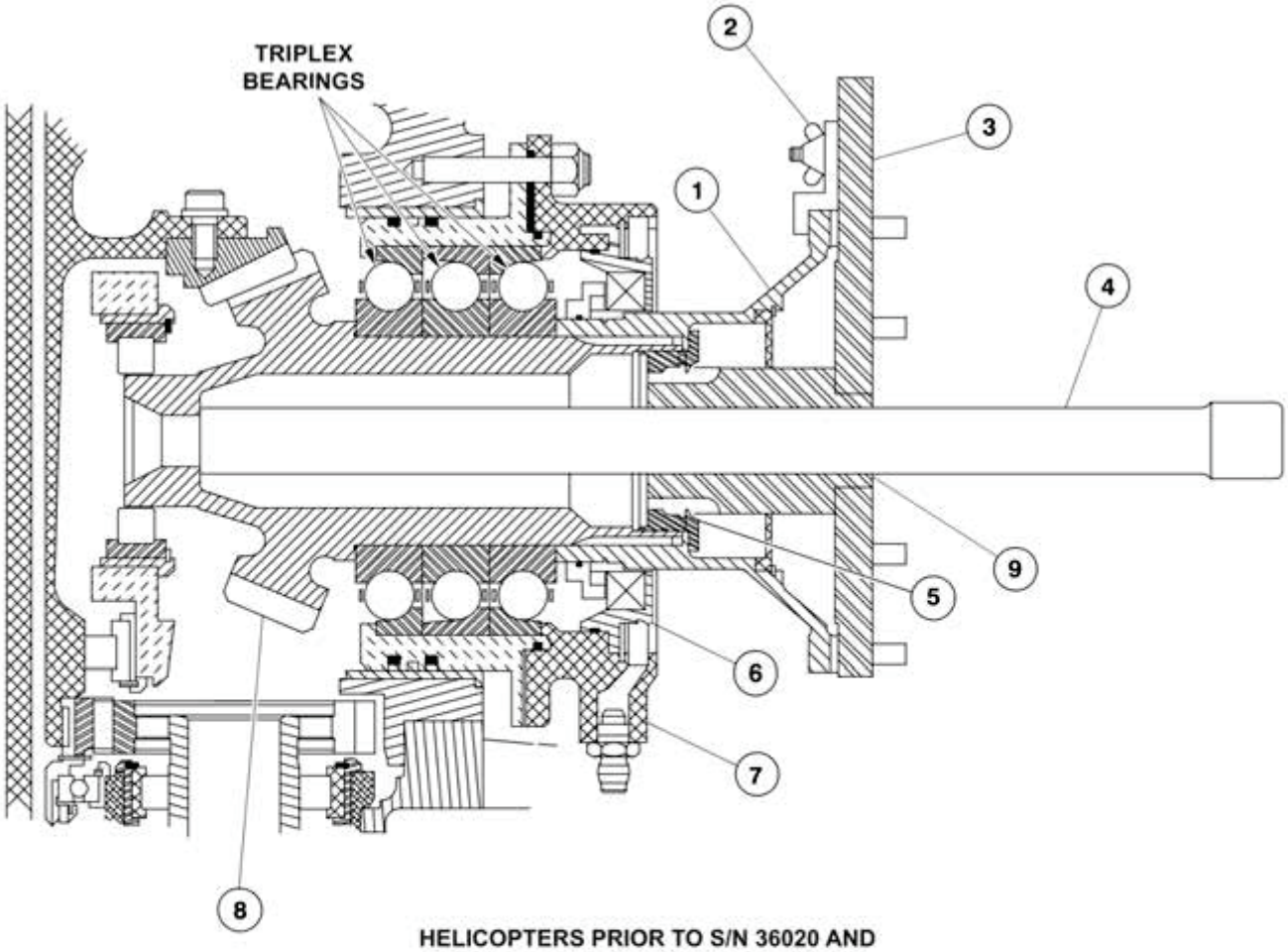
ICN-412-A-632102-B-97499-00014-A-002-01

Figure 1. Main Input Quill Magnetic Seal (Sheet 2 of 2)



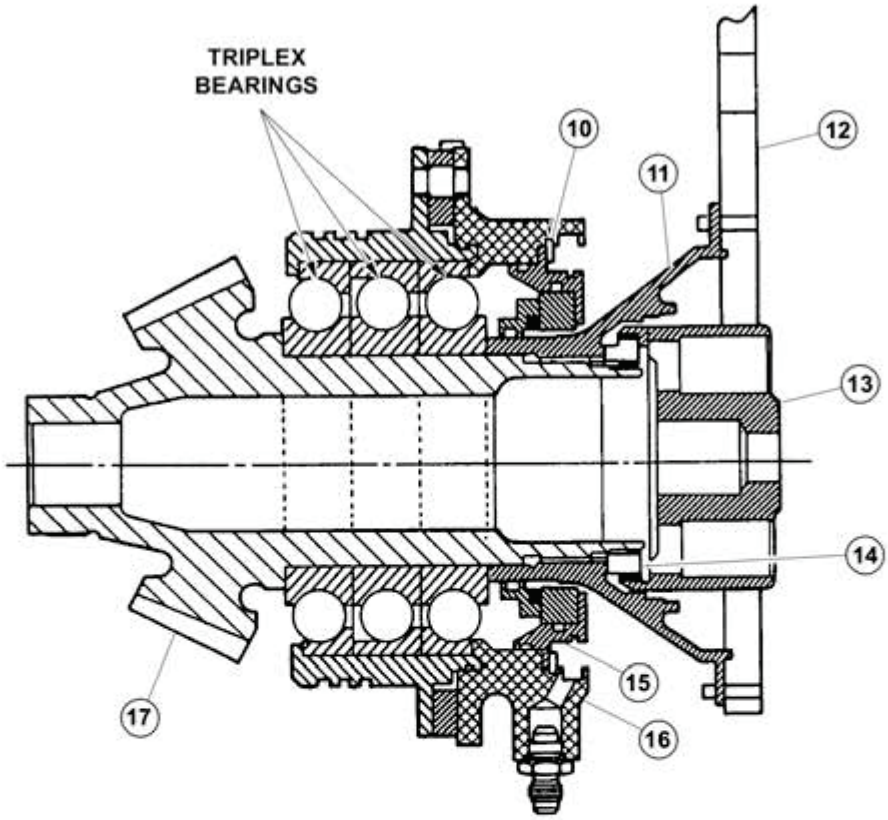
ICN-412-A-632102-B-97499-00017-A-001-01

Figure 2. Main Input Quill Curvic Coupling Adapter and Magnetic Seal Assembly (Sheet 1 of 1)



ICN-412-A-632102-B-97499-00015-A-002-01

Figure 3. Tool Application (Sheet 1 of 2)



HELICOPTERS S/N 36020 AND SUBSEQUENT AND
HELICOPTERS MODIFIED BY 412-570-001-103 OR POST BHT-412-SI-74 (412SP TO 412HP UPGRADE)
OR POST BHT-412-SI-97 (412CLASSIC TO 412HP UPGRADE)

1. Main driveshaft curvic coupling adapter
2. Wing nut (MS35426-14)
3. Wrench set (T101588)
4. Extension bar (L-122)
5. Nut
6. Seal retainer
7. Input quill cap assembly
8. Input quill pinion
9. Tool (T101306)
10. Retainer ring
11. Main driveshaft adapter
12. Wrench set 412-240-002-101
13. Wrench 412-240-001-101
14. Nut
15. Seal retainer
16. Input quill cap assembly
17. Input quill pinion

ICN-412-A-632102-B-97499-00016-A-002-01

Figure 3. Tool Application (Sheet 2 of 2)

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Main Input Quill Lip Seal - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Wrench	T101306	1	
Wrench Set	T101588	1	
Grinder, Handheld	Commercial	1	

Procedure

1. Remove retaining ring (12, Figure 1) and plate (11) with packing (10). Discard packing.

2. Remove lock-spring (9). Remove nut (8) using wrench (T101306) and wrench set (T101588) (412-A-63-21-02-02A-520A-A / 01140).

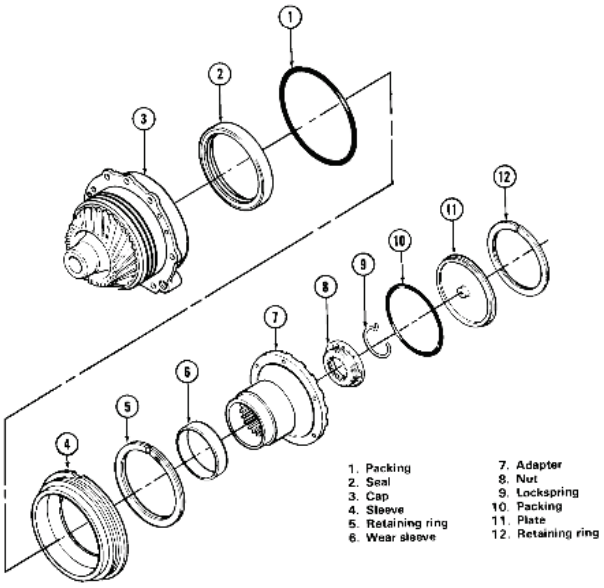


WHEN CUTTING THROUGH OLD WEAR SLEEVE IN THE FOLLOWING STEP, USE EXTREME CARE TO AVOID DAMAGING ADAPTER (7).

3. Remove adapter (7). Using a grinder, handheld, cut a groove almost through wear sleeve (6). Strike wear sleeve with a rubber mallet to break uncut portion of wear sleeve. Remove and discard wear sleeve.

4. Remove retaining ring (5) and sleeve (4) from cap (3). Remove and discard packing (1).

5. Remove seal (2) from sleeve (4) by pressing inboard.



ICN-412-A-632102-B-97499-00018-A-001-01

Figure 1. Main Input Quill Lip Seal (28203-3655) (Sheet 1 of 1)

Main Input Quill Lip Seal - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	
Abrasive pad	C-407	AR	

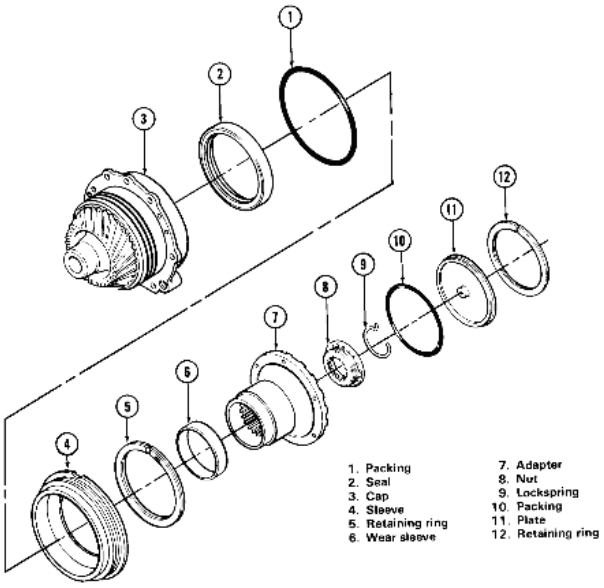
Procedure

1. Clean disassembled components with an abrasive pad (C-407).

2. Clean adapter (7, Figure 1) and sleeve (4) with a plastic scraper and an abrasive pad (C-407) to remove foreign matter.

3. Rinse adapter (7) and sleeve (4) with drycleaning solvent (C-304).

4. Dry parts using filtered, compressed air.



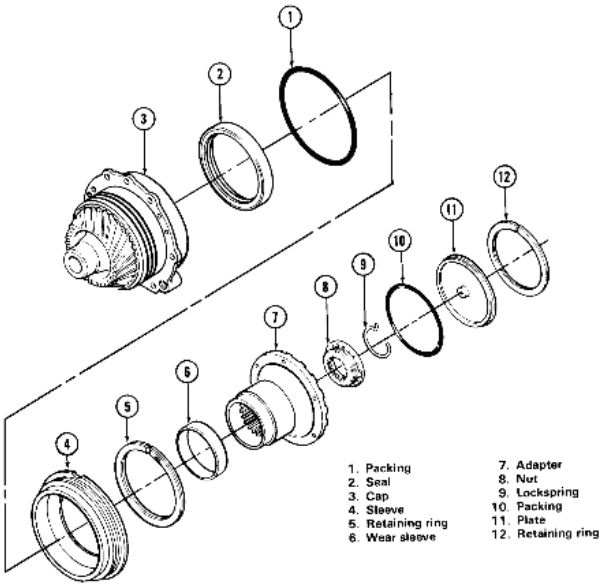
ICN-412-A-632102-B-97499-00018-A-001-01

Figure 1. Main Input Quill Lip Seal (28203-3655) (Sheet 1 of 1)

Main Input Quill Lip Seal - Inspection

Procedure

1. Inspect parts for excessive wear and corrosion.
2. Inspect parts for cracks. No cracks are allowed.
3. Inspect packing grooves in cap (3, Figure 1) and plate (11) for sharp nicks and distortion.
4. Inspect threaded components for thread damage. No thread damage is allowed.
5. Inspect retaining rings (5 and 12) for distortion. No distortion is allowed.



ICN-412-A-632102-B-97499-00018-A-001-01

Figure 1. Main Input Quill Lip Seal (28203-3655) (Sheet 1 of 1)

Main Input Quill Lip Seal - Repair

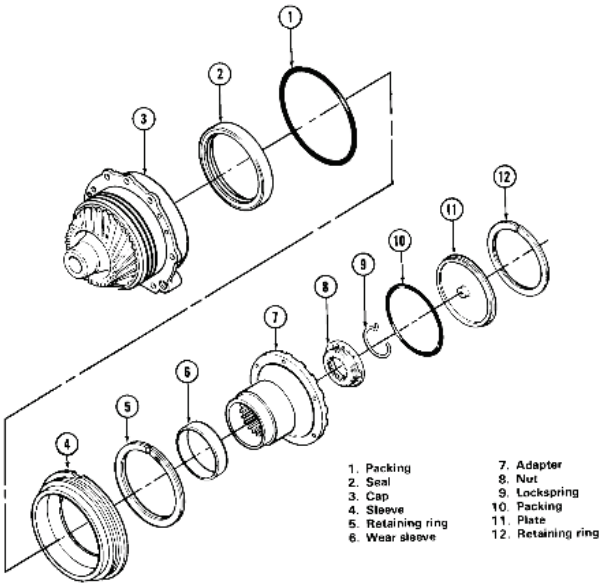
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Abrasive pad	C-407	AR	

Procedure

1. Remove light surface corrosion from adapter (7, Figure 1) and sleeve (4) with a plastic scraper and abrasive pad (C-407).



ICN-412-A-632102-B-97499-00018-A-001-01

Figure 1. Main Input Quill Lip Seal (28203-3655) (Sheet 1 of 1)

Main Input Quill Lip Seal - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Wrench	T101306	1	
Wrench Set	T101588	1	
Brush, Hog Bristle, Unpainted Handle	Commercial	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Dry film lubricant	C-028	AR	
Adhesive	C-317	AR	
Adhesion promoter	C-327	AR	
Sealant	C-328	AR	
Aliphatic naphtha	C-305	AR	
Abrasive pad	C-407	AR	
Barrier material	C-427	AR	

Procedure



BONDING SHALL BE ACCOMPLISHED WITHOUT DELAY BETWEEN STEPS. THE MAXIMUM TIME INTERVAL BETWEEN SANDBLASTING AND THE NEXT STEP IS 4 HOURS.

NOTE

Precautions shall be taken when accomplishing the following steps to keep wear sleeve and adapter clean. Wear clean white gloves when handling parts and change gloves when soiled. Cover work bench with clean white paper. Avoid any contact with mold release agents such as silicone, oil, waxes, grease, etc. Area where work is performed shall be maintained at 64 to 82°F (18 to 28°C) with relative humidity below 75%. Area shall be dust-free to the extent that bonding operations are not affected.

1. Mask outer diameter of new wear sleeve (6, Figure 1).

2. Mask all surfaces of adapter (7) except those surfaces which mate with wear sleeve (6).

3. Sandblast unmasked surfaces of wear sleeve (6) and adapter with 60 grit garnet at 5 to 10 PSI and remove residue. If there is to be any delay before accomplishing the next step, wrap parts in clean barrier material (C-427) after sandblasting.

4. Stir aliphatic naphtha (C-305) thoroughly in its container and apply to unmasked portion of adapter (7) only. Scrub primer into surface of adapter with abrasive pad (C-407). Repeat scrubbing as required to ensure adapter is clean.

5. Allow degreaser residue to dry to a white powder. Remove residue from adapter (7) with a clean, dry cheesecloth.



KEEP adhesion promoter (C-327) IN A COVERED, NON-ABSORPTIVE CONTAINER AND KEEP LID TIGHTLY CLOSED WHEN NOT IN USE. EXPOSURE TO ATMOSPHERE WILL CAUSE DECOMPOSITION. SEDIMENT OR BROWN RESIDUE IN BOTTOM OF CONTAINER INDICATES DECOMPOSITION OF PROMOTER TO THE EXTENT IT SHALL BE DISCARDED. PROMOTER PH RANGE SHALL BE MAINTAINED BETWEEN 7.0 AND 8.0.

NOTE

If the next step is not to be accomplished immediately, wrap adapter (7) with clean barrier material (C-427).

6. Apply adhesion promoter (C-327) to unmasked portion of wear sleeve (6) and adapter (7) with a pure hog bristle brush with an unpainted handle. Apply only enough promoter to wet the surfaces.

7. Allow promoter to dry for a minimum of 30 minutes at 65 to 85°F (18 to 29°C) or for 15 minutes at 150 to 160°F (66 to 71°C).

NOTE

If the next step is not to be accomplished immediately, wrap parts with clean barrier material (C-427).

8. Mix adhesive (C-317) 33 parts B to 100 parts A. The pot life of mixed adhesive is 30 to 50 minutes.

9. Coat inside diameter of wear sleeve (6) and mating surface on adapter (7) with mixed adhesive.

10. Press wear sleeve (6) on adapter (7) until inner shoulder of wear sleeve contacts adapter.

11. Clean excess adhesive from wear sleeve (6) thoroughly and remove all protective masking material. Allow adhesive to cure at 65 to 85°F (18 to 29°C) for 24 hours.



ENSURE NO EXCESSIVE BONDING MATERIAL REMAINS BETWEEN WEAR SLEEVE (6) AND COUPLING RACE. BONDING MATERIAL WILL BECOME HARD AND FLAKE OFF. THIS FLAKING BETWEEN SEAL LIP AND WEAR RING WILL CAUSE LEAKAGE.

12. Apply a film of sealant (C-328) to outside diameter of new seal (2). Press seal into sleeve (4) until bottomed out. Remove excess sealant.

13. Lubricate packing (1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packing on sleeve (4). Align cutout in sleeve with pin in cap (3) and install sleeve in cap by pressing with hand pressure. Install retaining ring (5).

14. Spray wear sleeve (6) with dry film lubricant (C-028). Position adapter (7) on quill pinion.

15. Install wrench (T101306) and wrench set (T101588) as described in 412-A-63-21-02-02A-520A-A / 01140. Torque nut (8) to 350 to 400 foot-pounds (475 to 542 Nm).

16. Install lockspring (9) in hole of nut that is in line with slot in pinion.

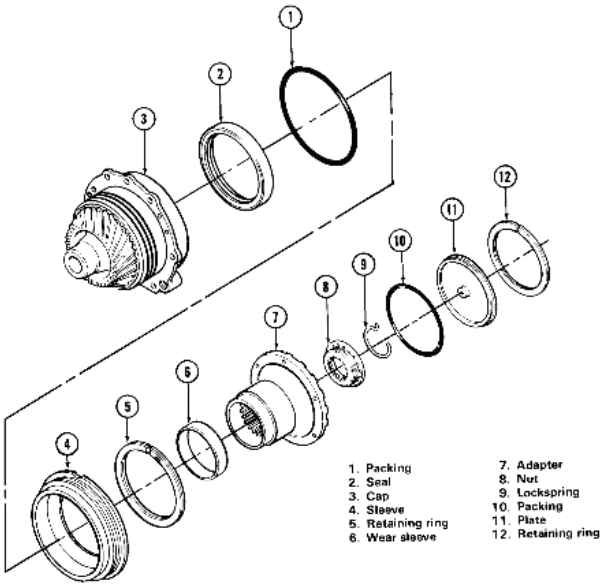
17. Lubricate new packing (10) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packing on plate (11). Install plate in adapter (7).

18. Install retaining ring (12) in adapter (7).

19. Install main driveshaft (412-A-63-10-01-01A-720A-A / 00999).

20. Close engine and transmission cowling.

21. Perform engine ground run (BHT-412-FM) and check for proper operation and leakage.



ICN-412-A-632102-B-97499-00018-A-001-01

Figure 1. Main Input Quill Lip Seal (28203-3655) (Sheet 1 of 1)

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No. 1 Hydraulic Pump Drive Quill - Description

The No. 1 hydraulic system pump and tachometer drive quill is located on the forward right side of the transmission sump assembly.

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No. 1 Hydraulic Pump Drive Quill - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Jackscrew Set	T101308	1	

Procedure

1. Remove access door on right side of pylon support in cabin area.

2. Using a sharp plastic scraper, cut sealant at mating flanges of tachometer generator (12, Figure 1), hydraulic pump (3), and quill (7). Remove sealant from jackscrew holes.

3. Remove tachometer generator (12) (412-A-63-21-02-06A-520A-A / 01165).

4. Remove hydraulic pump (3) (412-A-29-10-00-01A-520A-A / 00620).

5. Remove the nuts (5) and washers (6) from two remaining quill mounting studs (10).

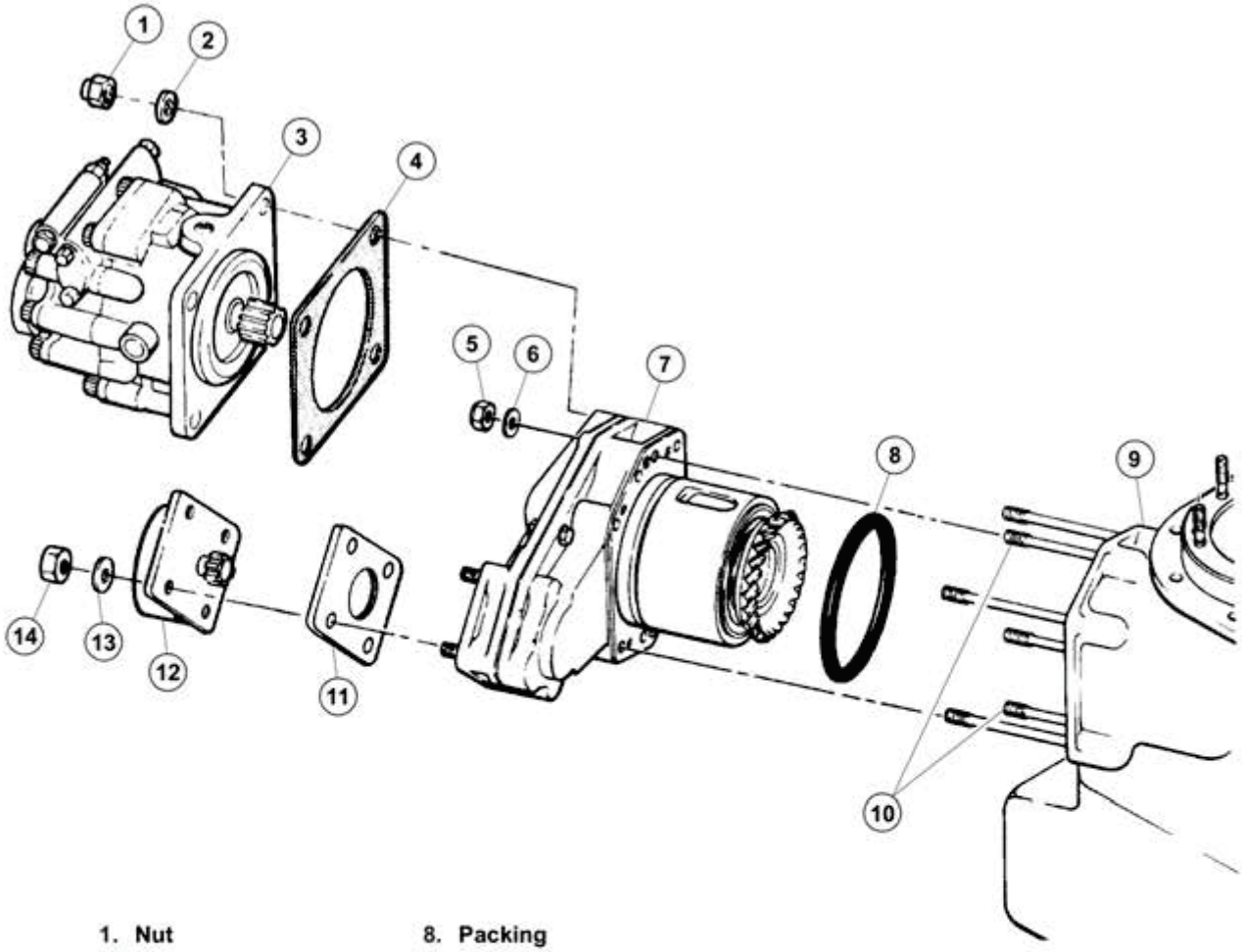
6. Install jackscrew set (T101308) on quill (7).



TIGHTEN JACKSCREWS CAREFULLY AND EVENLY TO KEEP QUILL ALIGNED DURING REMOVAL.

7. Use a heat lamp or a 250 watt light bulb to heat sump (9) in area of quill (7). Heat area until hot to the touch with bare hands. Remove quill by tightening jackscrews evenly. Cover open mounting port.

8. Remove and discard packing (8) from quill (7).



1. Nut
2. Steel washer
3. Hydraulic pump
4. Gasket
5. Nut
6. Aluminium washer
7. Quill
8. Packing
9. Sump
10. Drive quill mounting studs
11. Gasket
12. Tachometer generator
13. Steel washer
14. Nut

ICN-412-A-632102-B-97499-00019-A-001-01

Figure 1. No. 1 Hydraulic Pump Drive Quill (Sheet 1 of 1)

No. 1 Hydraulic Pump Drive Quill - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean sealant residue from both quill (7) and transmission mounting port using a sharp plastic scraper.
2. Clean external surfaces of parts using drycleaning solvent (C-304).
3. Dry with filtered, compressed air.

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No. 1 Hydraulic Pump Drive Quill - Inspection

Procedure

1. Visually inspect all accessible parts for damage.
2. Rotate quill gear and check bearings for smooth operation. Roughness that can be detected by feel is unacceptable.
3. Inspect gear teeth for nicks, dents, cracks, and abnormal wear pattern. Any such damage that can be detected visually is unacceptable.
4. Inspect seals for evidence of leakage, damage, and abnormal wear. Leaking, damaged, or worn seals are unacceptable.
5. For additional, detailed inspection criteria, refer to the BHT-412-CR&O.

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No. 1 Hydraulic Pump Drive Quill - Repair

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Jackscrew Set	T101308	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. The quill (7, Figure 1) shall be replaced as an assembly when inspection requirements are not met. Seals, however, may be replaced as required.

2. Replace seals (2 and 9, Figure 2) as follows:

2.1. Remove nuts (12), washers (11 and 10), and bolt (5) with washers (6 and 8) and bracket (7).

NOTE

As sleeve assembly (4) and cover (1) separate, tap splined end of gear shafts (not shown) where they protrude through seals (2 and 9) to prevent binding of shaft bearing in sleeve assembly.

2.2. Using jackscrew set (T101308) carefully and evenly remove cover (1) from sleeve assembly (4).

2.3. Remove and discard gasket (3) from sleeve assembly (4) studs.

2.4. Press seals (2 and 9) from cover (1).

2.5. Clean seal housing in cover (1) with drycleaning solvent (C-304) and dry using filtered, compressed air.

2.6. Install seals (2 and 9) on cover (1) with numbered side of seals outboard.

2.7. Coat lip of seals (2 and 9) with transmission oil (412-A-12-01-00-00A-040A-A / 00141).

2.8. Position new gasket (3) on sleeve assembly (4) studs.

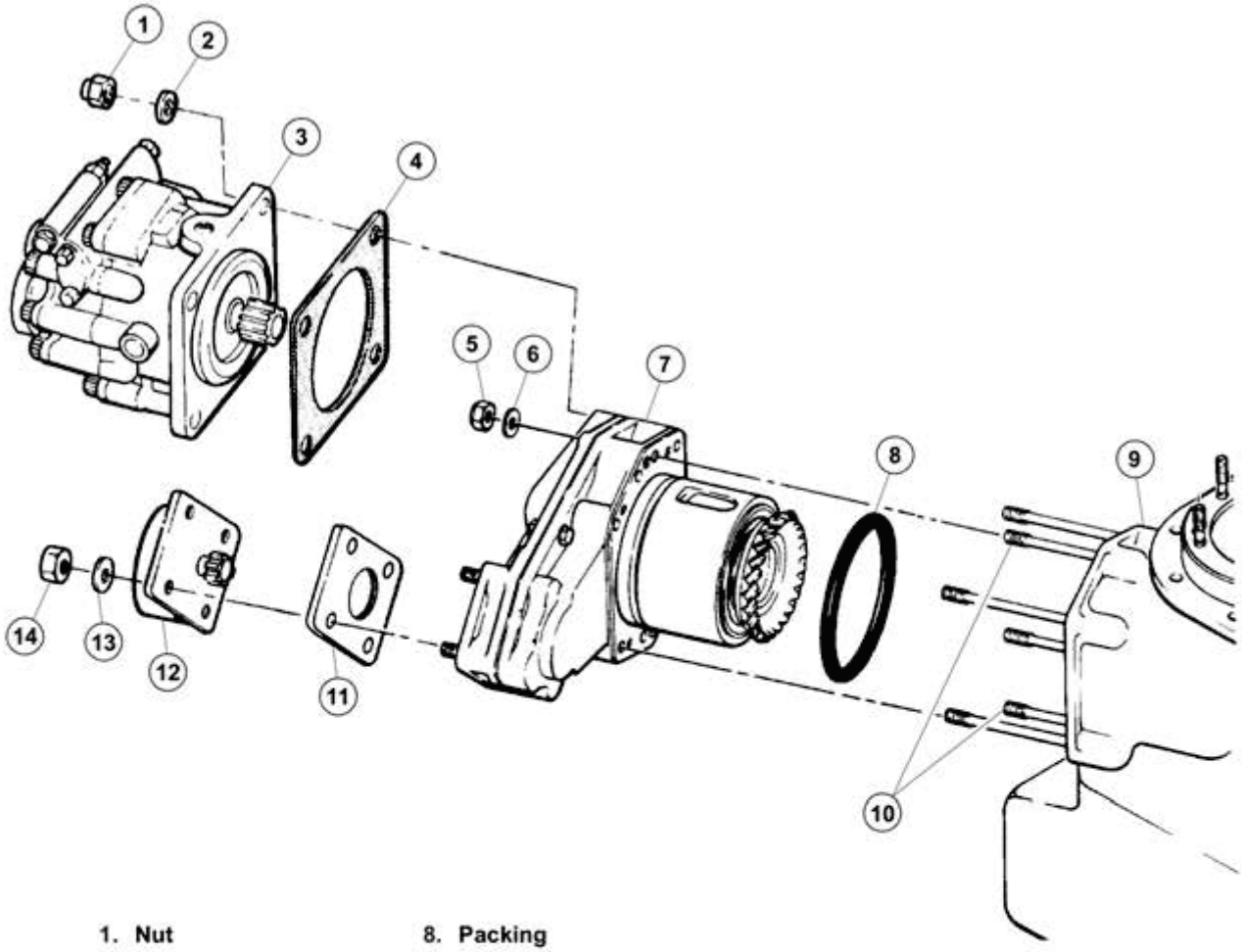
2.9. Install cover (1) on sleeve assembly (4).

NOTE

Bracket (7) replaces steel washer (6) at 12 o'clock position only.

2.10. Install bolt (5) with aluminum washers (8) next to quill housing and steel washers (6), or bracket (7), next to bolt head.

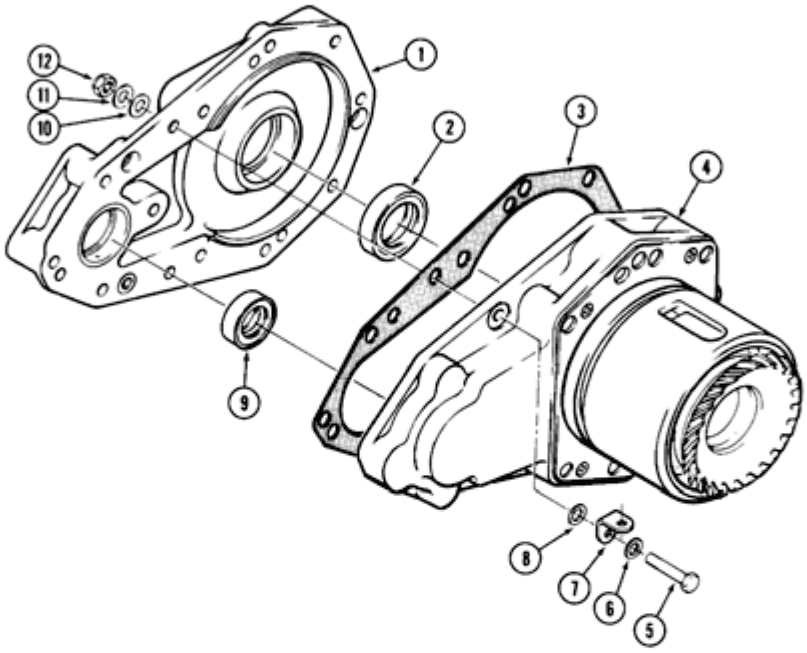
2.11. Install nuts (12) with aluminum washers (10) next to cover (1) and steel washers (11) next to nut.



1. Nut
2. Steel washer
3. Hydraulic pump
4. Gasket
5. Nut
6. Aluminium washer
7. Quill
8. Packing
9. Sump
10. Drive quill mounting studs
11. Gasket
12. Tachometer generator
13. Steel washer
14. Nut

ICN-412-A-632102-B-97499-00019-A-001-01

Figure 1. No. 1 Hydraulic Pump Drive Quill (Sheet 1 of 1)



1. Cover
2. Seal
3. Gasket
4. Sleeve assembly
5. Bolt
6. Steel washer
7. Bracket
8. Aluminum washer
9. Seal
10. Aluminum washer
11. Steel washer
12. Nut

ICN-412-A-632102-B-97499-00020-A-001-01

Figure 2. No. 1 Hydraulic Pump Drive Quill Assembly (Sheet 1 of 1)

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Applicable to: ALL
ECCN EAR99

No. 1 Hydraulic Pump Drive Quill - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

Procedure

1. Lubricate new packing (8, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packing in groove around quill (7) sleeve.

2. Use a heat lamp or 250 watt light bulb to heat sump mounting port until unit is hot to the touch.

3. Coat mating surfaces of quill (7) and sump (9) with transmission oil (412-A-12-01-00-00A-040A-A / 00141).

4. Position quill (7) on mounting port with tachometer generator pad forward.

5. Insert quill (7) into sump (9). Mesh gears carefully and guide sleeve flange over mounting studs.

6. Install aluminum washers (6), and nuts (5) on two drive quill mounting studs (10). Torque nuts to 100 to 140 inch-pounds (11.3 to 11.7 Nm).

7. Check quill (7) to verify presence of backlash.

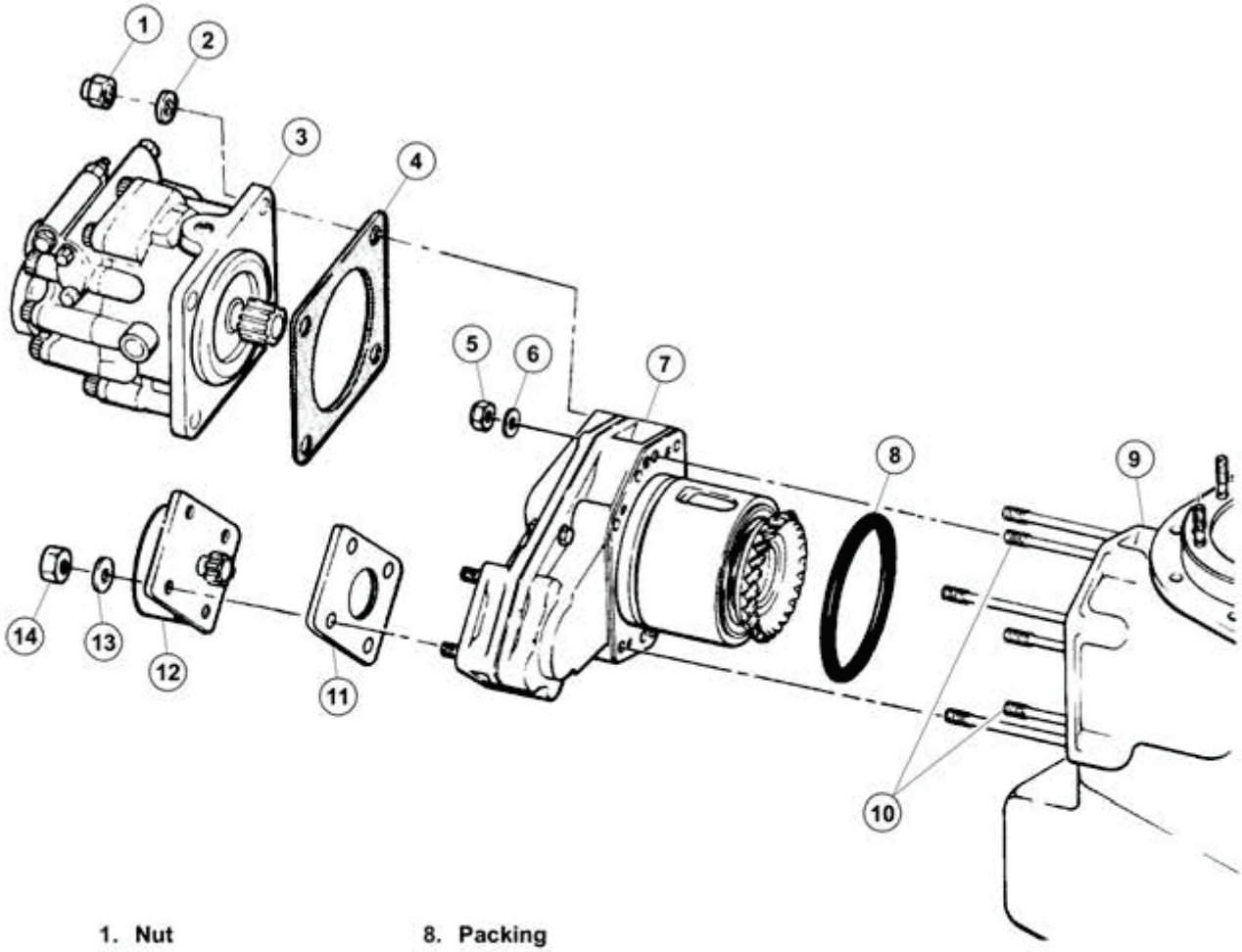
8. Install hydraulic system No. 1 pump (412-A-29-10-00-01A-720A-A / 00623).

9. Install rotor tachometer generator (12) (412-A-63-21-02-06A-720A-A / 01169).

10. Apply a bead of sealant (C-308) around mating surfaces of quill (7), hydraulic pump (3), and tachometer generator (12).

11. Install access door on right side of pylon support in cabin area.

12. Perform engine ground run (BHT-412-FM , Flight Manual) and check for proper operation of hydraulic system No. 1 and for leakage.



1. Nut
2. Steel washer
3. Hydraulic pump
4. Gasket
5. Nut
6. Aluminium washer
7. Quill
8. Packing
9. Sump
10. Drive quill mounting studs
11. Gasket
12. Tachometer generator
13. Steel washer
14. Nut

ICN-412-A-632102-B-97499-00019-A-001-01

Figure 1. No. 1 Hydraulic Pump Drive Quill (Sheet 1 of 1)

No. 2 Hydraulic Pump Drive Quill - Description

The No. 2 hydraulic system pump is driven by a quill located on forward side of transmission.

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No. 2 Hydraulic Pump Drive Quill - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Jackscrew Set	T101338	1	

Procedure

1. Open transmission forward cowling.

2. Remove hydraulic system No. 2 pump (412-A-29-10-00-02A-520A-A / 00625).

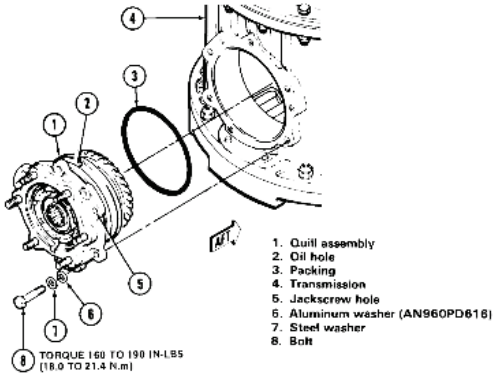
3. Using a sharp plastic scraper, cut sealant from around flange of quill assembly (1, Figure 1). Clean sealant from jackscrew holes (5).

4. Remove bolts (8) and washers (6 and 7).

5. Install jackscrew set (T101338) in holes on quill assembly (1). Tighten jackscrews carefully and evenly to remove quill assembly from transmission (4) case.

6. Remove and discard packing (3).

7. Cover open mounting port to prevent foreign objects from entering transmission (4) case.



ICN-412-A-632102-B-97499-00021-A-001-01

Figure 1. No. 2 Hydraulic Pump Drive Quill (Sheet 1 of 1)

No. 2 Hydraulic Pump Drive Quill - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean sealant residue from both quill and transmission mounting port using a sharp plastic scraper.
2. Clean external surfaces of parts using drycleaning solvent (C-304).
3. Dry with filtered, compressed air.

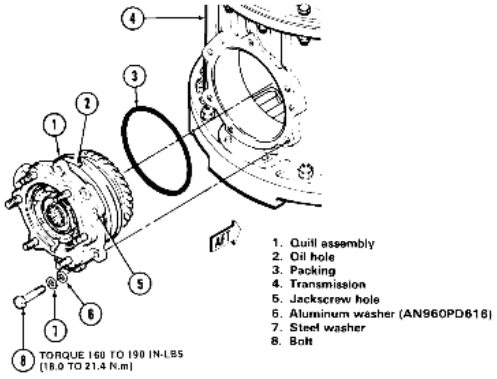
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No. 2 Hydraulic Pump Drive Quill - Inspection

Procedure

1. Inspect quill assembly (1, Figure 1) for obvious mechanical and corrosion damage (BHT-412-CR&O).

2. Rotate pinion in sleeve and check for rough bearings. Bearing roughness that can be detected by feel is not acceptable.



ICN-412-A-632102-B-97499-00021-A-001-01

Figure 1. No. 2 Hydraulic Pump Drive Quill (Sheet 1 of 1)

No. 2 Hydraulic Pump Drive Quill - Repair

Procedure

1. For repair and overhaul instructions, refer to the BHT-412-CR&O.

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No. 2 Hydraulic Pump Drive Quill - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

Procedure

1. Lubricate packing (3, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141) and install on quill assembly (1).

2. Lubricate mating surfaces of quill assembly (1) and transmission mounting port with transmission oil.

3. Use a heat lamp or 250 watt light bulb to heat sump mounting port until unit is hot to the touch.

4. Position quill assembly (1) with oil hole (2) straight up. Insert quill assembly into transmission (4) while simultaneously rotating pinion to permit gear meshing.

NOTE

Install aluminum washers AN960JD616 (6) next to quill sleeve. Do not substitute any other type washer (including NAS1197).

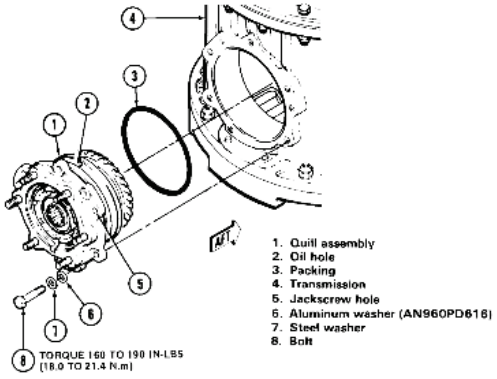
5. Install bolts (8) with steel washers (7) under bolt heads and aluminum washers adjacent to quill flange. Torque bolts to 160 to 190 inch-pounds (18.1 to 21.5 Nm).

6. Check quill pinion for freedom of rotation and existence of backlash.

7. Fill gap between flange on quill assembly (1) and flange on transmission (4) with sealant (C-308). Also fill jackscrew holes (5) with same sealant. Allow to cure.

8. Install hydraulic system No. 2 pump (412-A-29-10-00-02A-720A-A / 00628).

9. Perform engine ground run (BHT-412-FM , Flight Manual) and check for proper operation of hydraulic system No. 2 and for leakage.

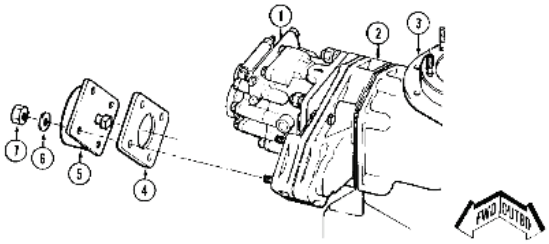


ICN-412-A-632102-B-97499-00021-A-001-01

Figure 1. No. 2 Hydraulic Pump Drive Quill (Sheet 1 of 1)

Rotor Tachometer Generator - Description

The rotor tachometer generator (5, Figure 1) is driven by a chain-and-sprocket offset drive from the hydraulic pump and tachometer generator drive quill (2), which also drives the system No. 1 hydraulic pump (1). The hydraulic pump and tachometer generator drive quill are located on the transmission sump case (3).



1. Hydraulic pump (System No. 1)
2. Hydraulic pump and tachometer generator drive quill
3. Transmission sump case
4. Gasket
5. Rotor tachometer generator
6. Steel washer
7. Nut

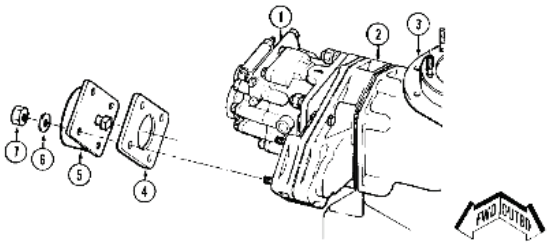
ICN-412-A-632102-B-97499-00022-A-001-01

Figure 1. Rotor Tachometer Generator (Sheet 1 of 1)

Rotor Tachometer Generator - Removal

Procedure

1. Disconnect electrical connector from rotor tachometer generator (5, Figure 1).
2. Remove nuts (7) and steel washers (6).
3. Using a sharp plastic scraper, cut sealant from flange of rotor tachometer generator (5).
4. Remove rotor tachometer generator (5) and gasket (4). Discard gasket.



1. Hydraulic pump (System No. 1)
2. Hydraulic pump and tachometer generator drive quill
3. Transmission sump case
4. Gasket
5. Rotor tachometer generator
6. Steel washer
7. Nut

ICN-412-A-632102-B-97499-00022-A-001-01

Figure 1. Rotor Tachometer Generator (Sheet 1 of 1)

Rotor Tachometer Generator - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean sealant residue from both quill and mating surface of offset drive using a sharp plastic scraper.
2. Clean external surfaces of parts using drycleaning solvent (C-304).
3. Dry with filtered, compressed air.

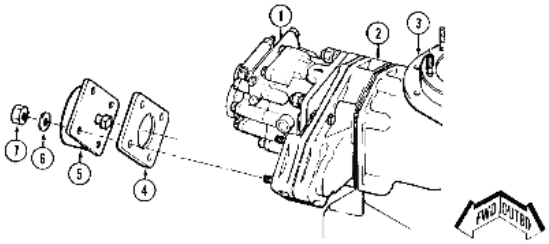
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Rotor Tachometer Generator - Inspection

Procedure

1. Inspect rotor tachometer generator (5, Figure 1) for mechanical and corrosion damage. Superficial damage is acceptable providing all corrosion is removed.

2. Inspect electrical connector for damaged threads and distorted or missing pins.



1. Hydraulic pump (System No. 1)
2. Hydraulic pump and tachometer generator drive quill
3. Transmission sump case
4. Gasket
5. Rotor tachometer generator
6. Steel washer
7. Nut

ICN-412-A-632102-B-97499-00022-A-001-01

Figure 1. Rotor Tachometer Generator (Sheet 1 of 1)

Rotor Tachometer Generator - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	
Abrasive pad	C-407	AR	

Procedure

1. Remove any surface corrosion using abrasive pad (C-407).
2. Remove residue by wiping area with a clean cloth dampened with drycleaning solvent (C-304).
3. Dry with filtered, compressed air.

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Rotor Tachometer Generator - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-007	AR	
Drycleaning solvent	C-304	AR	
Sealant	C-308	AR	

Procedure

1. Clean mating surfaces of rotor tachometer generator (5, Figure 1) and hydraulic pump and tachometer generator drive quill (2) with drycleaning solvent (C-304).

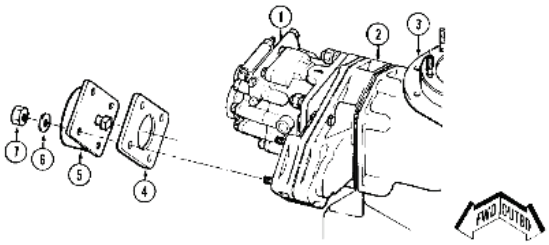
2. Lubricate rotor tachometer generator (5) shaft splines with grease (C-007).

3. Position gasket (4) on hydraulic pump and tachometer generator drive quill (2).

4. Position rotor tachometer generator (5) on hydraulic pump and tachometer generator drive quill (2) with electrical connector positioned for attachment of wiring. Install steel washers (6) and nuts (7).

5. Apply a bead of sealant (C-308) around mating flange of rotor tachometer generator (5) and hydraulic pump and tachometer generator drive quill (2). Allow sealant to cure.

6. Perform helicopter ground run (BHT-412-FM). Check for oil leaks and perform operational check of triple tachometer indicating system (412-A-96-11-00-08A-320A-A / 01967).



1. Hydraulic pump (System No. 1)
2. Hydraulic pump and tachometer generator drive quill
3. Transmission sump case
4. Gasket
5. Rotor tachometer generator
6. Steel washer
7. Nut

ICN-412-A-632102-B-97499-00022-A-001-01

Figure 1. Rotor Tachometer Generator (Sheet 1 of 1)

Tail Rotor Drive Quill - Description

Applicable: SN: 33001-33213 or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed):

Helicopters have a tail rotor drive quill equipped with a flexible coupling and located on aft side of transmission sump case. The tail rotor driveshaft is connected to the drive quill by a V-band clamp set.

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

Helicopters have a tail rotor drive quill equipped with a rigid adapter rather than the flexible coupling noted previously and is also located on aft side of transmission sump case. The tail rotor driveshaft is bolted to an independent flex coupling, which is attached to the rigid adapter.

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Tail Rotor Drive Quill - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Jackscrew Set	T101338	1	

Procedure

1. Remove access doors from left and right side of pylon support in cabin area.

2. Disconnect and remove forward section of tail rotor driveshaft (412-A-65-10-00-00A-520A-A / 01259).

3. Drain transmission oil (412-A-12-10-03-00A-040A-A / 00135) as necessary to a level below quill mounting port.

4. Remove nuts (1, Figure 1), steel washers (2), spacers (3), steel washers (4), and washers (5).

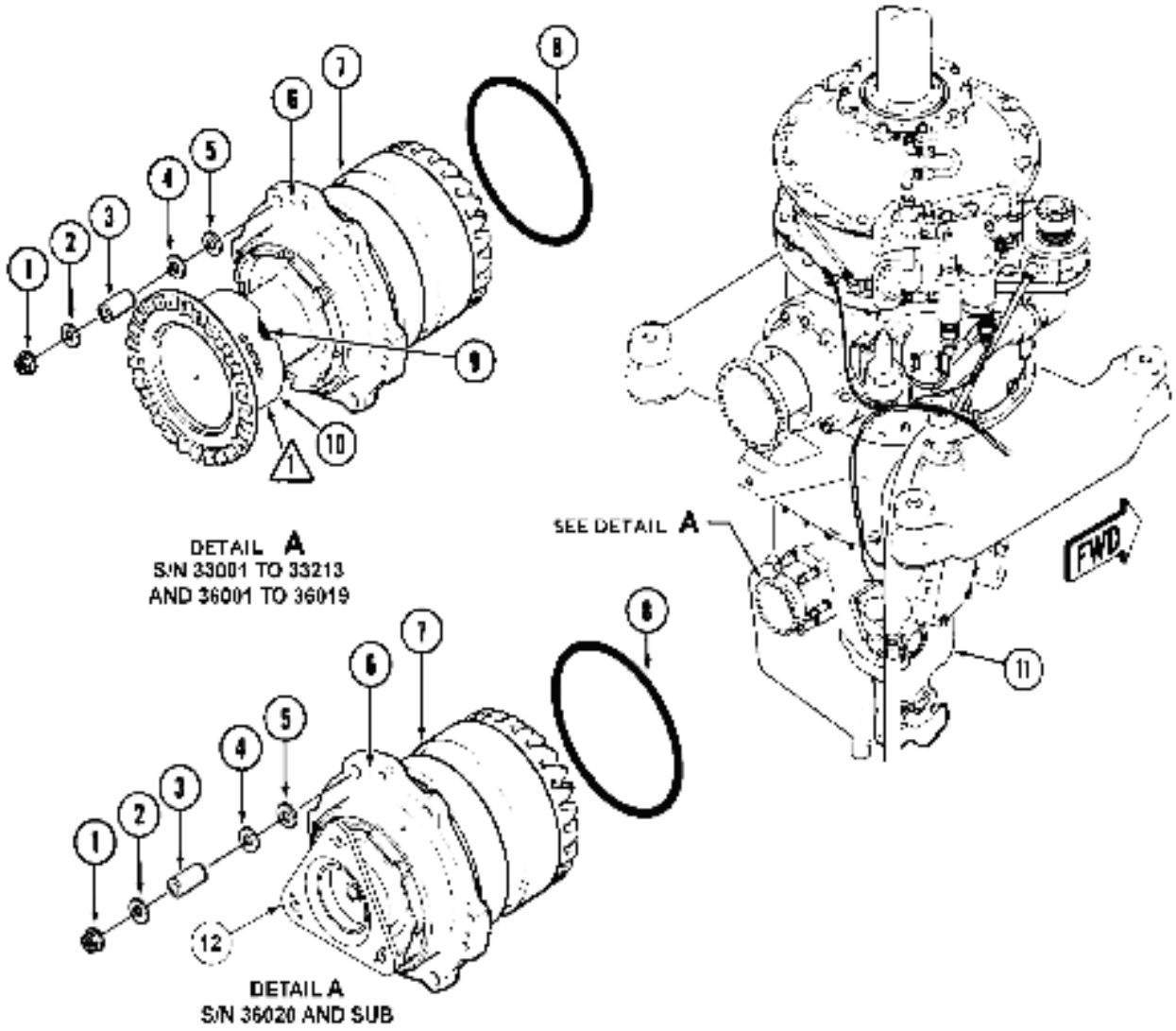
5. Using a sharp plastic scraper, cut sealant between quill mounting flange and transmission case. Remove sealant from jackscrew holes (6).

6. Using a heat lamp, heat sump mounting port until it is hot to the touch.

7. Install jackscrew set (T101338) in holes on quill. Tighten jackscrews carefully and evenly to remove tail rotor drive quill (7) from sump assembly (11).

8. Remove and discard packing (8).

9. Cover open mounting port to prevent foreign objects from entering transmission case.



1. Nut
2. Steel washer
3. Spacer
4. Steel washer
5. Washer
6. Jackscrew holes
7. Tail rotor drive quill
8. Packing
9. Seal
10. Coupling
11. Sump assembly
12. Adapter

NOTE: (1) Location of over temperature indicators (TEMP PLATE).

ICN-412-A-632102-B-97499-00023-A-001-01

Figure 1. Tail Rotor Drive Quill (Sheet 1 of 1)

Tail Rotor Drive Quill - Inspection

NOTE

The following inspection procedures shall be accomplished when tail rotor drive quill (7, Figure 1) is removed for normal maintenance.

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

Applicable: SN: 33001-33213 or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed):

1. *Inspect coupling (10) for grease leakage. Wetting of adjacent areas by grease is cause for replacement of defective parts with the following exception: A small amount of grease expelled from around lip of seal (9) and/or other quill parts, provided the following evaluation of grease leakage is accomplished.*



DO NOT CLEAN OR SPRAY COUPLING WITH ANY TYPE OF SOLVENT DURING INSPECTION. USE ONLY CLEAN CLOTH DAMPENED WITH drycleaning solvent (C-304) TO CLEAN EXTERIOR OF QUILL.

1.1. *Wipe grease from coupling (10) and adjacent parts with clean cloths.*

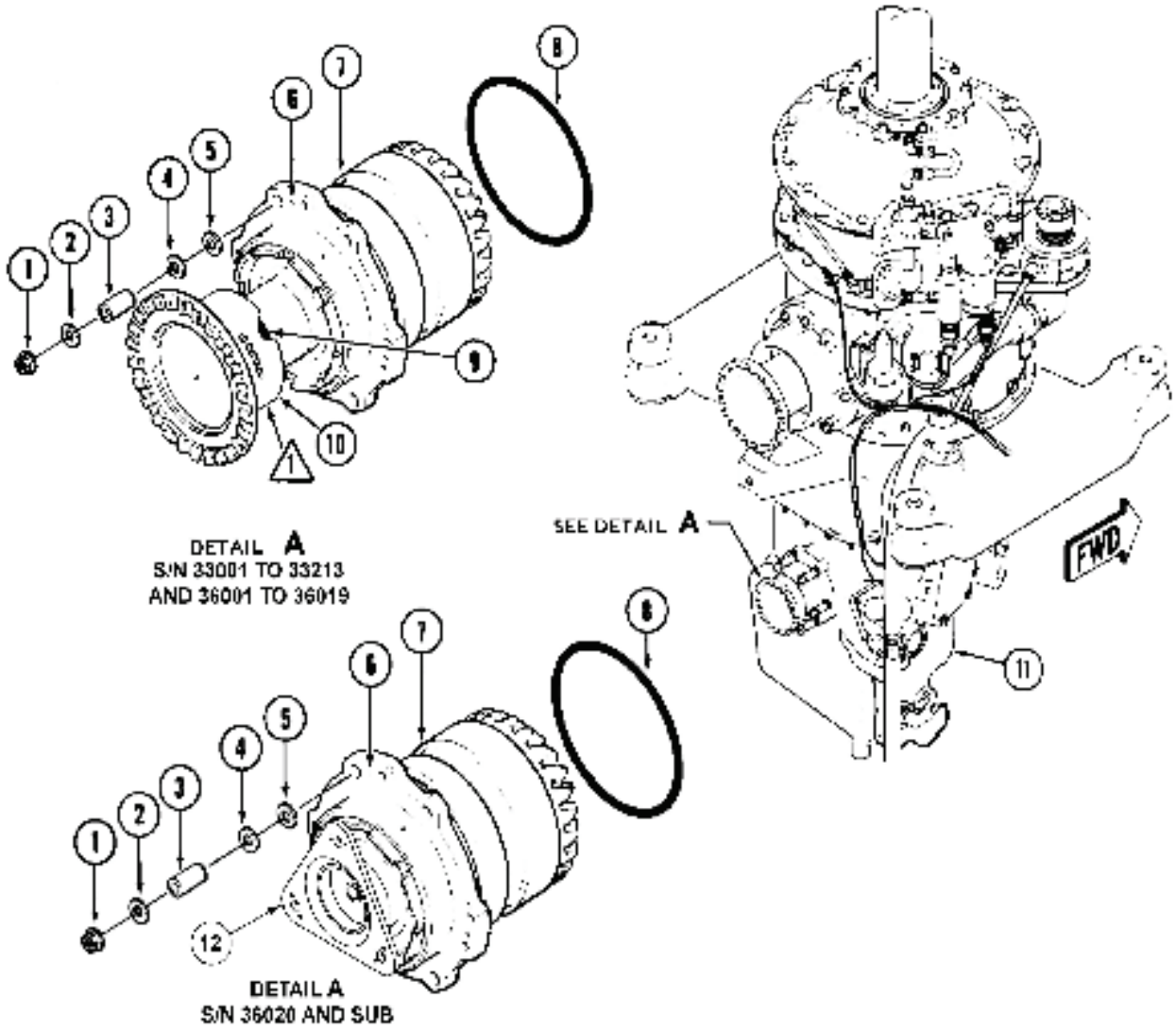
1.2. *Inspect coupling (10) over-temperature indicator (TEMP-PLATE) for discoloration and overheat condition. A change in color of over-temperature indicator (TEMP-PLATE) (from white to black) indicates a possible overheat condition and/or component degradation. Cause of discoloration shall be determined and corrected prior to continued operation. Refer to the BHT-412-CR&O for inspection procedures.*

1.3. *Inspect coupling (10) for grease leakage and for discoloration due to overheating at frequent intervals during the next 10 flight hours. If grease leakage and/or overheating occurs, replace tail rotor drive quill (7).*

Applicable: SN: 33001-33213 or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed):

2. *Inspect seal (9) for protrusion and other evidence of damage. Protruding and/or damaged seals are not acceptable.*

3. Inspect tail rotor drive quill (7) in accordance with the BHT-412-CR&O.



1. Nut
2. Steel washer
3. Spacer
4. Steel washer
5. Washer
6. Jackscrew holes
7. Tail rotor drive quill
8. Packing
9. Seal
10. Coupling
11. Sump assembly
12. Adapter

NOTE: (1) Location of over temperature indicators (TEMP PLATE).

ICN-412-A-632102-B-97499-00023-A-001-01

Figure 1. Tail Rotor Drive Quill (Sheet 1 of 1)

Tail Rotor Drive Quill - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure



DO NOT IMMERSE TAIL ROTOR DRIVE QUILL IN SOLVENT OR SPRAY SOLVENT ON QUILL. USE ONLY CLEAN CLOTHS DAMPENED WITH drycleaning solvent (C-304) TO EXTERIOR OF QUILL.

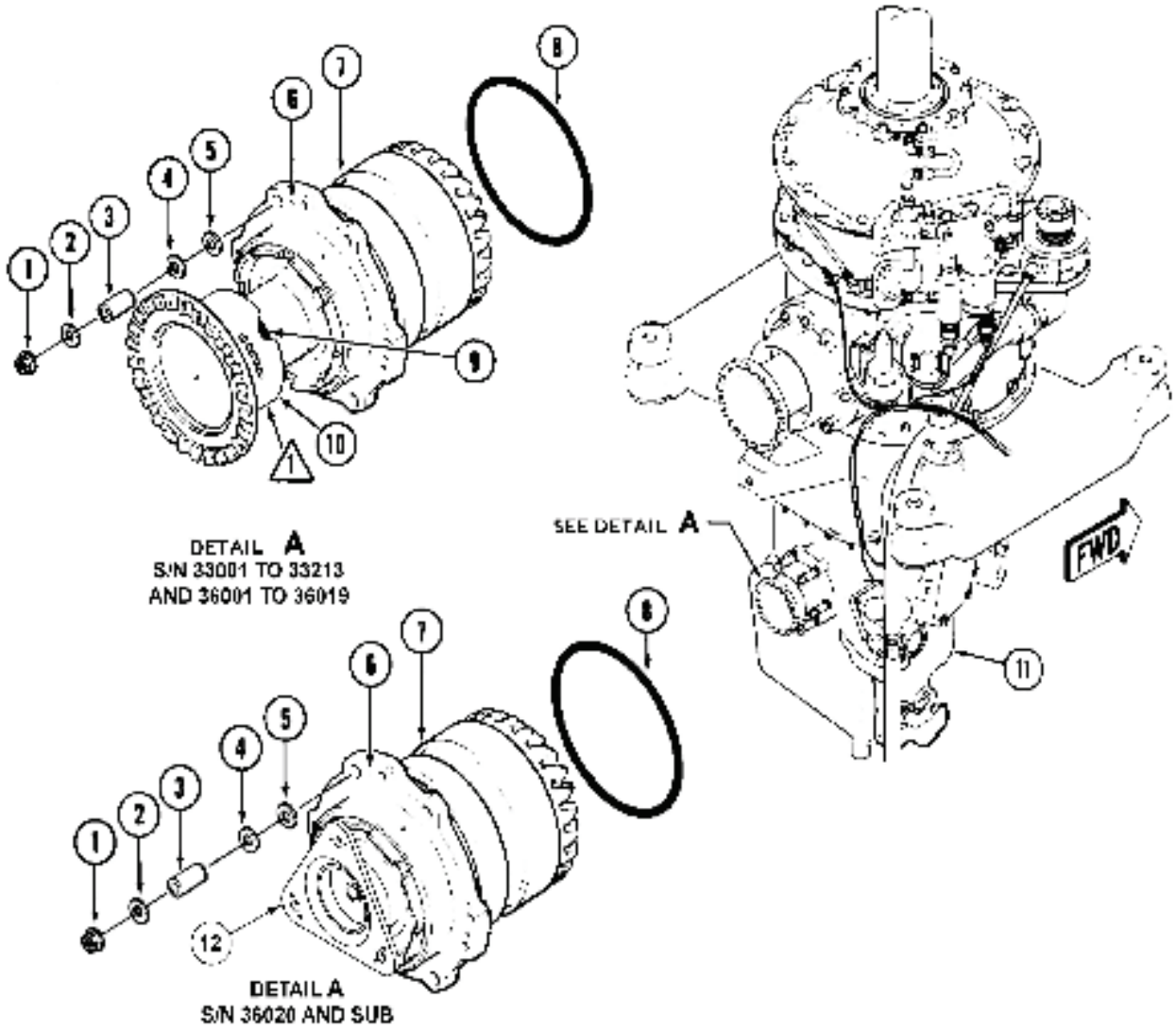
1. Clean sealant residue from both quill and transmission mounting port using a sharp plastic scraper.
2. Clean external surfaces of parts using drycleaning solvent (C-304).
3. Dry with filtered, compressed air or clean, dry, lint-free cloths.

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Tail Rotor Drive Quill - Repair

Procedure

1. Repair mechanical and corrosion damage in accordance with the BHT-412-CR&O.
2. Replace seal (9, Figure 1) in accordance with the BHT-412-CR&O.
3. Lubricate the tail rotor coupling (10) (412-A-65-20-01-11A-240A-A / 01318).
4. Replace over-temperature indicator (TEMP-PLATE) if required (BHT-412-CR&O).



1. Nut
2. Steel washer
3. Spacer
4. Steel washer
5. Washer
6. Jackscrew holes
7. Tail rotor drive quill
8. Packing
9. Seal
10. Coupling
11. Sump assembly
12. Adapter

NOTE: (1) Location of over temperature indicators (TEMP PLATE).

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Figure 1. Tail Rotor Drive Quill (Sheet 1 of 1)

Tail Rotor Drive Quill - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

Procedure

1. Lubricate new packing (8, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packing in groove around tail rotor drive quill (7) sleeve.

2. Verify mating surfaces of tail rotor drive quill (7) sleeve and transmission case are free of all foreign material.

3. Lubricate mating surfaces of tail rotor drive quill (7) and transmission mounting port with transmission oil (412-A-12-01-00-00A-040A-A / 00141).

4. Use a heat lamp or 250 watt light bulb to heat sump mounting port until unit is hot to the touch.

5. Insert tail rotor drive quill (7) into transmission while rotating pinion simultaneously to permit gear meshing. Position mounting holes in sleeve over transmission studs.

NOTE

Install aluminum washers AN960JD616 (5) next to quill sleeve. Do not substitute any other type washer (including NAS1197).

6. Install washers (5), steel washers (4), spacers (3), steel washers (2), and nuts (1). Torque nuts to 50 to 70 inch-pounds (5.45 to 7.91 Nm).

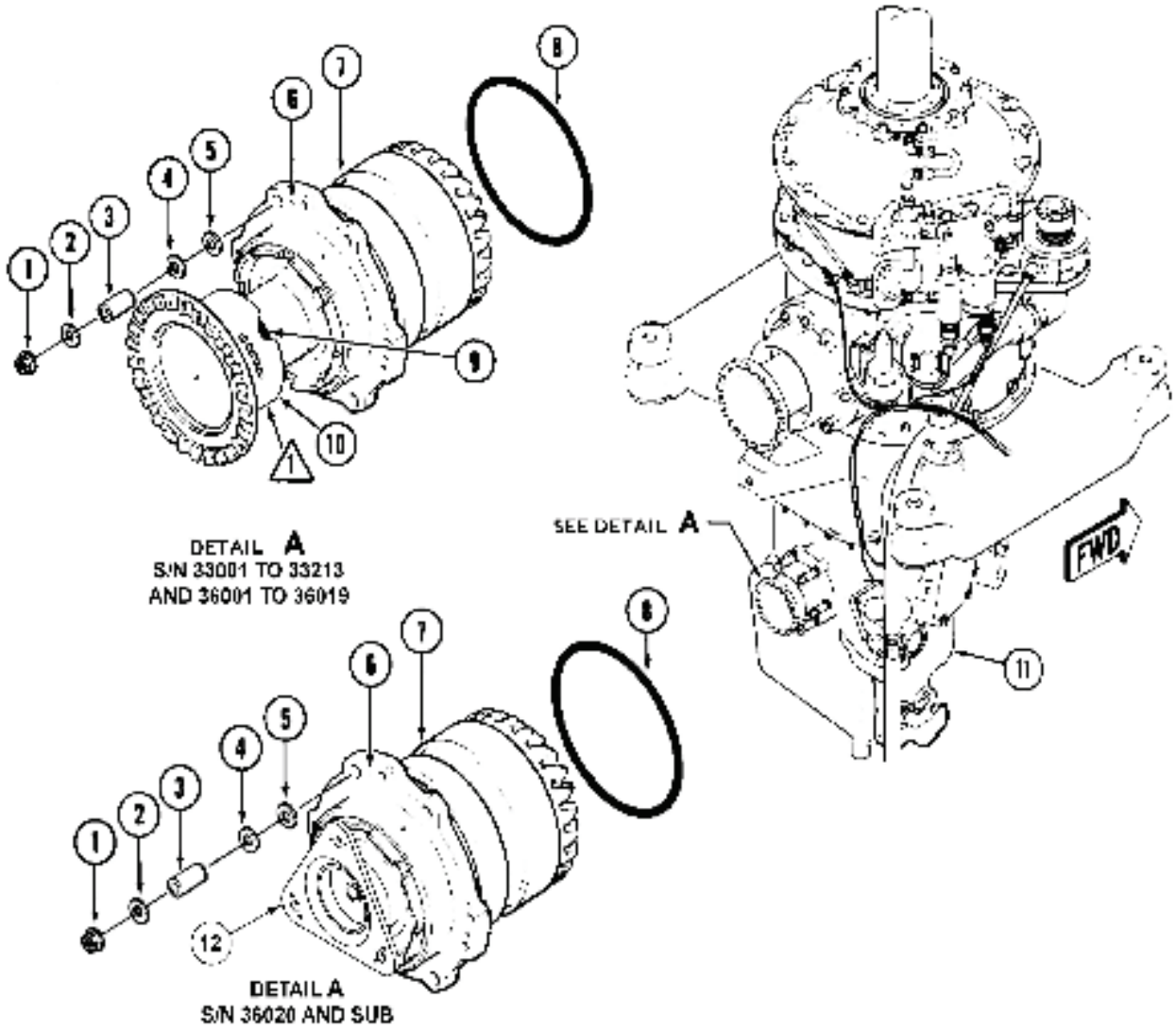
7. Check quill pinion for freedom of rotation and existence of backlash.

8. Fill gap between flange on tail rotor drive quill (7) and flange on transmission sump assembly (11) with sealant (C-308). Also fill jackscrew holes (6) with same sealant. Allow to cure.

9. Install forward section of tail rotor driveshaft (412-A-65-10-00-00A-720A-A / 01263).

10. Ground run helicopter (BHT-412-FM , Flight Manual) and check for leaks at quill flange and loss of lubricant at coupling. If lubricant loss is evident, refer to 412-A-65-10-00-00A-280A-A / 01261.

11. Install access doors on both sides of pylon support in cabin area.



1. Nut
2. Steel washer
3. Spacer
4. Steel washer
5. Washer
6. Jackscrew holes
7. Tail rotor drive quill
8. Packing
9. Seal
10. Coupling
11. Sump assembly
12. Adapter

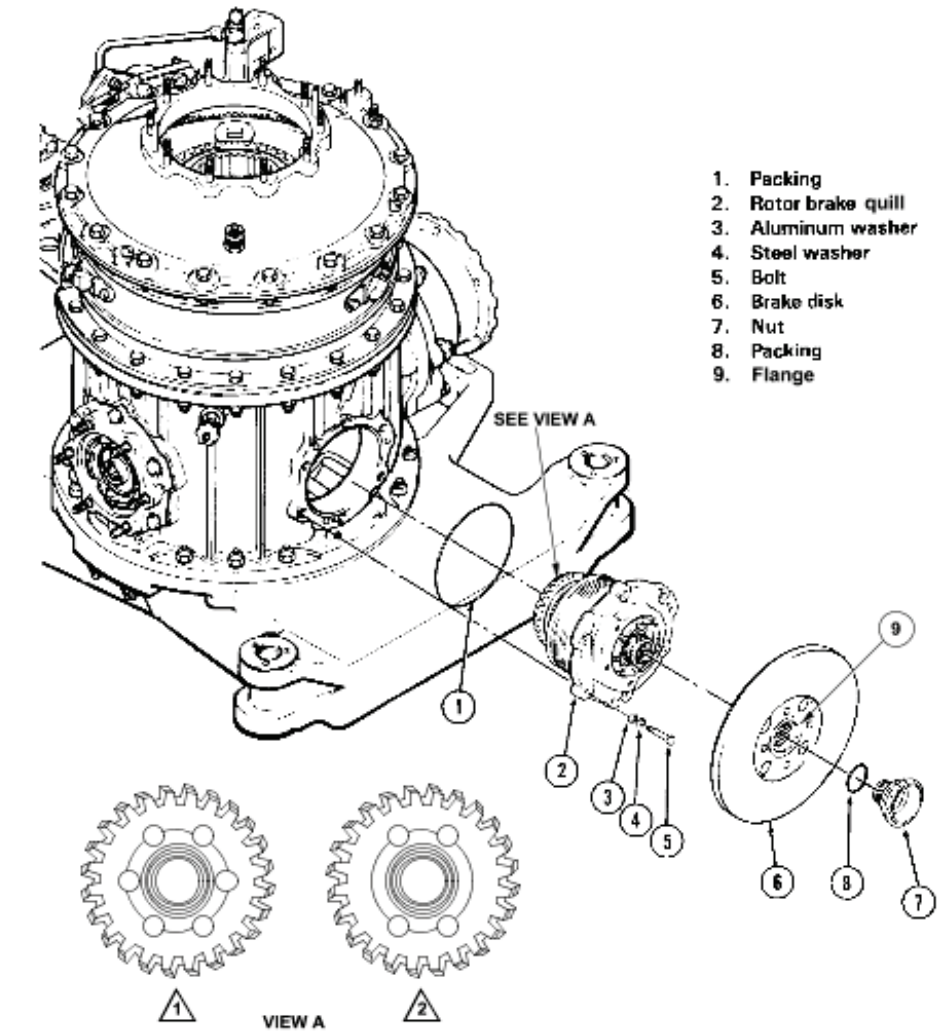
NOTE: (1) Location of over temperature indicators (TEMP PLATE).

ICN-412-A-632102-B-97499-00023-A-001-01

Figure 1. Tail Rotor Drive Quill (Sheet 1 of 1)

Rotor Brake Quill - Description

For helicopters so equipped, the rotor brake quill is installed on left side of transmission main case (Figure 1).



NOTES

1. The 412-040-123-101 quill assembly contains a look alike gear. Ensure pinion has six holes before installation in 412-040-002-ALL transmission and mast assembly (S/N 33001 to 33213 and 36001 to 36019).
2. The 412-040-123-103 quill assembly contains a look alike gear. Ensure pinion has four holes before installation in 412-040-004-ALL transmission and mast assembly (S/N 36020 and sub).

ICN-412-A-632102-B-97499-00024-A-001-01

Figure 1. Rotor Brake Quill (Sheet 1 of 1)

Rotor Brake Quill - Removal

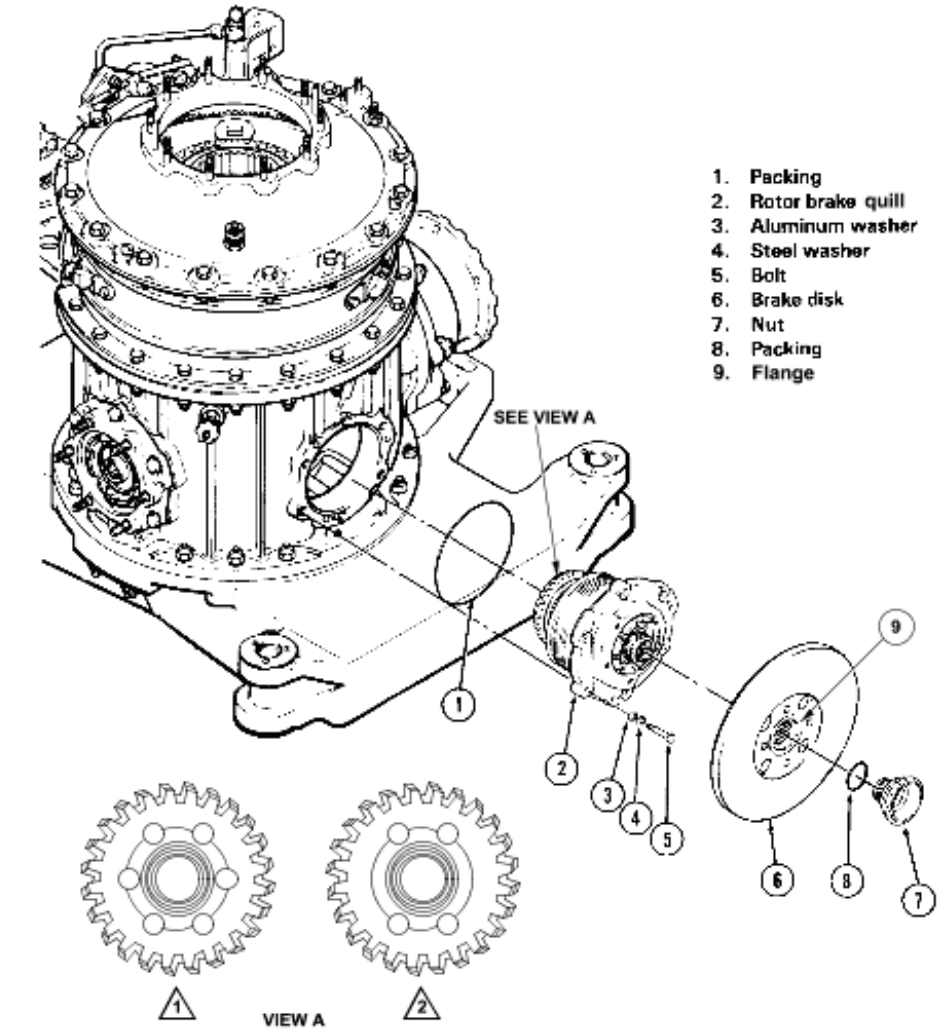
Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Jackscrew Set	T101308	1	

Procedure

1. Open transmission cowling to gain access to transmission.
2. Remove outboard brake assembly half (412-A-63-23-00-03A-042A-A / 01213).
3. If required, remove brake disc (412-A-63-23-00-04A-520A-A / 01220).
4. Using a sharp plastic scraper, cut sealant from between quill mounting flange and transmission case. Clean sealant from jackscrew holes.
5. Remove bolts (5, Figure 1), steel washers (4), and aluminum washers (3).
6. Use a heat lamp or 250 watt light bulb to heat sump mounting port until it is hot to the touch.
7. Install jackscrew set (T101308) and tighten carefully and evenly to remove rotor brake quill (2) with packing (1) from transmission. Discard packing.
8. Cover mounting port in transmission to prevent entry of foreign material.



NOTES

1. The 412-040-123-101 quill assembly contains a look alike gear. Ensure pinion has six holes before installation in 412-040-002-ALL transmission and mast assembly (S/N 33001 to 33213 and 36001 to 36019).
2. The 412-040-123-103 quill assembly contains a look alike gear. Ensure pinion has four holes before installation in 412-040-004-ALL transmission and mast assembly (S/N 36020 and sub).

ICN-412-A-632102-B-97499-00024-A-001-01

Figure 1. Rotor Brake Quill (Sheet 1 of 1)

Rotor Brake Quill - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean sealant residue from both quill and mating surface of transmission using a sharp plastic scraper and drycleaning solvent (C-304).
2. Clean external surfaces of parts using drycleaning solvent (C-304).
3. Dry with filtered, compressed air.

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Rotor Brake Quill - Inspection

Procedure

1. Inspect quill for signs of leakage.
2. Inspect quill and pinion gear for obvious mechanical and corrosion damage (BHT-412-CR&O).
3. Rotate pinion and check for rough bearings. Bearing roughness which can be detected by feel is not acceptable.

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Rotor Brake Quill - Repair

Procedure

1. For complete repair procedures, refer to the BHT-412-CR&O.

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Rotor Brake Quill - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

1. Remove cover from mounting port in transmission.

2. Make sure mating surfaces of quill and transmission are free of all sealant and other foreign matter. Clean as required.

3. Lubricate new packing (1, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141). Install packing in groove on rotor brake quill (2).

4. Apply a light coat of transmission oil to bore of transmission quill port and to mating diameter of quill assembly.

5. Use a heat lamp or 250 watt light bulb to heat sump mounting port until it is hot to the touch.



ROTOR BRAKE QUILL ASSEMBLIES HAVE LOOKALIKE GEARS. INSTALLATION OF INCORRECT QUILL ASSEMBLY WILL RESULT IN INTERNAL TRANSMISSION DAMAGE.

ENSURE SIX-HOLE PINION IS ONLY INSTALLED IN TRANSMISSION AND MAST ASSEMBLY 412-040-002-ALL.

SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: *ENSURE FOUR-HOLE PINION IS ONLY INSTALLED IN TRANSMISSION AND MAST ASSEMBLY 412-040-004-ALL.*

NOTE

During installation of quill assembly, lightly tap quill inward uniformly around face of quill flange near outer edge with a plastic mallet. Do not strike end of quill shaft.

6. Insert rotor brake quill (2) into transmission while simultaneously rotating pinion to permit gear meshing.

7. Install aluminum washers (3) against flange of quill, steel washers (4) under head of bolts (5), and install bolts. Torque bolts to 160 to 190 inch-pounds (18.1 to 21.47 Nm).

8. Check quill pinion for freedom of rotation and existence of backlash.

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

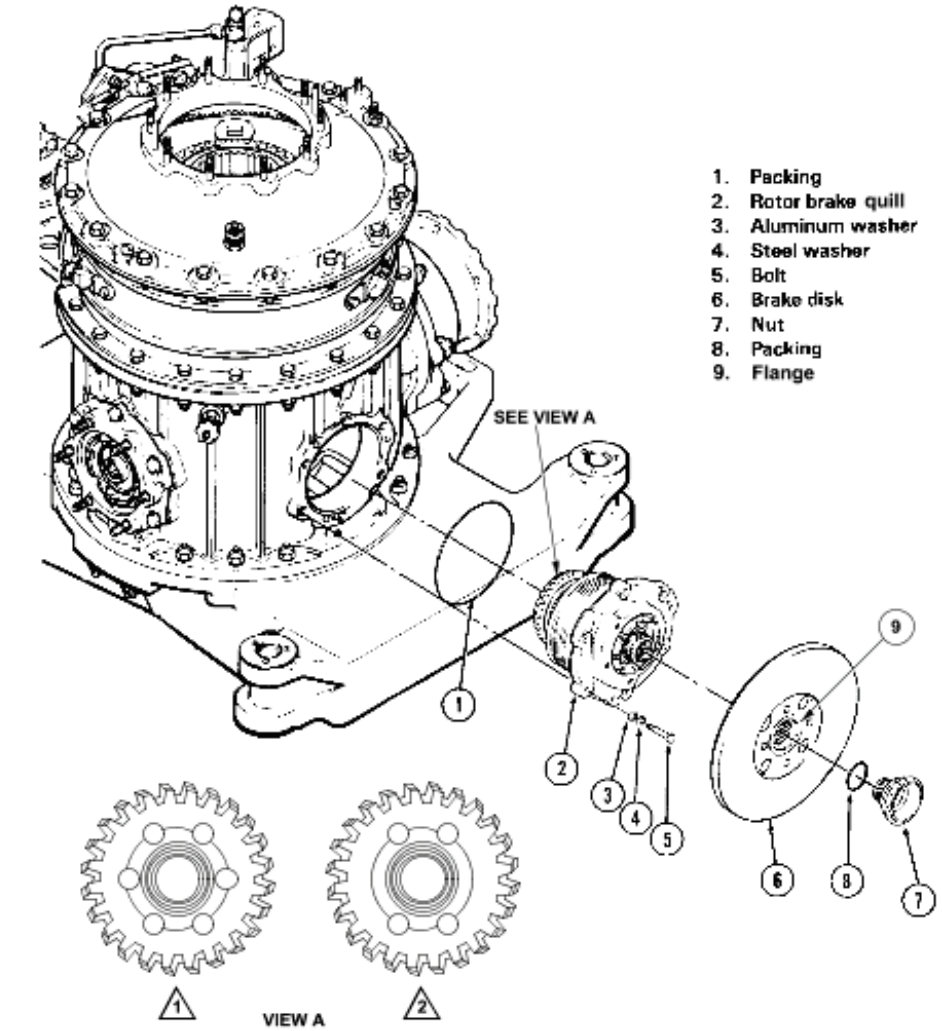
9. *Secure bolts (5) in pairs with* lockwire (C-405).

10. Fill gap between flange on rotor brake quill (2) and flange on transmission sump with sealant (C-308). Also fill jackscrew holes with same sealant. Allow to cure.

11. Install inboard brake assembly half (412-A-63-23-00-03A-720A-A / 01218).

12. If not previously accomplished, install brake disc (412-A-63-23-00-03A-720B-A / 01219).

13. Install outboard brake assembly half (412-A-63-23-00-03A-720A-A / 01218).



NOTES

1. The 412-040-123-101 quill assembly contains a look alike gear. Ensure pinion has six holes before installation in 412-040-002-ALL transmission and mast assembly (S/N 33001 to 33213 and 36001 to 36019).
2. The 412-040-123-103 quill assembly contains a look alike gear. Ensure pinion has four holes before installation in 412-040-004-ALL transmission and mast assembly (S/N 36020 and sub).

ICN-412-A-632102-B-97499-00024-A-001-01

Figure 1. Rotor Brake Quill (Sheet 1 of 1)

Transmission Quill Port Covers - Description

NOTE

Removal and installation procedures for both left and right quill port covers are identical. 412-A-63-21-03-00A-520A-A / 01183 and 412-A-63-21-03-00A-720A-A / 01184 details removal and installation of rotor brake quill port cover which is used when there is no rotor brake quill installed.

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Transmission Quill Port Covers - Removal

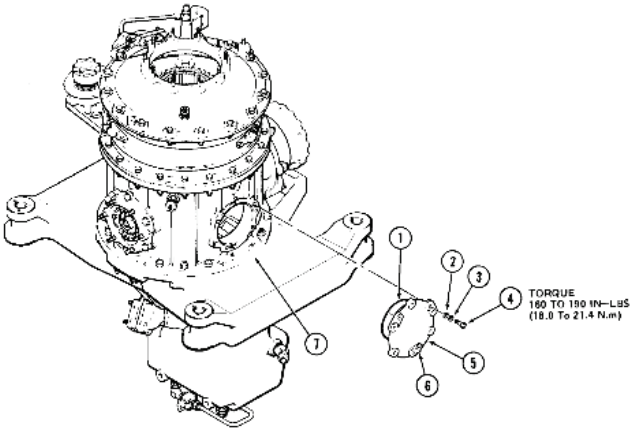
Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Jackscrew Set	T100929	1	

Procedure

1. Remove bolts (4, Figure 1), steel washers (3), and aluminum washers (2).
2. Cut sealant from around edge of cover (5) using a sharp plastic scraper. Clean sealant from jackscrew holes.
3. Clean sealant from tapped holes (6).
4. Install jackscrew set (T100929) in three tapped holes (6).
5. Carefully and evenly, tighten all three jackscrews and remove cover (5).
6. Remove and discard packing (1).



1. Packing
2. Aluminum washer
3. Steel washer
4. Bolt
5. Cover
6. Tapped hole
7. Transmission case

ICN-412-A-632103-B-97499-00001-A-001-01

Figure 1. Quill Port Covers (Sheet 1 of 1)

Transmission Quill Port Covers - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

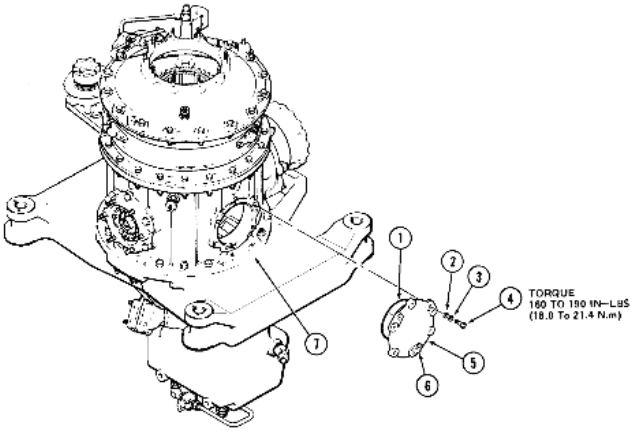
Procedure

1. Lubricate new packing (1, Figure 1) with transmission oil (412-A-12-01-00-00A-040A-A / 00141).

2. Install packing (1) on cover (5).

3. Place cover (5) over transmission quill port and install bolts (4) with steel washers (3) under bolt heads and aluminum washers (2) next to cover. Torque bolts to 160 to 190 inch-pounds (18.1 to 21.4 Nm).

4. Apply a bead of sealant (C-308) around perimeter of cover (5) where it meets transmission case (7).



1. Packing
2. Aluminum washer
3. Steel washer
4. Bolt
5. Cover
6. Tapped hole
7. Transmission case

ICN-412-A-632103-B-97499-00001-A-001-01

Figure 1. Quill Port Covers (Sheet 1 of 1)

Transmission Mounts - Description

The transmission mounts consists of forward and aft isolation mounts, aft friction dampers, and a lift link.

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Aft Isolation Mount and Friction Dampers - Description

NOTE

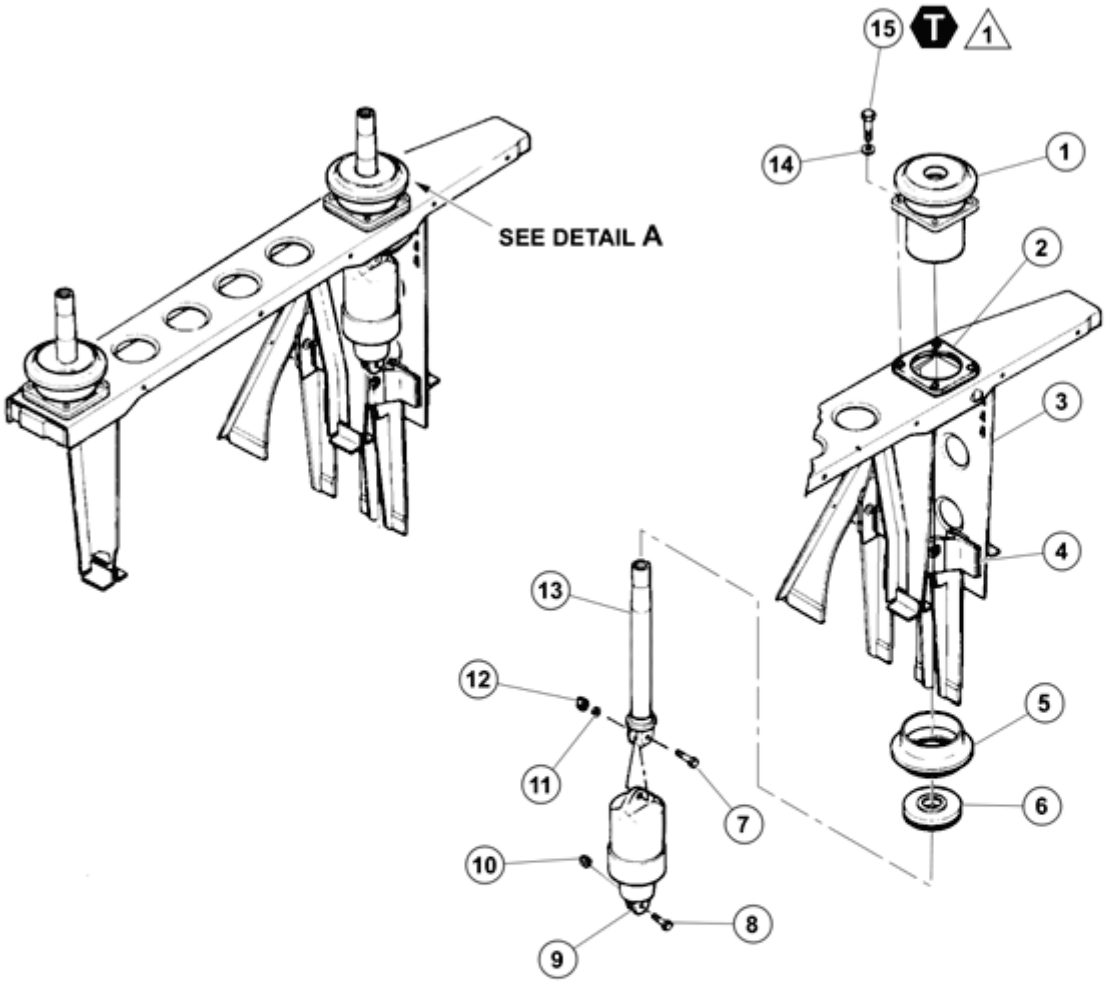
Maintenance procedures for left and right mounts are identical.

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Aft Isolation Mount and Friction Dampers - Removal

Procedure

1. Remove transmission and mast assembly (412-A-63-21-02-00A-520A-A / 01013).
2. Remove bolts (15, Figure 1) and thin steel washers (14).
3. *Disengage boot (5) from isolation mount assembly (1). Remove isolation mount assembly (1) from spacer (2).*
4. Remove nut (10) and bolt (8). Remove friction damper (9), bolt (13), bushing (6), and boot (5) as a unit.



1. Isolation mount assembly
2. Spacer
3. Pylon support
4. Damper fitting
5. Boot
6. Bushing
7. Bolt
8. Bolt
9. Friction damper
10. Nut
11. Thin steel washer
12. Nut
13. Bolt
14. Thin steel washer
15. Bolt

T 100 TO 140 IN-LBS (11.30 TO 15.82 Nm)

NOTE

1 Install longer bolts in forward outboard and aft inboard mounting holes. Install shorter bolts in remaining holes.

ICN-412-A-632104-B-97499-00001-A-002-01

Figure 1. Transmission Aft Isolation Mounts and Friction Dampers (Sheet 1 of 1)

Transmission Mounts - Inspection

Procedure

NOTE

Inspection procedures for all four mounts are the same.

NOTE

When you replace one or more transmission mounts, it is recommended to replace all four transmission mounts at the same time. This will be only applicable to the transmission mounts with more than 3000 hours/5 years time in service.

1. *Do the spring rate check as follows:*

1.1. *Fabricate a steel support sleeve and pressing button (Figure 1). Top and bottom of sleeve must be square in relation to sides so the mount will be deflected in vertical plane only.*

1.2. *Put the mount, long end down, in the support sleeve.*

1.3. *Put the support and the mount in a suitable press. Press indicator should be marked in 5 pound increments and should have a range of 0 to 5000 pounds (or equivalent).*

1.4. *Align the steel pressing button at the center above the inner sleeve of mount.*

1.5. *Apply pressure to the button and the mount with press ram so as to deflect mount in a vertical plane to 0.4 inch (10.16 mm) at approximately one cycle per minute. Do the same cycle again for three times.*

1.6. *Install a dial indicator to the press to make sure that the finger of indicator touches the pressing button.*

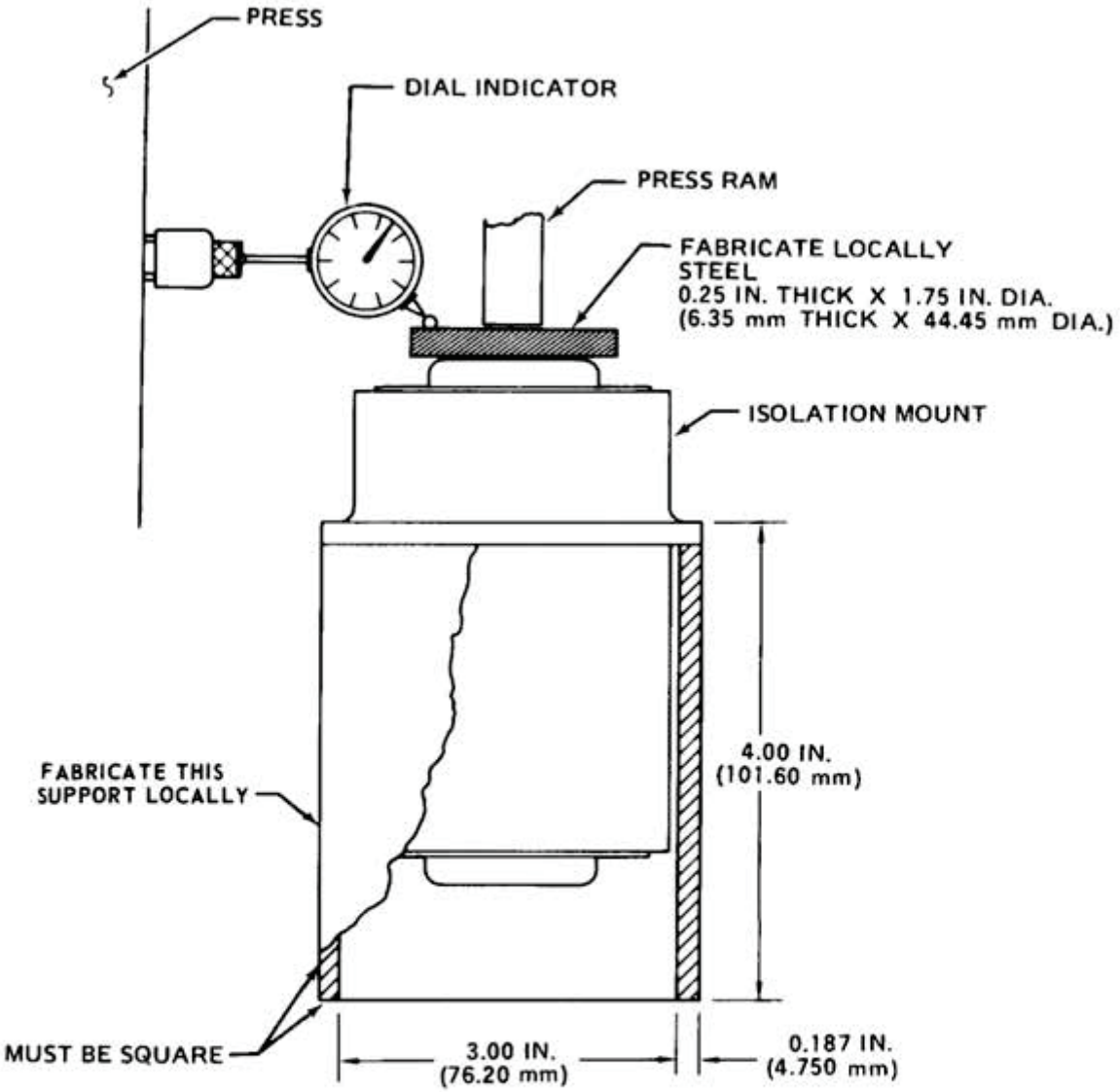
1.7. *Apply slight pressure to the button with ram to make sure that the button is seated.*

1.8. *Apply sufficient pressure to the button to deflect mount 0.1 inch (2.54 mm) and record load. Apply sufficient pressure to the pressing button to deflect mount 0.3 inch (7.62 mm) and record the load. Subtract load noted at 0.1 inch (2.54 mm) from the load at 0.3 inch (7.62 mm). If the value is less than 1320 pounds (5872 N) for -105 mount or 1800 pounds (8007 N) for -107 mount, replace the mount.*

2. *Examine the elastomer for cracks, tears, or debonded areas. Replace if a 0.005 inch (0.127 mm) feeler gauge can be inserted more than 0.25 inch (6.35 mm). Normal elastomer fretting (eraser type dust) is not cause for replacement.*

3. *Examine the bolts and the bushings for cracks, corrosion, scoring, wear, and scratches. Replace damaged parts. Replace damaged boots.*

4. *Replace the isolation mounts when too much vibration is believed to be caused by defective mounts or when there is other evidence of malfunction.*



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Figure 1. Isolation Mount Inspection Work Aid - Tool Application (Sheet 1 of 1)

Aft Mount Friction Dampers - Inspection

Procedure

1. *For inspection procedure, refer to* BHT-412-CR&O, Chapter 63.

2. *DELETED*

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Aft Isolation Mount and Friction Dampers - Installation

Procedure

NOTE

When you replace one or more transmission mounts, it is recommended to replace all four transmission mounts at the same time. This will be only applicable to the transmission mounts with more than 3000 hours/5 years time in service.

1. Put the bolt (13, Figure 1) on the friction damper (9), and install the bolt (7), thin steel washer (11), and the nut (12).

2. Insert the bushing (6) and the boot (5) on the bolt (13).

3. Put the friction damper (9) and the bolt (13) in pylon support (3) from the inner side of pylon. Fit bottom of friction damper (9) on the damper fitting (4). Install the bolt (8) and the nut (10).

4. *Put the isolation mount assembly (1) in its position on spacer (2) and bolt (13) and make sure that the bushing (6) and boot (5) are installed on bolt (13).*

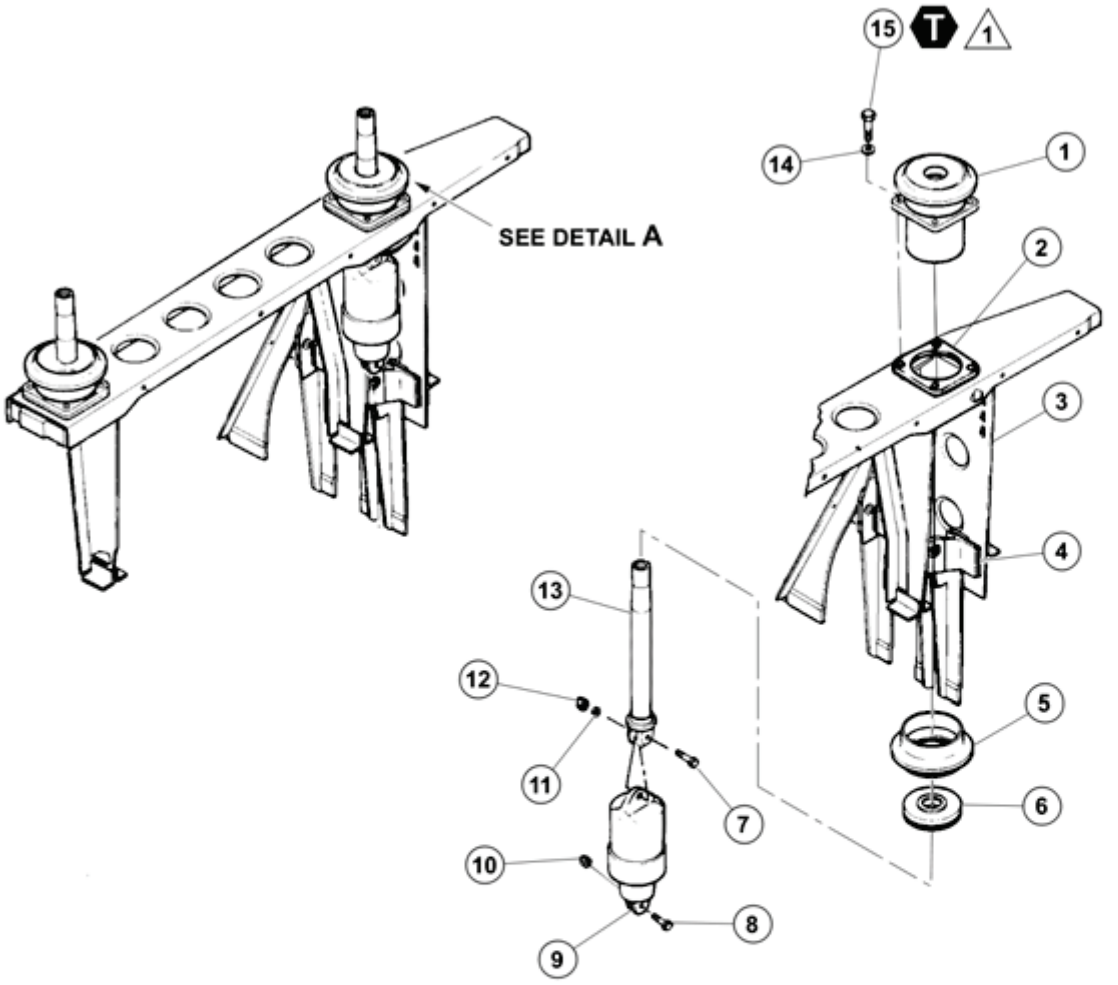


BOLTS (15) ARE IN DIFFERENT LENGTHS. INSTALL THE TWO LONGER BOLTS IN THE FORWARD OUTBOARD AND AFT INBOARD MOUNTING HOLES. INSTALL THE TWO SHORTER BOLTS IN REMAINING MOUNTING HOLES.

5. Install the bolts (15) and the thin steel washers (14). Make sure that the bolts (15) are the correct grip length. Torque the bolts (15) 100 to 140 inch-pounds (11.30 to 15.82 Nm).

6. *Install the boot (5) on the isolation mount assembly (1).*

7. Install the transmission and mast assembly (412-A-63-21-02-00A-720B-A / 01016).



1. Isolation mount assembly
2. Spacer
3. Pylon support
4. Damper fitting
5. Boot
6. Bushing
7. Bolt
8. Bolt
9. Friction damper
10. Nut
11. Thin steel washer
12. Nut
13. Bolt
14. Thin steel washer
15. Bolt

T 100 TO 140 IN-LBS (11.30 TO 15.82 Nm)

NOTE

1 Install longer bolts in forward outboard and aft inboard mounting holes. Install shorter bolts in remaining holes.

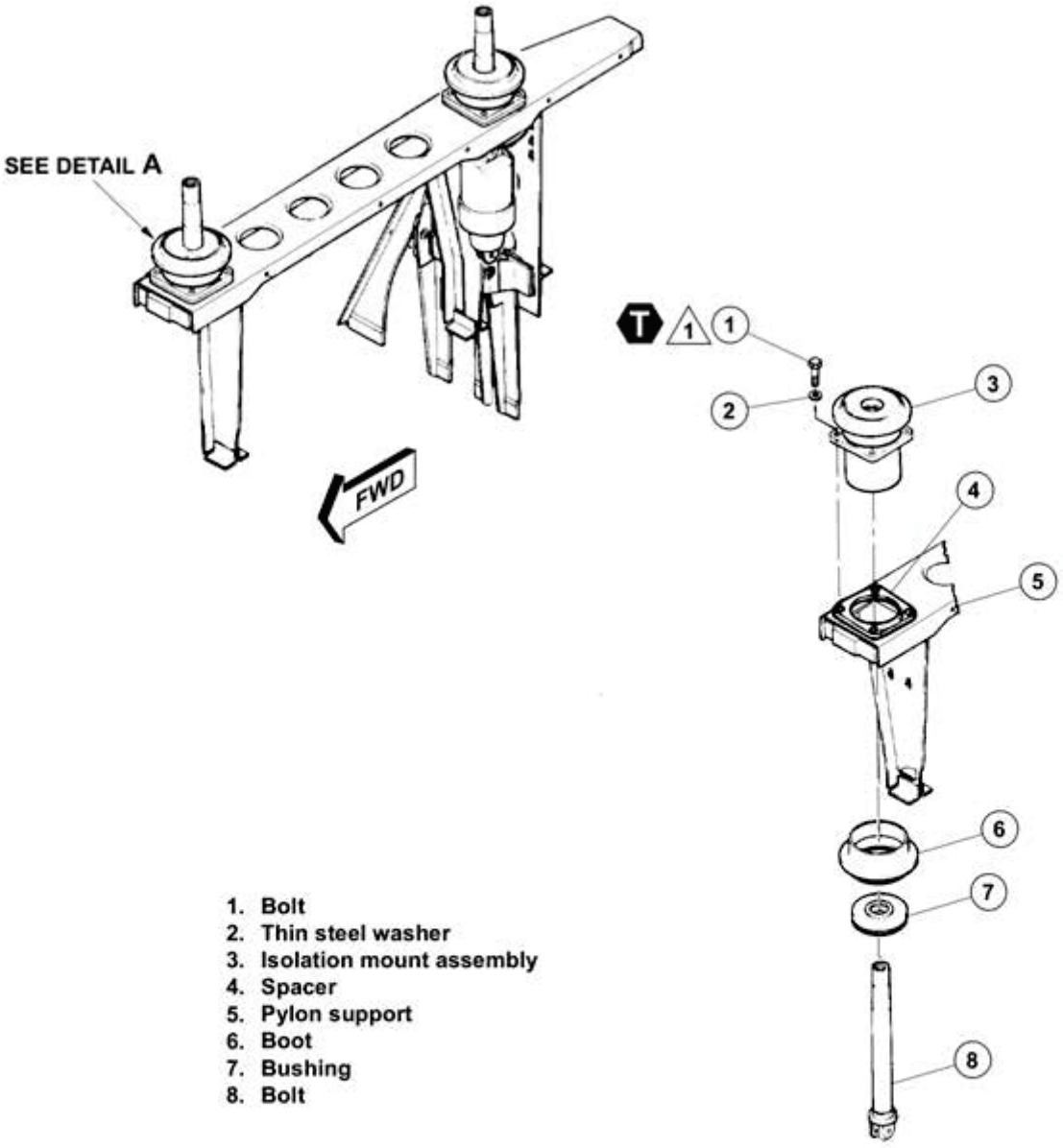
ICN-412-A-632104-B-97499-00001-A-002-01

Figure 1. Transmission Aft Isolation Mounts and Friction Dampers (Sheet 1 of 1)

Forward Isolation Mounts - Removal

Procedure

1. Remove transmission and mast assembly (412-A-63-21-02-00A-520A-A / 01013).
2. Remove bolt (8, Figure 1).
3. Disengage boot (6) from isolation mount assembly (3). Remove boot and bushing (7).
4. *Remove bolts (1) and thin steel washer (2). Remove isolation mount assembly (3) from spacer (4).*



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Figure 1. Forward Isolation Mounts (Sheet 1 of 1)

Forward Isolation Mounts - Inspection

Procedure

1. Refer to 412-A-63-21-04-00A-280A-A / 01188 for inspection procedures.

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Forward Isolation Mounts - Installation

Procedure

NOTE

When you replace one or more transmission mounts, it is recommended to replace all four transmission mounts at the same time. This will be only applicable to the transmission mounts with more than 3000 hours/5 years time in service.

1. *Put the isolation mount assembly (3,* Figure 1*) on the spacer (4).*

2. Insert the bushing (7) and the boot (6) on the bolt (8). Install the bolt assembly in the isolation mount assembly (3) from the bottom. Install the boot (6) to the bushing (7) and isolation mount assembly (3).

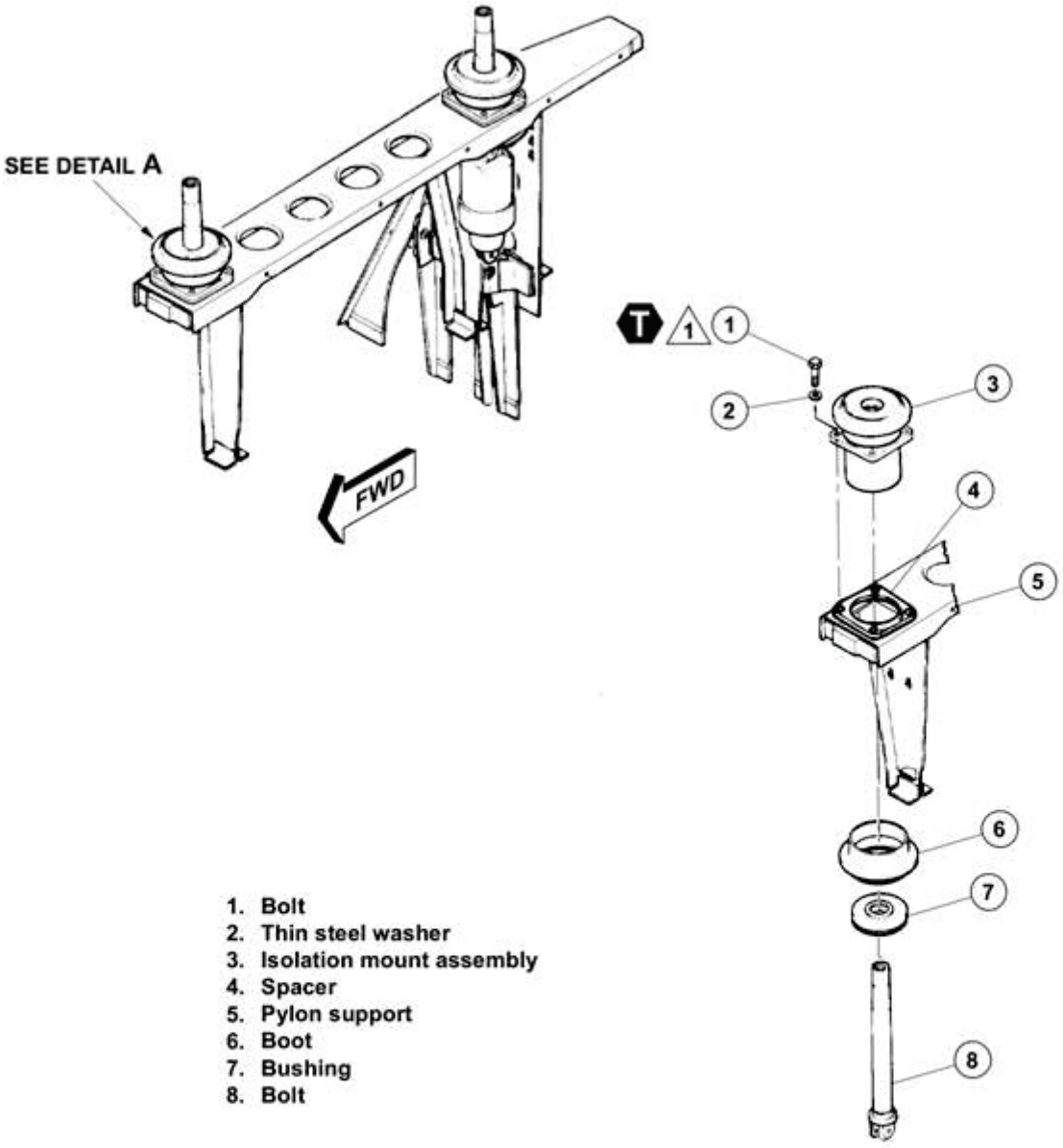
3. Attach the bolt (8) in its position until the transmission installation is completed.



BOLTS (1) ARE IN DIFFERENT LENGTHS. INSTALL THE TWO LONGER BOLTS IN THE FORWARD MOUNTING HOLES. INSTALL THE TWO SHORTER BOLTS IN THE AFT MOUNTING HOLES.

4. Install the bolts (1) and the thin steel washers (2). Make sure that the bolts (1) are the correct grip length. Torque the bolts (1) 100 to 140 inch-pounds (11.30 to 15.82 Nm).

5. Install the transmission and mast assembly (412-A-63-21-02-00A-720B-A / 01016).



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Figure 1. Forward Isolation Mounts (Sheet 1 of 1)

Lift Link - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Hoist	T101581	1	
Clevis	204-011-128-001	1	

Procedure

1. Open transmission cowling. Remove right or left access doors on pylon support in cabin.

2. Erect hoist (T101581) (or equivalent) to steady transmission while link is removed.

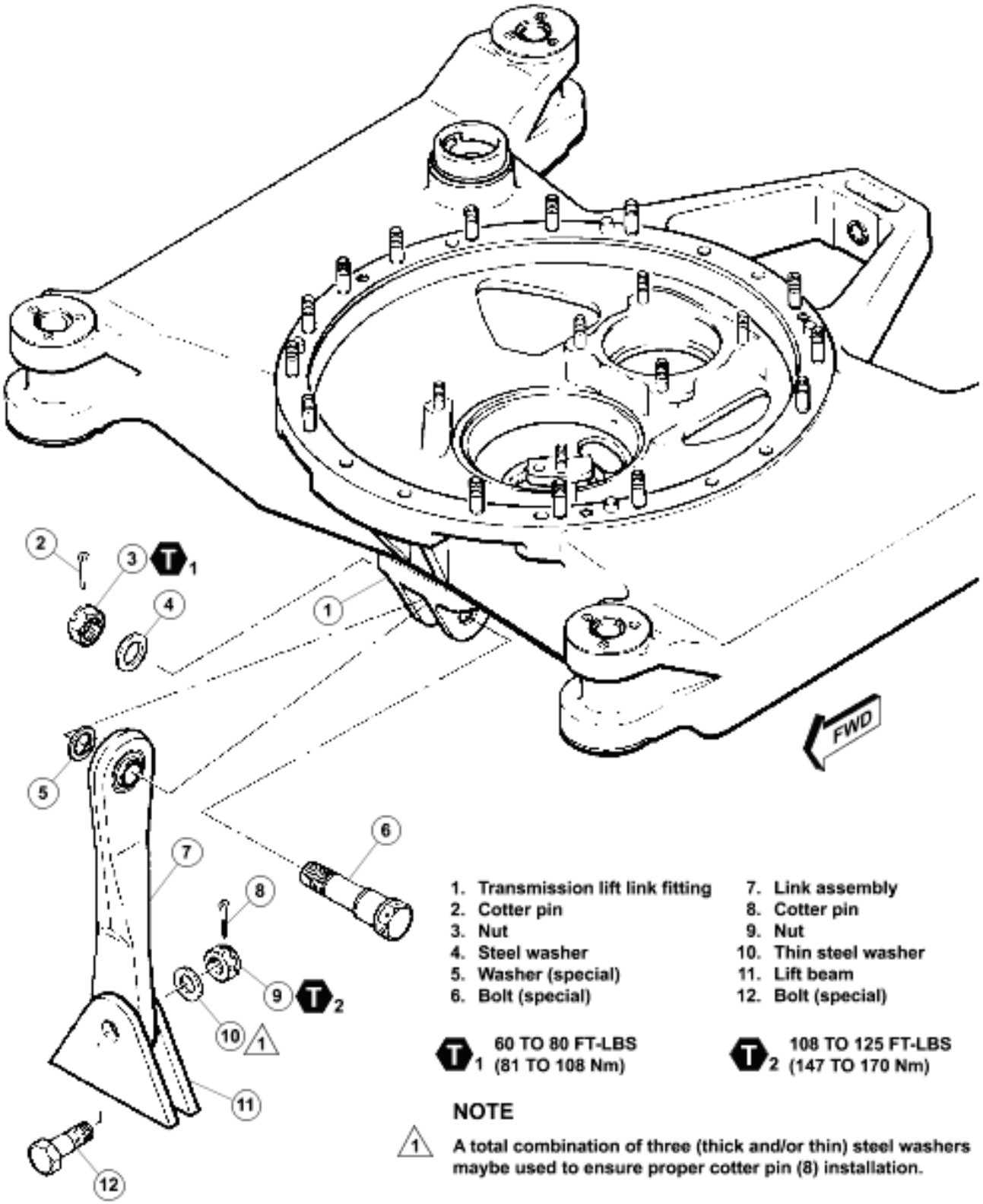


MAST NUT (CAP ASSEMBLY RETENTION NUT) (412-010-160-103) SHALL NOT BE USED TO LIFT HELICOPTER OR HELICOPTER COMPONENTS EXCEPT WHEN MAIN ROTOR HUB IS INSTALLED. IF MAIN ROTOR HUB IS NOT INSTALLED, USE MAST NUT (204-010-481-001) ON MAST.

3. Install clevis (204-011-128-001) on mast nut. Attach hoist to clevis. Raise transmission sufficiently to remove bolt (12, Figure 1) by hand.

4. Remove cotter pin (8), nut (9), thin steel washer (10), and remove bolt (12) from lift beam (11).

5. Remove cotter pin (2), nut (3), steel washer (4), bolt (special) (6), washer (special) (5), and remove link assembly (7).



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Figure 1. Transmission Lift Link (Sheet 1 of 1)

Lift Link - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure



DO NOT IMMERSE BEARING END OF LINK IN drycleaning solvent (C-304).

1. Wipe link clean with a clean cloth dampened with drycleaning solvent (C-304).
2. Dry with a clean, dry cloth or filtered, compressed air.

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Lift Link - Inspection and Repair

Procedure

1. Refer to the BHT-412-CR&O for inspection criteria and repair limits and procedures.

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Lift Link - Installation

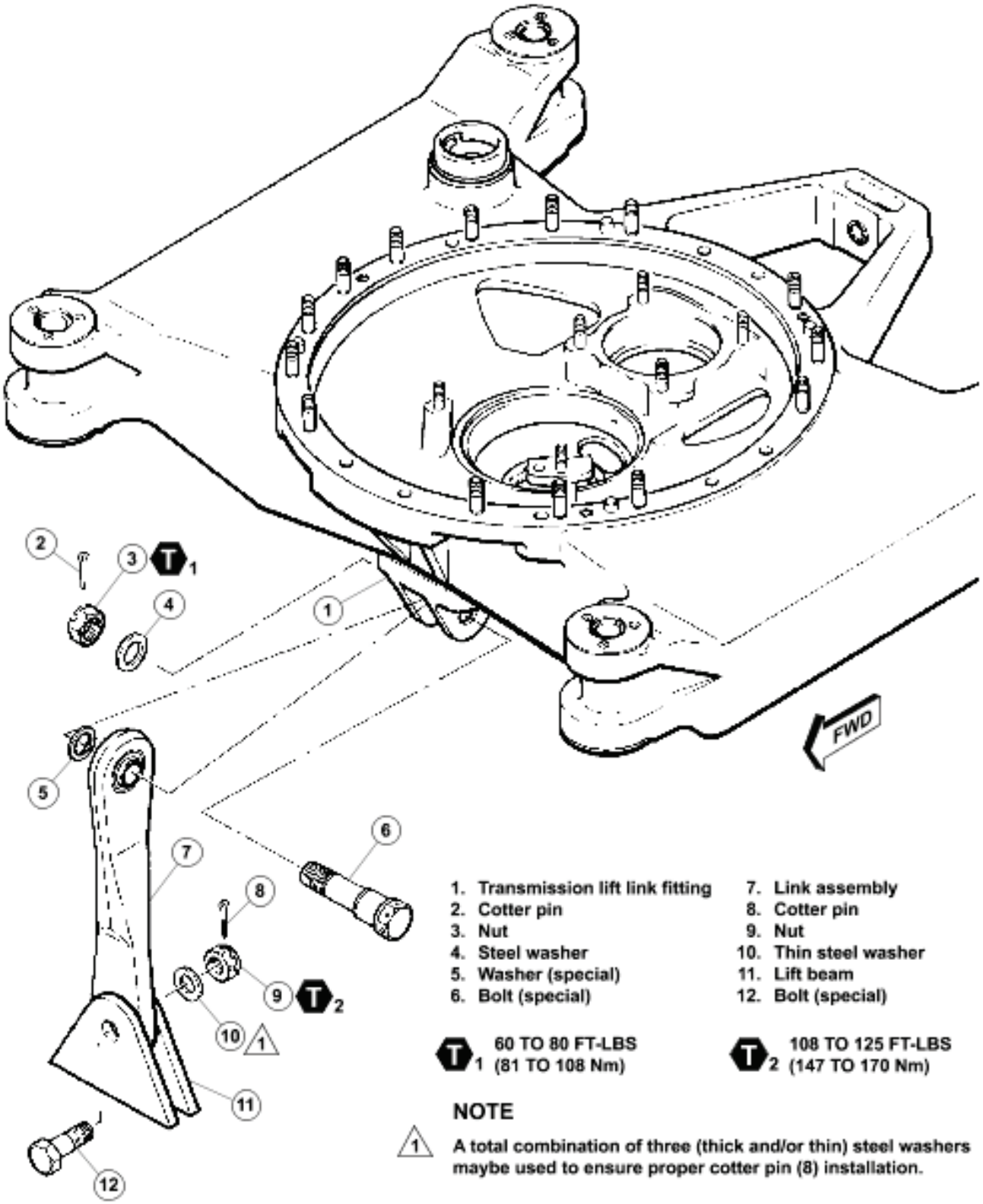
Procedure

1. Install hoist and clevis (412-A-63-21-04-03A-520A-A / 01194).
2. Position washer (special) (5, Figure 1) and link assembly (7) on transmission lift link fitting (1).
3. Install bolt (special) (6), steel washers (4), and nut (3).
4. Torque nut (3) to 60 to 80 foot-pounds (81 to 108 Nm). Install cotter pin (2).
5. Raise transmission sufficiently to allow insertion of bolt (special) (12) by hand pressure only.

NOTE

Large diameter of lift beam (11) fitting for bolt (special) (12) shall be forward.

6. Position link assembly (7) on lift beam (11) fitting.
7. Install bolt (special) (12), thin steel washer (10), and nut (9). Torque nut (9) to 108 to 125 foot-pounds (146.8 to 170 Nm).
8. Install cotter pin (8).
9. Remove hoist and clevis.
10. Close cowling and install access doors as required.



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Figure 1. Transmission Lift Link (Sheet 1 of 1)

Lift Link Fitting - Removal

Procedure

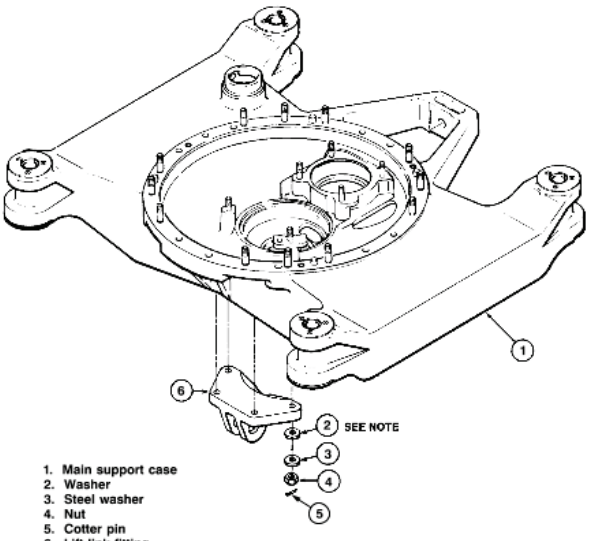
NOTE

TB 412-89-82 adds a washer (2, Figure 1 between lift link fitting (6) and existing steel washer (3). It is recommended this washer (2) be added any time fitting is removed for any reason.

1. Remove lift link (412-A-63-21-04-03A-520A-A / 01194).

2. Remove cotter pins (5), nuts (4), steel washers (3), and washers (2) if installed.

3. Remove lift link fitting (6) from mounting studs of main support case (1).



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Figure 1. Lift Link Fitting (Sheet 1 of 1)

Lift Link Fitting - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Wipe fitting clean with a clean cloth dampened with drycleaning solvent (C-304).
2. Dry with a clean, dry cloth or filtered, compressed air.

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Lift Link Fitting - Inspection and Repair

Procedure

1. Refer to the BHT-412-CR&O for inspection criteria and repair limits and procedures.

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Lift Link Fitting - Installation

Procedure

1. Position lift link fitting (6, Figure 1) on mounting studs of main support case (1).

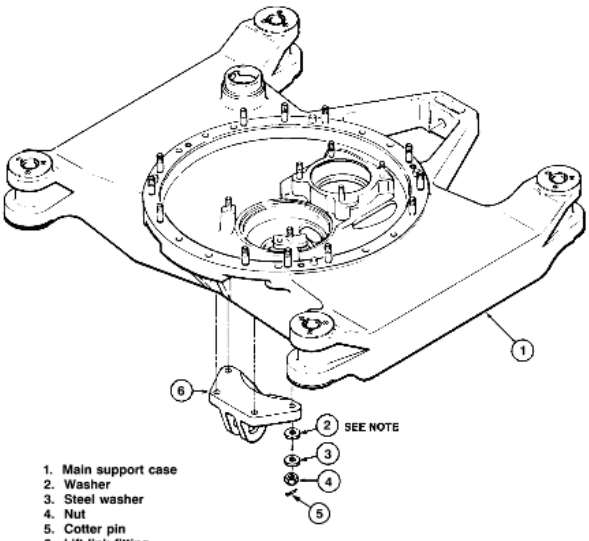
Applicable: TB 412-89-82: Pre:

2. Install washers (3). Install nuts (4). Torque nuts to 37 to 42 foot-pounds (50 to 56 Nm). Install cotter pins (5).

Applicable: TB 412-89-82: Post:

3. Install washers (2 and 3). Install nuts (4). Torque nuts to 40 to 58 foot-pounds. (54 to 79 Nm). Install cotter pins (5).

4. If desired, install lift link (412-A-63-21-04-03A-720A-A / 01197).



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Figure 1. Lift Link Fitting (Sheet 1 of 1)

Serviceability Checks - Transmission, Tail Rotor Gearbox, and Intermediate Gearbox - Description

Perform the maintenance procedures in 412-A-63-00-00-00A-280A-A / 01204 and 412-A-63-00-00-00A-280B-A / 01203; when any doubt exists as to the serviceability of a transmission, tail rotor gearbox, or intermediate gearbox after finding metal particles in operating oil.

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Intermediate or Tail Rotor Gearbox - Serviceability Check

Procedure

1. Remove, inspect, and clean appropriate chip detector.
2. Drain gearbox oil into a clean container. Inspect oil for chips.
3. Flush gearbox with clean, approved oil (412-A-12-10-04-00A-292A-A / 00137). Inspect for chips.
4. Service gearbox with clean, approved oil.
5. Accomplish ground run in accordance with 412-A-63-00-00-00A-280A-A / 01204, step 6.
6. Repeat Step 1., Step 2., and Step 4..
7. If number of particles has increased, or if particles can be visually identified as chips or flakes from a gear or bearing, replace gearbox. If number of particles has decreased and only minute particles are found, continue gearbox in service.
8. After 5 hours of service, repeat Step 1., Step 2., and Step 4..
9. If the number of particles has increased, or particles can be visually identified as chips or flakes from a gear or bearing, replace gearbox. If the number of particles has decreased and only minute particles are found, continue gearbox in service.

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Transmission - Serviceability Check

Procedure

1. Drain transmission, oil cooler, and connecting lines.
2. Insert a magnet into pump screen boss of plug to check for additional contamination.
3. Flush oil cooler and connecting lines with clean approved oil (412-A-12-10-03-00A-040A-A / 00135).
4. Clean and install chip detector plug, pump inlet screen, and sump oil filter. Replace external oil filter element.



NOTE CONDITION OF PACKINGS, SEALS, AND GASKETS BEFORE INSTALLATION. REPLACE IF DAMAGED.

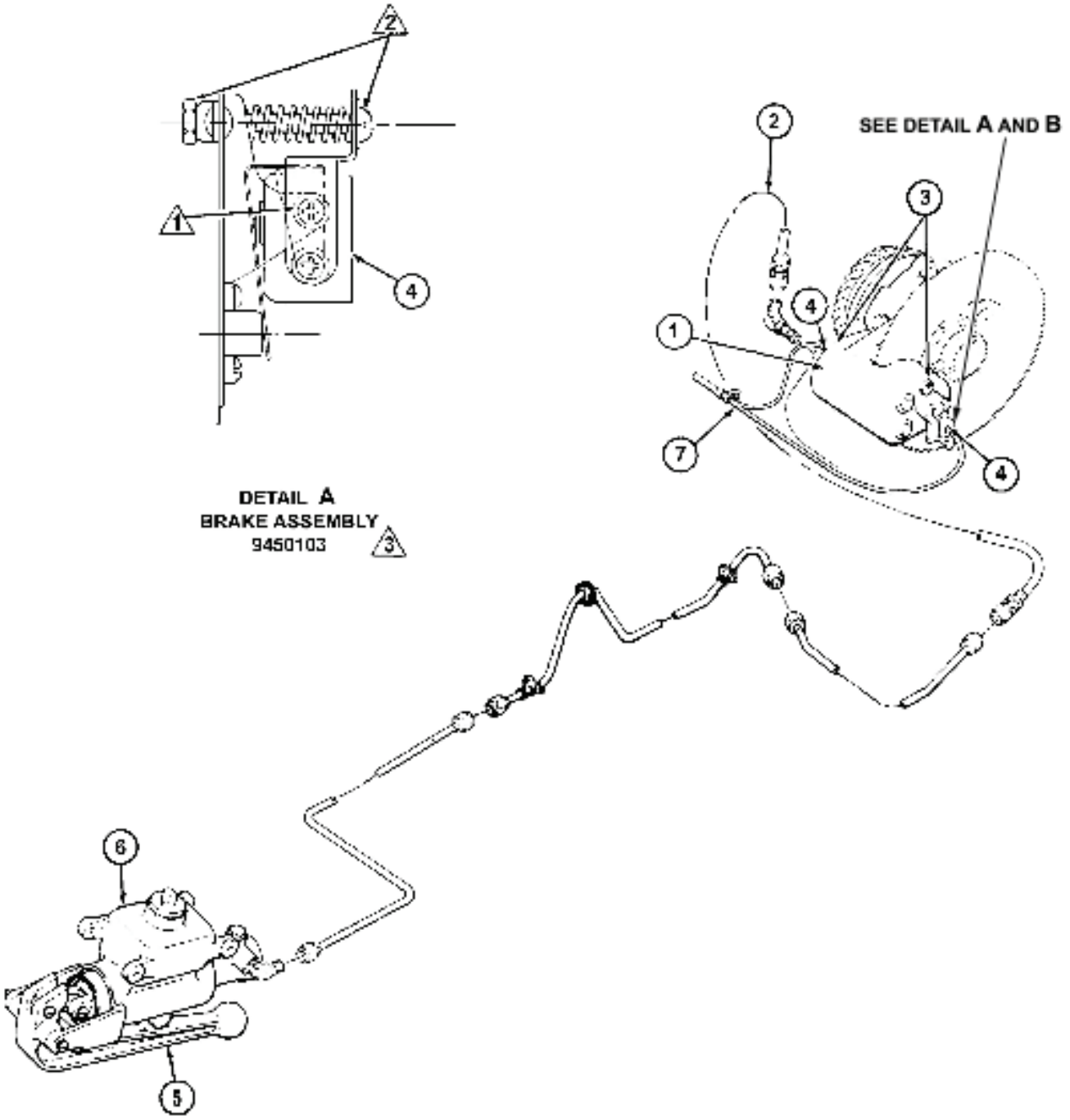
5. Service transmission with oil (412-A-12-10-03-00A-040A-A / 00135).
6. Ground run helicopter (BHT-412-FM) for 1 hour at 100% N_R with maximum pitch which can be applied without becoming airborne and tail rotor pedal equivalent to flight position.
7. Drain transmission oil into a clean container and inspect for chips.
8. Inspect sump internal filter and all chip detectors.
9. If number of particles has increased, or if there are particles which can be visually identified as chip or flakes from a gear or bearing, replace transmission, mast assembly, and transmission oil cooler. Flush oil lines. If number of particles has decreased and only minute particles are found, continue transmission in service.
10. After 5 hours of service, inspect sump internal filter and three chip detectors for particles.
11. If the number of particles has increased, or if there are particles which can visually be identified as chips or flakes from a gear or bearing, replace transmission, mast assembly, and transmission oil cooler. Flush oil lines. If the number of particles has decreased and only minute particles are found, continue transmission in service.

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Rotor Brake System - Description

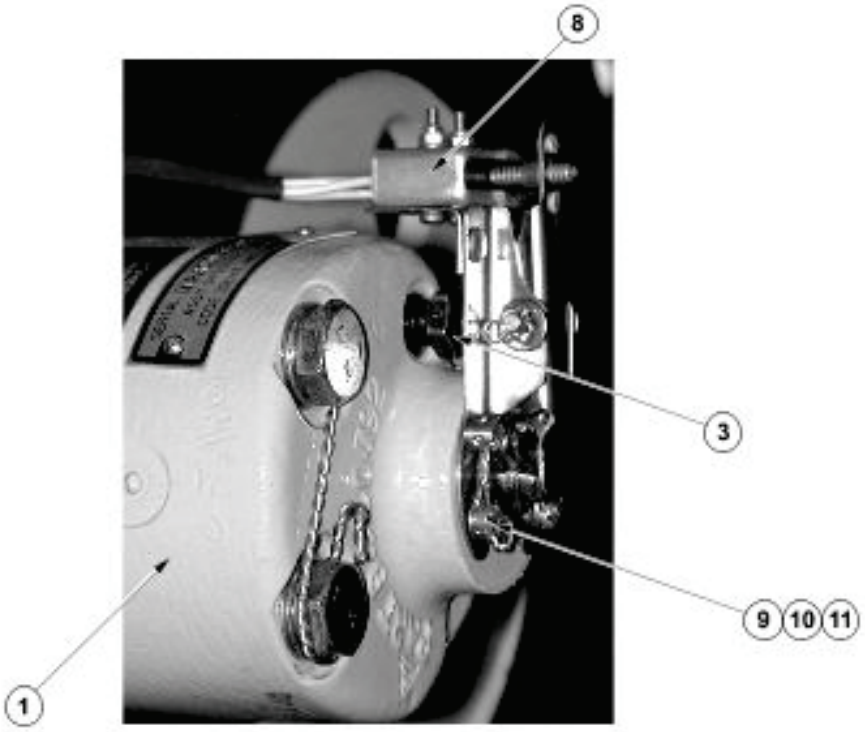
The rotor brake system is an optional equipment kit. For additional information on initial kit installation, refer to the BHT-412-SI-12.

Hydraulic pressure to operate the rotor brake system is provided by actuation of handle (5, Figure 1) on master cylinder (6) located in the cabin roof.



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Figure 1. Rotor Brake System (Sheet 1 of 2)



DETAIL B
BRAKE ASSEMBLY (9450103-1)

1. Rotor brake assembly
2. Hydraulic line
3. Bleed valve
4. Microswitch
5. Handle
6. Master cylinder
7. Microswitch electrical harness
8. Microswitch
9. Screws
10. Washers
11. Lockwire

NOTES

1. Loosen screw while making adjustment.
2. Loosen jam nut, turn screw to adjust switch, tighten jam nut.
3. Do not mix brake assembly 9450103-1 and 9450103 in any installation.

ICN-412-A-632300-B-97499-00002-A-001-01

Figure 1. Rotor Brake System (Sheet 2 of 2)

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Rotor Brake System - Troubleshooting

Procedure

1. Table 1 lists probable causes, isolation procedures, and remedies, and is intended to aid in rotor system troubleshooting.

Table 1. Rotor Brake System - Troubleshooting

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
No braking action.	Insufficient fluid in system.	Fill and bleed system.
	Foreign substance in fluid.	Purge system.
	Primary valve open.	Replace master cylinder.
	Relief valve open.	Replace master cylinder.
	Excessive brake lining wear.	Replace lining.
Spongy action of control.	Air in system.	Fill and bleed system.
Loss of braking action.	Worn seal in master cylinder.	Replace master cylinder.
	Worn seal in brake assembly.	Replace brake assembly.
	Excessive brake lining wear.	Replace lining.
Leaks at master cylinder.	Worn or damaged seal in cylinder.	Replace master cylinder seals.
Leaks at brake assembly.	Worn or damaged seals in brake assembly.	Replace brake assembly seals.
Indicator light inoperative.	Burned out bulbs.	Replace bulbs.
	Loose electrical connection.	Tighten connections.
	Broken or disconnected wire.	Replace or connect wire.
	Improper microswitch adjustment.	Adjust microswitches.
Indicator light illuminates when master cylinder not applied.	Faulty microswitch.	Replace switch.
	Improper microswitch adjustment.	Adjust microswitches.
	Lining binding.	Clean brake cylinder.

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Brake Assembly (P/N 9450103) - Bleeding

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Rubber Hose	Commercial	1	

Procedure

NOTE

Bleeding is invalid if hydraulic quick disconnect is removed.

NOTE

Drain a minimum of one-half pint (236.6 mL) of fluid from each valve before starting bleeding operation.

1. Check master cylinder (6, Figure 1) for proper fluid level (412-A-12-10-06-00A-210A-A / 00139).

NOTE

Ensure rubber hose used in Step 2. is clean.

2. Place one end of a length of clean rubber hose on bleed valve (3) extension.

3. Route other end of hose to a suitable container in order to capture brake fluid discharged during bleeding operation.

4. Open one bleed valve (3).

NOTE

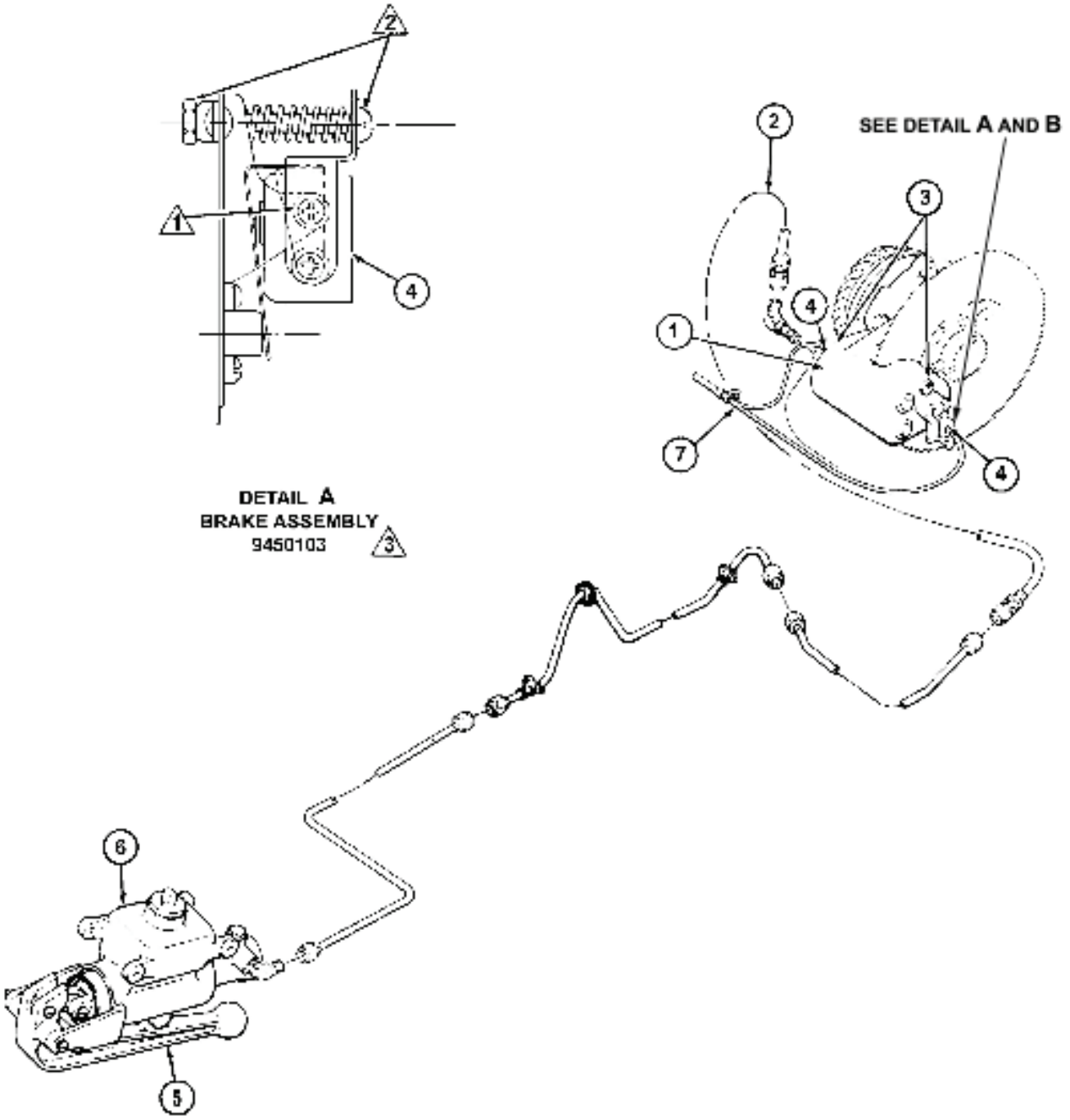
Do not allow handle to pass center into fully deployed position.

5. Slowly pull handle (5) down and hold before the fully deployed position.

6. Close bleed valve (3).

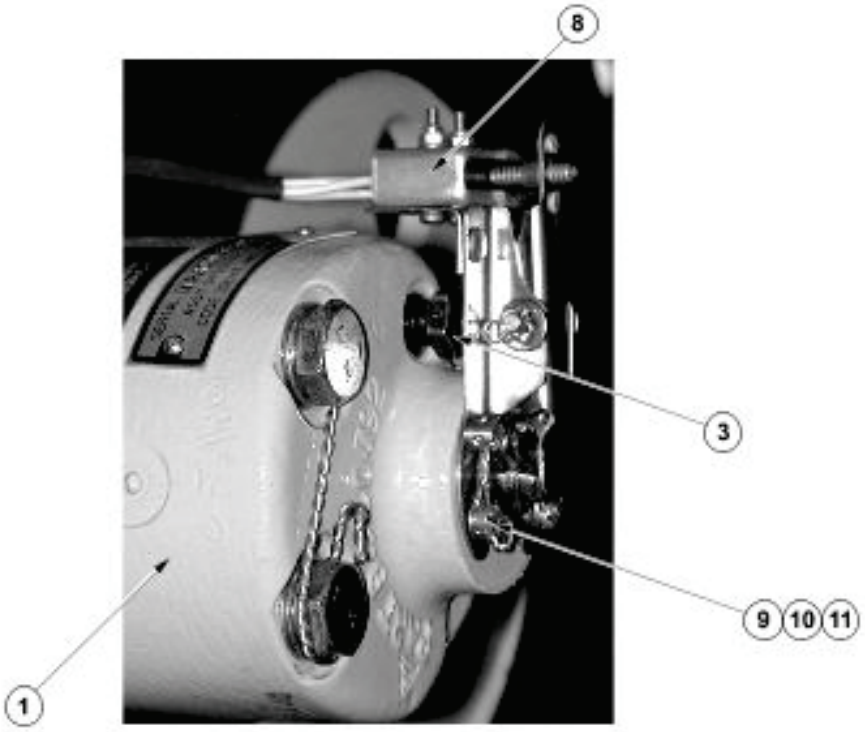
7. Service fluid in master cylinder (6) (412-A-12-10-06-00A-210A-A / 00139).

8. Repeat Step 2. through Step 6. until discharged fluid is free of air.



ICN-412-A-632300-B-97499-00001-A-001-01

Figure 1. Rotor Brake System (Sheet 1 of 2)



DETAIL B
BRAKE ASSEMBLY (9450103-1)

1. Rotor brake assembly
2. Hydraulic line
3. Bleed valve
4. Microswitch
5. Handle
6. Master cylinder
7. Microswitch electrical harness
8. Microswitch
9. Screws
10. Washers
11. Lockwire

NOTES

1. Loosen screw while making adjustment.
2. Loosen jam nut, turn screw to adjust switch, tighten jam nut.
3. Do not mix brake assembly 9450103-1 and 9450103 in any installation.

ICN-412-A-632300-B-97499-00002-A-001-01

Figure 1. Rotor Brake System (Sheet 2 of 2)

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Brake Assembly (P/N 9450103-1) - Bleeding

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

NOTE

Bleeding is invalid if hydraulic quick disconnect is removed.

NOTE

Drain a minimum of one-half pint (236.6 mL) of fluid from each valve before starting bleeding operation.

1. Check master cylinder (6, Figure 1) for proper fluid level (412-A-12-10-06-00A-210A-A / 00139).

2. Place rags under bleed valves (3) to catch fluid during bleeding.

3. Slowly open one bleed valve.

NOTE

Do not allow handle to pass center into fully deployed position.

4. Slowly pull handle (5) down and hold before the fully deployed position.

5. Close bleed valve (3).

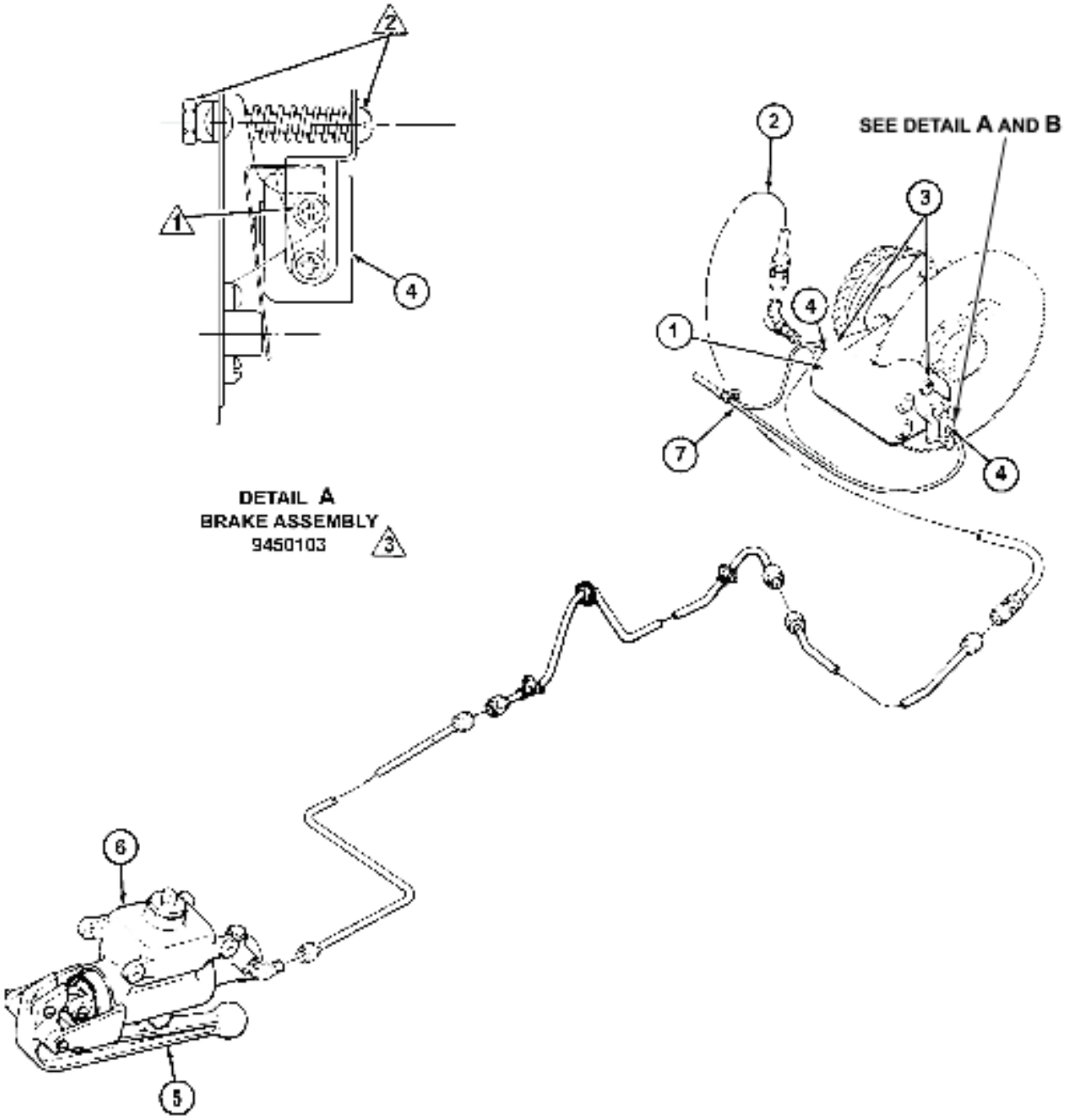
6. Service fluid in master cylinder (6) (412-A-12-10-06-00A-210A-A / 00139).

7. Return handle (5) to stowed position.

8. Repeat Step 2. through Step 7. until discharged fluid is free of air.

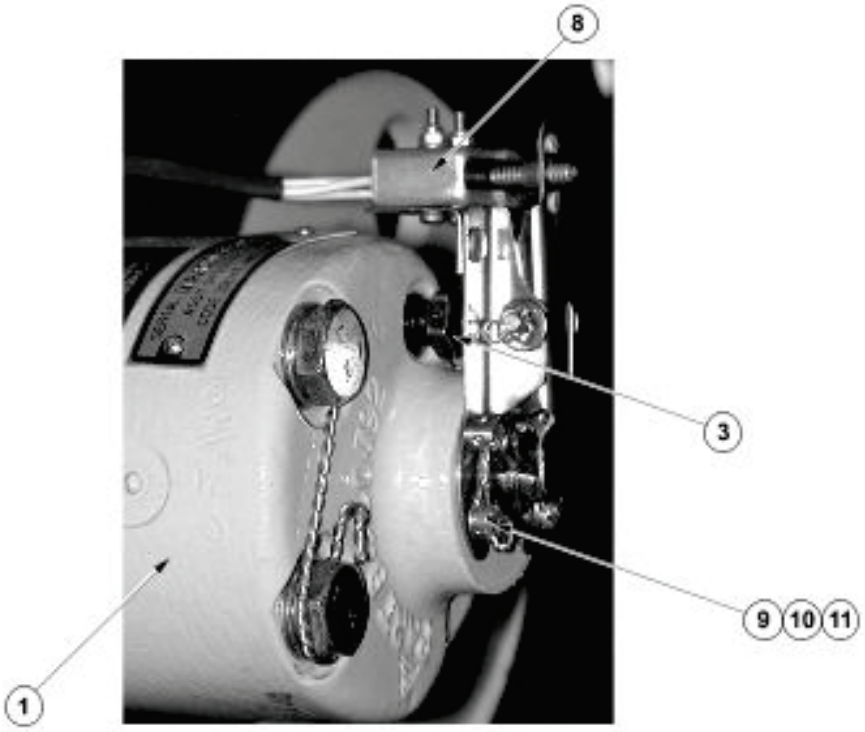
9. Repeat Step 2. through Step 8. for other bleed valve (3).

10. Clean area of any spilled fluid with drycleaning solvent (C-304).



ICN-412-A-632300-B-97499-00001-A-001-01

Figure 1. Rotor Brake System (Sheet 1 of 2)



DETAIL B
BRAKE ASSEMBLY (9450103-1)

1. Rotor brake assembly
2. Hydraulic line
3. Bleed valve
4. Microswitch
5. Handle
6. Master cylinder
7. Microswitch electrical harness
8. Microswitch
9. Screws
10. Washers
11. Lockwire

NOTES

1. Loosen screw while making adjustment.
2. Loosen jam nut, turn screw to adjust switch, tighten jam nut.
3. Do not mix brake assembly 9450103-1 and 9450103 in any installation.

ICN-412-A-632300-B-97499-00002-A-001-01

Figure 1. Rotor Brake System (Sheet 2 of 2)

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Master Cylinder - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure



CLEANING HYDRAULIC COMPONENTS SHALL BE ACCOMPLISHED WITH drycleaning solvent (C-304) ONLY. DO NOT USE ALCOHOL.

1. Open transmission fairing (412-A-52-40-00-02A-520A-A / 00837).
2. On cabin roof, remove sealing compound from around top of master cylinder reservoir (6, Figure 1).
3. Gain access to master cylinder (6) located inside cabin above pilot seat.

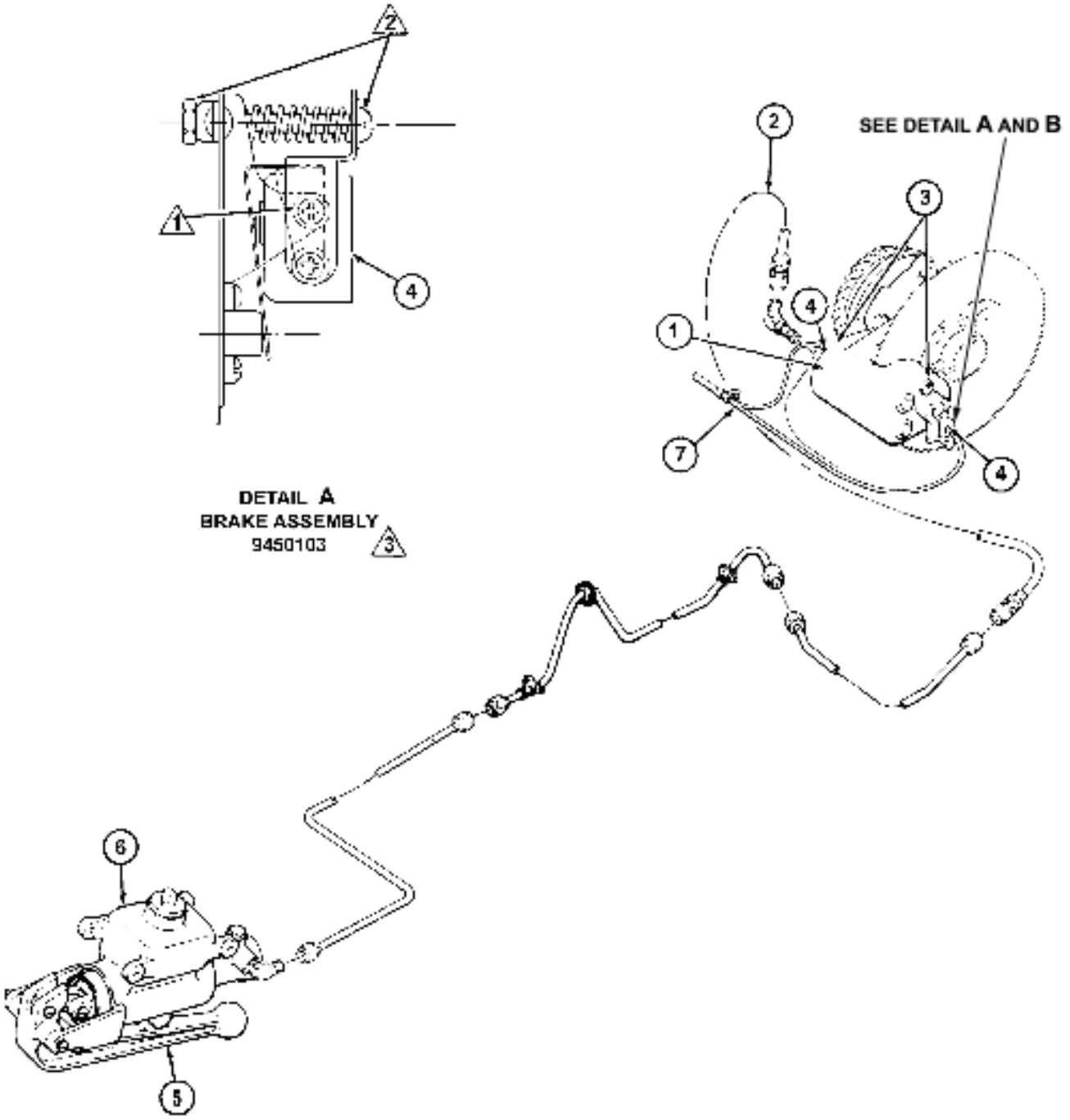


USE SUITABLE PRECAUTIONS TO PROTECT PILOT SEAT AND CABIN INTERIOR FROM HYDRAULIC FLUID WHEN LOOSENING HYDRAULIC LINE.

NOTE

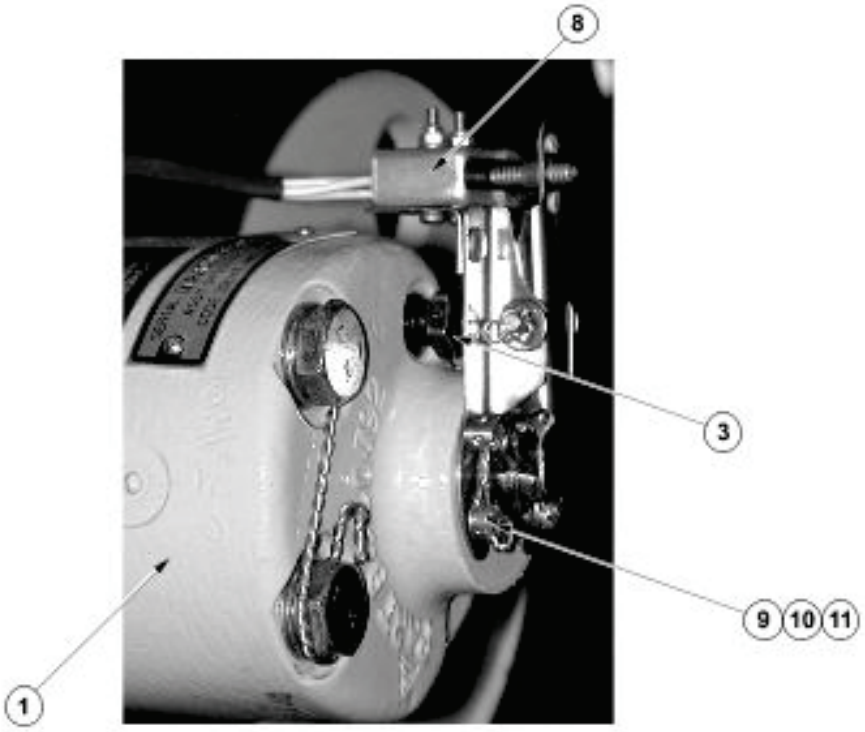
Drain hydraulic fluid into suitable container.

4. Loosen hydraulic line (2) from end of master cylinder (6) and allow fluid to drain into suitable container.
5. Disconnect hydraulic line (2) from master cylinder.
6. Plug openings to prevent entry of foreign materials.
7. Remove bolts, nuts, and washers attaching master cylinder (6) to roof structure.
8. Remove master cylinder (6).



ICN-412-A-632300-B-97499-00001-A-001-01

Figure 1. Rotor Brake System (Sheet 1 of 2)



DETAIL B
BRAKE ASSEMBLY (9450103-1)

1. Rotor brake assembly
2. Hydraulic line
3. Bleed valve
4. Microswitch
5. Handle
6. Master cylinder
7. Microswitch electrical harness
8. Microswitch
9. Screws
10. Washers
11. Lockwire

NOTES

1. Loosen screw while making adjustment.
2. Loosen jam nut, turn screw to adjust switch, tighten jam nut.
3. Do not mix brake assembly 9450103-1 and 9450103 in any installation.

ICN-412-A-632300-B-97499-00002-A-001-01

Figure 1. Rotor Brake System (Sheet 2 of 2)

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Master Cylinder - Inspection

Procedure

1. Refer to the BHT-412-CR&O.

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Master Cylinder - Repair

Procedure

1. Refer to the BHT-412-CR&O.

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Master Cylinder - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

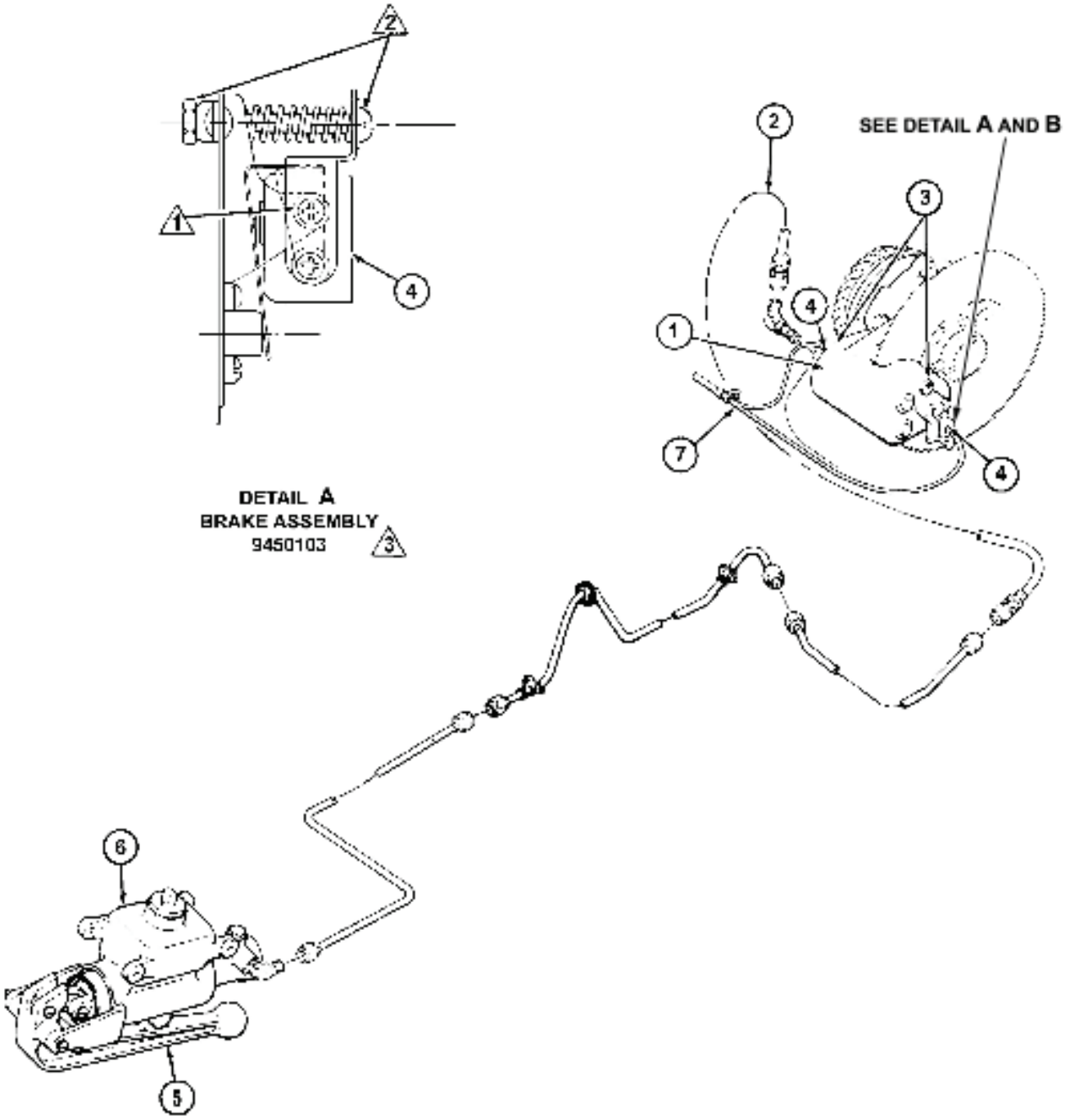
Procedure

1. Locate master cylinder (6, Figure 1) to the right of overhead console inside cabin and attach to roof structure with four bolts, four nuts, and eight washers.

2. Apply bead of sealant (C-308) to top of cabin roof surface around master cylinder (6) fill port.

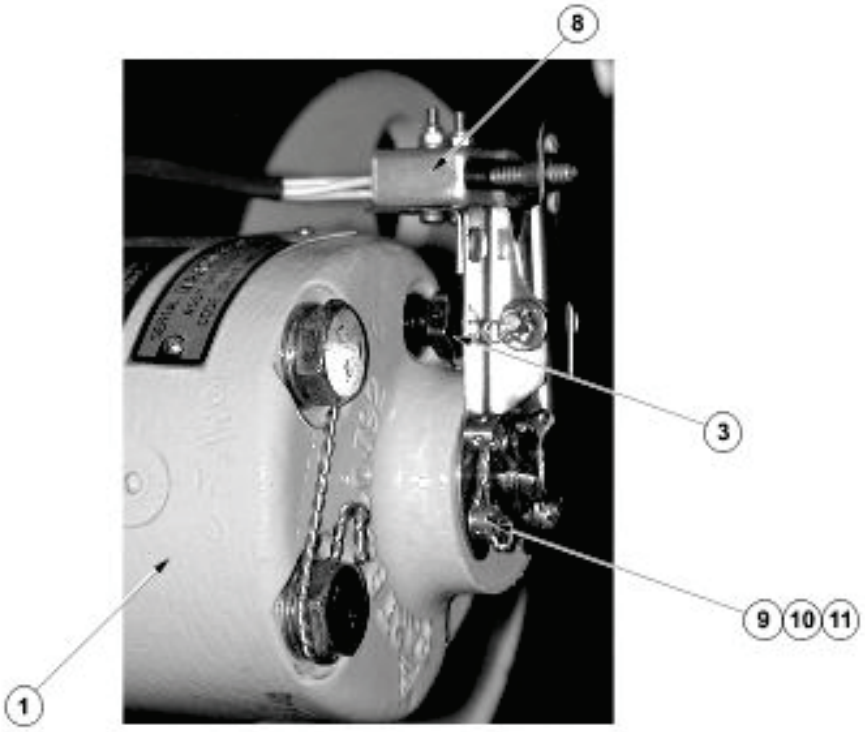
3. Attach hydraulic line (2) to master cylinder.

4. Bleed the rotor brake system (412-A-63-23-00-01A-231A-A / 01207 or 412-A-63-23-00-01A-231B-A / 01208).



ICN-412-A-632300-B-97499-00001-A-001-01

Figure 1. Rotor Brake System (Sheet 1 of 2)



DETAIL B
BRAKE ASSEMBLY (9450103-1)

1. Rotor brake assembly
2. Hydraulic line
3. Bleed valve
4. Microswitch
5. Handle
6. Master cylinder
7. Microswitch electrical harness
8. Microswitch
9. Screws
10. Washers
11. Lockwire

NOTES

1. Loosen screw while making adjustment.
2. Loosen jam nut, turn screw to adjust switch, tighten jam nut.
3. Do not mix brake assembly 9450103-1 and 9450103 in any installation.

ICN-412-A-632300-B-97499-00002-A-001-01

Figure 1. Rotor Brake System (Sheet 2 of 2)

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Brake Assembly - Description

NOTE

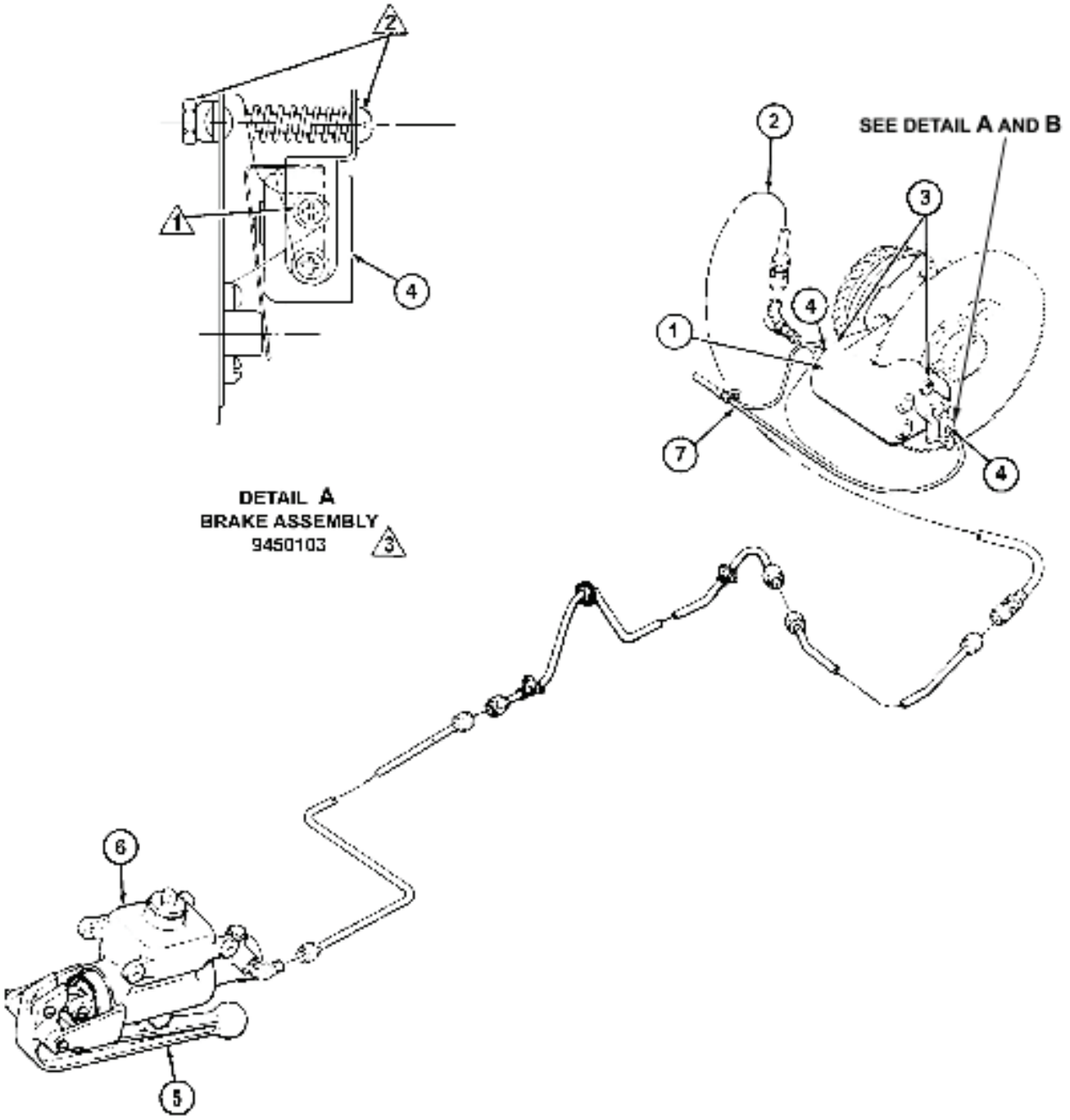
Maintenance practices are the same for brake assembly P/N 9450103 and 9450103-1, unless otherwise noted.

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Brake Assembly (P/N 9450103) - Removal

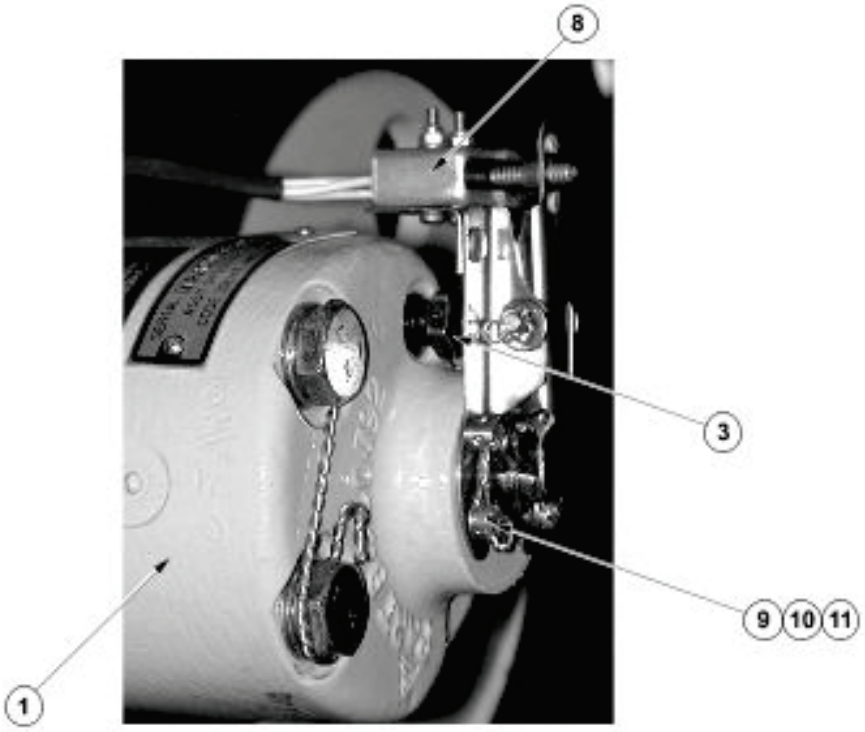
Procedure

1. Remove microswitches (4, Figure 1) from rotor brake assembly (1).
2. Disconnect hydraulic line (2) from rotor brake assembly (1).
3. Remove bolts (12, Figure 2), washers, and outboard brake assembly half (11).
4. Remove brake assembly spacer (23) from oil transfer tube (14) in inboard brake assembly half (5).
5. Remove brake oil transfer tube (14) from inboard brake assembly half (5).
6. Remove and discard packings (22) from counterbored cavities in mating surfaces of inboard brake assembly (5) and outboard brake assembly (11).
7. Remove rotor brake disc (412-A-63-23-00-04A-520A-A / 01220)
8. Cut lockwire (4), remove bolts (2), washers (3), and inboard brake assembly half (5) from quill assembly (1).



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Figure 1. Rotor Brake System (Sheet 1 of 2)



DETAIL B
BRAKE ASSEMBLY (9450103-1)

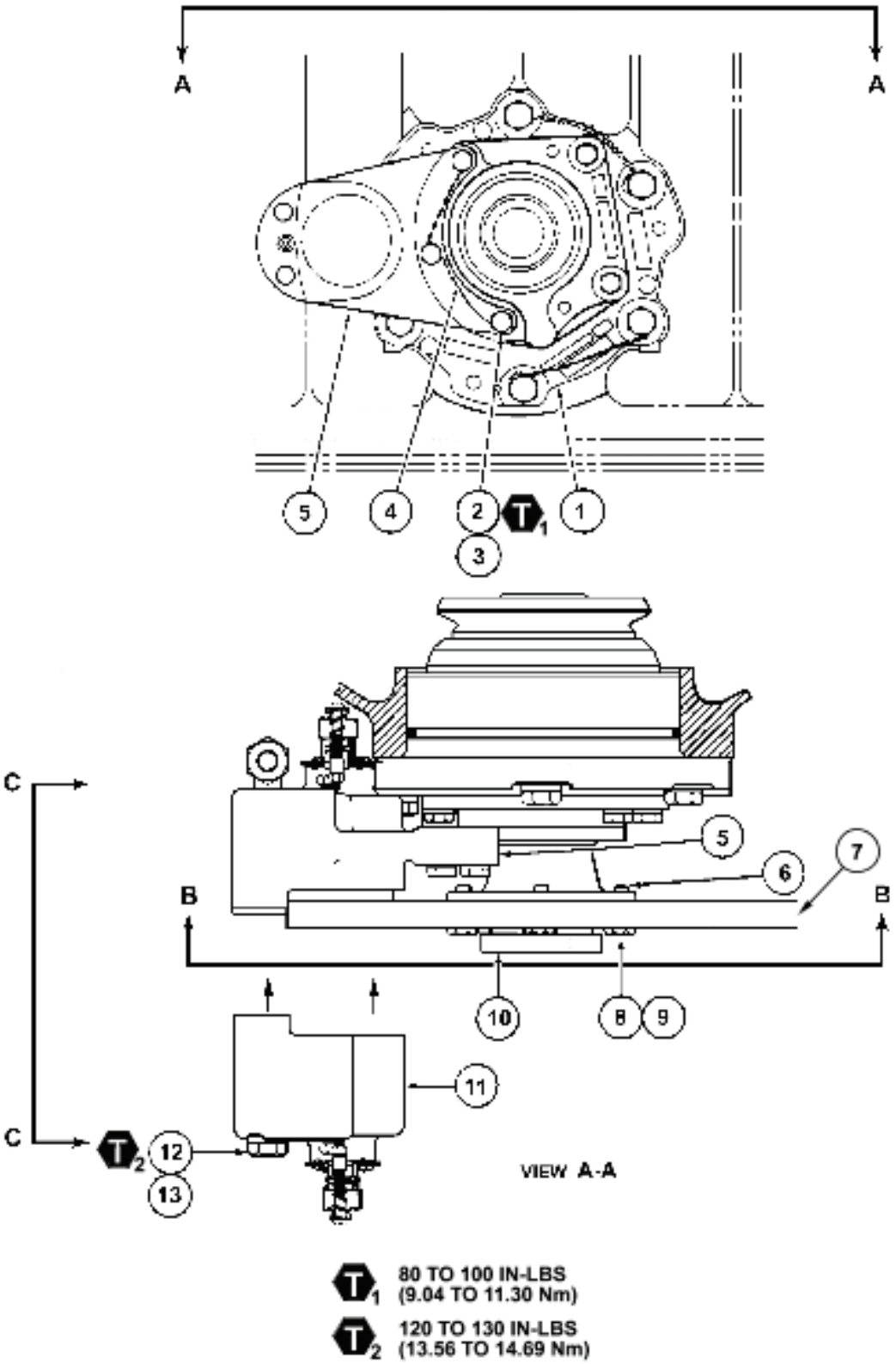
1. Rotor brake assembly
2. Hydraulic line
3. Bleed valve
4. Microswitch
5. Handle
6. Master cylinder
7. Microswitch electrical harness
8. Microswitch
9. Screws
10. Washers
11. Lockwire

NOTES

1. Loosen screw while making adjustment.
2. Loosen jam nut, turn screw to adjust switch, tighten jam nut.
3. Do not mix brake assembly 9450103-1 and 9450103 in any installation.

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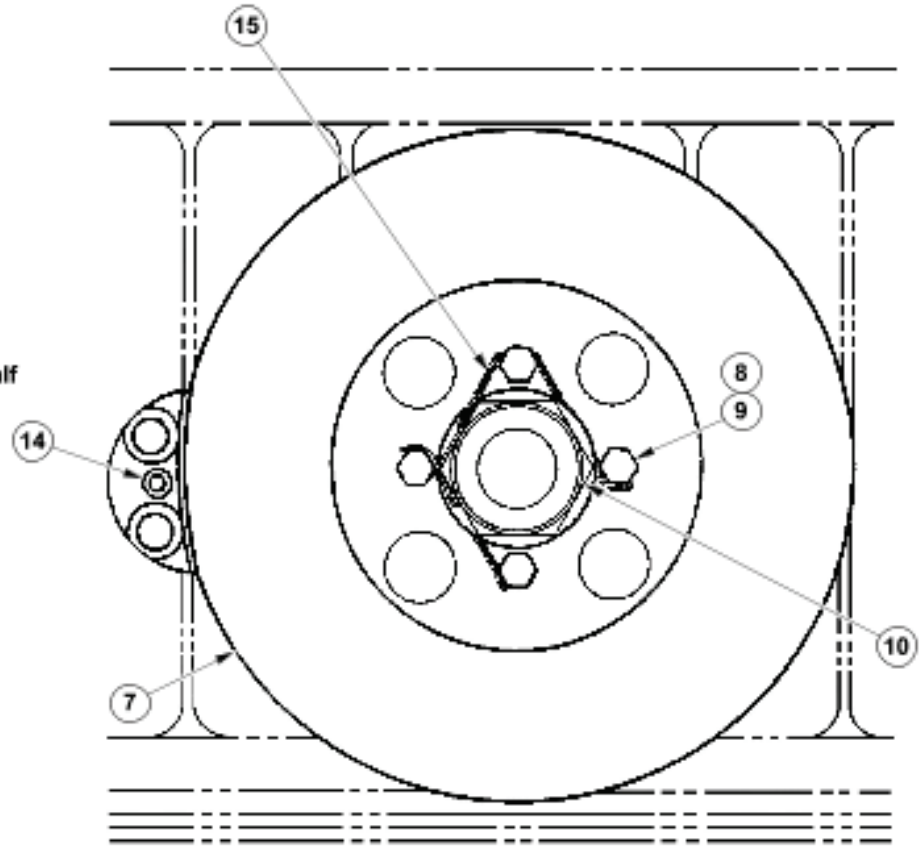
Figure 1. Rotor Brake System (Sheet 2 of 2)



ICN-412-A-632300-B-97499-00003-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 1 of 3)

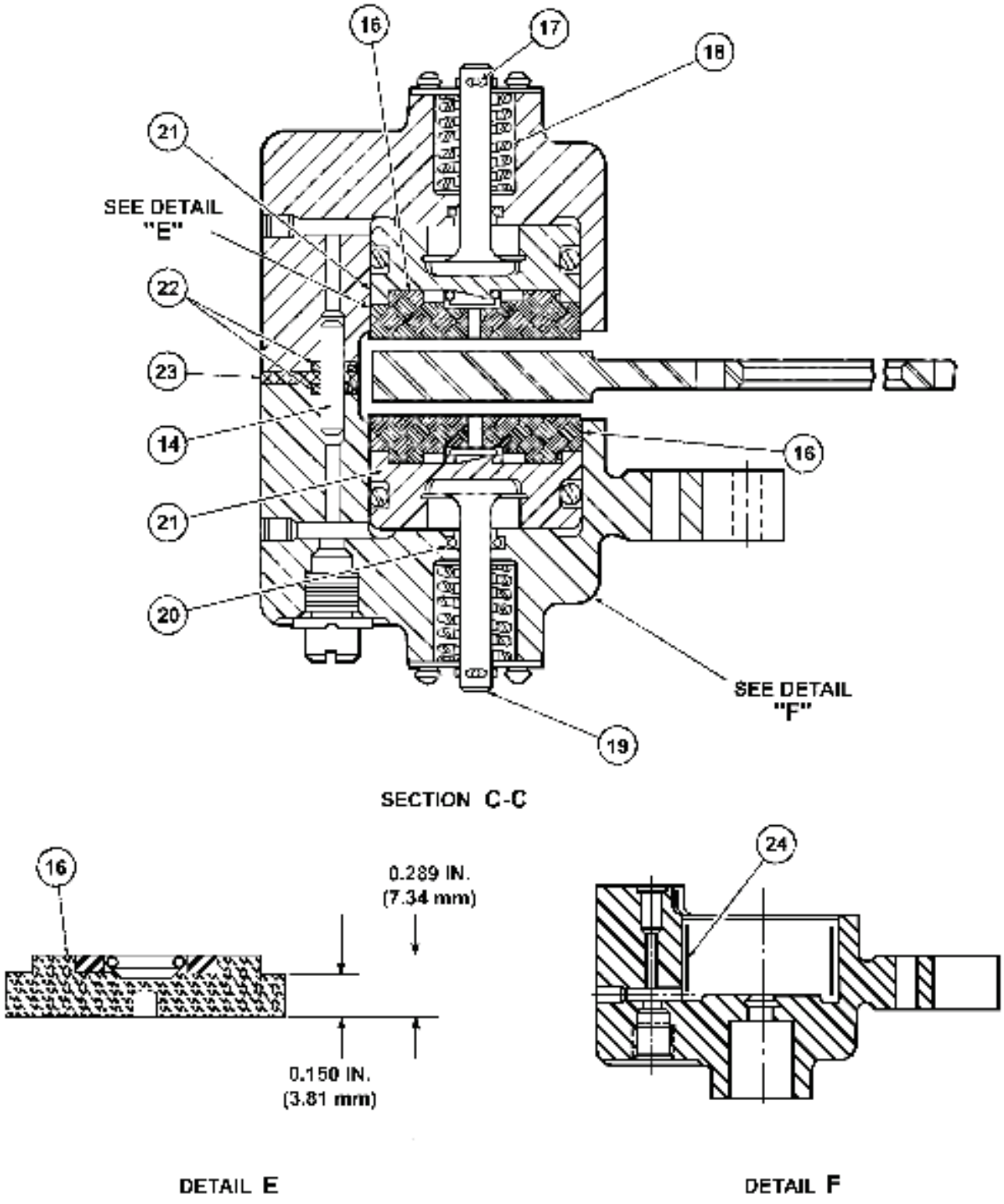
1. Quill assembly
2. Bolt
3. Washer
4. Lockwire
5. Inboard brake assembly half
6. Flange adapter
7. Brake disc
8. Bolt
9. Washer
10. Retaining nut
11. Outboard brake assembly half
12. Bolt
13. Washer
14. Oil transfer tube
15. Lockwire
16. Brake lining
17. Cotter pin
18. Return spring
19. Piston return pin
20. Packing
21. Piston
22. Packing
23. Spacer
24. Piston cavity



VIEW B-B

ICN-412-A-632300-B-97499-00004-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 2 of 3)



ICN-412-A-632300-B-97499-00005-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 3 of 3)

Brake Assembly (P/N 9450103-1) - Removal

Procedure

1. Disconnect hydraulic line (2, Figure 1) from rotor brake assembly (1).

NOTE

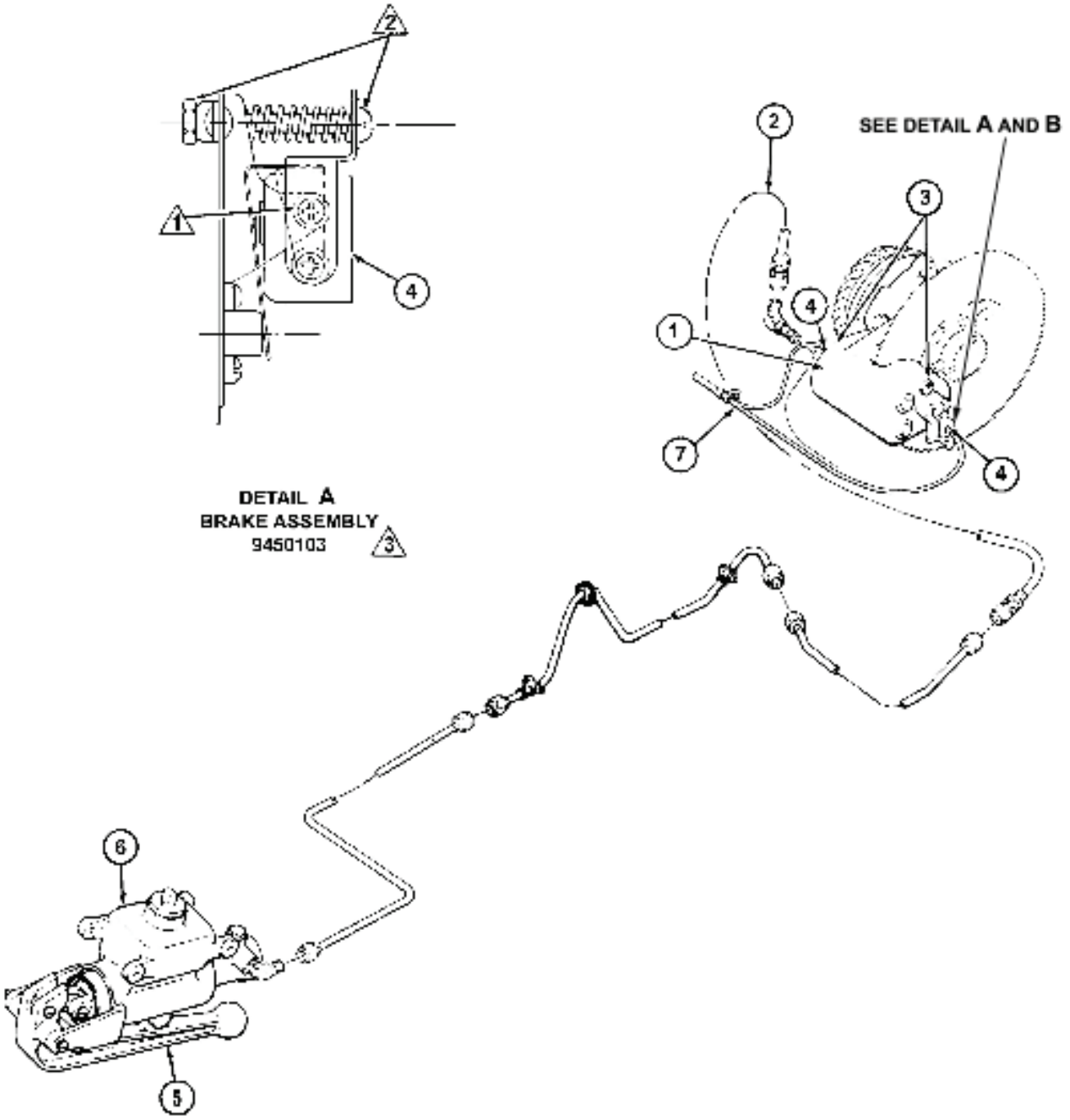
Access to bolts (2, Figure 2) can be achieved by turning rotor brake disc until holes in disc align with bolts.

2. Remove lockwire (4), bolts (2), and washers (3).



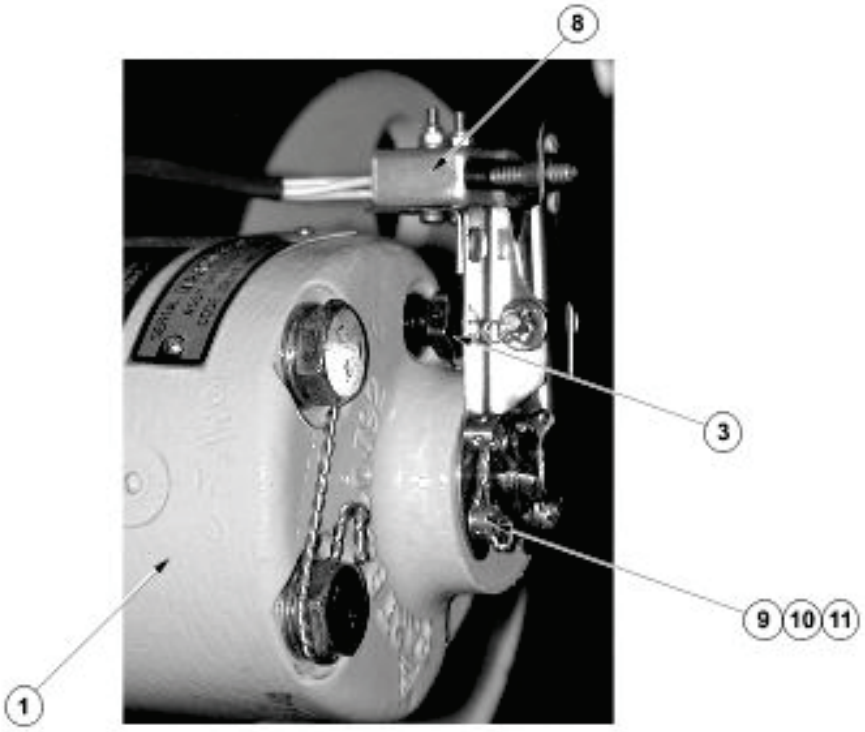
USE CARE SO AS NOT TO DAMAGE MICROSWITCH WIRING WHEN REMOVING BRAKE ASSEMBLY.

3. Remove rotor brake assembly (1, Figure 1) as a unit.
4. Remove lockwire (11), screws (9), and washers (10).
5. Remove microswitches (8).
6. Remove bolts (12, Figure 2), washers (13), and outboard brake assembly half (11).
7. Remove brake assembly spacer (23) from oil transfer tube (14) in inboard brake assembly half (5).
8. Remove brake oil transfer tube (14) from inboard brake assembly half (5).
9. Remove and discard packings (22) from counterbored cavities in mating surfaces of inboard brake assembly half (5) and outboard brake assembly half (11).
10. Remove rotor brake disc (412-A-63-23-00-04A-520A-A / 01220).



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Figure 1. Rotor Brake System (Sheet 1 of 2)



DETAIL B
BRAKE ASSEMBLY (9450103-1)

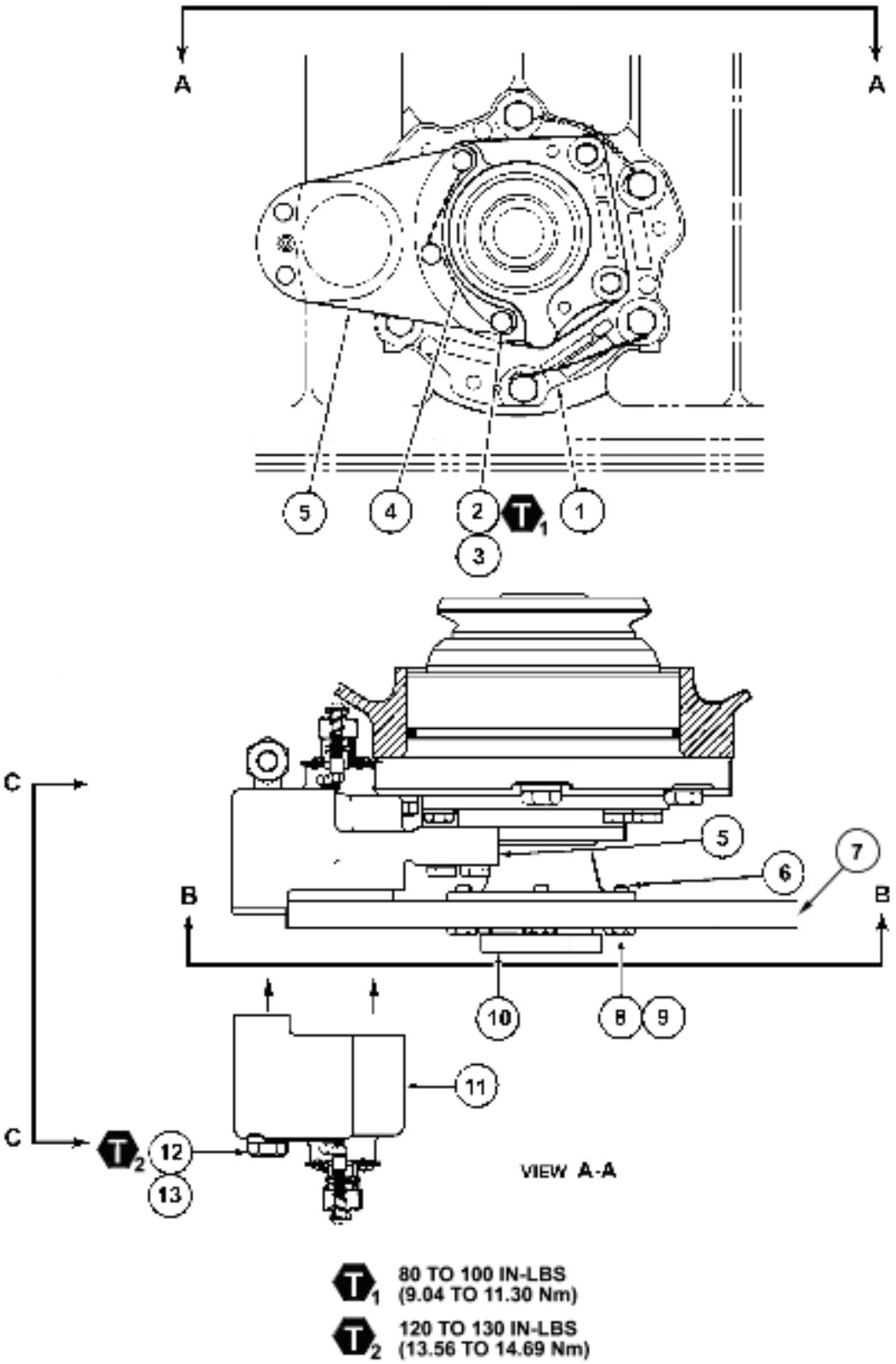
1. Rotor brake assembly
2. Hydraulic line
3. Bleed valve
4. Microswitch
5. Handle
6. Master cylinder
7. Microswitch electrical harness
8. Microswitch
9. Screws
10. Washers
11. Lockwire

NOTES

1. Loosen screw while making adjustment.
2. Loosen jam nut, turn screw to adjust switch, tighten jam nut.
3. Do not mix brake assembly 9450103-1 and 9450103 in any installation.

ICN-412-A-632300-B-97499-00002-A-001-01

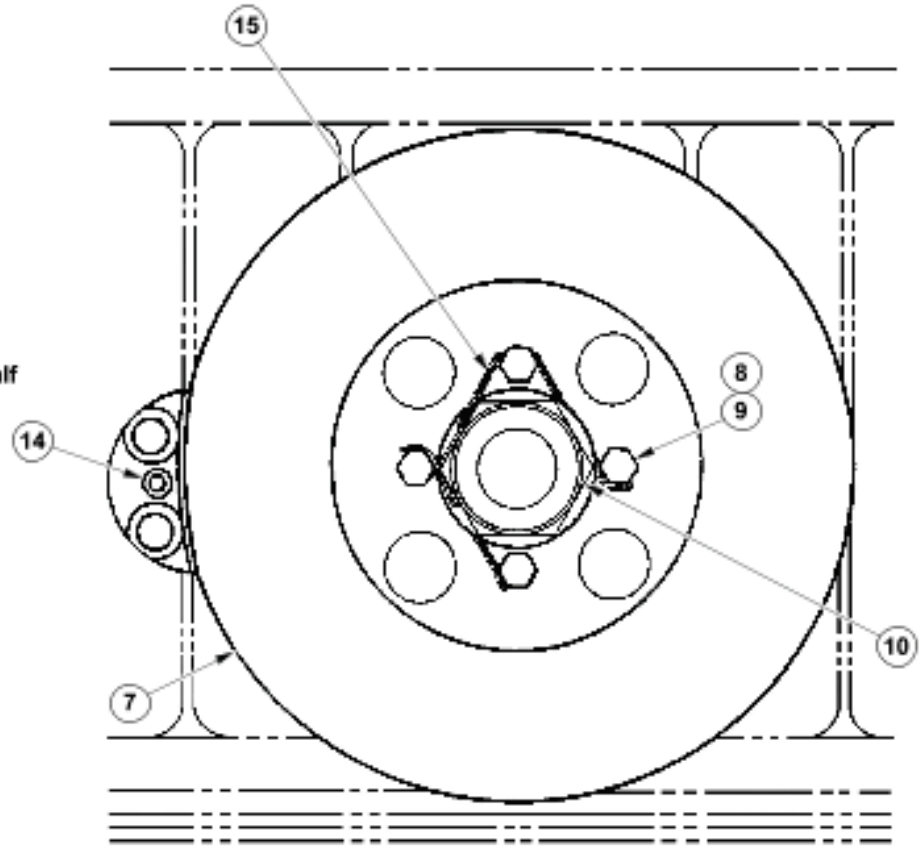
Figure 1. Rotor Brake System (Sheet 2 of 2)



ICN-412-A-632300-B-97499-00003-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 1 of 3)

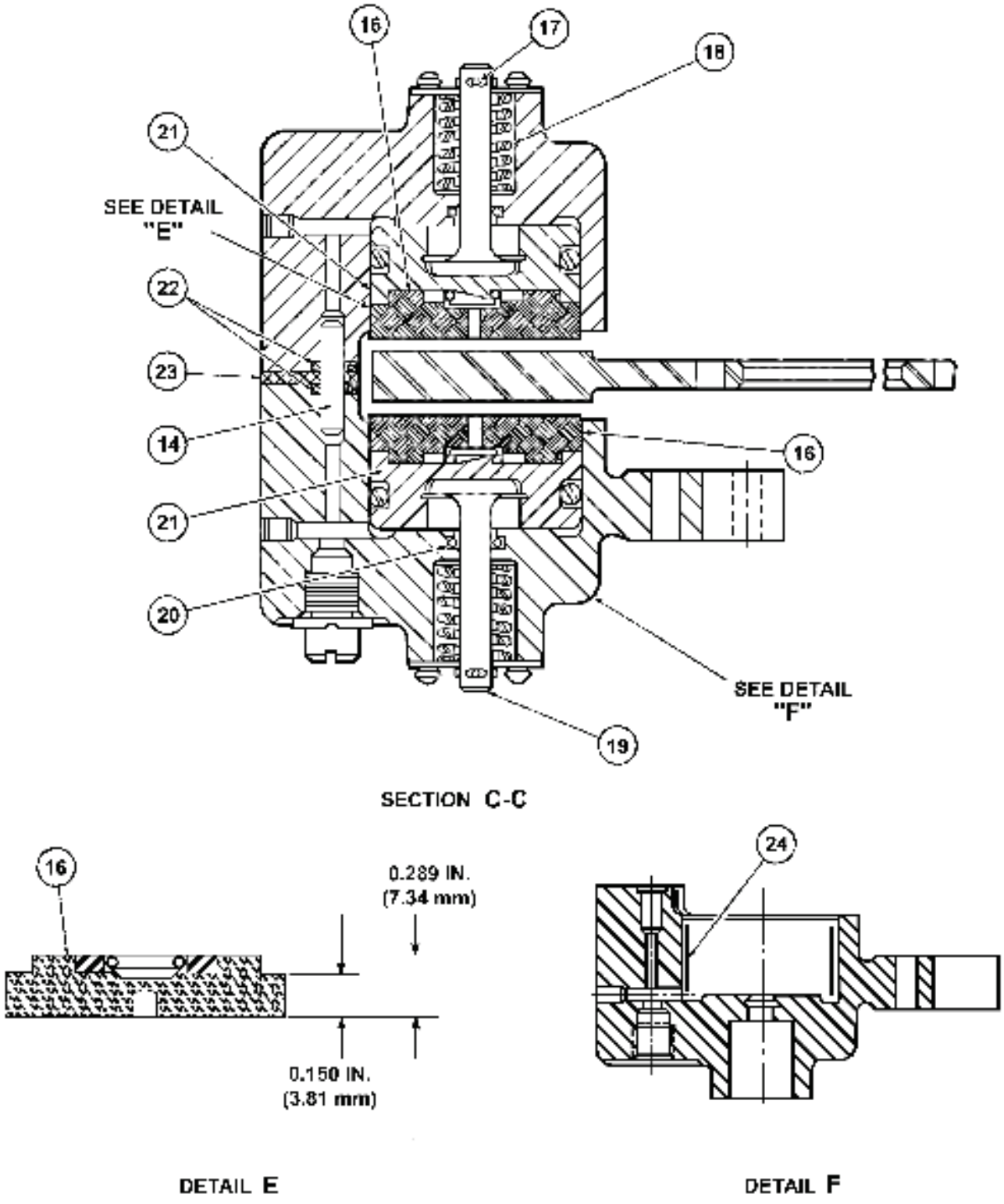
1. Quill assembly
2. Bolt
3. Washer
4. Lockwire
5. Inboard brake assembly half
6. Flange adapter
7. Brake disc
8. Bolt
9. Washer
10. Retaining nut
11. Outboard brake assembly half
12. Bolt
13. Washer
14. Oil transfer tube
15. Lockwire
16. Brake lining
17. Cotter pin
18. Return spring
19. Piston return pin
20. Packing
21. Piston
22. Packing
23. Spacer
24. Piston cavity



VIEW B-B

ICN-412-A-632300-B-97499-00004-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 2 of 3)



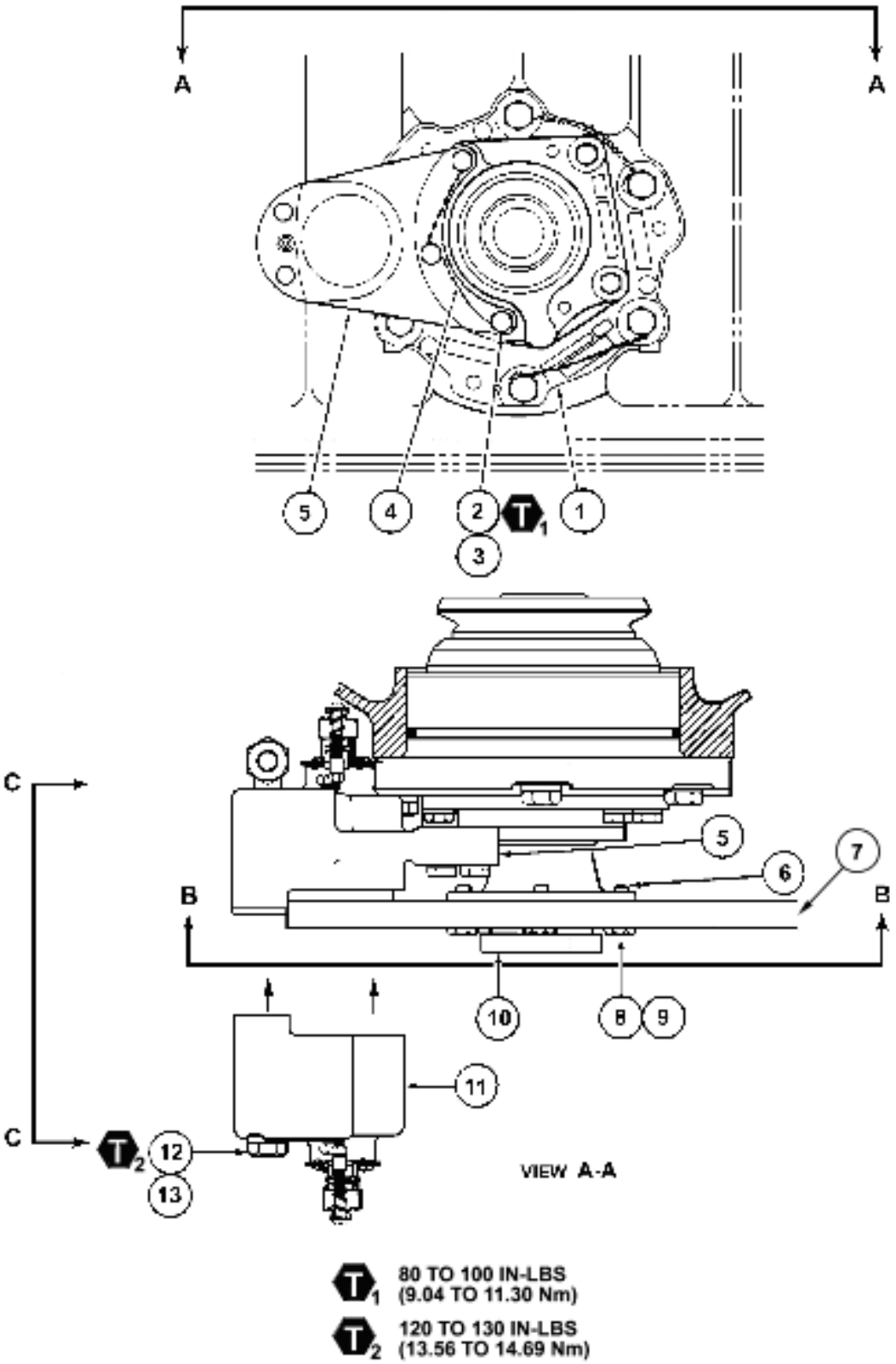
ICN-412-A-632300-B-97499-00005-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 3 of 3)

Brake Assembly - Inspection

Procedure

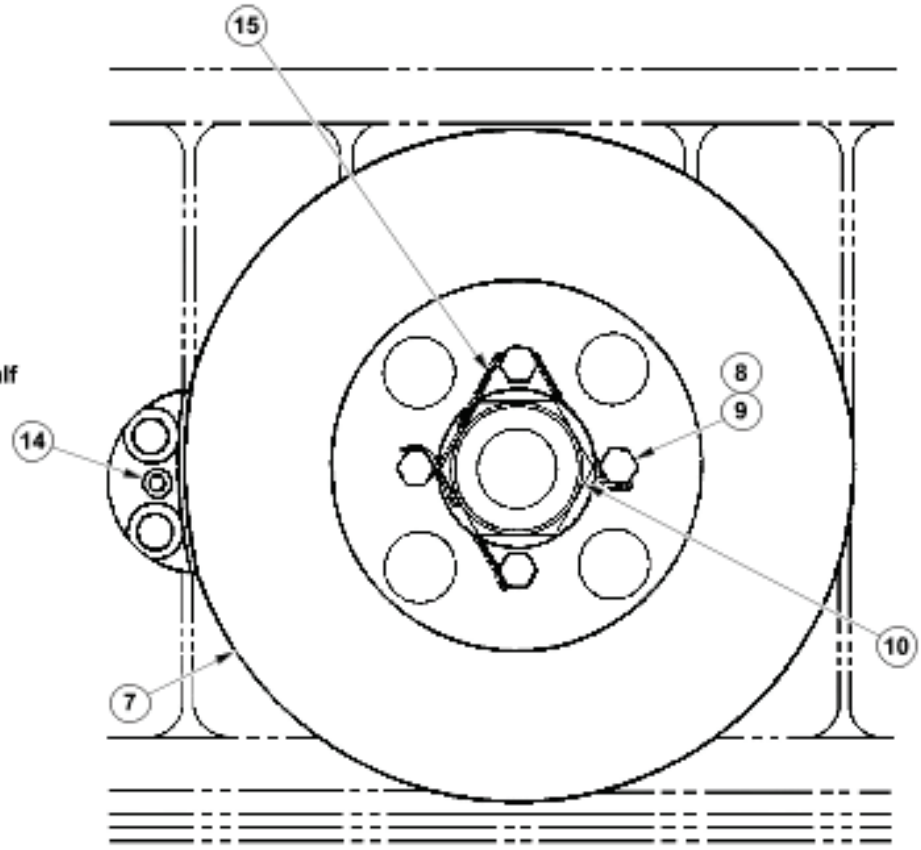
1. Replace brake lining (16, Figure 1, Detail E) when worn within 0.150 inch (3.81 mm) or less of piston (21) or when overall thickness of brake lining (16) measures 0.289 inch (7.34 mm) or less.
2. Inspect brake assembly in accordance with Figure 2.
3. Inspect bolts (12) for cleanliness, damage, and cracks.
4. Inspect piston return pin (19) for condition and nicks and burrs at cotter pin hole, which might damage return pin packing (20) on installation.
5. Inspect return spring (18) for rust and condition.
6. Inspect microswitches (8, Figure 3) for loose or cracked potting compound at electrical leads.



ICN-412-A-632300-B-97499-00003-A-001-01

Figure 1. Rotor Brake Assembly (Sheet 1 of 3)

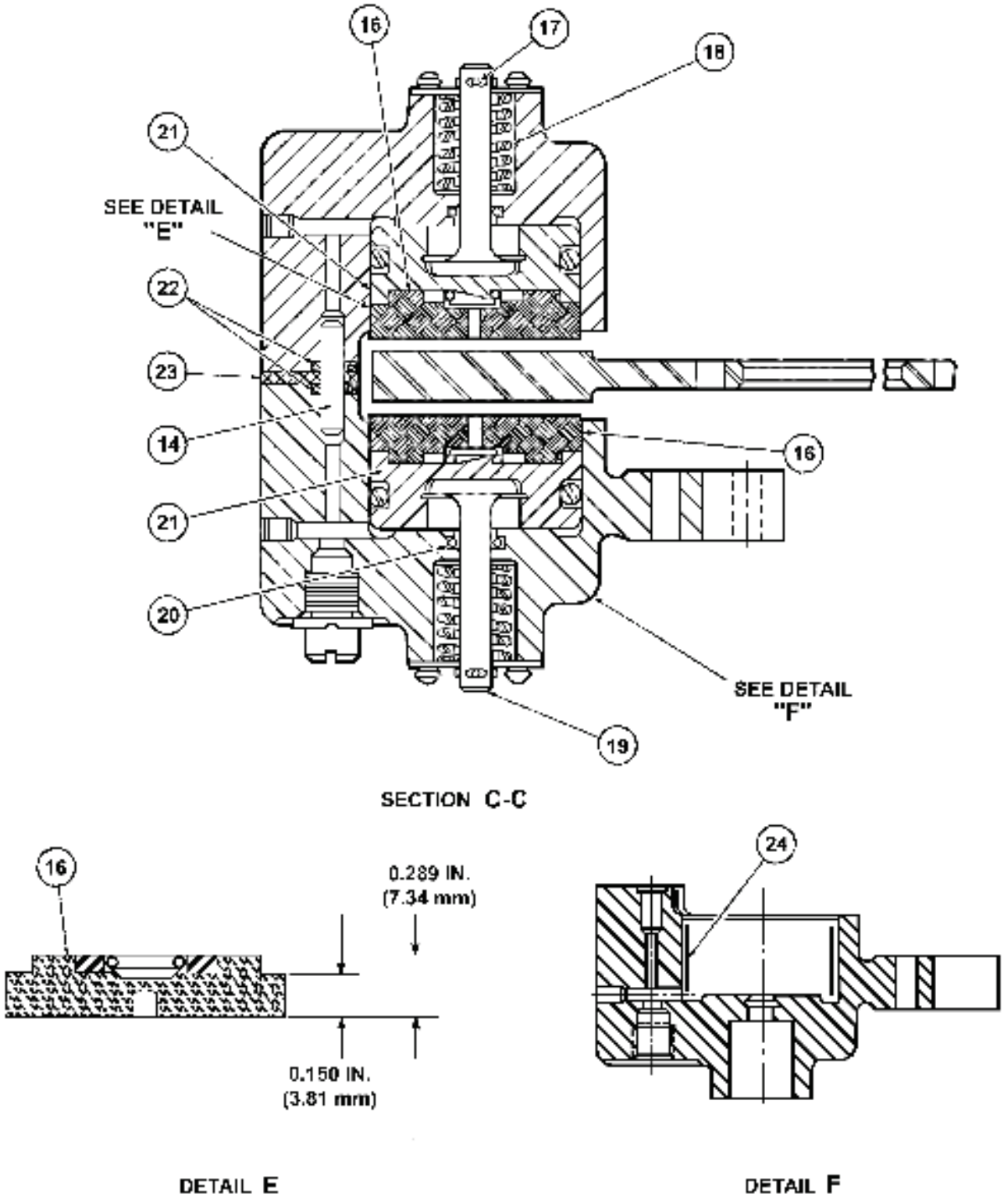
1. Quill assembly
2. Bolt
3. Washer
4. Lockwire
5. Inboard brake assembly half
6. Flange adapter
7. Brake disc
8. Bolt
9. Washer
10. Retaining nut
11. Outboard brake assembly half
12. Bolt
13. Washer
14. Oil transfer tube
15. Lockwire
16. Brake lining
17. Cotter pin
18. Return spring
19. Piston return pin
20. Packing
21. Piston
22. Packing
23. Spacer
24. Piston cavity



VIEW B-B

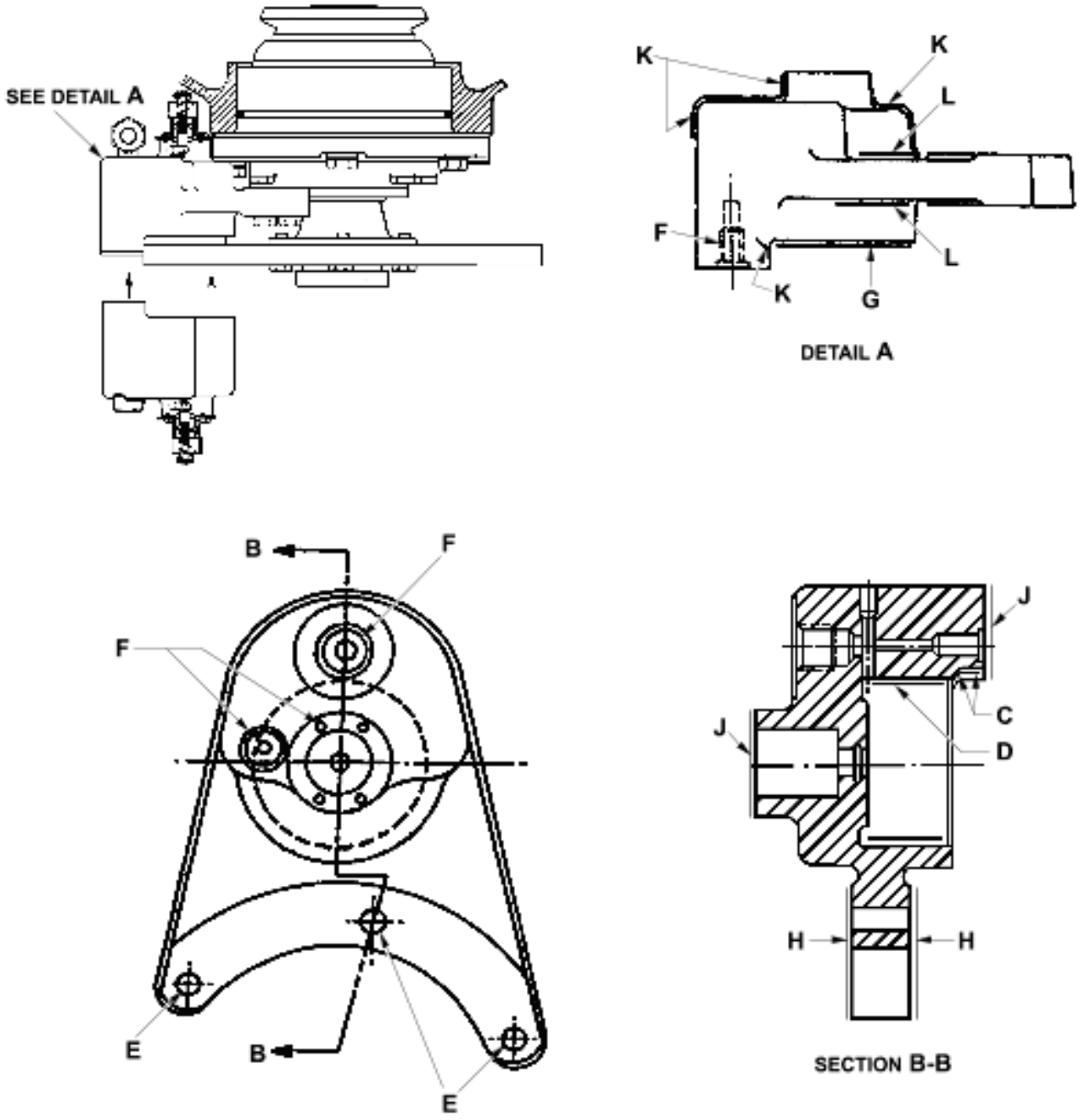
ICN-412-A-632300-B-97499-00004-A-001-01

Figure 1. Rotor Brake Assembly (Sheet 2 of 3)



ICN-412-A-632300-B-97499-00005-A-001-01

Figure 1. Rotor Brake Assembly (Sheet 3 of 3)



ICN-412-A-632300-B-97499-00006-A-001-01

Figure 2. Rotor Brake Assembly - Damage Limits (Sheet 1 of 2)

NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT
1.	C	Wear	Measure	0.015 inch (0.38 mm) deep 1.0 inch (25.4 mm) long maximum two places. (2)
2.	D	Mechanical/ corrosion damage	Visual/ measure	0.002 inch (0.051 mm) deep in piston cavity area, provided sealing qualities are maintained. Surface finish should be 15/30 microinches RHR. (2)
3.	E	Wear	Visual/ measure	0.015 inch (0.38 mm) maximum oversize on hole diameter. Round off sharp edges (for P/N 9421067 and 9431108).
4.	F	Thread	Visual	Remove housing from service if threads are stripped or badly damaged.
5.	G	Wear	Measure	0.030 inch (0.76 mm) deep, maximum on piston boss face.
6.	H	Wear	Measure	0.030 inch (0.76 mm) deep, maximum on mounting flange face. (2) 0.010 inch (0.25 mm) deep, maximum on flange face area covered by mounting bolt washers (for P/N 9421067 and 9431108).
7.	J	Wear	Measure	0.010 inch (0.25 mm) deep, maximum on return spring boss face and housing interface surface. (2)
8.	K	Wear	Measure	0.030 inch (0.76 mm) deep, maximum on housing exterior.
9.	L	Wear	Visual/ measure	0.030 inch (0.76 mm) deep, maximum on housing mounting flange; blend on both surfaces is permitted as long as they are not directly opposite each other (for P/N 9421067 and 9431108).

NOTES

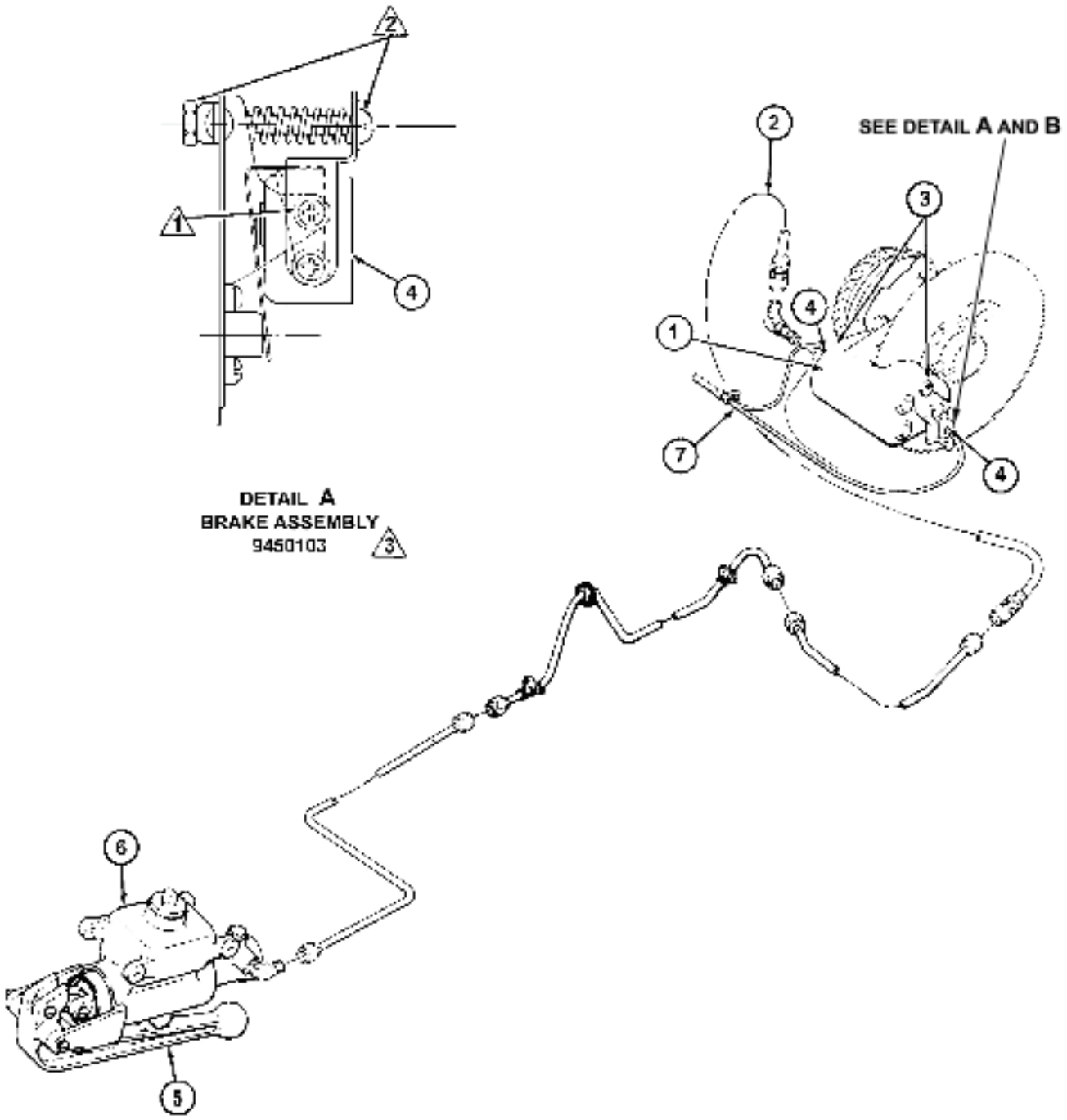
1. Inboard brake assembly half shown, outboard identical.

(2) Since removal of material in areas C, D, H, and J can shorten the life of the part, it is recommended that if removal of metal in these areas is necessary, it be kept to a minimum.

3. Scratches and/or pits in the areas designated, may be blended or polished out within the dimensional limits shown.

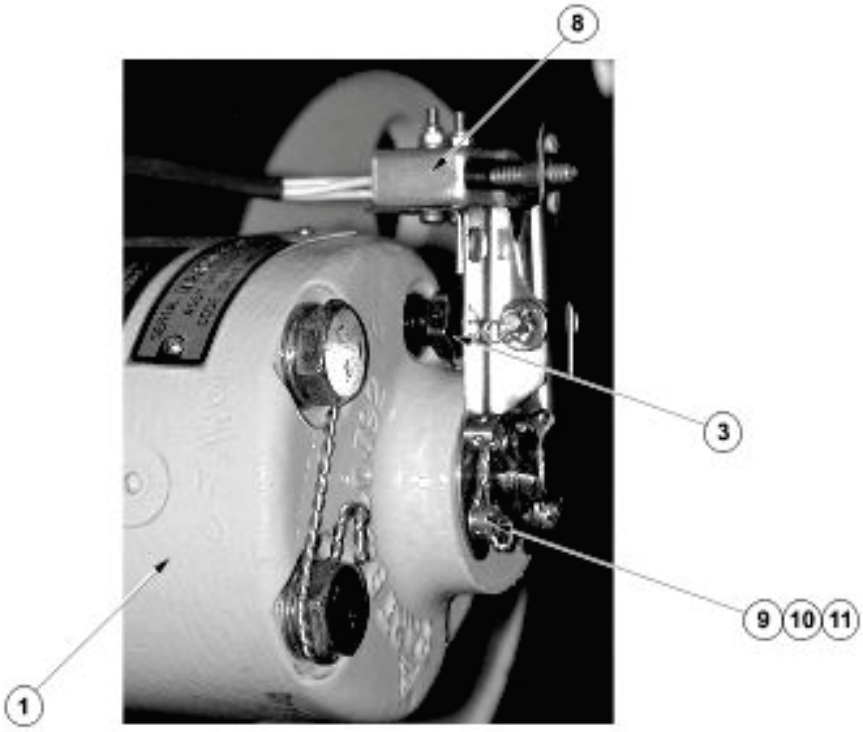
ICN-412-A-632300-B-97499-00007-A-001-01

Figure 2. Rotor Brake Assembly - Damage Limits (Sheet 2 of 2)



ICN-412-A-632300-B-97499-00001-A-001-01

Figure 3. Rotor Brake System (Sheet 1 of 2)



DETAIL B
BRAKE ASSEMBLY (9450103-1)

1. Rotor brake assembly
2. Hydraulic line
3. Bleed valve
4. Microswitch
5. Handle
6. Master cylinder
7. Microswitch electrical harness
8. Microswitch
9. Screws
10. Washers
11. Lockwire

NOTES

1. Loosen screw while making adjustment.
2. Loosen jam nut, turn screw to adjust switch, tighten jam nut.
3. Do not mix brake assembly 9450103-1 and 9450103 in any installation.

ICN-412-A-632300-B-97499-00002-A-001-01

Figure 3. Rotor Brake System (Sheet 2 of 2)

Brake Assembly - Repair

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
400 Grit Aluminum Oxide Cloth	Commercial	1	
Chromic Acid Solution MIL-C-5541	Commercial	1	

Procedure

NOTE

Repair of the brake is limited to the replacement of damaged parts and to the specific repairs listed in this section and Figure 1, Detail A.

1. Clean nicks on pistons (21, Figure 2), piston cavity (24), and throat with 400 grit aluminum oxide cloth.

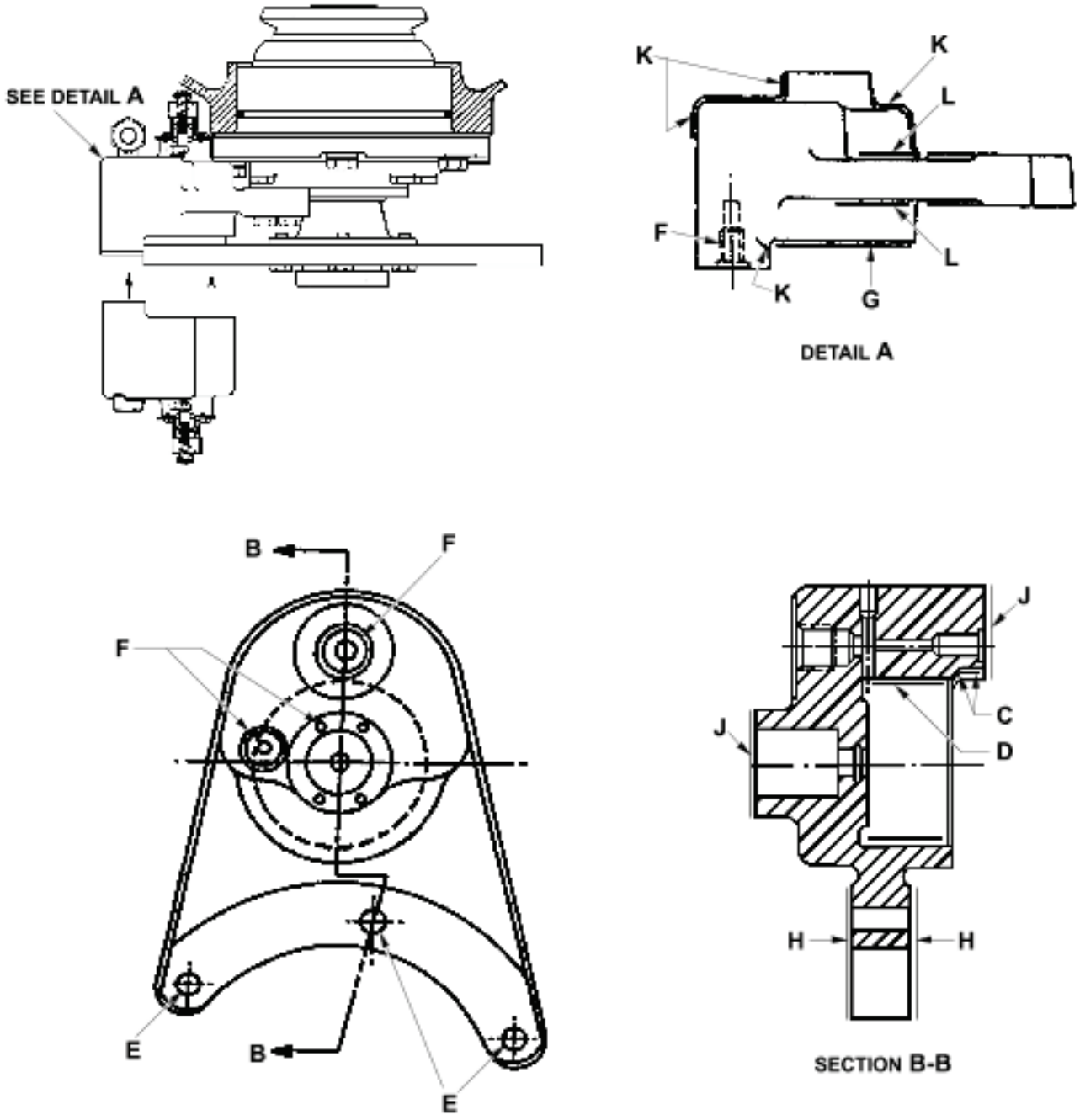
2. Replace any damaged fasteners.

3. Bend out nicks and burrs on shaft of piston return pin (19). Pay particular attention to the area around the cotter pin hole.



CHROMIC ACID SOLUTIONS ARE CORROSIVE AND POISONOUS. PROTECT THE EYES AND HANDS. DO NOT BREATHE IN THE FUMES. OBEY MANUFACTURER'S INSTRUCTIONS.

4. Treat reworked surfaces of piston (21) and exterior surfaces of brake assembly halves (5 and 11) with chromic acid solution mil-c-5541.



ICN-412-A-632300-B-97499-00006-A-001-01

Figure 1. Rotor Brake Assembly - Damage Limits (Sheet 1 of 2)

NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT
1.	C	Wear	Measure	0.015 inch (0.38 mm) deep 1.0 inch (25.4 mm) long maximum two places. (2)
2.	D	Mechanical/ corrosion damage	Visual/ measure	0.002 inch (0.051 mm) deep in piston cavity area, provided sealing qualities are maintained. Surface finish should be 15/30 microinches RHR. (2)
3.	E	Wear	Visual/ measure	0.015 inch (0.38 mm) maximum oversize on hole diameter. Round off sharp edges (for P/N 9421067 and 9431108).
4.	F	Thread	Visual	Remove housing from service if threads are stripped or badly damaged.
5.	G	Wear	Measure	0.030 inch (0.76 mm) deep, maximum on piston boss face.
6.	H	Wear	Measure	0.030 inch (0.76 mm) deep, maximum on mounting flange face. (2) 0.010 inch (0.25 mm) deep, maximum on flange face area covered by mounting bolt washers (for P/N 9421067 and 9431108).
7.	J	Wear	Measure	0.010 inch (0.25 mm) deep, maximum on return spring boss face and housing interface surface. (2)
8.	K	Wear	Measure	0.030 inch (0.76 mm) deep, maximum on housing exterior.
9.	L	Wear	Visual/ measure	0.030 inch (0.76 mm) deep, maximum on housing mounting flange; blend on both surfaces is permitted as long as they are not directly opposite each other (for P/N 9421067 and 9431108).

NOTES

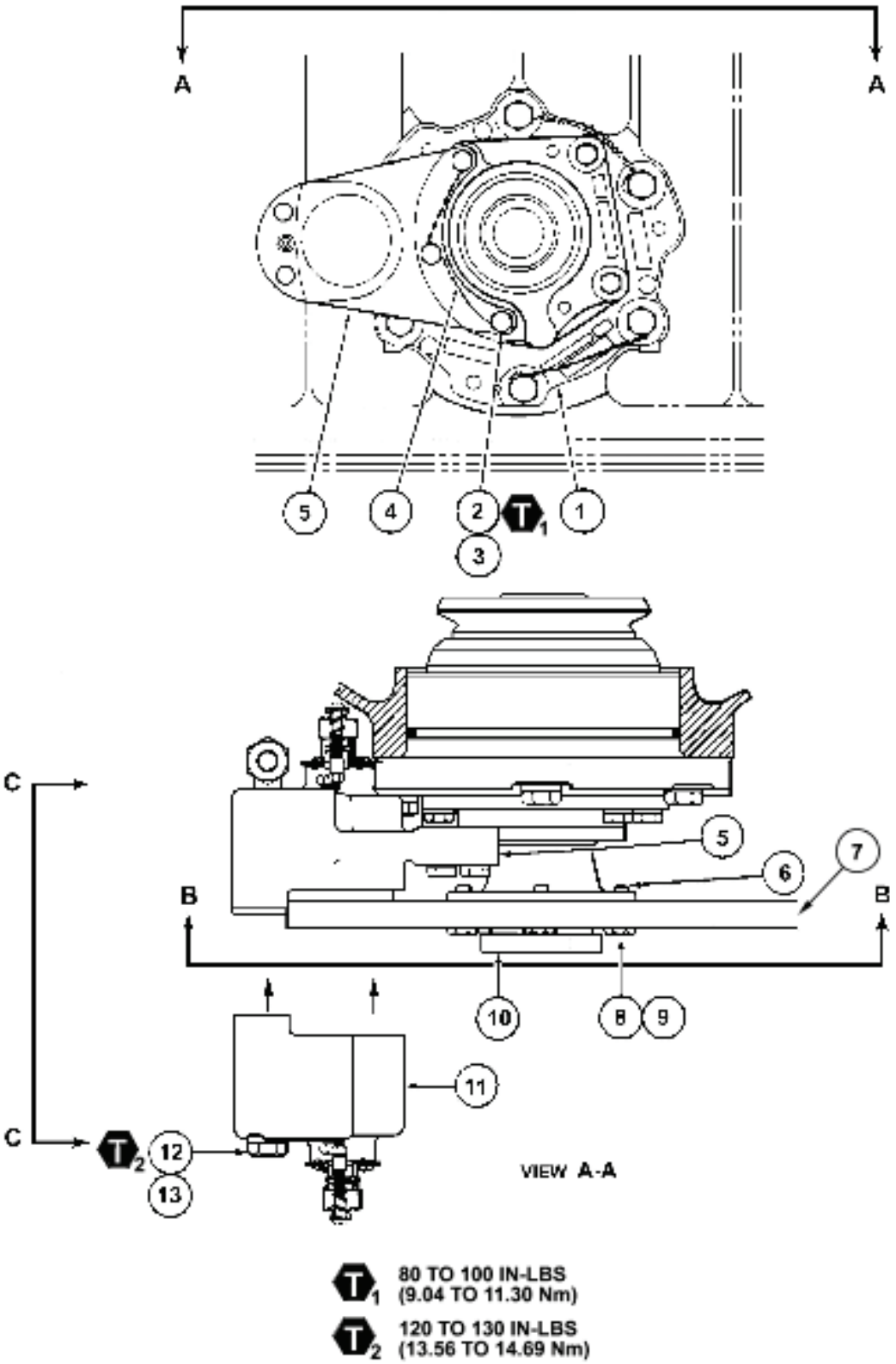
1. Inboard brake assembly half shown, outboard identical.

(2) Since removal of material in areas C, D, H, and J can shorten the life of the part, it is recommended that if removal of metal in these areas is necessary, it be kept to a minimum.

3. Scratches and/or pits in the areas designated, may be blended or polished out within the dimensional limits shown.

ICN-412-A-632300-B-97499-00007-A-001-01

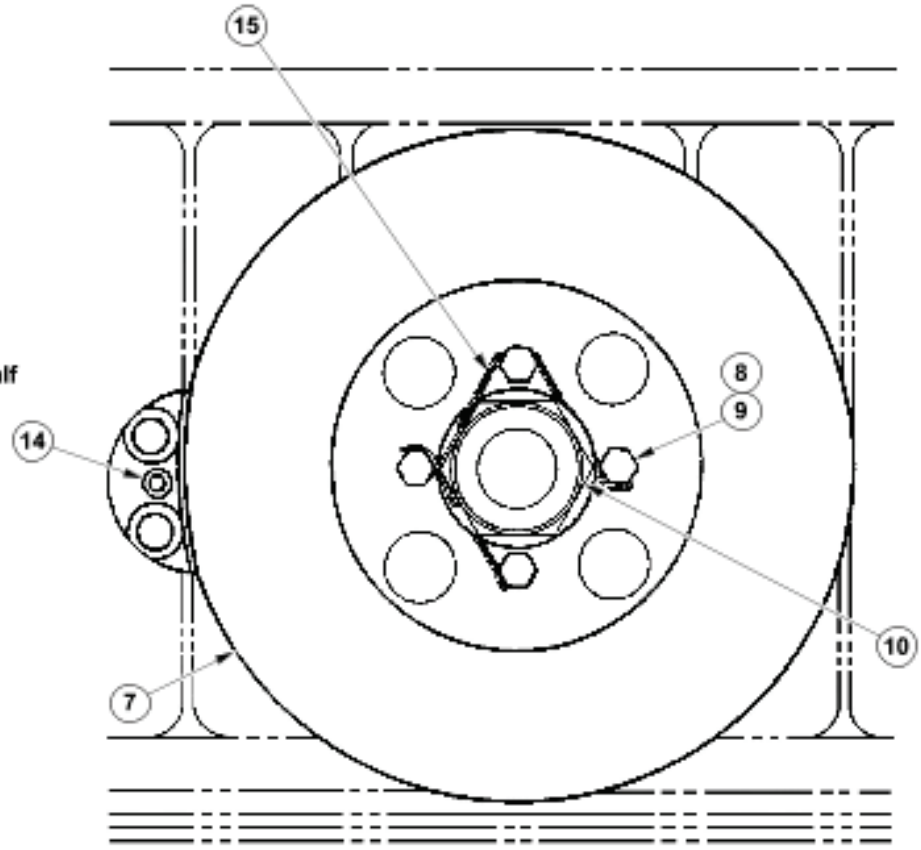
Figure 1. Rotor Brake Assembly - Damage Limits (Sheet 2 of 2)



ICN-412-A-632300-B-97499-00003-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 1 of 3)

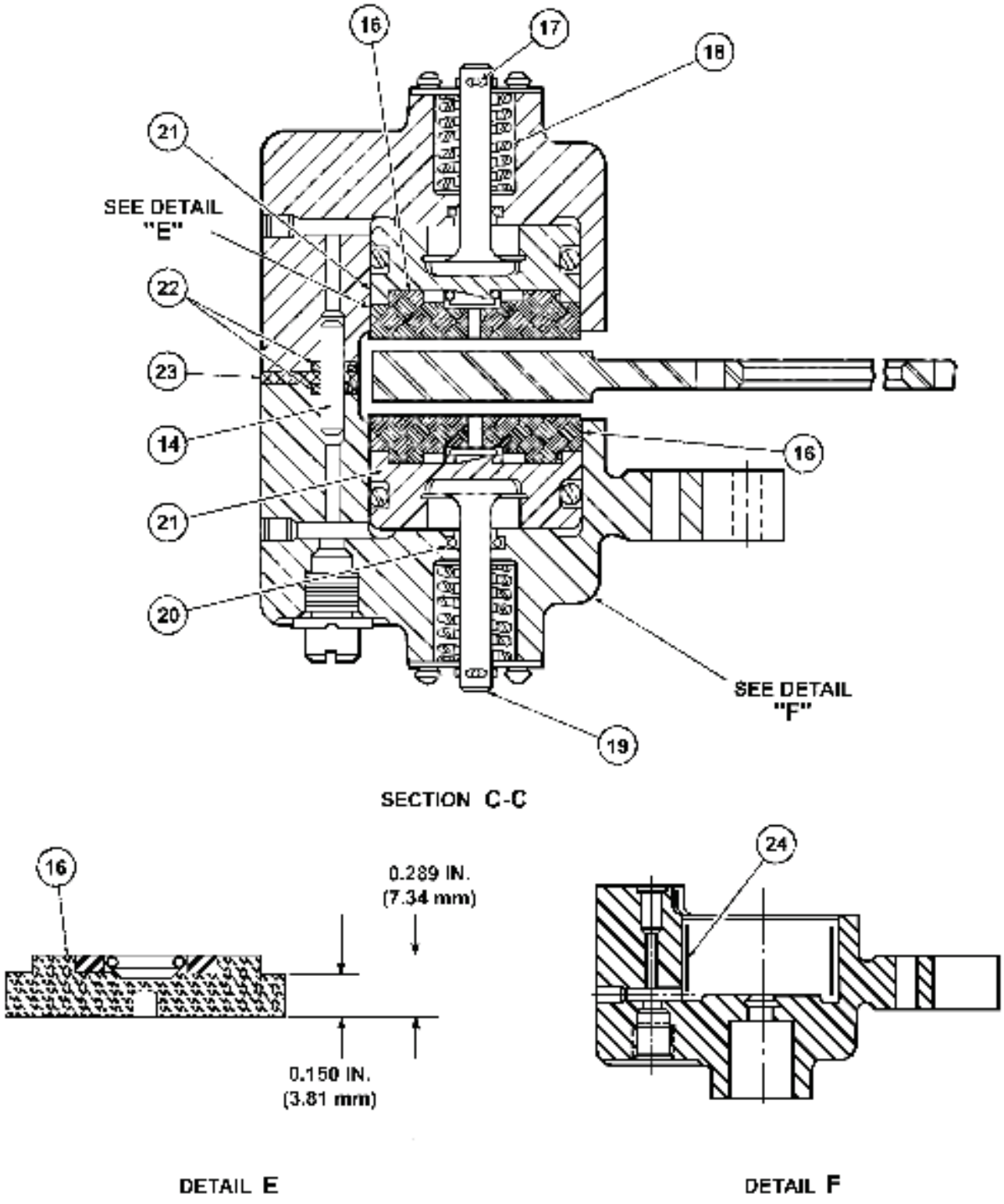
1. Quill assembly
2. Bolt
3. Washer
4. Lockwire
5. Inboard brake assembly half
6. Flange adapter
7. Brake disc
8. Bolt
9. Washer
10. Retaining nut
11. Outboard brake assembly half
12. Bolt
13. Washer
14. Oil transfer tube
15. Lockwire
16. Brake lining
17. Cotter pin
18. Return spring
19. Piston return pin
20. Packing
21. Piston
22. Packing
23. Spacer
24. Piston cavity



VIEW B-B

ICN-412-A-632300-B-97499-00004-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 2 of 3)



ICN-412-A-632300-B-97499-00005-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 3 of 3)

Brake Assembly (P/N 9450103) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

Ensure that microswitch is adjusted so that ROTOR BRAKE light illuminates when brake piston has moved 0.030 inch (0.76 mm) from the original position.

1. Install microswitch (8, Figure 1) on inboard brake assembly half (5, Figure 2) using screws (9, Figure 1) and washers (10).

2. Secure with lockwire (C-405).

3. Install inboard brake assembly half (5, Figure 2) on rotor brake quill assembly (1) using bolts (2) and washers (3).

4. Torque bolts (2) to 80 to 100 inch-pounds (9.04 to 11.30 Nm) and secure with lockwire (C-405).

5. Install rotor brake disc (7) (412-A-63-23-00-04A-720A-A / 01222).

6. Install packings (22) in counterbored cavities in mating surfaces of inboard brake assembly half (5) and outboard brake assembly half (11).

7. Insert brake oil transfer tube (14) through packing (22) in inboard brake assembly half (5) until it bottoms out in hole in inboard brake assembly half (5).

NOTE

Ensure that microswitch is adjusted so that ROTOR BRAKE light illuminates when brake piston has moved 0.030 inch (0.76 mm) from original position.

8. Install microswitch (8, Figure 1) on outboard brake assembly half (11, Figure 2) using screws (9, Figure 1) and washers (10).

9. Secure with lockwire (C-405).

10. Place brake assembly spacer (23, Figure 2) over protruding oil transfer tube (14) in inboard brake assembly half (5).

11. Ensure holes for bolts (12) in brake assembly spacer (23) align with corresponding holes in inboard brake assembly half (5).

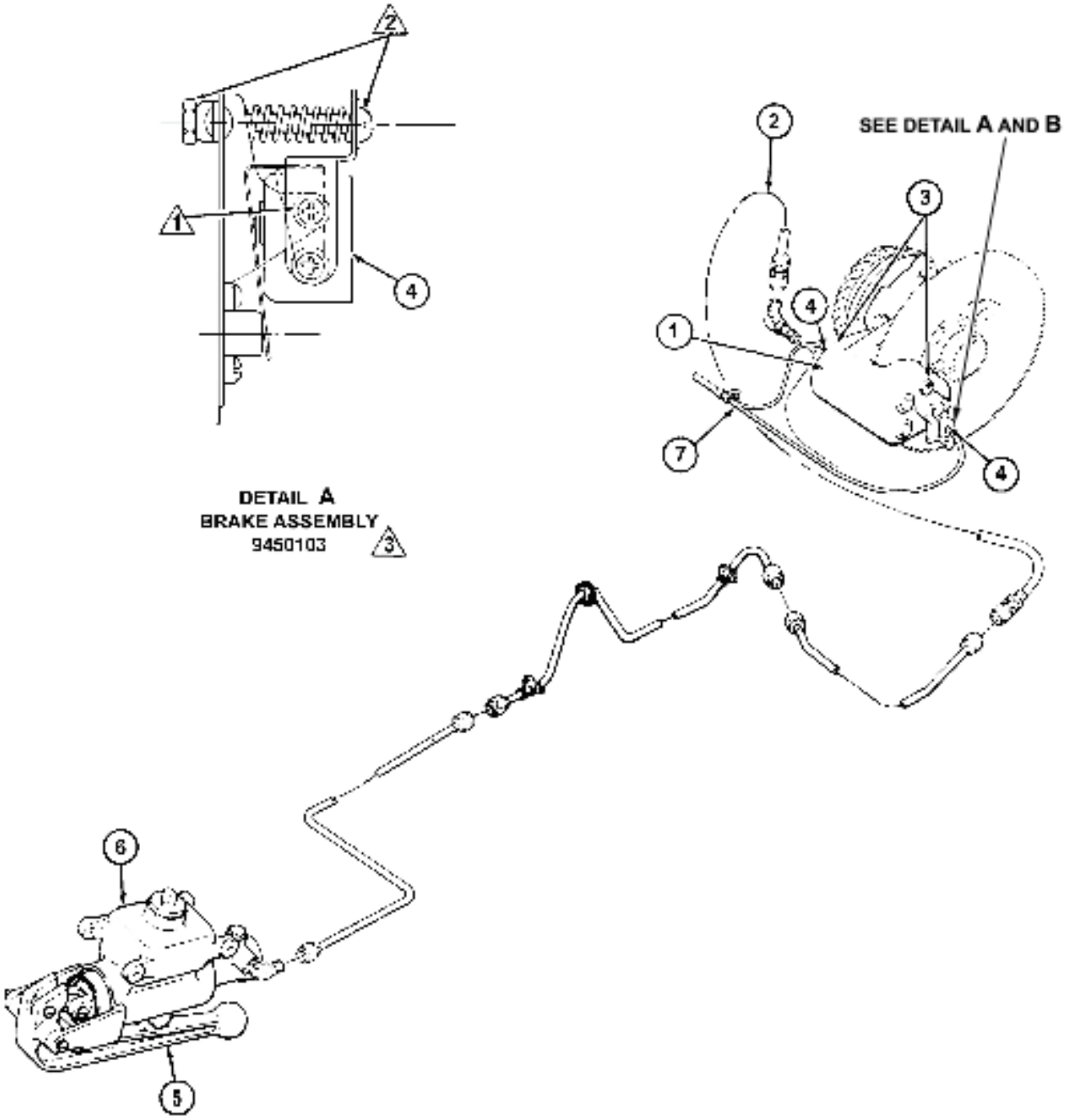
12. Install outboard brake assembly half (11) on inboard brake assembly half (5) using bolts (12) and washers (13).

13. Torque bolts (12) to 120 to 130 inch-pounds (13.56 to 14.69 Nm) and secure with lockwire (C-405).

14. Install hydraulic hose (2, Figure 1) on rotor brake assembly (1).

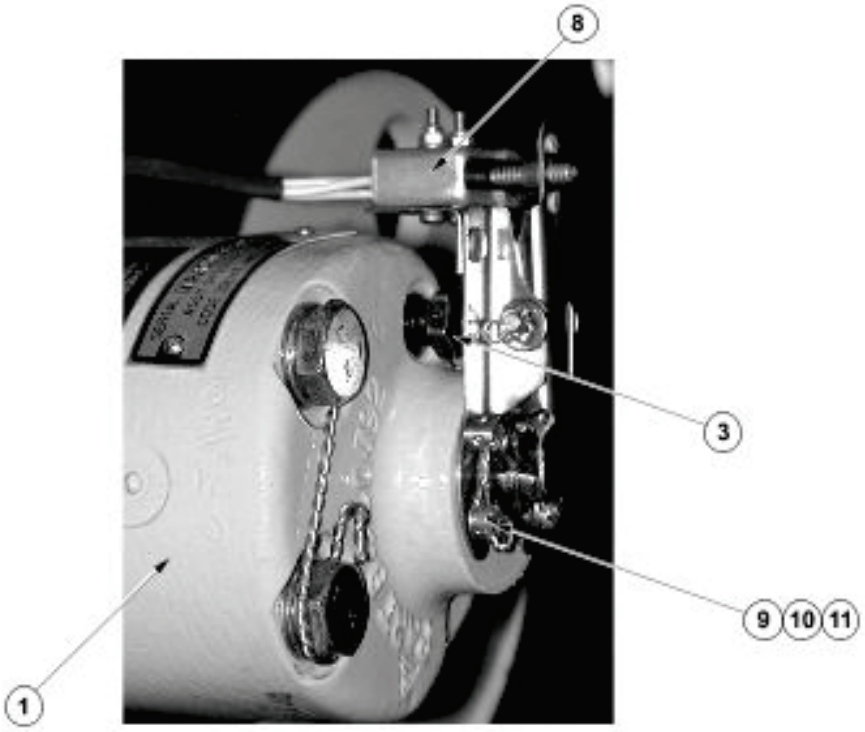
15. Bleed system (412-A-63-23-00-01A-231A-A / 01207).

16. Check for proper system operation including illumination of ROTOR BRAKE light.



ICN-412-A-632300-B-97499-00001-A-001-01

Figure 1. Rotor Brake System (Sheet 1 of 2)



DETAIL B
BRAKE ASSEMBLY (9450103-1)

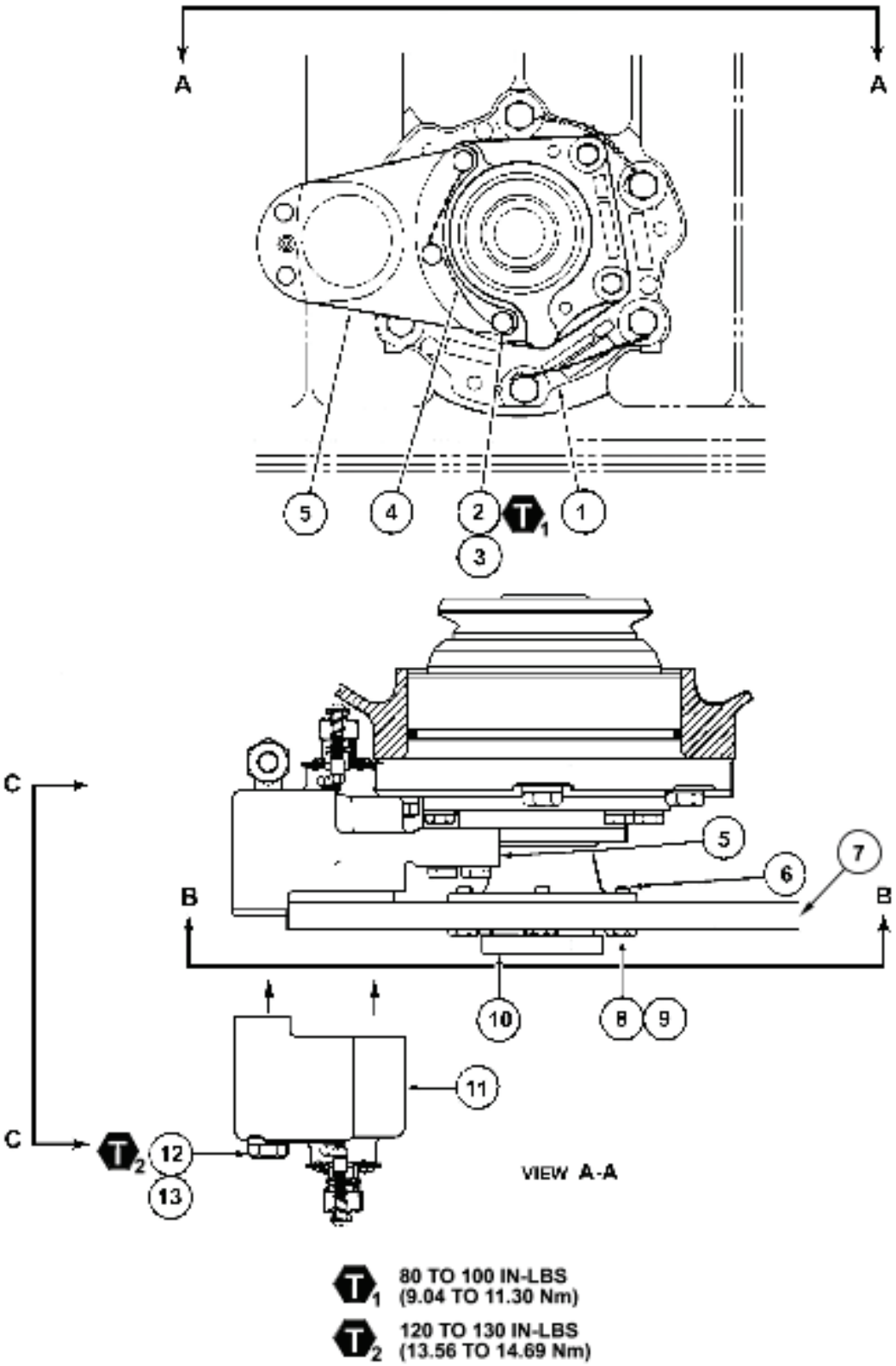
1. Rotor brake assembly
2. Hydraulic line
3. Bleed valve
4. Microswitch
5. Handle
6. Master cylinder
7. Microswitch electrical harness
8. Microswitch
9. Screws
10. Washers
11. Lockwire

NOTES

1. Loosen screw while making adjustment.
2. Loosen jam nut, turn screw to adjust switch, tighten jam nut.
3. Do not mix brake assembly 9450103-1 and 9450103 in any installation.

ICN-412-A-632300-B-97499-00002-A-001-01

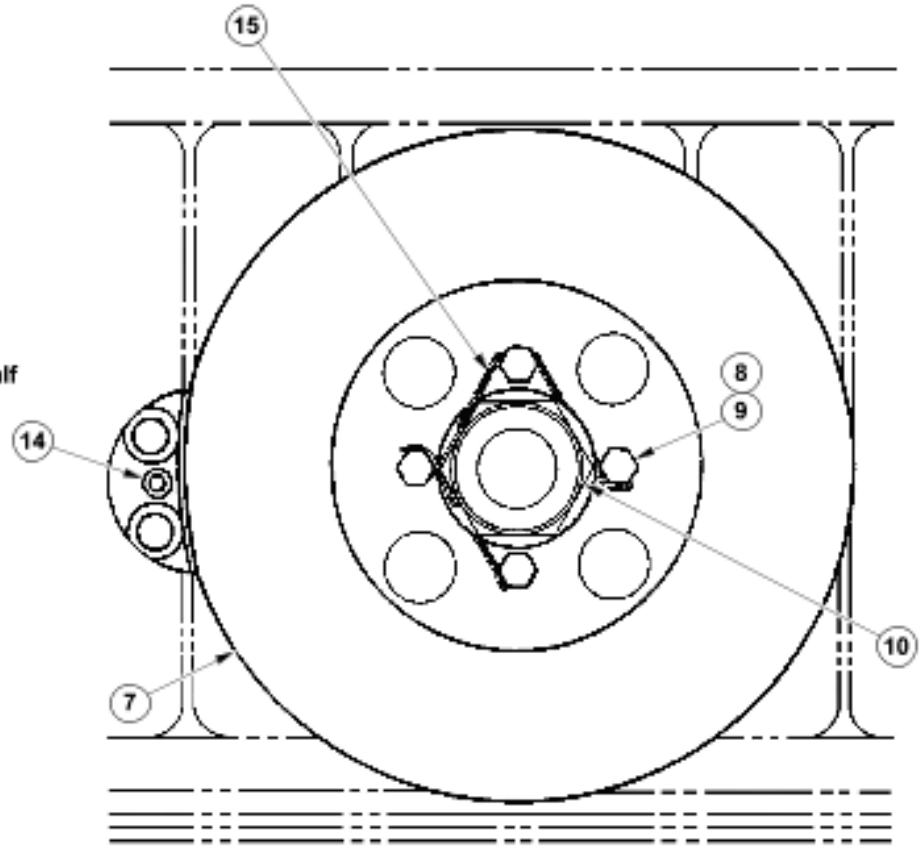
Figure 1. Rotor Brake System (Sheet 2 of 2)



ICN-412-A-632300-B-97499-00003-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 1 of 3)

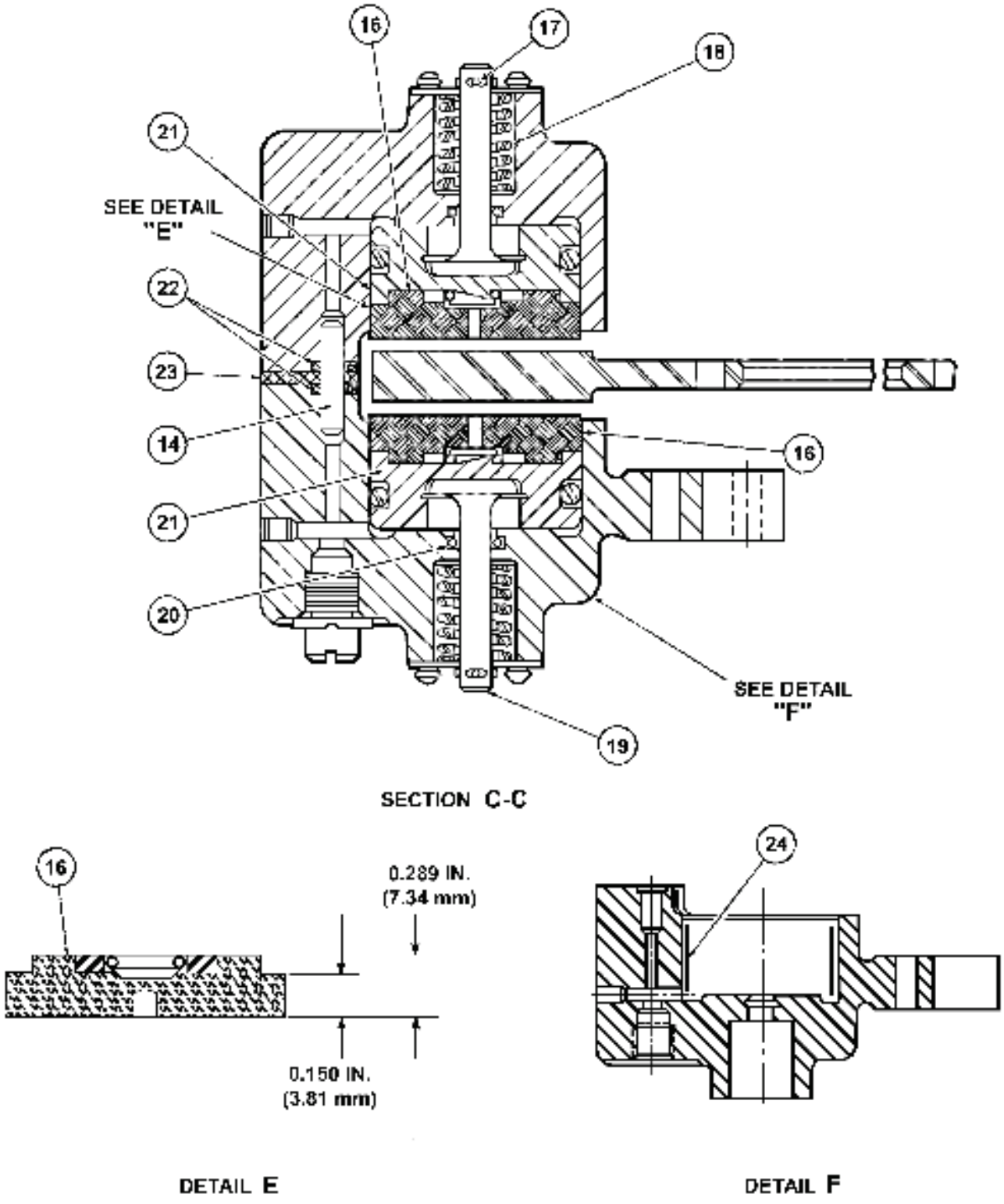
1. Quill assembly
2. Bolt
3. Washer
4. Lockwire
5. Inboard brake assembly half
6. Flange adapter
7. Brake disc
8. Bolt
9. Washer
10. Retaining nut
11. Outboard brake assembly half
12. Bolt
13. Washer
14. Oil transfer tube
15. Lockwire
16. Brake lining
17. Cotter pin
18. Return spring
19. Piston return pin
20. Packing
21. Piston
22. Packing
23. Spacer
24. Piston cavity



VIEW B-B

ICN-412-A-632300-B-97499-00004-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 2 of 3)



ICN-412-A-632300-B-97499-00005-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 3 of 3)

Brake Assembly (P/N 9450103-1) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

Ensure that microswitch is adjusted so that ROTOR BRAKE light illuminates when brake piston has moved 0.030 inch (0.76 mm) from the original position.

1. Install microswitch (8, Figure 1) on inboard brake assembly half (5, Figure 2) using screws (9, Figure 1) and washers (10).

2. Secure with lockwire (C-405).

3. Install inboard brake assembly half (5, Figure 2) on rotor brake quill assembly (1) using bolts (2) and washers (3).

4. Torque bolts (2) to 80 to 100 inch-pounds (9.04 to 11.30 Nm) and secure with lockwire (C-405).

5. Install rotor brake disc (7) (412-A-63-23-00-04A-720A-A / 01222).

6. Install packings (22) in counterbored cavities in mating surfaces of inboard brake assembly half (5) and outboard brake assembly half (11).

7. Insert brake oil transfer tube (14) through packing (22) in inboard brake assembly half (5) until it bottoms out in hole in inboard brake assembly half (5).

NOTE

Ensure that microswitch is adjusted so that ROTOR BRAKE light illuminates when brake piston has moved 0.030 inch (0.76 mm) from original position.

8. Install microswitch (8, Figure 1) on outboard brake assembly half (11, Figure 2) using screws (9, Figure 1) and washers (10).

9. Secure with lockwire (C-405).

10. Place brake assembly spacer (23, Figure 2) over protruding oil transfer tube (14) in inboard brake assembly half (5).

11. Ensure holes for bolts (12) in brake assembly spacer (23) align with corresponding holes in inboard brake assembly half (5).

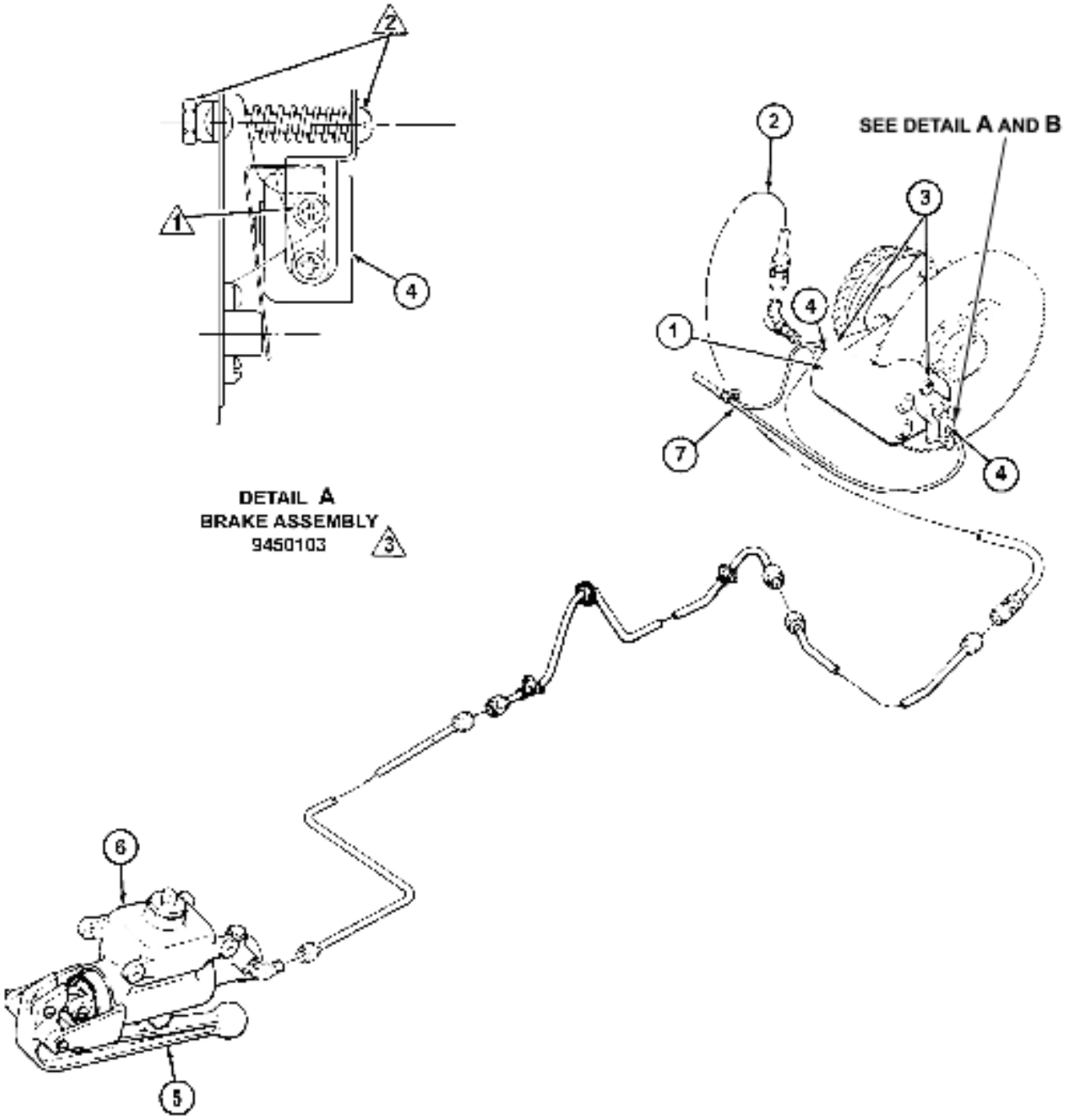
12. Install outboard brake assembly half (11) on inboard brake assembly half (5) using bolts (12) and washers (13).

13. Torque bolts (12) to 120 to 130 inch-pounds (13.56 to 14.69 Nm) and secure with lockwire (C-405).

14. Install hydraulic hose (2, Figure 1) on rotor brake assembly (1).

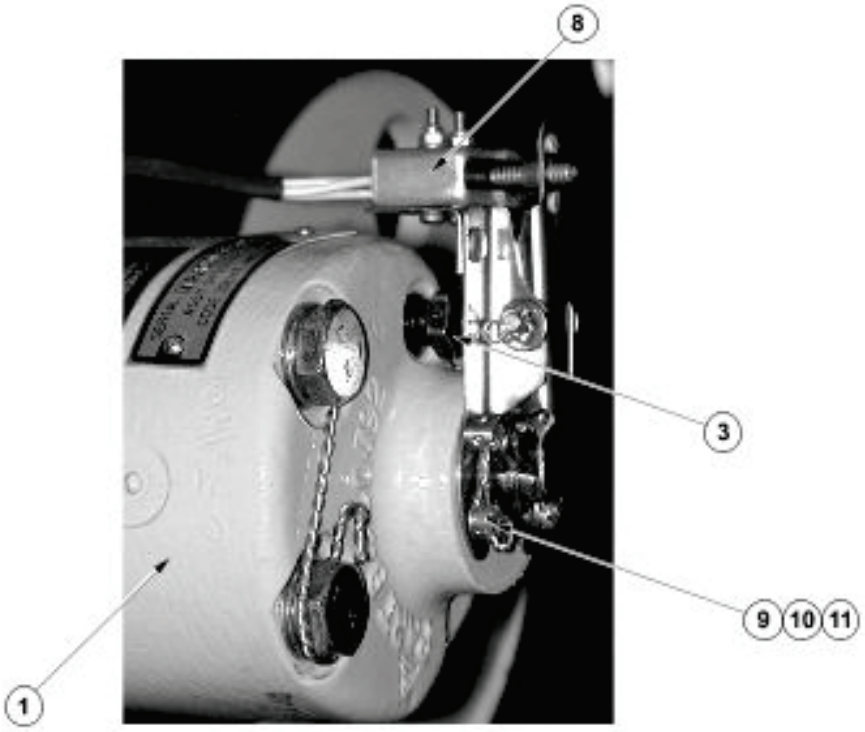
15. Bleed system (412-A-63-23-00-01A-231B-A / 01208).

16. Check for proper system operation including illumination of ROTOR BRAKE light.



ICN-412-A-632300-B-97499-00001-A-001-01

Figure 1. Rotor Brake System (Sheet 1 of 2)



DETAIL B
BRAKE ASSEMBLY (9450103-1)

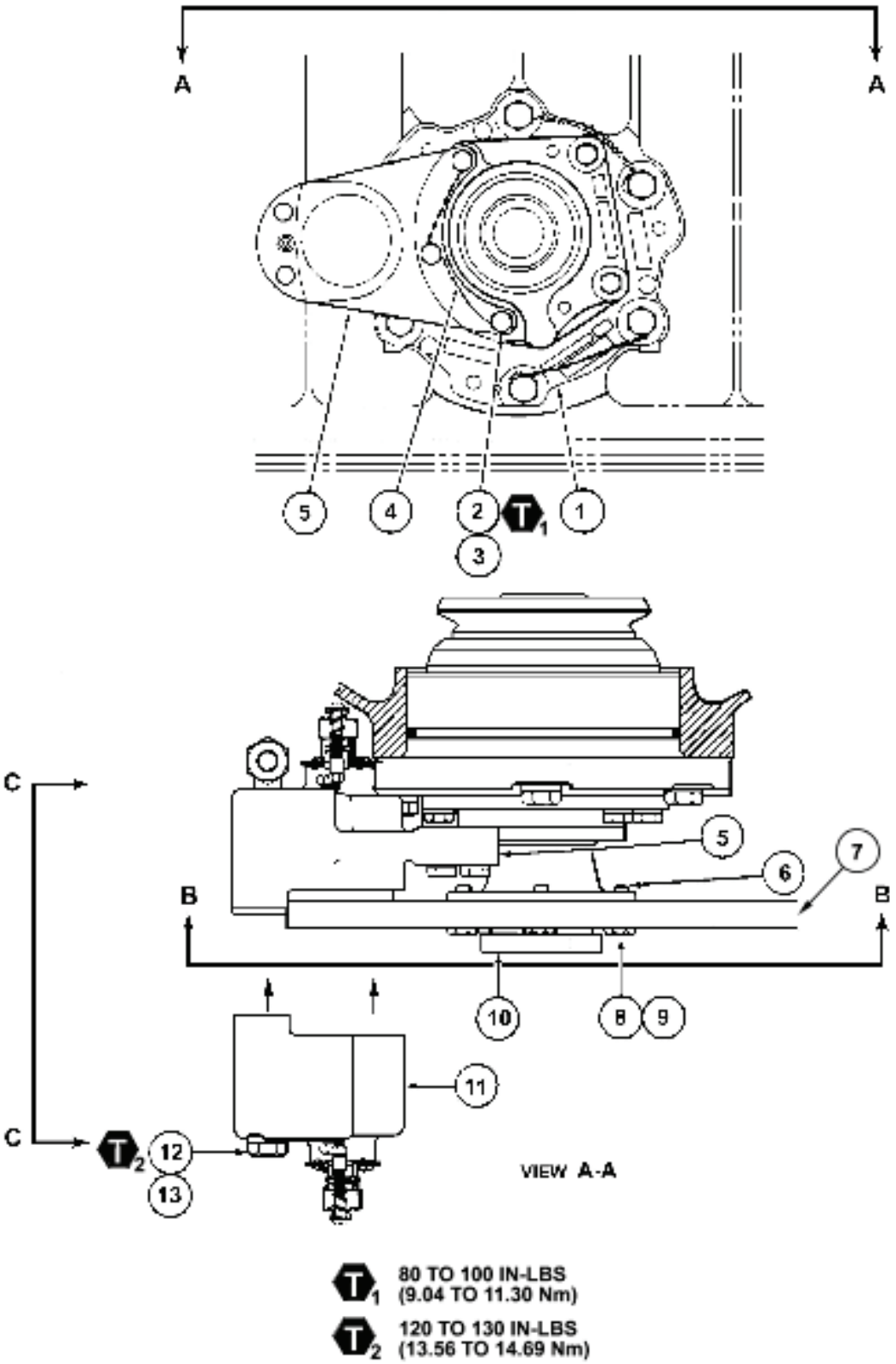
1. Rotor brake assembly
2. Hydraulic line
3. Bleed valve
4. Microswitch
5. Handle
6. Master cylinder
7. Microswitch electrical harness
8. Microswitch
9. Screws
10. Washers
11. Lockwire

NOTES

1. Loosen screw while making adjustment.
2. Loosen jam nut, turn screw to adjust switch, tighten jam nut.
3. Do not mix brake assembly 9450103-1 and 9450103 in any installation.

ICN-412-A-632300-B-97499-00002-A-001-01

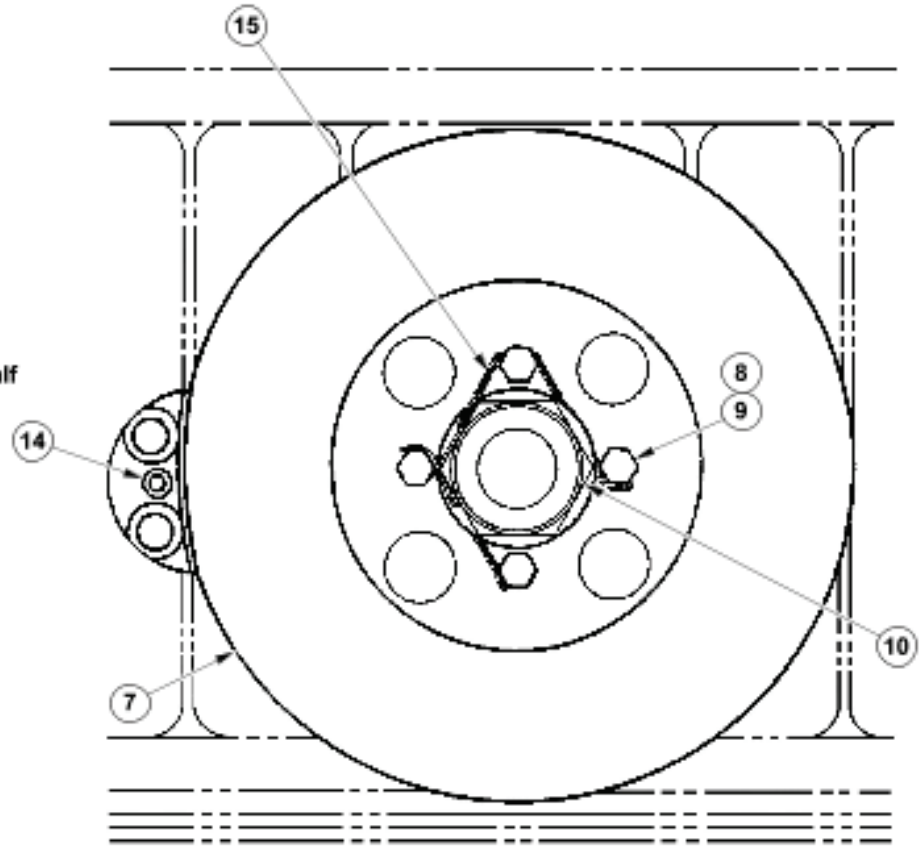
Figure 1. Rotor Brake System (Sheet 2 of 2)



ICN-412-A-632300-B-97499-00003-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 1 of 3)

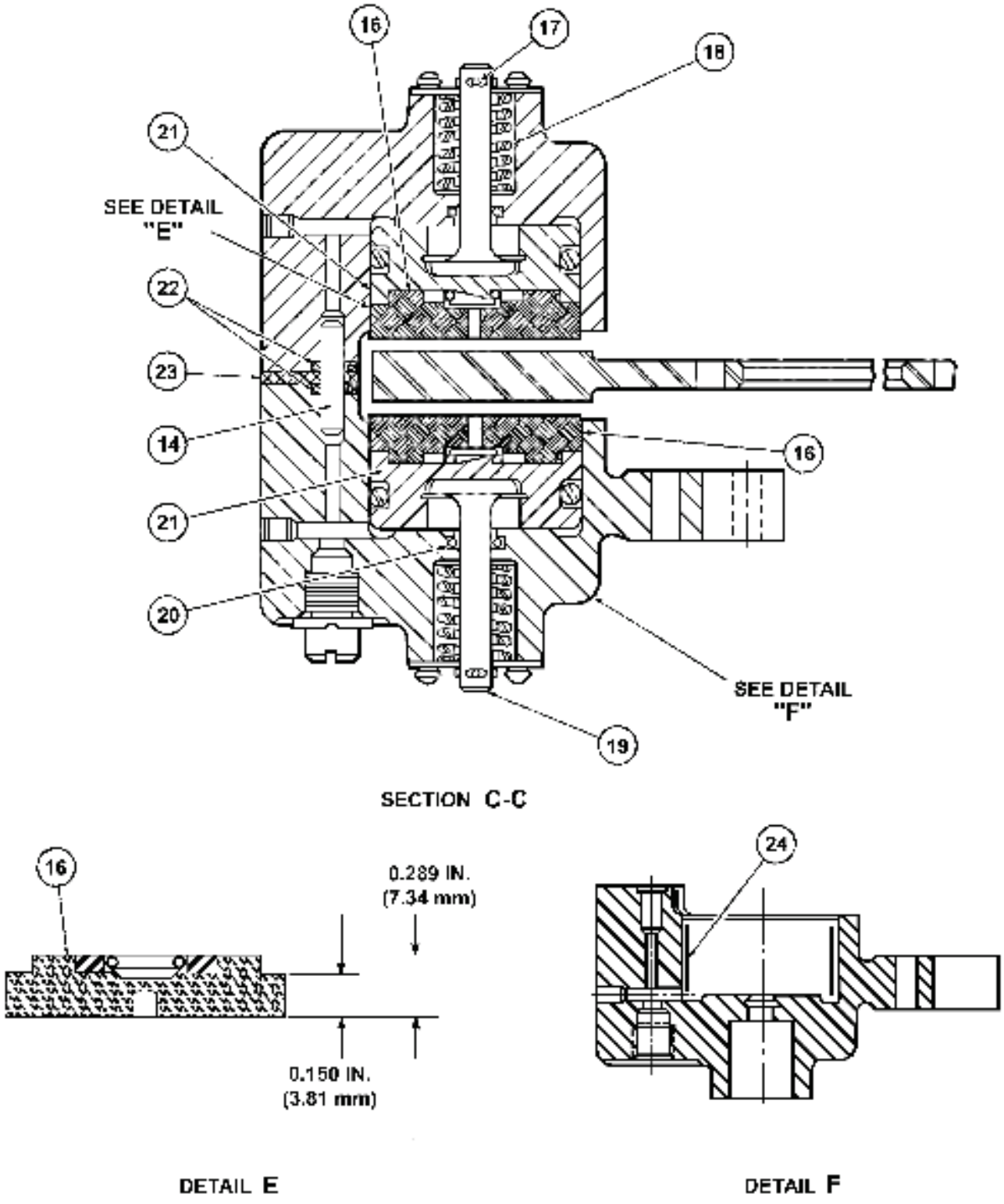
1. Quill assembly
2. Bolt
3. Washer
4. Lockwire
5. Inboard brake assembly half
6. Flange adapter
7. Brake disc
8. Bolt
9. Washer
10. Retaining nut
11. Outboard brake assembly half
12. Bolt
13. Washer
14. Oil transfer tube
15. Lockwire
16. Brake lining
17. Cotter pin
18. Return spring
19. Piston return pin
20. Packing
21. Piston
22. Packing
23. Spacer
24. Piston cavity



VIEW B-B

ICN-412-A-632300-B-97499-00004-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 2 of 3)



ICN-412-A-632300-B-97499-00005-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 3 of 3)

Rotor Brake Disc - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Jackscrew Set	T101308	1	
Brake Disc Holding Tool	T101649-101	1	

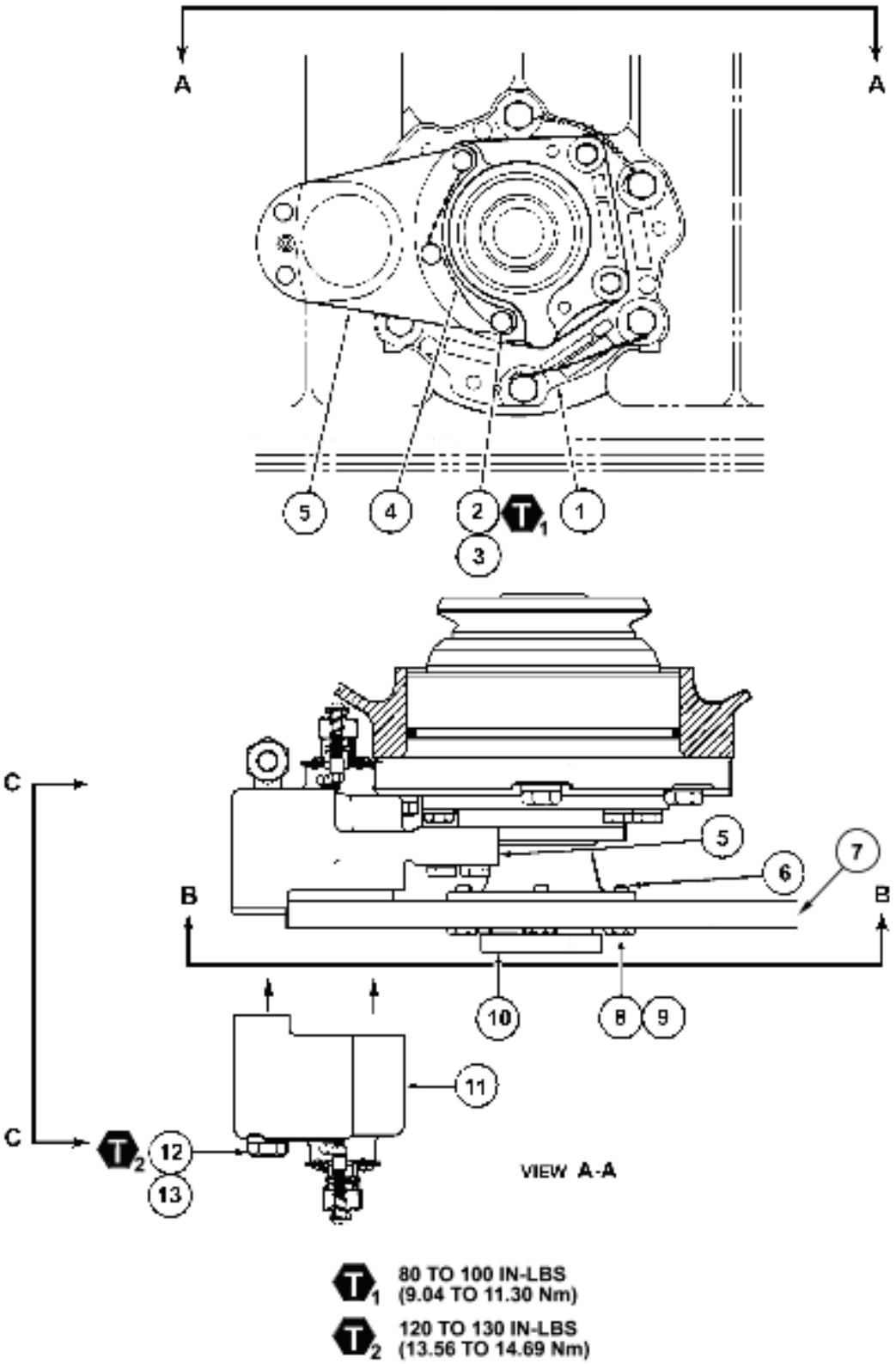
Procedure

1. Open the left side engine and transmission cowling to gain access to the transmission.
2. Remove the outboard brake assembly half (11, Figure 1) (412-A-63-23-00-02A-520A-A / 01209).
3. Cut the lockwire and remove the retaining nut (5, Figure 2) using the brake disc holding tool (T101649-101) (3) tool to restrain the brake disc (2).
4. Remove and discard the packing (4).



THE ROTOR BRAKE DISC AND FLANGE ASSEMBLY SHALL BE REMOVED AS AN ASSEMBLY. DO NOT ATTEMPT TO REMOVE THE DISC FROM THE FLANGE ADAPTER WHILE THE ASSEMBLY IS INSTALLED ON THE QUILL. DO NOT SEPARATE THE DISC FROM THE FLANGE IF THE DISC IS NOT BEING REPLACED.

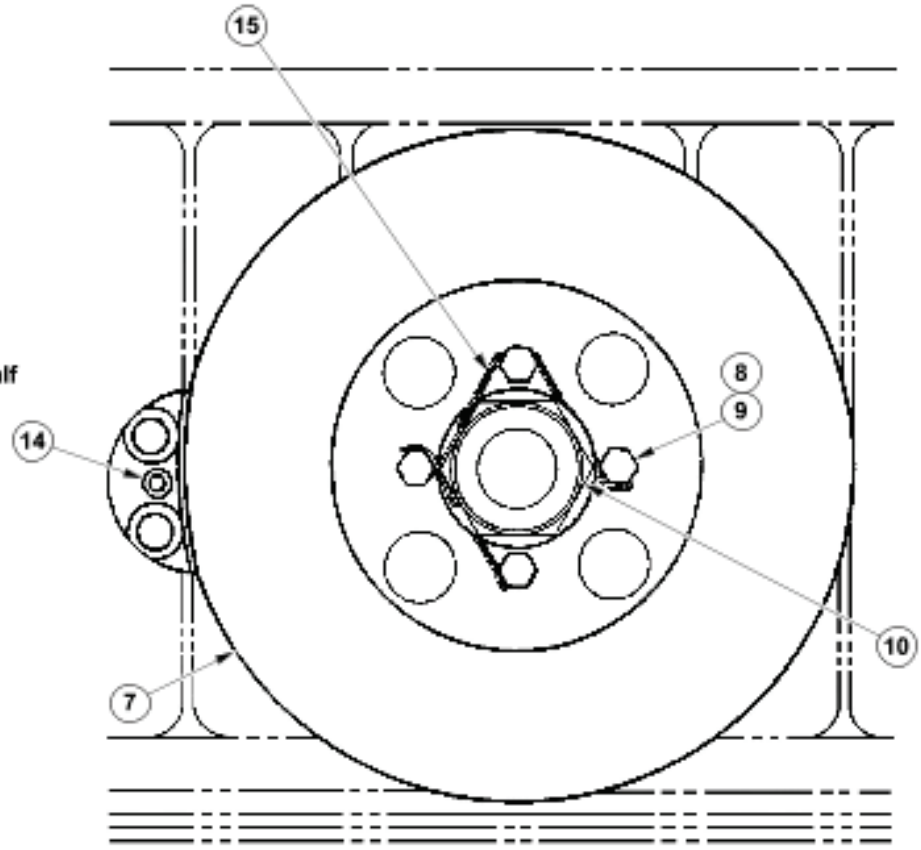
5. Remove the disc and the flange adapter from the quill assembly (1, Figure 1).
6. Remove the rotor brake disc (2, Figure 2) from the flange adapter (1) (412-A-63-23-00-05A-520A-A / 01223 and 412-A-63-23-00-05A-720A-A / 01224).



ICN-412-A-632300-B-97499-00003-A-001-01

Figure 1. Rotor Brake Assembly (Sheet 1 of 3)

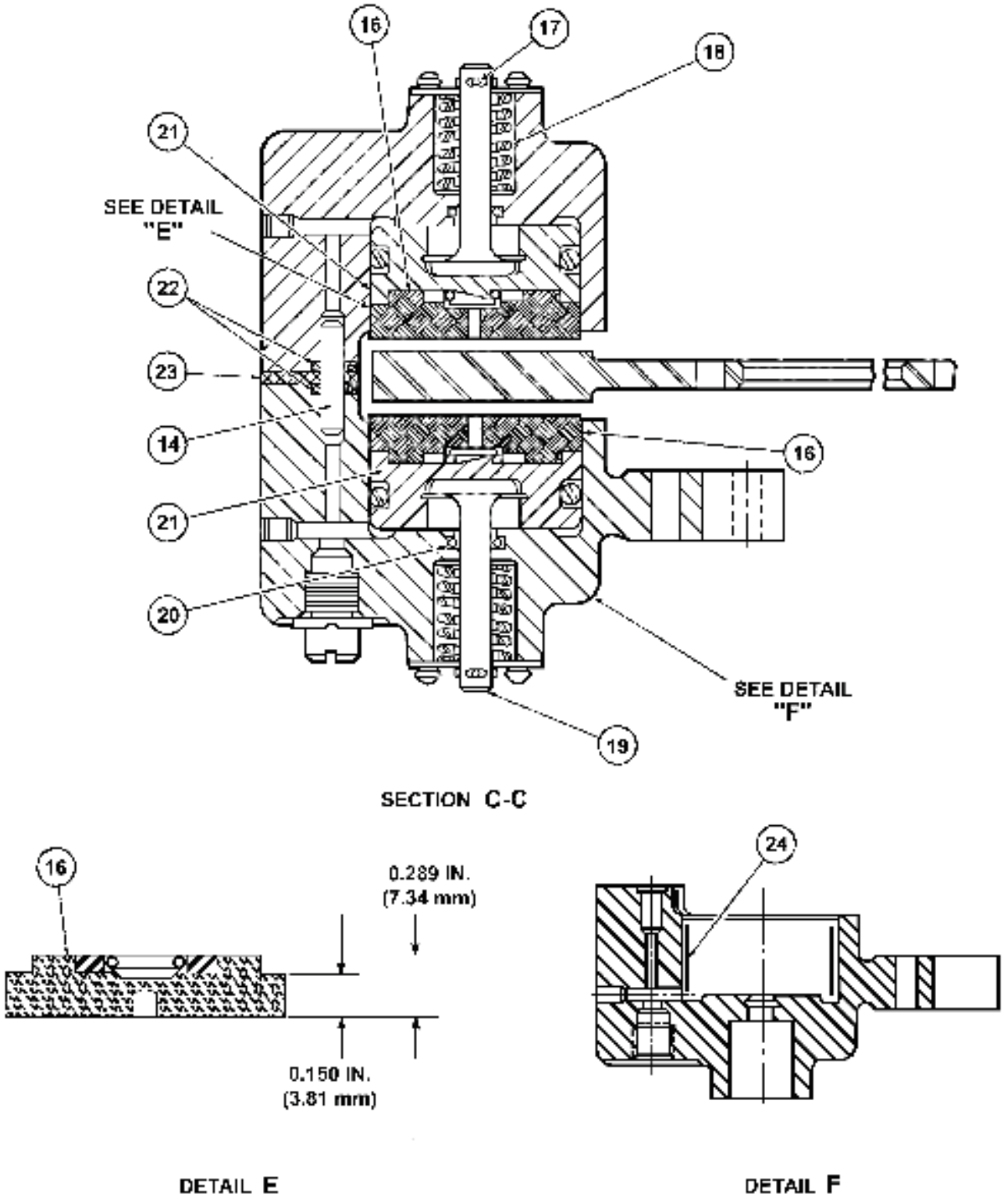
1. Quill assembly
2. Bolt
3. Washer
4. Lockwire
5. Inboard brake assembly half
6. Flange adapter
7. Brake disc
8. Bolt
9. Washer
10. Retaining nut
11. Outboard brake assembly half
12. Bolt
13. Washer
14. Oil transfer tube
15. Lockwire
16. Brake lining
17. Cotter pin
18. Return spring
19. Piston return pin
20. Packing
21. Piston
22. Packing
23. Spacer
24. Piston cavity



VIEW B-B

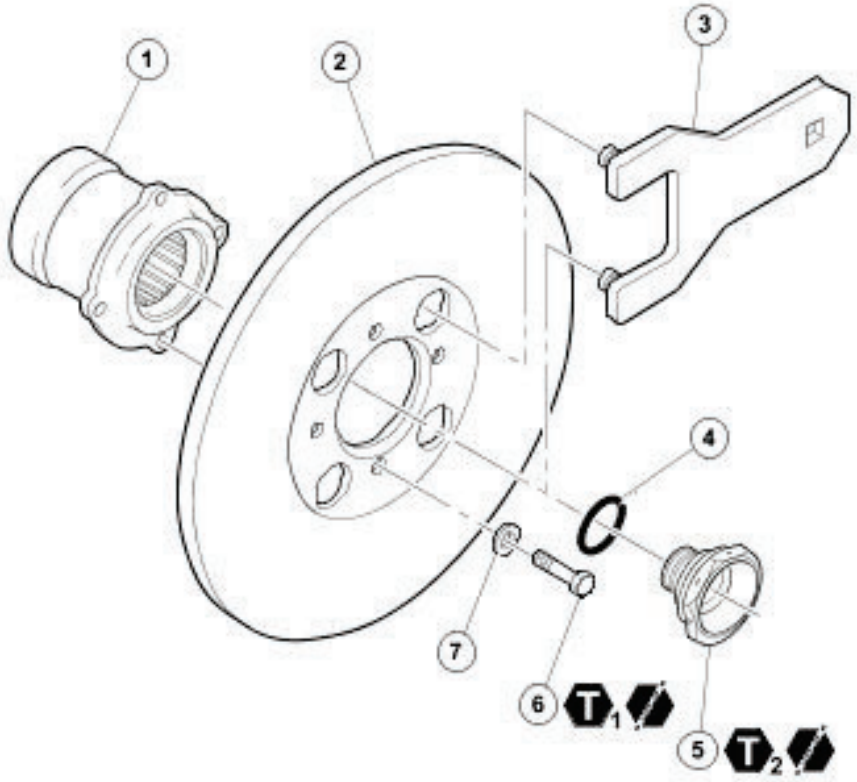
ICN-412-A-632300-B-97499-00004-A-001-01

Figure 1. Rotor Brake Assembly (Sheet 2 of 3)



ICN-412-A-632300-B-97499-00005-A-001-01

Figure 1. Rotor Brake Assembly (Sheet 3 of 3)



1. Flange adapter
2. Brake disc
3. Brake disc holding tool (T101649-101)
4. Packing
5. Retaining nut
6. Bolt
7. Washer

LOCKWIRE (C-405)

T1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T2 125 TO 150 FT-LBS (169.5 TO 203.4 Nm)

ICN-412-A-632300-B-97499-00008-A-001-01

Figure 2. Rotor Brake Disc (Sheet 1 of 1)

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Rotor Brake Disc - Inspection

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Bolt	AN4-5A	1	
Washer	AN960-416	1	
Dial Indicator	Commercial	1	
Lathe	Commercial	1	

Procedure

NOTE

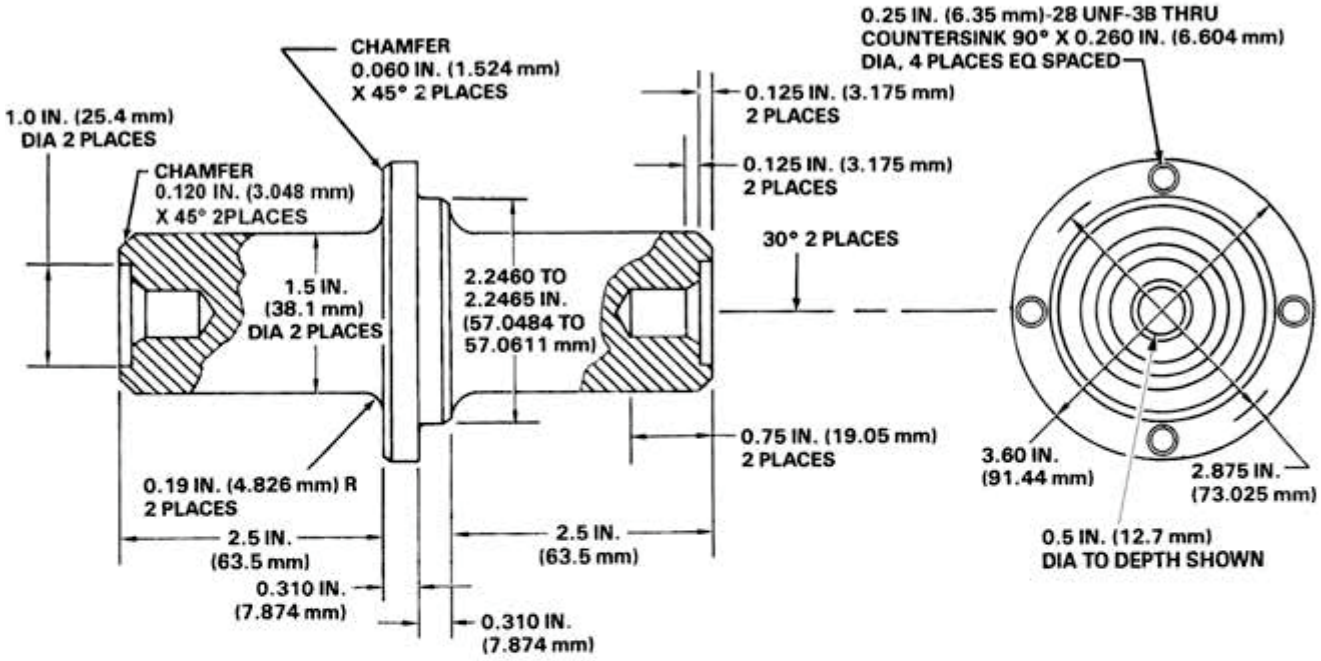
If the rotor brake disc has been removed using an arbor press (or equivalent), or warpage is suspected, the following disc inspection is required.

1. Install the inspection arbor (Figure 1) in the pilot hole of the disc. Install four bolt (AN4-5A) and washer (AN960-416).

1.1. Install the arbor and disc assembly between the lathe centers.

1.2. While rotating the disc, measure the runout as the indicator probe is swept across the full radial width of each friction surface (front and back).

1.3. Total indicator deflection shall not exceed 0.010 inch (0.254 mm).



NOTES

1. Material: AISI 4340 steel per MIL-S-5000. Heat treat to R_C 36-40 in accordance with MIL-H-6875.
2. Surface finish 125 microinches unless otherwise noted.
3. Break edges 0.015 in. (0.381 mm) radius or 0.015 in. (0.381 mm) x 40 to 50° chamfer unless otherwise noted.

ICN-412-A-632300-B-97499-00009-A-002-01

Figure 1. Rotor Brake Disc Inspection Arbor (Sheet 1 of 1)

Rotor Brake Disc - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Brake Disc Holding Tool	T101649-101	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. If not previously accomplished, install the brake disc (2, Figure 1) on the flange adapter (1) (412-A-63-23-00-05A-720A-A / 01224).

2. Apply a light coat of transmission oil (412-A-12-01-00-00A-040A-A / 00141) to the splines in the rotor brake quill assembly (1, Figure 2) and the mating splines on the flange adapter (6).

3. Install the brake disc (7) and the flange adapter (6) and install the assembly on the rotor brake quill assembly (1).

4. Lubricate a new packing (4, Figure 1) and install on the retaining nut (5).

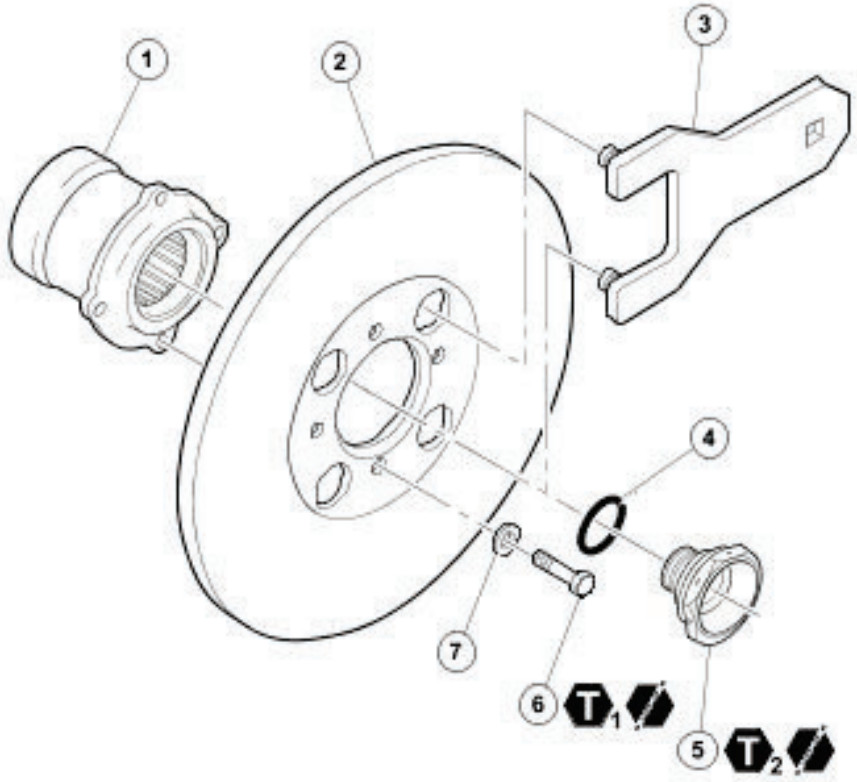
5. Install the retaining nut (5) in the center of the brake disc (2) and the flange adapter (1). Using the brake disc holding tool (T101649-101) (3) to hold the disc, tighten the retaining nut 125 to 150 foot-pounds (169.5 to 203.4 Nm) and secure the retaining nut with lockwire (C-405).

NOTE

Do a torque check to make sure a torque of 50 to 70 inch-pounds (5.65 to 7.91 Nm) has been applied to the bolts (6).

6. Secure the bolts (8, Figure 2, View B-B) to the retaining nut (10) with lockwire (C-405).

7. Install the outboard brake assembly half (412-A-63-23-00-03A-042A-A / 01213).



1. Flange adapter
2. Brake disc
3. Brake disc holding tool (T101649-101)
4. Packing
5. Retaining nut
6. Bolt
7. Washer

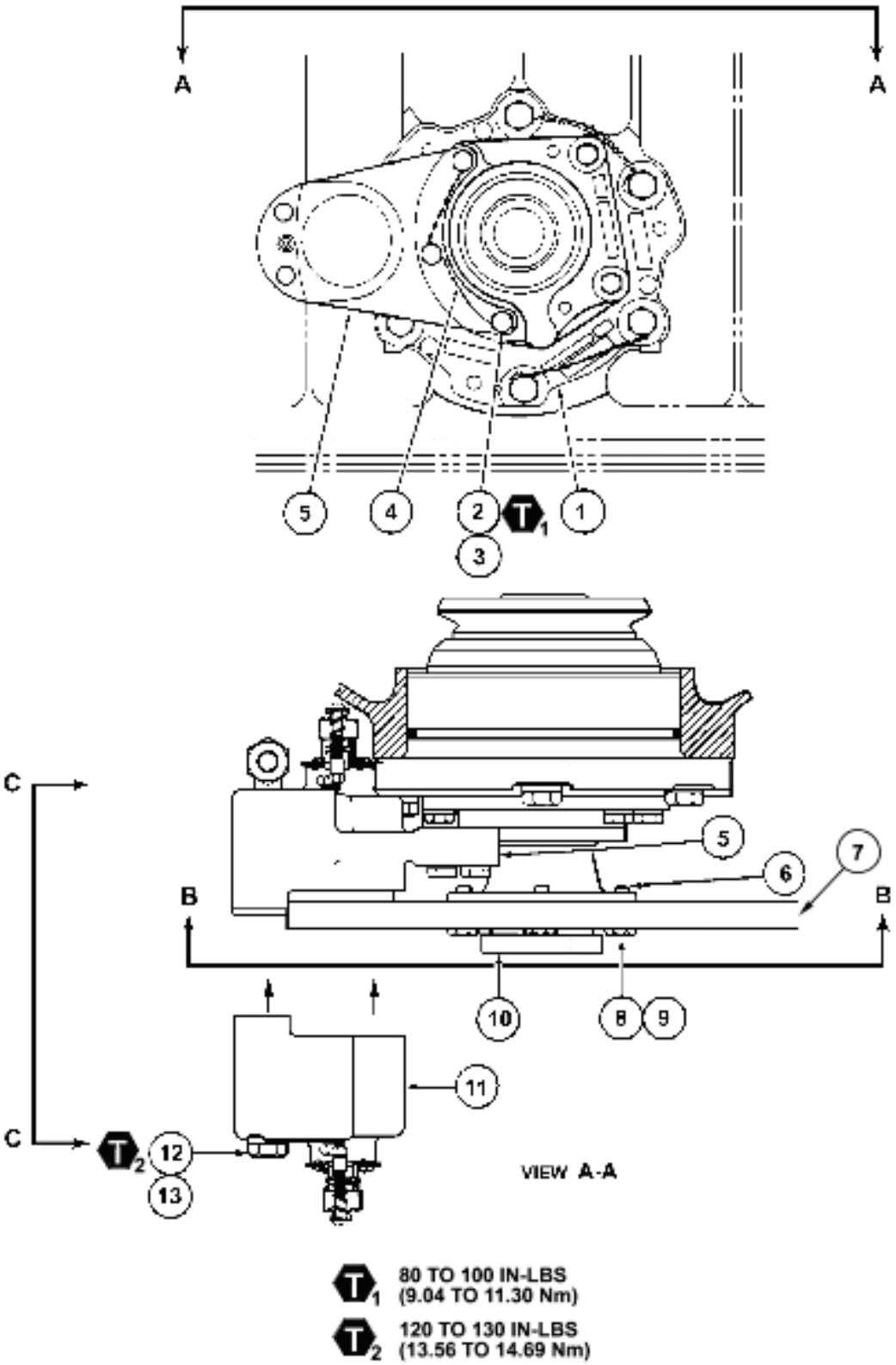
LOCKWIRE (C-405)

T1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T2 125 TO 150 FT-LBS (169.5 TO 203.4 Nm)

ICN-412-A-632300-B-97499-00008-A-001-01

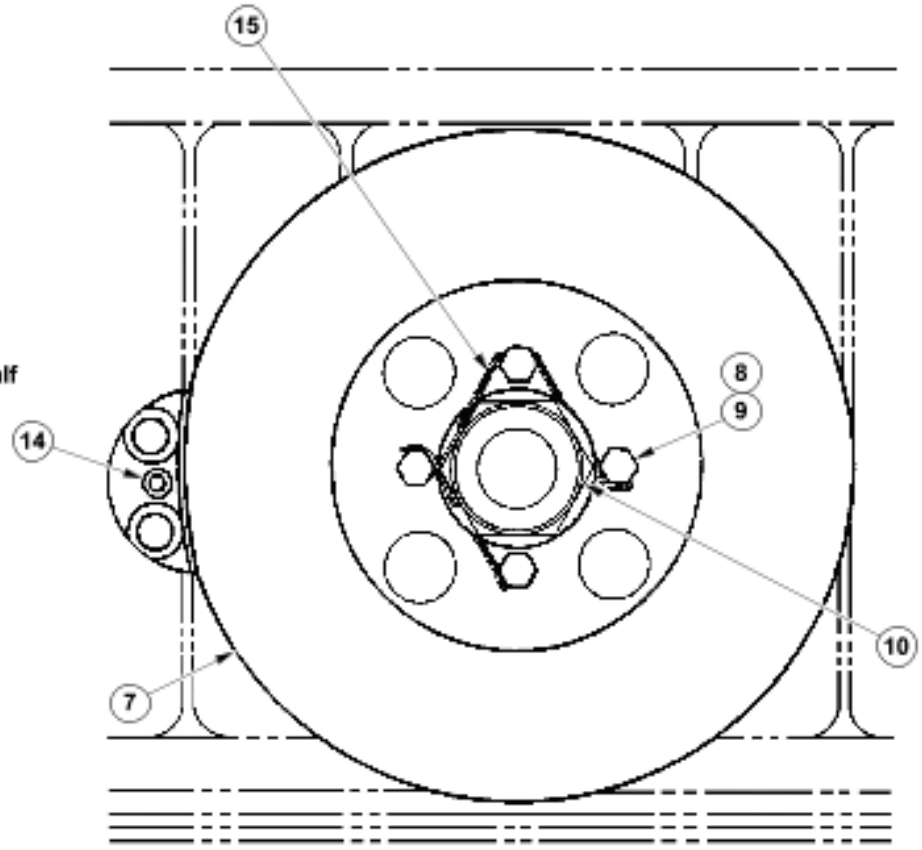
Figure 1. Rotor Brake Disc (Sheet 1 of 1)



ICN-412-A-632300-B-97499-00003-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 1 of 3)

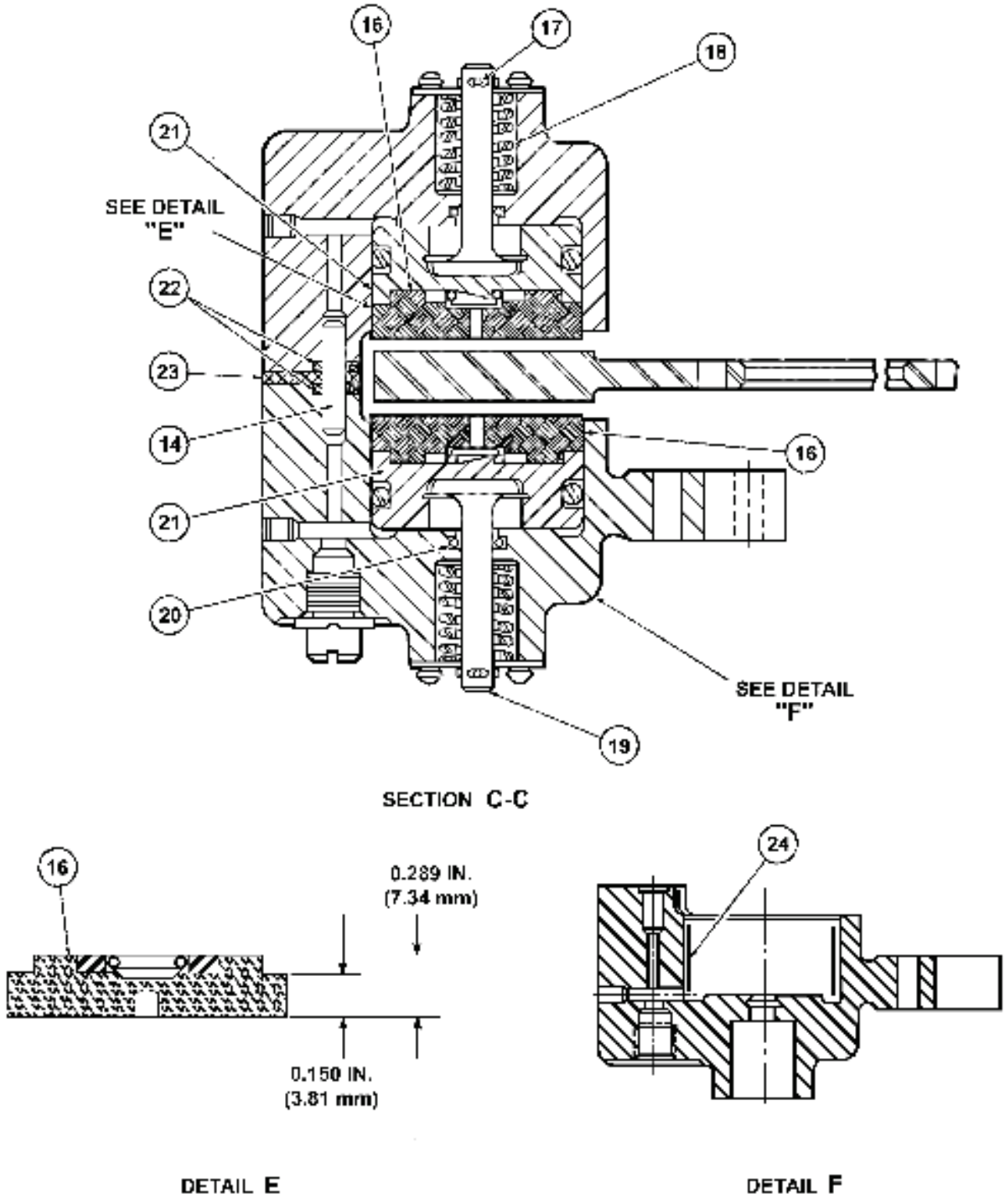
1. Quill assembly
2. Bolt
3. Washer
4. Lockwire
5. Inboard brake assembly half
6. Flange adapter
7. Brake disc
8. Bolt
9. Washer
10. Retaining nut
11. Outboard brake assembly half
12. Bolt
13. Washer
14. Oil transfer tube
15. Lockwire
16. Brake lining
17. Cotter pin
18. Return spring
19. Piston return pin
20. Packing
21. Piston
22. Packing
23. Spacer
24. Piston cavity



VIEW B-B

ICN-412-A-632300-B-97499-00004-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 2 of 3)



ICN-412-A-632300-B-97499-00005-A-001-01

Figure 2. Rotor Brake Assembly (Sheet 3 of 3)

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Flange Adapter - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Brake Disc Holding Tool	T101649-101	1	

Procedure

1. Cut the lockwire. Using the brake disc holding tool (T101649-101) (3, Figure 1), restrain the disc (2) and remove the bolts (6) and the washers (7).

2. Using hand pressure only, separate the flange adapter (1) from the brake disc (2).

3. If the brake disc (2) and the flange adapter (1) cannot be separated by hand pressure, proceed as follows:

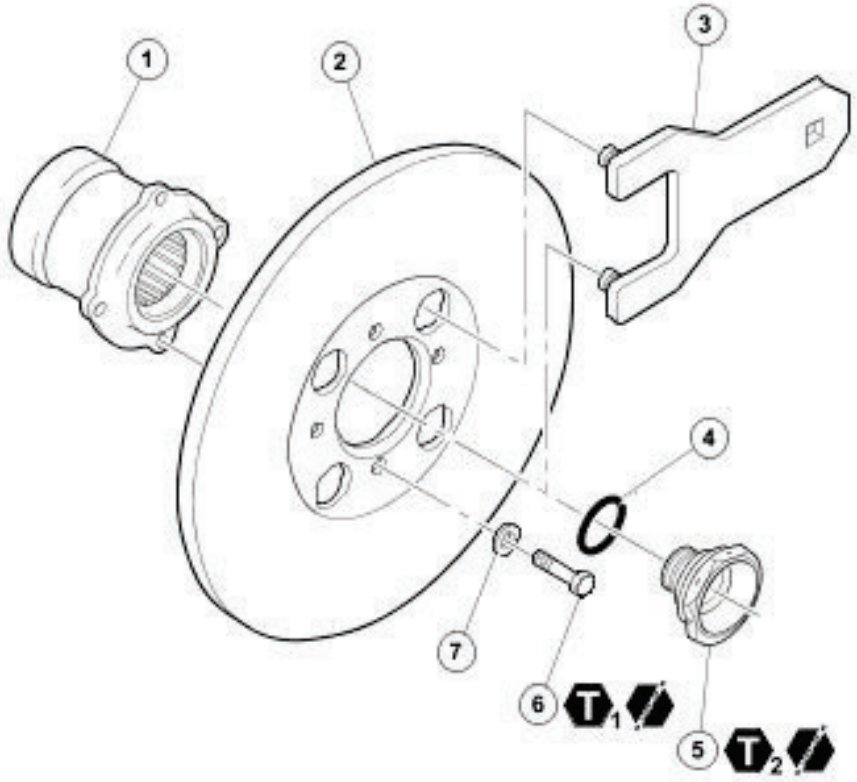
NOTE

The use of an arbor press, or equivalent, to separate the flange adapter (1) from the brake disc (2) requires the brake disc to be inspected for warpage (412-A-63-23-00-04A-280A-A / 01221).

3.1. To prevent inadvertent separation of the assembly, reinstall the bolts (6) and the washers (7).

3.2. Place the assembly in an arbor press. Remove the bolts (6) and the washers (7).

3.3. Apply sufficient pressure to separate the flange adapter (1) from the brake disc (2).



1. Flange adapter
2. Brake disc
3. Brake disc holding tool (T101649-101)
4. Packing
5. Retaining nut
6. Bolt
7. Washer

LOCKWIRE (C-405)

T1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T2 125 TO 150 FT-LBS (169.5 TO 203.4 Nm)

ICN-412-A-632300-B-97499-00008-A-001-01

Figure 1. Rotor Brake Disc (Sheet 1 of 1)

Flange Adapter - Installation

Preliminary requirements

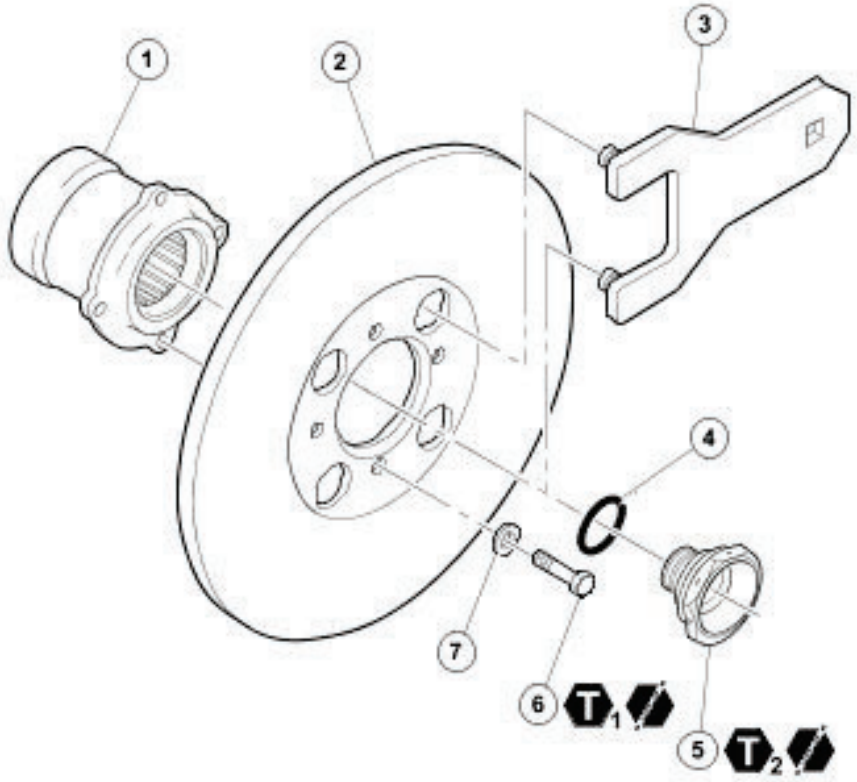
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Install the brake disc (2, Figure 1) on the flange (1) using the bolts (6) and the washers (7).

2. Torque the bolts (6) 50 to 70 inch-pounds (5.65 to 7.91 Nm) and secure with lockwire (C-405).



1. Flange adapter
2. Brake disc
3. Brake disc holding tool (T101649-101)
4. Packing
5. Retaining nut
6. Bolt
7. Washer

LOCKWIRE (C-405)

T1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T2 125 TO 150 FT-LBS (169.5 TO 203.4 Nm)

ICN-412-A-632300-B-97499-00008-A-001-01

Figure 1. Rotor Brake Disc (Sheet 1 of 1)

CHAPTER 64
TAIL ROTOR

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number		Issue date	Applicable to
Tail Rotor Hub and Blade Assembly - Description	DMC-412-A-64-00-00-00A-042A-A / 01226		2023-05-31	ALL
Tail Rotor Hub and Blade Assembly - Troubleshooting	DMC-412-A-64-00-00-00A-422A-A / 01227		2023-05-31	ALL
Tail Rotor Hub and Blade Assembly - Inspection - Static Stop (212-011-713) Yielding	DMC-412-A-64-00-00-00A-280A-A / 01228		2023-05-31	ALL
Tail Rotor Hub and Blade Assembly - Removal	DMC-412-A-64-00-00-00A-520A-A / 01229		2024-05-09	ALL
Tail Rotor Hub and Blade Assembly - Inspection	DMC-412-A-64-00-00-00A-280B-A / 01230	C	2025-11-18	ALL
Tail Rotor Hub and Blade Assembly - Repair	DMC-412-A-64-00-00-00A-685A-A / 01231		2023-05-31	ALL
Tail Rotor Hub and Blade Assembly - Installation	DMC-412-A-64-00-00-00A-720A-A / 01232		2024-12-19	ALL
Tail Rotor Controls - Description	DMC-412-A-64-30-00-00A-042A-A / 01233		2023-05-31	ALL
Counterweight Link - Removal	DMC-412-A-64-30-00-01A-520A-A / 01234		2023-05-31	ALL
Counterweight Link - Installation	DMC-412-A-64-30-00-01A-720A-A / 01235		2023-05-31	ALL
Pitch Control Links - Removal	DMC-412-A-64-30-00-02A-520A-A / 01236		2023-05-31	ALL
Pitch Control Links - Installation	DMC-412-A-64-30-00-02A-720A-A / 01237		2023-05-31	ALL
Control Tube - Removal	DMC-412-A-64-30-00-03A-520A-A / 01238		2024-05-09	ALL
Control Tube - Installation	DMC-412-A-64-30-00-03A-720A-A / 01239		2024-05-09	ALL
Control Housing - Removal	DMC-412-A-64-30-00-04A-520A-A / 01240		2023-05-31	ALL
Control Housing - Installation	DMC-412-A-64-30-00-04A-720A-A / 01241		2023-05-31	ALL
Tail Rotor Lever - Removal	DMC-412-A-64-30-00-05A-520A-A / 01242		2023-05-31	ALL
Tail Rotor Lever - Installation	DMC-412-A-64-30-00-05A-720A-A / 01243		2023-05-31	ALL
Control Link - Removal	DMC-412-A-64-30-00-06A-520A-A / 01244		2023-05-31	ALL
Control Link - Installation	DMC-412-A-64-30-00-06A-720A-A / 01245		2023-05-31	ALL
Idler - Removal	DMC-412-A-64-30-00-07A-520A-A / 01246		2023-05-31	ALL
Idler - Installation	DMC-412-A-64-30-00-07A-720A-A / 01247		2023-05-31	ALL
Crosshead Controls - Disassembly	DMC-412-A-64-30-00-08A-530A-A / 01248		2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Crosshead Controls - Assembly	DMC-412-A-64-30-00-08A-710A-A / 01249	2023-05-31	ALL
Tail Rotor Blades - Description	DMC-412-A-64-10-00-00A-042A-A / 01250	2023-05-31	ALL
Tail Rotor Blades - Maintenance	DMC-412-A-64-10-01-02A-913A-A / 01251	2023-05-31	ALL
Tail Rotor Blades - Cleaning	DMC-412-A-64-10-01-02A-251A-A / 01252	2024-05-09	ALL
Tail Rotor Blades - Inspection	DMC-412-A-64-10-01-02A-280A-A / 01253	2024-10-16	ALL
Tail Rotor Blades - Leak Check	DMC-412-A-64-10-01-02A-364A-A / 01254	2023-05-31	ALL
Tail Rotor Blades - Minor Repair	DMC-412-A-64-10-01-02A-685A-A / 01255	2023-05-31	ALL
Tail Rotor Blades - Paint Touch-up	DMC-412-A-64-10-01-02A-257A-A / 01256	2024-10-16	ALL

Tail Rotor Hub and Blade Assembly - Description

The tail rotor installation consists of the two-bladed tail rotor hub and blade assembly and control linkage. The rotor turns clockwise, as viewed from the gearbox, and counteracts main rotor torque to provide directional control.

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Tail Rotor Hub and Blade Assembly - Troubleshooting

NOTE

For the removal, inspection, repair, and installation of the hub and blade as an assembly, refer to 412-A-64-00-00-00A-520A-A / 01229, 412-A-64-00-00-00A-280B-A / 01230, 412-A-64-00-00-00A-685A-A / 01231 and 412-A-64-00-00-00A-720A-A / 01232. To separate the blades from the hub, refer to the 412-CRO, Chapter 64.

Potential troubles that may occur in the tail rotor are listed in Table 1, with probable causes indicated and corrective action recommended.

Table 1. Tail Rotor Troubleshooting

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
High frequency vibration	Tail rotor out of track	Check tracking of tail rotor (412-A-18-12-01-01A-373A-A / 00186)
	Tail rotor out of balance	Check and balance tail rotor (412-A-18-12-00-02A-271A-A / 00165)
	Loose tail rotor retaining nut	Torque retaining nut 900 inch-pounds (102 Nm)
	Loose counterweight bellcrank retention nuts	Replace bellcrank
	Tail rotor pitch change links bent	Replace damaged links
	Worn or loose pitch change tube bearing	Replace bearing and/or wear sleeve on tube
	90° gearbox mounting loose	Inspect gearbox mounting for loose rivets or studs or elongated bolt holes. Repair mounting.
Inability to make normal right and left turns in flight	Faulty rigging	Rig tail rotor controls (412-A-67-01-00-11A-276A-A / 01551)
Controls creeping	Hydraulic cylinder in tail rotor controls leaking internally	Replace faulty hydraulic cylinder

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Tail Rotor Hub and Blade Assembly - Inspection - Static Stop (212-011-713) Yielding

Procedure

NOTE

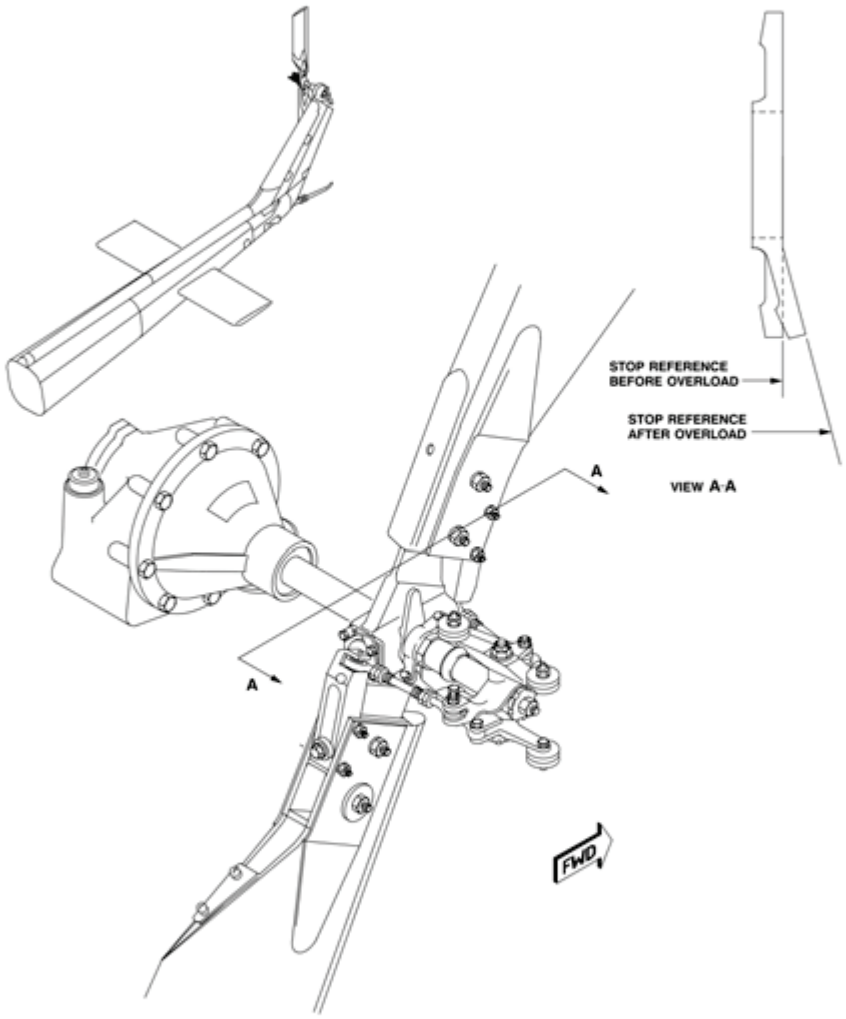
This inspection is required by the 25 Hour Special Inspection and certain Conditional Inspections (412-A-05-47-00-05A-283A-A / 00037, 412-A-05-50-00-00A-284A-A / 00063). This inspection is also required if damaging/excessive bending loads are suspected to have been sustained such as high wind gusts, improper ground handling (tail rotor blade used as a hand hold), or static ground strike that could cause yielding.

1. Gain access to the tail rotor assembly using a maintenance stand or other means to allow close viewing of the trunnion assembly. Perform a visual inspection of the flapping stop ears for deformation (Figure 1).

1.1. Gently place the tail rotor yoke against one static stop ear, to allow full view of the opposite stop.

1.2. Repeat in the opposite direction to allow viewing of the remaining stop.

2. If either static stop ear is deformed or bent, as shown in Figure 1, the yoke assembly has been yielded. Both the yoke assembly and static stop (ASB 412-96-89) are no longer serviceable and must be removed and discarded. If any doubt exists about serviceability of these parts, contact Product Support Engineering (PSE) .



ICN-412-A-640000-B-97499-00025-A-001-01

Figure 1. Tail Rotor Static Stop (212-011-713) Yielding (Sheet 1 of 1)

Tail Rotor Hub and Blade Assembly - Removal

Procedure

1. Cut and remove the lockwire from the bolt (2, Figure 1). Loosen the bolt.

2. Loosen the nuts (37). Remove the nuts (33) and the steel washers (34) from the bolts (36). Remove the bolts (36) and the steel washers (35) and slide the pitch horn (24) out so the bolt (2) will clear the bracket on the hub and blade assembly (29).

3. Remove the bolt (2) and separate the pitch link (19) from the pitch horn (24).

4. Reinstall the hardware to prevent loss. Disconnect the opposite pitch link in the same manner.

5. Remove the cotter pin (7), nut (6), and steel washer (5) from the bolt (22). Discard the cotter pin (7). Remove the bolt (22) and the steel washer (21). Separate the counterweight link (20) from the support (3). Disconnect the opposite counterweight link in the same manner.

6. Cut and remove the lockwire from the screws (18). Remove the screws (18), steel washers (17), lock (16), and retainer (15).

7. Push the crosshead assembly (8) inboard to expose the cotter pin (14). Remove the cotter pin (14), nut (13), and steel washer (12). Discard the cotter pin (14).

8. Remove the crosshead assembly (8), pitch links (19), and counterweight links (20) as a single unit.

9. Remove the bearing (11) and the special washer (9).

NOTE

Remove the sleeve (10) only if the control tube (4) is to be removed.

10. Pull the sleeve (10) outboard and rotate to engage the threads on the control tube (4), rotate the sleeve until clear of the threads, and remove the sleeve from the tube.

11. Cut and remove the lockwire from the nut (31). Remove the nut (31) and the shield (30) as a single unit. Remove the shield (30) from the nut (31).

12. Remove the support (3) and the static stop (32). Remove the tail rotor hub and blade assembly (29).

13. Remove the cone set (27). Remove and discard the packing (28) from the cone set (27). Keep two halves of the cone set (27) together.

NOTE

Accomplish Step 14. through Step 20. only if the control linkage must be removed.

14. Remove the counterweight links (20) from the crosshead assembly (8) (412-A-64-30-00-01A-520A-A / 01234).

15. Remove the pitch links (19) from the crosshead assembly (8) (412-A-64-30-00-02A-520A-A / 01236).

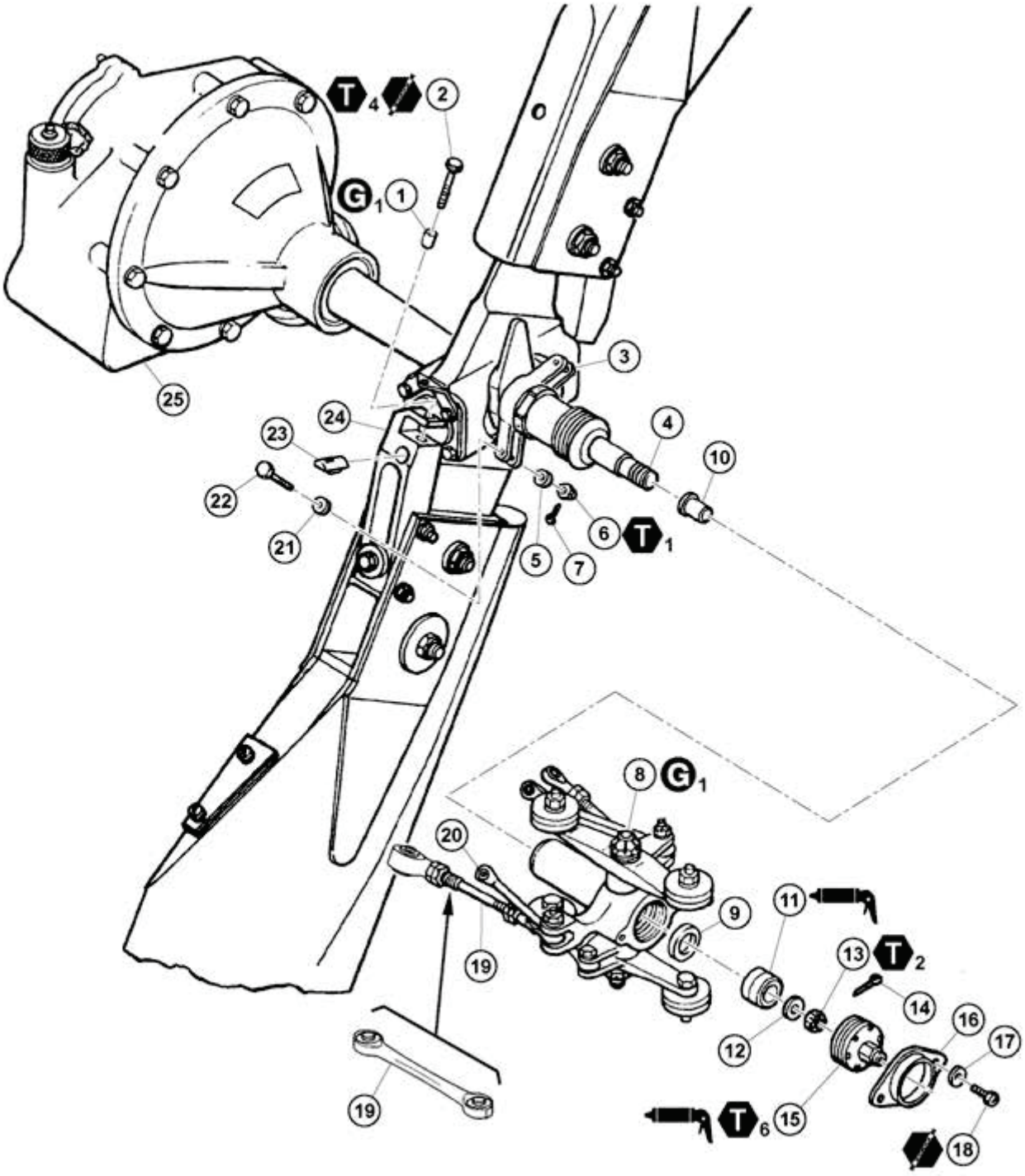
16. Remove the tail rotor control tube (4) (412-A-64-30-00-03A-520A-A / 01238).

17. Remove the control housing (2, Figure 2) (412-A-64-30-00-04A-520A-A / 01240).

18. Remove the lever (8, Figure 3) (412-A-64-30-00-05A-520A-A / 01242).

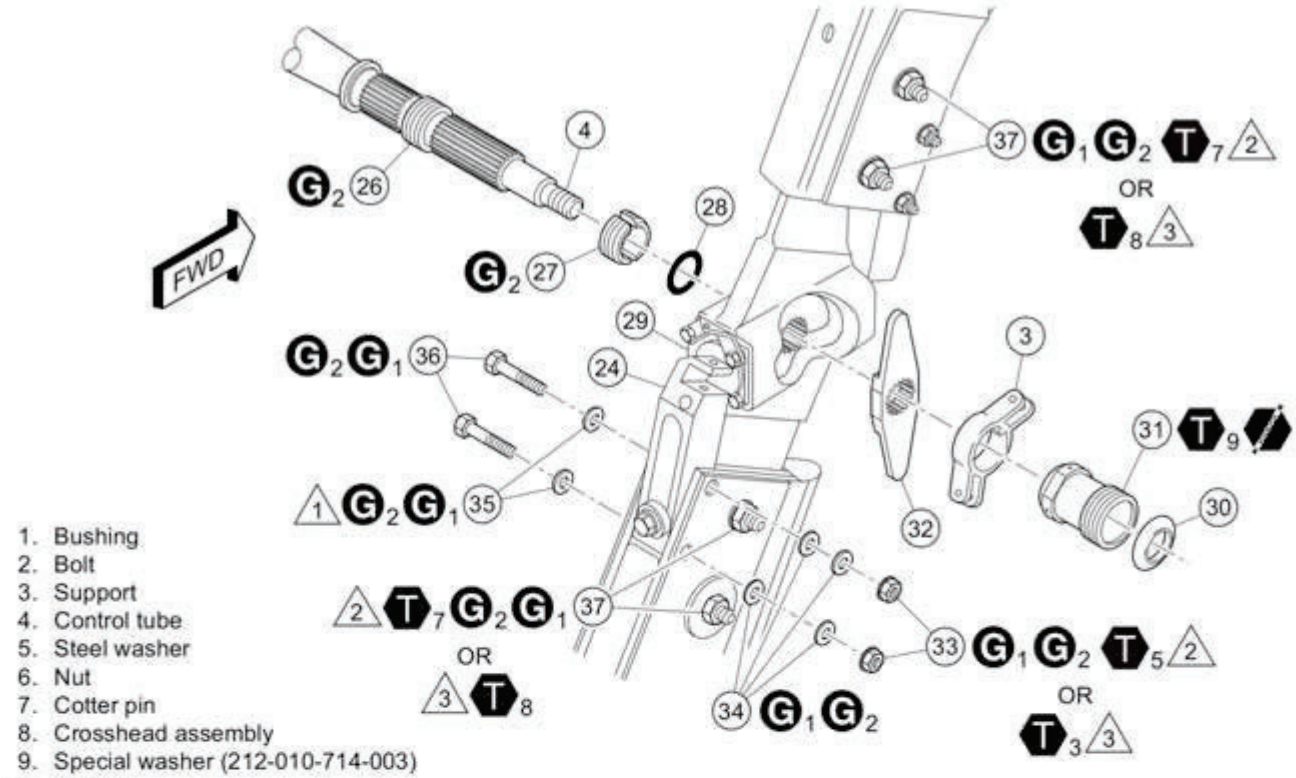
19. Remove the tail rotor control link (5, Figure 4) (412-A-64-30-00-06A-520A-A / 01244).

20. Remove the tail rotor control idler (12, Figure 5) (412-A-64-30-00-07A-520A-A / 01246).



ICN-412-A-640000-B-97499-00026-A-002-01

Figure 1. Tail Rotor Hub and Blade Assembly - Removal and Installation (Sheet 1 of 2)



1. Bushing
2. Bolt
3. Support
4. Control tube
5. Steel washer
6. Nut
7. Cotter pin
8. Crosshead assembly
9. Special washer (212-010-714-003)
10. Sleeve
11. Bearing
12. Steel washer
13. Nut
14. Cotter pin
15. Retainer
16. Lock
17. Steel washer
18. Screw
19. Pitch link
20. Counterweight link
21. Steel washer
22. Bolt
23. Barrel nut and retainer
24. Pitch horn
25. Tail rotor gearbox
26. Shaft
27. Cone set
28. Packing
29. Hub and blade assembly
30. Shield
31. Nut
32. Static stop
33. Nut
34. Steel washer
35. Steel washer
36. Bolt
37. Nut

G_1 CORROSION PREVENTIVE COMPOUND (C-101)

G_2 CORROSION PREVENTIVE COMPOUND (C-104)

GREASE (C-001)

LOCKWIRE (C-405)

T_1 60 TO 110 IN-LBS (6.8 TO 12 Nm)

T_2 70 TO 100 IN-LBS (7.9 TO 11 Nm)

3 T_3 130 TO 150 IN-LBS (15 TO 17 Nm)

T_4 135 IN-LBS (15 Nm)

2 T_5 150 IN-LBS (17 Nm)

T_6 300 IN-LBS (34 Nm)

2 T_7 500 TO 550 IN-LBS (57 TO 62 Nm)

3 T_8 45 TO 54 FT-LBS (61 TO 73 Nm)

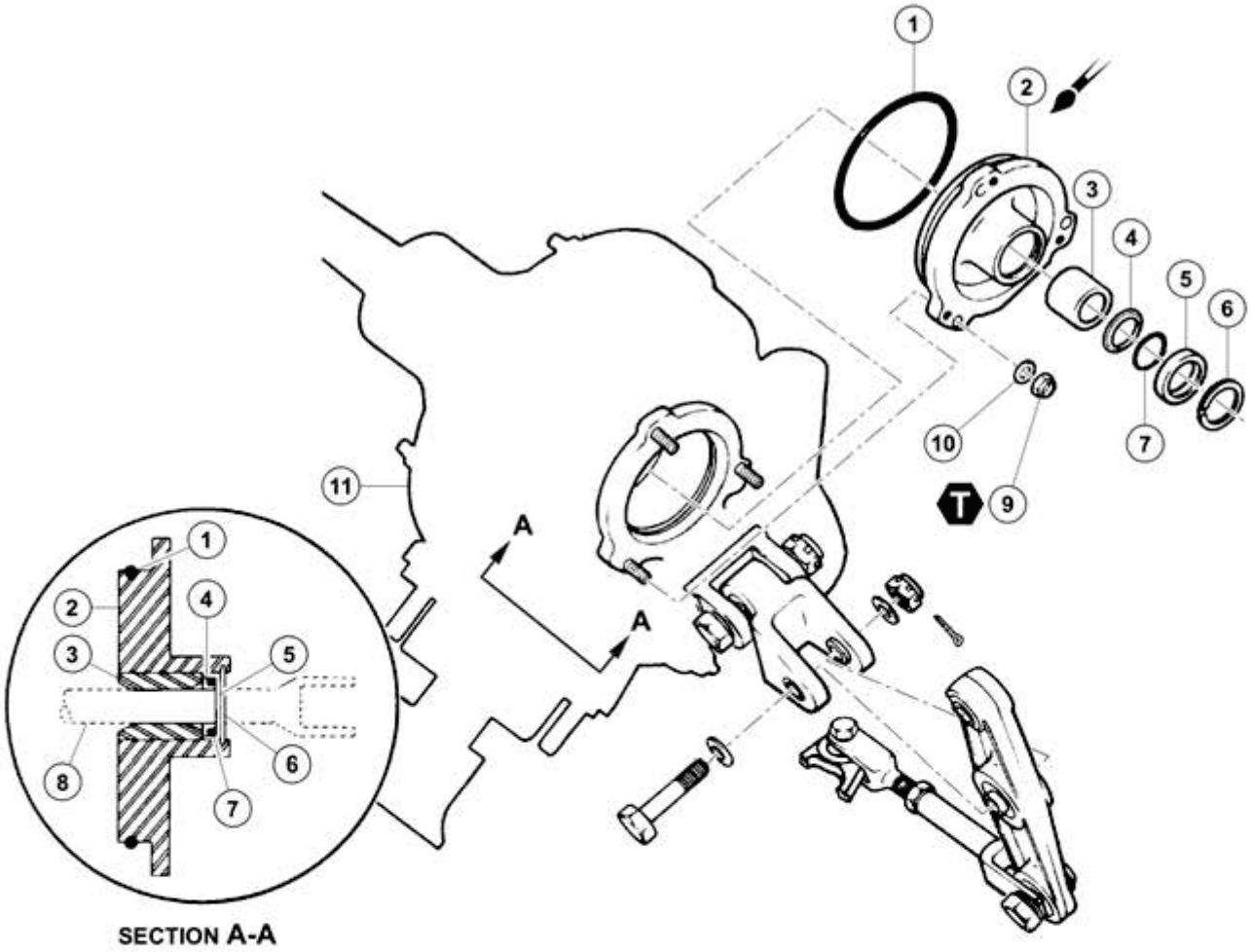
T_9 900 IN-LBS (102 Nm)

NOTES

1. Chamfered side of washers (35) should face toward head of bolts (36).
2. Torque applicable when tail rotor blades 212-010-750 installed.
3. Torque applicable when tail rotor blades 412-016-100 installed.

ICN-412-A-640000-B-97499-00027-A-001-01

Figure 1. Tail Rotor Hub and Blade Assembly - Removal and Installation (Sheet 2 of 2)



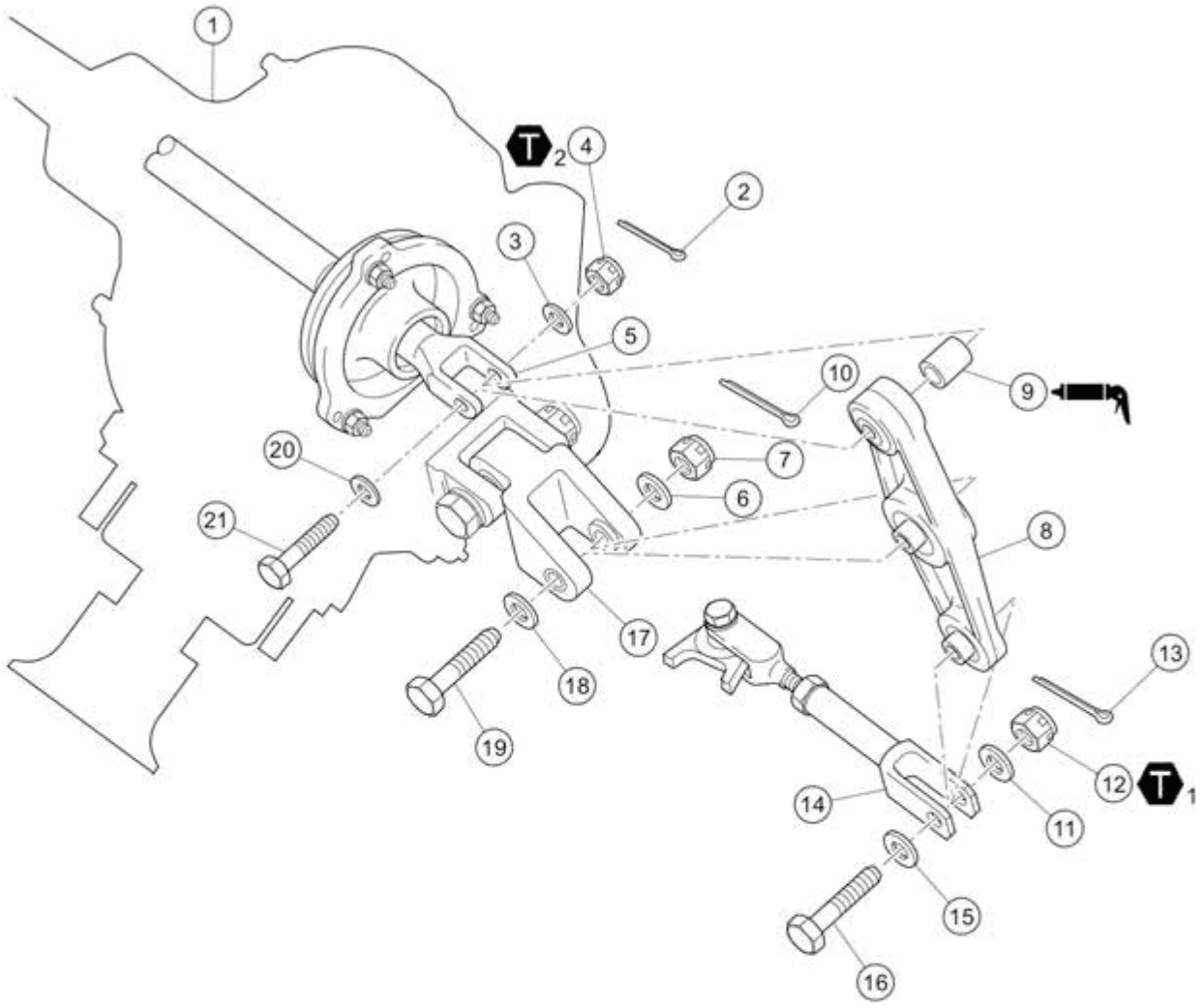
1. Preformed packing
2. Housing
3. Bearing
4. Excluder
5. Excluder housing
6. Retainer ring
7. Seal
8. Control tube
9. Nut
10. Steel washer
11. Tail rotor gearbox

PRIMER (C-204)

60 IN-LBS (6.8 Nm)

ICN-412-A-640000-B-97499-00028-A-001-01

Figure 2. Tail Rotor Controls - Control Housing - Removal and Installation (Sheet 1 of 1)



1. Tail rotor gearbox (ref)
2. Cotter pin
3. Steel washer
4. Nut
5. Control tube
6. Steel washer
7. Nut
8. Lever
9. Race
10. Cotter pin
11. Steel washer
12. Nut
13. Cotter pin
14. Link
15. Steel washer
16. Bolt
17. Idler
18. Steel washer
19. Bolt
20. Steel washer
21. Bolt

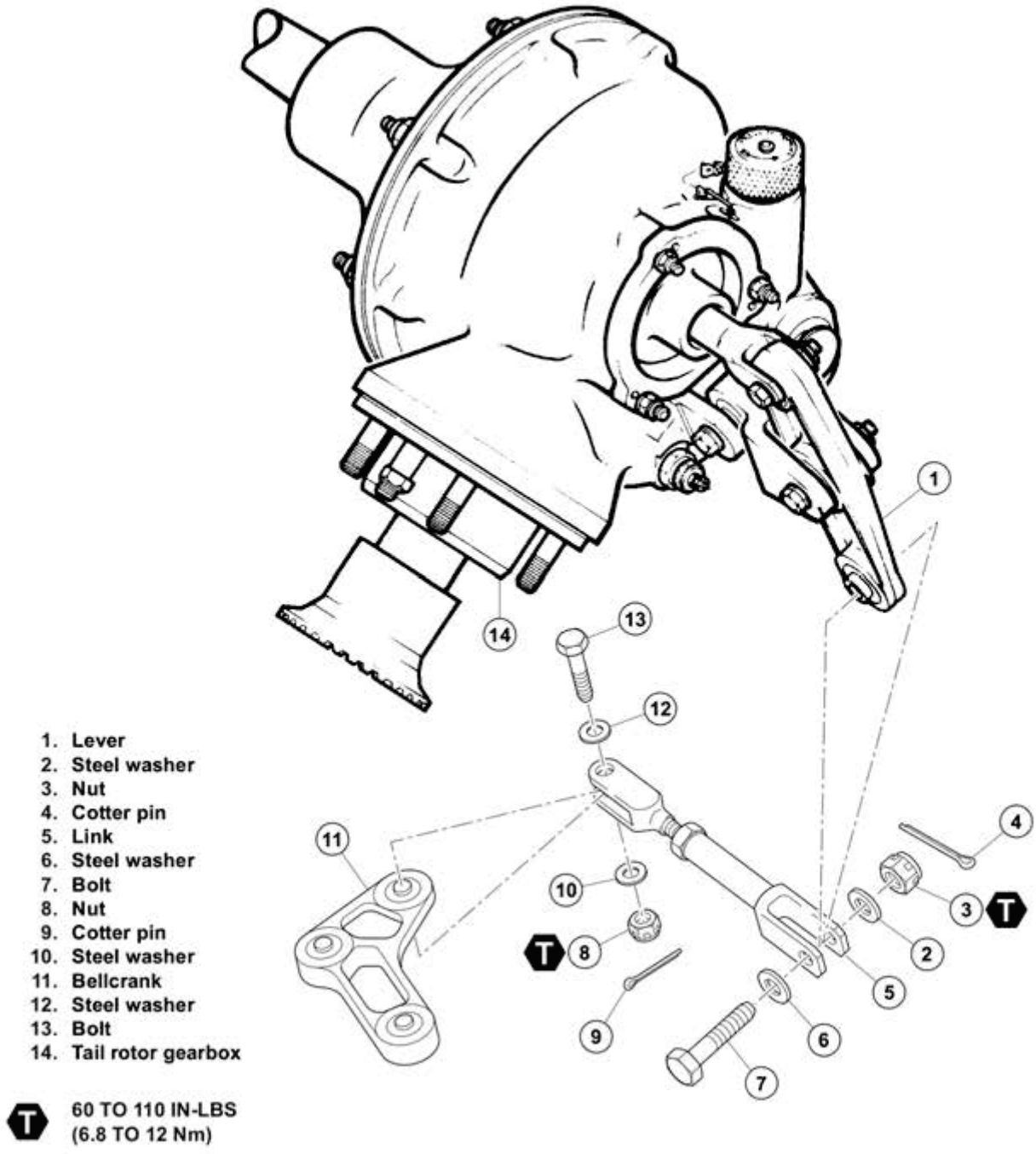
T1 60 TO 110 IN-LBS (6.8 TO 12 Nm)

T2 110 TO 165 IN-LBS (12 TO 19 Nm)

GREASE (C-001)

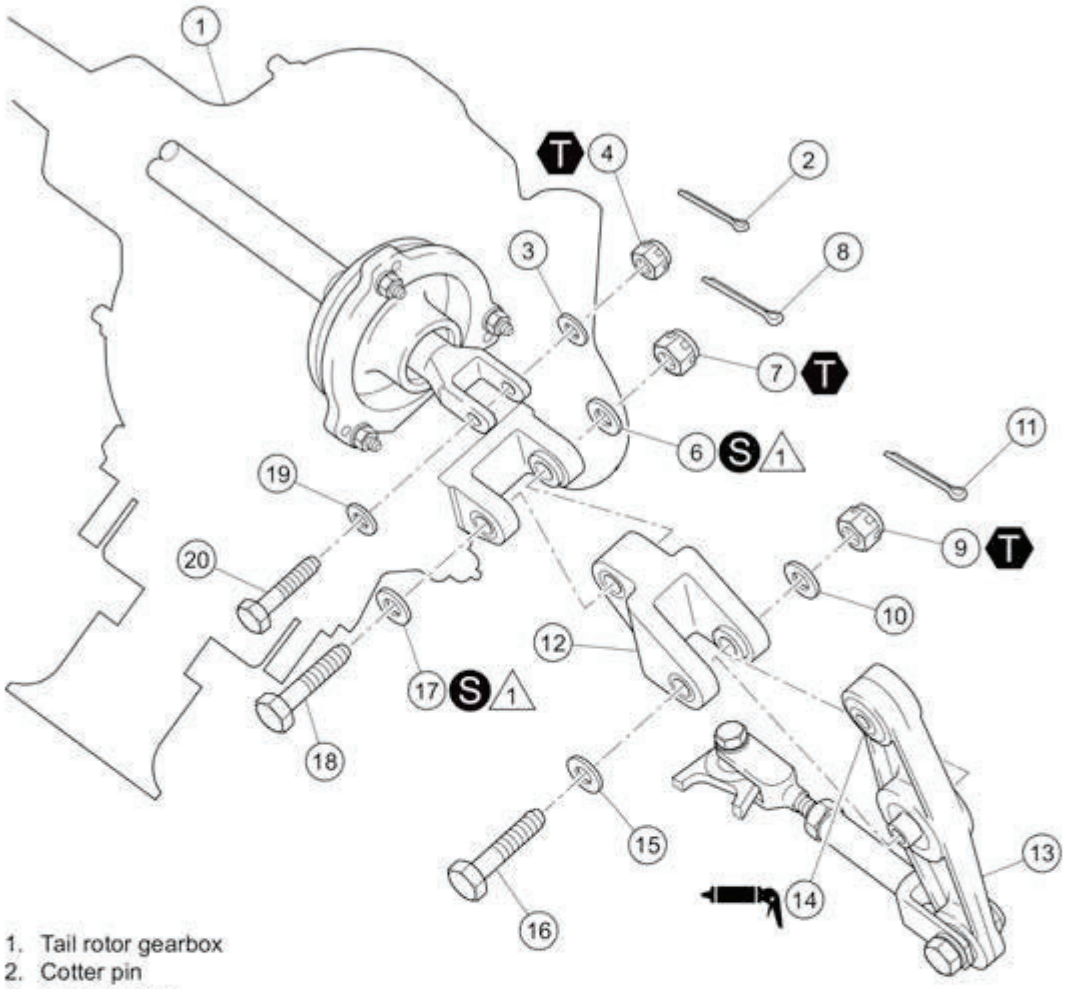
ICN-412-A-640000-B-97499-00029-A-001-01

Figure 3. Tail Rotor Controls - Lever - Removal and Installation (Sheet 1 of 1)



ICN-412-A-640000-B-97499-00030-A-001-01

Figure 4. Tail Rotor Controls - Control Link - Removal and Installation (Sheet 1 of 1)



1. Tail rotor gearbox
2. Cotter pin
3. Steel washer
4. Nut
5. Control tube
6. Steel washer
7. Nut
8. Cotter pin
9. Nut
10. Steel washer
11. Cotter pin
12. Idler
13. Lever
14. Race
15. Steel washer
16. Bolt
17. Steel washer
18. Bolt
19. Steel washer
20. Bolt

GREASE (C-001)

S SEALANT (C-328)

T 110 TO 165 IN-LBS (12 TO 19 Nm)

NOTE

1 Bond washers (6 and 17) to tail rotor gearbox (1) fitting with sealing compound (C-328).

ICN-412-A-640000-B-97499-00031-A-001-01

Figure 5. Tail Rotor Controls - Control Idler - Removal and Installation (Sheet 1 of 1)

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Tail Rotor Hub and Blade Assembly - Inspection

Procedure

NOTE

If records or physical appearance indicate the tail rotor has been involved in an accident or incident, refer to 412-A-05-50-00-00A-284A-A / 00063 for conditional inspection requirements.

1. Inspect the tail rotor blades (412-A-64-10-01-02A-280A-A / 01253).

2. Inspect the retainer (15, Figure 1) for damaged threads. Damage that can be detected visually is not acceptable.

3. Inspect the crosshead assembly (8) for damaged threads. Damage that can be detected visually is not acceptable.

4. Inspect the bearing (11) for contamination, lack of grease, and separation of the outer race. Check the bearing for roughness by feel. Rough bearings and/or bearings with loose races are not acceptable.

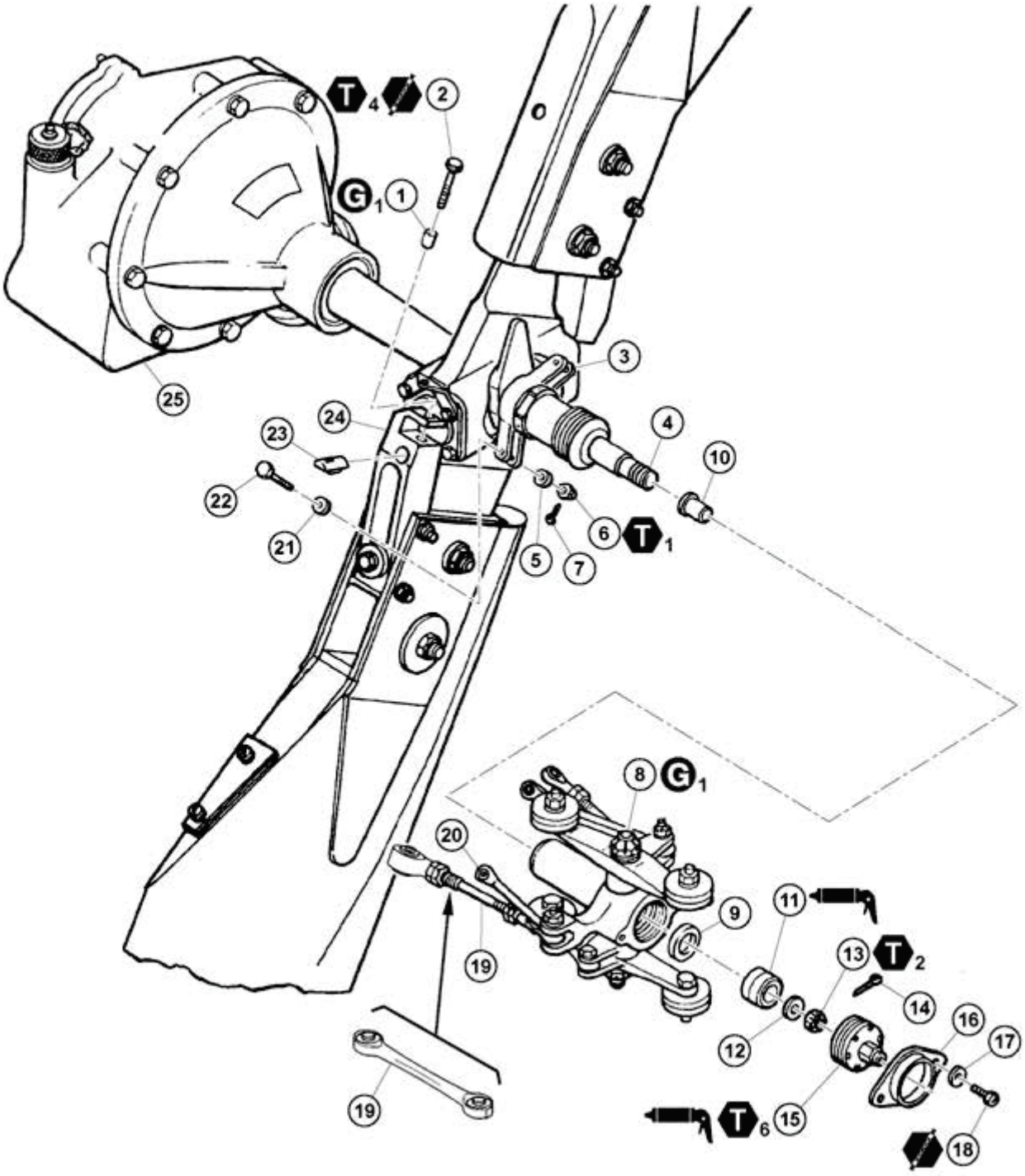
5. Inspect the sleeve (10) for mechanical damage. Superficial scratches are acceptable.

6. Inspect the special washer (9) for scratches, nicks, and distortion. Superficial scratches are acceptable.

7. Inspect the shield (30) for nicks and cuts. Superficial damage is acceptable.

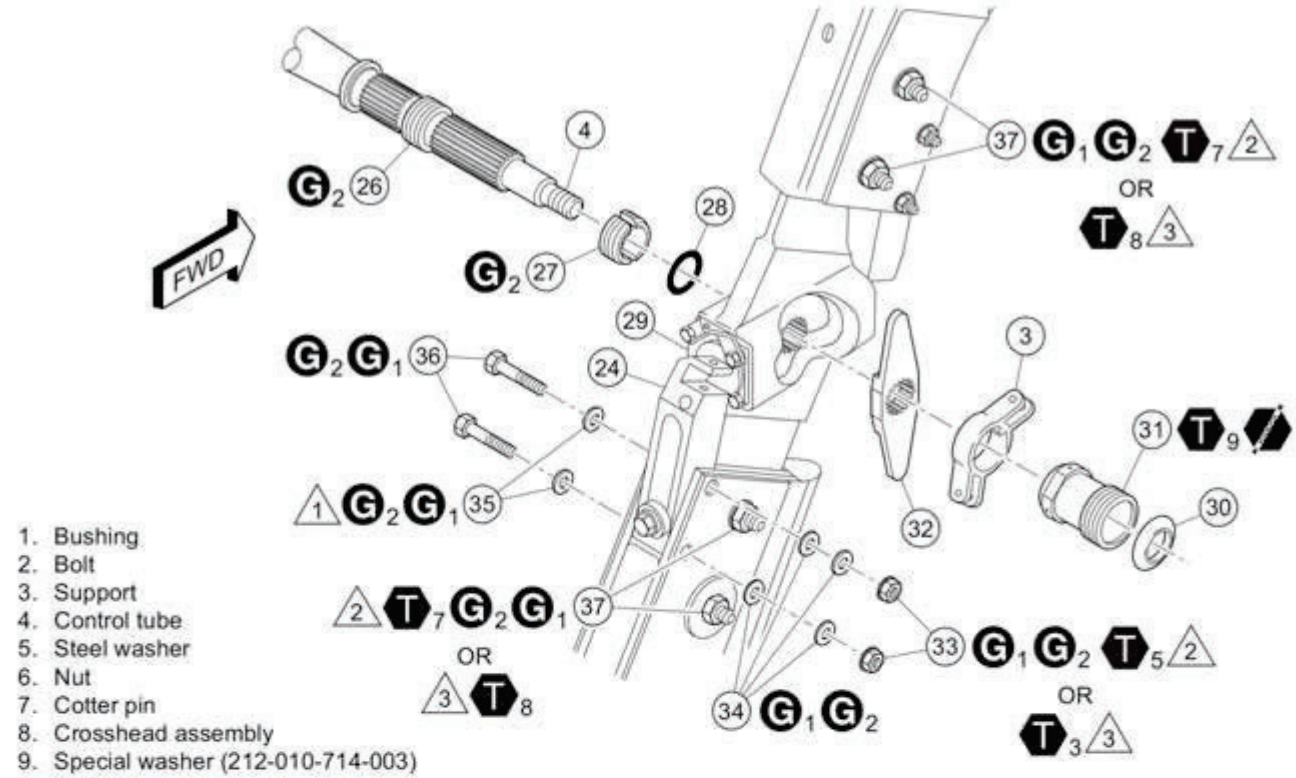
8. Inspect the nut (31) for damaged threads. No damaged threads are acceptable.

9. *For the remainder of the tail rotor components, refer to the* 412-CR&O, Chapter 64.



ICN-412-A-640000-B-97499-00026-A-002-01

Figure 1. Tail Rotor Hub and Blade Assembly - Removal and Installation (Sheet 1 of 2)



1. Bushing
2. Bolt
3. Support
4. Control tube
5. Steel washer
6. Nut
7. Cotter pin
8. Crosshead assembly
9. Special washer (212-010-714-003)
10. Sleeve
11. Bearing
12. Steel washer
13. Nut
14. Cotter pin
15. Retainer
16. Lock
17. Steel washer
18. Screw
19. Pitch link
20. Counterweight link
21. Steel washer
22. Bolt
23. Barrel nut and retainer
24. Pitch horn
25. Tail rotor gearbox
26. Shaft
27. Cone set
28. Packing
29. Hub and blade assembly
30. Shield
31. Nut
32. Static stop
33. Nut
34. Steel washer
35. Steel washer
36. Bolt
37. Nut

G1 CORROSION PREVENTIVE COMPOUND (C-101)

G2 CORROSION PREVENTIVE COMPOUND (C-104)

GREASE (C-001)

LOCKWIRE (C-405)

T1 60 TO 110 IN-LBS (6.8 TO 12 Nm)

T2 70 TO 100 IN-LBS (7.9 TO 11 Nm)

3 T3 130 TO 150 IN-LBS (15 TO 17 Nm)

T4 135 IN-LBS (15 Nm)

2 T5 150 IN-LBS (17 Nm)

T6 300 IN-LBS (34 Nm)

2 T7 500 TO 550 IN-LBS (57 TO 62 Nm)

3 T8 45 TO 54 FT-LBS (61 TO 73 Nm)

T9 900 IN-LBS (102 Nm)

NOTES

1. Chamfered side of washers (35) should face toward head of bolts (36).
2. Torque applicable when tail rotor blades 212-010-750 installed.
3. Torque applicable when tail rotor blades 412-016-100 installed.

ICN-412-A-640000-B-97499-00027-A-001-01

Figure 1. Tail Rotor Hub and Blade Assembly - Removal and Installation (Sheet 2 of 2)

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Tail Rotor Hub and Blade Assembly - Repair

Procedure

1. For the repair of the tail rotor components, refer to the BHT-412-CR&O.

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Tail Rotor Hub and Blade Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-001	AR	
Corrosion preventive compound	C-101	AR	
Corrosion preventive compound	C-104	AR	
Drycleaning solvent	C-304	AR	
Lockwire	C-405	AR	

Procedure



APPLY corrosion preventive compound (C-104) TO ALL BOLT SHANKS (DO NOT APPLY TO THREADS), WASHER FAYING SURFACES, AND NUT FAYING SURFACES (DO NOT APPLY TO THREADS) PRIOR TO INSTALLATION. AFTER FINAL INSTALLATION, COAT EXPOSED SURFACES OF NUTS, WASHERS, AND BOLTS (INCLUDING EXPOSED THREADS) WITH corrosion preventive compound (C-101).

1. Apply a thin coat of corrosion preventive compound (C-104) to the inner and mating surfaces of the cone set (27, Figure 1) and the shaft (26). Place the cone set, with bevel outboard, in the groove between the splines and the shoulder on the shaft. Install the new packing (28) on the cone set with equal gaps at the ends of the cone set. Seat the cone set in position on the shaft.

2. Place the hub and blade assembly (29) on the shaft (26) with the open side of the pitch horns (24) outboard. Align the master spline on the trunnion with the master spline on the shaft and press the hub and blade assembly inboard to seat against the cone set (27).

3. Align the master spline on the static stop (32) with the master spline on the shaft (26) and install against the trunnion with the flat side outboard.

4. Align the tooth on the support (3) with the master tooth on the shaft (26) and install against the static stop (32) with the arms angled outboard.

5. Install the shield (30) on the nut (31). Install the nut (31) on the shaft (26). Torque the nut (31) 900 inch-pounds (102 Nm). Secure the nut (31) to the support (3) with lockwire (C-405).

6. If not previously accomplished, thread the sleeve (10) onto the control tube (4) with the large diameter end inboard. Continue turning until the sleeve disengages from the threads.

7. If not previously accomplished, assemble the crosshead controls (412-A-64-30-00-08A-710A-A / 01249).

8. Apply a thin film of grease (C-001) to the splines in the crosshead assembly (8). Align the master splines in the crosshead assembly and the shaft (26). Engage the splines and press the crosshead inboard, carefully guiding the crosshead into the shield (30).

9. Clean the bearing (11) thoroughly with drycleaning solvent (C-304). Dry with clean dry air. Hand pack the bearing with grease (C-001).



MAKE SURE THE SPECIAL WASHER (9) SEATS AROUND THE LARGEST OUTSIDE DIAMETER OF THE SLEEVE (10).

10. Install the special washer (9) inside the crosshead assembly (8), around the largest outside diameter of the sleeve (10). Install the bearing (11), steel washer (12), and nut (13). Torque the nut (13) 70 to 100 inch-pounds (7.9 to 11 Nm) and install the cotter pin (14).

WARNING

MAKE SURE THE NUT (13) AND THE COTTER PIN (14) ARE PROPERLY INSTALLED IN ACCORDANCE WITH Step 10. PRIOR TO INSTALLATION OF THE RETAINER (15).

11. *Fill the cavity of the retainer (15) with grease (C-001) and install in the crosshead assembly (8). Torque the retainer (15) 300 inch-pounds (34 Nm). Apply two shots of grease (C-001) to the fitting with a hand-held grease gun.*

12. Position the lock (16) over the retainer (15) and secure with the screws (18) and the steel washers (17). Secure the screws (18) with lockwire (C-405). Deform the lock (16) at two places into the retainer (15).

13. If not previously accomplished, loosen the bolts (36). Remove the assembly hardware from the pitch horn (24) and slide the pitch horn out from the hub and blade. Apply a thin coat of corrosion preventive compound (C-101) to the bushing (1), mating surfaces of the pitch horn, and tail rotor blade.

14. With the pitch horn (24) removed, install the pitch link (19) in the pitch horn with the bolt (2), bushing (1), and barrel nut and retainer (23). Do not torque the bolt (2) at this time.

NOTE

Step 15. and Step 16. are only applicable to tail rotor hub and blade assemblies with tail rotor blades 212-010-750 installed. If tail rotor blades 412-016-100 are installed, refer to Step 17. and Step 18..

NOTE

Make sure the chamfered sides of the steel washers (35) are toward the heads of the bolts (36).

15. Position the pitch horn (24) in the hub and blade assembly (29). Install the bolts (36) with the steel washers (35 and 34) and the nuts (33). Torque the nuts (33) 150 inch-pounds (17 Nm).

16. Torque the nuts (37) 500 to 550 inch-pounds (57 to 62 Nm).

NOTE

Step 17. and Step 18. are only applicable to tail rotor hub and blade assemblies with tail rotor blades 412-016-100 installed.

NOTE

Make sure the chamfered sides of the steel washers (35) are toward the heads of the bolts (36).

17. Position the pitch horn (24) in the hub and blade assembly (29). Install the bolts (36) with the steel washers (35 and 34) and the nuts (33). Torque the nuts (33) 130 to 150 inch-pounds (15 to 17 Nm).

18. Torque the nuts (37) 45 to 54 foot-pounds (61 to 73 Nm).

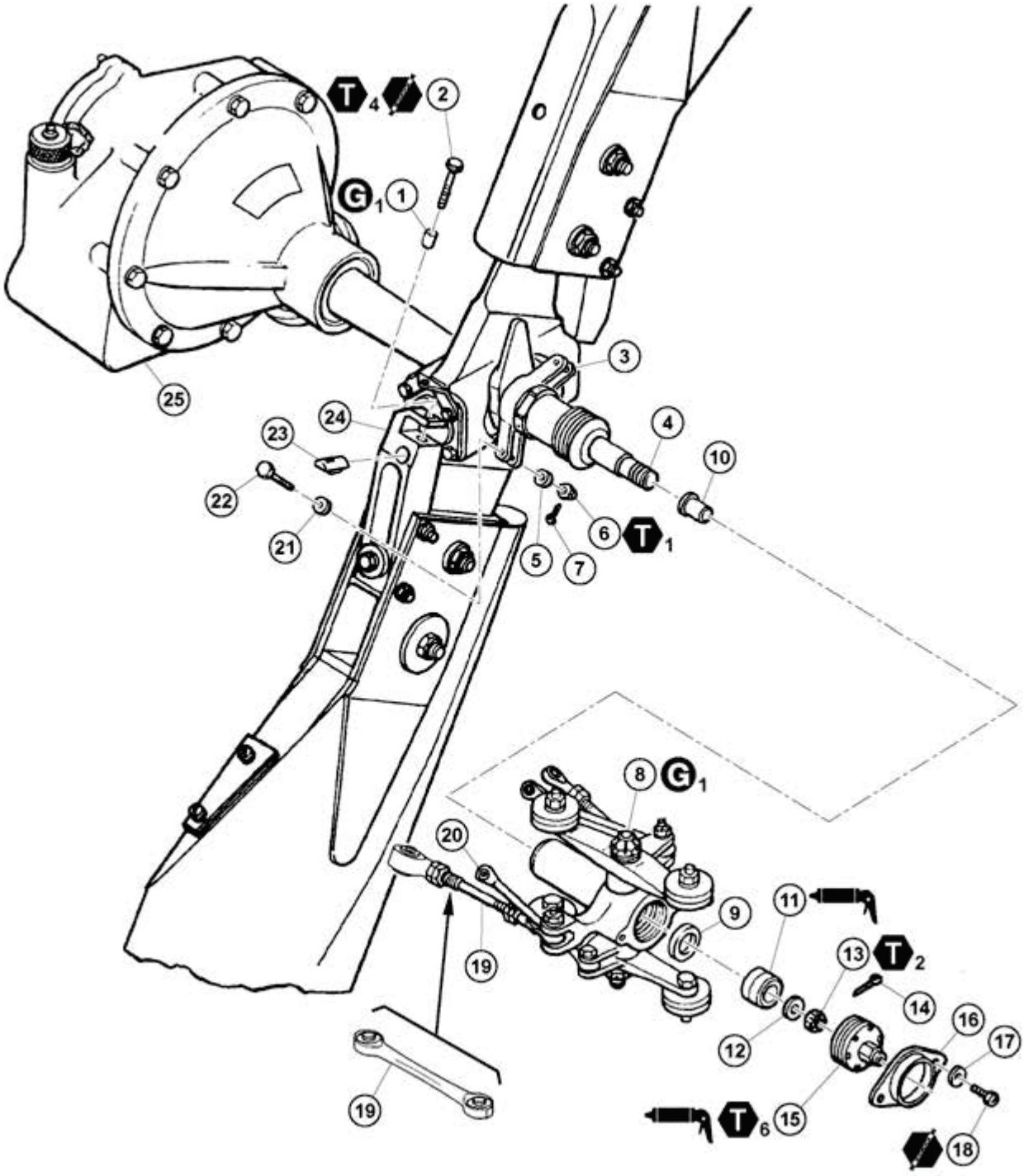
19. Torque the bolt (2) 135 inch-pounds (15 Nm) and secure with lockwire (C-405).

20. Install the opposite pitch link in the same manner.

21. Connect the counterweight links (20) to the support (3) with the bolts (22), steel washers (21), steel washers (5), and nuts (6). Torque the nuts (6) 60 to 110 inch-pounds (6.8 to 12 Nm). Install the new cotter pins (7).

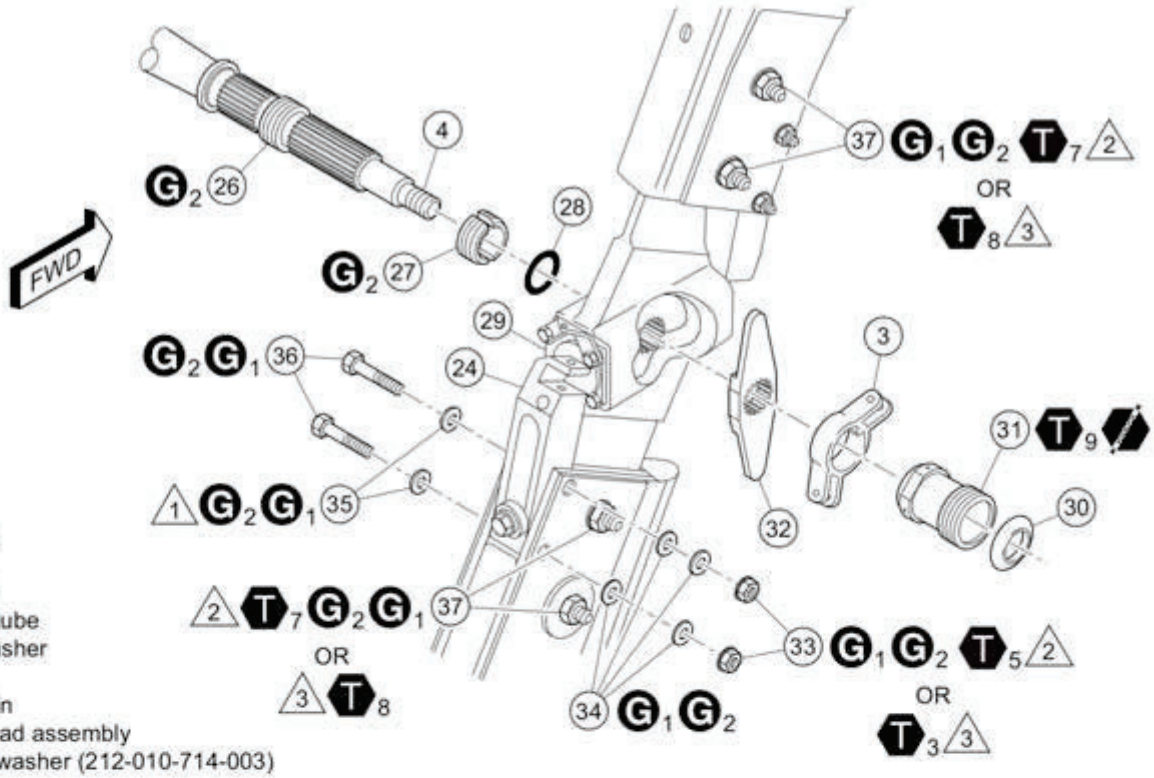
22. Perform the rigging check of the tail rotor (412-A-67-01-00-11A-276A-A / 01551).

23. Perform the operational check (412-A-18-12-01-00A-320A-A / 00183).



ICN-412-A-640000-B-97499-00026-A-002-01

Figure 1. Tail Rotor Hub and Blade Assembly - Removal and Installation (Sheet 1 of 2)



1. Bushing
2. Bolt
3. Support
4. Control tube
5. Steel washer
6. Nut
7. Cotter pin
8. Crosshead assembly
9. Special washer (212-010-714-003)
10. Sleeve
11. Bearing
12. Steel washer
13. Nut
14. Cotter pin
15. Retainer
16. Lock
17. Steel washer
18. Screw
19. Pitch link
20. Counterweight link
21. Steel washer
22. Bolt
23. Barrel nut and retainer
24. Pitch horn
25. Tail rotor gearbox
26. Shaft
27. Cone set
28. Packing
29. Hub and blade assembly
30. Shield
31. Nut
32. Static stop
33. Nut
34. Steel washer
35. Steel washer
36. Bolt
37. Nut

G1 CORROSION PREVENTIVE COMPOUND (C-101)

G2 CORROSION PREVENTIVE COMPOUND (C-104)

GREASE (C-001)

LOCKWIRE (C-405)

T1 60 TO 110 IN-LBS (6.8 TO 12 Nm)

T2 70 TO 100 IN-LBS (7.9 TO 11 Nm)

3 T3 130 TO 150 IN-LBS (15 TO 17 Nm)

T4 135 IN-LBS (15 Nm)

2 T5 150 IN-LBS (17 Nm)

T6 300 IN-LBS (34 Nm)

2 T7 500 TO 550 IN-LBS (57 TO 62 Nm)

3 T8 45 TO 54 FT-LBS (61 TO 73 Nm)

T9 900 IN-LBS (102 Nm)

NOTES

1. Chamfered side of washers (35) should face toward head of bolts (36).
2. Torque applicable when tail rotor blades 212-010-750 installed.
3. Torque applicable when tail rotor blades 412-016-100 installed.

ICN-412-A-640000-B-97499-00027-A-001-01

Figure 1. Tail Rotor Hub and Blade Assembly - Removal and Installation (Sheet 2 of 2)

Tail Rotor Controls - Description

The tail rotor controls consist of the counterweight links, control links, control tube, housing and seal installation, control lever, links and idler, and crosshead controls.

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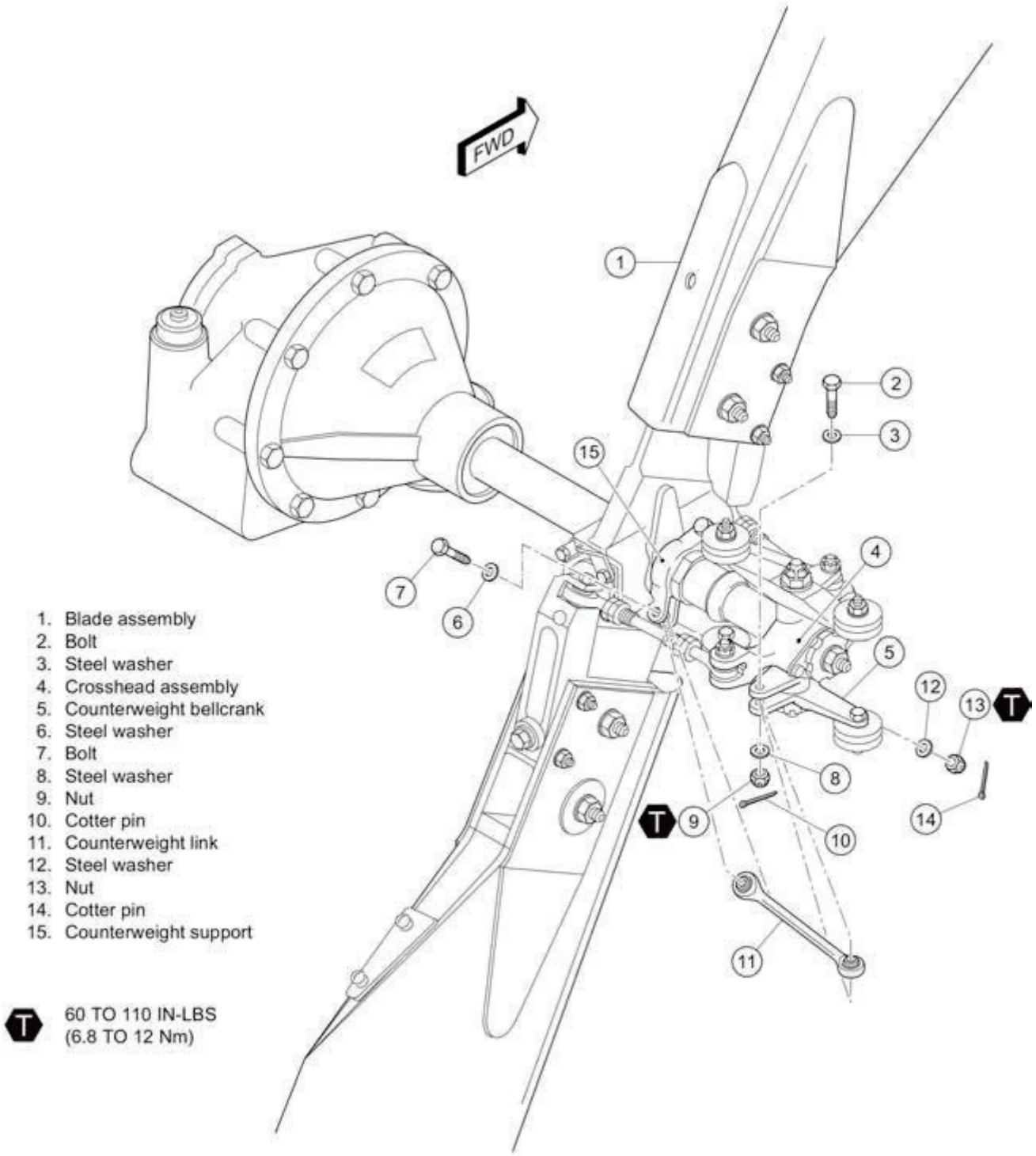
Counterweight Link - Removal

Procedure

1. Remove the cotter pin (10, Figure 1), nut (9), and steel washer (8). Discard the cotter pin (10). Remove the bolt (2) and the steel washer (3) to disconnect the counterweight link (11) from the counterweight bellcrank (5). Loosely reinstall the bolt, washers, and nut to prevent loss.

2. Remove the cotter pin (14), nut (13), and steel washer (12). Discard the cotter pin (14). Remove the bolt (7) and the steel washer (6) to disconnect the counterweight link (11) from the counterweight support (15). Remove the counterweight link (11) from the helicopter.

3. Loosely install the bolt (7), steel washers (6 and 12), and nut (13) to prevent loss.



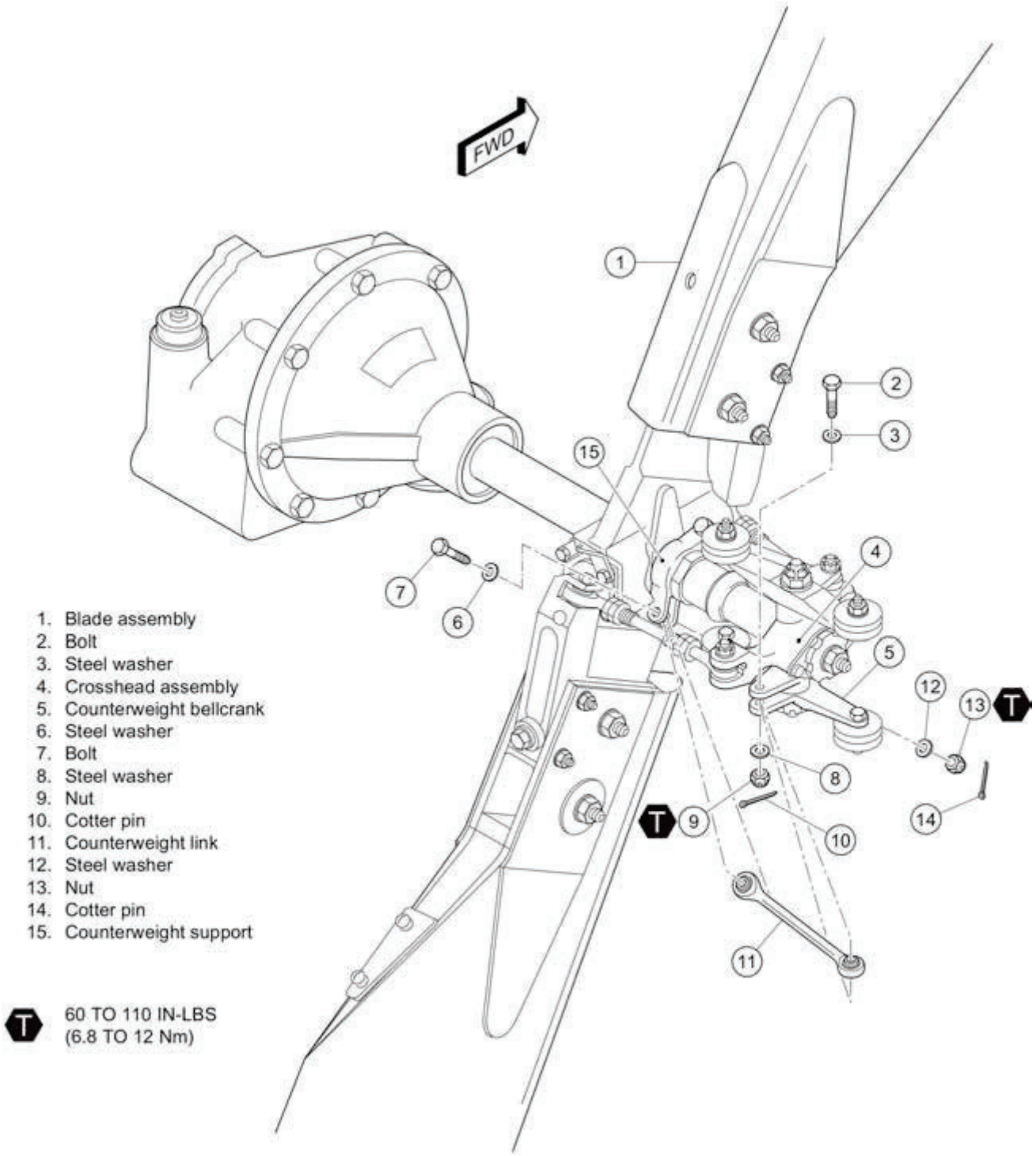
ICN-412-A-643000-B-97499-00001-A-001-01

Figure 1. Tail Rotor Controls - Counterweight Link - Removal and Installation (Sheet 1 of 1)

Counterweight Link - Installation

Procedure

1. Place the counterweight link (11, Figure 1) into the counterweight support (15) and loosely install the bolt (7), steel washers (6 and 12), and nut (13). Make sure the head of the bolt is installed in the opposite direction of rotation.
2. Secure the counterweight link (11) to the counterweight bellcrank (5) with the bolt (2), steel washers (3 and 8), and nut (9). Make sure the head of the bolt is installed in the opposite direction of rotation.
3. Torque the nut (9) 60 to 110 inch-pounds (6.8 to 12 Nm) and install the new cotter pin (10).
4. Torque the nut (13) 60 to 110 inch-pounds (6.8 to 12 Nm) and install the new cotter pin (14).
5. Move the counterweight bellcrank (5) to check for freedom of movement.



ICN-412-A-643000-B-97499-00001-A-001-01

Figure 1. Tail Rotor Controls - Counterweight Link - Removal and Installation (Sheet 1 of 1)

Pitch Control Links - Removal

Procedure

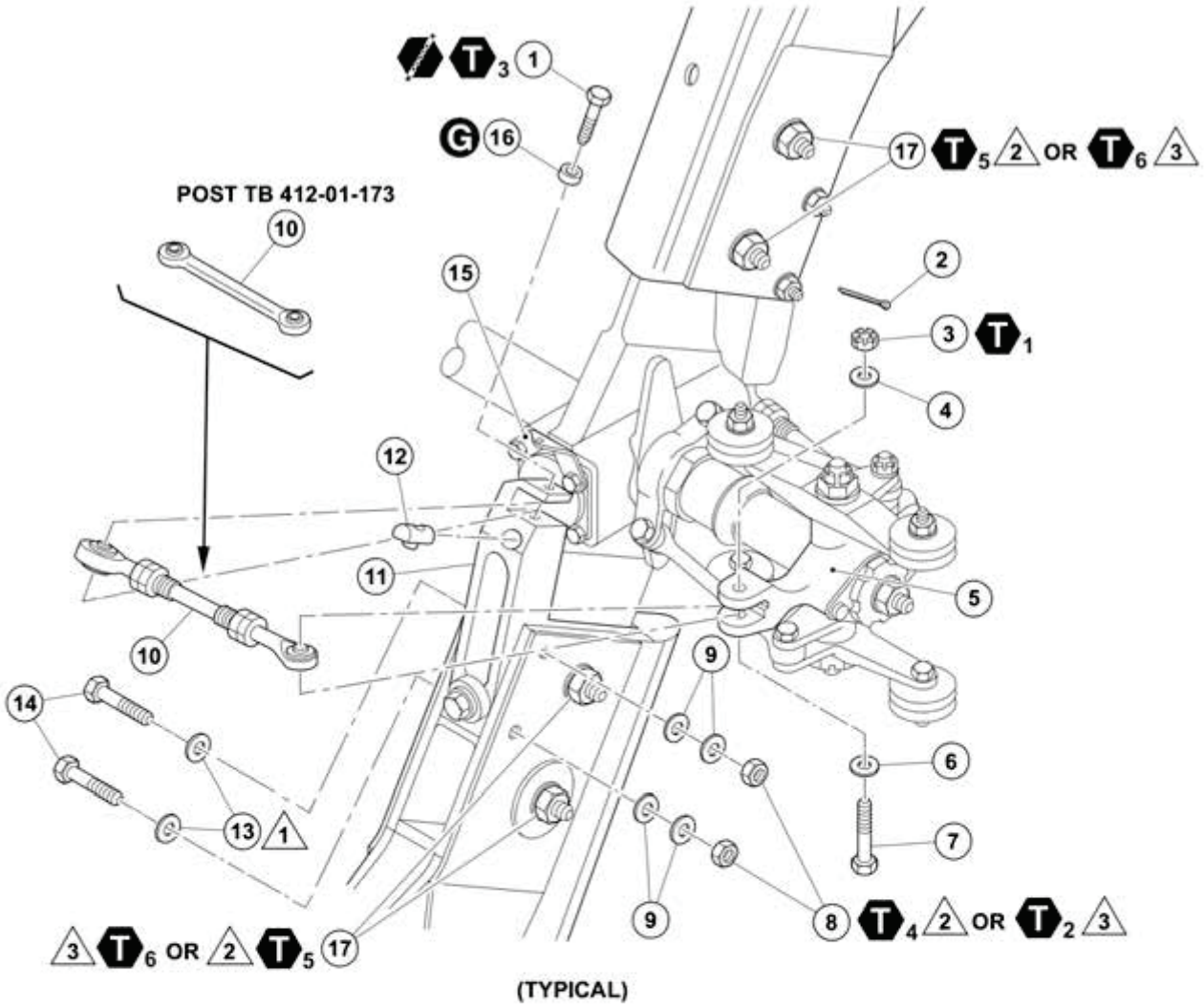
1. Remove the cotter pin (2, Figure 1), nut (3), and steel washer (4). Discard the cotter pin (2). Remove the bolt (7) and the steel washer (6) to disconnect the pitch link (10) from the crosshead (5). Loosely install the bolt, steel washers, and nut to prevent loss.

2. Loosen the nuts (17). Cut and remove lockwire from the bolt (1). Loosen the bolt (1). Remove the nuts (8) and the steel washers (9) from the two bolts (14). Remove the two bolts (14) and the steel washers (13) and slide the pitch horn (11) out so the bolt will clear the bracket (15) on the hub.

3. Remove the bolt (1) and separate the pitch link (10) from the pitch horn (11). Remove the floating bushing (16) and the barrel nut and retainer (12) from the pitch horn (11).

4. Loosely reinstall the bolt (1), floating bushing (16), and barrel nut and retainer (12) in the pitch horn (11) and reinstall the pitch horn (11) on the hub and blade with the bolts (14), steel washers (9 and 13), and nuts (8) to prevent loss.

5. Remove the opposite pitch link in the same manner.



1. Bolt
2. Cotter pin
3. Nut
4. Steel washer
5. Crosshead
6. Steel washer
7. Bolt
8. Nut
9. Steel washer
10. Pitch link
11. Pitch horn
12. Barrel nut and retainer
13. Steel washer
14. Bolt
15. Bracket
16. Floating bushing
17. Nut

G CORROSION PREVENTIVE COMPOUND (C-101)

LOCKWIRE (C-405)

T1 110 TO 165 IN-LBS (12 TO 19 Nm)

3 T2 130 TO 150 IN-LBS (15 TO 17 Nm)

T3 135 IN-LBS (15 Nm)

2 T4 150 IN-LBS (17 Nm)

2 T5 500 TO 550 IN-LBS (57 TO 62 Nm)

3 T6 45 TO 54 FT-LBS (61 TO 73 Nm)

NOTES

1. Chamfered side of washers (13) should face toward head of bolts (14).
2. Torque applicable when tail rotor blades 212-010-750 installed.
3. Torque applicable when tail rotor blades 412-016-100 installed.

ICN-412-A-643000-B-97499-00002-A-001-01

Figure 1. Tail Rotor Controls - Pitch Control Link - Removal and Installation (Sheet 1 of 1)

Pitch Control Links - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Lockwire	C-405	AR	

Procedure

NOTE

TB 412-01-173: Post SN: 33214-Subsequent or 34037-Subsequent or 36225-Subsequent:

Helicopters have fixed length control pitch links (10, Figure 1) and do not require adjustment or tracking of the tail rotor.

1. Adjust the length of the pitch link (10) to obtain a length between the hole centers of 6.170 to 6.190 inches (156.72 to 157.23 mm). Lightly tighten the jam nut on the adjustable end. Refer to 412-A-18-12-01-01A-373A-A / 00186 for tail rotor tracking.

NOTE

Replacement of the adjustable pitch link (10) 212-011-710-001 with the fixed length pitch link 212-310-701-101 requires the replacement of the tail rotor pitch horn (11) 212-011-708-001 with the pitch horn 212-011-708-105.

The configuration of both pitch horns (11) and both pitch links (10) must be the same (TB 212-01-185).

2. Remove the loosely installed pitch horn (11) and the attaching hardware. Apply a thin coat of corrosion preventive compound (C-101) to the floating bushings (16), mating surfaces of the pitch horn, and tail rotor blade.

NOTE

TB 412-01-173: Pre:

The riveted or the non-riveted end of the pitch link (10) can be put in the pitch horn (11), as long as both pitch links are installed the same (applicable to pitch links only).

3. Position the pitch link (10) in the pitch horn (11) and install the floating bushing (16) and the bolt (1). Do not torque the bolt (1) at this time.

NOTE

Make sure the chamfered sides of the steel washers (13) are installed toward the head of the bolt (14).

Applicable: : Installed:

4. Install the pitch horn (11) on the hub and blade with the bolts (14), steel washers (9 and 13), and nuts (8). Torque the nuts (8) 150 inch-pounds (17 Nm).

Applicable: : Installed:

5. Torque the nuts (17) 500 to 550 inch-pounds (57 to 62 Nm).

NOTE

Make sure the chamfered sides of the steel washers (13) are installed toward the head of the bolt (14).

Applicable: : Installed:

6. Install the pitch horn (11) on the hub and blade with the bolts (14), steel washers (9 and 13), and nuts (8). Torque the nuts (8) 130 to 150 inch-pounds (15 to 17 Nm).

Applicable: : Installed:

7. Torque the nuts (17) 45 to 54 foot-pounds (61 to 73 Nm).

8. Torque the bolt (1) 135 inch-pounds (15 Nm) and secure with lockwire (C-405).

NOTE

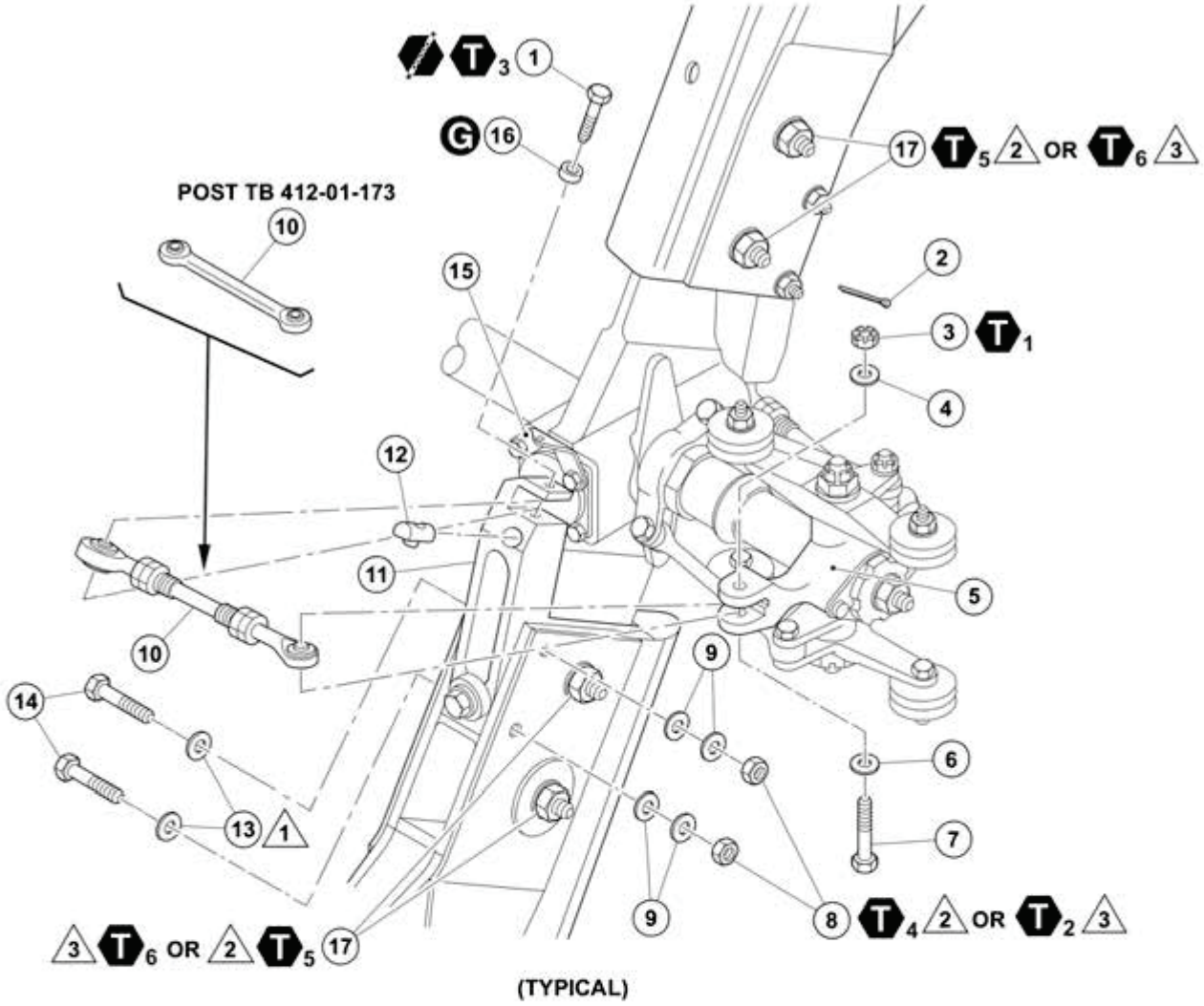
Make sure the head of the bolt (7) is installed in the direction of rotation.

NOTE

Helicopters that have TB 412-01-173 incorporated or S/N 33214 and subsequent, S/N 34037 and subsequent, and 36225 and subsequent do not require tightening of the link jam nut.

9. Place the end of the pitch link (10) into the crosshead (5) and tighten the link jam nut. Install the bolt (7), steel washers (4 and 6), and nut (3). Torque the nut (3) 110 to 165 inch-pounds (12 to 19 Nm) and install the new cotter pin (2).

10. Install the opposite pitch link in the same manner.



1. Bolt
2. Cotter pin
3. Nut
4. Steel washer
5. Crosshead
6. Steel washer
7. Bolt
8. Nut
9. Steel washer
10. Pitch link
11. Pitch horn
12. Barrel nut and retainer
13. Steel washer
14. Bolt
15. Bracket
16. Floating bushing
17. Nut

G CORROSION PREVENTIVE COMPOUND (C-101)

LOCKWIRE (C-405)

T1 110 TO 165 IN-LBS (12 TO 19 Nm)

3 T2 130 TO 150 IN-LBS (15 TO 17 Nm)

T3 135 IN-LBS (15 Nm)

2 T4 150 IN-LBS (17 Nm)

2 T5 500 TO 550 IN-LBS (57 TO 62 Nm)

3 T6 45 TO 54 FT-LBS (61 TO 73 Nm)

NOTES

1. Chamfered side of washers (13) should face toward head of bolts (14).
2. Torque applicable when tail rotor blades 212-010-750 installed.
3. Torque applicable when tail rotor blades 412-016-100 installed.

ICN-412-A-643000-B-97499-00002-A-001-01

Figure 1. Tail Rotor Controls - Pitch Control Link - Removal and Installation (Sheet 1 of 1)

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Control Tube - Removal

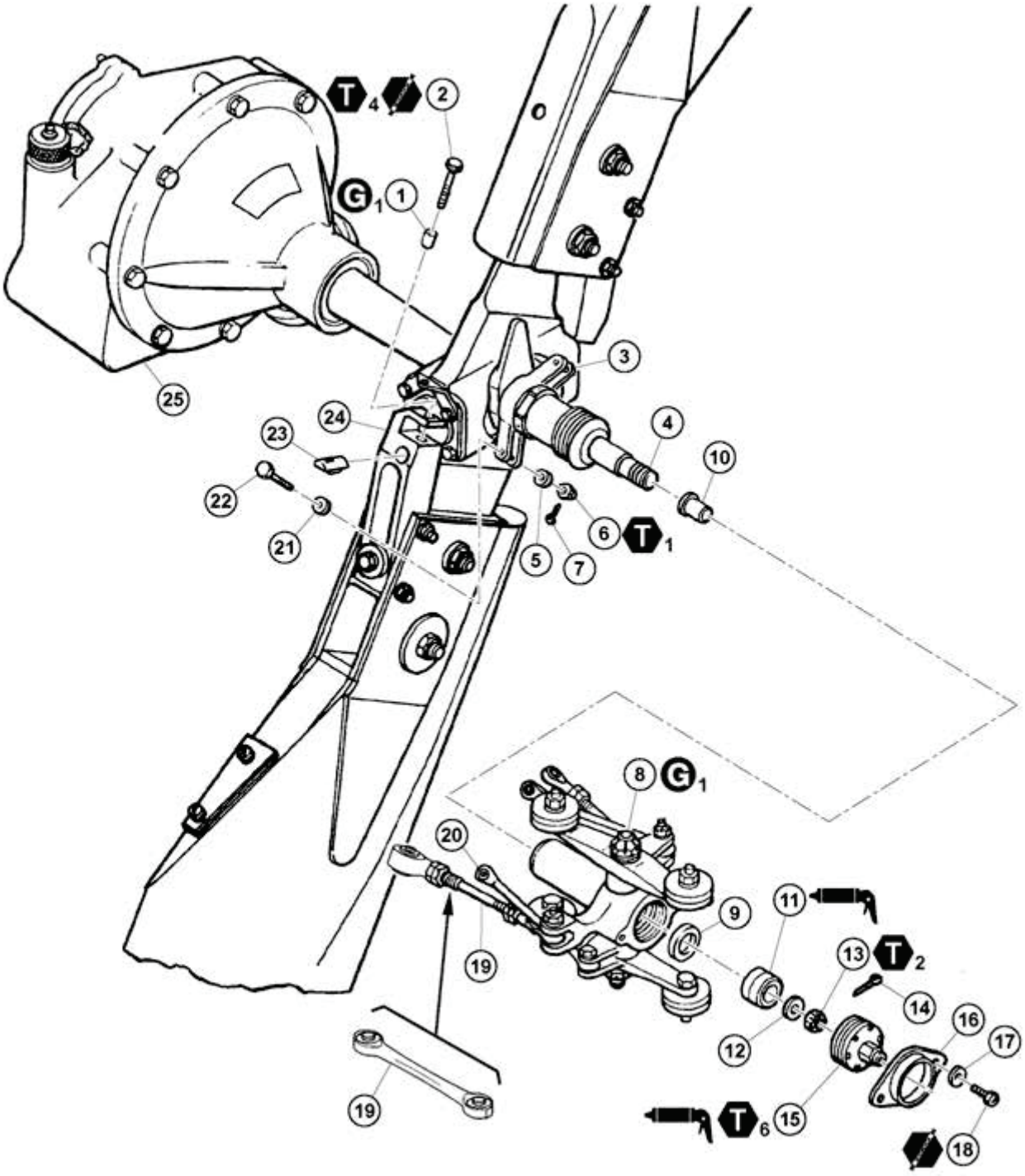
Procedure

1. Remove the crosshead assembly (8, Figure 1) and the sleeve (10) from the control tube (4) (412-A-64-00-00-00A-520A-A / 01229).

2. Remove the cotter pin (10, Figure 2), nut (11), and steel washer (12). Discard the cotter pin (10). Remove the bolt (14) and the steel washer (13) to disconnect the lever (5) from the control tube (15). Loosely reinstall the bolt, steel washers, and nut to prevent loss.

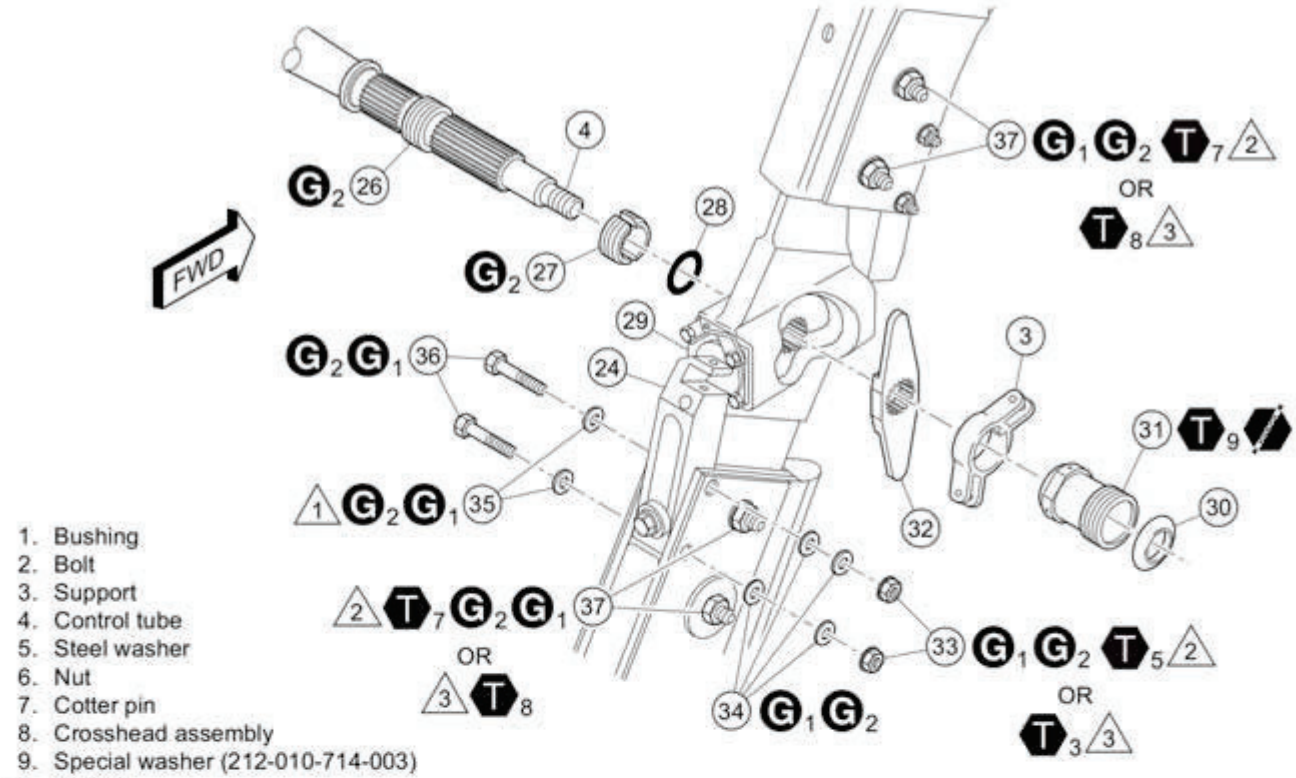
3. Remove the cotter pin (4), nut (3), and steel washer (6). Discard the cotter pin (4). Remove the bolt (9) and the steel washer (8) to disconnect the lever (5) from the idler (7). Loosely reinstall the bolt, steel washers, and nut to prevent loss.

4. Pull the control tube (15) from the housing (2).



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Figure 1. Tail Rotor Hub and Blade Assembly - Removal and Installation (Sheet 1 of 2)



1. Bushing
2. Bolt
3. Support
4. Control tube
5. Steel washer
6. Nut
7. Cotter pin
8. Crosshead assembly
9. Special washer (212-010-714-003)
10. Sleeve
11. Bearing
12. Steel washer
13. Nut
14. Cotter pin
15. Retainer
16. Lock
17. Steel washer
18. Screw
19. Pitch link
20. Counterweight link
21. Steel washer
22. Bolt
23. Barrel nut and retainer
24. Pitch horn
25. Tail rotor gearbox
26. Shaft
27. Cone set
28. Packing
29. Hub and blade assembly
30. Shield
31. Nut
32. Static stop
33. Nut
34. Steel washer
35. Steel washer
36. Bolt
37. Nut

G1 CORROSION PREVENTIVE COMPOUND (C-101)

G2 CORROSION PREVENTIVE COMPOUND (C-104)

GREASE (C-001)

LOCKWIRE (C-405)

T1 60 TO 110 IN-LBS (6.8 TO 12 Nm)

T2 70 TO 100 IN-LBS (7.9 TO 11 Nm)

3 T3 130 TO 150 IN-LBS (15 TO 17 Nm)

T4 135 IN-LBS (15 Nm)

2 T5 150 IN-LBS (17 Nm)

T6 300 IN-LBS (34 Nm)

2 T7 500 TO 550 IN-LBS (57 TO 62 Nm)

3 T8 45 TO 54 FT-LBS (61 TO 73 Nm)

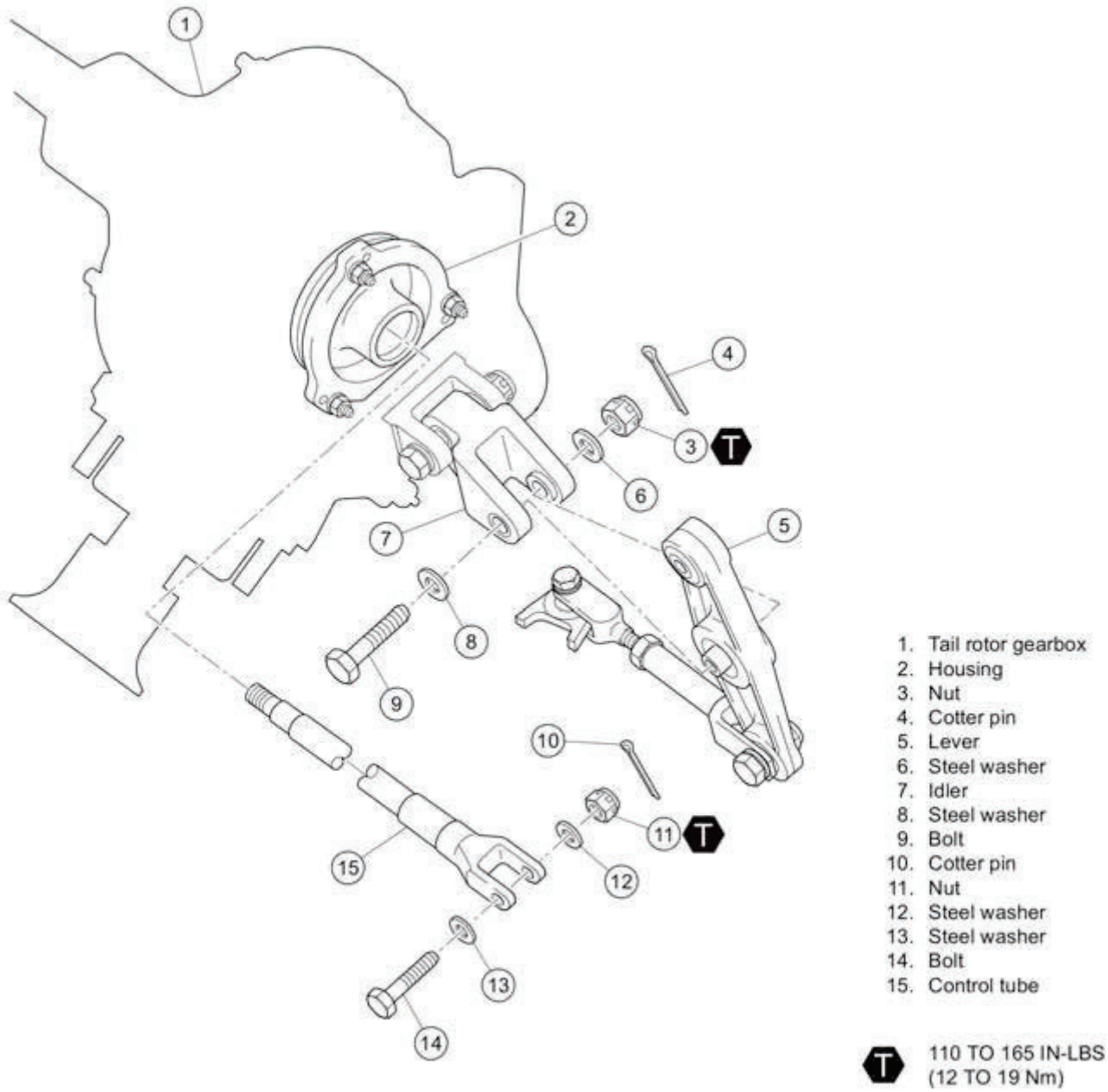
T9 900 IN-LBS (102 Nm)

NOTES

1. Chamfered side of washers (35) should face toward head of bolts (36).
2. Torque applicable when tail rotor blades 212-010-750 installed.
3. Torque applicable when tail rotor blades 412-016-100 installed.

ICN-412-A-640000-B-97499-00027-A-001-01

Figure 1. Tail Rotor Hub and Blade Assembly - Removal and Installation (Sheet 2 of 2)



ICN-412-A-643000-B-97499-00003-A-001-01

Figure 2. Tail Rotor Controls - Control Tube - Removal and Installation (Sheet 1 of 1)

Control Tube - Installation

Procedure

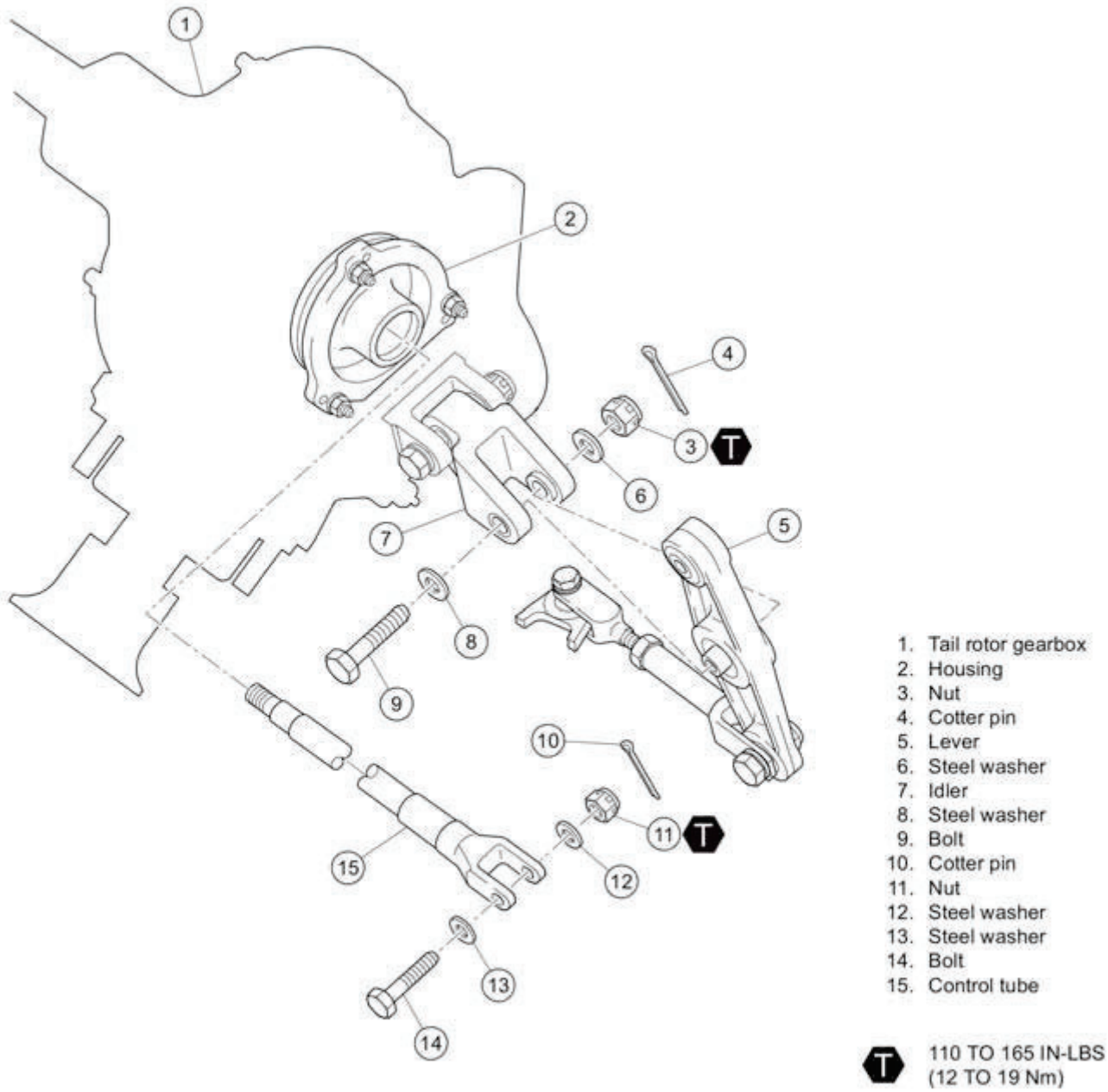
1. Thoroughly clean the bearing inside the housing (2, Figure 1) with a clean, dry cloth. Lubricate the bearing and the control tube (15) with gearbox oil (412-A-12-01-00-00A-040A-A / 00141).

2. Install the control tube (15) into the tail rotor gearbox (1) while carefully guiding it through the housing (2).

3. Rotate the lever (5) into the idler (7) and install the bolt (9), steel washers (6 and 8), and nut (3). Torque the nut (3) 110 to 165 inch-pounds (12 to 19 Nm) and install the new cotter pin (4).

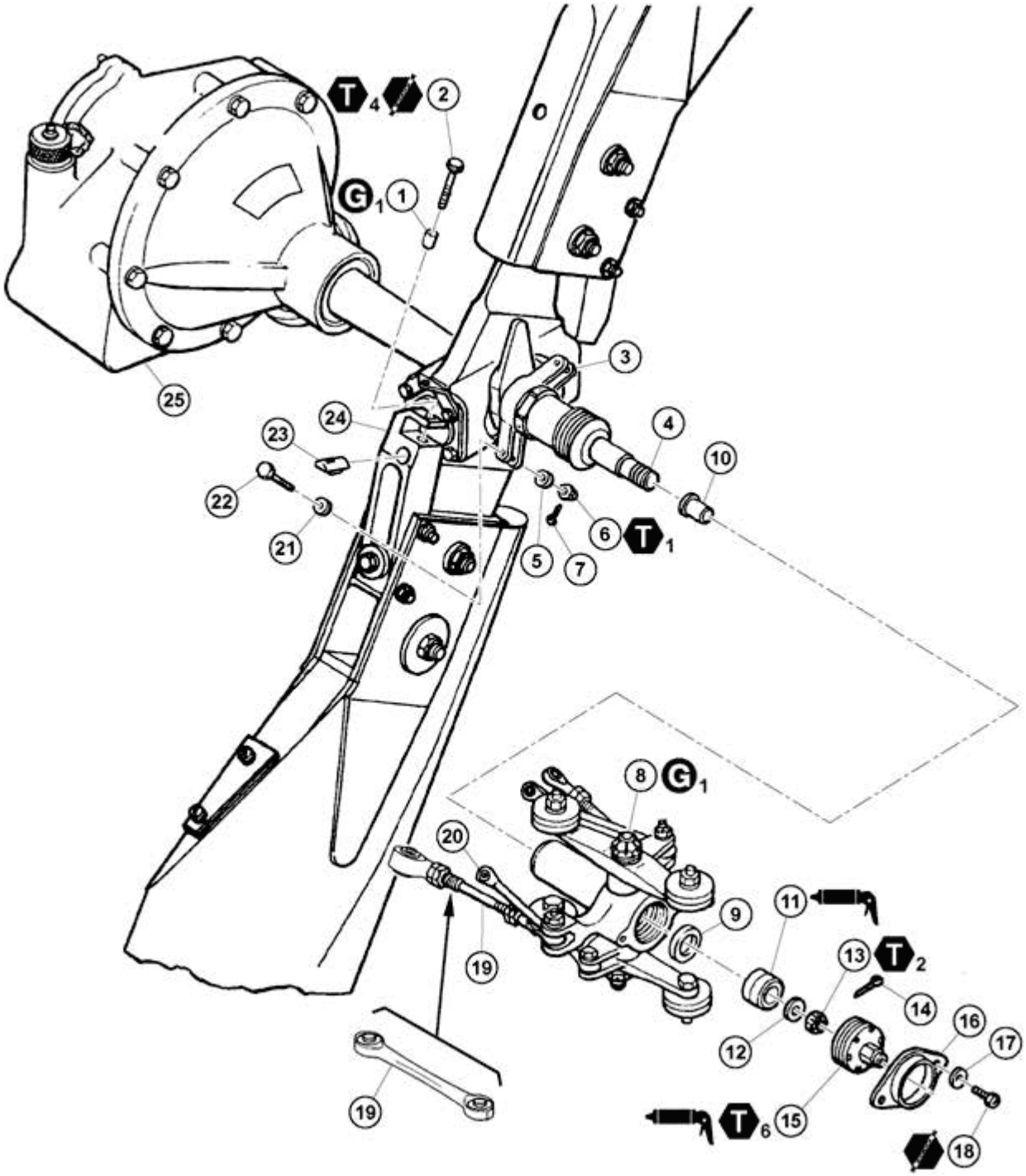
4. Position the lever (5) into the clevis of the control tube (15) and install the bolt (14), steel washers (12 and 13), and nut (11). Torque the nut (11) 110 to 165 inch-pounds (12 to 19 Nm) and install the new cotter pin (10).

5. Install the sleeve (10, Figure 2) and the crosshead assembly (8) onto the control tube (4) (412-A-64-00-00-00A-720A-A / 01232).



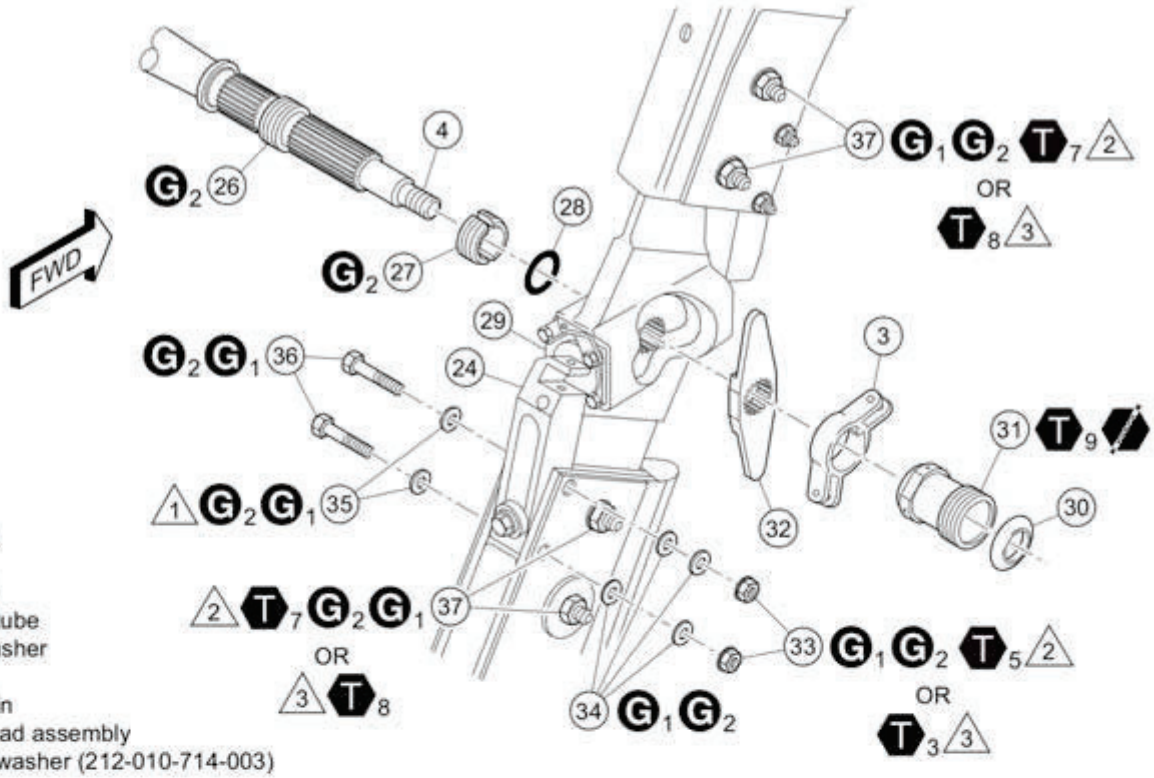
ICN-412-A-643000-B-97499-00003-A-001-01

Figure 1. Tail Rotor Controls - Control Tube - Removal and Installation (Sheet 1 of 1)



ICN-412-A-640000-B-97499-00026-A-002-01

Figure 2. Tail Rotor Hub and Blade Assembly - Removal and Installation (Sheet 1 of 2)



1. Bushing
2. Bolt
3. Support
4. Control tube
5. Steel washer
6. Nut
7. Cotter pin
8. Crosshead assembly
9. Special washer (212-010-714-003)
10. Sleeve
11. Bearing
12. Steel washer
13. Nut
14. Cotter pin
15. Retainer
16. Lock
17. Steel washer
18. Screw
19. Pitch link
20. Counterweight link
21. Steel washer
22. Bolt
23. Barrel nut and retainer
24. Pitch horn
25. Tail rotor gearbox
26. Shaft
27. Cone set
28. Packing
29. Hub and blade assembly
30. Shield
31. Nut
32. Static stop
33. Nut
34. Steel washer
35. Steel washer
36. Bolt
37. Nut

G_1 CORROSION PREVENTIVE COMPOUND (C-101)

G_2 CORROSION PREVENTIVE COMPOUND (C-104)

GREASE (C-001)

LOCKWIRE (C-405)

T_1 60 TO 110 IN-LBS (6.8 TO 12 Nm)

T_2 70 TO 100 IN-LBS (7.9 TO 11 Nm)

3 T_3 130 TO 150 IN-LBS (15 TO 17 Nm)

T_4 135 IN-LBS (15 Nm)

2 T_5 150 IN-LBS (17 Nm)

T_6 300 IN-LBS (34 Nm)

2 T_7 500 TO 550 IN-LBS (57 TO 62 Nm)

3 T_8 45 TO 54 FT-LBS (61 TO 73 Nm)

T_9 900 IN-LBS (102 Nm)

NOTES

1. Chamfered side of washers (35) should face toward head of bolts (36).
2. Torque applicable when tail rotor blades 212-010-750 installed.
3. Torque applicable when tail rotor blades 412-016-100 installed.

ICN-412-A-640000-B-97499-00027-A-001-01

Figure 2. Tail Rotor Hub and Blade Assembly - Removal and Installation (Sheet 2 of 2)

Control Housing - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Jackscrew Set	T101308	1	

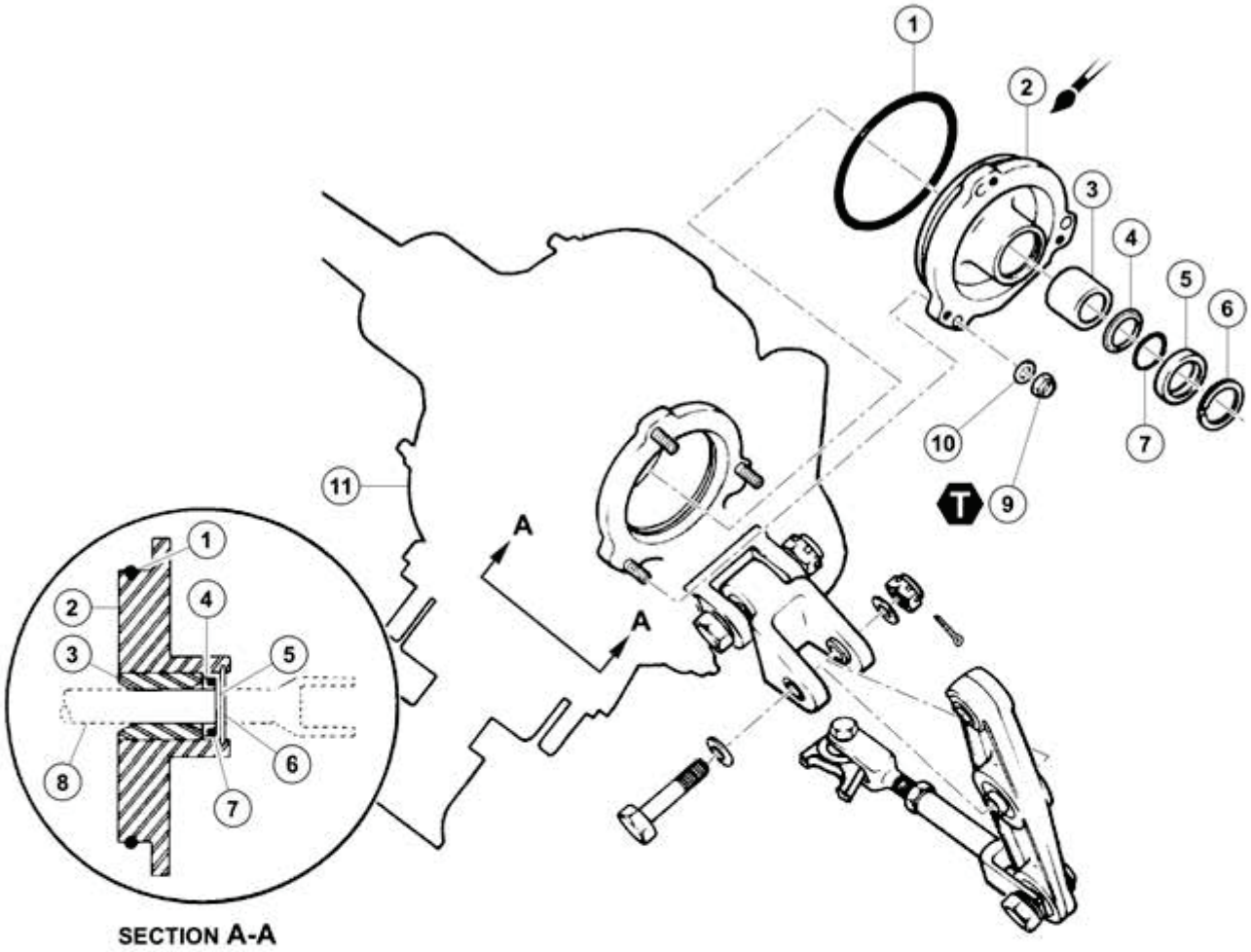
Procedure

1. Remove the tail rotor control tube (412-A-64-30-00-03A-520A-A / 01238).

2. Remove the nuts (9, Figure 1) and the steel washers (10). Remove the housing (2) with the jackscrew set (T101308). Remove and discard the preformed packing (1).

3. Remove the retainer ring (6) from the housing (2).

4. Press the bearing (3), excluder (4), excluder housing (5), and seal (7) out of the housing (2) from the inboard side. Discard the seal (7).



ICN-412-A-640000-B-97499-00028-A-001-01

Figure 1. Tail Rotor Controls - Control Housing - Removal and Installation (Sheet 1 of 1)

Control Housing - Installation

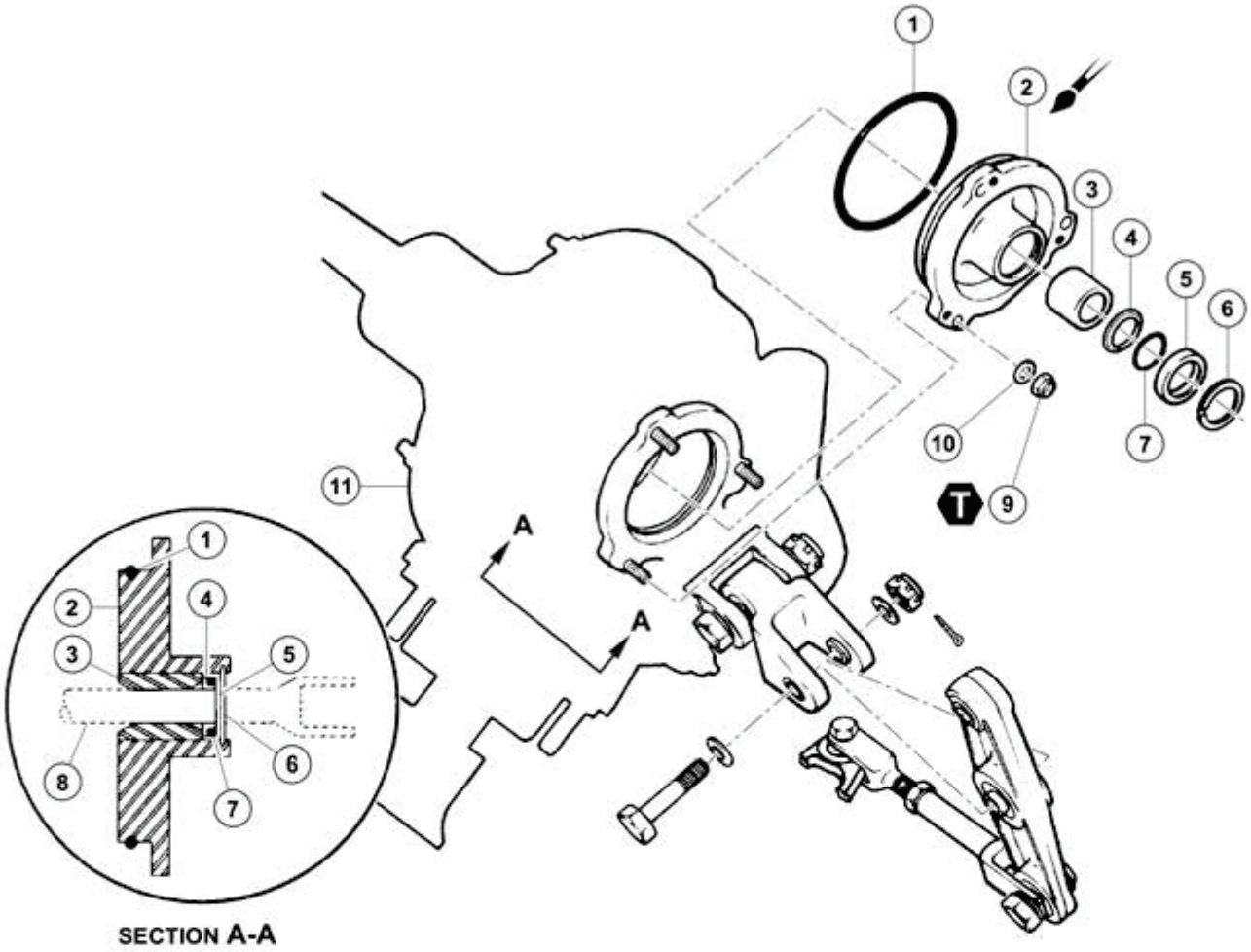
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	

Procedure

1. Inspect all components for wear and damage (BHT-412-CR&O).
2. Press the bearing (3, Figure 1) into the housing (2).
3. Assemble the excluder (4), new seal (7), and excluder housing (5) and install into the housing (2) (Section A-A).
4. Install the retainer ring (6).
5. Lubricate the new preformed packing (1) with the gearbox oil (412-A-12-01-00-00A-040A-A / 00141) and install on the housing (2).
6. Apply a film of gearbox oil (412-A-12-00-00-00A-040A-A / 00126) to the inside diameter of the bearing (3).
7. Apply a film of wet epoxy polyamide primer (C-204) to the mating surface of the flange of the housing (2).
8. Install the housing (2) over the studs in the case and install the steel washers (10) and the nuts (9). Alternately and evenly torque the nuts (9) 60 inch-pounds (6.78 Nm).
9. Install the tail rotor control tube (412-A-64-30-00-03A-720A-A / 01239).



1. Preformed packing
2. Housing
3. Bearing
4. Excluder
5. Excluder housing
6. Retainer ring
7. Seal
8. Control tube
9. Nut
10. Steel washer
11. Tail rotor gearbox

PRIMER (C-204)

60 IN-LBS (6.8 Nm)

ICN-412-A-640000-B-97499-00028-A-001-01

Figure 1. Tail Rotor Controls - Control Housing - Removal and Installation (Sheet 1 of 1)

Tail Rotor Lever - Removal

Procedure

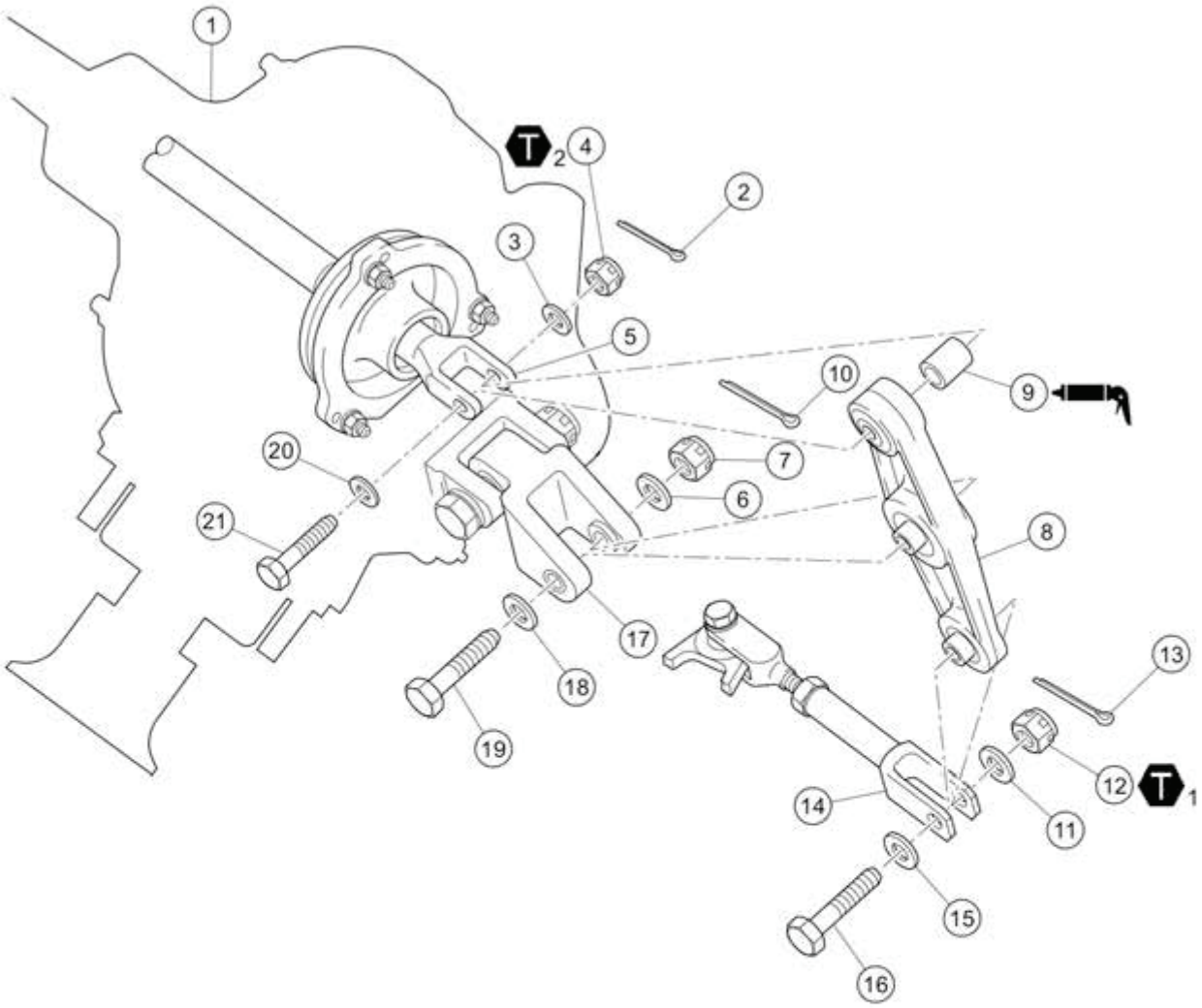
1. Remove the cotter pin (2, Figure 1), nut (4), and steel washer (3). Discard the cotter pin (2). Remove the bolt (21) and the steel washer (20) to disconnect the lever (8) from the control tube (5). Loosely reinstall the bolt, steel washers, and nut to prevent loss.

2. Remove the cotter pin (10), nut (7), and steel washer (6). Discard the cotter pin (10). Remove the bolt (19) and the steel washer (18) to disconnect the lever (8) from the idler (17). Loosely reinstall the bolt, steel washers, and nut to prevent loss.

3. Rotate the lever (8) around the lower end bearing to check for freedom of movement. If the bearing is tight, use a spring scale to measure the force required to rotate the bearing (install the scale on the lever upper bearing bolt (21)). If the force required exceeds 10 pounds (44.5 N), replace the lever (8).

4. Remove the cotter pin (13), nut (12), and steel washer (11). Discard the cotter pin (13). Remove the bolt (16) and the steel washer (15) to disconnect the lever (8) from the link (14). Loosely reinstall the bolt, steel washers, and nut to prevent loss.

5. Remove the lever (8) from the helicopter. Remove the race (9) from the lever (8).



1. Tail rotor gearbox (ref)
2. Cotter pin
3. Steel washer
4. Nut
5. Control tube
6. Steel washer
7. Nut
8. Lever
9. Race
10. Cotter pin
11. Steel washer
12. Nut
13. Cotter pin
14. Link
15. Steel washer
16. Bolt
17. Idler
18. Steel washer
19. Bolt
20. Steel washer
21. Bolt

T1 60 TO 110 IN-LBS (6.8 TO 12 Nm)

T2 110 TO 165 IN-LBS (12 TO 19 Nm)

GREASE (C-001)

ICN-412-A-640000-B-97499-00029-A-001-01

Figure 1. Tail Rotor Controls - Lever - Removal and Installation (Sheet 1 of 1)

Tail Rotor Lever - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-001	AR	
Drycleaning solvent	C-304	AR	

Procedure

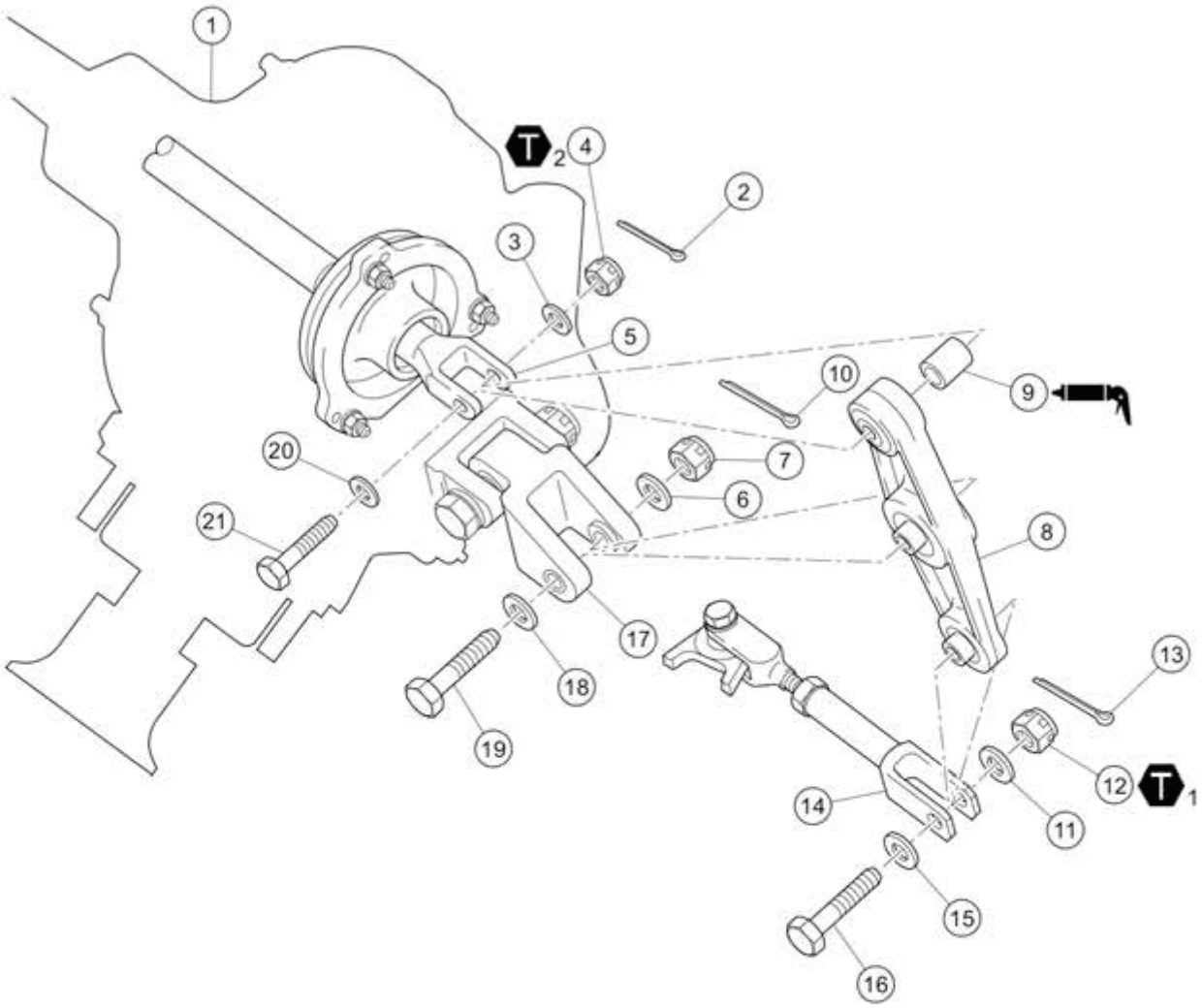
1. Clean the race (9, Figure 1) and the upper end of the lever (8) thoroughly with drycleaning solvent (C-304) and dry with filtered, compressed air. Hand pack the bearing and the race (9) with grease (C-001). Install the race into the bearing in the lever (8).

2. Place the upper end of the lever (8) into the control tube (5) and install the bolt (21), steel washers (3 and 20), and nut (4). Torque the nut (4) 110 to 165 inch-pounds (12 to 19 Nm) and install the new cotter pin (2).

3. Position the lever (8) into the idler (17) and install the bolt (19), steel washers (6 and 18), and nut (7). Torque the nut (7) 110 to 165 inch-pounds (12 to 19 Nm) and install the new cotter pin (10).

4. Position the lever (8) into the link (14) and install the bolt (16), steel washers (11 and 15), and nut (12). Torque the nut (12) 60 to 110 inch-pounds (6.8 to 12 Nm) and install the new cotter pin (13).

5. Move the tail rotor controls through full range of travel to check for freedom of operation.



1. Tail rotor gearbox (ref)
2. Cotter pin
3. Steel washer
4. Nut
5. Control tube
6. Steel washer
7. Nut
8. Lever
9. Race
10. Cotter pin
11. Steel washer
12. Nut
13. Cotter pin
14. Link
15. Steel washer
16. Bolt
17. Idler
18. Steel washer
19. Bolt
20. Steel washer
21. Bolt

T1 60 TO 110 IN-LBS (6.8 TO 12 Nm)

T2 110 TO 165 IN-LBS (12 TO 19 Nm)

GREASE (C-001)

ICN-412-A-640000-B-97499-00029-A-001-01

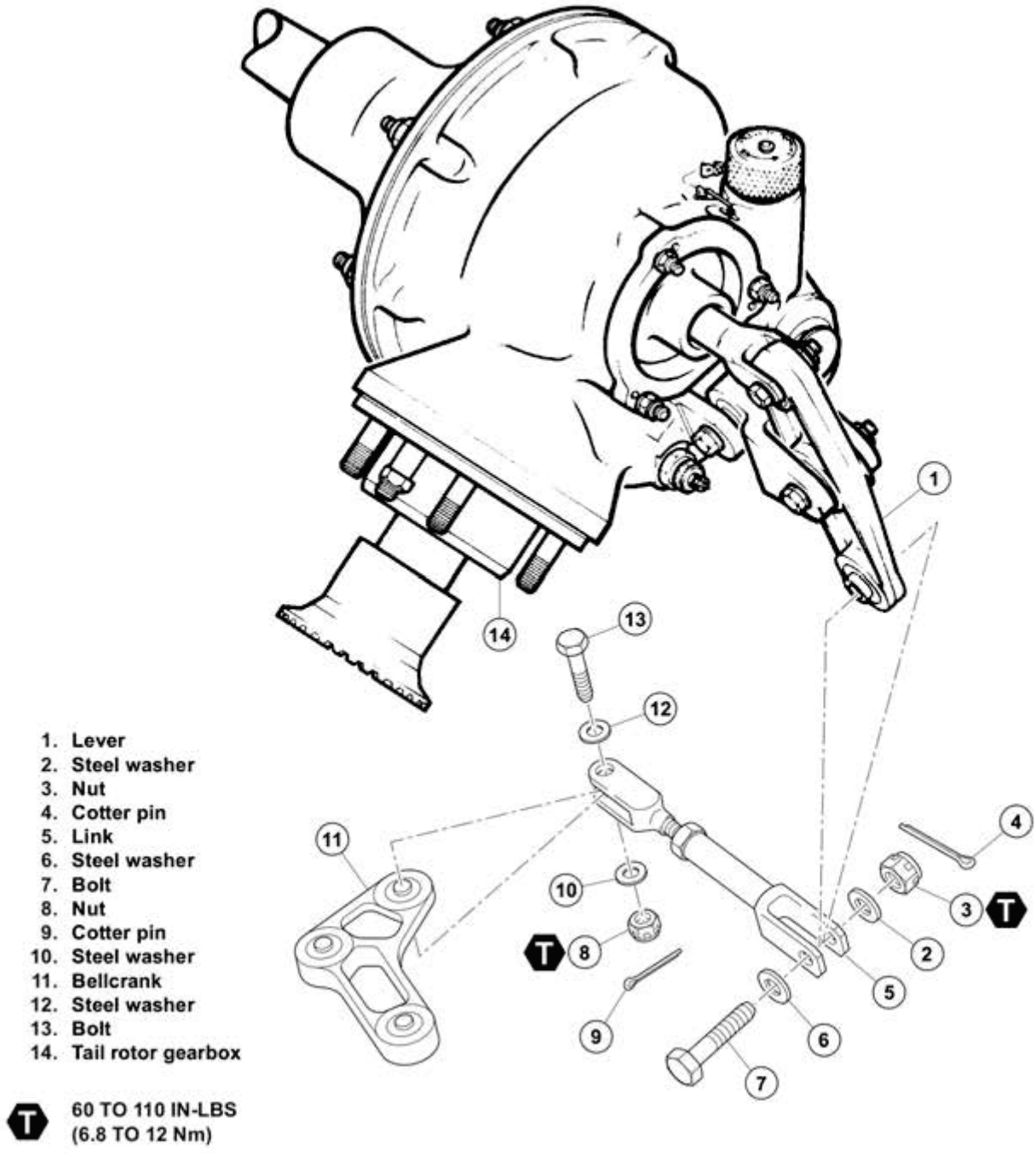
Figure 1. Tail Rotor Controls - Lever - Removal and Installation (Sheet 1 of 1)

Control Link - Removal

Procedure

1. Remove the cotter pin (9, Figure 1), nut (8), and steel washer (10). Discard the cotter pin (9). Remove the bolt (13) and the steel washer (12) to disconnect the link (5) from the bellcrank (11). Loosely reinstall the bolt, steel washers, and nut to prevent loss.

2. Remove the cotter pin (4), nut (3), and steel washer (2). Discard the cotter pin (4). Remove the bolt (7) and the steel washer (6) to disconnect the link (5) from the lever (1). Remove the link from the helicopter. Loosely reinstall the bolt, steel washers, and nut to prevent loss.



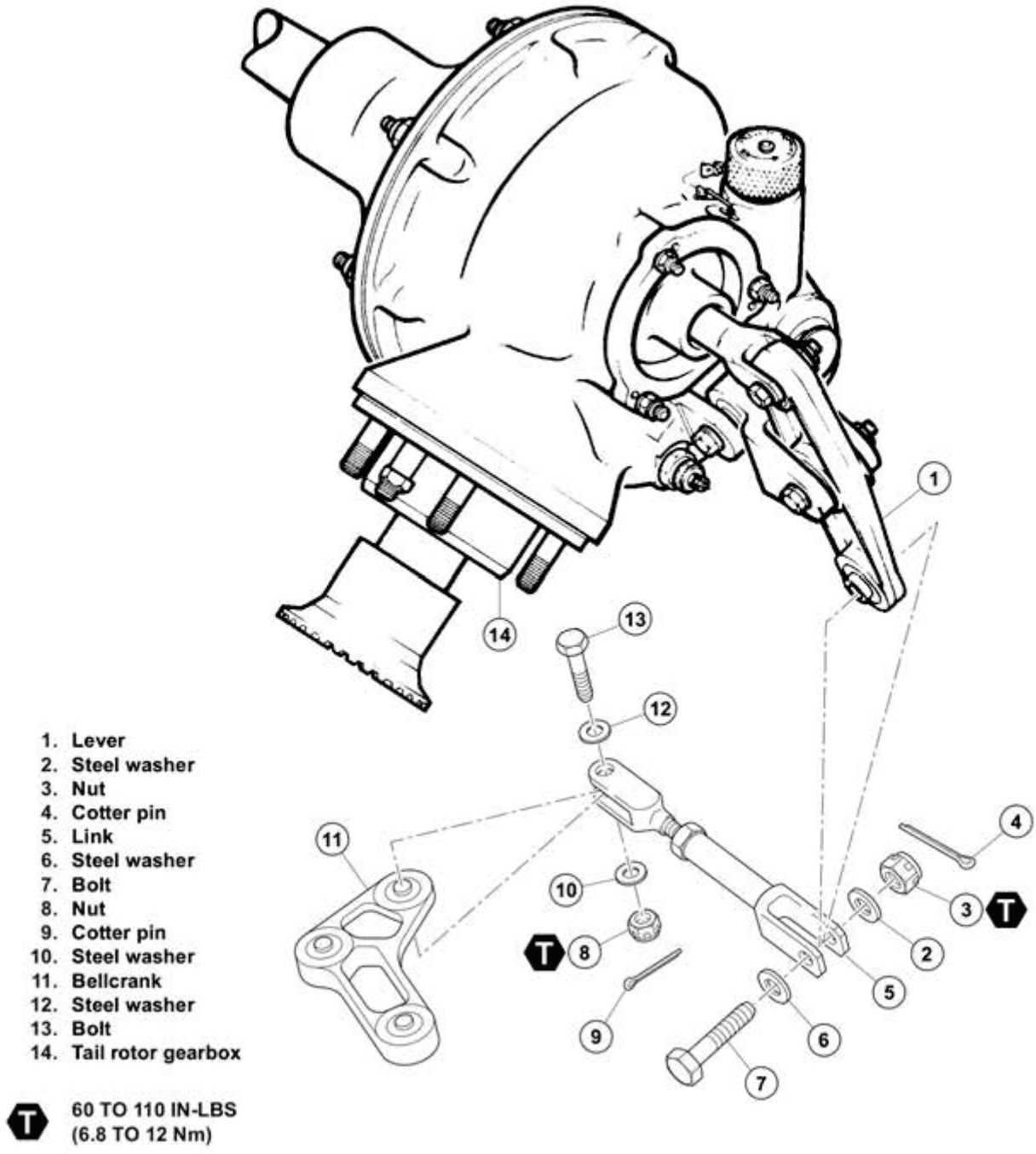
ICN-412-A-640000-B-97499-00030-A-001-01

Figure 1. Tail Rotor Controls - Control Link - Removal and Installation (Sheet 1 of 1)

Control Link - Installation

Procedure

1. Adjust the length of the link (5, Figure 1) to 6.90 inches (175.3 mm), center to center of the bolt holes in either end.
2. Place the link (5) in position and secure to the bellcrank (11) with the bolt (13), steel washers (12 and 10), and nut (8). Do not torque the nut (8) at this time.
3. Secure the link (5) to the lever (1) with the bolt (7), steel washers (6 and 2), and nut (3). Torque the nut (3) 60 to 110 inch-pounds (6.8 to 12 Nm) and install the cotter pin (4).
4. Rig the tail rotor controls as necessary (412-A-67-01-00-11A-276A-A / 01551).
5. Torque the nut (8) 60 to 110 inch-pounds (6.8 to 12 Nm) and install the new cotter pin (9).



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Figure 1. Tail Rotor Controls - Control Link - Removal and Installation (Sheet 1 of 1)

Idler - Removal

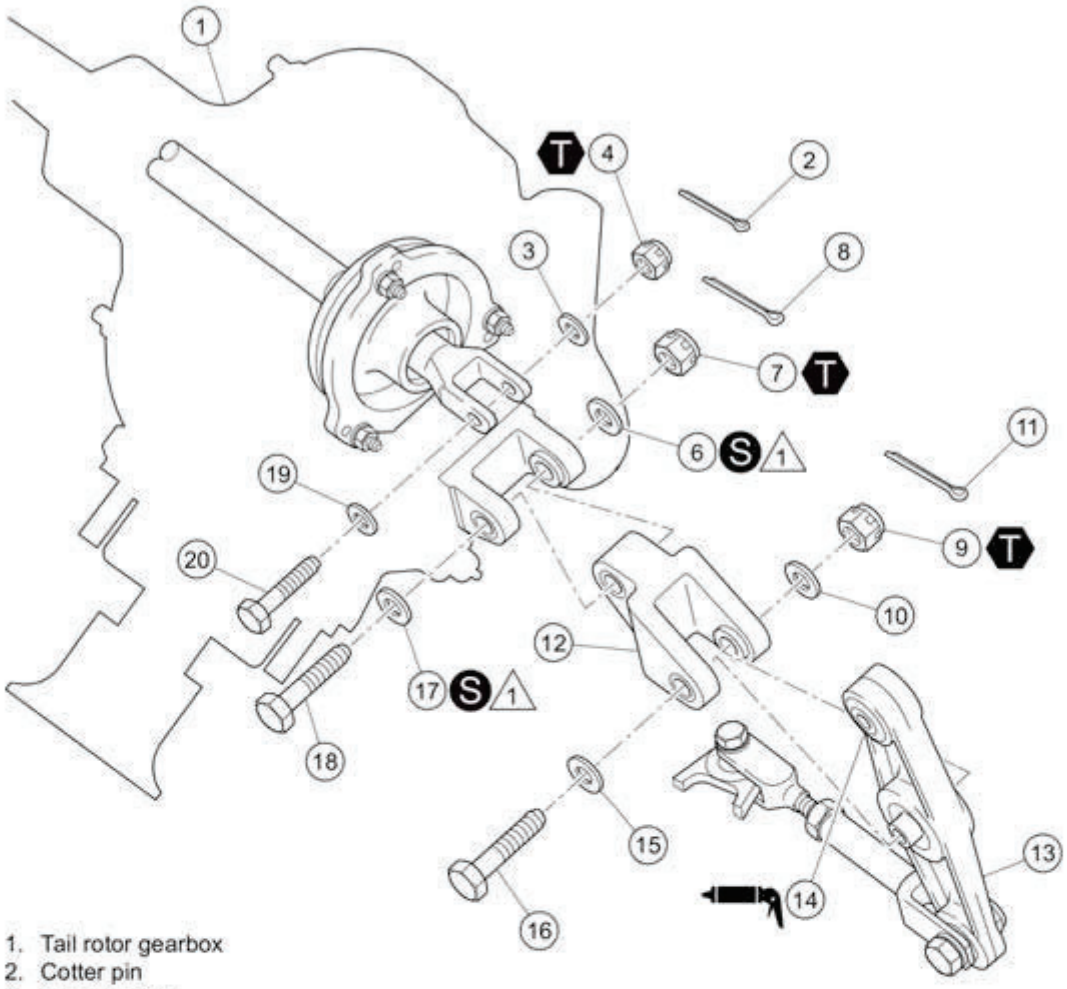
Procedure

1. Remove the cotter pin (2, Figure 1), nut (4), and steel washer (3). Discard the cotter pin (2). Remove the bolt (20) and the steel washer (19) and disconnect the lever (13) from the control tube (5). Loosely reinstall the bolt, steel washers, and nut to prevent loss.

2. Tie the race (14) into the upper end of the lever (13) to prevent loss.

3. Remove the cotter pin (11), nut (9), and steel washer (10). Discard the cotter pin (11). Remove the bolt (16) and the steel washer (15) to disconnect the lever (13) from the idler (12). Loosely reinstall the bolt, steel washers, and nut to prevent loss.

4. Remove the cotter pin (8) and the nut (7). Discard the cotter pin (8). Remove the bolt (18) to remove the idler (12) from the helicopter. Loosely reinstall the bolt and the nut to prevent loss. The steel washers (6 and 17) should be bonded in place.



1. Tail rotor gearbox
2. Cotter pin
3. Steel washer
4. Nut
5. Control tube
6. Steel washer
7. Nut
8. Cotter pin
9. Nut
10. Steel washer
11. Cotter pin
12. Idler
13. Lever
14. Race
15. Steel washer
16. Bolt
17. Steel washer
18. Bolt
19. Steel washer
20. Bolt

GREASE (C-001)

S SEALANT (C-328)

T 110 TO 165 IN-LBS (12 TO 19 Nm)

NOTE

1 Bond washers (6 and 17) to tail rotor gearbox (1) fitting with sealing compound (C-328).

ICN-412-A-640000-B-97499-00031-A-001-01

Figure 1. Tail Rotor Controls - Control Idler - Removal and Installation (Sheet 1 of 1)

Idler - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-001	AR	
Sealant	C-328	AR	

Procedure

1. Bond the steel washers (6 and 17, Figure 1) to the gearbox (1) with sealant (C-328).

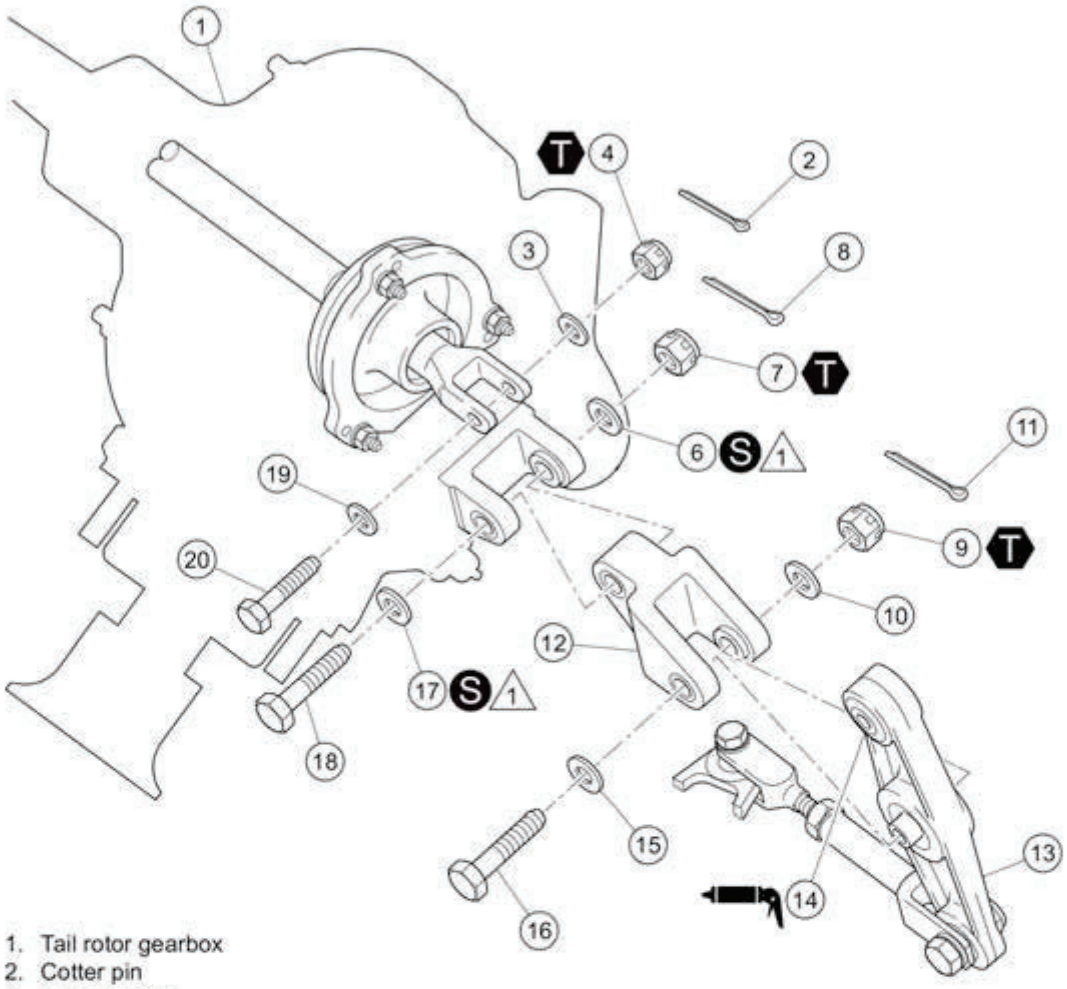
2. Place the idler (12) in position and install the bolt (18) and the nut (7). Torque the nut (7) 110 to 165 inch-pounds (12 to 19 Nm) and install the new cotter pin (8).

3. Place the lever (13) into the idler (12) and install the bolt (16), steel washers (15 and 10), and nut (9). Torque the nut (9) 110 to 165 inch-pounds (12 to 19 Nm) and install the new cotter pin (11).

4. Apply a thin film of grease (C-001) to the race (14) in the upper end of the lever (13).

5. Place the lever (13) into the clevis of the control tube (5) and install the bolt (20), steel washers (3 and 19), and nut (4). Torque the nut (4) 110 to 165 inch-pounds (12 to 19 Nm) and install the new cotter pin (2).

6. Move the tail rotor control through full range of travel to check for freedom of movement.



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Figure 1. Tail Rotor Controls - Control Idler - Removal and Installation (Sheet 1 of 1)

Crosshead Controls - Disassembly

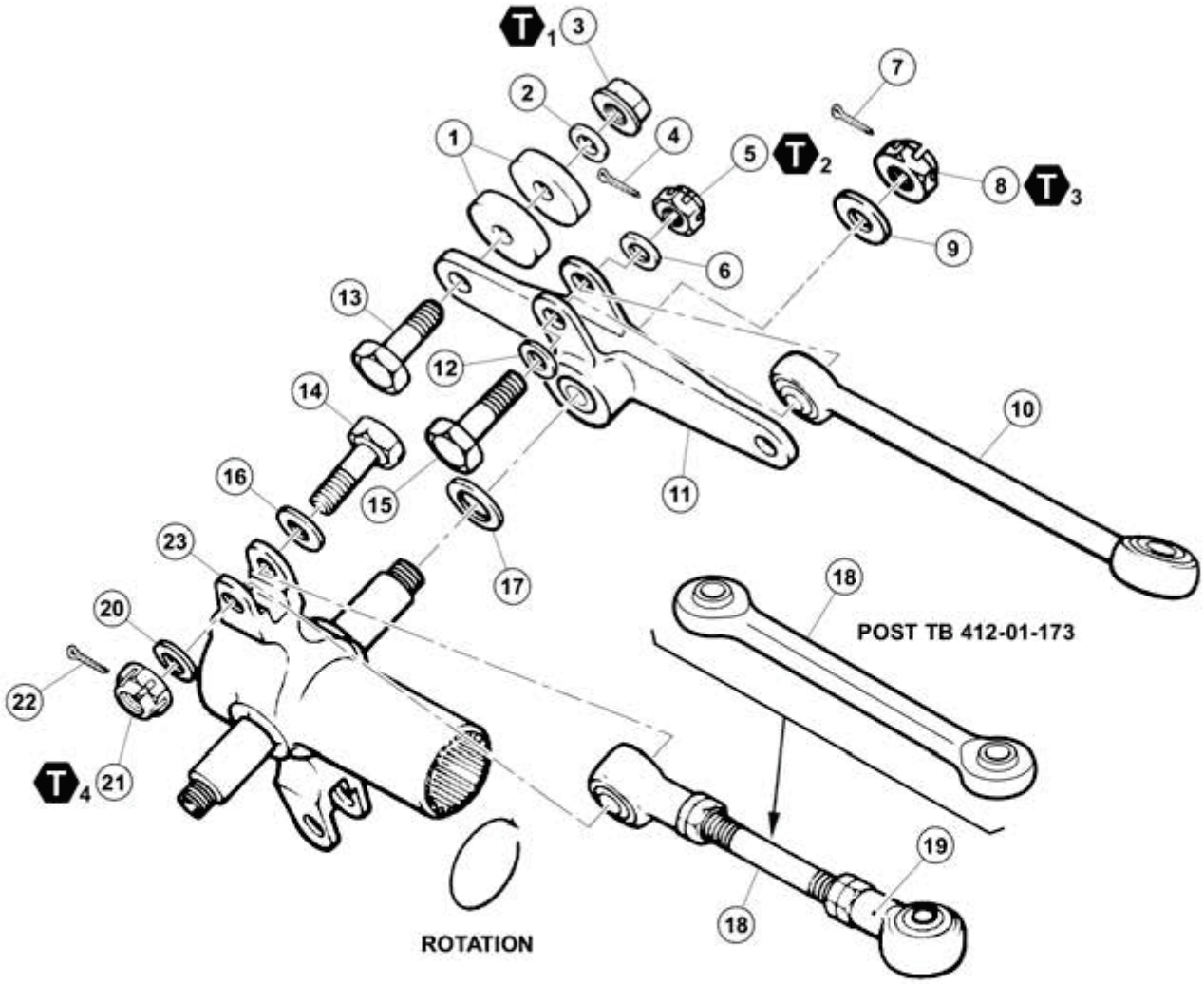
Procedure

1. Remove the cotter pin (4, Figure 1), nut (5), and steel washer (6). Discard the cotter pin (4). Remove the bolt (15) and the steel washer (12) and remove the counterweight link (10) from the counterweight bellcrank (11). Remove the opposite link in the same manner.

2. Remove the cotter pin (22), nut (21), and steel washer (20). Discard the cotter pin (22). Remove the bolt (14) and the steel washer (16) and remove the pitch control link (18) from the crosshead (23). Remove the opposite link in the same manner.

3. Remove the cotter pin (7), nut (8), and special washer (9), and pull the counterweight bellcrank (11) from the crosshead (23). Discard the nut (8) and the cotter pin (7). Secure the special washer (17) to prevent loss. Remove the opposite bellcrank in the same manner..

4. Remove the nut (3), steel washer (2), and bolt (13) and remove the counterweights (1) from the counterweight bellcrank (11). Remove the opposite weights in the same manner.



1. Counterweight
2. Steel washer
3. Nut
4. Cotter pin
5. Nut
6. Steel washer
7. Cotter pin
8. Nut
9. Special washer (212-010-728-001)
10. Counterweight link
11. Counterweight bellcrank
12. Steel washer
13. Bolt
14. Bolt
15. Bolt
16. Steel washer
17. Special washer (212-010-749-001)
18. Pitch control link
19. Rivet
20. Steel washer
21. Nut
22. Cotter pin
23. Crosshead

T1 60 IN-LBS (6.8 Nm)
T2 60 TO 110 IN-LBS (6.8 TO 12 Nm)
T3 100 TO 170 IN-LBS (11 TO 19 Nm)
T4 110 TO 165 IN-LBS (12 TO 19 Nm)

ICN-412-A-643000-B-97499-00004-A-001-01

Figure 1. Tail Rotor Crosshead Controls - Removal and Installation (Sheet 1 of 1)

Crosshead Controls - Assembly

Procedure

1. Install the two counterweights (1, Figure 1) on one end of the counterweight bellcrank (11) and secure with the bolt (13), steel washer (2), and nut (3). Torque the nut (3) 60 inch-pounds (6.78 Nm). Install the two counterweights on the opposite end of the bellcrank in the same manner. Assemble the counterweights on the bellcrank on the opposite side of the crosshead in the same manner.



COUNTERWEIGHT BELLCRANK RETENTION NUT (MS14145L6 OR ITS EQUIVALENT MS17826-6) IS DESIGNATED AS ONE-TIME USE ONLY. IF MAINTENANCE ACTION REQUIRES THAT IT BE REMOVED, THE COUNTERWEIGHT BELLCRANK NUT IS TO BE DISCARDED AND REPLACED WITH A NEW NUT. OVERTORQUING OR INCORRECT APPLICATION OF THE NUT COULD CAUSE MECHANICAL FAILURE RESULTING IN EXCESSIVE VIBRATION AND/OR DAMAGE TO THE TAIL ROTOR BLADES.

2. Install the special washer (17), counterweight bellcrank (11), special washer (9), and new nut (8) on the crosshead (23). Torque the nut (8) 100 to 170 inch-pounds (11 to 19 Nm) and install the new cotter pin (7). Install the opposite bellcrank in the same manner.

3. Position the counterweight bellcrank (11) with the mounting lugs for the counterweight links (10) in the opposite direction of rotation (trailing studs).

4. Place the counterweight link (10) into the lugs of the counterweight bellcrank (11) and install the bolt (15), steel washers (12 and 6), and nuts (5). Torque the nuts (5) 60 to 110 inch-pounds (6.8 to 12 Nm) and install the new cotter pin (4). Install the opposite counterweight link (10) in the same manner.

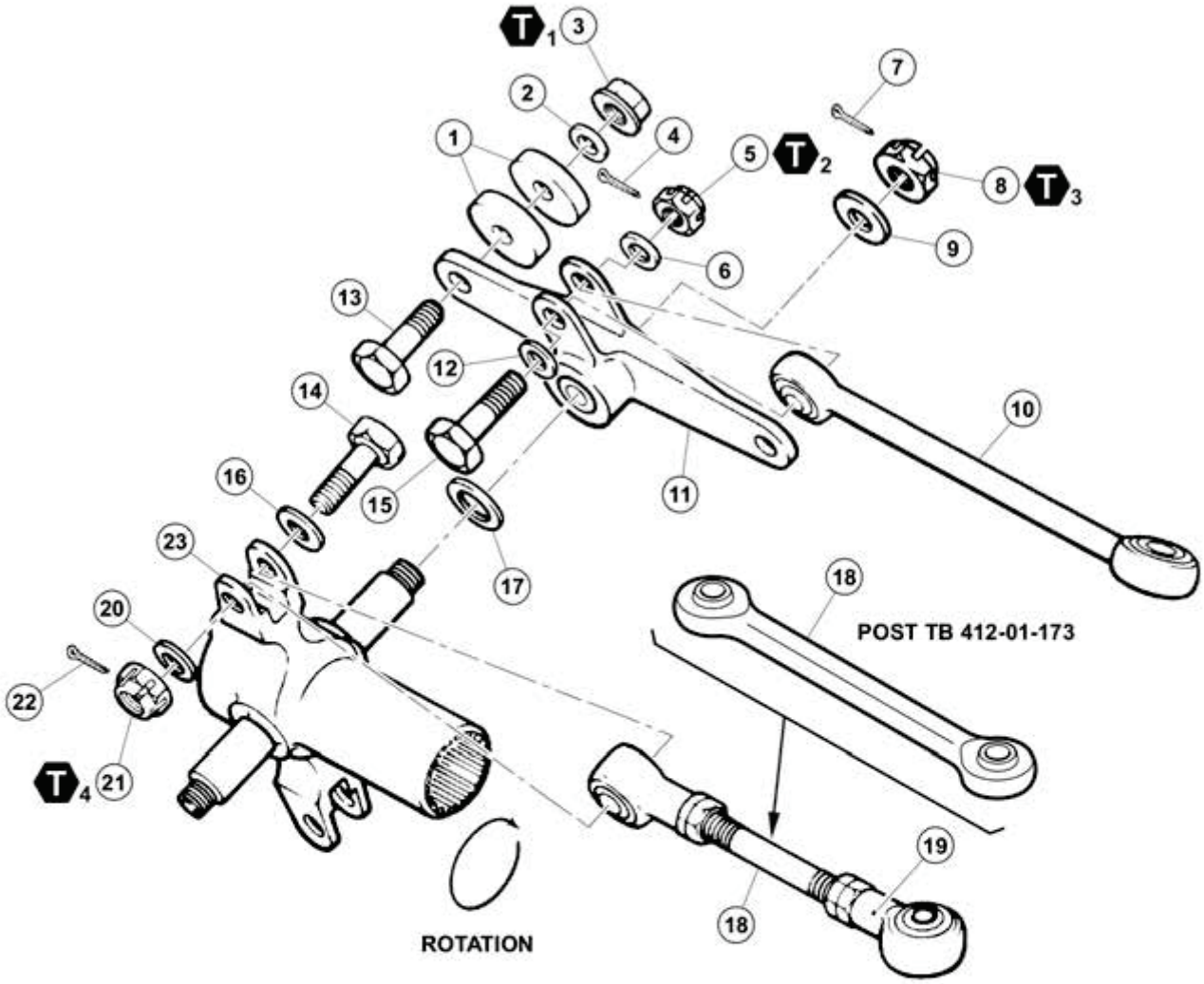
NOTE

TB 412-01-173: Pre:

Either the riveted or the non-riveted ends of the pitch control link (18) can be placed in the crosshead (23), as long as both links are installed the same way (applicable to pitch links only).

5. Place one end of the pitch control link (18) into the crosshead (23). Install the bolt (14), steel washers (16 and 20), and nut (21). Torque the nut (21) 110 to 165 inch-pounds (12 to 19 Nm) and install the new cotter pin (22). Install the opposite pitch control link in the same manner.

6. Check all bearings for freedom of movement.



1. Counterweight
2. Steel washer
3. Nut
4. Cotter pin
5. Nut
6. Steel washer
7. Cotter pin
8. Nut
9. Special washer (212-010-728-001)
10. Counterweight link
11. Counterweight bellcrank
12. Steel washer
13. Bolt
14. Bolt
15. Bolt
16. Steel washer
17. Special washer (212-010-749-001)
18. Pitch control link
19. Rivet
20. Steel washer
21. Nut
22. Cotter pin
23. Crosshead

T1 60 IN-LBS (6.8 Nm)
T2 60 TO 110 IN-LBS (6.8 TO 12 Nm)
T3 100 TO 170 IN-LBS (11 TO 19 Nm)
T4 110 TO 165 IN-LBS (12 TO 19 Nm)

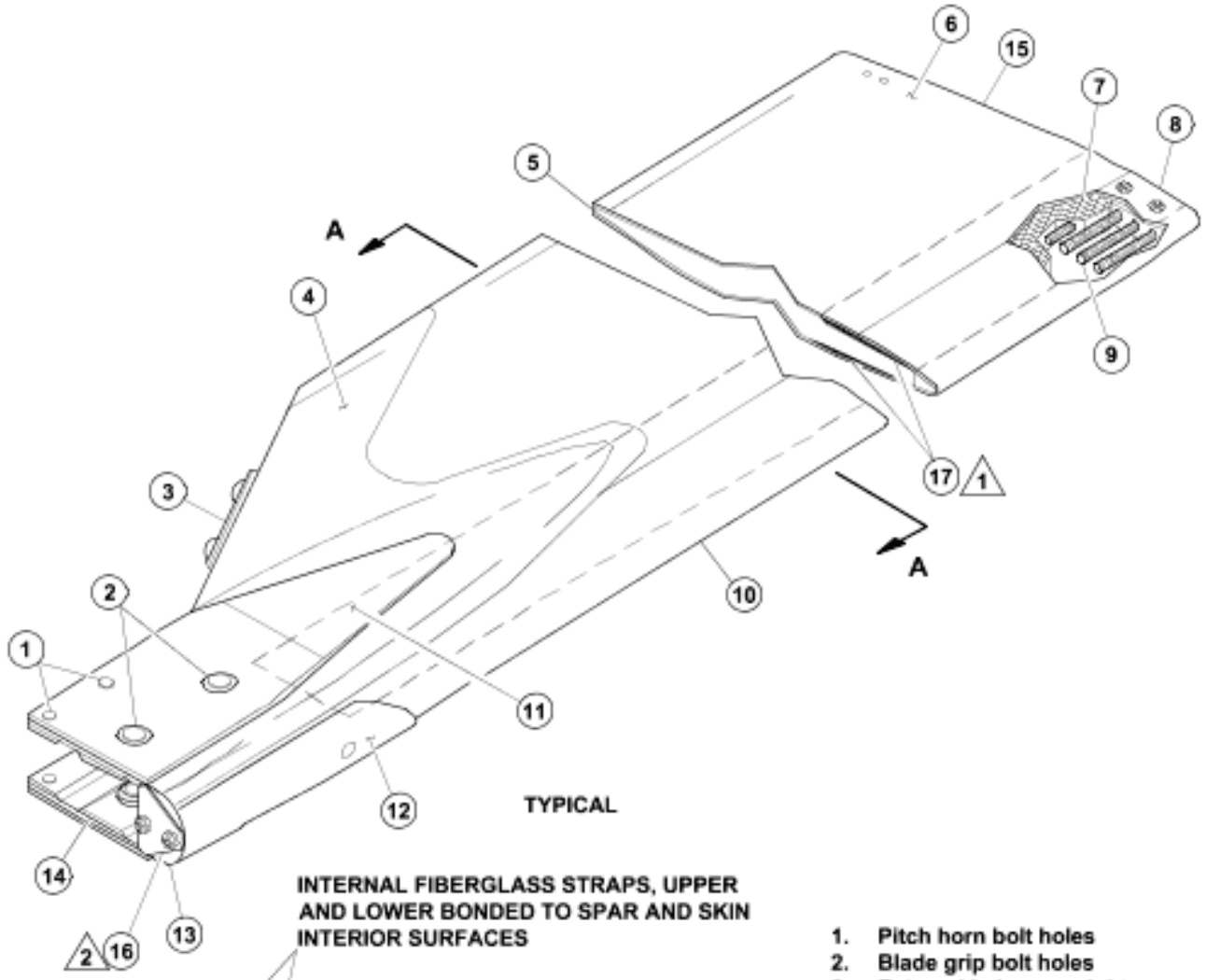
ICN-412-A-643000-B-97499-00004-A-001-01

Figure 1. Tail Rotor Crosshead Controls - Removal and Installation (Sheet 1 of 1)

Tail Rotor Blades - Description

This paragraph provides information on tail rotor blades 212-010-750 and “glass strap” tail rotor blades 412-016-100. Tail rotor blade 212-010-750 (Figure 1) is a bonded assembly consisting of a shaped stainless steel spar forming the leading edge, an aluminum alloy honeycomb core, and aluminum alloy skins. A flat trailing edge is formed by the extension of skins over a narrow aluminum strip. Aluminum alloy doublers and grip plates and butt blocks reinforce root end of blade. This section extends inboard to form a box, open toward hub and trailing edge. Two bolt holes through upper and lower surfaces provide for attachment of blade to hub. Externally attached balance weights and balance screws inside blade tip facilitate blade balancing.

Tail rotor blade 412-016-100 is a modified version of tail rotor blade 212-010-750. The modification incorporates two composite straps, constructed of unidirectional glass fiber reinforced tape, bonded to the interior upper and lower spar walls. The purpose of the composite straps is to slow any crack propagation in the blade to the point that it can be detected by visual inspection at recurring maintenance intervals. As shown in Figure 1, the straps extend chordwise across the spar fillet step over the spar trailing edge, and are tapered onto the skin. Spanwise, the straps start at station 10.92 and extend over the interior grip plates down the length of the spar, ending at the blade tip block. Slots are cut into opposite sides of the aluminum honeycomb core to allow space for the straps.



1. Pitch horn bolt holes
2. Blade grip bolt holes
3. External balance weights
4. Doubler
5. Trailing edge
6. Skin
7. Honeycomb core
8. Tip block
9. Balance screws
10. Spar
11. Grip plate
12. Drain hole doubler
13. Butt block
14. Inner grip plate
15. Tip closure
16. External balance weight
17. Fiberglass strap

NOTE

1 Fiberglass straps (17) are only appplicable to tail rotor blades 412-016-100.

2 External balance weight (16), installed if required.

ICN-412-A-641000-B-97499-00006-A-001-01

Figure 1. Tail Rotor Blade Assembly (Sheet 1 of 1)

Tail Rotor Blades - Maintenance

Procedure

NOTE

Removal and installation of the tail rotor blades is not provided in this manual. If a requirement exists to replace or repair any component of the hub and blade assembly, the tail rotor hub and blade must be statically balanced (BHT-412-CR&O).

1. For the maintenance procedures provide inspection, minor repair, and touch-up painting procedures only, refer to 412-A-64-10-01-02A-280A-A / 01253, 412-A-64-10-01-02A-364A-A / 01254, 412-A-64-10-01-02A-685A-A / 01255 and 412-A-64-10-01-02A-257A-A / 01256. For more extensive maintenance (i.e., removal, balancing, and installation), refer to the BHT-412-CR&O. For buffer pad replacement and blade butt sealing, refer to the BHT-412-CR&O.

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Tail Rotor Blades - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Cleaning compound	*C-318*	*AR*	
Clean cloth	C-516	AR	

Procedure

1. *Wash the tail rotor blades with a mild solution of* cleaning compound (C-318) *and water.*

2. Rinse the tail rotor blades with clean water and wipe dry with a clean cloth (C-516).

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Tail Rotor Blades - Inspection

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Fairing compound	C-323	AR	
Adhesive	C-331	AR	
Abrasive cloth or paper	C-406	AR	
Clean cloth	C-516	AR	

Procedure

1. Inspect the tail rotor blade records. If records or physical appearance indicate the tail rotor blades have been involved in an accident or incident, refer to 412-A-05-50-00-00A-284A-A / 00063 for conditional inspection requirements.



DESCRIBING AND DEFINING ALL DEFECTS THAT CAN OCCUR ON A TAIL ROTOR BLADE IS VIRTUALLY IMPOSSIBLE. THEREFORE, USE THE FOLLOWING INSPECTION CRITERIA ONLY AS A GUIDE. CONSULT PRODUCT SUPPORT ENGINEERING BEFORE SCRAPPING ANY BLADE EXCEPT IN CASES WHERE BLADES ARE OBVIOUSLY NON-REPAIRABLE.

2. Inspect the tail rotor blades and record whether damage is within repair limits noted in Step 2.1. through Step 2.6..



IF A FATIGUE CRACK EXISTS AT ANY LOCATION, FORWARD THE BLADE TO AN AUTHORIZED BLADE REPAIR FACILITY FOR EVALUATION. INSPECT DENTS CLOSELY FOR NICKS, SCRATCHES, AND CRACKS. IF NICKS OR SCRATCHES EXIST IN DENTS AND THE TOTAL DEPTH EXCEEDS THAT PERMITTED FOR DENTS ALONE, REPLACE THE BLADE.

NOTE

Dents shall be separated into two classifications: non-sharp dents and sharp dents. A dent is sharp if a 0.030 inch (0.76 mm) radius probe bridges across the bottom of the dent. If the probe makes a clicking sound or does not have a smooth feel when held perpendicular to the blade contour and dragged across the dent surface, then it is a sharp dent.

2.1. Nicks, scratches, dents, and corrosion on the skins, doublers, grip plates, and stainless steel spar (leading edge) that fall within the following limits are considered repairable damage and are acceptable if repaired in accordance with 412-A-64-10-01-02A-685A-A / 01255 and 412-A-64-10-01-02A-257A-A / 01256. Blades with more extensive damage may be sent to an authorized Bell Helicopter blade repair facility for evaluation.

2.1.1. Skins:

2.1.1.1. Inboard of blade station 40.50 (Figure 1):

Nicks and scratches on the aluminum skin 0.003 inch (0.08 mm) deep or less are acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255.

Sharp dents are not acceptable.

Non-sharp dents 0.015 inch (0.38 mm) deep or less are permitted, provided all other repair limits are met.

Corrosion: Only superficial corrosion that can be removed with aluminum wool or very fine grade Scotch-Brite abrasive pad is permitted. Repair in accordance with 412-A-64-10-01-02A-685A-A / 01255.

No damage permitted within 0.750 inch (19.05 mm) of a doubler edge.

2.1.1.2. Outboard of blade station 40.50:

Nicks and scratches on the aluminum skin 0.005 inch (0.13 mm) deep or less are acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255.

Sharp dents 0.005 inch (0.13 mm) deep or less are acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255.

Non-sharp dents 0.030 inch (0.76 mm) deep or less are permitted, provided all other repair limits are met.

Corrosion: Only superficial corrosion that can be removed with aluminum wool or very fine grade Scotch-Brite abrasive pad is permitted. Repair in accordance with 412-A-64-10-01-02A-685A-A / 01255.

2.1.2. Spar:

2.1.2.1. Inboard of blade station 40.50:

Nicks and scratches on the steel spar 0.003 inch (0.08 mm) deep or less are acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255.

Sharp dents on the steel spar 0.005 inch (0.13 mm) deep or less are acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255.

Non-sharp dents 0.015 inch (0.38 mm) deep or less are permitted, provided all other repair limits are met.

Corrosion: Corrosion on the steel spar 0.003 inch (0.08 mm) deep or less is acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255.

No damage permitted within 0.750 inch (19.05 mm) of a doubler edge.

2.1.2.2. Outboard of blade station 40.50:

Nicks and scratches on the steel spar 0.005 inch (0.13 mm) deep or less are acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255.

Sharp dents on the steel spar 0.015 inch (0.38 mm) deep or less are acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255.

Non-sharp dents 0.030 inch (0.76 mm) deep or less are permitted, provided all other repair limits are met.

Corrosion: Corrosion on the steel spar 0.005 inch (0.13 mm) deep or less is acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255.

2.1.3. Doublers and Grip Plates:

Damage to edges: No damage or repair permitted on an edge of a doubler or grip plate, or within 0.50 inch (12.7 mm) of an edge.

Nicks and scratches on the aluminum doublers and grip plates 0.005 inch (0.13 mm) deep or less are acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255.

Sharp dents are not acceptable.

Non-sharp dents 0.010 inch (0.25 mm) deep or less are permitted.

Corrosion: Corrosion on the aluminum doublers and grip plates 0.0025 inch (0.064 mm) deep or less is acceptable when polished out in accordance with 412-A-64-10-01-02A-685A-A / 01255. Repair must be twice the depth of the visible corrosion to ensure complete removal.

2.1.4. Limits - Trailing Edge:

2.1.4.1. Nicks and scratches in trailing edge up to 0.030 inch (0.76 mm) in depth (chordwise) are acceptable if polished out over a distance of at least 3.0 inches (76 mm) each side of damage.

2.1.4.2. Refer to 412-A-64-10-01-02A-685A-A / 01255 for instructions to polish out damage.

2.1.5. *Requirement to smooth and fair dented areas: Non-sharp dents that are within limits, but are more than 0.010 inch (0.25 mm) deep shall be filled and faired with* fairing compound (C-323).

2.2. Void limits:

NOTE

Voids are defined as unbonded areas that are supposed to be bonded. Many subdefinitions of voids are often given, such as lack of adhesion, gas pockets, misfits, etc. The general term void, as used herein, makes no distinction between these definitions. Where two voids of different limitations (e.g., void between core and skin adjacent to a void between skin and doubler) are closer together than 1.0 inch (25.4 mm), consider the defect as one void and apply the stricter limitation.

2.2.1. Edge voids (Figure 2): Acceptable edge voids must be sealed in accordance with 412-A-64-10-01-02A-685A-A / 01255.

2.2.1.1. Edge voids of 0.250 inch (6.35 mm) maximum in depth by 1.50 inches (38.1 mm) maximum in length between the butt block (13) and the spar (10), or between the butt block and the inner grip plates (14).

2.2.1.2. Edge voids of 0.060 inch (1.52 mm) maximum in depth by 2.0 inches (51 mm) maximum in length between the grip plates (11) and the doublers (4), doublers and skins (6) or spar (10), inner grip plates (14) and skins (6) or spar, edge of butt block (13) and skins (6). Edge voids between the skin and spar are not acceptable.

2.2.1.3. Edge voids of 0.120 inch (3.048 mm) maximum in depth by 3.0 inches (76 mm) maximum in length between the skins (6) and trailing edge (5).

2.2.1.4. Edge voids of 0.50 inch (12.7 mm) maximum in width (chordwise) between the spar (10) and the tip block (8).



NO VOIDS ARE ACCEPTABLE WITHIN 0.50 INCH (12.7 MM) OF THE EDGES OF THE DRAIN HOLE DOUBLER. NO VOIDS ARE ACCEPTABLE BETWEEN THE DRAIN HOLE DOUBLER AND THE SPAR WITHIN 0.50 INCH (12.7 MM) OF THE EDGE OF THE DRAIN HOLE. NO VOIDS ARE ALLOWED BETWEEN BLADE STATION 25.50 AND 40.50.

2.2.2. Voids, other than edge voids:

2.2.2.1. No voids are acceptable between grip plates (11) and doublers (4), or between inner grip plates (14) and spar (10), except as described in Step 2.2.1.2..

2.2.2.2. A void of 0.50 inch (12.7 mm) maximum chordwise by 2.0 inches (51 mm) maximum spanwise between doublers (4) and skin (6), or spar (10), is acceptable provided it is not within 0.50 inch (12.7 mm) of edge of doubler.

2.2.2.3. Inboard of blade station 25.50, a void 0.50 inch (12.7 mm) maximum chordwise by 2.0 inches (51 mm) maximum spanwise, is acceptable between skin (6) and spar (10), provided it is not within 0.250 inch (6.35 mm) of edge of skin. Outboard of blade station 40.50 where the skin overlaps the spar, the maximum size for acceptable voids is 0.250 inch (6.35 mm) chordwise by 2.0 inches (51 mm) spanwise, provided it is not within 0.250 inch (6.35 mm) of edge of skin.

2.2.2.4. A void of 0.50 inch (12.7 mm) maximum chordwise by 2.0 inches (51 mm) maximum spanwise is acceptable between skin (6) and honeycomb core (7) or the fiberglass strap (17) and honeycomb core.

2.2.2.5. A void of 0.38 inch (9.652 mm) maximum spanwise by 3.0 inches (76 mm) maximum chordwise is acceptable between the skin and tip closure in either the top or bottom or both, provided it is not within 0.12 inch (3.048 mm) from the edge of skin. If edge distance is less than 0.12 inch (3.048 mm), void may be repaired by carefully lifting the skin and rebonding with adhesive (C-331).

2.2.2.6. Voids at Trailing Edge to Core Interface. A 0.38-inch (9.6 mm) maximum gap/void between the trailing edge of the honeycomb core and the trailing edge strip is acceptable.

NOTE

Step 2.2.2.7. *is applicable to tail rotor blade 412-016-100.*

2.2.2.7. Voids between the fiberglass strap (17) and spar (10) or the glass strap and skin (6). The following void allowables apply to a single strap. There shall be no void indications within 0.35 inch (8.9 mm) of the edge of the fiberglass straps.

Void indications shall not exceed 0.50 inch (12.7 mm) in chordwise width and the area shall not exceed 1.0 square inch (645 mm^2) maximum. Maximum length shall not exceed 4.0 inches (102 mm). The combined area of all void indications shall not exceed 4.0 square inches (2580 mm^2) total. No two voids shall be within 1.0 inch (25.4 mm) of each other. No more than one void on a chord line per strap. No voids allowed between blade stations 25.50 and 40.50.

CAUTION

IF A CRACK EXISTS IN ANY LOCATION, THE BLADE IS NOT ACCEPTABLE AND MUST BE REMOVED FROM SERVICE.

2.3. Cracks defined in the following steps are not acceptable.

2.3.1. Visually check for bond line cracks between phenolic blocks and skin, spar inside drain hole, inner grip plates or joint between phenolic blocks.

2.3.2. Inspect all butt area bond lines placing emphasis on those areas illustrated in Figure 3.

2.3.3. If suspected cracks are found, do a tail rotor blade leak check (412-A-64-10-01-02A-364A-A / 01254).

2.3.4. If no cracks are found, touch up paint after inspection is complete (412-A-64-10-01-02A-257A-A / 01256).

2.4. *Inspect buffer pads around blade bolt holes for wear, tears, or debonding. For repair, refer to the* 412-CR&O, Chapter 64.

2.5. Inspect bushings in tail rotor blade bolt holes (2) for secure installation. Any looseness is cause for blade replacement. Measure bushing inside diameter. If dimension exceeds 0.5005 inch (12.71 mm), replace blade bushing at an authorized blade overhaul facility.

2.6. Inspect bushings for cracks. Any crack is cause to remove the blade from service.

2.7. Damage defined in the following steps is cause to remove the tail rotor blade from service. The blade may be sent to an authorized repair facility for evaluation or possible repair.

2.7.1. Damage in excess of limits specified in inspection Step 2.1. through Step 2.3..

2.7.2. One or more holes in the skin.

2.7.3. Edge voids between the skin and the spar.

2.7.4. Evidence of corrosion at any edge void.

2.8. Damage defined in the following steps is cause to scrap tail rotor blade.

2.8.1. Water or corrosion in rotor blade honeycomb core.

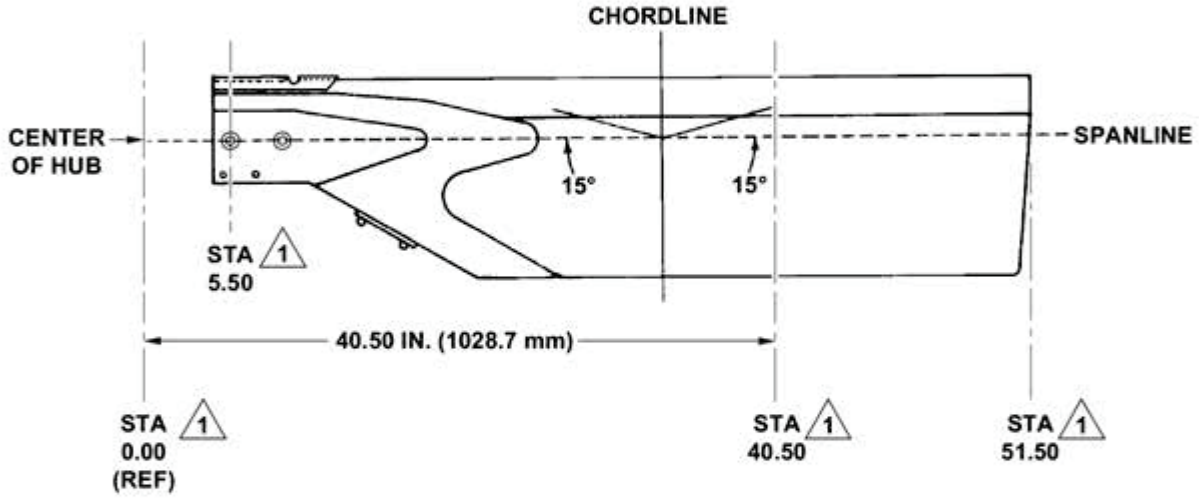
2.8.2. Damage in excess of limits defined in inspection Step 2.1. through Step 2.3. and/or in the 412-CR&O, Chapter 64 on a rotor blade that has reached maximum airworthiness life (hours) (412-A-04-00-00-00A-288A-A / 00013), or has less than 200 hours remaining airworthiness life (hours).

2.8.3. One or more cracks extending from a previously repaired area.

2.8.4. Wear completely through blade spar at tip.

2.8.5. Loose balance screws (9).

2.8.6. Loose bushings at blade grip bolt holes (2).

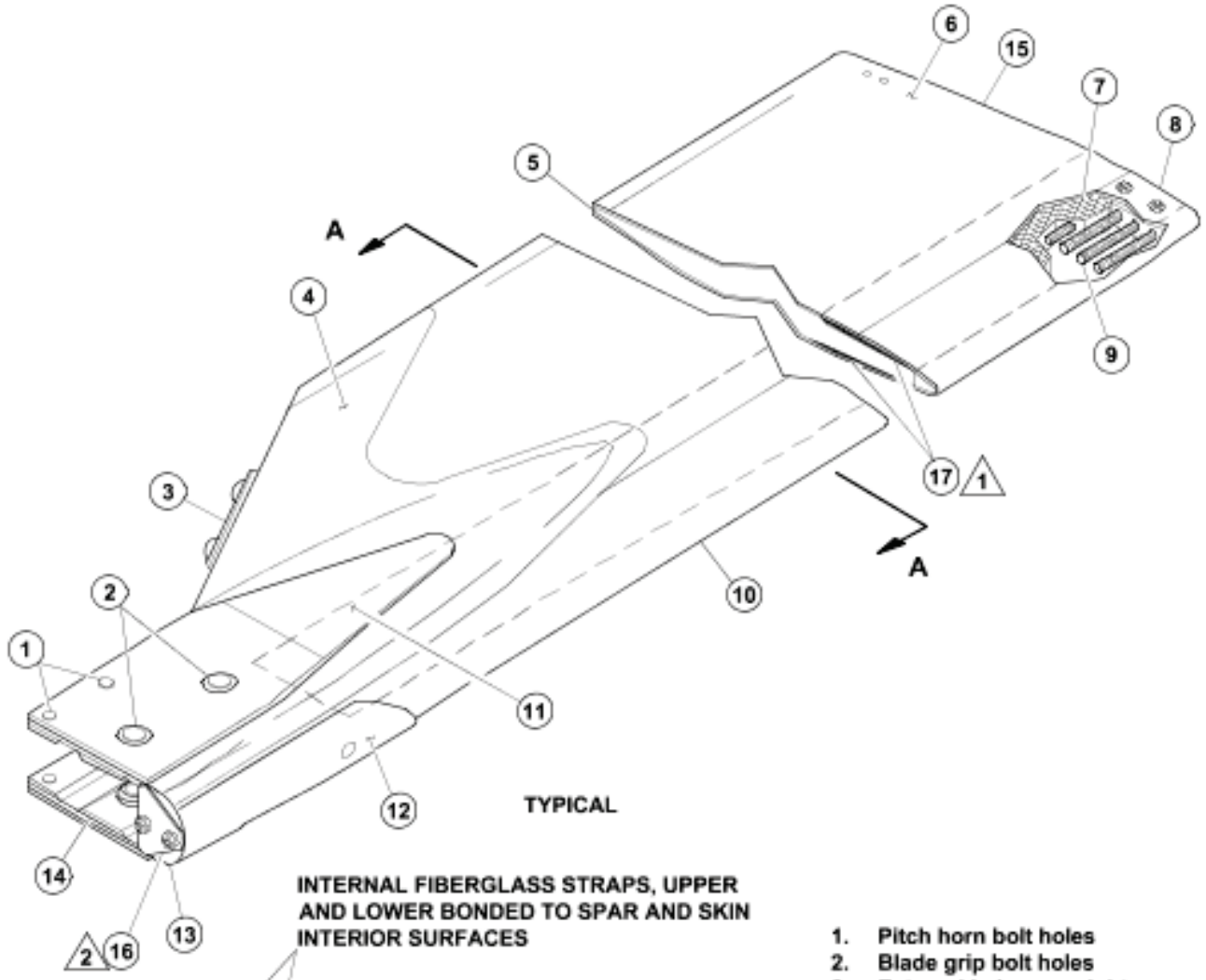


NOTE

1 Tail rotor blade stations (212-011-702 yoke installed).

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Figure 1. Tail Rotor Blade (Sheet 1 of 1)



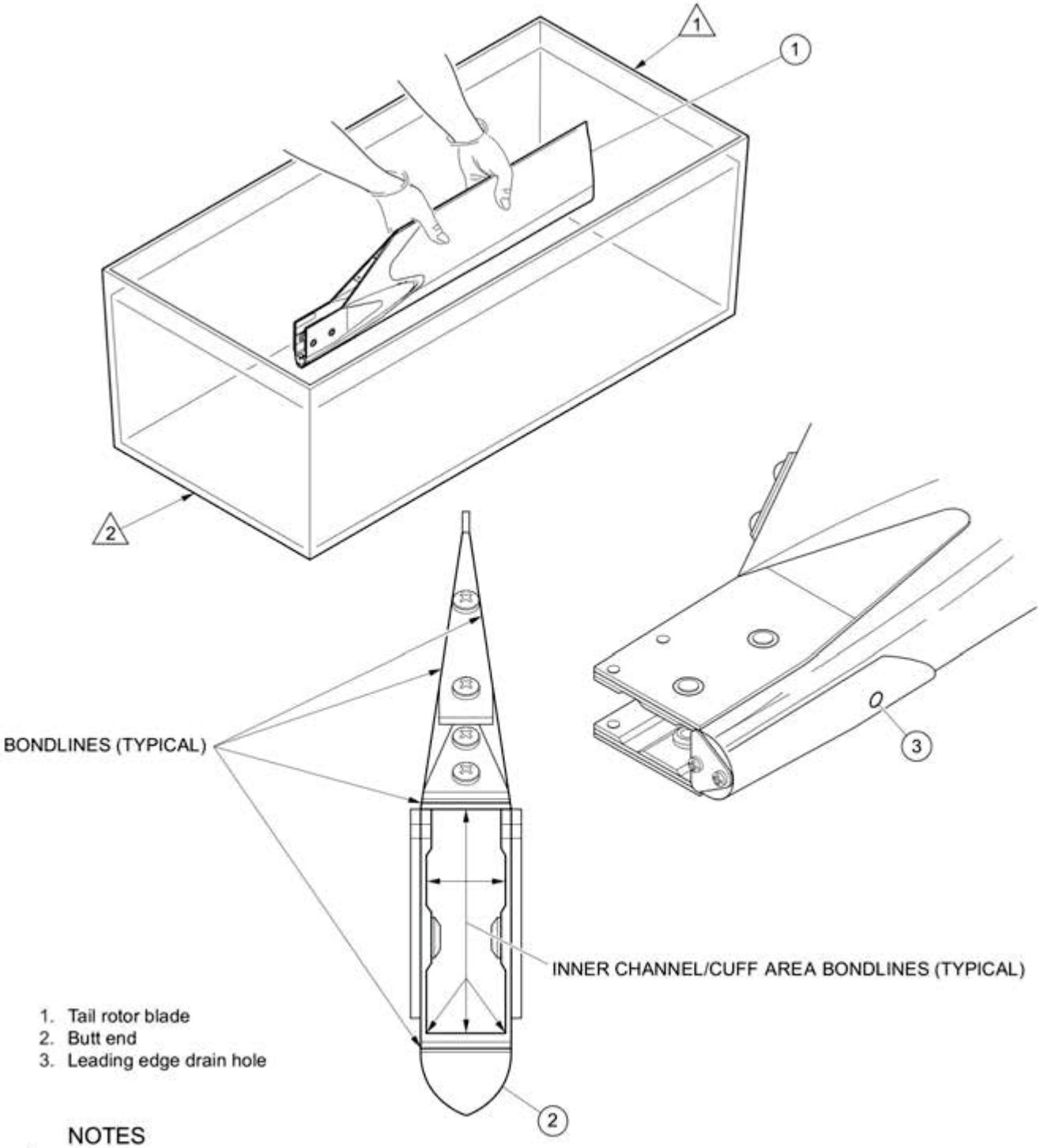
1. Pitch horn bolt holes
2. Blade grip bolt holes
3. External balance weights
4. Doubler
5. Trailing edge
6. Skin
7. Honeycomb core
8. Tip block
9. Balance screws
10. Spar
11. Grip plate
12. Drain hole doubler
13. Butt block
14. Inner grip plate
15. Tip closure
16. External balance weight
17. Fiberglass strap

NOTE

1. Fiberglass straps (17) are only appplicable to tail rotor blades 412-016-100.
2. External balance weight (16), installed if required.

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Figure 2. Tail Rotor Blade Assembly (Sheet 1 of 1)



1. Tail rotor blade
2. Butt end
3. Leading edge drain hole

NOTES

1. Water temperature to be maintained between 140 to 176 °F (60 to 80 °C)
2. Water tank should be 20.0 inches (508 mm) wide X 52.0 inches (1321 mm) long by 20.0 inches (508 mm) deep.

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Figure 3. Tail Rotor Blade - Leak Check (Sheet 1 of 1)

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Tail Rotor Blades - Leak Check

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Cleaning compound	C-318	AR	

Procedure

NOTE

The following inspection should be performed whenever signs of delamination/ separation are visible in the bond lines at the butt end area of the blade (Figure 1), as well as other tail rotor blade locations, or if excessive tail rotor vibrations are reported with no apparent causes.

1. Thoroughly clean the tail rotor blade (1), including the surfaces and bond lines of the butt end (2), with a weak solution of cleaning compound (C-318). Rinse with clear water.

NOTE

The intent is to obtain an approximate 100°F or 55.6°C temperature differential between the tail rotor blade (1) and the water in the tank.

2. Cool the tail rotor blade (1) in a refrigerator for a minimum of 1 hour at a temperature of 30 to 45°F (-1 to 7°C)..



DO NOT EXCEED A WATER TEMPERATURE OF 176°F (80°C). OTHERWISE DAMAGE TO THE TAIL ROTOR BLADE MAY OCCUR.

3. Fill the water tank with clean clear water. For best results, the water should be maintained at 140 to 176°F (60 to 80°C)..



DO NOT SUBMERGE THE TAIL ROTOR BLADE IN WATER FOR MORE THAN 2 MINUTES. MAINTAINING THE BLADE IN WATER FOR MORE THAN 2 MINUTES COULD ALLOW WATER TO ENTER THE BLADE IF A LEAK PATH IS PRESENT.

NOTE

The tail rotor blade (1) should be left in the heated water for a minimum of 1 minute and a maximum of 2 minutes.

4. Wear protective gloves and place the cool tail rotor blade (1) into the heated water. The blade will float so it will be necessary to manually submerge the blade.

NOTE

Be careful not to misinterpret random trapped surface air bubbles as a leak point.

5. Slightly rotate the tail rotor blade (1) back and forth about the blade pitch change axis to allow any air trapped on the blade surfaces to be expelled.

NOTE

It may take several seconds before the air is forced out through any existing leak path.

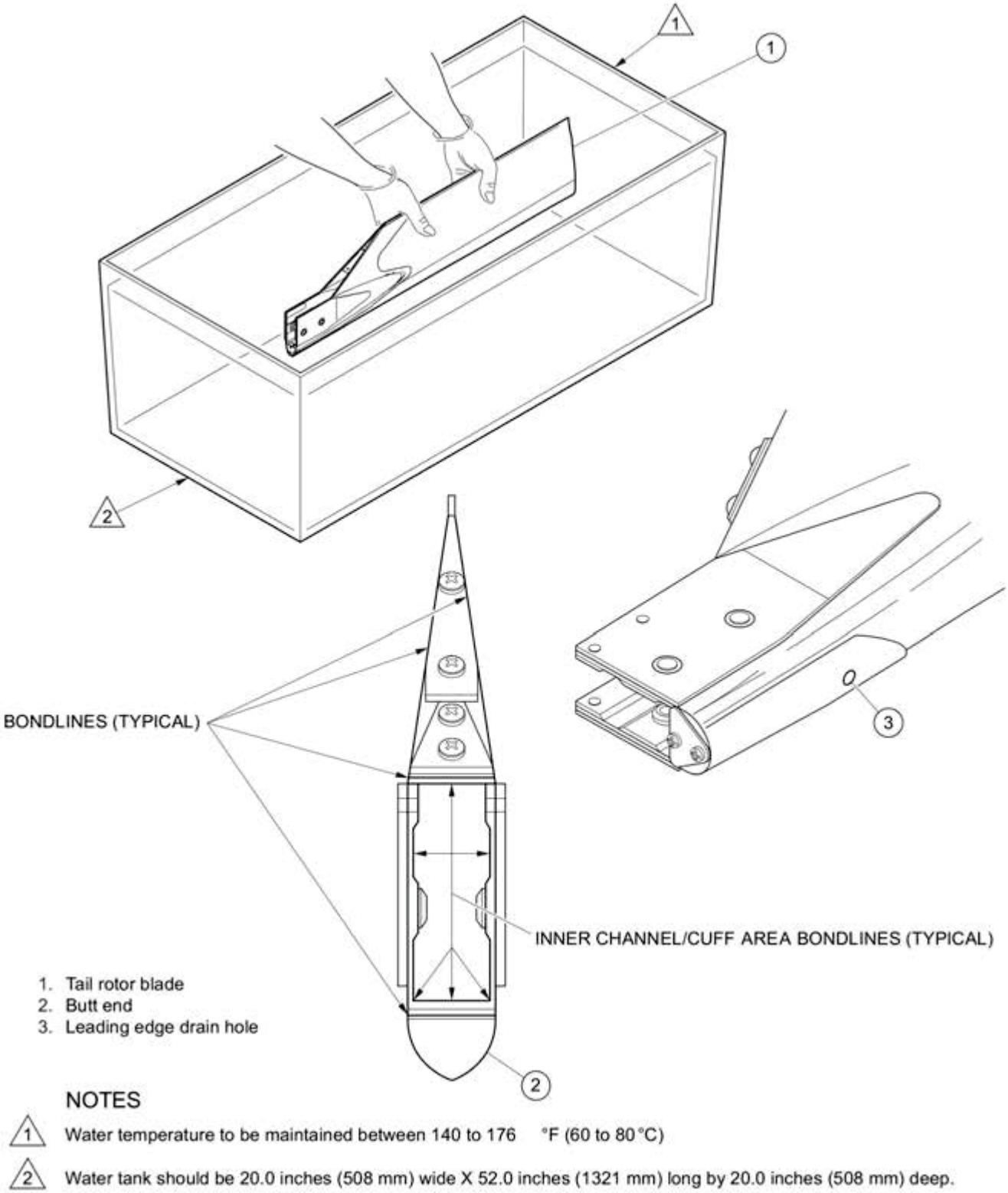
6. With the tail rotor blade (1) under water, check for air leaks, which will be indicated by a steady stream of fine air bubbles originating from a single point at a rate exceeding one every five seconds. Occasional air bubbles is not criteria for rejection. If in doubt, contact Product Support Engineering (PSE).

7. Slightly rotate the tail rotor blade (1) back and forth to allow a thorough inspection of all the blade surfaces. Pay particular attention to the butt end (2) and around the leading edge drain hole (3).

8. If an unacceptable air leak is found on tail rotor blade 212-010-750 as defined in Step 6., the blade is to be removed from service and scrapped (notify PSE). If an unacceptable air leak is found on tail rotor blade 412-016-100 as defined in step 6, contact PSE for disposition.

9. If no air leak is found, but the tail rotor assembly has a history of balance change or balance difficulties, additional maintenance is required (412-A-18-12-00-02A-271A-A / 00165).

10. If no air leak is found based on Step 6. criteria, the tail rotor blade (1) can be returned to service, after repair per 412-A-64-10-01-02A-685A-A / 01255.



1. Tail rotor blade
2. Butt end
3. Leading edge drain hole

NOTES

1. Water temperature to be maintained between 140 to 176 °F (60 to 80 °C)
2. Water tank should be 20.0 inches (508 mm) wide X 52.0 inches (1321 mm) long by 20.0 inches (508 mm) deep.

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Figure 1. Tail Rotor Blade - Leak Check (Sheet 1 of 1)

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Tail Rotor Blades - Minor Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Abrasive cloth or paper	C-406	AR	
Abrasive pad	C-407	AR	
Cleaner	C-344	AR	
Adhesive	C-363	AR	
Clean cloth	C-516	AR	
Aliphatic naphtha	C-305	AR	
Sealant	C-308	AR	

Procedure

1. All nicks, scratches, sharp dents, and corrosion that are within the repair limits in 412-A-64-10-01-02A-280A-A / 01253 must be removed by hand polishing in a spanwise direction (±15°) only.

NOTE

Repair to the blade by material removal (polishing) is permitted only once at any local area. Overlapping repairs are not acceptable. Any permitted repair by polishing should be accomplished by removing the minimum amount of material required to remove the damage.

1.1. Nicks, Scratches, and Sharp Dents:

Polish out all repairable nicks and scratches with aluminum wool or a very fine grade abrasive pad (C-407) when repairing aluminum. On steel surfaces, fine abrasive cloth or paper (C-406) may be used (180 grit or finer may be used to remove damage, but must be followed by polishing with 400 grit or finer to improve the surface finish). Nicks in the trailing edge may be polished out with very fine abrasive cloth or paper (C-406) if followed by polishing with aluminum wool or very fine grade abrasive pad (C-407). Hand polish in a spanwise direction.

1.2. Non-sharp Dents:

The bottom of a dent must be smooth and clearly non-sharp to be classified as a non-sharp dent. A dent with any crease in it is considered to be a sharp dent. Non-sharp dents that are within the limits in 412-A-64-10-01-02A-280A-A / 01253 are not to be polished out. Carefully inspect for any surface irregularities, such as a nick, scratch, or burr, in and around a non-sharp dent. Any additional damage must be within the limits in 412-A-64-10-01-02A-280A-A / 01253.

1.3. Corrosion:

Polish out all repairable corrosion with aluminum wool or a very fine grade abrasive pad (C-407) on all aluminum surfaces, or very fine abrasive cloth or paper (C-406) (400 grit or finer) may be used on the stainless steel spar. 180 grit abrasive cloth or paper (C-406) may be used to polish out corrosion on the steel spar if followed with polishing using 400 grit abrasive cloth or paper (C-406) to improve the surface finish. Repair depth on aluminum must be to twice the depth of the visible corrosion to ensure complete removal. Hand polish in a spanwise direction. Do not over-polish. On all surfaces where corrosion has been polished out, apply cleaner (C-344) using a clean cloth (C-516) or brush. Rub solution briskly into surface for approximately 40 to 60 seconds. Carefully check for pits with a 10X magnifying glass following cleanup. No pitting is permitted.

1.4. Sealing Edge Voids:

Seal edge voids by thoroughly cleaning affected area with alcohol and clean wipes. A thin plastic spatula may be used to solvent wipe the bond line. Do not stretch or gouge blade skins, spars, or doublers. Allow to dry. Seal area with adhesive (C-363) or equivalent. Adhesive and blade may be warmed to 120°F (49°C) to promote adhesive flow. A thin plastic spatula may be used to work adhesive into bond lines. Allow to cure.

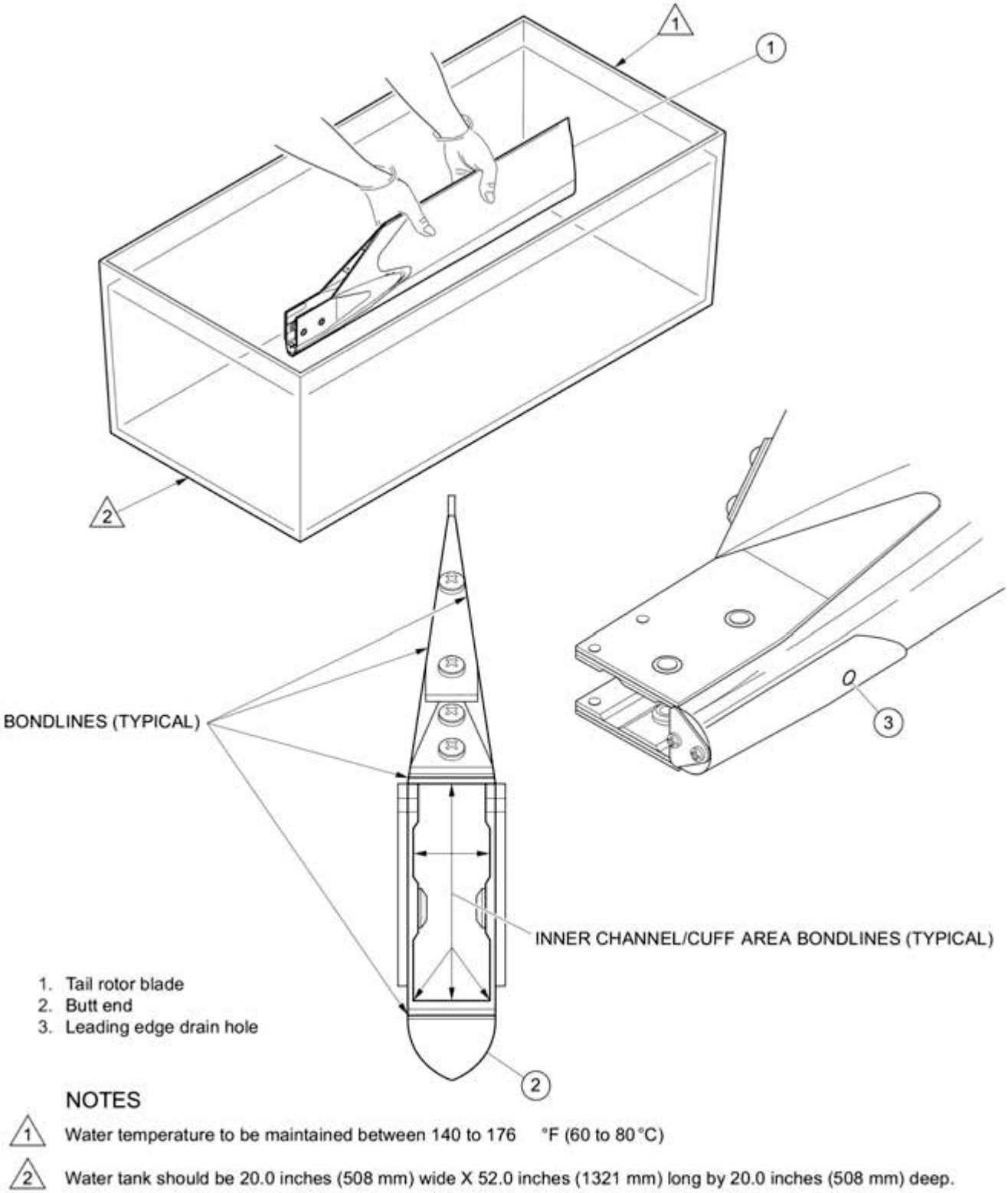
1.5. Butt end area (Figure 1) adhesive cracks and voids repair.

1.5.1. Sand some of the cracked adhesive in the direction of the crack with a 400 grit abrasive cloth or paper (C-406). The intent is to remove only sufficient cracked adhesive to adequately seal the crack with sealant (C-308).

1.5.2. Remove the sanding residue with a clean cloth (C-516) moistened with aliphatic naphtha (C-305).

1.5.3. Apply a bead of sealant (C-308) to seal the crack and/or voids in the adhesive.

1.5.4. Allow the sealant (C-308) to cure and refinish the reworked area of the tail rotor blade (412-A-64-10-01-02A-257A-A / 01256).



1. Tail rotor blade
2. Butt end
3. Leading edge drain hole

NOTES

1. Water temperature to be maintained between 140 to 176 °F (60 to 80 °C)
2. Water tank should be 20.0 inches (508 mm) wide X 52.0 inches (1321 mm) long by 20.0 inches (508 mm) deep.

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Figure 1. Tail Rotor Blade - Leak Check (Sheet 1 of 1)

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Tail Rotor Blades - Paint Touch-up

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Chemical film material	C-100	AR	
Tape	C-165	AR	
Epoxy primer coating	C-246	AR	
Coating	C-265	AR	Polyurethane coating (C-245) or lacquer (C-594) may be used as an alternate for touch-up.
Cleaning compound	C-318	AR	
Solvent	C-386	AR	
Abrasive cloth or paper	C-406	AR	
Abrasive pad	C-407	AR	
Masking paper	C-420	AR	
Tape	C-434	AR	
Cheesecloth	C-486	AR	

Procedure

NOTE

Touch up tail rotor blades when warranted by paint condition, or when paint is removed to repair scratches, nicks, gouges, dents, or corrosion.

1. Thoroughly wipe the surface to be repainted with solvent (C-386). Allow the solvent to wet the area for a few minutes, then wipe dry with clean cheesecloth (C-486).



DO NOT USE ROTARY TYPE SANDING DEVICES. OTHERWISE DAMAGE TO THE TAIL ROTOR BLADE CAN OCCUR.

2. Thoroughly sand the area to be painted with 240 grit, or finer, abrasive cloth or paper (C-406) until a uniform matte finish is obtained.

3. Wash the sanded surface with an abrasive pad (C-407) and cleaning compound (C-318) to obtain a water break free surface. Rinse with water. RepeatStep 1. through Step 3. as required to obtain a water break free surface.

NOTE

From completion of this step through final paint, the surface of the blades that are to be painted shall not be handled with bare hands.

4. Apply chemical film material (C-100) on all bare aluminum surfaces (BHT-ALL-SPM, Chapter 3, Standard Practices Manual).

5. Mask the tail rotor blade, pitch horn bolt holes, blade grip bolt holes and buffer pads, identification plate and leading edge area, as applicable, with tape (C-434) and masking paper (C-420).

6. Apply one coat of epoxy primer coating (C-246) to touch-up areas only.

NOTE

Primer shall be overcoated not less than 1 nor more than 8 hours after application. If 8 hours is exceeded, repeat Step 2. through Step 6..

NOTE

Refer to the 412-CR&O, Chapter 64 *for the required paint color to match the area being painted.*

7. Using the required paint color to match the area being painted, apply one to two covering coats of coating (C-265). Allow 30 minutes to 8 hours drying time between coats. If 8 hours is exceeded, the initial coat must be sanded and reprimed with epoxy primer coating (C-246) prior to overcoating.

8. Remove masking tape and marking paper from the blade.

9. After air drying for a minimum of 8 hours and if touch-up was applied to leading edge, feather out paint lines on leading edge as follows:

9.1. Mask approximately 1/4 inch (6.35 mm) back from the forward edge of the paint using 2.0 inches (51 mm) wide tape (C-165) or equivalent.

9.2. Hand sand, overlapping the tape and the edge to be feathered, in a spanwise direction with 240 grit or finer abrasive cloth or paper (C-406). Do not sand through the tape.

9.3. A correctly feathered edge should show a uniform primer line of about 1/16 inch (1.58 mm) in width in the striation of the feathered paint. This indicates a smooth transition from metal to paint. A fingernail, when dragged over the feathered edge, should not hang on the edge.

10. Static balancing of the tail rotor hub and blade assembly is required (412-CR&O, Chapter 64).

CHAPTER 65
TAIL ROTOR DRIVE SYSTEM

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number		Issue date	Applicable to
Tail Rotor Driveshaft - Troubleshooting	DMC-412-A-65-10-00-00A-422A-A / 01258	C	2025-11-18	SN: 33001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Tail Rotor Driveshaft - Removal	DMC-412-A-65-10-00-00A-520A-A / 01259	C	2025-11-18	SN: 33001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Tail Rotor Driveshaft - Cleaning	DMC-412-A-65-10-00-00A-251A-A / 01260	C	2025-11-18	SN: 33001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Tail Rotor Driveshaft - Inspection	DMC-412-A-65-10-00-00A-280A-A / 01261	C	2025-11-18	SN: 33001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Tail Rotor Driveshaft - Repair	DMC-412-A-65-10-00-00A-685A-A / 01262	C	2025-11-18	SN: 33001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))

Document title	Data module code / Seq number		Issue date	Applicable to
Tail Rotor Driveshaft - Installation	DMC-412-A-65-10-00-00A-720A-A / 01263	C	2025-11-18	SN: 33001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Driveshaft Hanger Assemblies - Removal	DMC-412-A-65-10-01-00A-520A-A / 01264	C	2025-11-18	SN: 33001-33213 or 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Driveshaft Hanger Assemblies - Cleaning	DMC-412-A-65-10-01-00A-251A-A / 01265	C	2025-11-18	SN: 33001-33213 or 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Driveshaft Hanger Assemblies - Inspection	DMC-412-A-65-10-01-00A-280A-A / 01266	C	2025-11-18	SN: 33001-33213 or 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))

Document title	Data module code / Seq number		Issue date	Applicable to
Driveshaft Hanger Assemblies - Lubrication	DMC-412-A-65-10-01-00A-240A-A / 01267	C	2025-11-18	SN: 33001-33213 or 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Driveshaft Hanger Assemblies - Installation	DMC-412-A-65-10-01-00A-720A-A / 01268	C	2025-11-18	SN: 33001-33213 or 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Tail Rotor Drive System - Alignment	DMC-412-A-65-00-00-00A-272A-A / 01269		2023-05-31	ALL
Tail Rotor Driveshaft and Hangers - Removal	DMC-412-A-65-10-02-00A-520A-A / 01270	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Tail Rotor Driveshaft and Hangers - Cleaning	DMC-412-A-65-10-02-00A-251A-A / 01271	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Tail Rotor Driveshaft and Hangers - Inspection	DMC-412-A-65-10-02-00A-280A-A / 01272	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed

Document title	Data module code / Seq number		Issue date	Applicable to
Tail Rotor Driveshaft and Hangers - Repair	DMC-412-A-65-10-02-00A-685A-A / 01273	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Tail Rotor Driveshaft and Hangers - Lubrication	DMC-412-A-65-10-02-00A-240A-A / 01274	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Tail Rotor Driveshaft and Hangers - Installation	DMC-412-A-65-10-02-00A-720A-A / 01275	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Intermediate Gearbox - Description	DMC-412-A-65-20-00-00A-028A-A / 01276		2023-05-31	ALL
Intermediate Gearbox - Maintenance Procedures	DMC-412-A-65-20-00-00A-913A-A / 01277		2023-05-31	ALL
Intermediate Gearbox - Troubleshooting	DMC-412-A-65-20-00-00A-422A-A / 01278		2023-05-31	ALL
Intermediate Gearbox - Inspection (Installed)	DMC-412-A-65-20-00-00A-280A-A / 01279		2023-05-31	ALL
Intermediate Gearbox - Removal	DMC-412-A-65-20-00-00A-520A-A / 01280		2023-05-31	ALL
Intermediate Gearbox - Cleaning	DMC-412-A-65-20-00-00A-251A-A / 01281		2023-05-31	ALL
Intermediate Gearbox - Inspection (Not Installed)	DMC-412-A-65-20-00-00A-280B-A / 01282		2023-05-31	ALL
Intermediate Gearbox - Inspection (Conditional)	DMC-412-A-65-20-00-00A-280C-A / 01283		2023-05-31	ALL
Intermediate Gearbox - Repair	DMC-412-A-65-20-00-00A-685A-A / 01284		2023-05-31	ALL
Intermediate Gearbox - Installation	DMC-412-A-65-20-00-00A-720A-A / 01285		2024-12-19	ALL
Intermediate Gearbox - Alternate Run-In Test	DMC-412-A-65-20-00-00A-322A-A / 01286		2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Sight Glass - Removal	DMC-412-A-65-20-01-08A-520A-A / 01287		2023-05-31	ALL
Sight Glass - Installation	DMC-412-A-65-20-01-08A-720A-A / 01288		2023-05-31	ALL
Electrical Chip Detector - Removal	DMC-412-A-65-40-01-00A-520A-A / 01289		2023-05-31	ALL
Electrical Chip Detector - Inspection	DMC-412-A-65-40-01-00A-280A-A / 01290		2023-05-31	ALL
Electrical Chip Detector - Repair	DMC-412-A-65-40-01-00A-685A-A / 01291		2023-05-31	ALL
Electrical Chip Detector - Installation	DMC-412-A-65-40-01-00A-720A-A / 01292		2023-05-31	ALL
Oil Filler Cap - Removal	DMC-412-A-65-20-01-02A-520A-A / 01293		2023-05-31	ALL
Oil Filler Cap - Inspection	DMC-412-A-65-20-01-02A-280A-A / 01294		2023-05-31	ALL
Oil Filler Cap - Repair	DMC-412-A-65-20-01-02A-685A-A / 01295		2023-05-31	ALL
Oil Filler Cap - Installation	DMC-412-A-65-20-01-02A-720A-A / 01296		2023-05-31	ALL
Flexible Couplings - Inspection and Lubrication	DMC-412-A-65-20-01-11A-280A-A / 01297	C	2025-11-18	ALL
Tail Rotor Gearbox - Maintenance Procedures	DMC-412-A-65-20-01-00A-913A-A / 01298		2025-05-13	ALL
Tail Rotor Gearbox - Troubleshooting	DMC-412-A-65-20-01-00A-422A-A / 01299		2023-05-31	ALL
Tail Rotor Gearbox - Cleaning	DMC-412-A-65-20-01-00A-251A-A / 01300		2023-05-31	ALL
Tail Rotor Gearbox - Inspection (Installed)	DMC-412-A-65-20-01-00A-280A-A / 01301		2023-05-31	ALL
Tail Rotor Gearbox - Removal	DMC-412-A-65-20-01-00A-520A-A / 01302	C	2025-11-18	ALL
Tail Rotor Gearbox - Inspection (Not Installed)	DMC-412-A-65-20-01-00A-280B-A / 01303		2023-05-31	ALL
Tail Rotor Gearbox - Inspection (Conditional)	DMC-412-A-65-20-01-00A-280C-A / 01304		2023-05-31	ALL
Tail Rotor Gearbox - Repair	DMC-412-A-65-20-01-00A-685A-A / 01305		2023-05-31	ALL
Tail Rotor Gearbox - Installation	DMC-412-A-65-20-01-00A-720A-A / 01306	C	2025-11-18	ALL
Tail Rotor Gearbox - Alternate Run-In Test	DMC-412-A-65-20-01-00A-322A-A / 01307		2023-05-31	ALL
Sight Glass - Removal	DMC-412-A-65-20-01-08A-520B-A / 01308		2023-05-31	ALL
Sight Glass - Installation	DMC-412-A-65-20-01-08A-720B-A / 01309		2023-05-31	ALL
Electrical Chip Detector - Removal	DMC-412-A-65-40-01-00A-520B-A / 01310		2023-05-31	ALL
Electrical Chip Detector - Inspection	DMC-412-A-65-40-01-00A-280B-A / 01311		2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Electrical Chip Detector - Repair	DMC-412-A-65-40-01-00A-685B-A / 01312	2023-05-31	ALL
Electrical Chip Detector - Installation	DMC-412-A-65-40-01-00A-720B-A / 01313	2023-05-31	ALL
Oil Filler Cap - Removal	DMC-412-A-65-20-01-02A-520B-A / 01314	2023-05-31	ALL
Oil Filler Cap - Inspection	DMC-412-A-65-20-01-02A-280B-A / 01315	2023-05-31	ALL
Oil Filler Cap - Repair	DMC-412-A-65-20-01-02A-685B-A / 01316	2023-05-31	ALL
Oil Filler Cap - Installation	DMC-412-A-65-20-01-02A-720B-A / 01317	2023-05-31	ALL
Flexible Coupling - Lubrication	DMC-412-A-65-20-01-11A-240A-A / 01318	2023-05-31	SN: 33001-36019 or 412-570-001-103: Not Installed

Tail Rotor Driveshaft - Troubleshooting

Procedure

1. Refer to Table 1.

Table 1. Tail Rotor Driveshaft - Troubleshooting

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
Vibration believed to be originating in tail rotor drive system.	Bolts in V-band couplings loose.	Inspect all V-band clamps for loose bolts.
		Replace or torque bolts as required.
	Broken hanger or hanger support.	Inspect hangers and supports for cracks. Replace defective parts.
	Tail rotor driveshaft out of balance.	Verify driveshaft balance weights are secure. Balance weights are small metal strips bonded to driveshaft (one empty bonded space should be open where a test strip was removed). If more than one space is open, a balance weight is missing. Replace driveshaft.
	Shaft hanger bearings rough or overheating.	Check visually for signs of overheated bearings. If no sign of overheating is noted, recheck after 5 hours of operation.
		Replace bearings if overheating. Replace bearings if rough, whether or not evidence of overheating is present.
Excessive loss of grease from hanger bearings.	Misalignment of bearing in hangers.	Check and correct alignment of bearings.
Binding or roughness when rotor and tail rotor driveshaft is turned by hand.	Dry or faulty hanger bearings.	Lubricate bearings.
		Remove driveshaft segments, if necessary, to reidentify faulty bearings.
	Faulty tail rotor gearbox.	Inspect electric chip detectors in intermediate and tail rotor gearboxes for evidence of internal failure. Replace faulty gearbox.

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Tail Rotor Driveshaft - Removal

Procedure

NOTE

The removal procedure for driveshaft (6, Figure 1) is given. The removal procedure for remaining tail rotor driveshafts is the same except for access provisions and component removal. The following is required access/component removal for each driveshaft in Figure 1.

1. To access driveshaft (2, Figure 1), proceed as follows:

1.1. Remove oil cooler fairing.

1.2. Open reduction gearbox side cowling.

1.3. Remove left and right access doors on pylon support.

1.4. Open hinged cover for driveshaft (4).

1.5. Remove driveshafts (3 and 4) (Step 7.) and hanger bearing (5) between driveshafts (2 and 3) (412-A-65-10-01-00A-520A-A / 01264).

2. To access driveshaft (3), proceed as follows:

2.1. Remove oil cooler fairing.

2.2. Open reduction gearbox side cowling.

2.3. Open hinged cover for driveshaft (4).

3. To access driveshaft (4), proceed as follows:

3.1. Remove oil cooler fairing.

3.2. Open hinged cover for driveshaft (4).

4. To access driveshaft (7), proceed as follows:

4.1. Open hinged cover for driveshaft (7).

4.2. Remove cowling from intermediate gearbox (11).

5. To access driveshaft (9), proceed as follows:

5.1. Remove cowling from intermediate gearbox (11).

5.2. Open vertical fin cover (8).

6. Open hinged covers to provide access to driveshaft (6).



DO NOT INTERMIX PARTS OF CLAMP SETS.

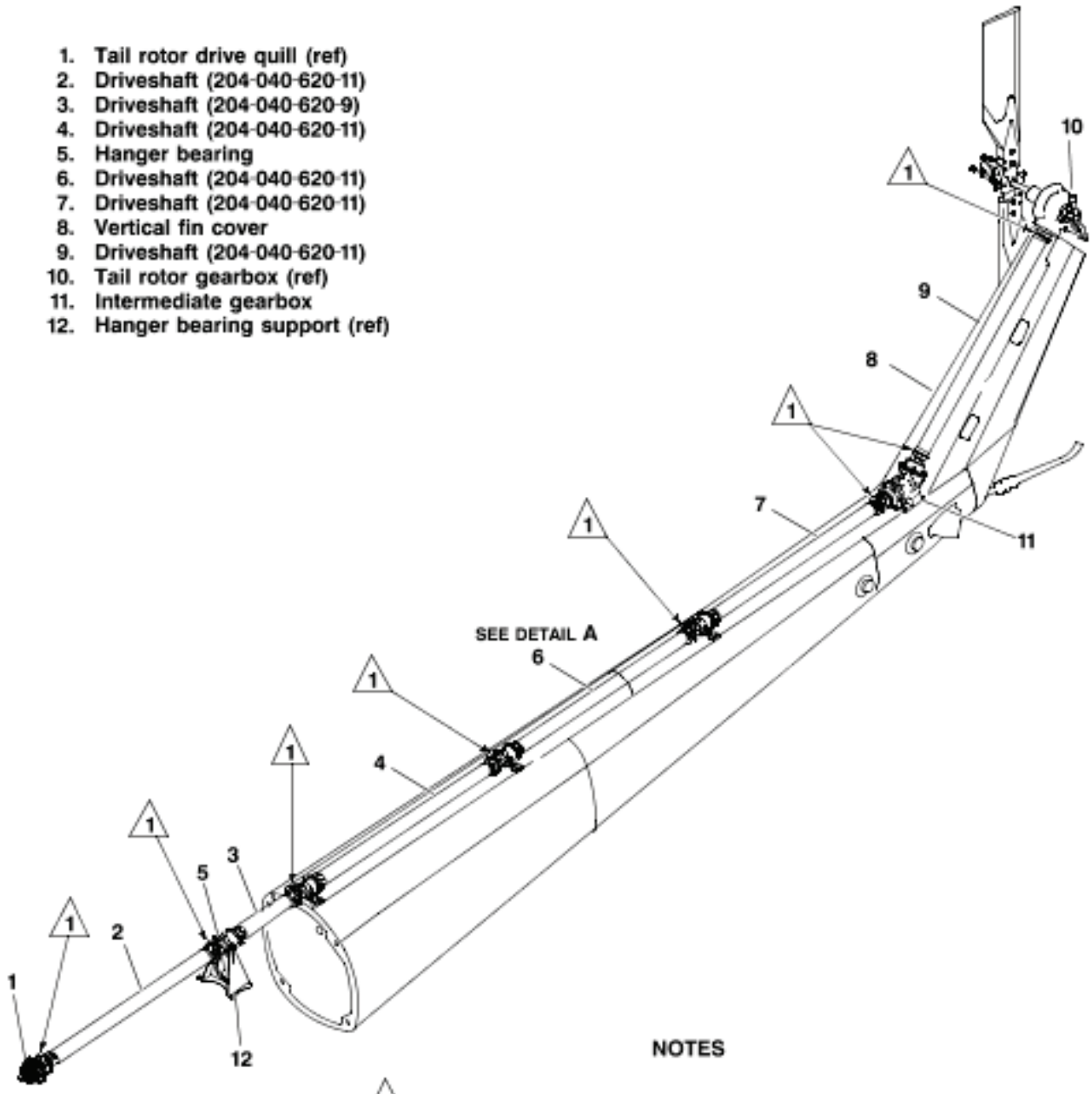


DO NOT DISPLACE EITHER END OF DRIVESHAFT UNTIL BOTH CLAMPS ARE REMOVED. FAILURE TO COMPLY MAY RESULT IN DAMAGE TO DRIVESHAFT HANGER BEARING AND/OR CURVIC COUPLINGS.

7. Remove nuts (14) and bolts (21). Remove clamp set (15). Reassemble clamp set to prevent intermixing of parts.

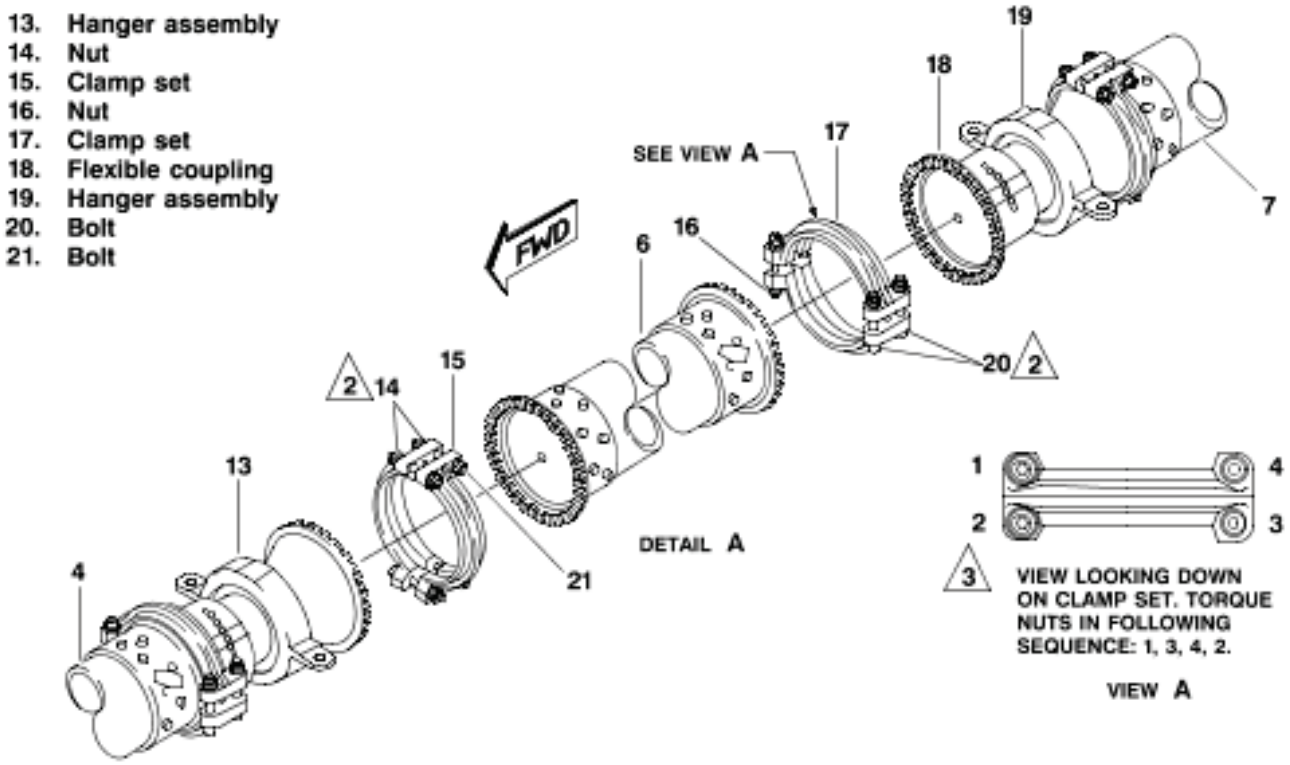
8. Remove nuts (16) and bolts (20). Remove clamp set (17). Reassemble clamp set to prevent intermixing parts.

9. Push flexible coupling (18) aft to disengage driveshaft (6) and remove driveshaft.



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Figure 1. Tail Rotor Driveshaft (Sheet 1 of 2)



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Figure 1. Tail Rotor Driveshaft (Sheet 2 of 2)

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Tail Rotor Driveshaft - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

NOTE

Do not allow solvent to enter driveshaft.

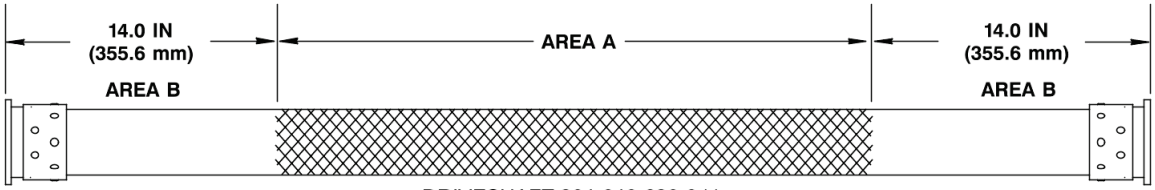
1. Clean driveshafts with clean cloth dampened with drycleaning solvent (C-304).
2. Dry with filtered, compressed air.

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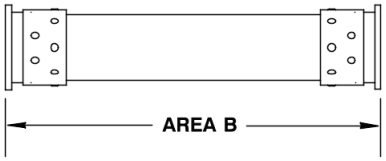
Tail Rotor Driveshaft - Inspection

Procedure

1. Inspect driveshafts in accordance with Figure 1 and Figure 2.
2. Inspect clamp sets for cracks, worn bolt holes, and distortion. Damage that can be detected visually is not acceptable.
3. Inspect clamp sets for gouges or wear pattern extending into fillet radius at bottom of internal groove. Damage that can be detected visually is not acceptable.
4. Inspect clamp sets for mechanical damage on spot face, lug fillets, and internal groove. Damage not exceeding 0.008 inch (0.203 mm) is acceptable without repair. Damage exceeding this limit is not acceptable and shall be repaired or the part replaced. Inspect remaining surfaces of clamp set for mechanical damage. Damage not exceeding 0.010 inch (0.254 mm) is acceptable without repair. Damage exceeding this limit is not acceptable and clamp set shall be repaired or replaced.



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412-040-620-101 AND -103
412-040-621-103 AND -107



DRIVESHAFT 204-040-620-009
412-040-621-101 AND -105

DAMAGE LOCATION SYMBOLS

AREA A **AREA B**

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED AFTER POLISHING OUT	
CRACKS ALLOWED	See Note 6	See Note 6
NICKS AND SCRATCHES	0.008 IN (0.203 mm)	0.012 IN (0.305 mm)
SHARP DENTS	0.010 IN (0.254 mm)	0.015 IN (0.381 mm)
NONSHARP DENTS	0.020 IN (0.508 mm)	0.030 IN (0.762 mm)

NOTES

1. **Damage defined on the chart above is acceptable if repaired as follows:**
   a. **Damage is polished out with a fine abrasive cloth with minimum radius of 0.5 inch (12.7 mm) and with surface finish of 32 RMS or better.**
   b. **Maximum deph after rework does not exceed limits shown in chart.**
   c. **Reworked area is treated for corrosion prevention.**
   d. **If rework area is eight square inches more on one side than the other, shaft balance must be checked.**

2. **Loss of one or more balance weights is cause to replace driveshaft. One empty bonding space should be open where bond test coupon was removed. If more than one empty space is observed, replace the driveshaft.**

3. **Damaged curvic coupling splines that result in radial play and/or backlash between assembled couplings when fully meshed without clamps are not acceptable.**

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Figure 1. Tail Rotor Driveshaft - Mechanical Damage Limits (Sheet 1 of 2)

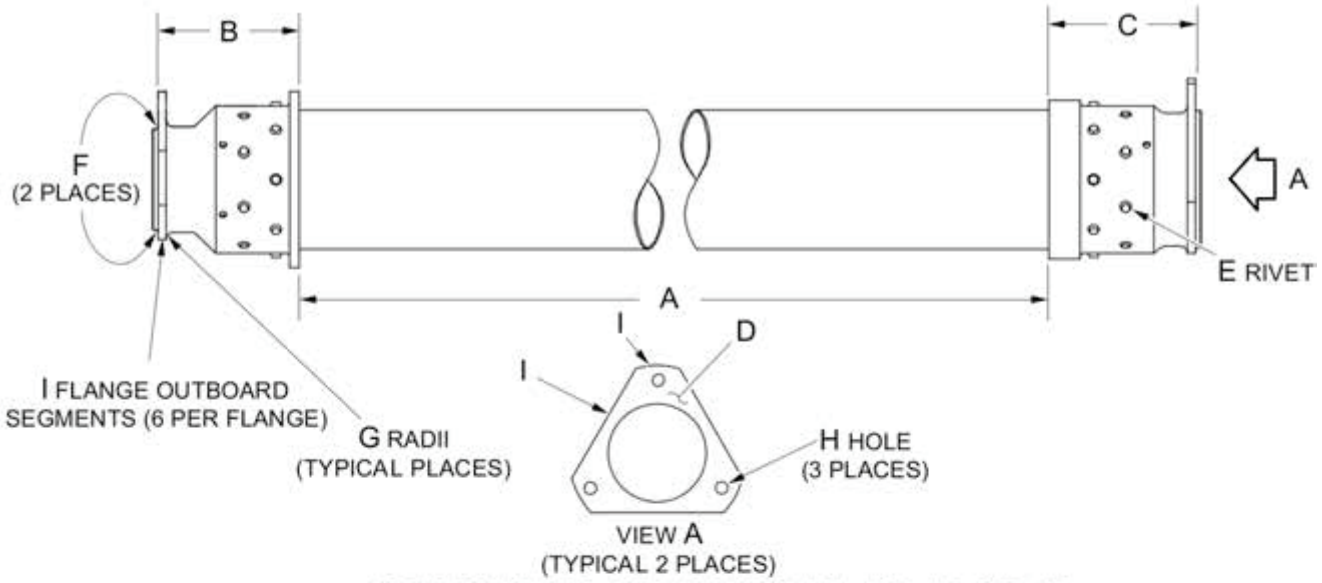
4. Grooves worn on shaft coupling by clamp to extent that wear prevents proper clamping are cause to replace driveshaft.

5. If tail rotor driveshaft distortion is suspected, support driveshaft on "V" blocks and measure runout. Maximum acceptable total indicator reading on long shafts is 0.050 inch (1.27 mm). Maximum acceptable total indicator reading on short shaft is 0.020 inch (0.508 mm). Do not attempt to straighten driveshaft.

6. Any crack in a driveshaft is cause to replace the affected driveshaft except cracks in rivet heads are acceptable if they do not exceed the following limits:
   a. Cracks are approximately radial when viewing top of shop head.
   b. No crack extends into an area with a diameter less than 1.25 diameters of rivet shank.
   c. Cracks do not intersect.
   d. The minimum distance between cracks is less than on shank diameter.
   e. A maximum of three such cracks.
   f. Crack width does not exceed 0.06 inch (1.52 mm) times rivet shank diameter.
   g. Maximum or 10% of rivet shop heads may have cracks within above limits.

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Figure 1. Tail Rotor Driveshaft - Mechanical Damage Limits (Sheet 2 of 2)



412-040-620-101 AND -103, 412-040-621-101, -103, -105 AND -107

NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT
1.	A	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth and 1.0 square inch (645 mm²) maximum area or 0.90 inch (22.8 mm) wide for 1/4 of circumference.
2.	A	Corrosion damage	Visual/ measure	0.002 inch (0.05 mm) maximum depth and 0.25 inch (6.3 mm) wide for a full circumference.
3.	B, C	Corrosion damage	Visual/ measure	0.012 inch (0.30 mm) maximum depth for 25% of any 1.0 square inch (645 mm²) and 25% of any surfaces. △1
4.	D	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth for 50% of any 1.0 square inch (645 mm²) and 0.002 inch (0.05 mm) maximum depth for the entire surface. △2
5.	E, F, G, H	Corrosion damage	Visual	None permitted.
6.	I	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth. Applicable to all segments, no maximum area. One repair per segment allowed.

NOTES

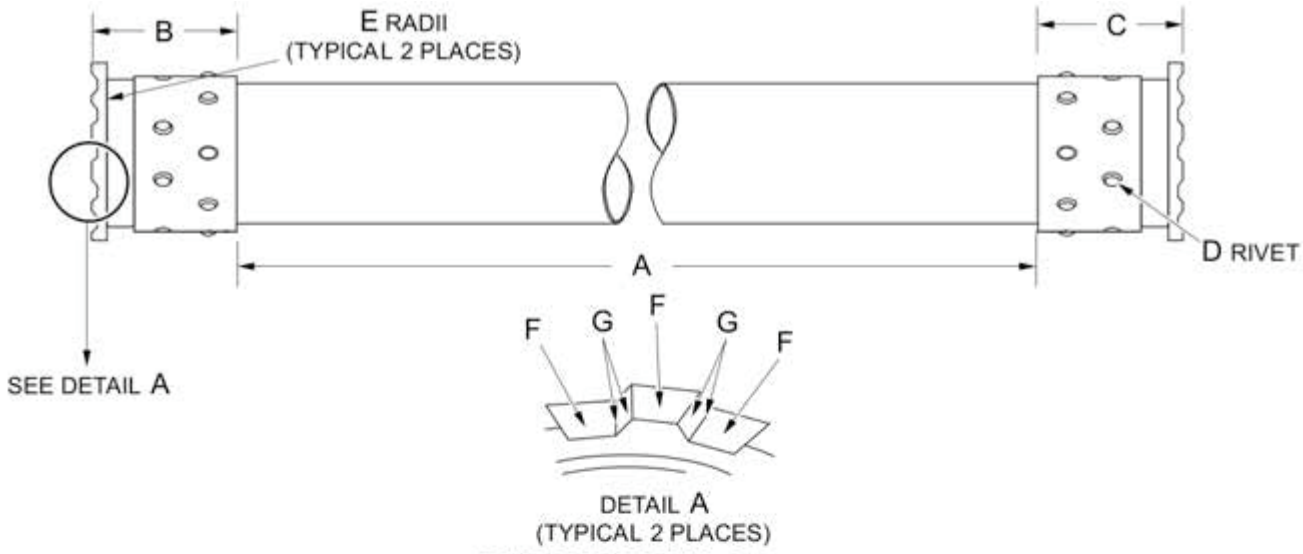
△1 No corrosion allowed within 0.20 inch (5.1 mm) of rivets or mounting holes.

△2 No corrosion allowed within 0.20 inch (5.1 mm) of mounting holes.

3. All dimensions, wea r, and damage limits provided are limits required after repair except where indicated.

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Figure 2. Tail Rotor Driveshaft - Corrosion Damage Limits (Sheet 1 of 2)



204-040-620-009 AND -011

NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT
1.	A	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth and 1.0 square inch (645 mm²) maximum area or 0.90 inch (22.8 mm) wide for 1/4 of circumference.
2.	A	Corrosion damage	Visual/ measure	0.002 inch (0.05 mm) maximum depth and 0.25 inch (6.3 mm) wide for a full circumference.
3.	B, C	Corrosion damage	Visual/ measure	0.012 inch (0.30 mm) maximum depth for 25% of any 1.0 square inch (645 mm²) and 25% of any surfaces. △1
4.	D, E	Corrosion damage	Visual	None permitted.
5.	F	Corrosion damage on curvic tooth top and bottom lands	Visual/ measure	0.012 inch (0.30 mm) maximum depth for 10% of any surfaces. △2
6.	G	Corrosion damage on curvic tooth working faces and fillets	Visual	None permitted.

NOTES

△1 No corrosion allowed within 0.20 inch (5.1 mm) of rivets or mounting holes.

△2 No corrosion or repair allowed within 0.030 inch (0.76 mm) of the curvic tooth fillets area.

3. All dimensions, wea r, and damage limits provided are limits required after repair except where indicated.

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Figure 2. Tail Rotor Driveshaft - Corrosion Damage Limits (Sheet 2 of 2)

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Tail Rotor Driveshaft - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Chemical film material	C-100	AR	
Epoxy primer	C-202	AR	
Abrasive cloth or paper	C-423	AR	

Procedure



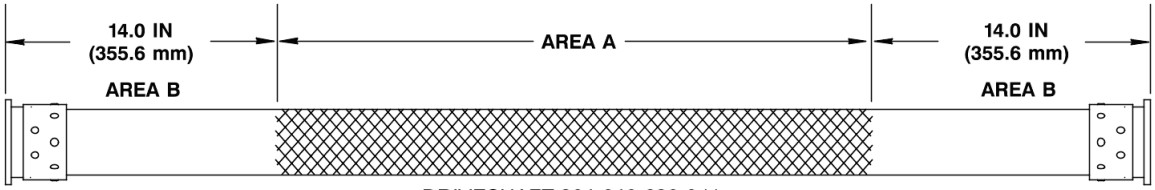
DO NOT USE GRINDING WHEEL TO POLISH OUT DAMAGE.

DRIVESHAFT MUST BE CHECKED FOR BALANCE IF REWORKED AREA IS 8.0 SQUARE INCHES (51.6 CM^2) MORE ON ONE SIDE THAN THE OTHER.

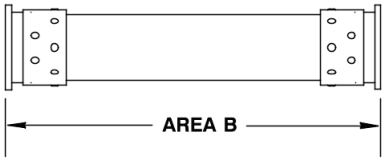
NOTE

Repair instructions for tail rotor driveshaft assemblies in this chapter are restricted to polishing out mechanical or corrosion damage.

1. Polish out mechanical and corrosion damage on driveshaft assemblies within limits shown (Figure 1 and Figure 2). Use 400 to 600 grit abrasive cloth or paper (C-423).
2. Make radius in rework areas 0.50 inch (12.7 mm) or greater. Surface finish shall be 32 RMS or better. When rework is complete, verify damage limits have not been exceeded (Figure 1 and Figure 2).
3. Treat repaired areas with chemical film material (C-100).
4. Apply one coat of epoxy primer (C-202).
5. Touch up paint to match existing finish (BHT-ALL-SPM, Standard Practices Manual).



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412-040-620-101 AND -103
412-040-621-103 AND -107



DRIVESHAFT 204-040-620-009
412-040-621-101 AND -105

DAMAGE LOCATION SYMBOLS

AREA A **AREA B**

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED AFTER POLISHING OUT	
CRACKS ALLOWED	See Note 6	See Note 6
NICKS AND SCRATCHES	0.008 IN (0.203 mm)	0.012 IN (0.305 mm)
SHARP DENTS	0.010 IN (0.254 mm)	0.015 IN (0.381 mm)
NONSHARP DENTS	0.020 IN (0.508 mm)	0.030 IN (0.762 mm)

NOTES

1. **Damage defined on the chart above is acceptable if repaired as follows:**
   a. **Damage is polished out with a fine abrasive cloth with minimum radius of 0.5 inch (12.7 mm) and with surface finish of 32 RMS or better.**
   b. **Maximum deph after rework does not exceed limits shown in chart.**
   c. **Reworked area is treated for corrosion prevention.**
   d. **If rework area is eight square inches more on one side than the other, shaft balance must be checked.**

2. **Loss of one or more balance weights is cause to replace driveshaft. One empty bonding space should be open where bond test coupon was removed. If more than one empty space is observed, replace the driveshaft.**

3. **Damaged curvic coupling splines that result in radial play and/or backlash between assembled couplings when fully meshed without clamps are not acceptable.**

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Figure 1. Tail Rotor Driveshaft - Mechanical Damage Limits (Sheet 1 of 2)

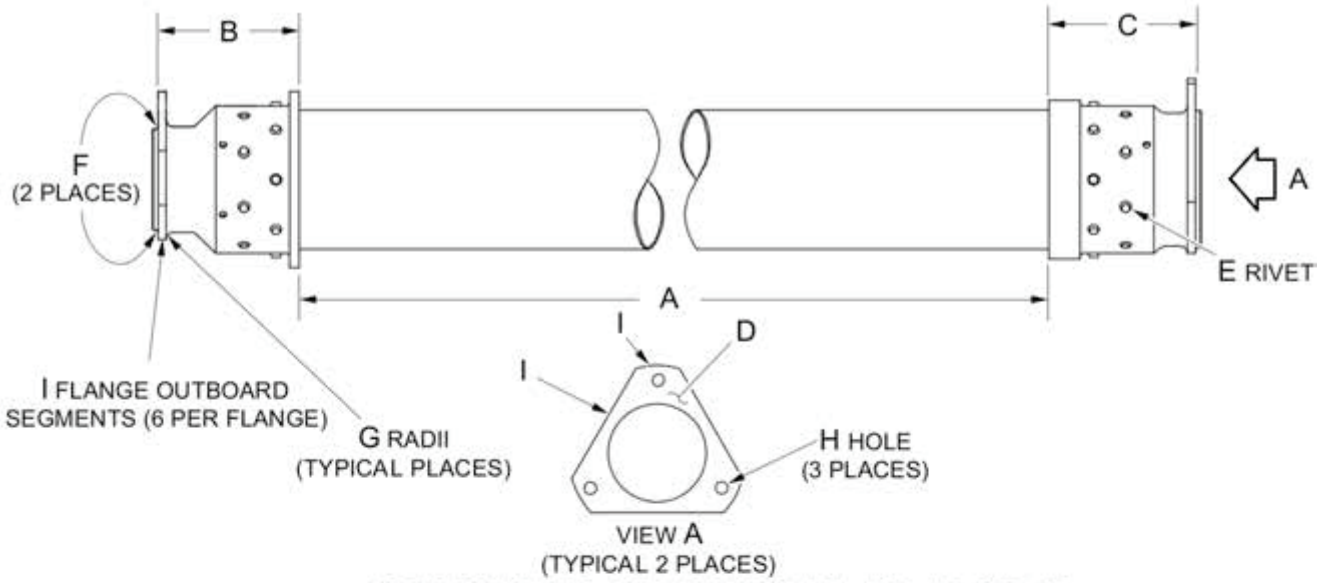
4. Grooves worn on shaft coupling by clamp to extent that wear prevents proper clamping are cause to replace driveshaft.

5. If tail rotor driveshaft distortion is suspected, support driveshaft on "V" blocks and measure runout. Maximum acceptable total indicator reading on long shafts is 0.050 inch (1.27 mm). Maximum acceptable total indicator reading on short shaft is 0.020 inch (0.508 mm). Do not attempt to straighten driveshaft.

6. Any crack in a driveshaft is cause to replace the affected driveshaft except cracks in rivet heads are acceptable if they do not exceed the following limits:
   a. Cracks are approximately radial when viewing top of shop head.
   b. No crack extends into an area with a diameter less than 1.25 diameters of rivet shank.
   c. Cracks do not intersect.
   d. The minimum distance between cracks is less than on shank diameter.
   e. A maximum of three such cracks.
   f. Crack width does not exceed 0.06 inch (1.52 mm) times rivet shank diameter.
   g. Maximum or 10% of rivet shop heads may have cracks within above limits.

ICN-412-A-651000-B-97499-00023-A-001-01

Figure 1. Tail Rotor Driveshaft - Mechanical Damage Limits (Sheet 2 of 2)



412-040-620-101 AND -103, 412-040-621-101, -103, -105 AND -107

NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT
1.	A	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth and 1.0 square inch (645 mm²) maximum area or 0.90 inch (22.8 mm) wide for 1/4 of circumference.
2.	A	Corrosion damage	Visual/ measure	0.002 inch (0.05 mm) maximum depth and 0.25 inch (6.3 mm) wide for a full circumference.
3.	B, C	Corrosion damage	Visual/ measure	0.012 inch (0.30 mm) maximum depth for 25% of any 1.0 square inch (645 mm²) and 25% of any surfaces. △1
4.	D	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth for 50% of any 1.0 square inch (645 mm²) and 0.002 inch (0.05 mm) maximum depth for the entire surface. △2
5.	E, F, G, H	Corrosion damage	Visual	None permitted.
6.	I	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth. Applicable to all segments, no maximum area. One repair per segment allowed.

NOTES

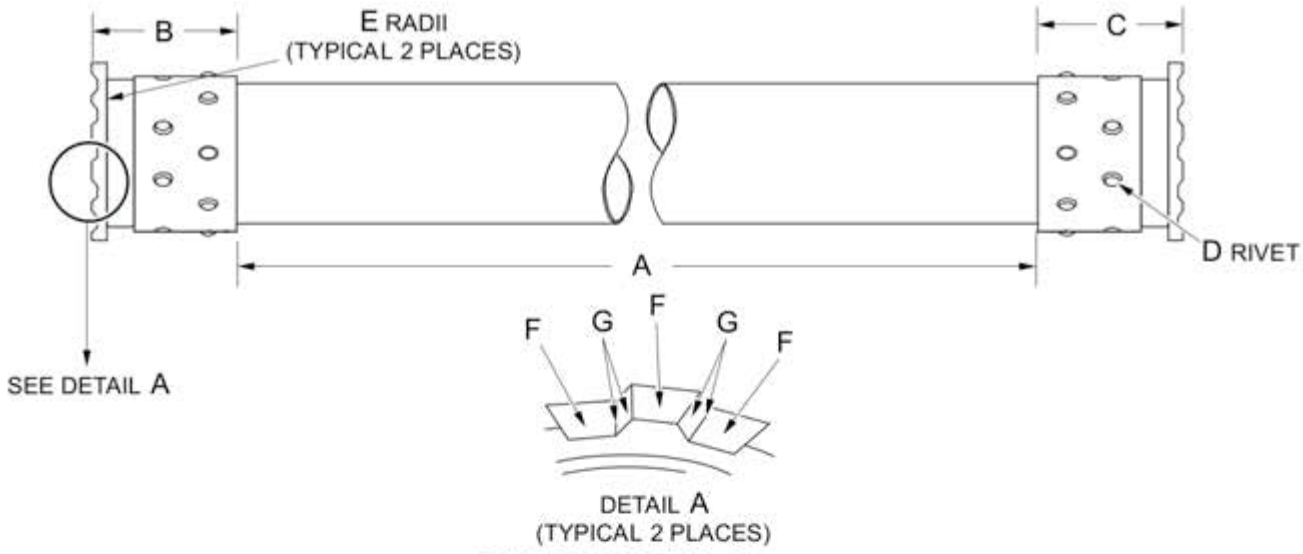
△1 No corrosion allowed within 0.20 inch (5.1 mm) of rivets or mounting holes.

△2 No corrosion allowed within 0.20 inch (5.1 mm) of mounting holes.

3. All dimensions, wea r, and damage limits provided are limits required after repair except where indicated.

ICN-412-A-651000-B-97499-01026-A-001-01

Figure 2. Tail Rotor Driveshaft - Corrosion Damage Limits (Sheet 1 of 2)



204-040-620-009 AND -011

NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT
1.	A	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth and 1.0 square inch (645 mm²) maximum area or 0.90 inch (22.8 mm) wide for 1/4 of circumference.
2.	A	Corrosion damage	Visual/ measure	0.002 inch (0.05 mm) maximum depth and 0.25 inch (6.3 mm) wide for a full circumference.
3.	B, C	Corrosion damage	Visual/ measure	0.012 inch (0.30 mm) maximum depth for 25% of any 1.0 square inch (645 mm²) and 25% of any surfaces. △1
4.	D, E	Corrosion damage	Visual	None permitted.
5.	F	Corrosion damage on curvic tooth top and bottom lands	Visual/ measure	0.012 inch (0.30 mm) maximum depth for 10% of any surfaces. △2
6.	G	Corrosion damage on curvic tooth working faces and fillets	Visual	None permitted.

NOTES

△1 No corrosion allowed within 0.20 inch (5.1 mm) of rivets or mounting holes.

△2 No corrosion or repair allowed within 0.030 inch (0.76 mm) of the curvic tooth fillets area.

3. All dimensions, wea r, and damage limits provided are limits required after repair except where indicated.

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Figure 2. Tail Rotor Driveshaft - Corrosion Damage Limits (Sheet 2 of 2)

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Tail Rotor Driveshaft - Installation

Procedure

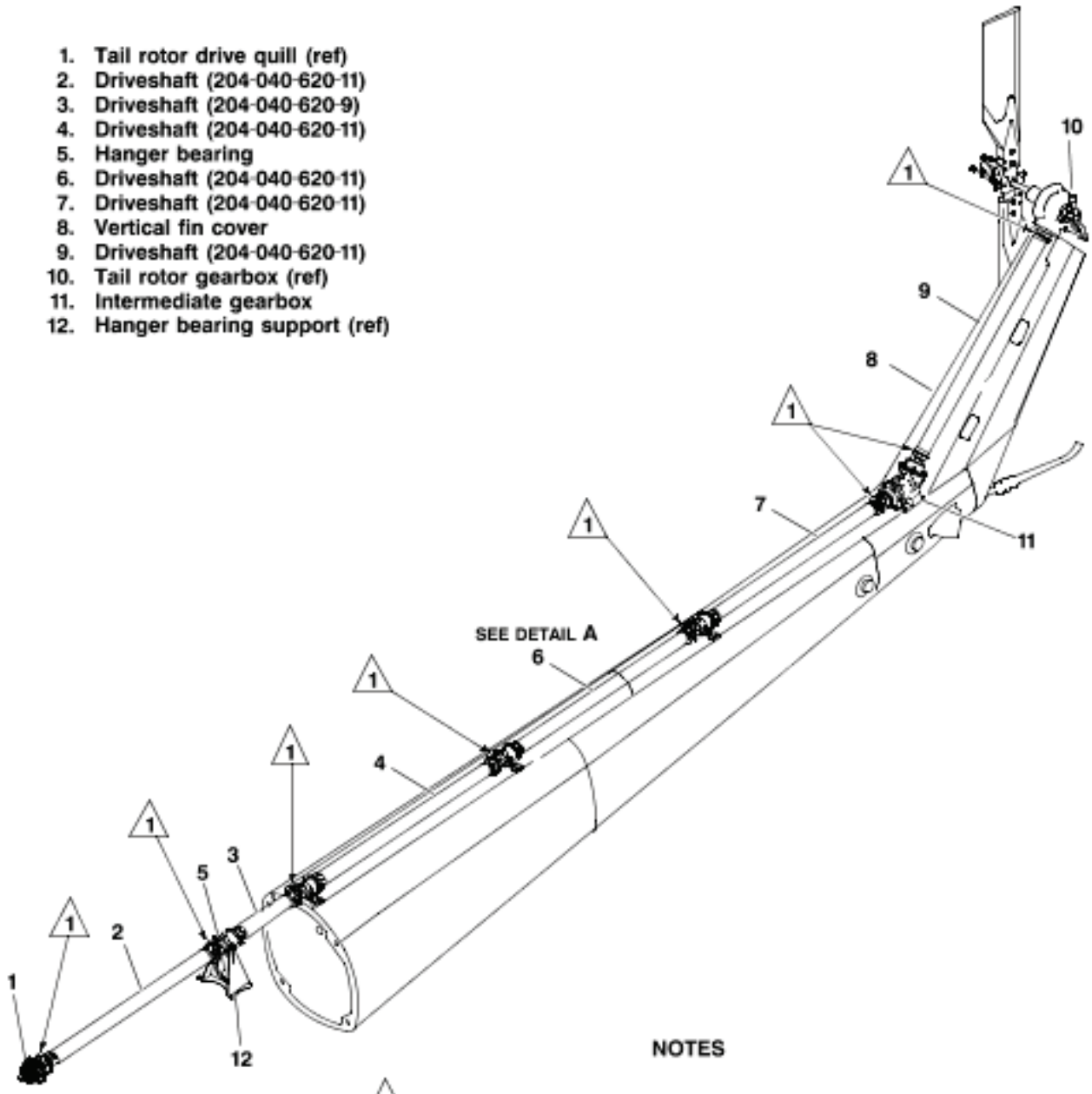


CLAMP SETS ARE MATCHED HALVES. IF IT IS NECESSARY TO REPLACE A CLAMP HALF, MATCH WITH ONE OF EQUAL WEIGHT WITHIN 1 GRAM.

NOTE

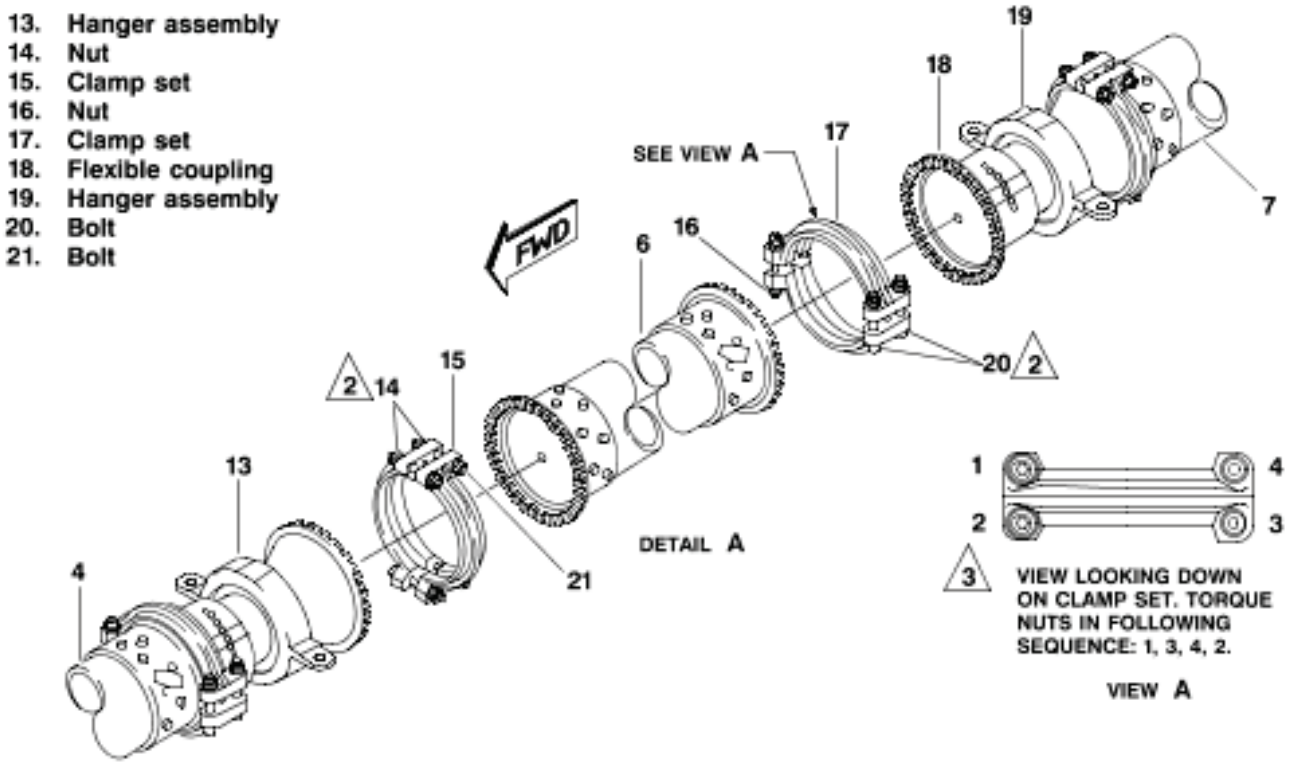
Installation procedure for driveshaft (6, Figure 1) is given. Installation procedures for remaining driveshafts are the same except for access and component removal provisions given for each driveshaft in note in 412-A-65-10-00-00A-520A-A / 01259.

1. Position driveshaft (6) between hanger assembly (13) and hanger assembly (19) with teeth meshed on each end.
2. Install clamp set (15) with heads of bolts (21) in direction of rotation. Do not torque nuts (14) at this time.
3. Install clamp set (17) indexed 90° to clamp set (15) with head of bolts (20) in direction of rotation. Do not torque nuts (16) at this time.
4. Adjust clamp sets (halves) (15 and 17) to obtain equal gaps at both ends within 0.030 inch (0.762 mm).
5. Torque nuts (16) evenly to 30 to 35 inch-pounds (3.39 to 3.95 Nm) in sequence (View A). Tap lightly around outside of clamp set (17) with a plastic mallet and retorque nuts.
6. Torque nuts (14) evenly to 30 to 35 inch-pounds (3.39 to 3.95 Nm) in sequence (View A). Tap lightly around outside of clamp set (15) with a plastic mallet and retorque nuts.
7. Close driveshaft hinged cover.
8. Install components and access doors or panels removed to facilitate removal of driveshaft other than driveshaft (6) (412-A-65-10-00-00A-520A-A / 01259).



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Figure 1. Tail Rotor Driveshaft (Sheet 1 of 2)



ICN-412-A-651000-B-97499-00021-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 2 of 2)

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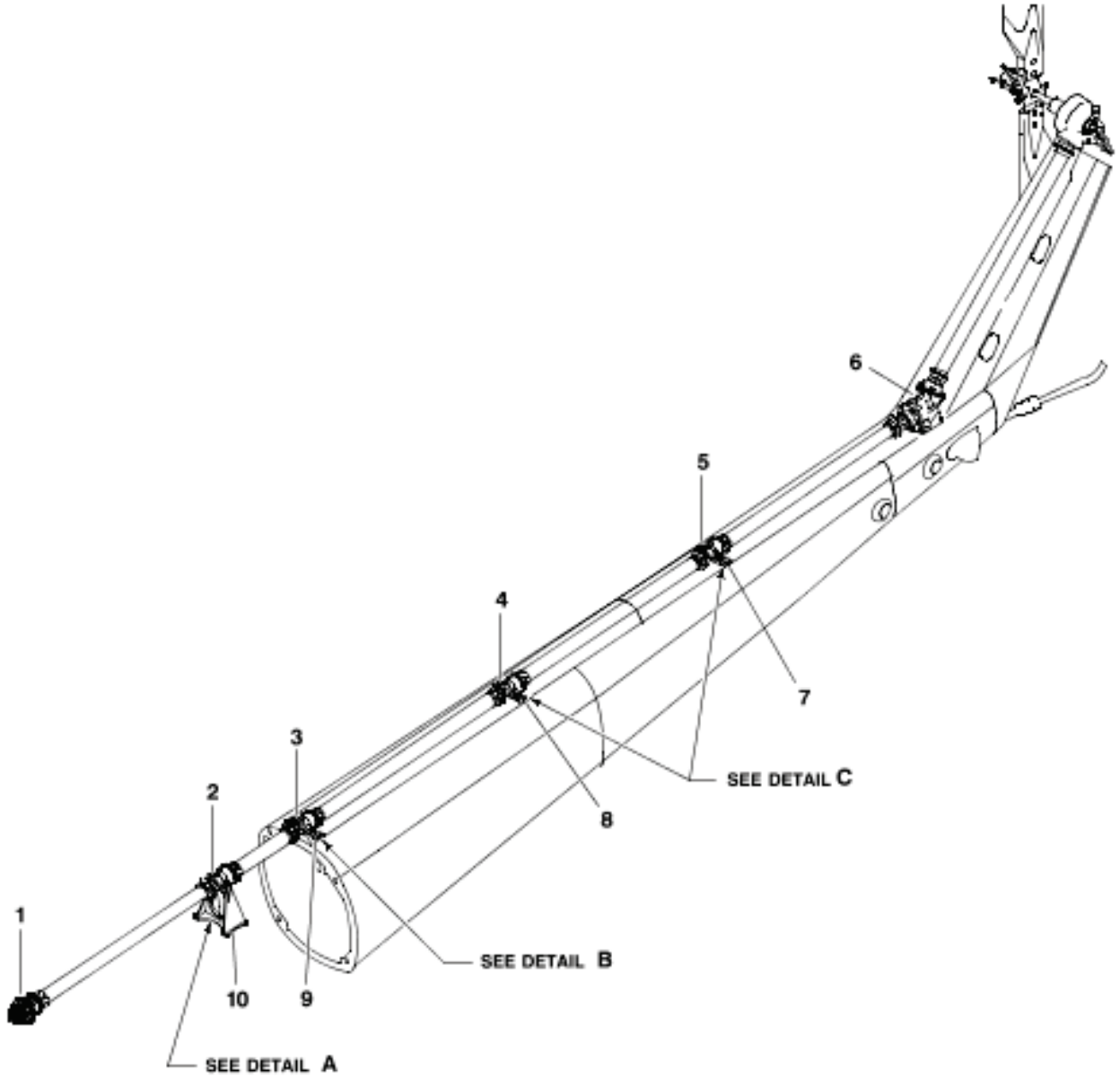
Driveshaft Hanger Assemblies - Removal

Procedure



DO NOT DISPLACE EITHER END OF DRIVESHAFT FROM MATING COUPLING UNLESS CLAMP SET IS REMOVED FROM BOTH ENDS OF DRIVESHAFT. FAILURE TO COMPLY MAY RESULT IN DAMAGE TO HANGER BEARING AND/OR CURVIC COUPLING.

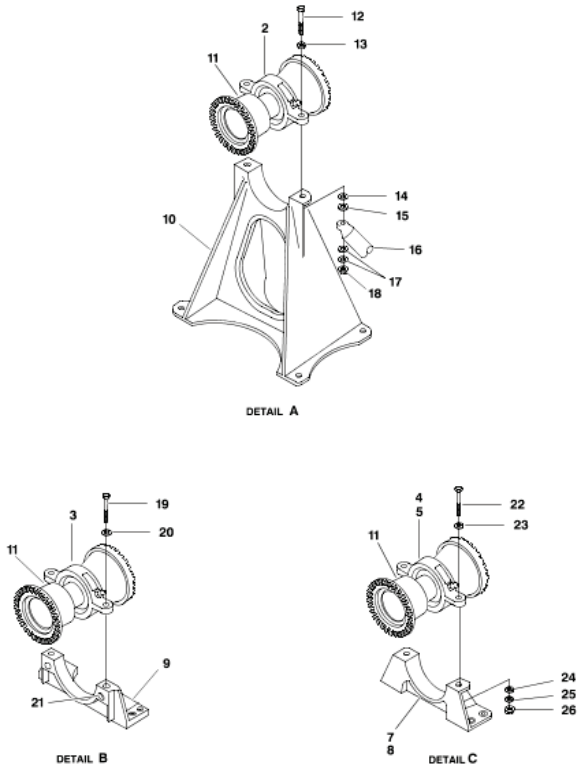
1. Remove tail rotor driveshaft sections from each end of driveshaft hanger to be removed (412-A-65-10-00-00A-520A-A / 01259).
2. Remove two bolts (12, Figure 1), washers (13, 14, 15, and 17), and nuts (18). Remove hanger assembly (2) from support (10).
3. Remove two bolts (19) and washers (20). Remove hanger assembly (3) from fitting (9).
4. Remove two bolts (22), washers (23, 24, and 25), and nuts (26). Remove hanger assembly (4) from fitting (8).
5. Remove hanger assembly (5) from fitting (7) same as previous step.



1. Drive quill (transmission) (ref)
2. Hanger assembly
3. Hanger assembly
4. Hanger assembly
5. Hanger assembly
6. Intermediate gearbox assembly (ref)
7. Fitting
8. Fitting
9. Fitting
10. Support
11. Flexible coupling
12. Bolt
13. Steel washer
14. Aluminum washer
15. Steel washer
16. Brace assembly
17. Steel washer
18. Nut
19. Bolt
20. Thin steel washer
21. Barrel nut
22. Bolt
23. Thin steel washer
24. Aluminum washer
25. Thin steel washer
26. Nut

ICN-412-A-651001-B-97499-00001-A-001-01

Figure 1. Tail Rotor Driveshaft Hanger (Sheet 1 of 2)



ICN-412-A-651001-B-97499-00002-A-001-01

Figure 1. Tail Rotor Driveshaft Hanger (Sheet 2 of 2)

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Driveshaft Hanger Assemblies - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure



DO NOT IMMERSE HANGER BEARING ASSEMBLIES IN SOLVENT. DO NOT ALLOW SOLVENT TO CONTACT INTERNAL SURFACES OF HANGER ASSEMBLIES.

1. Clean external surfaces of hanger assemblies with clean cloth dampened with drycleaning solvent (C-304).

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Driveshaft Hanger Assemblies - Inspection

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
India stone	C-464	AR	

Procedure

1. Inspect installed tail rotor driveshaft hanger assemblies for cracks, elongated bolt holes, or other visible damage.

NOTE

Procedures for hanger assembly (6, Figure 1) are typical for all hanger assemblies.

2. Inspect hanger assembly (6) exterior surfaces for discoloration and overheat condition. A change in color indicates a possible overheat condition and/or component degradation. The cause of exterior surface discoloration shall be determined and corrected prior to continued operation (412-CR&O, Chapter 65).

3. Inspect non-flexible couplings (9) for nicks, dents, and cracks. Minor damage that can be polished out with fine india stone (C-464) is acceptable.

4. Inspect outer coupling (3) for nicks, dents, and cracks. Minor damage that can be polished out with fine india stone (C-464) is acceptable. Inspect outer coupling overtemperature indicators TEMP-PLATES (10) for discoloration and overheat condition. Perform a serviceability check as follows:

4.1. Overtemperature indicators TEMP-PLATES (10) must not show evidence of overtemperature, deterioration, debonding, or discoloration of the epoxy coating that prevents interpretation of the indicating dots. If any of these conditions exist, proceed to Step 4.2..

4.2. If only one TEMP-PLATE is missing, and no dot on other TEMP-PLATE (on same coupling) is discolored or shows mechanical damage or degradation of the epoxy overcoating, the helicopter may be returned to service. The discrepant TEMP-PLATE should be replaced as soon as practical. If any indicator dot on the TEMP-PLATE has changed color to black, refer to Table 1 for probable cause and required corrective action.

5. Inspect seal (18) for protrusions, leakage, cuts, tears, and deterioration. Replace unserviceable seals (412-CR&O, Chapter 65).

6. Inspect bearing (4) in hanger assembly (6) and outer coupling for grease leakage. A small amount of grease expelled from around lip of bearing seals indicates a slight overservicing and is not cause for hanger replacement. If leakage is detected, perform an evaluation of grease leakage as follows:



DO NOT CLEAN OR SPRAY BEARING HANGER ASSEMBLY WITH ANY TYPE OF SOLVENT DURING INSPECTION. USE CLEAN, DRY CLOTHS ONLY TO CLEAN EXTERIOR OF HANGER ASSEMBLIES.

6.1. Wipe grease from exterior of inner coupling (17), outer coupling (3), and bearing (4) in hanger assembly (6).

6.2. Record which bearing is leaking and keep under observation for the next 10 flight hours. If amount of grease expelled from bearing seal does not decrease after 10 flight hours, replace hanger assembly.

7. Inspect bearing (4) in hanger assembly (6) for wear, roughness, and binding as follows:

7.1. Remove driveshaft from each side of hanger assembly (6).

7.2. Rotate bearing while pressing in axially on non-flexible coupling end of hanger bearing assembly. Bearing may feel smooth when turned with no load, but rough when loaded by pressing in with hand. Obvious roughness or binding when turned by hand is cause for replacement of hanger bearing assembly.

8. Inspect hanger bearing fittings (7, 8 and 9, Figure 2) and support (10) for mechanical and corrosion damage. Mechanical damage in excess of superficial and/or any corrosion damage that can be detected visually is not acceptable.

9. Replace driveshaft hanger assemblies for the following conditions:

9.1. Bearing is rough, binds when rotated, or exhibits excessive wear.

9.2. Hanger assembly (6, Figure 1) excessively worn, cracked, or exhibits signs of overheating.

9.3. Signs of metal particles and/or rust colored fretting debris near bearing.

9.4. Curvic face of couplings are excessively worn.

9.5. Couplings show signs of overheating or being run without lubricant.

10. Lubricate hanger bearing assembly as required (412-A-12-01-00-00A-040A-A / 00141).

11. Replace visual overtemperature indicators (TEMP-PLATES) if mutilated or discolored (412-CR&O, Chapter 65).

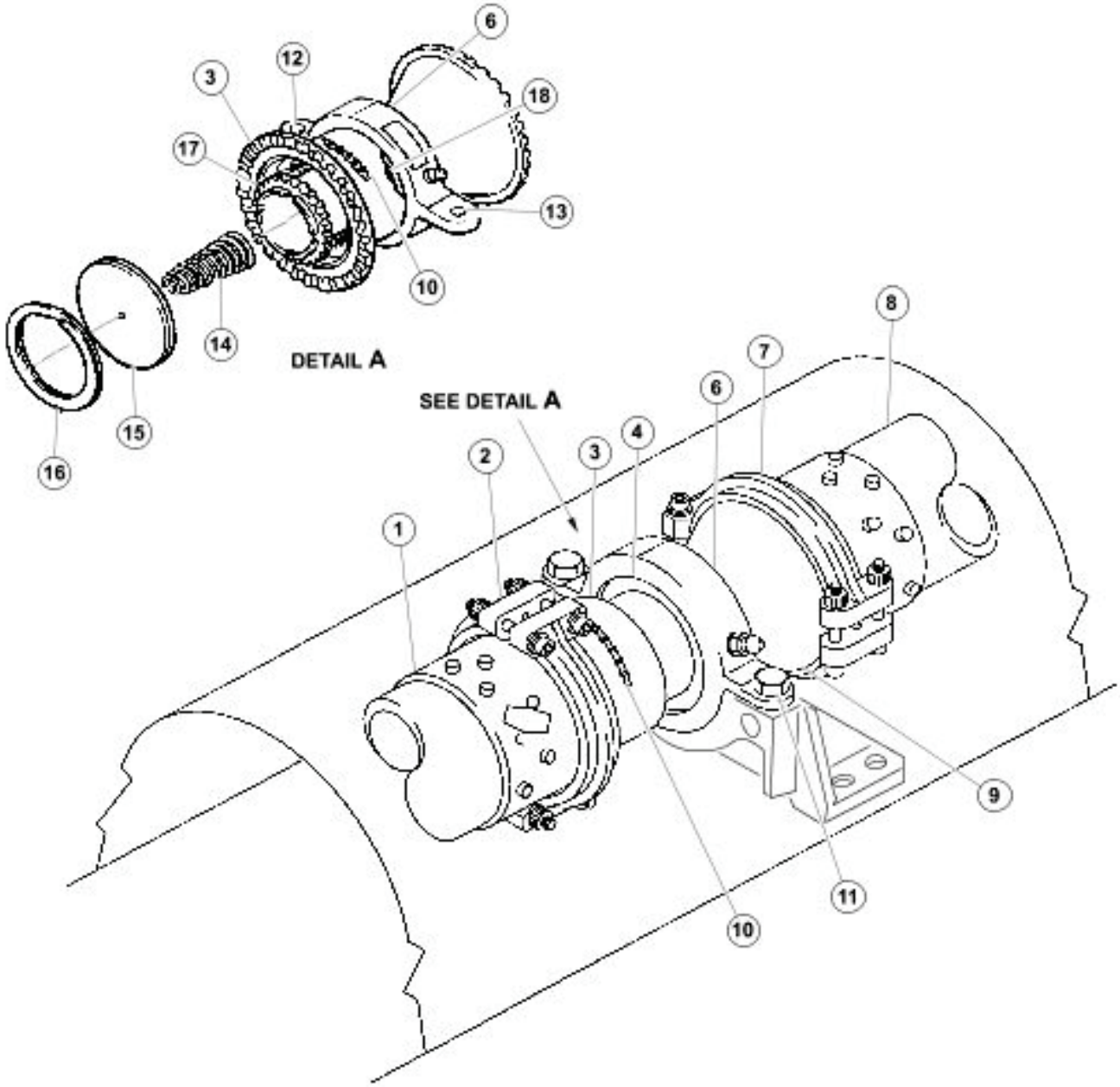
Table 1. Tail Rotor Driveshaft TEMP-PLATE Condition and Correction Action

ONE RED TEMP-PLATE	OTHER RED TEMP-PLATE	PROBABLE CAUSE	NOTES
Good	Black	Defect/Instl	1
Black	Black	Overtemp	2
Part Black	Good	Chem. Contamination	1
Missing	Good	Defect/Instl	1
Missing	Missing	Possible Overtemp	2

NOTES:

1. Resume operation. Defective TEMP-PLATE or improper installation, replace defective TEMP-PLATE as soon as practical in accordance with 412-CR&O, Chapter 65.

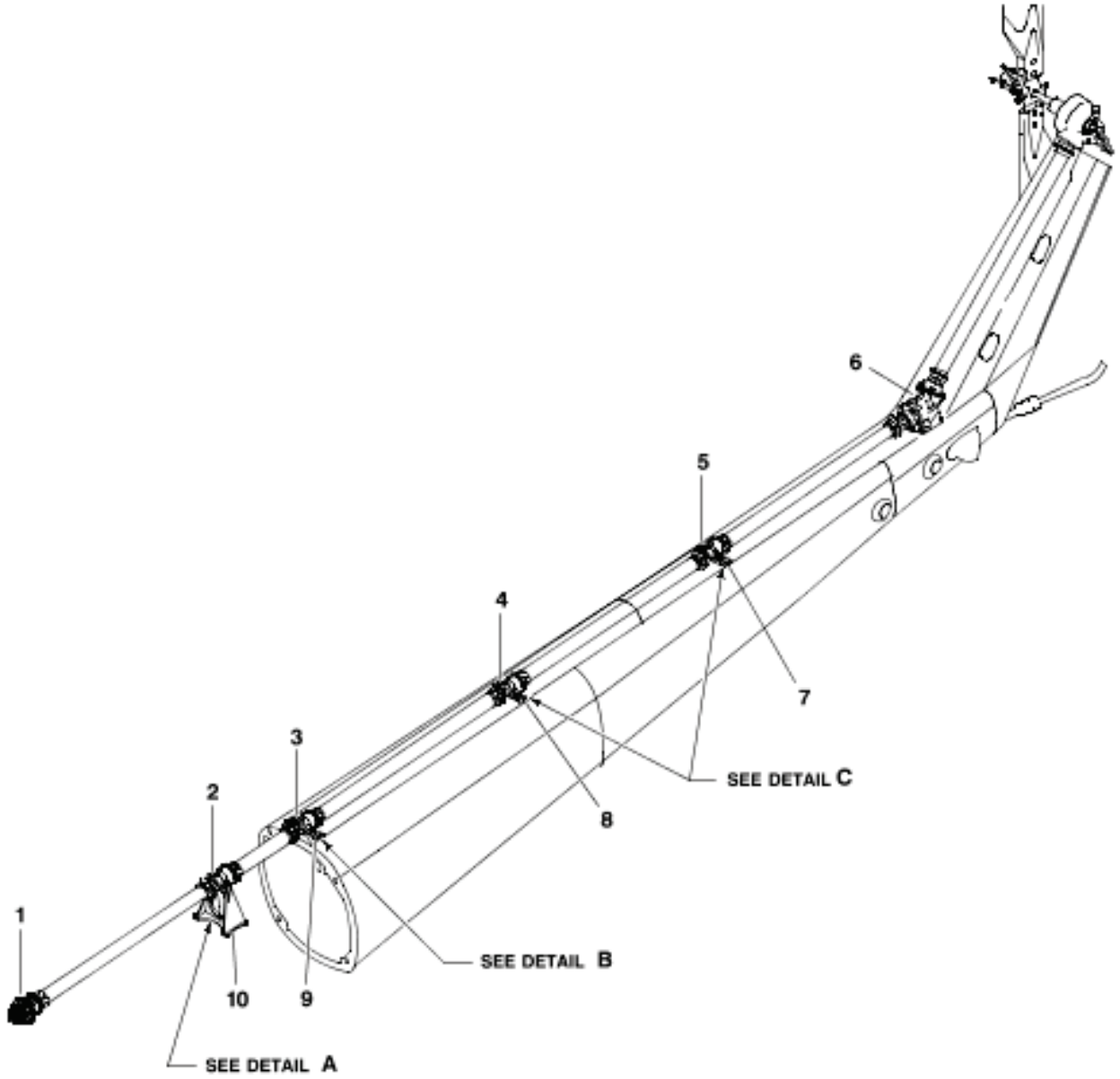
2. Coupling overtemperature condition is very likely. Remove coupling assembly and perform overtemperature inspection in accordance with 412-CR&O, Chapter 65.



1. Driveshaft
2. Clamp
3. Outer coupling
4. Bearing
5. Deleted
6. Hanger assembly
7. Clamp
8. Driveshaft
9. Coupling
10. Overtemperature indicator TEMP-PLATES
11. Bolt
12. Hanger bolt hole
13. Hanger bolt hole
14. Spring
15. Plate
16. Retainer ring
17. Inner coupling
18. Seal

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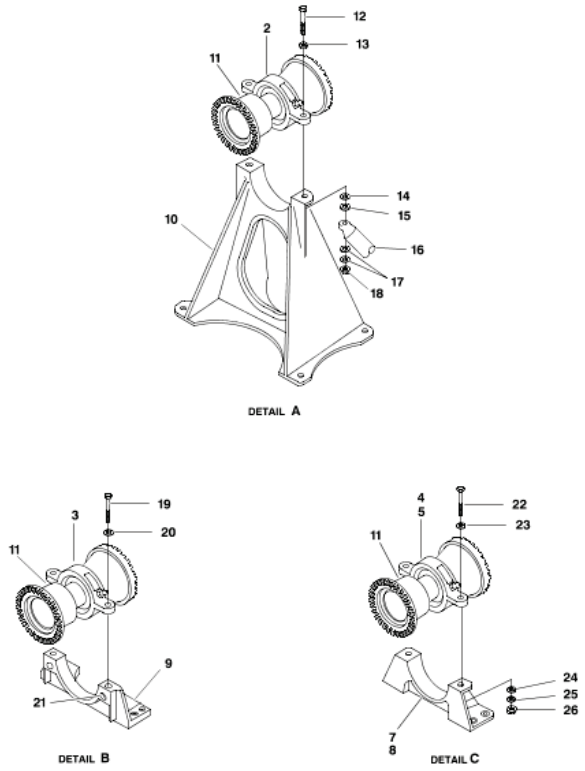
Figure 1. Tail Rotor Driveshaft Hanger Assembly (Sheet 1 of 1)



1. Drive quill (transmission) (ref)
2. Hanger assembly
3. Hanger assembly
4. Hanger assembly
5. Hanger assembly
6. Intermediate gearbox assembly (ref)
7. Fitting
8. Fitting
9. Fitting
10. Support
11. Flexible coupling
12. Bolt
13. Steel washer
14. Aluminum washer
15. Steel washer
16. Brace assembly
17. Steel washer
18. Nut
19. Bolt
20. Thin steel washer
21. Barrel nut
22. Bolt
23. Thin steel washer
24. Aluminum washer
25. Thin steel washer
26. Nut

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Figure 2. Tail Rotor Driveshaft Hanger (Sheet 1 of 2)



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Figure 2. Tail Rotor Driveshaft Hanger (Sheet 2 of 2)

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Driveshaft Hanger Assemblies - Lubrication

Procedure

1. Remove tail rotor driveshaft sections attached to hanger bearing to be lubricated (412-A-65-10-00-00A-520A-A / 01259).

2. Remove tail rotor driveshaft hanger assemblies (412-A-65-10-01-00A-520A-A / 01264).

3. Clean tail rotor drive shaft hanger assemblies (412-A-65-10-01-00A-251A-A / 01265).

4. Disassemble and lubricate tail rotor driveshaft hanger assemblies (412-CR&O, Chapter 65).

5. Install tail rotor driveshaft hanger assemblies (412-A-65-10-01-00A-720A-A / 01268).

NOTE

Ground run and leak check must be performed following installation of new, repaired, or relubricated flexible couplings.

6. Make an entry in helicopter log and in flex couplings lubrication log of date lubricated, date grease manufactured and helicopter hours (412-A-12-01-00-00A-040A-A / 00141).

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Driveshaft Hanger Assemblies - Installation

Procedure

1. Position hanger assembly (2, Figure 1) on support (10) with flexible coupling (11) forward. Install washers (14 and 15) between support and brace assembly (16) on each side. Install bolts (12) with steel washers (13) through hanger assembly, support, and brace assembly. Install steel washers (17) and nuts (18).



VERIFY BOLTS (19) FOR PROPER LENGTH. BOLTS THAT ARE TOO LONG COULD RESULT IN IMPROPER TORQUE APPLICATION.

2. Position hanger assembly (3) on fitting (9) with flexible coupling (11) forward. Make sure barrel nuts (21) are aligned on each side. Install two bolts (19) and thin steel washers (20).

NOTE

Use additional thin steel washers (23) if necessary to obtain proper bolt thread engagement when installing hanger assembly (4 or 5).

3. Position hanger assembly (4) on fitting (8) with flexible coupling (11) forward. Install bolts (22) and thin steel washers (23) through hanger assembly (4) and fittings. Install aluminum washer (24) next to fitting, thin steel washer (25), and nut (26).

4. Install hanger assembly (5) on fitting (7) using procedures in previous step.

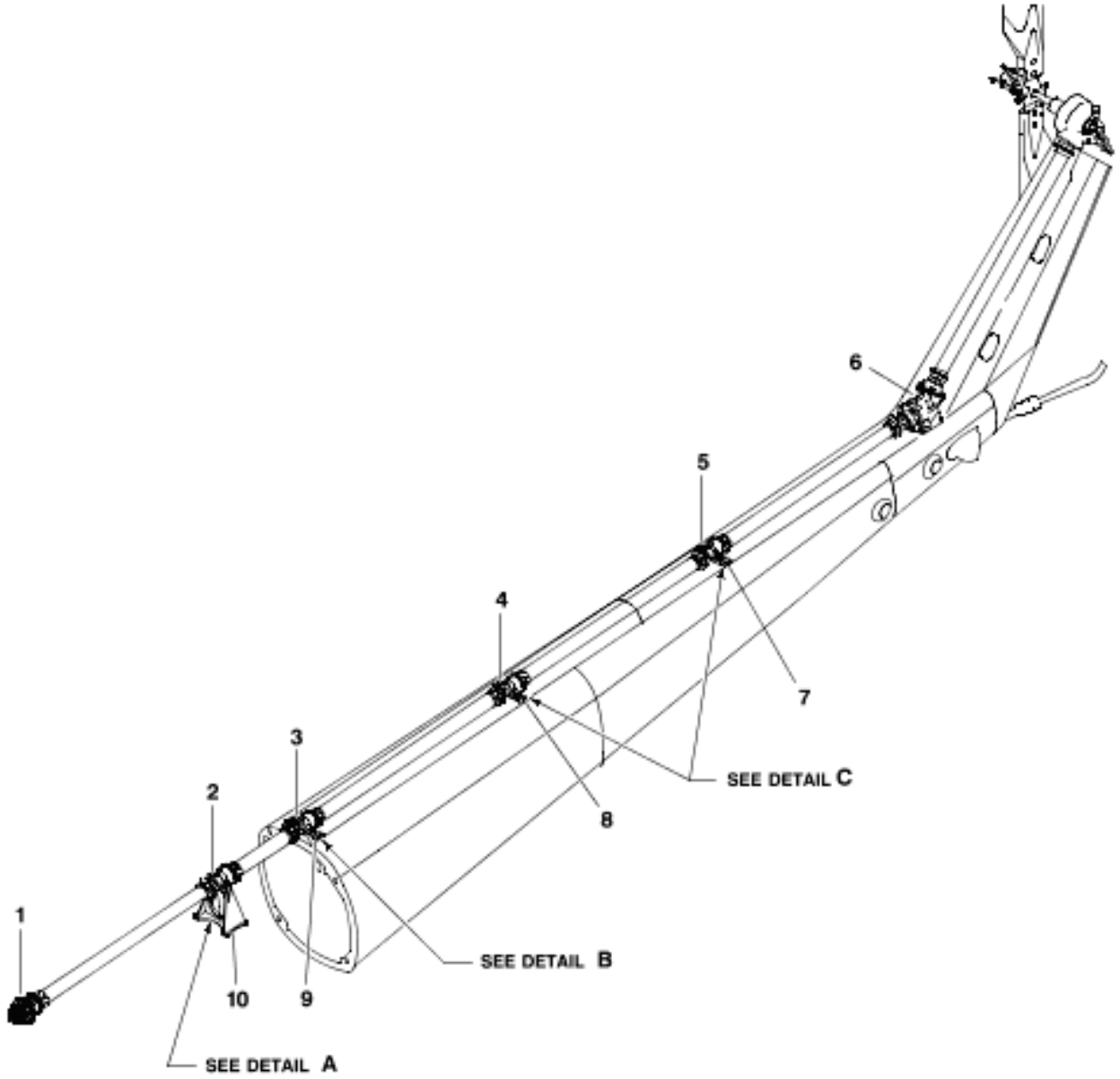
5. Install tail rotor driveshafts (412-A-65-10-00-00A-720A-A / 01263).

6. Operate helicopter for 5 minutes at 100% (BHT-412-FM-5, Flight Manual).

7. Shut down engine and inspect coupling and surrounding structure for evidence of grease leakage. If there is any evidence of grease leakage, remove hanger assembly and replace defective part(s).

8. Inspect hanger assemblies (2, 3, 4, and 5) exterior surfaces for discoloration and overheat condition. A change in color indicates a possible overheat condition and/or component degradation. The cause of exterior surface discoloration shall be determined and corrected prior to continued operation (412-CR&O, Chapter 65).

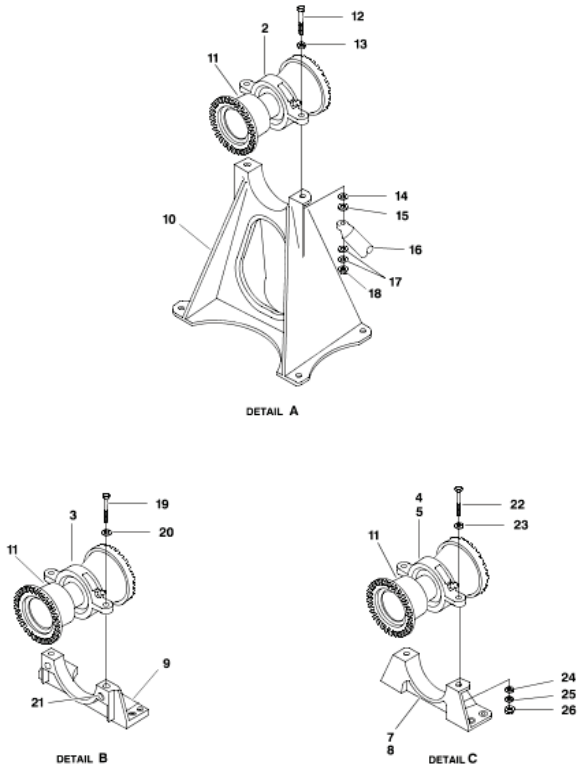
9. Inspect flexible couplings (11) overtemperature indicator (TEMP-PLATES) for discoloration and overheat condition (412-A-65-10-01-00A-280A-A / 01266).



1. Drive quill (transmission) (ref)
2. Hanger assembly
3. Hanger assembly
4. Hanger assembly
5. Hanger assembly
6. Intermediate gearbox assembly (ref)
7. Fitting
8. Fitting
9. Fitting
10. Support
11. Flexible coupling
12. Bolt
13. Steel washer
14. Aluminum washer
15. Steel washer
16. Brace assembly
17. Steel washer
18. Nut
19. Bolt
20. Thin steel washer
21. Barrel nut
22. Bolt
23. Thin steel washer
24. Aluminum washer
25. Thin steel washer
26. Nut

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Figure 1. Tail Rotor Driveshaft Hanger (Sheet 1 of 2)



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Figure 1. Tail Rotor Driveshaft Hanger (Sheet 2 of 2)

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Tail Rotor Drive System - Alignment

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Hanger Simulator	412-240-033-101	2	
Hanger Simulator	412-240-034-101	1	
Hanger Simulator	412-240-035-101	1	
Intermediate Gearbox Simulator	T103224-101	1	
Bushing	T103225-105	1	Part of Tool Assembly T103225-119
Bushing	T103225-109	1	Part of Tool Assembly T103225-119
Alignment Plate	T103225-119	1	
Tail Rotor Gearbox Simulator	T103226-101	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-001	AR	
Sealant	C-251	AR	
Aliphatic naphtha	C-305	AR	
MEK	C-309	AR	
Adhesive	C-317	AR	
Lockwire	C-508	AR	

Procedure

1. A tail rotor drive system alignment check shall be made whenever driveshaft misalignment is suspected, when damage to airframe structure in area of driveshaft hangers requires structural repair or parts replacement, or when major structural repairs are performed in any area of tailboom or in aft section of forward fuselage. Alignment of driveshaft hanger supports and intermediate gearbox shall be accomplished as follows:

1.1. Open tail rotor driveshaft covers and intermediate gearbox cover.

1.2. Remove tail rotor driveshafts and hangers (412-A-65-10-00-00A-520A-A / 01259 and 412-A-65-10-01-00A-520A-A / 01264).

NOTE

Do not remove driveshaft hanger support assemblies from tailboom or forward fuselage.

1.3. Remove intermediate gearbox (412-A-65-20-00-00A-520A-A / 01280). Ensure any shims between gearbox and tailboom remain in their original locations on tailboom.

1.4. Remove tail rotor gearbox (412-A-65-20-01-00A-520A-A / 01302).

1.5. Remove tailboom from helicopter and place in suitable support cradle (412-A-53-50-00-00A-520A-A / 00912).

1.6. Clean surfaces on tailboom and hanger supports where hangers and gearboxes were removed of primer and sealant. Use a plastic scraper and a clean cloth dampened with MEK (C-309) or aliphatic naphtha (C-305).

1.7. Install alignment plate (T103225-119) on forward tailboom bulkhead as follows:

NOTE

alignment plate (T103225-119) has two bolts going through bushings in lower part of plate that are secured with two barrel nuts. These barrel nuts have to be removed prior to installing alignment plate to forward tailboom bulkhead.

1.7.1. Position alignment plate (T103225-119) on forward tailboom bulkhead with bushing (T103225-109) on left side and bushing (T103225-105) on right side (Figure 1, View A, Detail A).

NOTE

Before fully tightening bolts AN9-31A and AN8-31A, centralize position of alignment plate (T103225-119) within limits of looseness on bolts.

1.7.2. Secure left side of alignment plate (T103225-119) to forward tailboom bulkhead with bolt AN9-31A going through washer AN960-916 and bushing (T103225-109).

1.7.3. Secure right side alignment plate (T103225-119) to forward tailboom bulkhead with bolt AN8-31A going through washer AN960-816 and bushing (T103225-105).

1.8. Install tail rotor gearbox simulator (T103226-101)/90° on top of vertical fin.



MAKE SURE THREE LEVELING SCREWS IN INTERMEDIATE GEARBOX SIMULATOR ARE BACKED OFF SO THEY DO NOT PROJECT BELOW THE MOUNTING SURFACE OF THE SIMULATOR BEFORE PROCEEDING.

1.9. Install intermediate gearbox simulator (T103224-101) using bolts AN4-13A and sufficient number of washers AN960-416 and AN960-416L under heads of bolts, so bolt grips do not bottom out in mating tailboom nutplates.



VERIFY tail rotor gearbox simulator (T103226-101) AND BULKHEAD alignment plate (T103225-119) ARE SECURED BEFORE PROCEEDING.

1.10. Install unkinked 0.020 steel lockwire (C-508) in hole of tail rotor gearbox simulator (T103226-101). Secure end of wire on upper side of plate.

1.11. Pass other end of wire through hole in top of plate of intermediate gearbox simulator (T103224-101) around V groove in pin and out through hole of forward (vertical) end plate (Figure 1, View E).

1.12. Pass wire through hole in tailboom bulkhead alignment plate (T103225-119) and tighten wire until all slack is removed. Secure end of wire with safety wire spinner pliers or other positive locking pliers.

NOTE

Make sure wire is still unkinked and is still positioned in V groove in the pin in intermediate gearbox simulator (T103224-101).

1.13. Check for alignment of intermediate gearbox by confirming wire clears the edges of holes in top and forward end plates by at least 0.005 inch (0.127 mm) (View B). Pluck wire and check to see if the wire continues to vibrate. If vibration continues, wire clearance is sufficient to proceed. Using a 6-inch ruler or equivalent, measure distance from center of wire to top surface of tailboom deck at forward face of intermediate gearbox simulator (T103224-101). Distance must measure 2.470 to 2.560 inches (62.738 to 65.024 mm).

NOTE

If either of the above conditions are not satisfactory, proceed to Step 1.13.1.. If both conditions are satisfactory, proceed directly to Step 1.14..

1.13.1. Loosen 0.020 inch diameter alignment wire and remove intermediate gearbox simulator (T103224-101).

1.13.2. Remove all shims on tailboom at intermediate gearbox location; note thickness of shim-stacks removed. Using a plastic scraper and a cloth damp with MEK (C-309) remove any adhesive or sealant residue around gearbox mounting hole locations on tailboom.

1.13.3. Adjust three levelling screws on intermediate gearbox simulator (T103224-101) so that they project below mounting surface an amount approximately equal to thickness of shim-stack established in Step 1.13.2. at corresponding mounting bolt location.



WHEN intermediate gearbox simulator (T103224-101) IS SUPPORTED ON ITS THREE LEVELING SCREWS. TIGHTEN FOUR SIMULATOR MOUNTING BOLTS TO ONLY 20 TO 30 INCH-POUNDS (2.26 TO 3.39 NM).

1.13.4. Install intermediate gearbox simulator (T103224-101) on tailboom in accordance with Step 1.9. and reinstall alignment wire in accordance with Step 1.11. through Step 1.12.. Ensure alignment wire is positioned in V groove in simulator pin.

1.13.5. Loosen or tighten each of four simulator mounting bolts while adjusting each of three levelling bolts until an acceptable simulator position is obtained (Step 1.13.).

1.13.6. Make sure simulator mounting bolts are torqued to only 20 to 30 inch-pounds (2.26 to 3.39 Nm); then, using a thickness gauge, measure and record gaps between lower surface of simulator and tailboom at each of the mounting bolt locations.

1.13.7. Size thickness of shim-stacks for each simulator mounting bolt location equal to the respective gaps measured in Step 1.13.6..

1.13.8. Loosen alignment wire and remove intermediate gearbox simulator (T103224-101). Retract each of three leveling bolts so they do not project below lower surface of simulator.

1.13.9. Reinstall simulator in accordance with Step 1.9. using shim-stacks manufactured in Step 1.13.7. and reinstall alignment wire in accordance with Step 1.11. and Step 1.12.. Tighten simulator mounting bolts to 50 to 70 inch-pounds (5.65 to 7.91 Nm) and confirm satisfactory wire positions in holes in top and forward end plates of simulator and satisfactory wire elevation above tailboom in accordance with Step 1.13.. If wire placement is satisfactory, proceed to Step 1.13.10.. Otherwise, repeat Step 1.13.1. through Step 1.13.9. until satisfactory alignment is obtained.

1.13.10. Loosen alignment wire and remove intermediate gearbox simulator (T103224-101). Measure and record final thickness of shims at each of four mounting bolt locations.

1.13.11. Clean the bonding surfaces of the shims and the tailboom with aliphatic naphtha (C-305).

1.13.12. Bond the solid portion of the shims on the tailboom with adhesive (C-317). Do not put adhesive within 0.250 inch (6.35 mm) of the bolt holes.

1.13.13. Seal the edges of the shims with sealant (C-251).

NOTE

Coat bottom surface of intermediate gearbox simulator (T103224-101) and the threads and shank of each simulator mounting bolt with a very light coat of petroleum jelly or grease (C-001) prior to installation.

1.13.14. Reinstall intermediate gearbox simulator and tighten four mounting bolts 50 to 70 inch-pounds (5.65 to 7.91 Nm). Do not remove simulator until shim adhesive has set.

1.13.15. Reinstall alignment wire in accordance with Step 1.11. and Step 1.12. and confirm satisfactory wire positions in holes in top and forward end plates of simulator in accordance with Step 1.13..

1.14. Remove alignment wire from alignment plate (T103225-119) and thread wire through forward and aft holes of a hanger simulator (412-240-033-101).

1.15. Install hanger simulator (412-240-033-101) at number four location on tailboom using two bolts AN4-10A, two nuts MS35650-3252, and sufficient number of washers AN960-416L under nut to keep nut from bottoming out in bolt grip (Detail B). Leave fasteners slightly loose at this point.

1.16. Using uniform hand pressure on extreme aft end of simulator, push hanger simulator forward and hold in this position while tightening the two bolts.

1.17. Tighten alignment wire in accordance with Step 1.11. and Step 1.12.. Ensure alignment wire is positioned in V groove in the intermediate gearbox simulator (T103224-101) pin.

1.18. Verify acceptable alignment of hanger simulator (412-240-033-101) by confirming alignment wire clears edges of holes in forward and aft end plates of simulator by at least 0.005 inch (0.127 mm) (View B). Pluck wire and watch wire closely to see if the wire continues to vibrate. If vibration continues, wire clearance in holes is sufficient to proceed. Also verify alignment wire is in acceptable positions in holes in forward and top end plates of intermediate gearbox simulator (T103224-101). If alignment of hanger simulator is acceptable proceed directly to Step 1.19.; otherwise proceed as follows:



DO NOT LOOSEN HANGER SUPPORT TO AIRFRAME BOLTS ONCE YAW, LATERAL POSITION, AND ELEVATION ADJUSTMENTS HAVE BEEN MADE.

1.18.1. If hanger simulator (412-240-033-101) position error is in yaw, loosen four bolts securing hanger support to tailboom and shift support to obtain acceptable alignment. Retighten bolts and verify acceptable alignment in accordance with Step 1.18..

1.18.2. If hanger simulator error is in pitch direction, remove four bolts securing hanger support to tailboom. To adjust hanger simulator pitch forward, add equal shims under both aft support mounting pads and remove same thickness of shims from under both forward mounting pads. Do the opposite to adjust hanger simulator pitch aft. Install four mounting bolts and verify acceptable alignment in accordance with Step 1.18.. It will likely be necessary to perform Step 1.18.1. to achieve acceptable yaw alignment.

NOTE

Elongation of the four mounting holes in the hanger support (20, Figure 2) on the forward fuselage is not permitted.

No more than two hanger supports on any tailboom may be reworked to facilitate lateral alignment.

1.18.3. If a lateral position error exists at any hanger (not simply a yaw error), remove alignment wire and hanger simulator. Remove four bolts securing hanger support to tailboom and remove support. Rework only the problem support in accordance with Figure 1, View F. Following rework, reinstall support. Reinstall hanger simulator and alignment wire in accordance with Step 1.14.. through Step 1.17.. Loosen four bolts slightly and adjust lateral position of support so as to obtain satisfactory lateral alignment of simulator. Verify acceptable alignment in accordance with Step 1.18.. It will likely be necessary to repeat Step 1.18.1. to restore acceptable yaw alignment.

1.18.4. If hanger simulator position error is in "elevation" (vertical position), remove four bolts securing hanger support to tailboom and add or remove same thickness of shims under each of four support mounting pads. Install four mounting bolts and verify acceptable hanger simulator alignment in accordance with Step 1.18.. It will likely be necessary to repeat Step 1.18.1. and Step 1.18.3. to achieve acceptable "yaw" and lateral position alignment.

1.18.5. When all three alignment conditions are satisfied proceed to Step 1.19..

NOTE

Ensure alignment wire remains free of kinks and is seated in "V" groove in intermediate gearbox simulator pin in the following operations.

1.19. Remove hanger simulator (412-240-033-101), from number four hanger location and install at location three. Perform Step 1.15. thru Step 1.18..

1.19.1. Loosen the alignment wire and remove hanger simulator (412-240-033-101), from hanger location three.

NOTE

hanger simulator (412-240-035-101) and hanger simulator (412-240-034-101), are very similar in appearance. Confirm the correct simulator is installed in the following operation.

1.19.2. Install hanger simulator (412-240-035-101), at hanger location two using two AN4H7A bolts and a sufficient number of AN960-416L washers to keep the bolts from bottoming out in the bolt grip. Leave fasteners slightly loose at this point. Perform Step 1.16. thru Step 1.18., except, in Step 1.18. the hanger simulator will be the hanger simulator (412-240-035-101).

1.20. Tail rotor driveshaft hanger assembly alignment on tailboom is now complete. Remove tailboom bulkhead alignment plate (T103225-119), alignment wire, and hanger simulator (412-240-035-101).

1.21. Install tailboom on helicopter (412-A-53-50-00-00A-720A-A / 00915).



THE FOLLOWING PROCEDURES FOR ALIGNMENT OF HANGER BEARING ASSEMBLY AT HANGER POSITION NUMBER ONE MUST BE PERFORMED USING A TAILBOOM KNOWN TO HAVE ACCEPTABLE TAIL ROTOR DRIVE SYSTEM ALIGNMENT.

1.22. Install hanger assemblies as follows:

1.22.1. Install hanger simulator (412-240-034-101), at number two hanger location in accordance with Step 1.19.2. for hanger simulator (412-240-035-101) except perform only referenced Step 1.16.. See view C.

1.22.2. Install hanger simulator (412-240-033-101), at number three and number four hanger locations in accordance with Step 1.15.. and Step 1.16.. See view C.

1.22.3. Install hanger simulator (412-240-035-101), at number one hanger position in accordance with instructions given for hanger simulator (412-240-033-101) in Step 1.16.. and Step 1.16.. See view C.



DO NOT LOOSEN BOLTS SECURING ANY HANGER SUPPORT TO TAILBOOM IN AN EFFORT TO ACHIEVE SATISFACTORY ALIGNMENT IN Step 1.23. THROUGH Step 1.28..

NOTE

In Step 1.23.., Step 1.24.., and Step 1.25.., the hanger simulator (412-240-035-101) at hanger number one location is used as a wire anchor. Alignment of number one hanger support is accomplished in accordance with Step 1.26.. through ab.

1.23. Install a length of unkinked 0.020 in. diameter wire in accordance with Step 1.10. and Step 1.11.. Route wire through each of three hanger simulators. Tighten wire until all slack is removed. Secure wire at forward end of hanger simulator (412-240-035-101), at number one hanger location. Secure end of wire with safety-wire spinner pliers or other positive-locking pliers. Ensure alignment wire is positioned in "V" groove in the intermediate gearbox simulator pin.

NOTE

A stable cushion such as shot bags or a sand bag should be placed between tailboom skin and support so as to distribute the load and prevent damage to tailboom structure. Tension on alignment wire may require adjustment during tailboom position operation to prevent wire sag.

1.24. Position an adjustable support under tailboom. This support should be located at a point approximately 12 in. (30.48 cm) forward of intermediate gearbox location where extension of canted vertical fin spar approaches adjacent tailboom bulkhead at lower surface of tailboom, see view C. Slowly adjust elevation of support as required to position alignment wire within acceptable limits in each of three hanger simulators, see view B. Once this position is obtained do not reposition helicopter or readjust tailboom support until tail rotor drive system alignment is complete.

1.25. Remove alignment wire and hanger simulator (412-240-035-101), from number one hanger location. Reposition hanger simulator (412-240-034-101), from number two hanger location to number one hanger location on forward fuselage. Install hanger simulator (412-240-034-101), in accordance with instructions given for the hanger simulator (412-240-033-101) in Step 1.15.. and Step 1.16.. See view D.

1.26. Remove hanger simulator (412-240-033-101), from number four hanger location. Install hanger simulator (412-240-035-101), at number two hanger location in accordance with Step 1.18.2..

1.27. Install a new length of unkinked 0.020 in. diameter wire in accordance with Step 1.10. and Step 1.11.. Route wire through hanger simulators at number three, two and one hanger locations. Tighten wire until all slack is removed and secure wire at the forward end of hanger simulator (412-240-034-101), at number one hanger location. Secure end of wire with safety-wire spinner pliers or other positive-locking pliers. Ensure alignment wire is seated in "V" groove in intermediate gearbox simulator (T103224-101) pin.

NOTE

As an alternate procedure the alignment wire may be anchored at forward face of hanger simulator (412-240-034-101), and tension/clamped at top end plate of intermediate gearbox simulator, T103224-101.

1.28. Verify acceptable alignment of hanger simulator by confirming alignment wire clears edges of holes in forward and aft end plates of hanger simulators on tailboom, in aft end plate of hanger simulator at number one hanger location and in forward and top end plates of intermediate gearbox simulator. See view B. If alignment of hanger simulator (412-240-034-101), is acceptable proceed directly to Step 1.29.; otherwise proceed as follows:

NOTE

Elongation of the four mounting holes in the No. 1 hanger support is not permitted.

1.28.1. If hanger simulator error is in "yaw" direction, loosen four bolts securing hanger support to airframe and shift support to obtain acceptable alignment. Tighten four bolts and verify acceptable alignment in accordance with Step 1.28..

1.28.2. If hanger simulator error is in "pitch" position, relax tension on alignment wire and remove four bolts securing hanger support to airframe. To "pitch" hanger simulator forward add equal shims under both aft support mounting pads and remove same thickness of shims from under both forward mounting pads. Do just the opposite to "pitch" support aft. Install four support mounting bolts, tighten alignment wire and verify acceptable hanger simulator alignment in accordance with Step 1.28.. It may be necessary to perform Step 1.28.1. to achieve acceptable "yaw" alignment.

1.28.3. If hanger simulator error is in "elevation" position (alignment wire hits top or bottom of holes in hanger simulator at number two hanger location), relax tension on alignment wire and remove four bolts securing hanger support to airframe. Add or remove equal shim thickness under each of four support mounting pads to obtain required elevation. Install four mounting bolts and verify acceptable hanger simulator alignment in accordance with Step 1.28.. It may be necessary to repeat Step 1.28.1. to achieve acceptable yaw alignment.

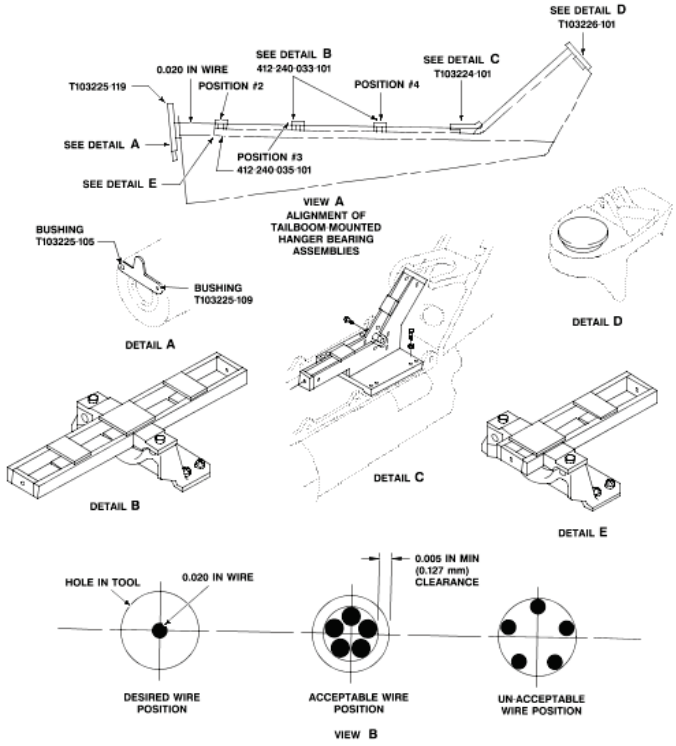
1.29. Tail rotor drive system alignment is complete. Remove all alignment tools and tailboom support. Verify bolts securing hanger supports to forward fuselage and to tailboom are properly torqued. Remove any grease remaining on uppermost shim at intermediate gearbox location.

2. Install tail rotor gearbox (412-A-65-20-01-00A-720A-A / 01306).

3. Install intermediate gearbox (412-A-65-20-00-00A-720A-A / 01285).

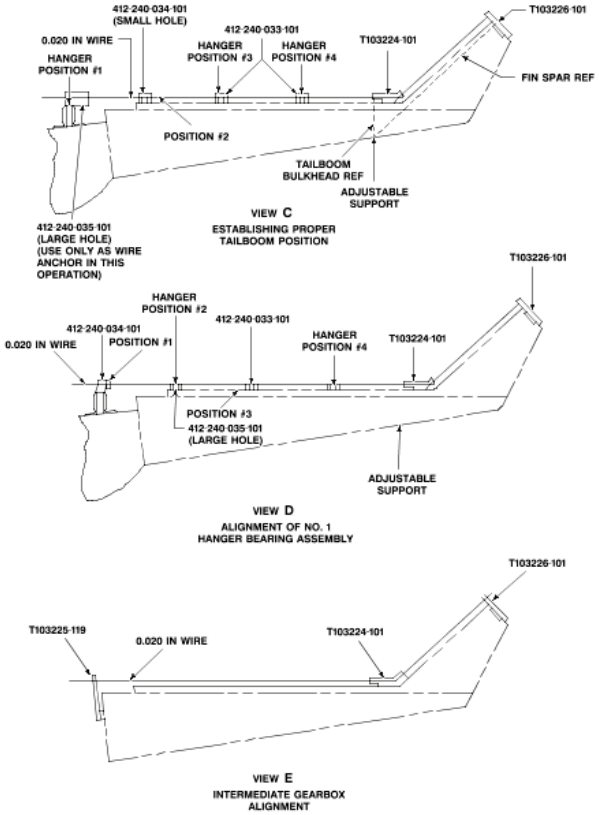
4. Install tail rotor hangers and tail rotor driveshafts (412-A-65-10-01-00A-720A-A / 01268 and 412-A-65-10-00-00A-720A-A / 01263).

5. Close and secure tail rotor driveshaft covers.



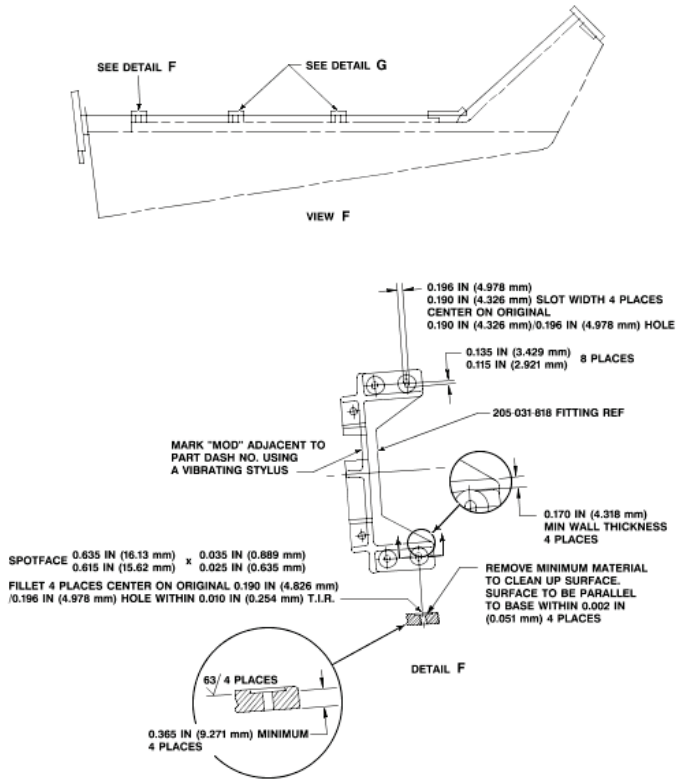
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Figure 1. Tail Rotor Driveshaft Alignment (Sheet 1 of 4)



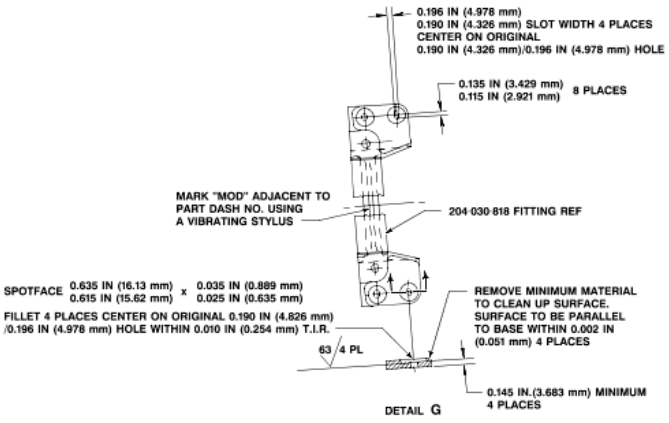
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Figure 1. Tail Rotor Driveshaft Alignment (Sheet 2 of 4)



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Figure 1. Tail Rotor Driveshaft Alignment (Sheet 3 of 4)

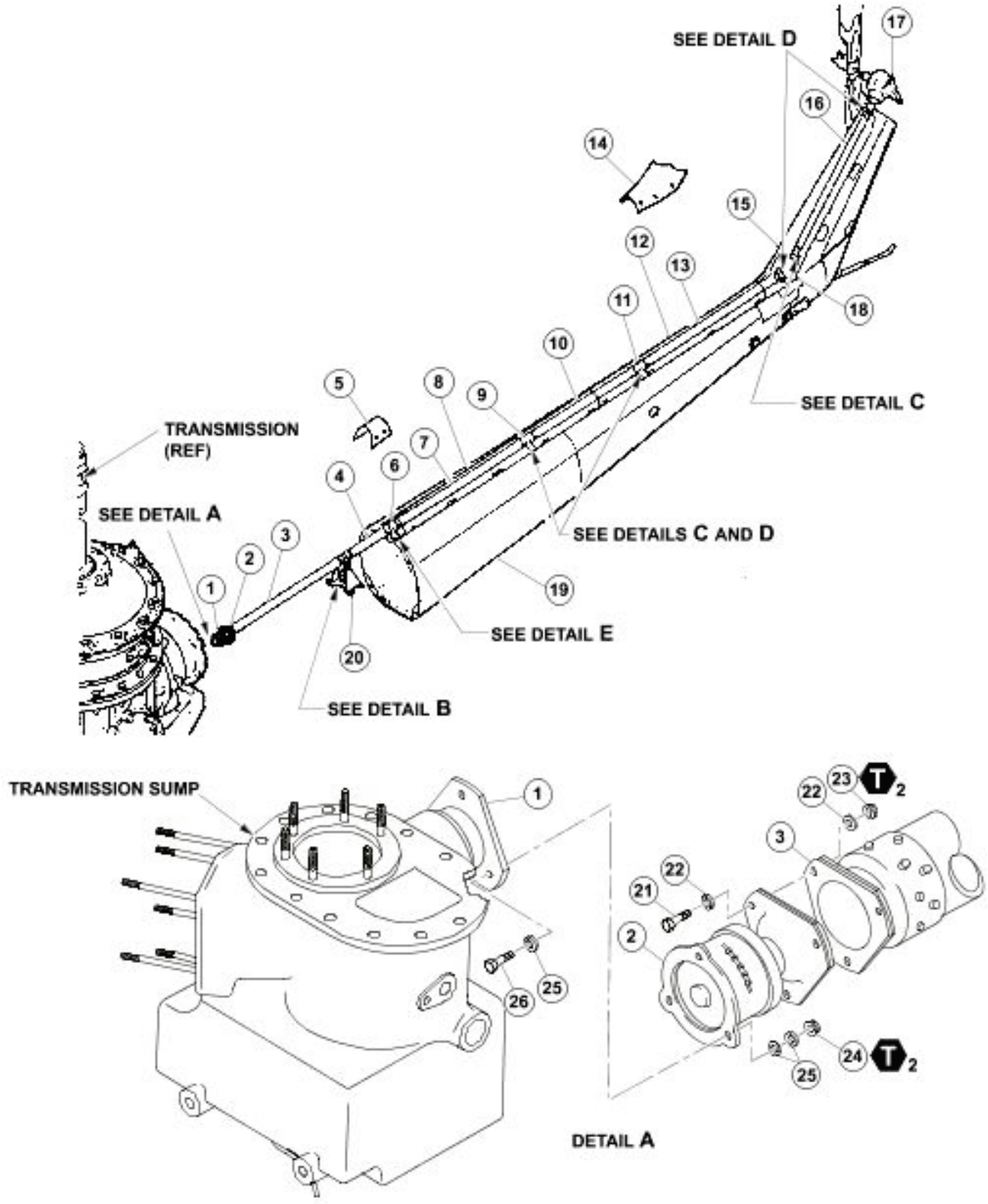


NOTES

1. Chemical-Film treat reworked areas of the P/N 204-030-818-013 and P/N 205-031-818-001 fittings in accordance with instructions given in BHT-ALL-SPM for magnesium parts.
2. Chemical-Film treat reworked areas of the P/N 204-030-818-103 and P/N 205 031-818-101 fittings in accordance with instructions given in BHT-ALL-SPM for aluminum parts.
3. Touch-up reworked areas of the fittings with primer (C-204).

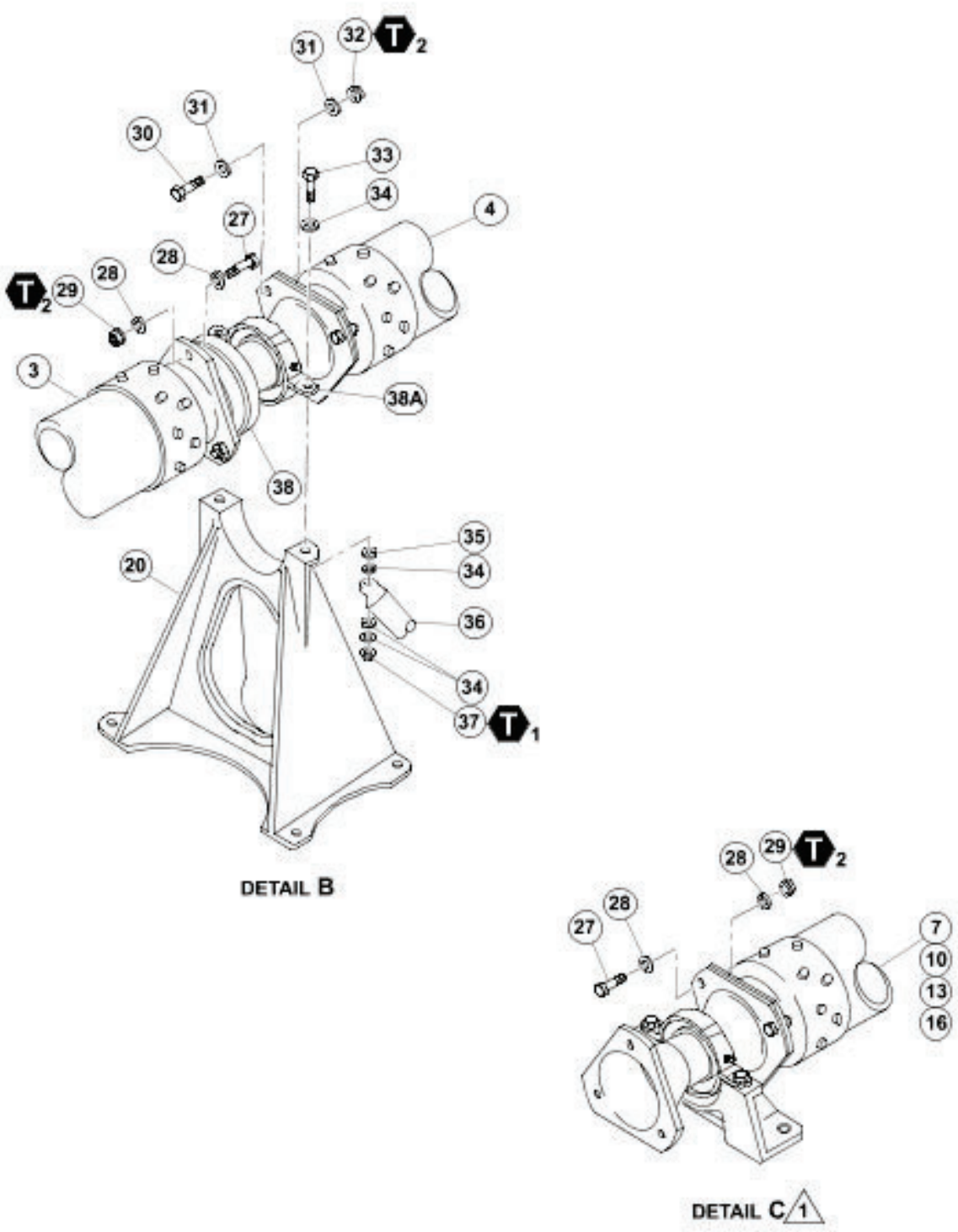
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Figure 1. Tail Rotor Driveshaft Alignment (Sheet 4 of 4)



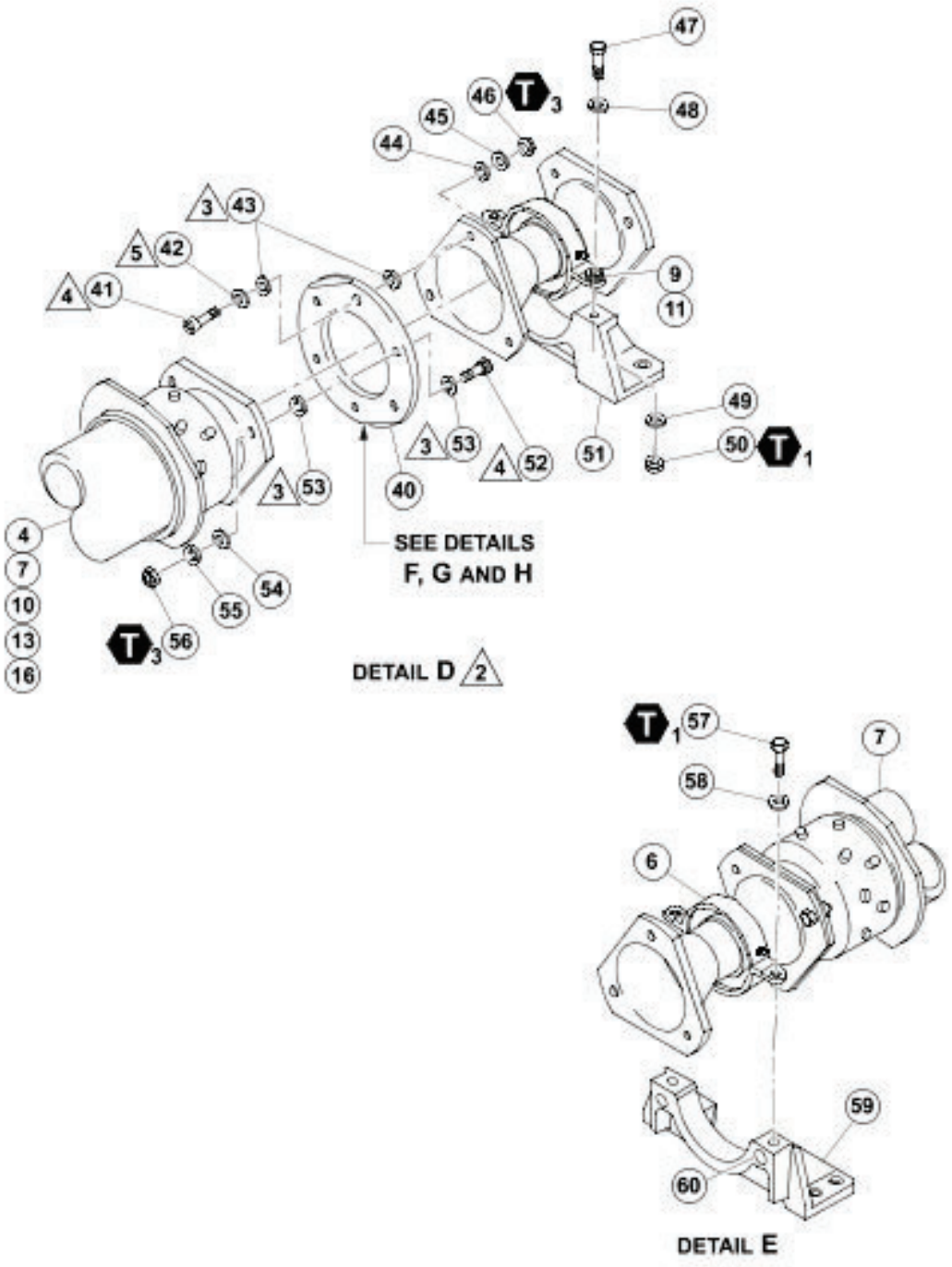
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Figure 2. Tail Rotor Driveshaft (Sheet 1 of 5)



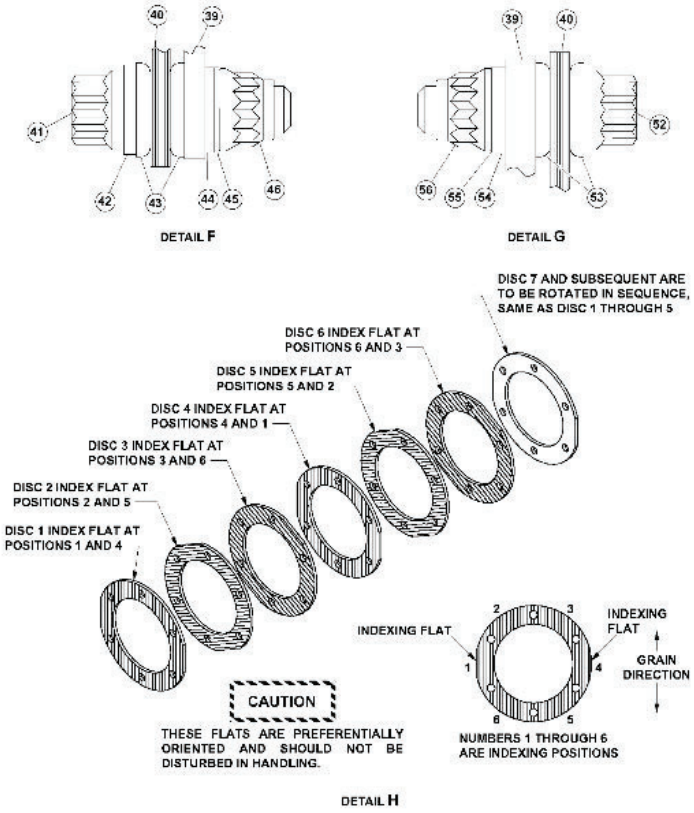
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Figure 2. Tail Rotor Driveshaft (Sheet 2 of 5)



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Figure 2. Tail Rotor Driveshaft (Sheet 3 of 5)



ICN-412-A-650000-B-97499-00008-A-001-01

Figure 2. Tail Rotor Driveshaft (Sheet 4 of 5)

1. Adapter (tail rotor drive quill)
2. Coupling assembly
3. Driveshaft
4. Driveshaft
5. Cover
6. Hanger bearing assembly
7. Driveshaft
8. Driveshaft cover
9. Hanger bearing assembly
10. Driveshaft
11. Hanger bearing assembly
12. Driveshaft cover
13. Driveshaft
14. Intermediate gearbox cover
15. Vertical fin cover
16. Driveshaft
17. Tail rotor gearbox
18. Intermediate gearbox
19. Tailboom
20. Hanger bearing support
21. Bolt
22. Steel washer
23. Nut
24. Nut
25. Steel washer
26. Bolt
27. Bolt
28. Steel washer
29. Nut
30. Bolt
31. Steel washer
32. Nut
33. Bolt
34. Steel washer
35. Aluminum washer
36. Brace assembly
37. Nut
38. Coupling assembly
38A. Hanger bearing assembly
39. Driveshaft adapter (typical)
40. Disc pack coupling
41. Bolt
42. Chamfered steel washer
43. Special washer
44. Thick steel washer
45. Thin steel washer
46. Nut
47. Bolt
48. Thin steel washer
49. Aluminum washer
50. Nut
51. Support
52. Bolt
53. Special washer
54. Thick steel washer
55. Thin steel washer
56. Nut
57. Bolt
58. Thin steel washer
59. Support
60. Barrel nut

T1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T2 120 TO 160 IN-LBS (13.56 TO 18.08 Nm)

T3 280 TO 300 IN-LBS (31.64 TO 33.90 Nm)

NOTES

1. Detail C shows typical installation of forward end of driveshafts (7, 10, 13 and 16).
2. Detail D shows typical installation of tailboom mounted hanger assembly and aft end installation of driveshaft assemblies (4, 7, 10, and 13).
3. Place radiused outer edge of washers adjacent to surface of disc pack (40).
4. Position heads of bolts (41 and 52), Detail D, 180° from each other, as shown.
5. Place chamfered bore of chamfered steel washer (42) adjacent to bolt head (41).

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Figure 2. Tail Rotor Driveshaft (Sheet 5 of 5)

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Tail Rotor Driveshaft and Hangers - Removal

Procedure

1. Open the following covers as necessary to gain access to driveshafts and hangers:

1.1. Remove oil cooler fairing.

1.2. Open reduction gearbox fairing.

1.3. Remove left and right access doors.

1.4. Open driveshaft covers (8 and 12, Figure 1).

1.5. Remove intermediate gearbox cover (14).

1.6. Open vertical fin cover (15).

1.7. Remove cover (5).

2. Remove driveshafts and adjacent disc pack coupling (40) as follows:



SUPPORT DRIVESHAFTS DURING DISCONNECTING AND REMOVAL PROCESSES TO PREVENT DAMAGE TO DISC PACK COUPLINGS. THE FREE END OF DRIVESHAFT (4) SHALL NOT BE ALLOWED TO DISPLACE MORE THAN 0.34 INCH (8.64 MM) WHEN OPPOSITE END IS SUPPORTED BY DISC PACK COUPLING. FREE ENDS OF DRIVESHAFTS (7, 10, 13, AND 16) SHALL NOT BE ALLOWED TO DISPLACE MORE THAN 1.0 INCH (25.4 MM) WHEN OPPOSITE ENDS ARE SUPPORTED BY DISC PACK COUPLINGS. THE DISASSEMBLY OPERATION SHALL BE PERFORMED IN SEQUENCE WITH THE FOLLOWING STEPS TO MINIMIZE THE CHANCE OF DAMAGE TO DRIVESHAFT HANGER BEARINGS.

NOTE

If the tailboom alignment is changed, if any damage has occurred that makes the tail rotor drive alignment suspect, or if any tail rotor driveshaft hanger support is disturbed (loosened) or replaced, the entire tail rotor drive system shall be aligned in accordance with step 7, 412-A-65-10-02-00A-720A-A / 01275. If driveshafts (3, 4, 7, 10, 13, and 16), disc pack couplings (40), special washers (43 and 53), or intermediate gearbox (18) are disturbed (loosened) or replaced, inspect disc pack coupling for gaps in accordance with step 5.7, 412-A-65-10-02-00A-720A-A / 01275.

2.1. Remove nuts (32), steel washers (31), and bolts (30) from forward end of driveshaft (4).

2.2. Remove nuts (29), steel washers (28), and bolts (27) from forward ends of driveshafts (7, 10, 13, and 16).

2.3. Remove uppermost nut (46), washers (42 through 45), and bolt (41) from disc pack coupling (40) and adapter of hanger bearing assembly (6, 9, and 11) or tail rotor gearbox (17) at aft end of driveshaft (16).

2.4. Remove nuts (56), washers (53 through 55), and bolts (52) from disc pack coupling (40) and driveshaft adapter (39).

2.5. Remove driveshaft and place in a suitable rack. Mark driveshaft with regard to original location in helicopter.

3. If required, remove disc pack coupling (40) as follows:



DO NOT DISASSEMBLE DISC PACK COUPLING.

3.1. Tie disc of disc pack couplings (40) together through at least two bolt holes with string or other nonscratching material.

3.2. Remove remaining nuts (46), washers (42 through 45), and bolts (41) from disc pack coupling and adapter of hanger bearing assembly (6, 9, or 11), tail rotor gearbox (17), or intermediate gearbox (18).

3.3. Remove disc pack coupling. Mark disc pack with regard to original location on helicopter.

4. Remove driveshaft (3), coupling assembly (2), and coupling assembly (38) as follows:



DRIVESHAFT MUST BE SUPPORTED DURING DISCONNECTING AND REMOVAL PROCESS TO PREVENT OVERLY DISTORTING SEAL IN COUPLING ASSEMBLIES. THE FREE END OF DRIVESHAFT SHALL NOT BE ALLOWED TO BE DISPLACED MORE THAN 2.0 INCHES (50.8 MM) WHEN OPPOSITE END IS SUPPORTED AT COUPLING.

4.1. Remove driveshaft (4) (Step 2.).

4.2. Remove nuts (24), steel washers (25), and bolts (26) from transmission adapter (1) and coupling assembly (2).

4.3. Remove nuts (29), steel washers (28), and bolts (27) from aft end of driveshaft (3) and coupling assembly (38).

NOTE

Before removing driveshaft, place suitable protective material on engine deck below driveshaft to prevent damage to driveshaft.

4.4. Force driveshaft (3) forward by hand. Compress coupling assembly (2) and separate aft end of driveshaft from coupling assembly (38).

4.5. Gently lower driveshaft with coupling assembly (2) onto engine deck within driveshaft enclosure.

NOTE

Do not remove washers (34 and 35) from between upper end of brace assembly (36) and bearing hanger support (20) unless brace or support requires repair or replacement.

4.6. Remove coupling assembly (38) by removing nuts (37), steel washers (34), and bolts (33) from hanger bearing support (20), brace assembly (36), and coupling assembly (38). Loosely reinstall bolts, steel washers, and nuts to prevent loss of washers (34 and 35) that are between upper end of brace and support.



USE CARE WHEN REMOVING DRIVESHAFT FROM ENCLOSURE TO PREVENT DAMAGE.

4.7. Carefully withdraw driveshaft (3) with coupling assembly (2) from aft end of driveshaft enclosure. Place driveshaft on suitable rack.

4.8. If required, remove coupling assembly (2) from driveshaft (3) by removing nuts (23), steel washers (22), and bolts (21).

4.9. Mark driveshaft to show original location in helicopter.

5. Remove hanger bearing assembly (6) as follows:

5.1. Remove driveshafts (4 and 7) and disc pack coupling (40) (Step 2.).

5.2. Remove bolts (57) with thin steel washers (58) from hanger bearing assembly (6) and support (59).

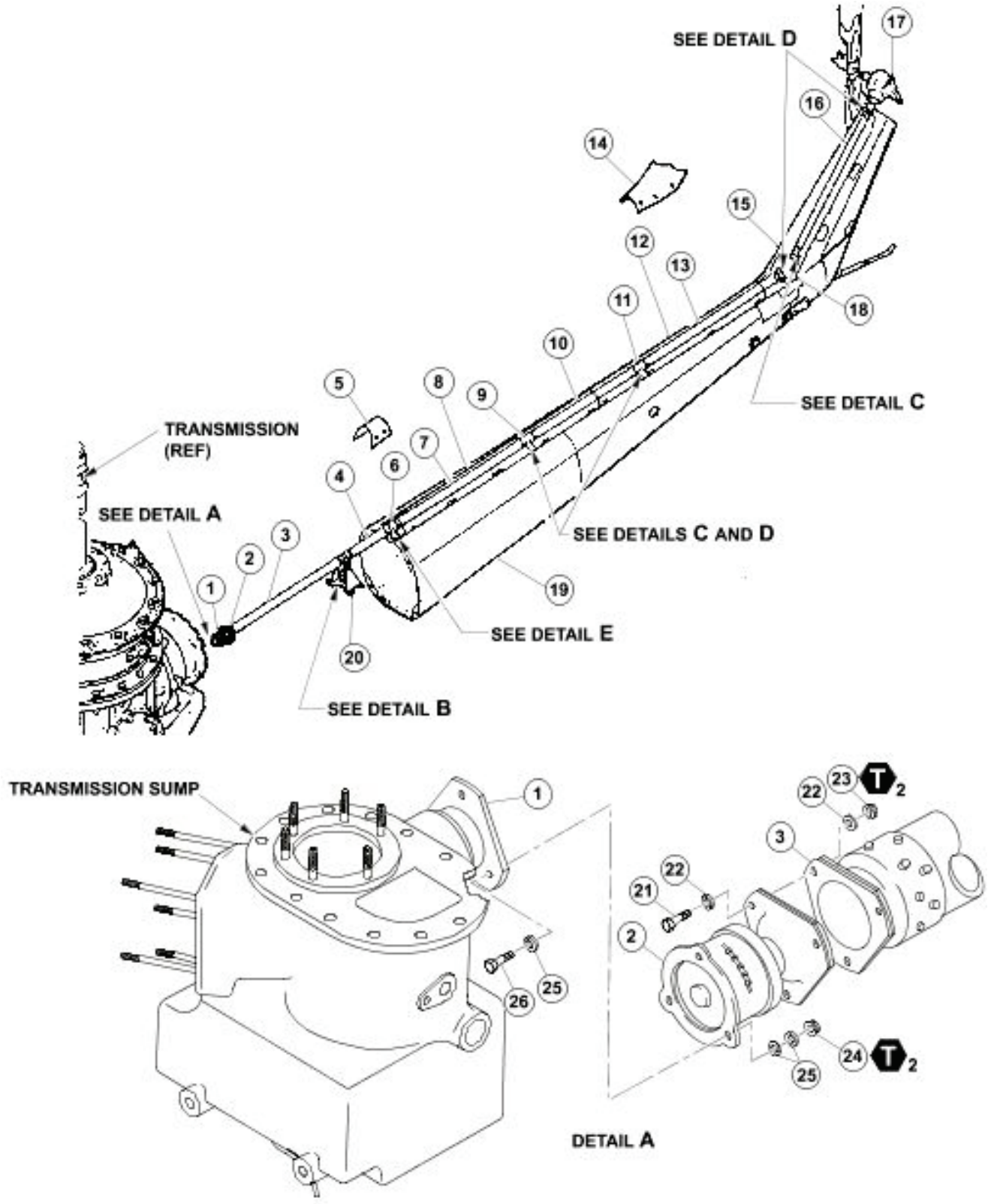
5.3. Remove hanger assembly from support. Mark hanger to show original location in helicopter.

6. Remove hanger bearing assemblies (9 and 11) as follows:

6.1. Remove driveshafts (7, 10, and 13) and disc pack couplings (40) (Step 2.).

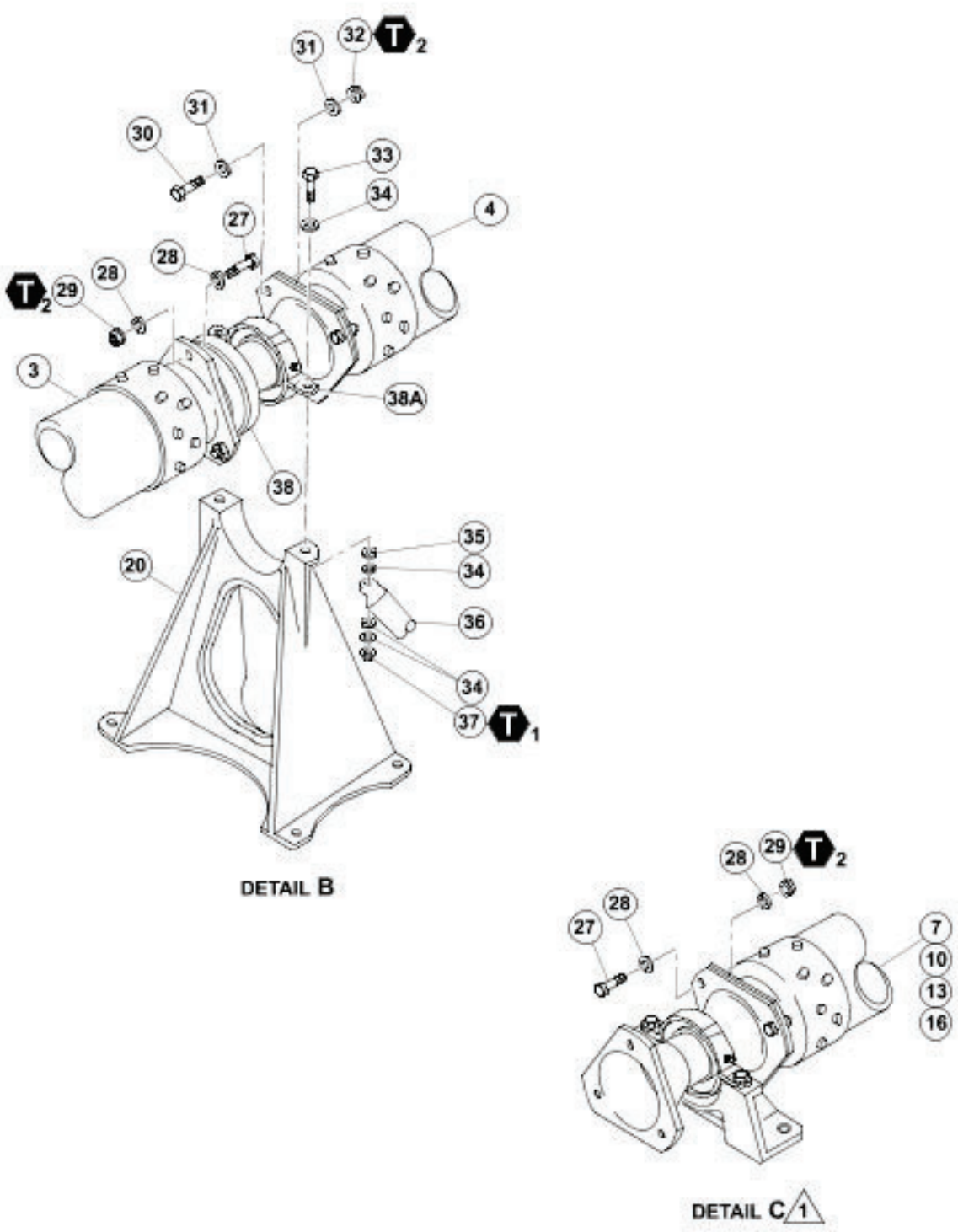
6.2. Remove nuts (50), washers (48 and 49), and bolts (47) from hanger bearing assembly (9 or 11) and support (51).

6.3. Remove hanger bearing assembly from support. Mark hanger to show original location in helicopter.



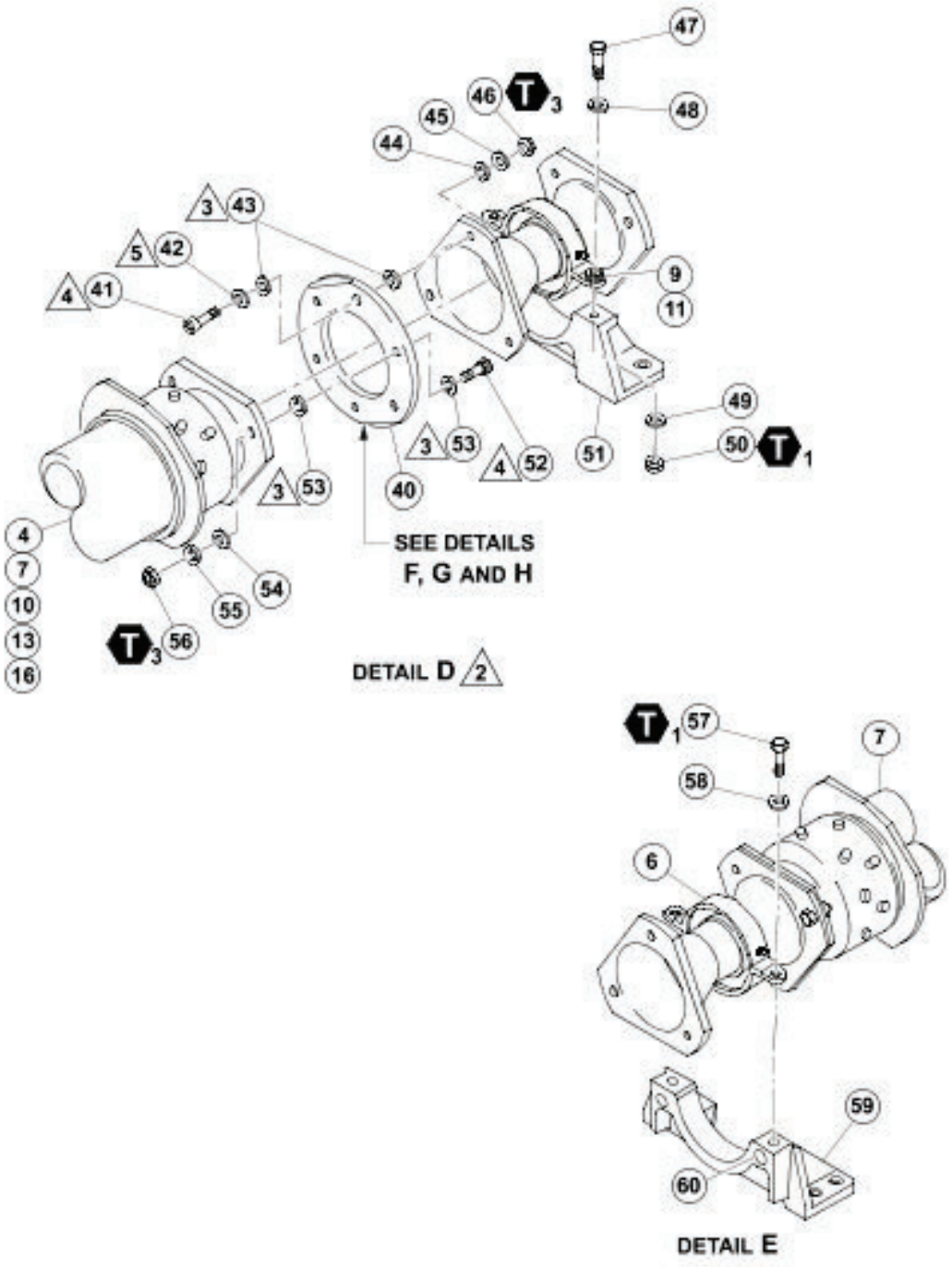
ICN-412-A-650000-B-97499-00005-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 1 of 5)



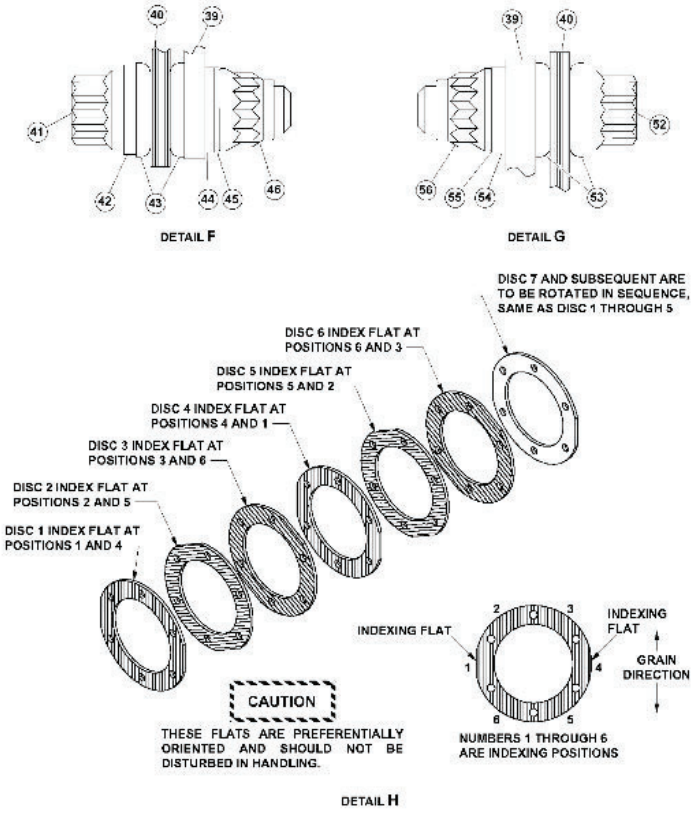
ICN-412-A-650000-B-97499-00006-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 2 of 5)



ICN-412-A-650000-B-97499-00007-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 3 of 5)



ICN-412-A-650000-B-97499-00008-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 4 of 5)

1. Adapter (tail rotor drive quill)
2. Coupling assembly
3. Driveshaft
4. Driveshaft
5. Cover
6. Hanger bearing assembly
7. Driveshaft
8. Driveshaft cover
9. Hanger bearing assembly
10. Driveshaft
11. Hanger bearing assembly
12. Driveshaft cover
13. Driveshaft
14. Intermediate gearbox cover
15. Vertical fin cover
16. Driveshaft
17. Tail rotor gearbox
18. Intermediate gearbox
19. Tailboom
20. Hanger bearing support
21. Bolt
22. Steel washer
23. Nut
24. Nut
25. Steel washer
26. Bolt
27. Bolt
28. Steel washer
29. Nut
30. Bolt
31. Steel washer
32. Nut
33. Bolt
34. Steel washer
35. Aluminum washer
36. Brace assembly
37. Nut
38. Coupling assembly
38A. Hanger bearing assembly
39. Driveshaft adapter (typical)
40. Disc pack coupling
41. Bolt
42. Chamfered steel washer
43. Special washer
44. Thick steel washer
45. Thin steel washer
46. Nut
47. Bolt
48. Thin steel washer
49. Aluminum washer
50. Nut
51. Support
52. Bolt
53. Special washer
54. Thick steel washer
55. Thin steel washer
56. Nut
57. Bolt
58. Thin steel washer
59. Support
60. Barrel nut

T1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T2 120 TO 160 IN-LBS (13.56 TO 18.08 Nm)

T3 280 TO 300 IN-LBS (31.64 TO 33.90 Nm)

NOTES

1. Detail C shows typical installation of forward end of driveshafts (7, 10, 13 and 16).
2. Detail D shows typical installation of tailboom mounted hanger assembly and aft end installation of driveshaft assemblies (4, 7, 10, and 13).
3. Place radiused outer edge of washers adjacent to surface of disc pack (40).
4. Position heads of bolts (41 and 52), Detail D, 180° from each other, as shown.
5. Place chamfered bore of chamfered steel washer (42) adjacent to bolt head (41).

ICN-412-A-650000-B-97499-00009-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 5 of 5)

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Tail Rotor Driveshaft and Hangers - Cleaning

Preliminary requirements

Consumables, materials and expendables

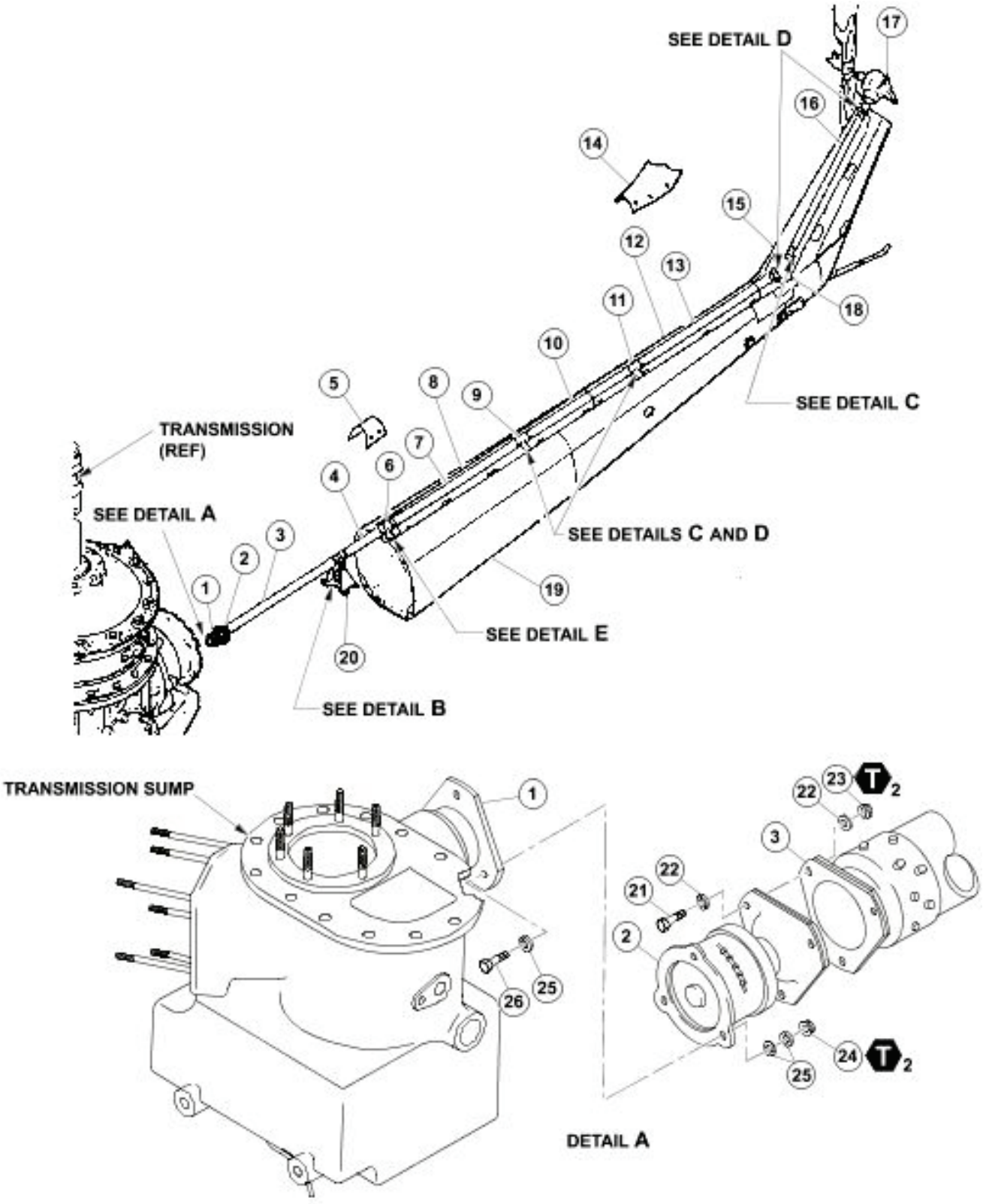
Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure



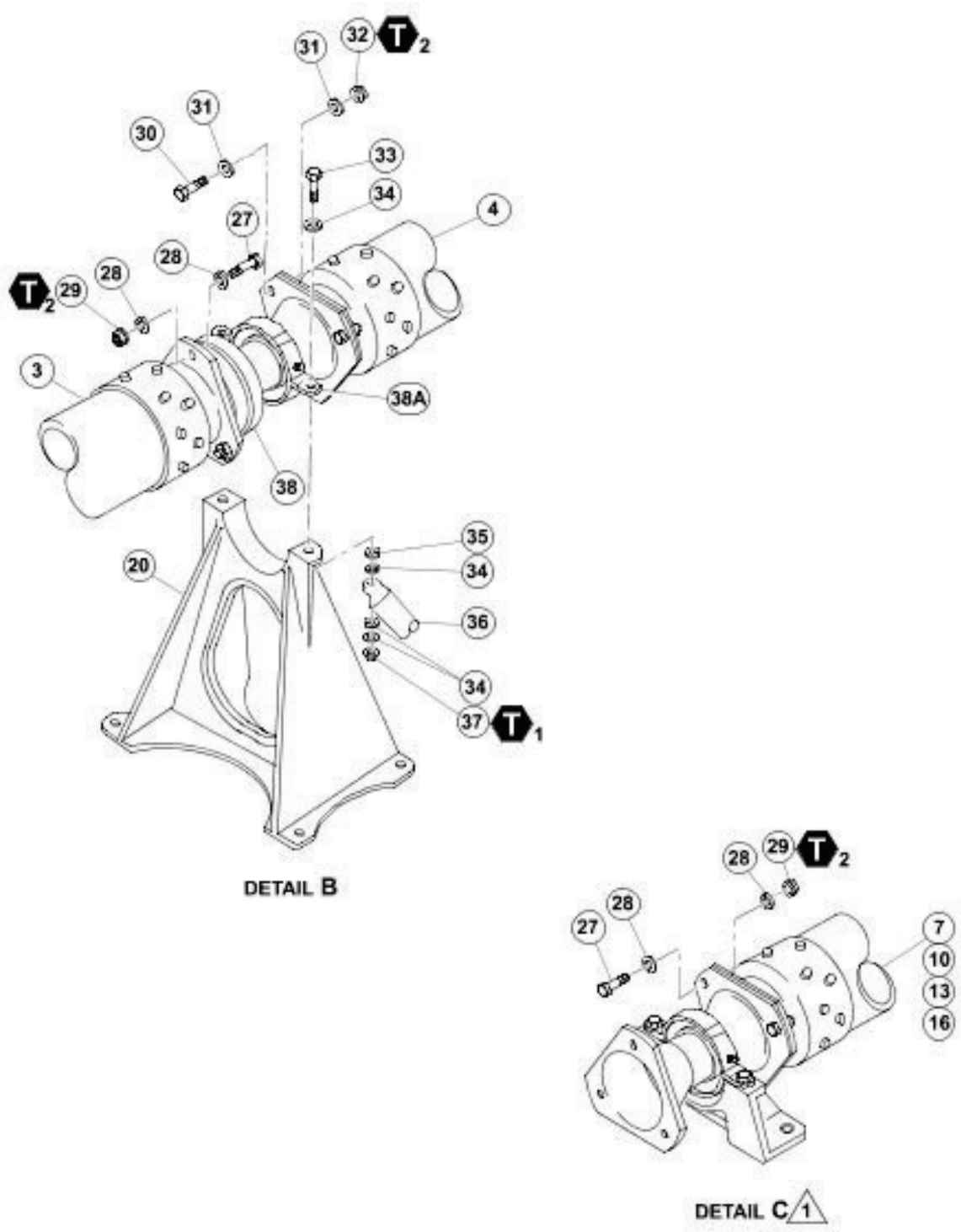
DO NOT SUBMERGE COUPLING ASSEMBLY (2, Figure 1) OR HANGER ASSEMBLIES (6, 9, AND 11) IN SOLVENT.

1. Wash parts with drycleaning solvent (C-304).
2. Dry clean with clean, filtered compressed air.



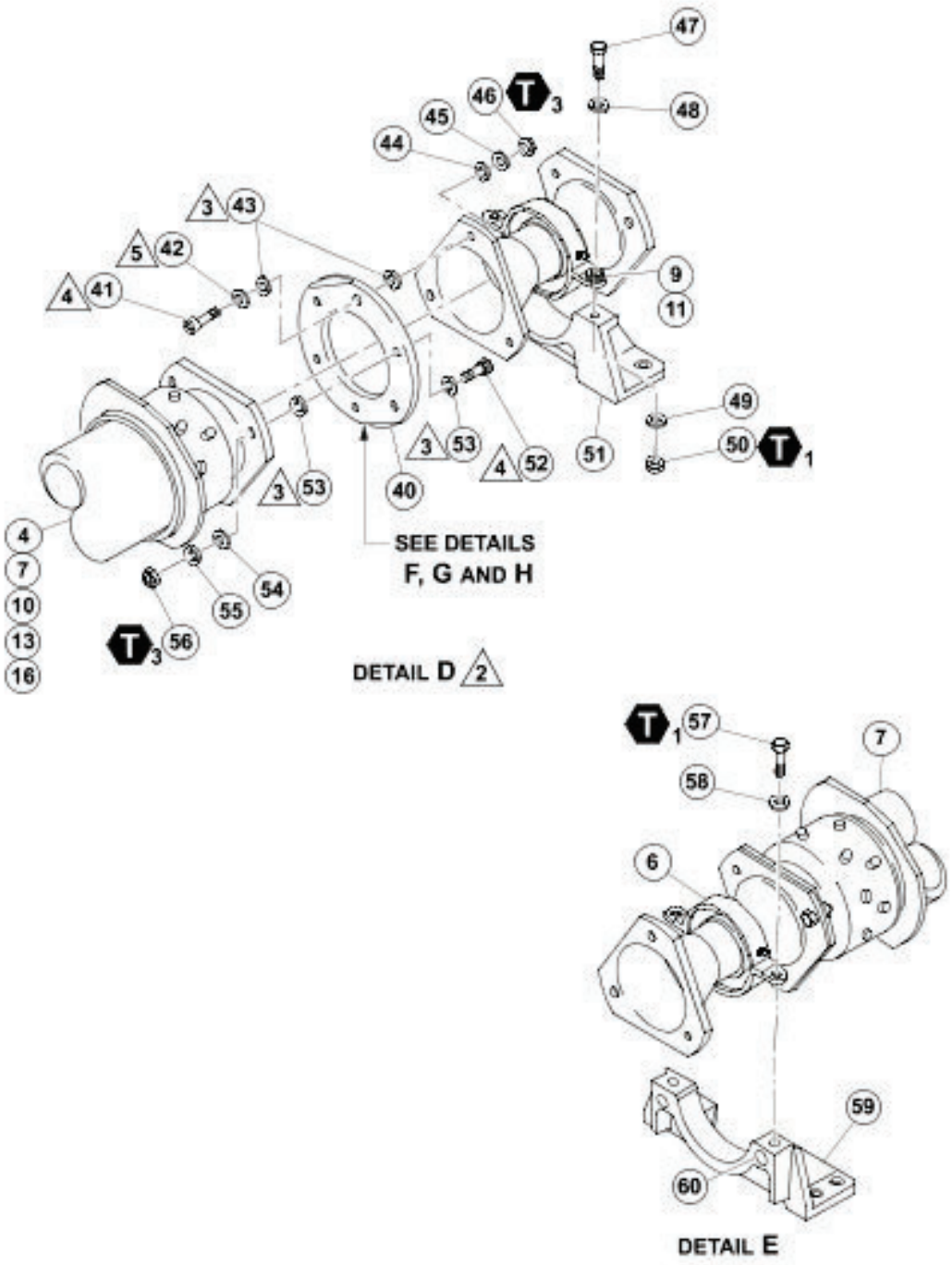
ICN-412-A-650000-B-97499-00005-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 1 of 5)



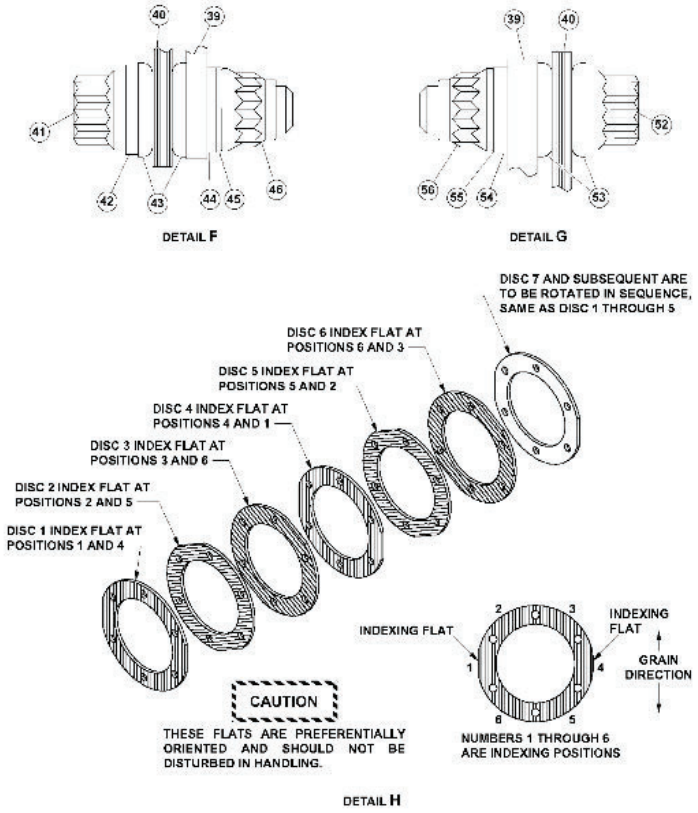
ICN-412-A-650000-B-97499-00006-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 2 of 5)



ICN-412-A-650000-B-97499-00007-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 3 of 5)



ICN-412-A-650000-B-97499-00008-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 4 of 5)

1. Adapter (tail rotor drive quill)
2. Coupling assembly
3. Driveshaft
4. Driveshaft
5. Cover
6. Hanger bearing assembly
7. Driveshaft
8. Driveshaft cover
9. Hanger bearing assembly
10. Driveshaft
11. Hanger bearing assembly
12. Driveshaft cover
13. Driveshaft
14. Intermediate gearbox cover
15. Vertical fin cover
16. Driveshaft
17. Tail rotor gearbox
18. Intermediate gearbox
19. Tailboom
20. Hanger bearing support
21. Bolt
22. Steel washer
23. Nut
24. Nut
25. Steel washer
26. Bolt
27. Bolt
28. Steel washer
29. Nut
30. Bolt
31. Steel washer
32. Nut
33. Bolt
34. Steel washer
35. Aluminum washer
36. Brace assembly
37. Nut
38. Coupling assembly
38A. Hanger bearing assembly
39. Driveshaft adapter (typical)
40. Disc pack coupling
41. Bolt
42. Chamfered steel washer
43. Special washer
44. Thick steel washer
45. Thin steel washer
46. Nut
47. Bolt
48. Thin steel washer
49. Aluminum washer
50. Nut
51. Support
52. Bolt
53. Special washer
54. Thick steel washer
55. Thin steel washer
56. Nut
57. Bolt
58. Thin steel washer
59. Support
60. Barrel nut

T1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T2 120 TO 160 IN-LBS (13.56 TO 18.08 Nm)

T3 280 TO 300 IN-LBS (31.64 TO 33.90 Nm)

NOTES

1. Detail C shows typical installation of forward end of driveshafts (7, 10, 13 and 16).
2. Detail D shows typical installation of tailboom mounted hanger assembly and aft end installation of driveshaft assemblies (4, 7, 10, and 13).
3. Place radiused outer edge of washers adjacent to surface of disc pack (40).
4. Position heads of bolts (41 and 52), Detail D, 180° from each other, as shown.
5. Place chamfered bore of chamfered steel washer (42) adjacent to bolt head (41).

ICN-412-A-650000-B-97499-00009-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 5 of 5)

Tail Rotor Driveshaft and Hangers - Inspection

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Cleaner	C-344	AR	

Procedure

1. Inspect driveshafts (3, 4, 7, 10, 13, and 16, Figure 1) in accordance with Figure 2 and Figure 3.

2. Inspect for damage to anodized finish. Damage to finish requires anticorrosion treatment (BHT-ALL-SPM, Chapter 3, Standard Practices Manual).

3. Inspect splined coupling of coupling assembly (2, Figure 1) visual overtemperature indicator (TEMP-PLATES) for discoloration indicating a possible overheat condition. A change in color of TEMP-PLATES dots (from white to black) indicates a possible overheat condition and/or component degradation. Cause of discoloration shall be determined and corrected prior to continued operation (412-A-65-10-01-00A-280A-A / 01266).

4. Replace supports if damaged.

5. Inspect brace assembly (36) for cracks, corrosion, and damaged threads.

6. Inspect bearing hanger bearing assemblies (6, 9, 11, and 38A) for excessive grease leakage. A small amount of grease expelled from around lip of bearing seals indicates a slight overservicing and is not cause for hanger replacement. If leakage is detected, perform an evaluation of grease leakage as follows:



DO NOT CLEAN OR SPRAY BEARING HANGER ASSEMBLY WITH ANY TYPE OF SOLVENT DURING INSPECTION. USE CLEAN, DRY CLOTHS ONLY TO CLEAN EXTERIOR OF HANGER ASSEMBLIES.

6.1. Wipe grease from bearing in hanger assemblies (6, 9, 11, and 38A).

6.2. Record which bearing is leaking and keep under observation for the next 10 flight hours. If amount of grease expelled from bearing seal does not decrease after 10 flight hours, replace hanger assembly.

7. Inspect hanger bearing assemblies (6, 9, 11, and 38A) and coupling assembly (38) for obvious damage and exterior surfaces for discoloration and overheat condition. A change in color indicates a possible overheat condition and/or component degradation. The cause of exterior surface discoloration shall be determined and corrected prior to continued operation (412-CR&O, Chapter 65).

8. Inspect disc assembly (Figure 4) as follows:

NOTE

Keep disc pack as an assembly.

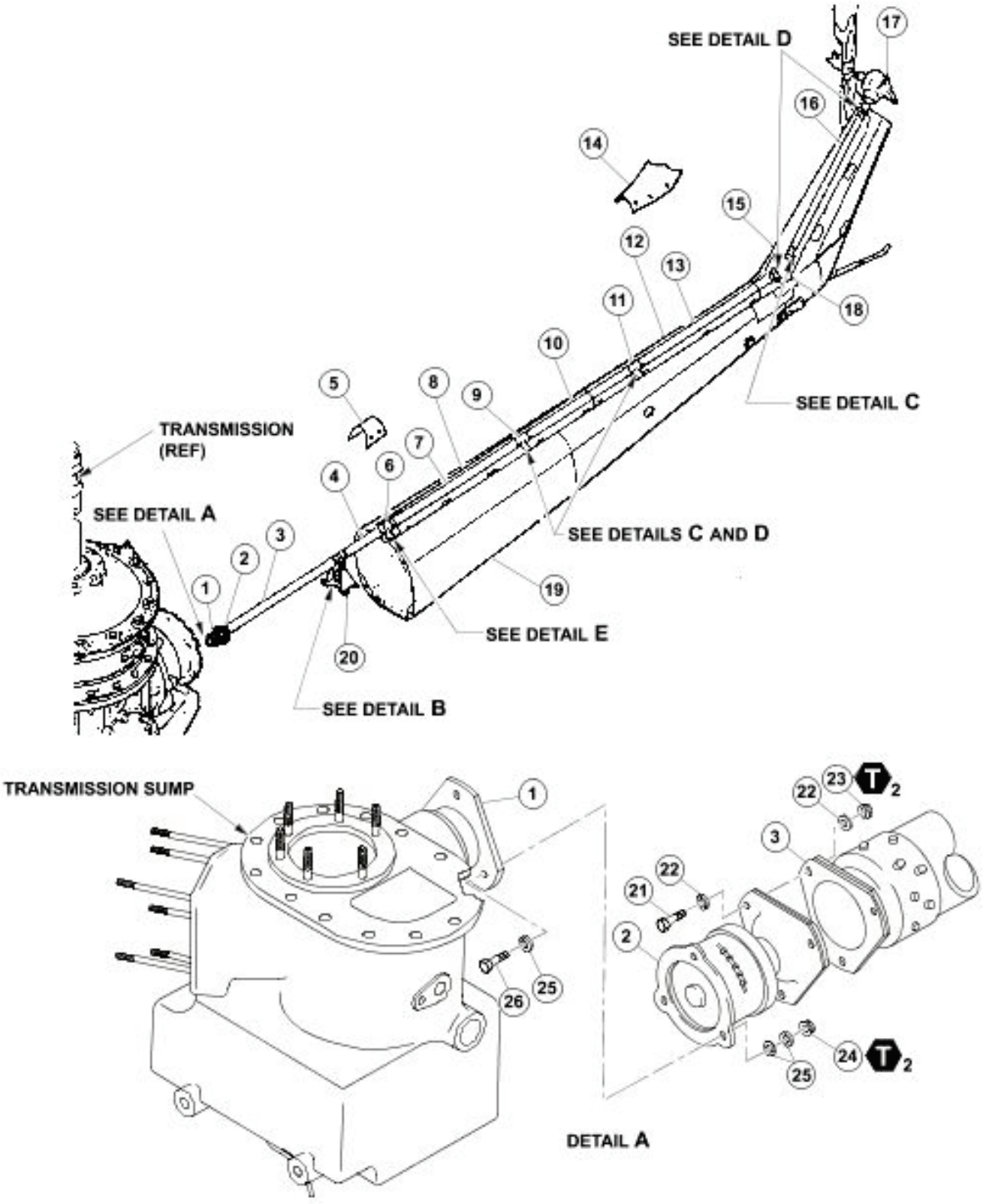
8.1. Clean the grease and oil from each disc.

8.2. Examine the disc for cracks, nicks, and scratches. None are permitted.

8.3. Examine the disc for fretting and corrosion.

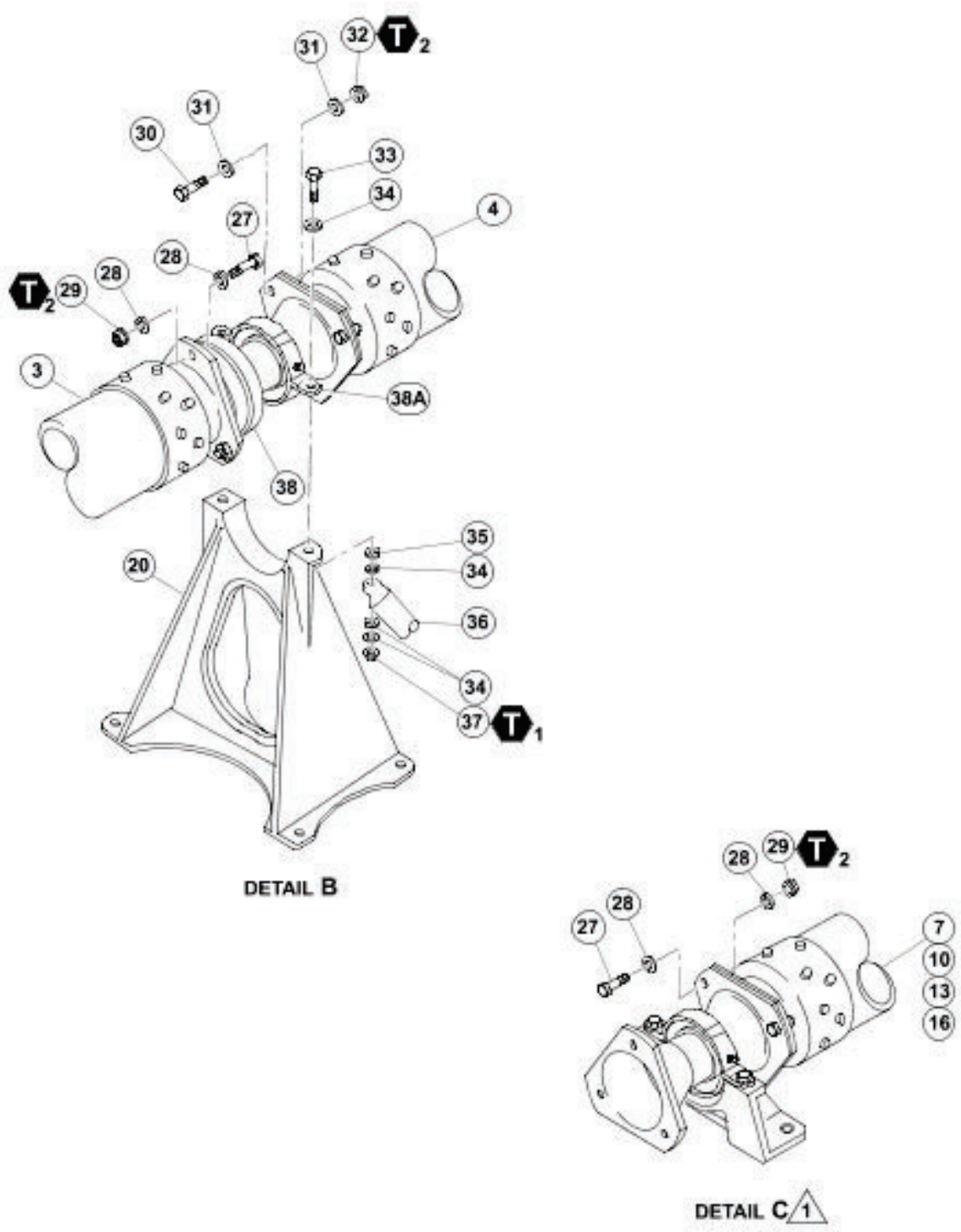
8.4. Clean the light fretting and corrosion with a soft rubber eraser. If light fretting and corrosion still exists, soak disc in cleaner (C-344) (BHT-ALL-SPM, Standard Practices Manual). If the disc cannot be cleaned, discard it.

8.5. Replace unserviceable laminates with new laminates as necessary. The disc assembly must be within 0.126 to 0.138 inch (3.200 to 3.505 mm) thick. A maximum of 11 laminates may be used to make a disc assembly. Each laminate is 0.010 to 0.014 inch (0.254 to 0.355 mm) thick.



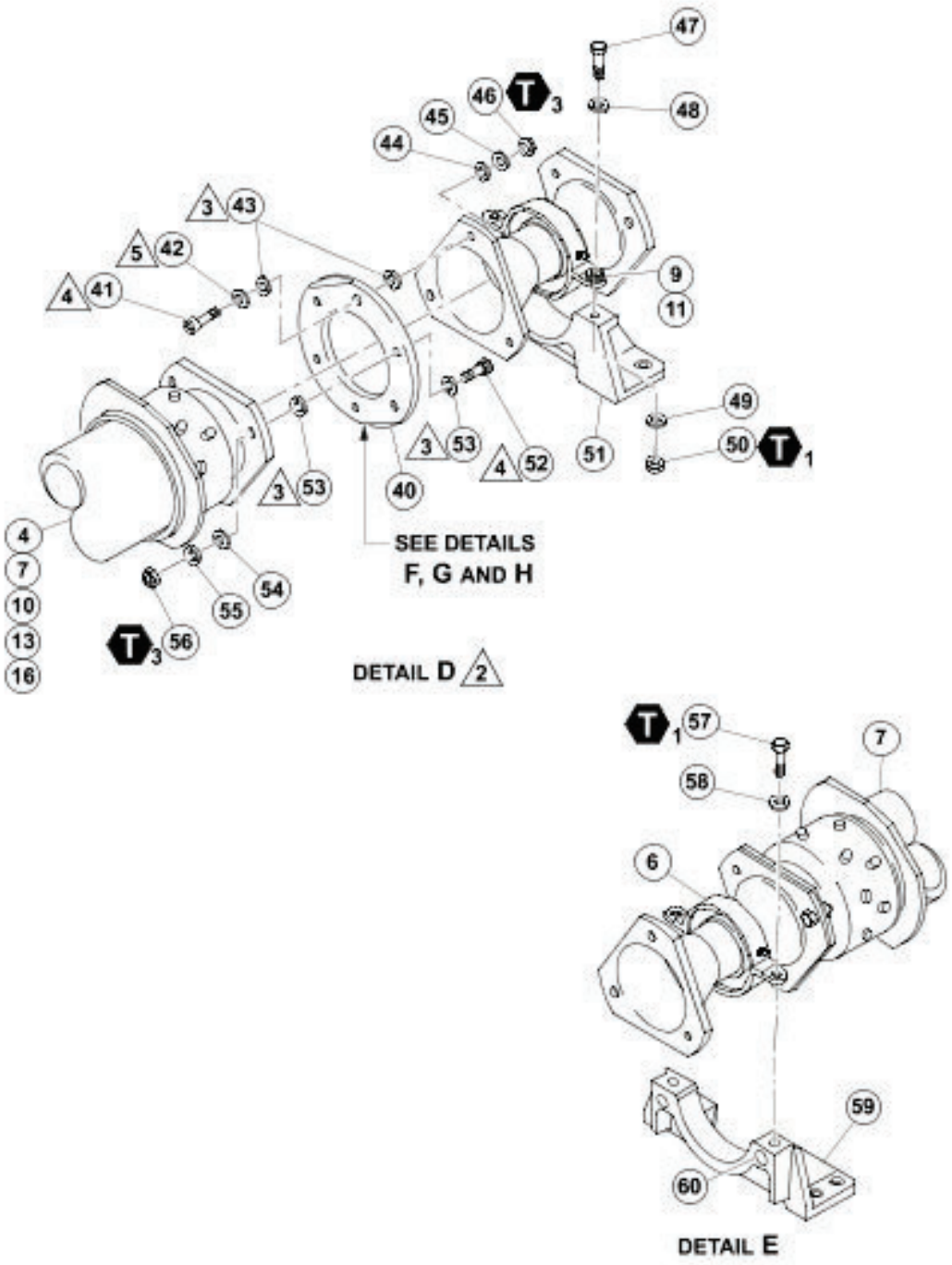
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Figure 1. Tail Rotor Driveshaft (Sheet 1 of 5)



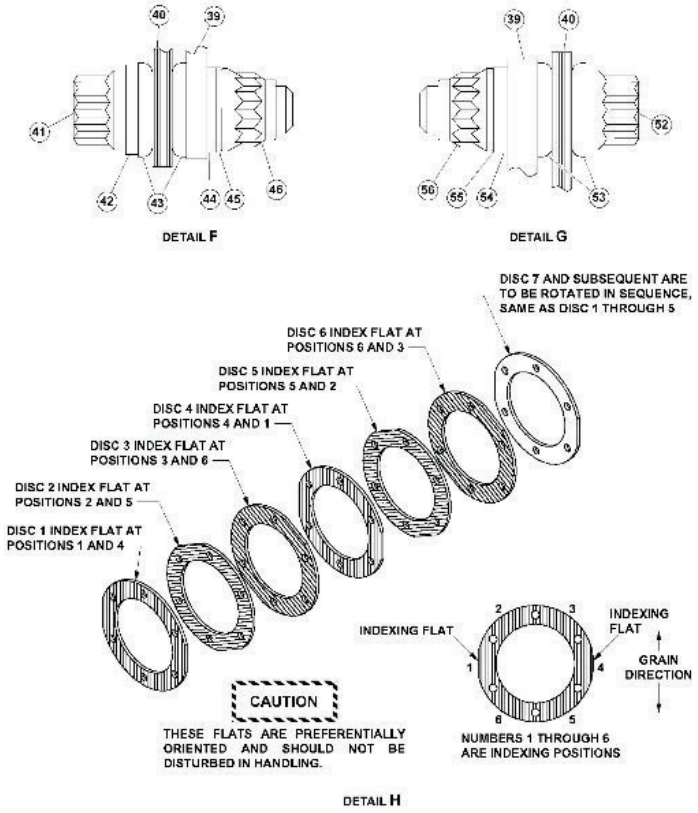
ICN-412-A-650000-B-97499-00006-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 2 of 5)



ICN-412-A-650000-B-97499-00007-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 3 of 5)



ICN-412-A-650000-B-97499-00008-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 4 of 5)

1. Adapter (tail rotor drive quill)
2. Coupling assembly
3. Driveshaft
4. Driveshaft
5. Cover
6. Hanger bearing assembly
7. Driveshaft
8. Driveshaft cover
9. Hanger bearing assembly
10. Driveshaft
11. Hanger bearing assembly
12. Driveshaft cover
13. Driveshaft
14. Intermediate gearbox cover
15. Vertical fin cover
16. Driveshaft
17. Tail rotor gearbox
18. Intermediate gearbox
19. Tailboom
20. Hanger bearing support
21. Bolt
22. Steel washer
23. Nut
24. Nut
25. Steel washer
26. Bolt
27. Bolt
28. Steel washer
29. Nut
30. Bolt
31. Steel washer
32. Nut
33. Bolt
34. Steel washer
35. Aluminum washer
36. Brace assembly
37. Nut
38. Coupling assembly
38A. Hanger bearing assembly
39. Driveshaft adapter (typical)
40. Disc pack coupling
41. Bolt
42. Chamfered steel washer
43. Special washer
44. Thick steel washer
45. Thin steel washer
46. Nut
47. Bolt
48. Thin steel washer
49. Aluminum washer
50. Nut
51. Support
52. Bolt
53. Special washer
54. Thick steel washer
55. Thin steel washer
56. Nut
57. Bolt
58. Thin steel washer
59. Support
60. Barrel nut

T1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T2 120 TO 160 IN-LBS (13.56 TO 18.08 Nm)

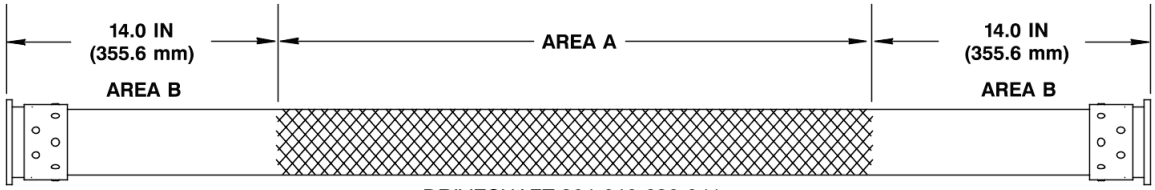
T3 280 TO 300 IN-LBS (31.64 TO 33.90 Nm)

NOTES

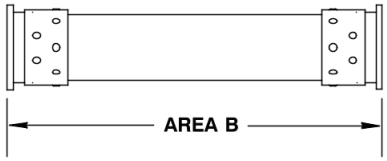
1. Detail C shows typical installation of forward end of driveshafts (7, 10, 13 and 16).
2. Detail D shows typical installation of tailboom mounted hanger assembly and aft end installation of driveshaft assemblies (4, 7, 10, and 13).
3. Place radiused outer edge of washers adjacent to surface of disc pack (40).
4. Position heads of bolts (41 and 52), Detail D, 180° from each other, as shown.
5. Place chamfered bore of chamfered steel washer (42) adjacent to bolt head (41).

ICN-412-A-650000-B-97499-00009-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 5 of 5)



DRIVESHAFT 204-040-620-011
412-040-620-101 AND -103
412-040-621-103 AND -107



DRIVESHAFT 204-040-620-009
412-040-621-101 AND -105

DAMAGE LOCATION SYMBOLS

AREA A **AREA B**

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED AFTER POLISHING OUT	
CRACKS ALLOWED	See Note 6	See Note 6
NICKS AND SCRATCHES	0.008 IN (0.203 mm)	0.012 IN (0.305 mm)
SHARP DENTS	0.010 IN (0.254 mm)	0.015 IN (0.381 mm)
NONSHARP DENTS	0.020 IN (0.508 mm)	0.030 IN (0.762 mm)

NOTES

1. **Damage defined on the chart above is acceptable if repaired as follows:**
   a. **Damage is polished out with a fine abrasive cloth with minimum radius of 0.5 inch (12.7 mm) and with surface finish of 32 RMS or better.**
   b. **Maximum deph after rework does not exceed limits shown in chart.**
   c. **Reworked area is treated for corrosion prevention.**
   d. **If rework area is eight square inches more on one side than the other, shaft balance must be checked.**

2. **Loss of one or more balance weights is cause to replace driveshaft. One empty bonding space should be open where bond test coupon was removed. If more than one empty space is observed, replace the driveshaft.**

3. **Damaged curvic coupling splines that result in radial play and/or backlash between assembled couplings when fully meshed without clamps are not acceptable.**

ICN-412-A-651000-B-97499-00022-A-001-01

Figure 2. Tail Rotor Driveshaft - Mechanical Damage Limits (Sheet 1 of 2)

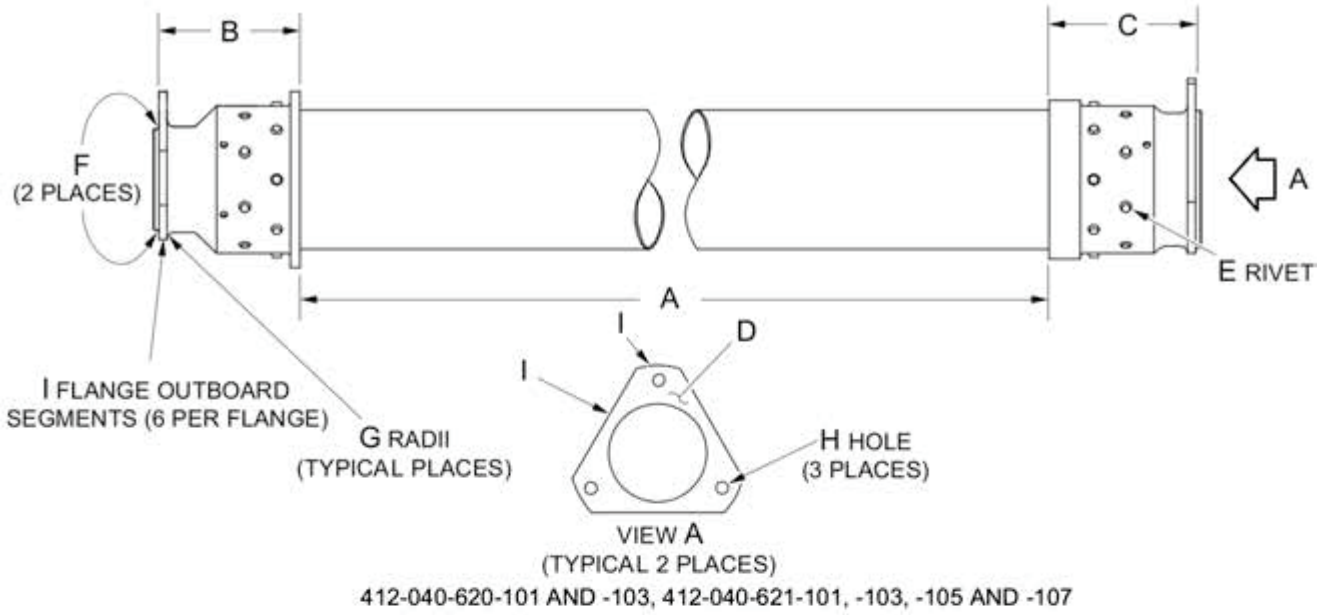
4. Grooves worn on shaft coupling by clamp to extent that wear prevents proper clamping are cause to replace driveshaft.

5. If tail rotor driveshaft distortion is suspected, support driveshaft on "V" blocks and measure runout. Maximum acceptable total indicator reading on long shafts is 0.050 inch (1.27 mm). Maximum acceptable total indicator reading on short shaft is 0.020 inch (0.508 mm). Do not attempt to straighten driveshaft.

6. Any crack in a driveshaft is cause to replace the affected driveshaft except cracks in rivet heads are acceptable if they do not exceed the following limits:
   a. Cracks are approximately radial when viewing top of shop head.
   b. No crack extends into an area with a diameter less than 1.25 diameters of rivet shank.
   c. Cracks do not intersect.
   d. The minimum distance between cracks is less than on shank diameter.
   e. A maximum of three such cracks.
   f. Crack width does not exceed 0.06 inch (1.52 mm) times rivet shank diameter.
   g. Maximum or 10% of rivet shop heads may have cracks within above limits.

ICN-412-A-651000-B-97499-00023-A-001-01

Figure 2. Tail Rotor Driveshaft - Mechanical Damage Limits (Sheet 2 of 2)



412-040-620-101 AND -103, 412-040-621-101, -103, -105 AND -107

NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT
1.	A	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth and 1.0 square inch (645 mm²) maximum area or 0.90 inch (22.8 mm) wide for 1/4 of circumference.
2.	A	Corrosion damage	Visual/ measure	0.002 inch (0.05 mm) maximum depth and 0.25 inch (6.3 mm) wide for a full circumference.
3.	B, C	Corrosion damage	Visual/ measure	0.012 inch (0.30 mm) maximum depth for 25% of any 1.0 square inch (645 mm²) and 25% of any surfaces. △1
4.	D	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth for 50% of any 1.0 square inch (645 mm²) and 0.002 inch (0.05 mm) maximum depth for the entire surface. △2
5.	E, F, G, H	Corrosion damage	Visual	None permitted.
6.	I	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth. Applicable to all segments, no maximum area. One repair per segment allowed.

NOTES

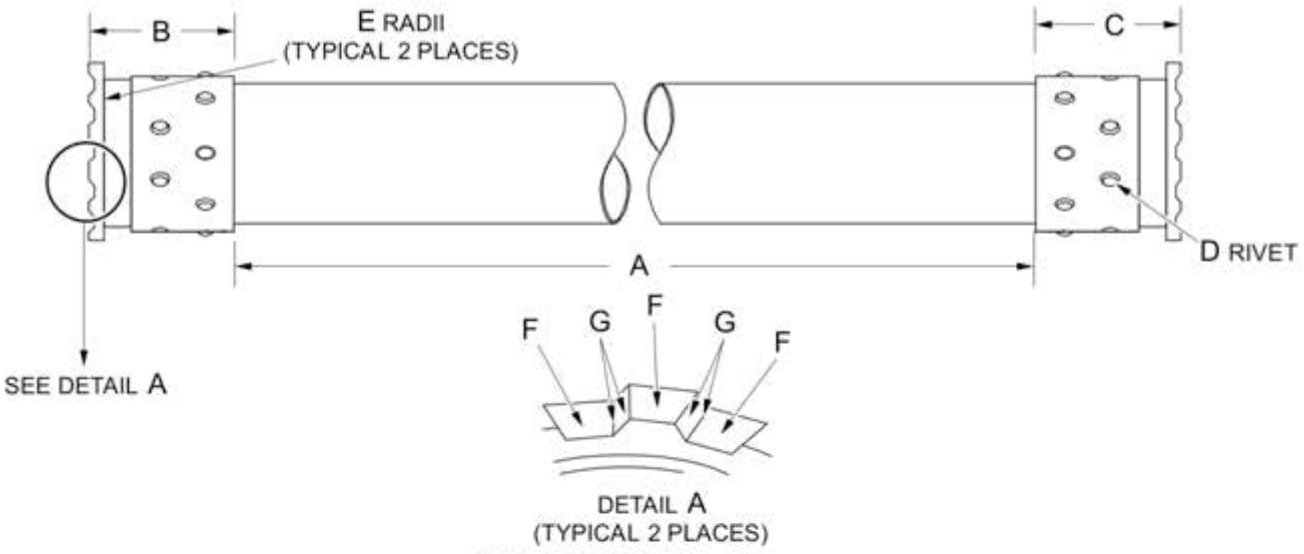
△1 No corrosion allowed within 0.20 inch (5.1 mm) of rivets or mounting holes.

△2 No corrosion allowed within 0.20 inch (5.1 mm) of mounting holes.

3. All dimensions, wea r, and damage limits provided are limits required after repair except where indicated.

ICN-412-A-651000-B-97499-01026-A-001-01

Figure 3. Tail Rotor Driveshaft - Corrosion Damage Limits (Sheet 1 of 2)



204-040-620-009 AND -011

NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT
1.	A	Corrosion damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth and 1.0 square inch (645 mm²) maximum area or 0.90 inch (22.8 mm) wide for 1/4 of circumference.
2.	A	Corrosion damage	Visual/ measure	0.002 inch (0.05 mm) maximum depth and 0.25 inch (6.3 mm) wide for a full circumference.
3.	B, C	Corrosion damage	Visual/ measure	0.012 inch (0.30 mm) maximum depth for 25% of any 1.0 square inch (645 mm²) and 25% of any surfaces. (1)
4.	D, E	Corrosion damage	Visual	None permitted.
5.	F	Corrosion damage on curvic tooth top and bottom lands	Visual/ measure	0.012 inch (0.30 mm) maximum depth for 10% of any surfaces. (2)
6.	G	Corrosion damage on curvic tooth working faces and fillets	Visual	None permitted.

NOTES

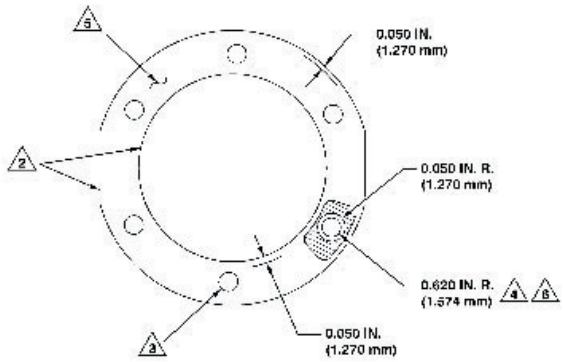
(1) No corrosion allowed within 0.20 inch (5.1 mm) of rivets or mounting holes.

(2) No corrosion or repair allowed within 0.030 inch (0.76 mm) of the curvic tooth fillets area.

3. All dimensions, wea r, and damage limits provided are limits required after repair except where indicated.

ICN-412-A-651000-B-97499-01027-A-001-01

Figure 3. Tail Rotor Driveshaft - Corrosion Damage Limits (Sheet 2 of 2)



NOTES

1. Cracks, nicks or scratches: none allowed.
2. No damage allowed within 0.050 inch (1.270 mm) of any edge.
3. Circular scratches are acceptable on outside faces of outer laminate adjacent to bolt hole.
4. Area of light fretting and corrosion - surfaces distress not greater than 0.001 inch (0.025 mm) - which can be repaired by soaking in alcohol phosphoric cleaner (C-344, BHT-ALL-SPM) are acceptable.
5. Random light fretting and corrosion pitting are permitted if not greater than 5 percent of each quadrant is affected, and any pitted areas do not exceed 0.050 inch (1.270 mm) diameter.
6. Light fretting and corrosion pitting is allowed if area affected is not greater than 40 percent (maximum four places on each side).

ICN-412-A-651002-B-97499-00001-A-001-01

Figure 4. Disc Pack Assembly Wear Limits (412-340-610-101) (Sheet 1 of 1)

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Tail Rotor Driveshaft and Hangers - Repair

Procedure

1. Refer to 412-A-65-10-00-00A-685A-A / 01262 and 412-A-65-10-01-00A-280A-A / 01266.

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Tail Rotor Driveshaft and Hangers - Lubrication

Procedure

1. Lubricate coupling assemblies (BHT-412-CR&O, Chapter 65).

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Tail Rotor Driveshaft and Hangers - Installation

Procedure

NOTE

If any of the following tail rotor drive train components is replaced or installed in a different location other than originally installed, entire tail rotor drive train shall be aligned in accordance with Step 7..

- Driveshafts (4, 7, 10, or 13, Figure 1),
- Coupling assembly (38),
- Hanger bearing assemblies (6, 9, or 11),
- Disc pack couplings (40),
- Special washers (43 and 53), or
- Intermediate gearbox (18).

If the tailboom alignment is changed, if any damage has occurred that makes the tail rotor drive alignment suspect, or if any tail rotor driveshaft hanger support is disturbed (loosened) or replaced, the entire tail rotor drive system shall be aligned in accordance with Step 7.. If driveshafts (3, 4, 7, 10, 13, and 16), disc pack couplings (40), special washers (43 and 53), or intermediate gearbox (18) are disturbed (loosened) or replaced, inspect disc pack coupling for gaps in accordance with Step 5.7..

1. Install hanger bearing assembly (9 or 11, Detail D) as follows:

1.1. Position hanger bearing assembly (9 or 11) on top of support (51) with grease fitting in hanger on left side of helicopter.

1.2. Install bolts (47), thin steel washers (48), aluminum washers (49) adjacent to support (51) and nuts (50). Torque nuts 50 to 70 inch-pounds (5.65 to 7.91 Nm), then loosen 1/2 turn.

1.3. Install disc pack coupling (40) (Step 3.).

1.4. Install adjacent driveshaft (Step 5.).

NOTE

If maintenance does not necessitate drive system alignment, perform Step 5.5.. through Step 5.7.. If alignment is required, proceed to Step 7..

1.5. Before torquing nuts (50) at hanger assembly, manually rotate (yaw) bearing hanger housing clockwise, then counterclockwise to limits of travel while looking down from above. Note the limits of travel. Position hanger approximately in center of range of motion and tighten hardware attaching hanger to support (51).

1.6. Torque nuts (50) between hanger assembly (9 or 11, Detail D) and support (51) 50 to 70 inch-pounds (5.65 to 7.91 Nm).

2. Install hanger assembly (6, Detail E) as follows:

2.1. Position hanger assembly (6) on top of support (59) with grease fitting in hanger on left side of helicopter.

2.2. Install bolts (57) with washers (58) through hanger and support into barrel nuts (60). Torque bolts 50 to 70 inch-pounds (5.65 to 7.91 Nm), then loosen bolts 1/2 turn.

2.3. Install disc pack couplings (40, Detail D). Refer to Step 3..

2.4. Install adjacent driveshafts. Refer to Step 5..

NOTE

If maintenance does not necessitate drivetrain alignment, perform Step 5.6.. and Step 5.7.. If alignment is required, proceed to Step 7..

2.5. Before torquing bolts (57) at hanger assembly, manually rotate yaw bearing hanger housing hanger clockwise, then counterclockwise to limits of travel while looking down from above. Note the limits of travel. Position hanger approximately in center of range of motion and tighten hardware attaching hanger to support (59).

2.6. Torque bolts (57) between hanger assembly (6) and support (59) 50 to 70 inch-pounds (5.65 to 7.91 Nm).

3. If disc pack couplings (40, Figure 2, Detail F and G) are removed, install as follows:

CAUTION

INDEXING FLATS ON DISCS OF DISC PACK COUPLINGS (40) ARE PREFERENTIALLY ORIENTED. DISCS SHOULD NOT BE DISTURBED IN HANDLING. IF ORIENTATION IS INADVERTENTLY DISTURBED, REINDEX DISCS BY ALTERNATELY INDEXING FLATS ON EACH DISC 60° FROM FLAT ON ADJACENT DISC (Figure 1, DETAIL H). IT IS NOT ACCEPTABLE TO CHANGE RELATIVE POSITION OF A DISC WITHIN A PACK EXCEPT AS NOTED ABOVE OR TO INTERCHANGE DISCS FROM ONE DISC PACK TO ANOTHER.

WHEN INSTALLING BOLTS (WITH SPECIAL WASHERS) THROUGH A DISC PACK COUPLING, WASHERS SHALL BE CAREFULLY INSPECTED TO ENSURE RADIUSED EDGES ON WASHERS ARE INSTALLED AGAINST DISC PACK. RELATIVE ORIENTATION OF BOLTS (HEADS FACING FORWARD OR AFT) SHALL BE AS SHOWN IN DETAIL D.

3.1. Install disc pack coupling on forward facing adapters of hanger assemblies (6, 9, or 11) or on input adapter of intermediate gearbox (18), or on tail rotor gearbox (17), using two lower-most bolts (41), countersunk washers (42) (with chamfers adjacent to boltheads), special washers (43) (with radiused outer edge of washers adjacent to surface of disc pack), thick washers (44), thin washers (45), and nuts (46). Tighten nuts until torque increases above tare value, then loosen one-fourth turn. The top bolt (41), washers (42, 43, 44 and 45) and nut (46) are installed when driveshafts are installed.

3.2. Install adjacent driveshaft (4, 7, 10, 13 or 16). Refer to Step 4.. and/or Step 5..

4. Install driveshaft (16) as follows:

NOTE

Tail rotor gearbox and intermediate gearbox shall be fully installed at time driveshaft (16) is installed.

4.1. Place driveshaft in position between disc pack coupling (40) that is installed on tail rotor gearbox (17) (Step 3.) and output coupling of intermediate gearbox (18).

4.2. Connect upper end of driveshaft (Detail D).

CAUTION

WHEN INSTALLING BOLTS, WITH SPECIAL WASHERS, THROUGH DISC PACK COUPLING, WASHER SHALL BE CAREFULLY INSPECTED TO ENSURE RADIUSED EDGE ON WASHERS ARE INSTALLED AGAINST DISC PACK (DETAIL F AND G). THE RELATIVE ORIENTATION OF BOLTS, HEADS FACING FORWARD OR AFT, SHALL BE AS SHOWN IN DETAIL D.

4.2.1. Align bolt holes in disc pack coupling (40) on tail rotor gearbox with holes in driveshaft flange. Install bolts (52), special washers (53) (with radiused edge adjacent to disc) under heads of bolts, special washers (53) (with radiused edge adjacent to disc), thick washers (54), thin washers (55), and nuts (56). Refer to Detail G for installation of hardware.

4.2.2. Torque nuts (56) evenly until torque increases above tare value, then loosen nuts 1/4 turn.

4.2.3. Loosen two remaining nuts (46) attaching disc pack coupling (40) to adapter of tail rotor gearbox one to two turns.

4.2.4. Install bolt (41), chamfered washer (42) (with chamfer adjacent to bolthead), washers (43) (with radiused edge adjacent to disc pack coupling), thick washer (44), thin washer (45), and nut (46) at disc pack coupling and adapter of tail rotor gearbox. Refer to Detail F for installation of hardware.

4.2.5. Torque nuts (46) evenly until the torque increases above tare value, then loosen 1/4 turn.

4.3. Connect lower end of driveshaft (Detail C).

4.3.1. Align bolt boles in driveshaft flange with those in coupling of intermediate gearbox (18). Install bolts (27), washers (28), and nuts (29).

4.3.2. Torque nuts evenly 120 to 160 inch-pounds (13.56 to 18.08 Nm).

4.4. Progressively torque three nuts (56) at upper end of driveshaft uniformly 280 to 300 inch-pounds (31.64 to 33.90 Nm).

4.5. Progressively torque three nuts (46) at tail rotor gearbox adapter uniformly 280 to 300 inch-pounds (31.64 to 33.90 Nm).

4.6. Inspect disc pack coupling for gaps between disc. Gaps in excess of 0.010 in. (0.254 mm) are not acceptable. If excessive gaps are present, proceed as follows:

4.6.1. Loosen three nuts (56) and three nuts (46) approximately 1/2 turn.

4.6.2. Rotate disc pack coupling (40) back and forth on bolts and repeat Step 4.4. through Step 4.6..

5. Install driveshafts (4, 7, 10, and 13) as follows:

NOTE

Intermediate gearbox (18) shall be fully installed at time driveshaft (13) is installed.

5.1. Position driveshaft(s) as follows:

NOTE

If more than one driveshaft is removed, install driveshafts beginning at the intermediate gearbox (18) (or the most aft driveshaft) and proceed forward.

Detail C shows typical installation of forward end of driveshafts (7, 10 and 13). Detail D shows typical installation of aft end of driveshaft assemblies (4, 7, 10, and 13). Aft end of driveshaft assemblies are mated to the disc pack coupling(s) (40) already installed on intermediate gearbox (18), hanger assemblies (6, 9, and 11) and coupling assembly (38) in Step 3..

5.1.1. Position driveshaft (13) between disc pack coupling (40) and hanger assembly (11).

5.1.2. Position driveshaft (10) between disc pack coupling (40) and hanger assembly (9).

5.1.3. Position driveshaft (7) between disc pack coupling (40) and hanger assembly (6).

NOTE

If driveshaft (3) is removed, do not install driveshaft (4) until driveshaft (3) is installed (Step 6.).

5.1.4. Position driveshaft (4) between disc pack coupling (40) installed on hanger assembly (6) and coupling assembly (38). Refer to detail B to show installation of forward end of driveshaft (4) to coupling (38).

5.2. Insert bolt (27) through uppermost bolt hole in forward end flange and in mating hanger assembly or coupling assembly adapter and bolt (52) through uppermost bolt hole in driveshaft aft end flange and in mating disc pack coupling to hold driveshaft in position until permanently installed.

5.3. Connect aft end of driveshaft (detail D).



WHEN INSTALLING BOLTS WITH SPECIAL WASHERS, THROUGH A DISC PACK COUPLING, ENSURE RADIUSED EDGES ON WASHERS ARE INSTALLED AGAINST DISC PACK. THE RELATIVE ORIENTATION OF BOLTS, HEADS FACING FORWARD OR AFT, SHALL BE AS SHOWN IN DETAIL D.

5.3.1. Align bolt holes in disc pack coupling (40) with holes in driveshaft flange. Install bolts (52) and washers (53) under heads of bolts (52) (with countersunk side adjacent to bolt heads). Install washers (53) (with radiused edge adjacent to disc), thick washers (54), thin washers (55), and nuts (56).

5.3.2. Torque nuts (56) evenly until torque increases above tare value, then loosen nuts onefourth turn.

5.3.3. Loosen two remaining nuts (46) attaching disc pack coupling (40) to adapter of hanger assembly or intermediate gearbox one to two turns.

5.3.4. Install bolt (41), chamfered washer (42) (with chamfer adjacent to bolthead), special washers (43) (with radiused edge adjacent to disc pack coupling), thick washer (44), thin washer (45), and nut (46) at disc pack coupling and adapter of hanger assembly or intermediate gearbox.

5.3.5. Torque nuts (46) evenly until torque increases above tare value, then loosen one-forth turn.

5.4. Connect forward end of driveshaft (details B and C).

5.4.1. Align bolt holes in driveshaft flange with holes in adapter of hanger assembly (6, 9, or 11) or of coupling assembly (38). Install bolts (27 or 30), washers (28 or 31), and nuts (29 or 32).

5.4.2. Torque nuts evenly 120 to 160 in.-lbs. (13.56 to 18.08 N-m).

NOTE

If maintenance requires drive train alignment (refer to note preceding Step 1..), proceed to Step 7.. If alignment is not required, proceed to Step 5.5.. below.

5.5. Progressively torque nuts (56) at aft end of driveshaft uniformly to 280 to 300 in.-lbs. (31.64 to 33.90 N-m).

5.6. Progressively torque three nuts (46) at intermediate gearbox, or aft end hanger assembly adapter uniformly 280 to 300 in.-lbs. (31.64 to 33.90 Nm).

5.7. Inspect each disc pack coupling for gaps between disc. Gaps in excess of 0.010 in. (0.254 mm) are not acceptable. If excessive gaps are present in any disc pack coupling, proceed as follows:

5.7.1. Loosen nuts (56) and nuts (46) approximately one-half turn.

5.7.2. Rotate disc pack coupling (40) back and forth on bolts, then repeat Step 5.5., Step 5.6., and Step 5.7..

6. Install driveshaft (3) and coupling assembly (2) and coupling assembly (38) as follows:



WHEN INSTALLING DRIVESHAFT INTO ENCLOSURE, PLACE SUITABLE PROTECTIVE MATERIAL ON ENGINE DECK WITHIN DRIVESHAFT ENCLOSURE TO PREVENT DAMAGE TO DRIVESHAFT.

6.1. Install coupling assembly (2) on forward end of driveshaft (3) using bolts (21), washers (22), and nuts (23). Torque nuts evenly 120 to 160 in.-lbs. (13.56 to 18.08 N-m).

6.2. Install driveshaft (3) with coupling assembly (2), coupling end first, into driveshaft enclosure. Gently lower driveshaft onto engine deck.

6.3. Install coupling assembly (38) on top of support (20) as follows:

6.3.1. Remove bolts (33), washers (34), and nuts (37) that were loosely installed in support (20). Inspect upper ends of braces (36) to confirm aluminum washers (35) and washers (34) are still in place between upper ends of braces (36) and lower surfaces of support (20). If washers are in place, proceed directly to Step 6.3.3.. If washers are not present, proceed to Step 6.3.2..

6.3.2. Install washers (35 and 34) between the upper end of brace(s) (36) and lower surface(s) of support (20). Washer(s) (35) shall be adjacent to the lower surface of support (20) and washer(s) (34) shall be adjacent to upper end of brace(s) (36). Refer to detail B.

6.3.3. Place coupling assembly (38) on top of support (20) with grease-packed coupling facing forward.

6.3.4. Install bolts (33), washers (34), and nuts (37) through coupling assembly (38), support (20), and braces (36) (refer to detail B). Temporarily torque nuts 50 to 70 in.-lbs. (5.65 to 7.91 N-m).

6.3.5. Check installed brace assemblies (36) for preload condition. Remove bolts attaching braces to service deck. If brace is properly adjusted, there will be no spring-back or hard contact with service deck when bolts are removed. If spring-back is in excess of 0.020 in. (0.508 mm), or hard contact is evident, adjust brace assemblies as follows:

6.3.5.1. Adjust thickness of peel shims between lower ends of brace assemblies (36) and service deck, so there is no excessive spring-back or hard contact (preload).

6.3.5.2. After proper brace shimming has been achieved, install bolts attaching braces to service deck. Torque bolts 50 to 70 in.-lbs. (5.65 to 7.91 N-m).

6.3.6. Loosen nuts (37) at coupling assembly (38) and support (20) one to one and one-half turns.

6.3.7. Install driveshaft (4). Refer to Step 5.., except do not perform Step 6.5.. through Step 6.7..

NOTE

If subject maintenance is such that tail rotor drive alignment is required (refer to notes preceding Step 1..), proceed to Step 7.. If alignment is not required, proceed with note and Step 6.3.8.. below.

Before tightening bolts at coupling assembly (38), manually rotate bearing hanger housing clockwise and then counterclockwise while looking down from above, to limits of travel. Position hanger approximately in center of range of motion and tighten hardware attaching hanger to fuselage support.

6.3.8. Tighten two nuts (37) at coupling assembly (38) and support (20). Torque nuts 50 to 70 in.-lbs. (5.65 to 7.91 N-m).

6.3.9. Progressively torque nuts (56) at aft end of driveshaft uniformly 280 to 300 in.-lbs. (31.64 to 33.90 N-m).

6.3.10. Progressively torque nuts (46) at intermediate gearbox or aft end hanger assembly uniformly 280 to 300 in.-lbs. (31.64 to 33.90 N-m).

6.3.11. Inspect disc pack coupling for gaps between disc. Gaps in excess of 0.010 in. (0.254 mm) are not acceptable. If excessive gaps are present, proceed as follows:

6.3.11.1. Loosen nuts (56) and nuts (46) approximately one-half turn.

6.3.11.2. Rotate disc pack coupling back and forth on bolts and repeat Step 6.3.9. through Step 6.3.11..

6.4. Position driveshaft (3) with coupling assembly (2) between transmission adapter (1) and grease packed coupling of coupling assembly (38).

6.5. Align bolt holes in coupling (2) with those in adapter (1). Install bolts (26), washers (25), and nuts (24). Torque nuts 120 to 160 in.-lbs. (13.56 to 18.08 N-m).

6.6. Align bolt holes in aft end flange of driveshaft (3) with those in grease packed coupling of coupling assembly (38). Install bolts (27), washers (28), and nuts (29). Torque nuts 120 to 160 in.-lbs. (13.56 to 18.08 N-m).

6.7. Remove protective material from driveshaft enclosure.

7. Align tail rotor drive train as follows:



USE CARE WHEN INSTALLING DRIVESHAFT INTO ENCLOSURE. PLACE SUITABLE PROTECTIVE MATERIAL ON ENGINE DECK WITHIN DRIVESHAFT ENCLOSURE TO PREVENT DAMAGE TO DRIVESHAFT DURING INSTALLATION.

NOTE

If the tailboom alignment is changed, if any damage has occurred that makes the tail rotor drive alignment suspect, or if any tail rotor driveshaft hanger support is disturbed (loosened) or replaced, the entire tail rotor drive system shall be aligned. If driveshafts (3, 4, 7, 10, 13 and 16), disc pack couplings (40), washers (43 and 53), or intermediate gearbox (18) are disturbed (loosened) or replaced, inspect disc pack coupling for gaps in accordance with Step 5.7..

NOTE

Certain steps in the following procedures, such as loosening of specific fasteners or partial removal of certain components may have already been accomplished during conditional installation of components in Step 1.., Step 2.., Step 3.., Step 5.., and Step 6..

7.1. Remove driveshaft (3) and coupling assembly (2) from between transmission adapter (1) and coupling assembly (38). Rest assembly on engine deck within driveshaft enclosure. Do not remove driveshaft from enclosure or coupling assembly (38) from support (20).

7.2. Starting at intermediate gearbox and working forward, loosen nuts (46) securing each disc pack coupling (40) to adapters of hanger assemblies (11, 9, and 6), and to adapter of intermediate gearbox. Loosen nuts (56) securing each disc packing to driveshaft (13, 10, 7, and 4). Loosen nuts one to one and one-half turns.

7.3. Starting at intermediate gearbox and working forward, tighten nuts (loosened in Step 7.2..) until torque increases above tare value, then loosen one-forth turn.

7.4. Starting at intermediate gearbox and working forward, loosen nuts (50) securing hanger assemblies (9 and 11) to supports (51) one to one and one-half turns.

7.5. Loosen bolts (57) securing hanger assembly (6) to support (59), one to one and one-half turns.

7.6. Loosen nuts (37) securing coupling assembly (38) to support (20), one to one and one-half turns.

7.7. Starting at intermediate gearbox and working forward, progressively torque nuts (46) and nuts (56), securing disc pack couplings (40) to adapter of intermediate gearbox, adapters of hanger assemblies (11, 9, and 6) and to driveshafts (13, 10, 7, and 4) uniformly to 280 to 300 inch-pounds (31.64 to 33.90 Nm).

7.8. Inspect each disc pack coupling (40) for gaps between the disc. Gaps in excess of 0.010 inch (0.254 mm) are not acceptable. If excessive gaps are present in any disc pack coupling, proceed as follows:

7.8.1. Loosen nuts (56) and nuts (46) approximately one-half turn.

7.8.2. Rotate disc pack coupling (40) back-and-forth on its bolts, then progressively retorque nuts (56) and nuts (46) uniformly to 280 to 300 inch-pounds (31.64 to 33.90 Nm).

7.9. Repeat Step 7.8..

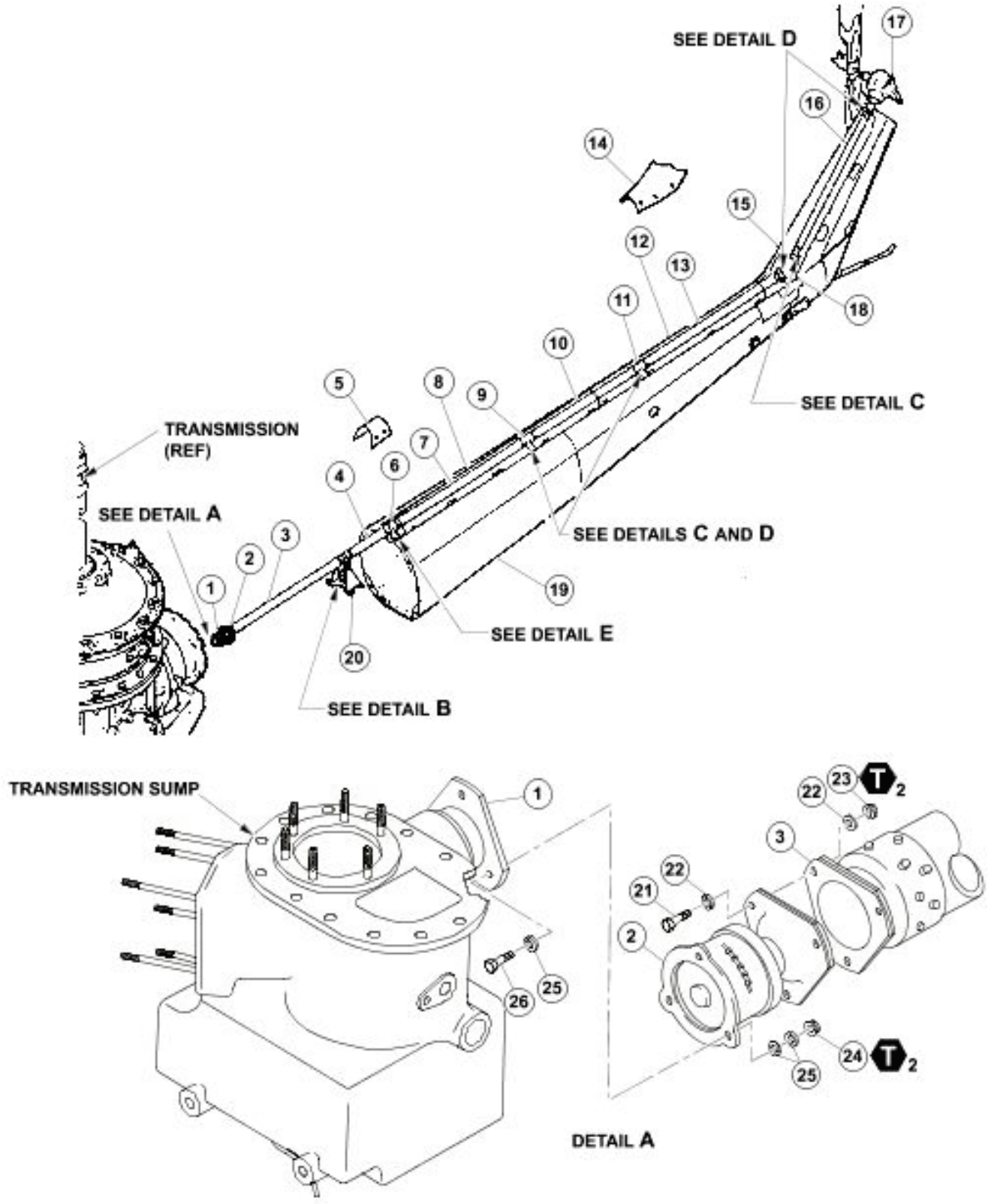
NOTE

Before tightening bolts at each hanger and coupling assembly, manually rotate bearing hanger housing clockwise, then counterclockwise, to limits of travel while looking down from above. Position hanger approximately in center of range of motion and tighten hardware attaching hanger to tailboom or fuselage support.

7.10. Starting at intermediate gearbox and working forward, torque nuts and bolts (loosened in Step 7.4. through Step 7.6.) to 50 to 70 inch-pounds (5.65 to 7.91 Nm).

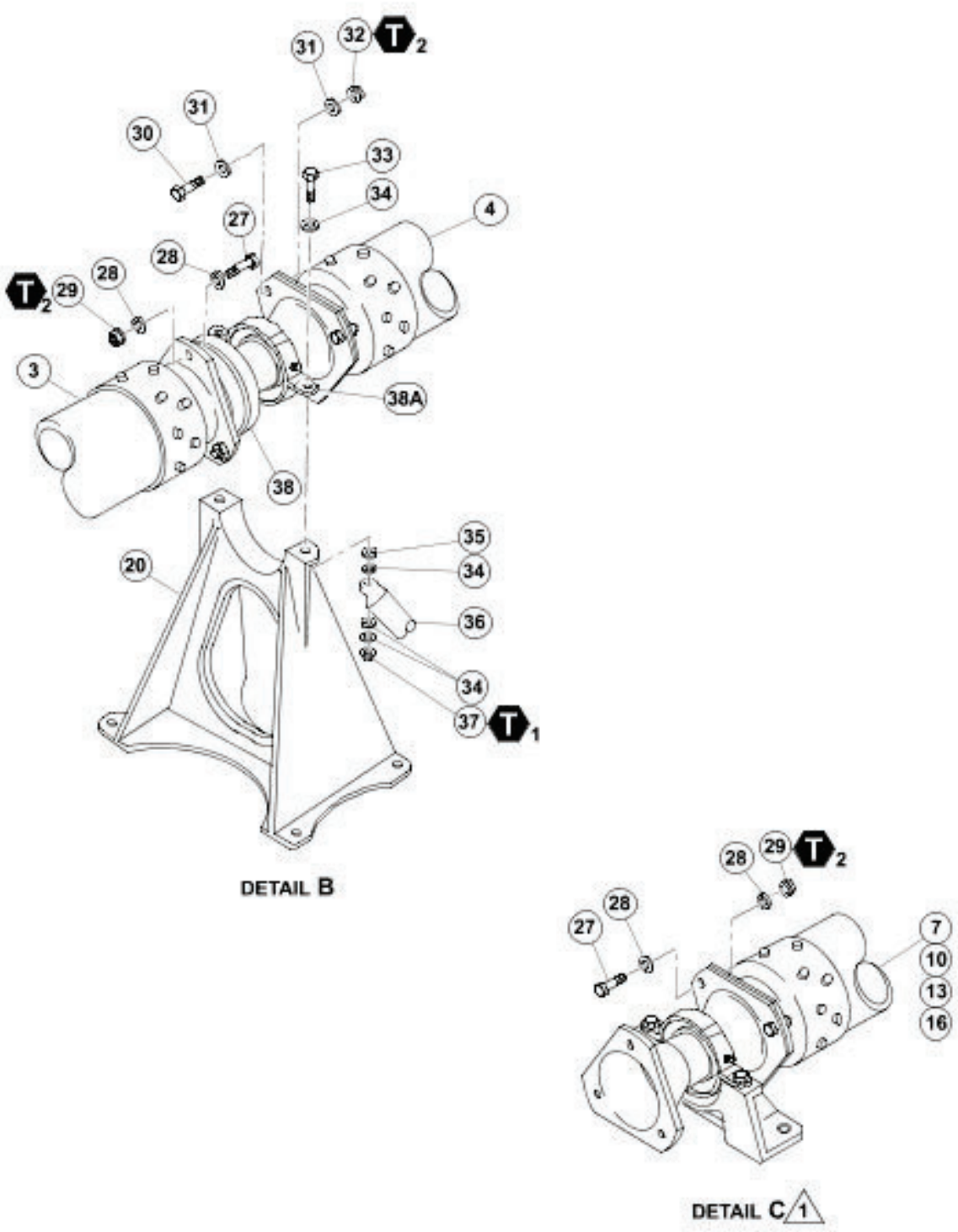
7.11. Install driveshaft (3) (Step 6.).

8. Install or close covers, fairings, and doors that were removed in 412-A-65-10-02-00A-520A-A / 01270.



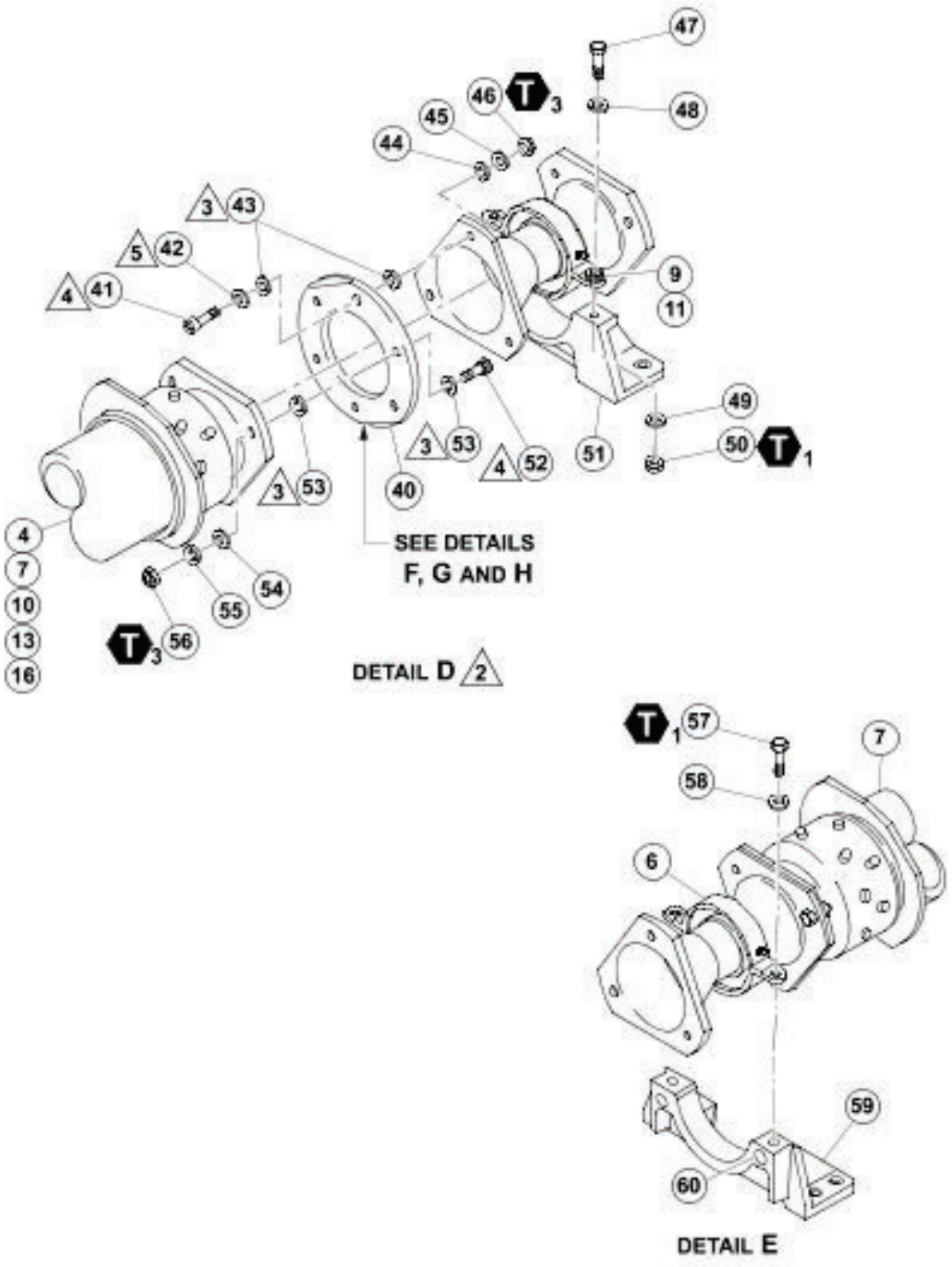
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Figure 1. Tail Rotor Driveshaft (Sheet 1 of 5)



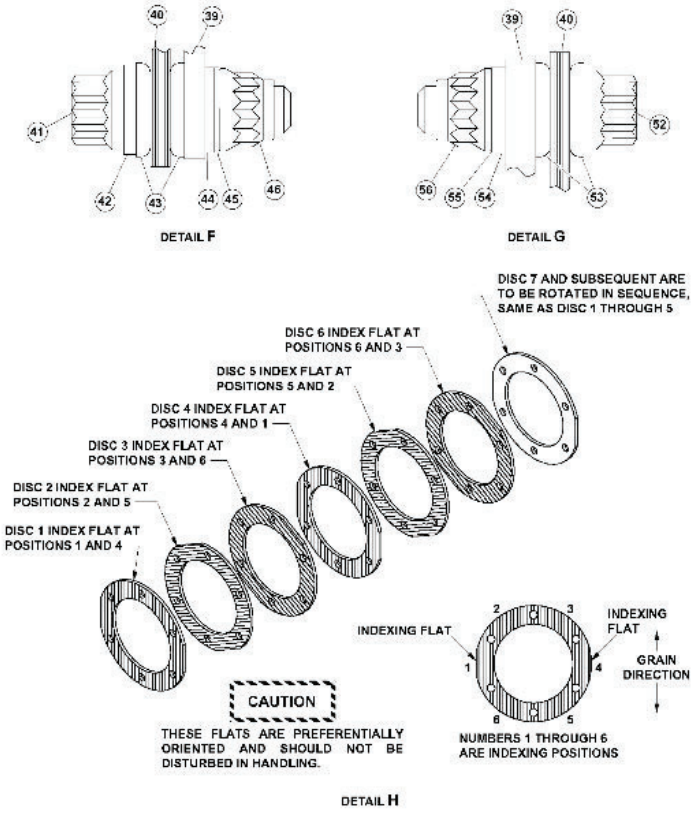
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Figure 1. Tail Rotor Driveshaft (Sheet 2 of 5)



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Figure 1. Tail Rotor Driveshaft (Sheet 3 of 5)



ICN-412-A-650000-B-97499-00008-A-001-01

Figure 1. Tail Rotor Driveshaft (Sheet 4 of 5)

1. Adapter (tail rotor drive quill)
2. Coupling assembly
3. Driveshaft
4. Driveshaft
5. Cover
6. Hanger bearing assembly
7. Driveshaft
8. Driveshaft cover
9. Hanger bearing assembly
10. Driveshaft
11. Hanger bearing assembly
12. Driveshaft cover
13. Driveshaft
14. Intermediate gearbox cover
15. Vertical fin cover
16. Driveshaft
17. Tail rotor gearbox
18. Intermediate gearbox
19. Tailboom
20. Hanger bearing support
21. Bolt
22. Steel washer
23. Nut
24. Nut
25. Steel washer
26. Bolt
27. Bolt
28. Steel washer
29. Nut
30. Bolt
31. Steel washer
32. Nut
33. Bolt
34. Steel washer
35. Aluminum washer
36. Brace assembly
37. Nut
38. Coupling assembly
38A. Hanger bearing assembly
39. Driveshaft adapter (typical)
40. Disc pack coupling
41. Bolt
42. Chamfered steel washer
43. Special washer
44. Thick steel washer
45. Thin steel washer
46. Nut
47. Bolt
48. Thin steel washer
49. Aluminum washer
50. Nut
51. Support
52. Bolt
53. Special washer
54. Thick steel washer
55. Thin steel washer
56. Nut
57. Bolt
58. Thin steel washer
59. Support
60. Barrel nut

T1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T2 120 TO 160 IN-LBS (13.56 TO 18.08 Nm)

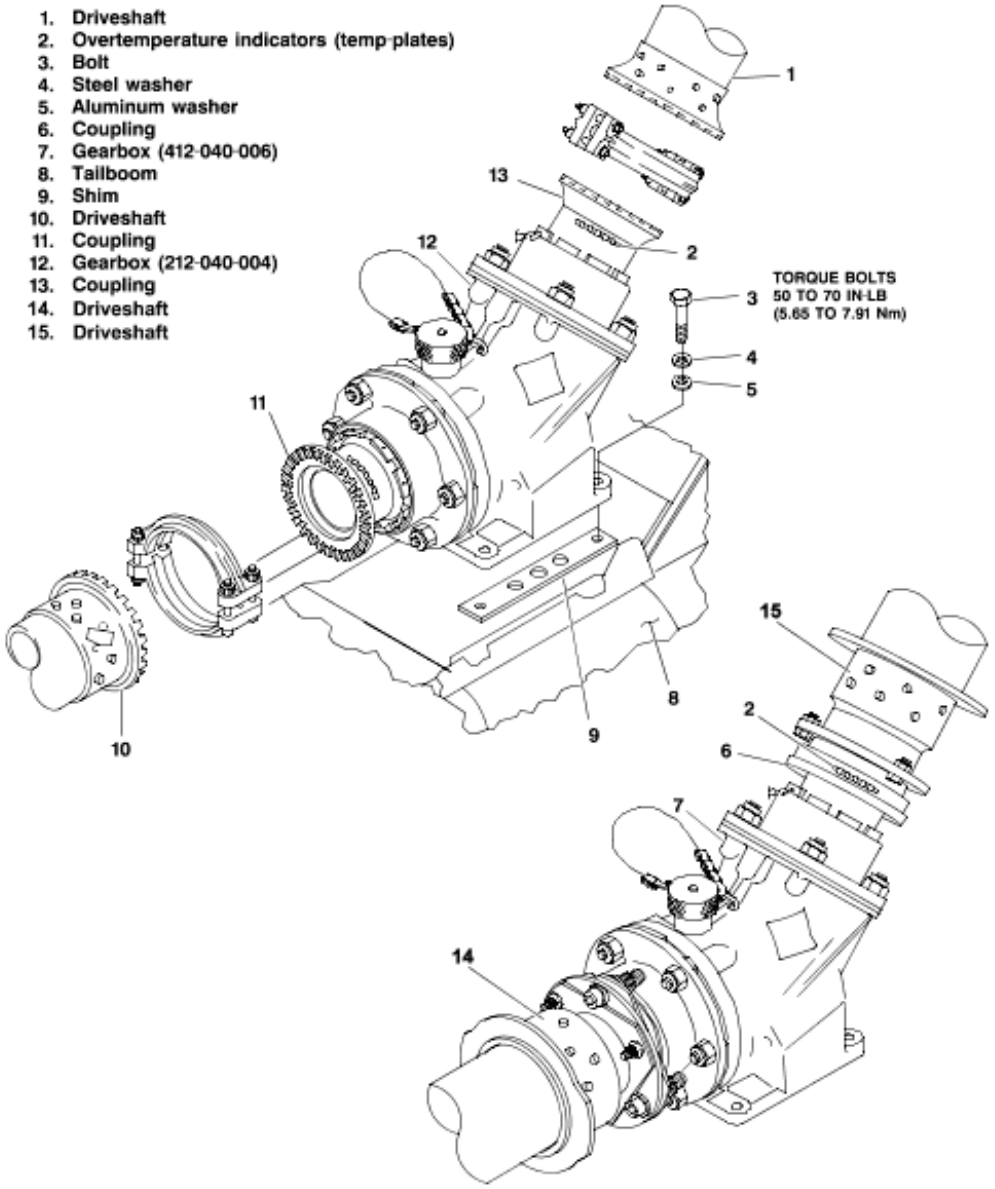
T3 280 TO 300 IN-LBS (31.64 TO 33.90 Nm)

NOTES

1. Detail C shows typical installation of forward end of driveshafts (7, 10, 13 and 16).
2. Detail D shows typical installation of tailboom mounted hanger assembly and aft end installation of driveshaft assemblies (4, 7, 10, and 13).
3. Place radiused outer edge of washers adjacent to surface of disc pack (40).
4. Position heads of bolts (41 and 52), Detail D, 180° from each other, as shown.
5. Place chamfered bore of chamfered steel washer (42) adjacent to bolt head (41).

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Figure 1. Tail Rotor Driveshaft (Sheet 5 of 5)



ICN-412-A-651002-B-97499-00002-A-001-01

Figure 2. Intermediate Gearbox (Sheet 1 of 1)

Intermediate Gearbox - Description

The intermediate gearbox is installed on the tailboom forward of the vertical fin. The gearbox is aligned by use of shims between the gearbox and the tailboom. The gearbox has an electric chip detector and an oil level sight gage located on the right side. A vented filler plug is installed on top of the case. A cover with cowl fasteners provides access to the gearbox.

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Intermediate Gearbox - Maintenance Procedures

Procedure

1. For the maintenance procedures consisting of troubleshooting, inspection, removal, cleaning, limited repair, and installation, refer to 412-A-65-20-00-00A-422A-A / 01278, 412-A-65-20-00-00A-280A-A / 01279, 412-A-65-20-00-00A-520A-A / 01280, 412-A-65-20-00-00A-251A-A / 01281, 412-A-65-20-00-00A-280B-A / 01282, 412-A-65-20-00-00A-280C-A / 01283, 412-A-65-20-00-00A-685A-A / 01284 and 412-A-65-20-00-00A-720A-A / 01285. When any doubt exists as to the serviceability of a gearbox, perform serviceability checks defined in 412-A-63-00-00-00A-028B-A / 01202.

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Intermediate Gearbox - Troubleshooting

Procedure

1. Refer to Table 1.

Table 1. Intermediate Gearbox - Troubleshooting

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
CHIP 42/90 BOX caution segment illuminated.	Metal chips have accumulated on chip detector.	Remove and inspect electric chip detector. If chips are found, retain in a clean container for subsequent investigation (412-A-65-40-01-00A-520A-A / 01289).
	Faulty wiring or faulty chip detector.	Check wiring to electric chip detector for continuity and grounding (412-A-98-00-00-20B-051A-A / 02580 and 412-A-98-00-00-21B-051A-A / 02581). Repair faulty wiring or replace electric chip detector as required (412-A-96-03-01-13A-320A-A / 02059).
Oil leakage from intermediate gearbox.	Faulty seals, packings, or wear sleeves.	Clean gearbox and recheck for oil leak both static and dynamic (engine operating). Measure and record rate of leakage. Maximum allowable static or dynamic leakage at input drive quill is two drops per minute. Total allowable leakage from all sources on gearbox is six drops per minute.
		Replace gearbox seals, packings, or wear sleeves, if maximum allowable leakage rate is exceeded (BHT-412-CR&O).
	Cracked gearbox housing.	Inspect gearbox for cracks in housing. Replace gearbox if cracks are found.

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Intermediate Gearbox - Inspection (Installed)

Procedure

1. Open tail rotor driveshaft covers and remove intermediate gearbox cover.

2. Shake gearbox (7 or 12, Figure 1) and check for looseness on tailboom. No looseness is acceptable.

3. Check for evidence of fretting corrosion at mating surface between gearbox and tailboom which could be caused by movement of gearbox on tailboom. A gray residue is an indication of fretting corrosion. If residue is present, remove gearbox and inspect in accordance with the BHT-412-CR&O.

NOTE

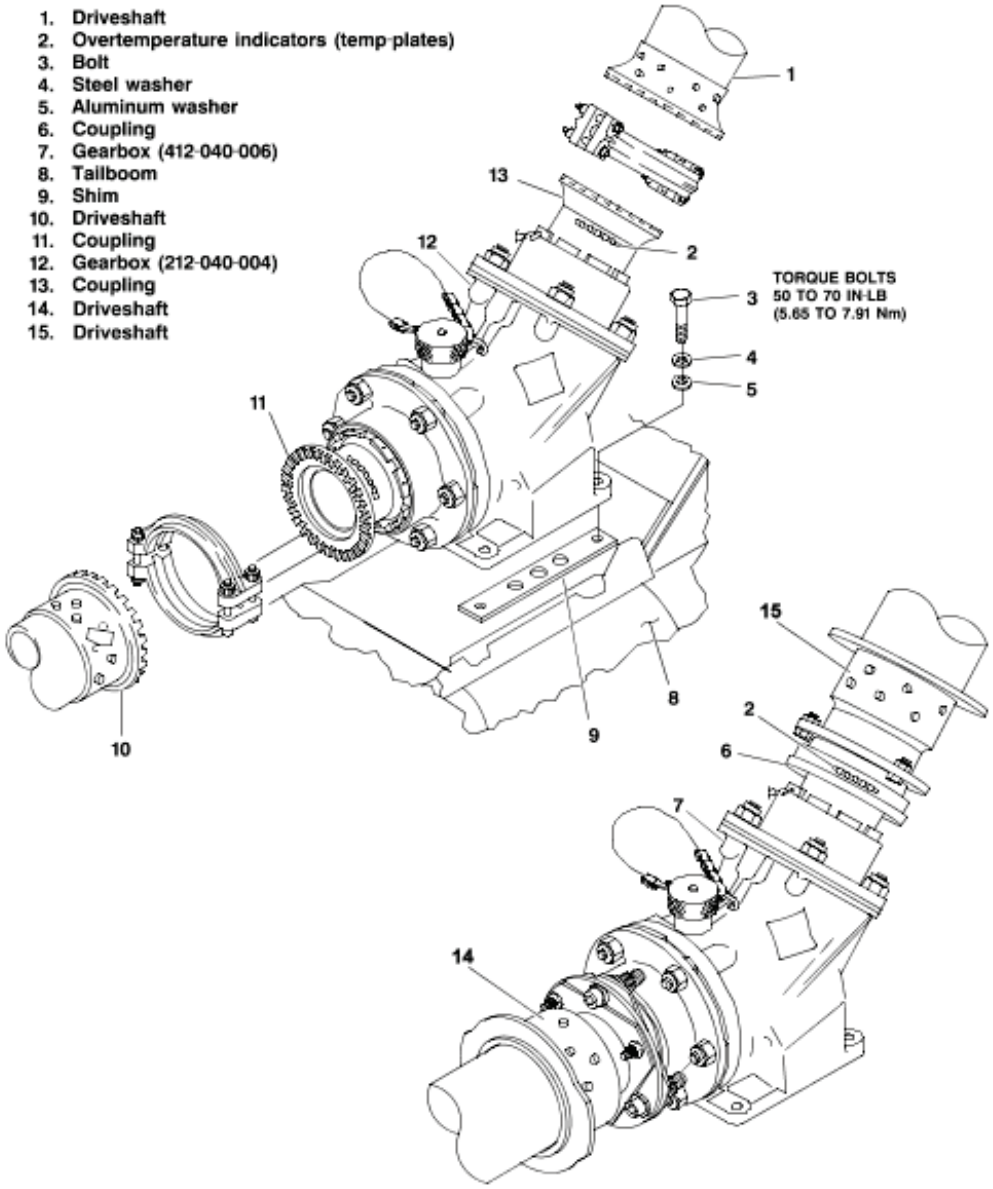
Inspection Step 4. through Step 7. are not applicable if gearbox is removed in Step 3..

4. Inspect bolts (3) for correct torque and correct thread engagement.

5. Inspect couplings (6, 11, and 13) for grease leakage and visual overtemperature indicators (TEMP-PLATE) for discoloration and overheat condition. A change in color of TEMP-PLATE dots (white to black) indicates a possible overheat condition and/or component degradation. Cause of TEMP-PLATE discoloration shall be determined and corrected prior to continued operation. Refer to 412-A-65-10-01-00A-280A-A / 01266. If coupling has overheated, remove gearbox for inspection of inner and outer couplings. Refer to the BHT-412-CR&O for inspection procedures and 412-A-65-20-01-11A-280A-A / 01297 for lubrication procedures.

6. Inspect oil sight glass for correct oil level. Inspect sight glass and indicator for staining and/or cracks and crazing. Damage which can cause oil leakage or make oil level difficult to determine is not acceptable.

7. Inspect electrical chip detector for metal particles (412-A-65-40-01-00A-280A-A / 01290). If metal particles are found, refer to 412-A-63-00-00-00A-280B-A / 01203 for serviceability check required.



ICN-412-A-651002-B-97499-00002-A-001-01

Figure 1. Intermediate Gearbox (Sheet 1 of 1)

Intermediate Gearbox - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Drain line	DB 3703-30	1	

Procedure

1. When intermediate gearbox is to be replaced, unless conditions prevent operation, perform a ten minute ground run-up and drain oil using drain line (DB 3703-30). If run-up is not practical, remove gearbox and flush with new lubrication oil of the same type being used in the gearbox (412-A-12-10-04-00A-292A-A / 00137). Attach tag to intermediate gearbox stating: PRESERVED WITH OPERATING LUBRICANT.

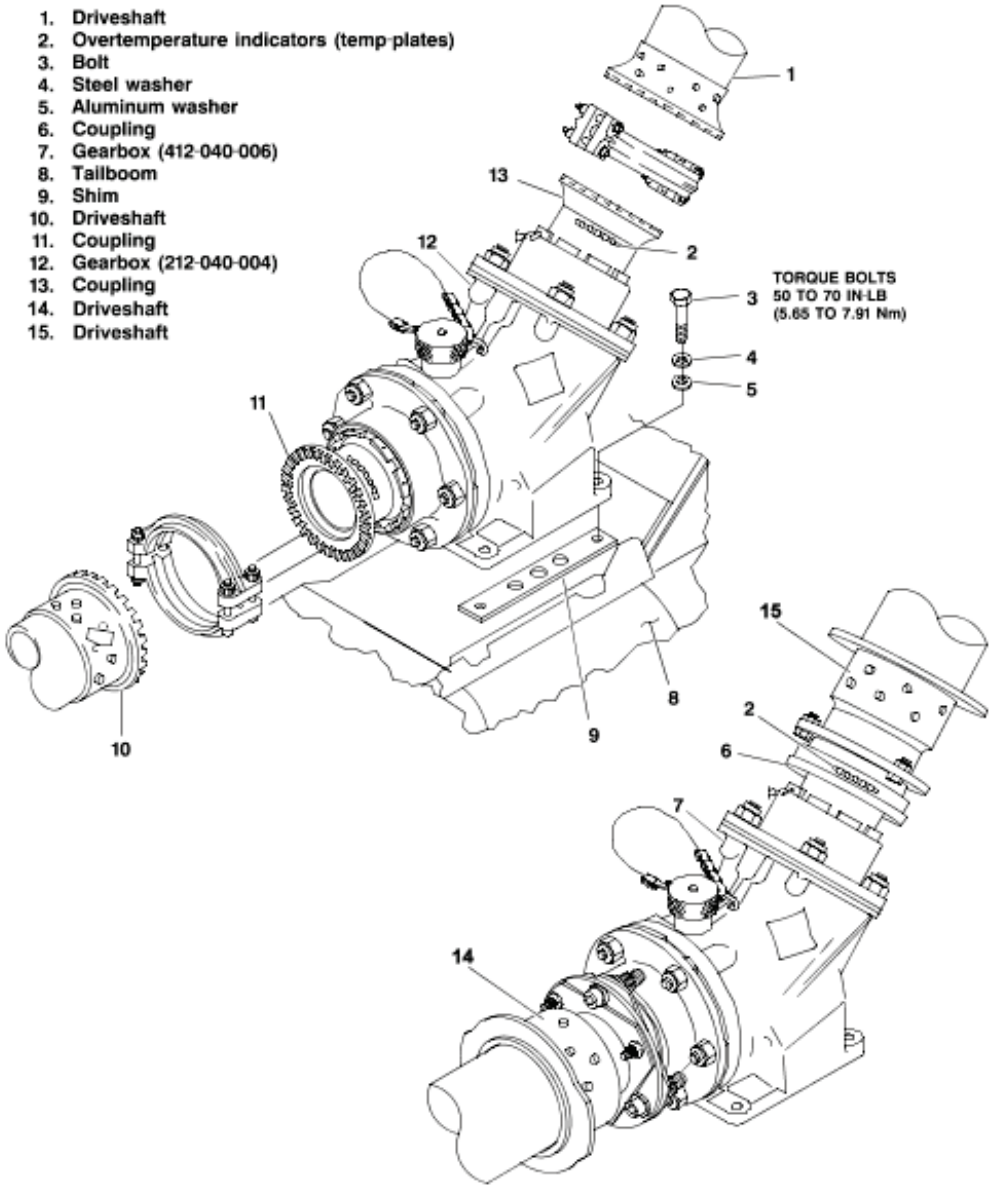
2. Open tail rotor driveshaft covers and remove intermediate gearbox cover.

3. Remove tail rotor driveshafts (1 and 10 or 14 and 15, Figure 1) (412-A-65-10-00-00A-520A-A / 01259 or 412-A-65-10-02-00A-520A-A / 01270).

4. Remove chip detector (412-A-65-40-01-00A-520A-A / 01289).

5. Cut and remove lockwire from bolts (3) (four places). Remove bolts (3) and washers (4 and 5).

6. Remove gearbox (7 or 12). Inspect shims (9) for secure installation on tailboom (8). If shims are not bonded to tailboom, index shims for installation in same location.



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Figure 1. Intermediate Gearbox (Sheet 1 of 1)

Intermediate Gearbox - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean gearbox with clean cloth dampened with drycleaning solvent (C-304).



DO NOT FORCE SOLVENT AND\OR DIRT INTO FLEXIBLE COUPLING WHEN USING COMPRESSED AIR.

2. Dry with filtered, compressed air.

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Intermediate Gearbox - Inspection (Not Installed)

Procedure

1. Refer to the BHT-412-CR&O.

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Intermediate Gearbox - Inspection (Conditional)

Procedure

1.

NOTE

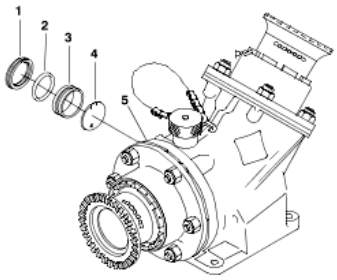
Conditional inspections of intermediate gearbox are required after tail rotor drive system overtorque, sudden stoppage, compressor stall, etc. (412-A-05-50-00-03A-284A-A / 00066).

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Intermediate Gearbox - Repair

Procedure

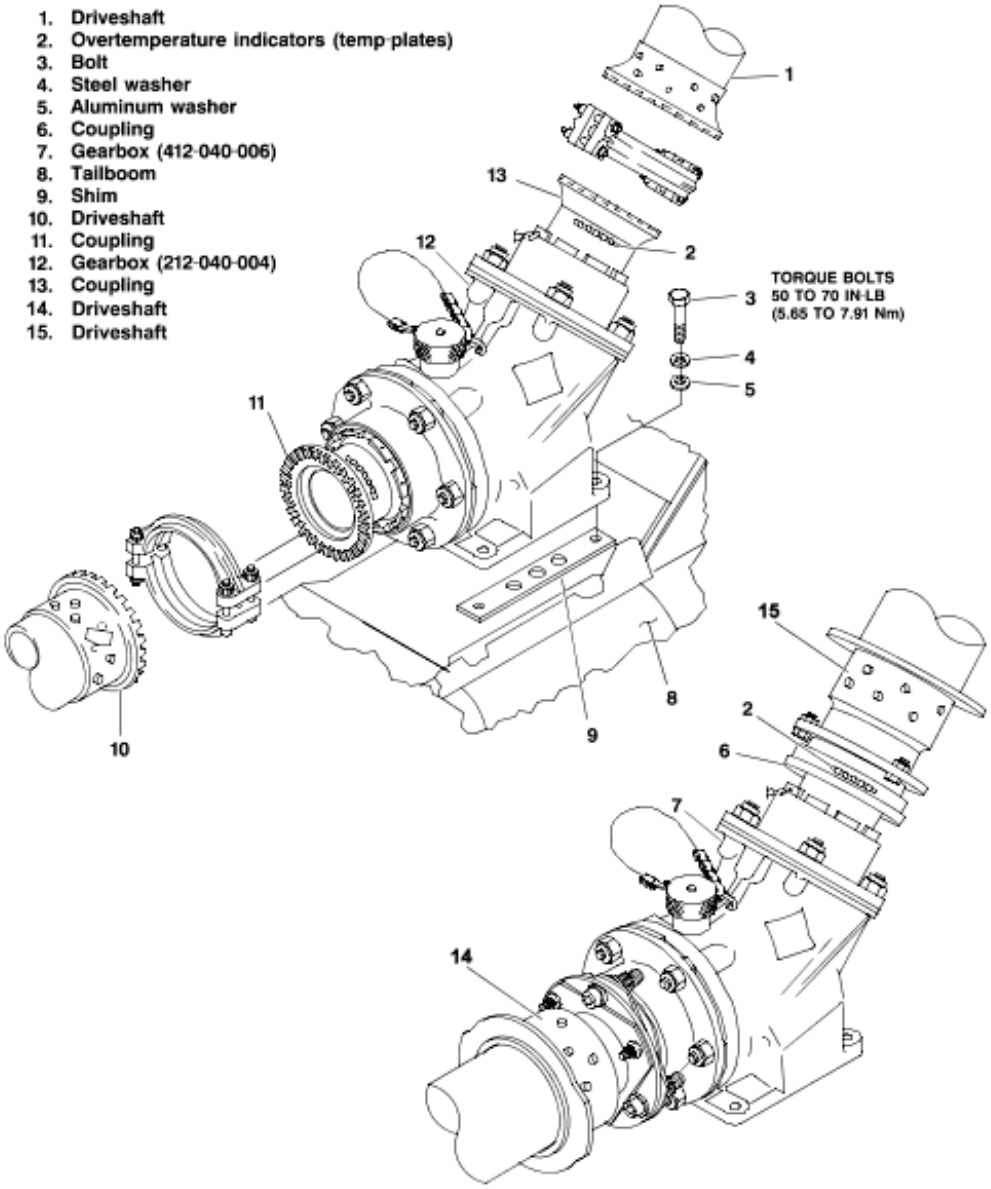
1. Repair the intermediate gearbox in accordance with the BHT-412-CR&O.
2. Repair the faulty filler cap by the same procedure outlined for the tail rotor gearbox filler cap (412-A-65-20-01-02A-685B-A / 01316).
3. Replace the sight glass (3, Figure 1) and/or oil level indicator (4) that fails to pass inspection.
4. Replace the chip detector and/or self-closing valve that fails to pass inspection (412-A-65-40-01-00A-280A-A / 01290).
5. Replace the overtemperature indicators (TEMP-PLATE) (2, Figure 2) on the couplings (6, 11, and 13) if mutilated or discolored (BHT-412-CR&O).



1. Retaining ring
2. Packing
3. Sight glass
4. Oil level indicator
5. Gearbox

ICN-412-A-652000-B-97499-00053-A-001-01

Figure 1. Intermediate Gearbox Sight Glass (Sheet 1 of 1)



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Figure 2. Intermediate Gearbox (Sheet 1 of 1)

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Intermediate Gearbox - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Zinc chromate primer	C-201	AR	
Epoxy polyamide primer	C-204	AR	
Lockwire	C-405	AR	

Procedure



PRIOR TO INSTALLATION OF INTERMEDIATE GEARBOX, ACCOMPLISH THE FOLLOWING:

MAKE SURE FLEXIBLE COUPLINGS IN INPUT AND OUTPUT QUILLS ARE PROPERLY LUBRICATED (412-A-65-20-01-11A-280A-A / 01297).

ROTATE FLEXIBLE COUPLING ON INPUT QUILL AND CHECK FOR BINDING AND PRESENCE OF BACKLASH BETWEEN PINION ON INPUT QUILL AND GEAR ON OUTPUT QUILL.

DRAIN OIL FROM INTERMEDIATE GEARBOX AND SERVICE WITH APPROVED LUBRICATING OIL (412-A-12-10-04-00A-292A-A / 00137).

REFER TO THE TAIL ROTOR DRIVE SYSTEM ALIGNMENT PROCEDURE (412-A-65-00-00-00A-272A-A / 01269) TO DETERMINE IF AN ALIGNMENT CHECK OR GEARBOX SHIMMING IS REQUIRED.

SHIMS (9, Figure 1) SHOULD BE BONDED TO TAILBOOM (8). IF SHIMS ARE NOT BONDED TO TAILBOOM, MAKE SURE THE SAME SHIMS THAT WERE REMOVED, OR SHIMS OF IDENTICAL THICKNESS ARE INSTALLED. FAILURE TO COMPLY COULD CAUSE MISALIGNMENT AND FAILURE OF TAIL ROTOR DRIVE COMPONENTS.

1. Clean the mating surfaces of the intermediate gearbox (7 or 12, Figure 1) and the tailboom (8).

2. Position the intermediate gearbox on the tailboom.

3. Check the gearbox for contact with the shim (9) at four points with the bolts (3) installed and torque 20 to 30 inch-pounds (2.26 to 3.39 Nm). Maximum acceptable gap at any of the four points is 0.001 inch (0.025 mm).

4. Remove the gearbox and coat all faying surfaces, bolts, washers, and bolt holes with unreduced zinc chromate primer (C-201) or unreduced epoxy polyamide primer (C-204). Do not apply primer to the bolt threads.

5. Position the gearbox on the tailboom and wet install the bolts (3) with the steel washer (4) next to the bolt head and the aluminum washer (5) next to the gearbox. Bottoming of the bolt grip is to be determined as follows:

5.1. Additional aluminum washers (5) may be required if the bolt (3) grip bottoms in the nutplate prior to securing the gearbox.

5.2. Torque each bolt (3) 30 to 40 inch-pounds (3.39 to 4.52 Nm) and check the steel washers (4) under the bolt heads for looseness. If any of the steel washers are loose, add aluminum washers (5) as required at each bolt location to ensure no looseness.

5.3. After completion of the bolt-grip-bottoming test, retorque the bolts (3) 50 to 70 inch-pounds (5.65 to 7.91 Nm). Apply lockwire (C-405) to the bolts in pairs at each side.

6. Install the driveshafts (1 and 10 or 14 and 15) (412-A-65-10-00-00A-720A-A / 01263 and 412-A-65-10-02-00A-720A-A / 01275).

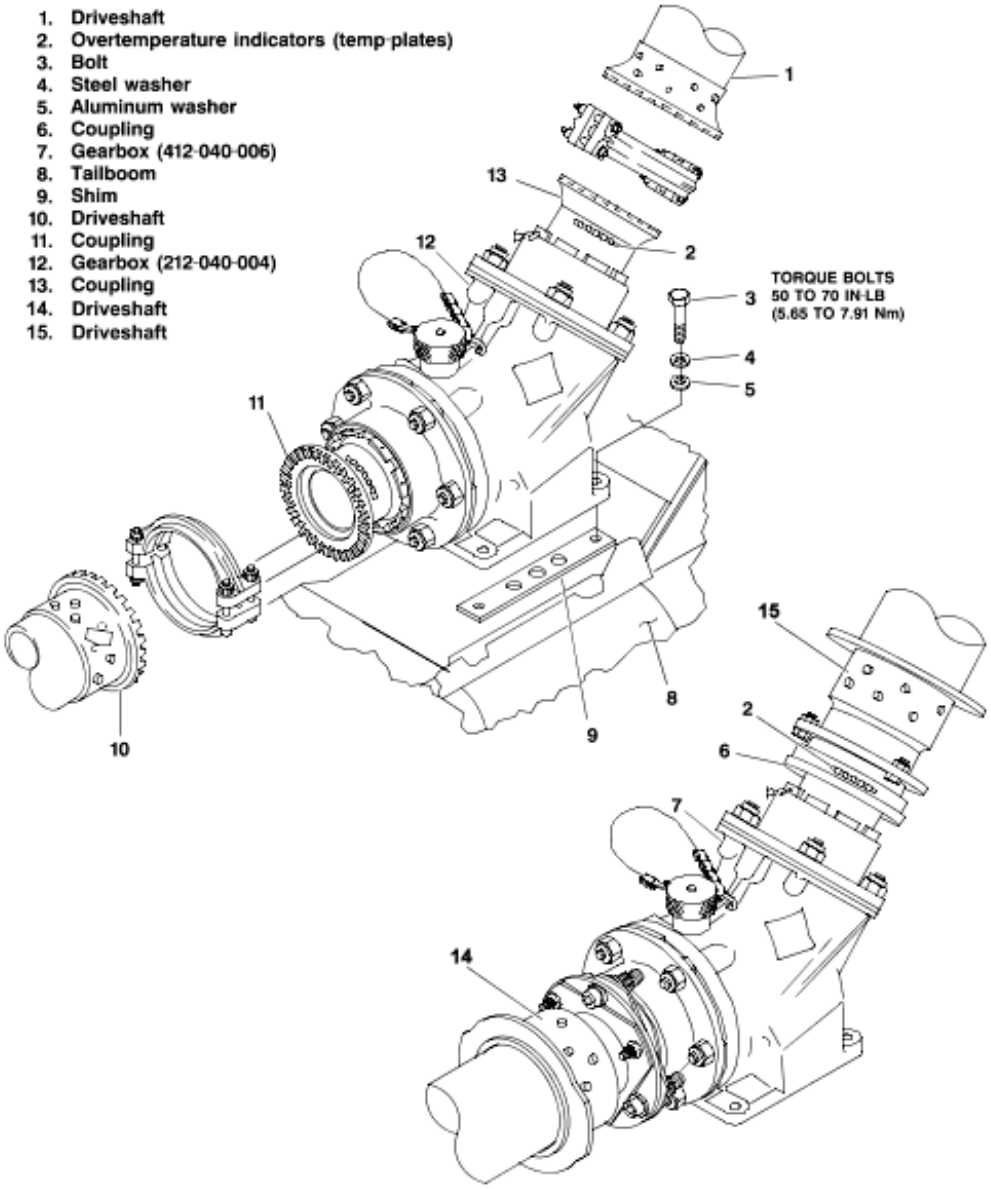
7. Verify actual presence of oil in the sight gauge and check oil level on the full mark on the indicator.

8. Install the chip detector (412-A-65-40-01-00A-720A-A / 01292).

9. If the intermediate gearbox has not previously been qualified on the run-in test stand, accomplish the alternate run-in test (412-A-65-20-00-00A-322A-A / 01286).

10. Install the intermediate gearbox cover and close the tail rotor driveshaft covers.

11. *Do the special inspection (412-A-05-47-00-32A-283A-A / 00033.1) Between 1 and 5 Hours of Flight After Each Installation.*



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Figure 1. Intermediate Gearbox (Sheet 1 of 1)

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Intermediate Gearbox - Alternate Run-In Test

Procedure

NOTE

It is desirable that intermediate gearbox be test run and inspected after overhaul in accordance with publication furnished with run-in test stand for qualification for flight. In the event a run-in test stand is not available, the following functional test may be accomplished as a minimum acceptable alternate run-in test.

1. Install sufficient ballast in the cabin to obtain maximum allowable gross weight (BHT-412-FM, Flight Manual).

2. Verify the helicopter is properly serviced and lubricated (412-A-12-01-00-00A-040A-A / 00141).

3. Start the engines (BHT-412-FM, Flight Manual) and operate at 60% RPM for 5 minutes.

4. Gradually increase RPM to 88% at a rate of 7% each 6 minutes.

5. Hold at 88% RPM for 2 minutes with collective pitch applied to the maximum that can be maintained without becoming airborne or exceeding 100% torque.

6. Shut down the engines.

7. Remove, inspect, clean, and install the chip detector on the intermediate gearbox.

8. Start the engines and slowly increase to 88% RPM. Run at 88% RPM for 6 minutes. Apply collective pitch to the maximum position that can be maintained without becoming airborne or exceeding 100% torque. Gradually increase RPM to 100% without becoming airborne or exceeding 100% torque, at a rate of 3% each 6 minutes.

9. Run at 100% RPM for 30 minutes with collective pitch in the maximum up position that can be maintained without becoming airborne or exceeding 100% torque.

10. Stop the engines and drain the oil from the intermediate gearbox. Check the gearbox oil and chip detector for an excessive amount of metal particles indicating internal failure (412-A-63-00-00-01A-280A-A / 00989). Service the intermediate gearbox (412-A-12-10-04-00A-040A-A / 00136).

11. Inspect the input and output quill couplings and surrounding area for evidence of grease leakage. If there is evidence of grease leakage, remove the couplings and replace the defective parts. Verify the servicing of the gearbox (412-A-12-10-04-00A-040A-A / 00136). Repeat the test run.

12. Check the visual overtemperature indicators (TEMP-PLATE) for discoloration and overheat condition. A change in color of TEMP-PLATE dots (from white to black) will indicate a possible overheat condition and/or component degradation. The cause of indicators discoloration shall be determined and corrected prior to continued operation (412-A-65-10-01-00A-280A-A / 01266).

13. Replace the visual overtemperature indicators (TEMP-PLATE) if mutilated or discolored (BHT-412-CR&O).

14. Install the chip detector and service the intermediate gearbox (412-A-12-10-04-00A-040A-A / 00136).

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Sight Glass - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Drain line	DB 3703-30	1	

Procedure

1. Remove the chip detector (412-A-65-40-01-00A-520A-A / 01289).

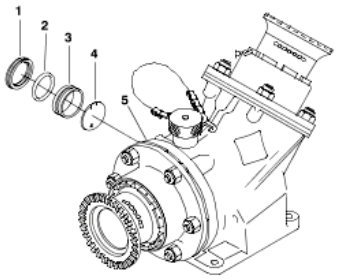
2. Insert one end of the drain line (DB 3703-30) into the container. Insert the other end in the hole where the detector was installed. Oil will automatically drain.

3. Remove the retaining ring (1, Figure 1) from the gearbox (5).

4. Remove the sight glass (3) and packing (2). Discard the packing.

5. Remove the oil level indicator (4).

6. Remove the drain line (DB 3703-30).



1. Retaining ring
2. Packing
3. Sight glass
4. Oil level indicator
5. Gearbox

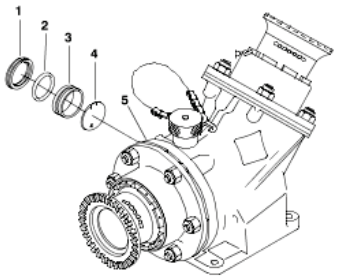
ICN-412-A-652000-B-97499-00053-A-001-01

Figure 1. Intermediate Gearbox Sight Glass (Sheet 1 of 1)

Sight Glass - Installation

Procedure

1. Install the oil level indicator (4, Figure 1) in the gearbox (5).
2. Install the packing (2) on the sight glass (3). Coat the packing with gearbox oil (412-A-12-10-04-00A-292A-A / 00137). Install the glass.
3. Install the retaining ring (1).
4. Install the chip detector (412-A-65-40-01-00A-720A-A / 01292).
5. Service the gearbox (412-A-12-10-04-00A-040A-A / 00136).



1. Retaining ring
2. Packing
3. Sight glass
4. Oil level indicator
5. Gearbox

ICN-412-A-652000-B-97499-00053-A-001-01

Figure 1. Intermediate Gearbox Sight Glass (Sheet 1 of 1)

Electrical Chip Detector - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Drain line	DB 3703-30	1	

Procedure

NOTE

Chip detector element (3, Figure 1) may be removed for inspection without draining gearbox oil.

1. Move the nipple (8) back on the electrical wire terminal (2) to expose the nut (1). Remove the nut and the electrical wire.

2. Press in on the knurled body of the chip detector element (3), turn counterclockwise to disengage locking detent, and withdraw chip detector element from the self-closing valve (5). Remove the packing (4) and discard.

NOTE

If the self-closing valve (5) is to be removed, install the oil drain line (DB 3703-30) in the self-closing valve and allow oil to drain from the gearbox.

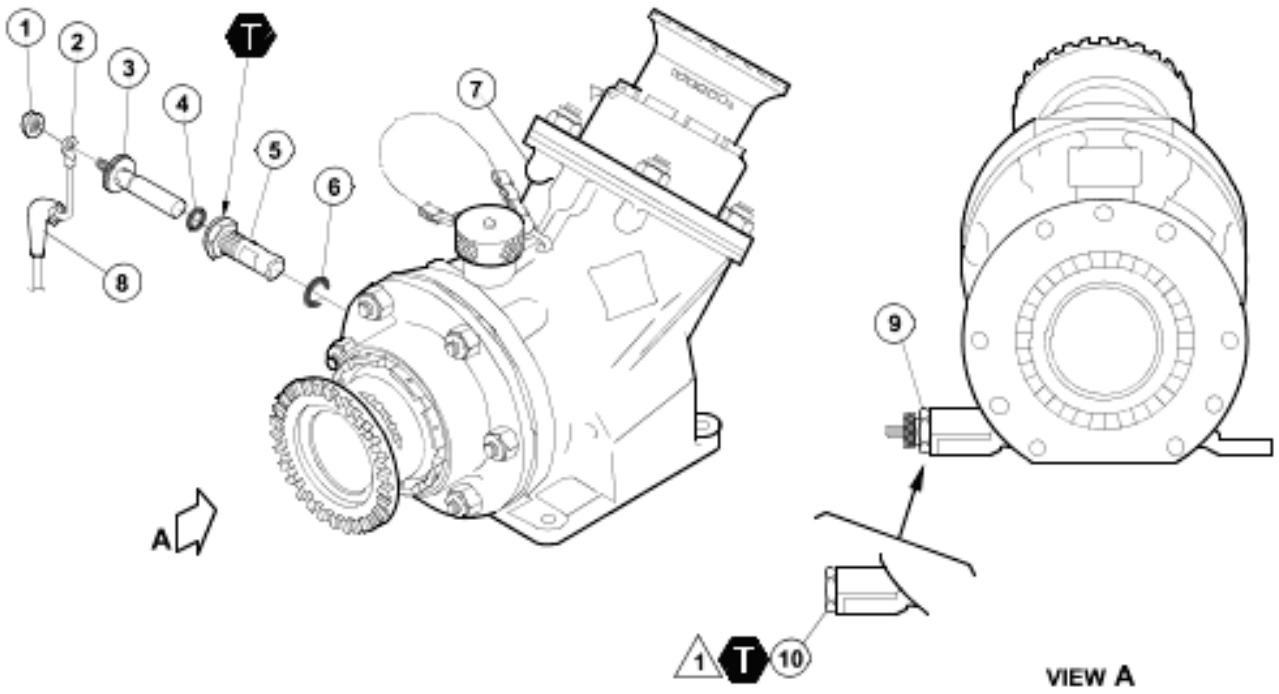
3. Cut and remove the lockwire from the self-closing valve (5). Remove the self-closing valve from the intermediate gearbox (7). Remove the gasket (6) from the self-closing valve. Discard the gasket.

NOTE

If the tail rotor gearbox will not be reinstalled in a timely manner, or shipped for service or repair, do the following steps.

4. Lubricate a new packing (2) with gearbox oil (412-A-12-10-04-00A-292A-A / 00137) and install on the plug (10).

5. Install the plug (10) into the gearbox. Torque the plug 120 to 150 inch-pounds (13.6 to 16.9 Nm).



1. Nut
2. Electrical wire terminal
3. Chip detector element
4. Packing
5. Self-closing valve
6. Gasket
7. Intermediate gearbox
8. Nipple
9. Chip detector assembly
10. Plug

T 120 TO 150 IN-LBS
(13.56 TO 16.95 Nm)

NOTE

1 For helicopters S/N 36694 and subsequent.

ICN-412-A-654001-B-97499-00001-A-001-01

Figure 1. Intermediate Gearbox Electrical Chip Detector (Sheet 1 of 1)

Electrical Chip Detector - Inspection

Procedure

1. Refer to 412-A-63-00-00-00A-009A-A / 00987 for the procedure to inspect the electrical chip detector for metal particles.

2. Inspect the electrical chip detector.

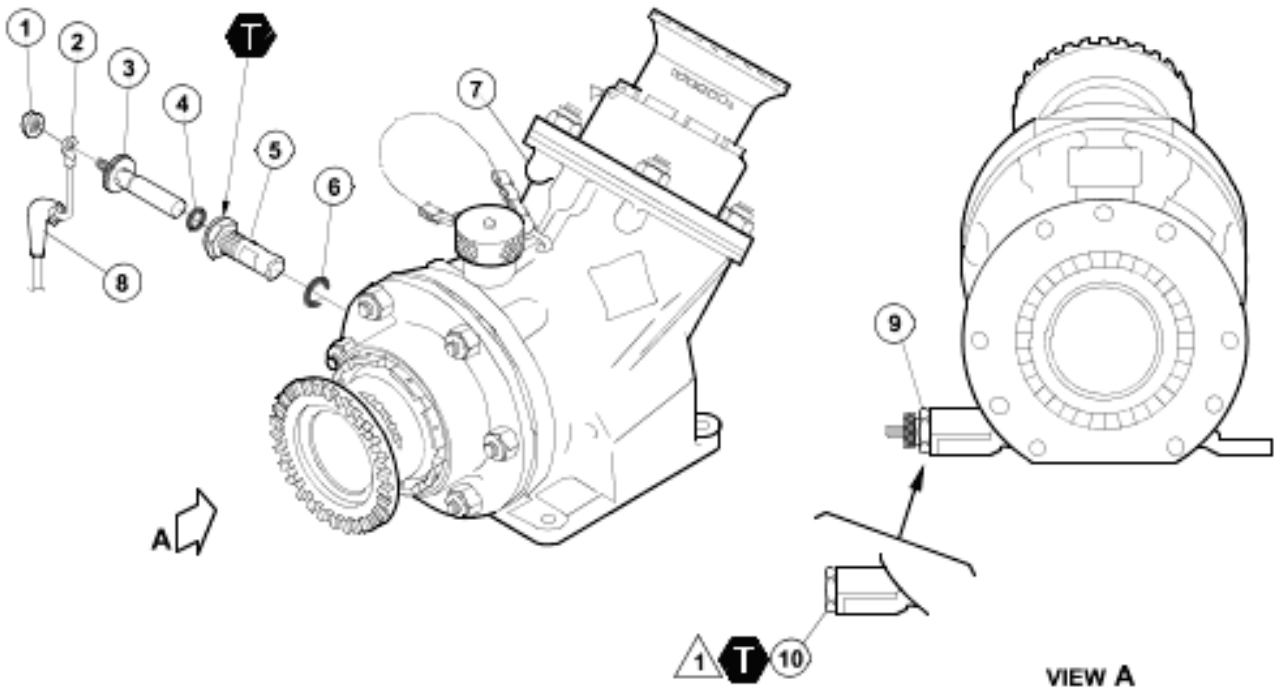
3. Perform an operational check on the chip detector (412-A-96-00-00-00A-009A-A / 01870).

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Electrical Chip Detector - Repair

Procedure

1. Repair of the electrical chip detector assembly (9, Figure 1) is restricted to the replacement of defective parts.



1. Nut
2. Electrical wire terminal
3. Chip detector element
4. Packing
5. Self-closing valve
6. Gasket
7. Intermediate gearbox
8. Nipple
9. Chip detector assembly
10. Plug

T 120 TO 150 IN-LBS
(13.56 TO 16.95 Nm)

NOTE

1 For helicopters S/N 36694 and subsequent.

ICN-412-A-654001-B-97499-00001-A-001-01

Figure 1. Intermediate Gearbox Electrical Chip Detector (Sheet 1 of 1)

Electrical Chip Detector - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

If the tail rotor gearbox is new or returned from service or repair, remove the plug (10, Figure 1) and discard the packing.

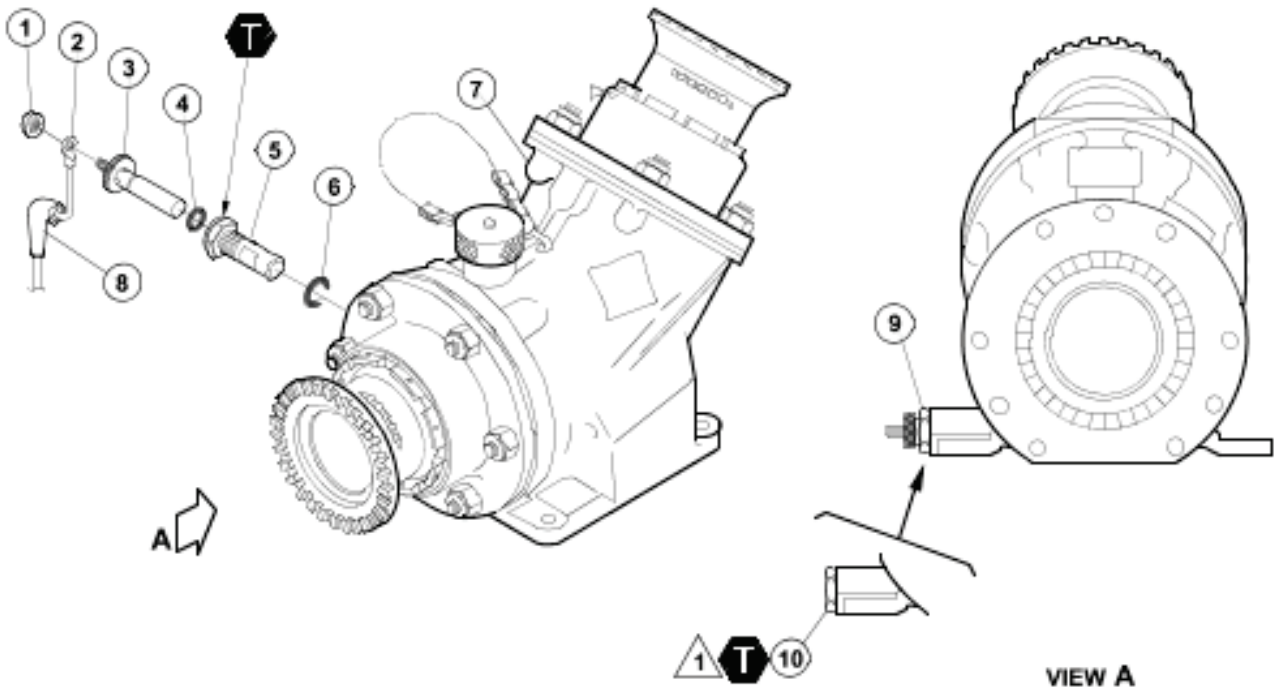
1. Lubricate the new gasket (6) with gearbox lubricating oil (412-A-12-01-00-00A-040A-A / 00141) and install the gasket on the self-closing valve (5).

2. Install the self-closing valve (5) on the intermediate gearbox case (7). Torque the valve 120 to 150 inch-pounds (13.56 to 16.95 Nm). Apply lockwire (C-405) to secure the valve to the gearbox case.

3. Lubricate a new packing (4) with gearbox lubricating oil (412-A-12-01-00-00A-040A-A / 00141) and install the packing on the chip detector element (3). Insert the chip detector into the self-closing valve (5). Push and twist clockwise to engage the locking detents.

4. Position the electrical wire terminal (2) on the chip detector element (3) and install the nut (1). Do not exceed 4 inch-pounds (0.45 Nm) torque on the nut. Position the nipple (8) over the nut.

5. Service the gearbox (412-A-12-10-04-00A-040A-A / 00136) if the self-closing valve (5) was removed.



1. Nut
2. Electrical wire terminal
3. Chip detector element
4. Packing
5. Self-closing valve
6. Gasket
7. Intermediate gearbox
8. Nipple
9. Chip detector assembly
10. Plug

T 120 TO 150 IN-LBS
(13.56 TO 16.95 Nm)

NOTE

1 For helicopters S/N 36694 and subsequent.

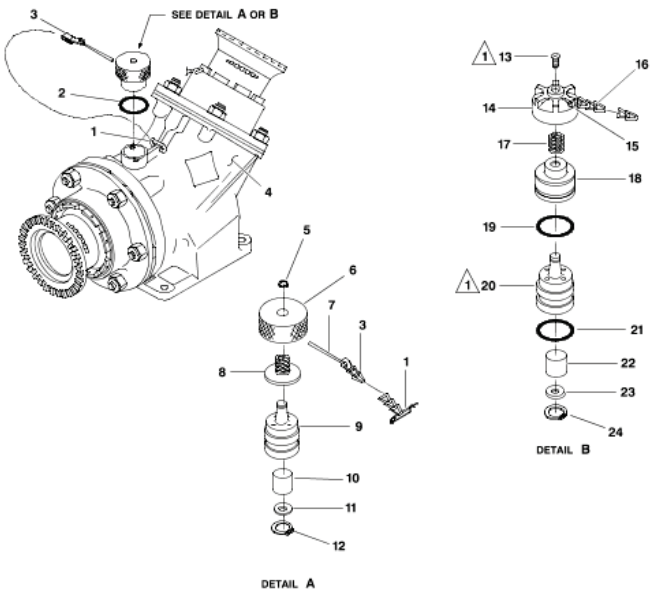
ICN-412-A-654001-B-97499-00001-A-001-01

Figure 1. Intermediate Gearbox Electrical Chip Detector (Sheet 1 of 1)

Oil Filler Cap - Removal

Procedure

1. Push down on the cap (6, Figure 1) and twist to disengage the pin (7) from the gearbox (4).
2. Unsnap the safety pin (1) from the gearbox (4).



NOTE

1 Bonded to cap.

1. Safety pin
2. Packing
3. Chain
4. Gearbox
5. Ring
6. Cap
7. Pin
8. Spring assembly
9. Plug
10. Packing (aluminum wool)
11. Washer
12. Ring (spiralox)
13. Screw
14. Cap
15. Ring
16. Chain
17. Spring
18. Slider
19. Packing
20. Plug
21. Packing
22. Filter (aluminum wool)
23. Washer
24. Ring (retainer)

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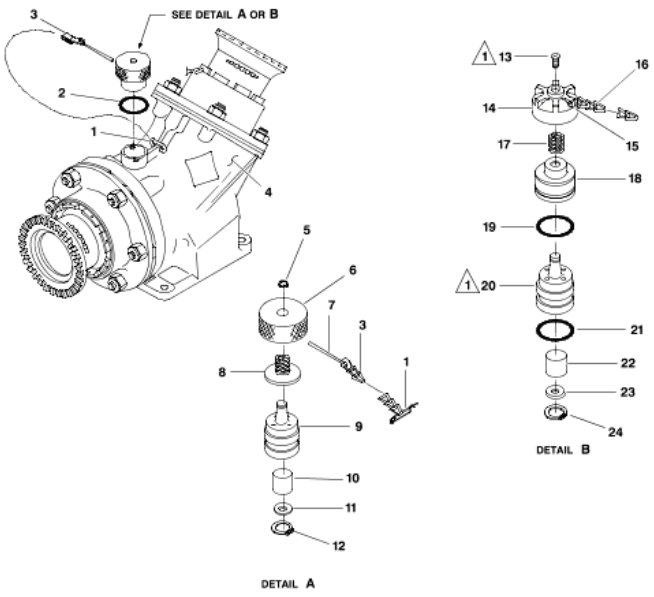
Figure 1. Intermediate Gearbox Oil Filler Cap (Sheet 1 of 1)

Oil Filler Cap - Inspection

Procedure

1. Inspect the cap for distortion of the pin (7, Figure 1) or other damage that may affect function.

2. Invert the filler cap to expose the washer (11). Press the washer approximately 0.06 inch (1.52 mm) and release. The washer should spring back against the ring (12). If the washer does not return to the original position, the aluminum wool packing is dirty, or an insufficient amount of aluminum wool is installed.



NOTE

1 Bonded to cap.

1. Safety pin
2. Packing
3. Chain
4. Gearbox
5. Ring
6. Cap
7. Pin
8. Spring assembly
9. Plug
10. Packing (aluminum wool)
11. Washer
12. Ring (spiralox)
13. Screw
14. Cap
15. Ring
16. Chain
17. Spring
18. Slider
19. Packing
20. Plug
21. Packing
22. Filter (aluminum wool)
23. Washer
24. Ring (retainer)

ICN-412-A-652001-B-97499-00001-A-001-01

Figure 1. Intermediate Gearbox Oil Filler Cap (Sheet 1 of 1)

Oil Filler Cap - Repair

Procedure

1. Repair the intermediate gearbox filler cap assembly using the same procedure outlined for the tail rotor gearbox filler cap (412-A-65-20-01-02A-685B-A / 01316).

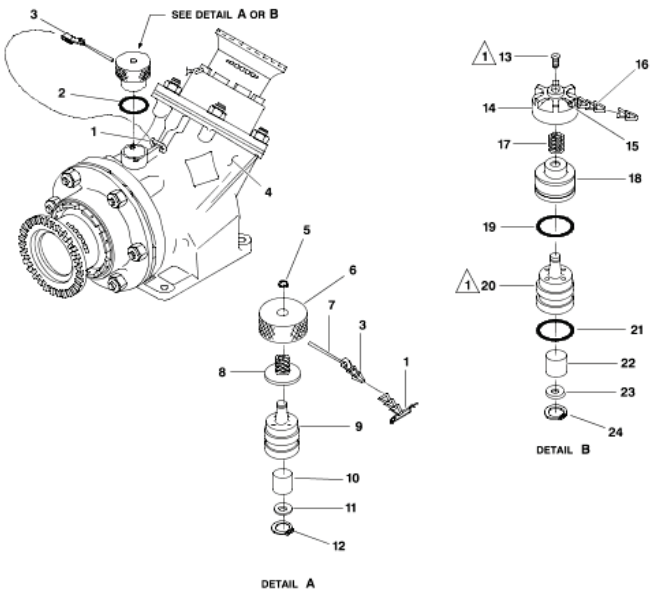
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Oil Filler Cap - Installation

Procedure

1. Position the cap assembly on the gearbox (4, Figure 1). Push down on the cap (6) and twist clockwise to engage the pin (7) in the gearbox adapter.

2. Secure the chain (3) to the gearbox (4) with the safety pin (1).



NOTE

1 Bonded to cap.

1. Safety pin
2. Packing
3. Chain
4. Gearbox
5. Ring
6. Cap
7. Pin
8. Spring assembly
9. Plug
10. Packing (aluminum wool)
11. Washer
12. Ring (spiralox)
13. Screw
14. Cap
15. Ring
16. Chain
17. Spring
18. Slider
19. Packing
20. Plug
21. Packing
22. Filter (aluminum wool)
23. Washer
24. Ring (retainer)

ICN-412-A-652001-B-97499-00001-A-001-01

Figure 1. Intermediate Gearbox Oil Filler Cap (Sheet 1 of 1)

Flexible Couplings - Inspection and Lubrication

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lubricant	C-015	AR	

Procedure

NOTE

SN: 33001-33213 or (SN: 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))):

Refer to Step 1. *through* Step 11. *and* Step 21. *through* Step 25..

1. Remove tail rotor driveshaft sections (1 and 12, Figure 1) attached to intermediate gearbox (412-A-65-10-00-00A-520A-A / 01259).

NOTE

Instructions to lubricate the intermediate gearbox input quill flexible coupling assembly (13) are given. The procedure to lubricate the output quill flexible coupling (10) is the same.

2. While holding coupling seal plate assembly (4) against spring (5), remove retainer ring (3). Remove coupling seal plate assembly and spring.



DO NOT USE SOLVENT TO CLEAN COUPLING SPLINES. SOLVENT MAY LEAVE RESIDUE THAT PREVENTS LUBRICANT FROM ADHERING TO COUPLING.

3. Hold outer coupling (7) at full inboard position and use clean, lint-free cloths to remove all old grease. Clean splines thoroughly.

NOTE

For detailed disassembly, inspection, and assembly instructions, refer to the BHT-412-CR&O.

4. Visually inspect splines of inner coupling (6) and outer coupling (7) for unusual wear patterns, nicks, dents, and cracks. Maximum acceptable wear is 0.005 inch (0.127 mm). Refer to the BHT-412-CR&O for detailed damage limits for coupling splines.

5. Inspect coupling seal plate assembly (4) for damage, which could cause leakage of lubricant.

6. Inspect seal (8) in outer coupling (7) for protrusion or other damage, which could cause leakage of lubricant.

7. Move outer coupling (7) on inner coupling (6) forward and aft with clockwise and counterclockwise preload and feel for roughness. If any roughness or resistance is felt, reinspect coupling splines for damage.

8. Hold outer coupling (7) at full outboard position. Hand pack lubricant (C-015) 0.12 inch (3.5 mm) over top of coupling spline teeth. A workaid may be fabricated to obtain depth of grease (412-A-65-20-01-11A-240A-A / 01318).

9. Hold outer coupling (7) at full outboard position. Install spring (5), coupling seal plate assembly (4), and retainer ring (3).

10. Lubricate output quill flexible coupling (10) in same same manner outlined in Step 2. through Step 9..

11. Install tail rotor driveshaft sections (1 and 12) (412-A-65-10-00-00A-720A-A / 01263). Proceed to Step 21..

NOTE

SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

Refer to Step 12. *through* Step 25..

12. Remove tail rotor driveshaft (15, Figure 2) from gearbox output quill (412-A-65-10-02-00A-520A-A / 01270).

13. Remove retainer ring (14, Figure 1), plate (15), and packing (16). Discard packing.



DO NOT USE SOLVENT TO CLEAN COUPLING SPLINES. SOLVENT MAY LEAVE RESIDUE THAT PREVENTS LUBRICANT FROM ADHERING TO COUPLING.

14. Hold outer coupling (17) at full inboard position and use clean, lint-free cloths to remove all old grease. Clean splines thoroughly.

NOTE

For detailed disassembly, inspection, and assembly instructions, refer to the BHT-412-CR&O.

15. Visually inspect splines of inner coupling (18) for unusual wear patterns, nicks, cracks, or dents. Maximum acceptable wear is 0.005 inch (0.13 mm).

16. Inspect plate (15) for damage that could cause leakage of lubricant.

17. Inspect the nut and seal assembly (19) in outer coupling (17) for damage that could cause leakage of lubricant.

18. Hold outer coupling (17) at full outboard position. Hand pack lubricant (C-015) over top of outer coupling to a depth of 0.20 inch (5.1 mm) over top of spline teeth.

19. Hold outer coupling (17) at full outboard position. Install new packing (16) on plate (15). Install plate and retainer ring (14) in outer coupling.

20. Install tail rotor driveshaft (15, Figure 2) (412-A-65-10-02-00A-720A-A / 01275).

NOTE

Ground run and leak check must be performed following installation of new, repaired, or relubricated flexible couplings.

21. After flexible couplings are lubricated, perform the following test to ensure coupling is not leaking:

21.1. Clean all lubricant from exterior surfaces of coupling and clamp set securing tail rotor driveshaft to coupling. Clean all lubricant from surrounding helicopter structure.

21.2. Operate helicopter for 5 minutes at 100% RPM.

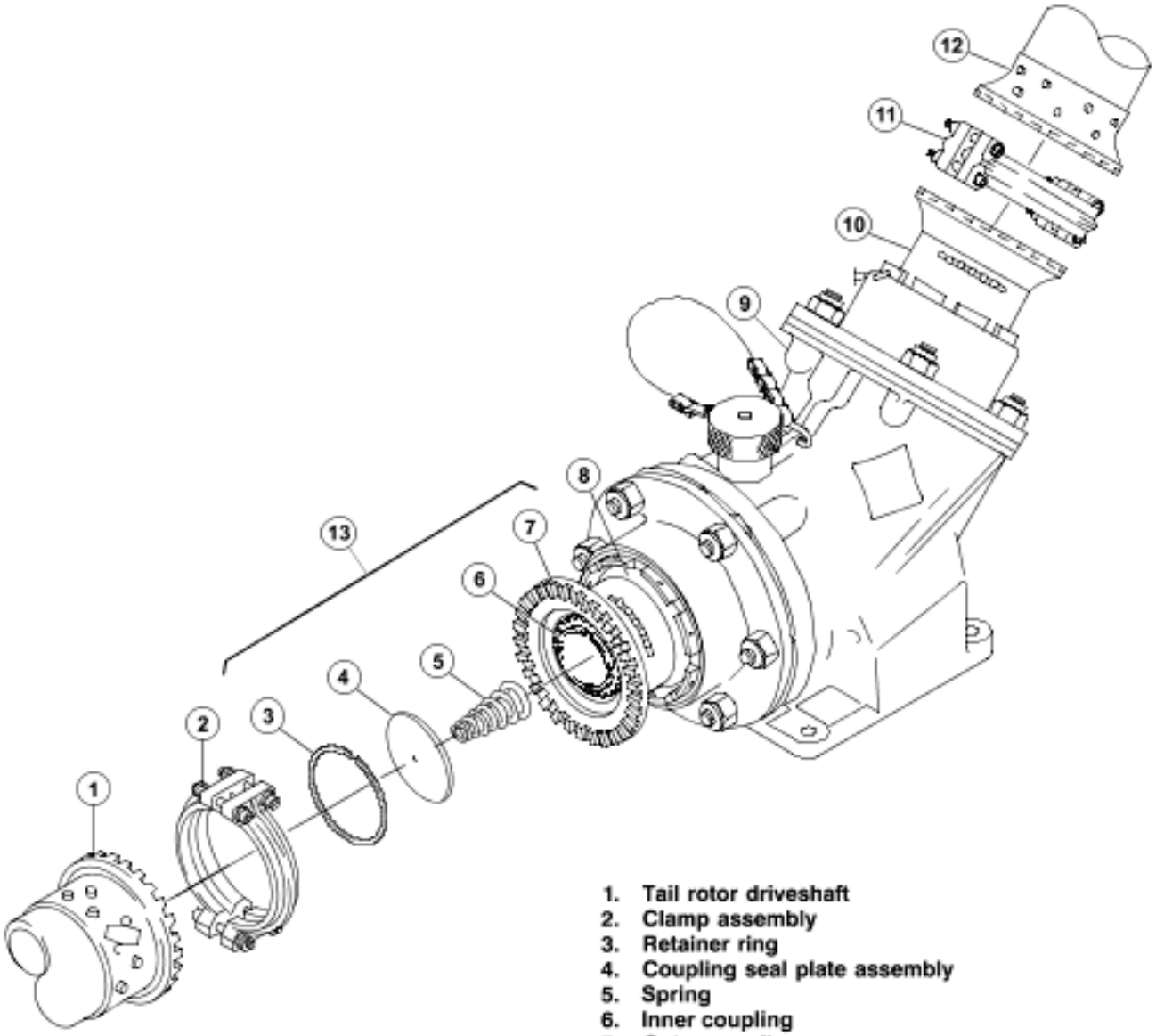
21.3. After shutdown, inspect coupling and surrounding structure for evidence of grease leakage. If there is evidence of grease leakage, remove coupling and replace defective parts. Repeat leak test run.

22. Inspect input quill coupling and surrounding area for grease leakage (BHT-412-CR&O). If there is evidence of grease leakage, remove coupling and replace defective parts. Repeat leak test run.

23. Check visual overtemperature indicators (TEMP-PLATES) for discoloration and overheat condition. A change in color of TEMP-PLATES dots (from white to black) indicates a possible overheat condition and/or component degradation. The cause of overtemperature indicator discoloration shall be determined and corrected prior to continued operation (412-A-65-10-01-00A-280A-A / 01266).

24. Replace visual overtemperature indicators if mutilated or discolored (BHT-412-CR&O).

25. Make an entry in helicopter log and in flexible couplings lubrication log of date lubricated, date grease manufactured, and helicopter hours (412-A-12-01-00-00A-240B-A / 00143).



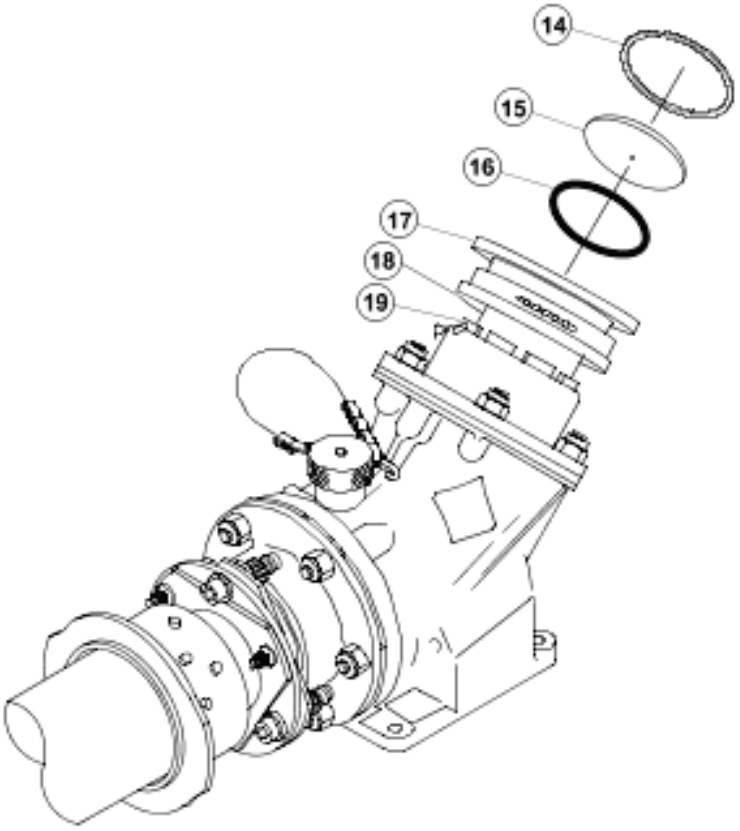
1. Tail rotor driveshaft
2. Clamp assembly
3. Retainer ring
4. Coupling seal plate assembly
5. Spring
6. Inner coupling
7. Outer coupling
8. Seal
9. Intermediate gearbox
10. Output quill flexible coupling
11. Clamp assembly
12. Tail rotor driveshaft
13. Input quill flexible coupling assembly

NOTE

Applicable to helicopters S/N 33001 through 33213 and 36001 through 36019 not modified by 412-570-001-103.

ICN-412-A-652001-B-97499-00002-A-001-01

Figure 1. Intermediate Gearbox Flexible Couplings (Sheet 1 of 2)



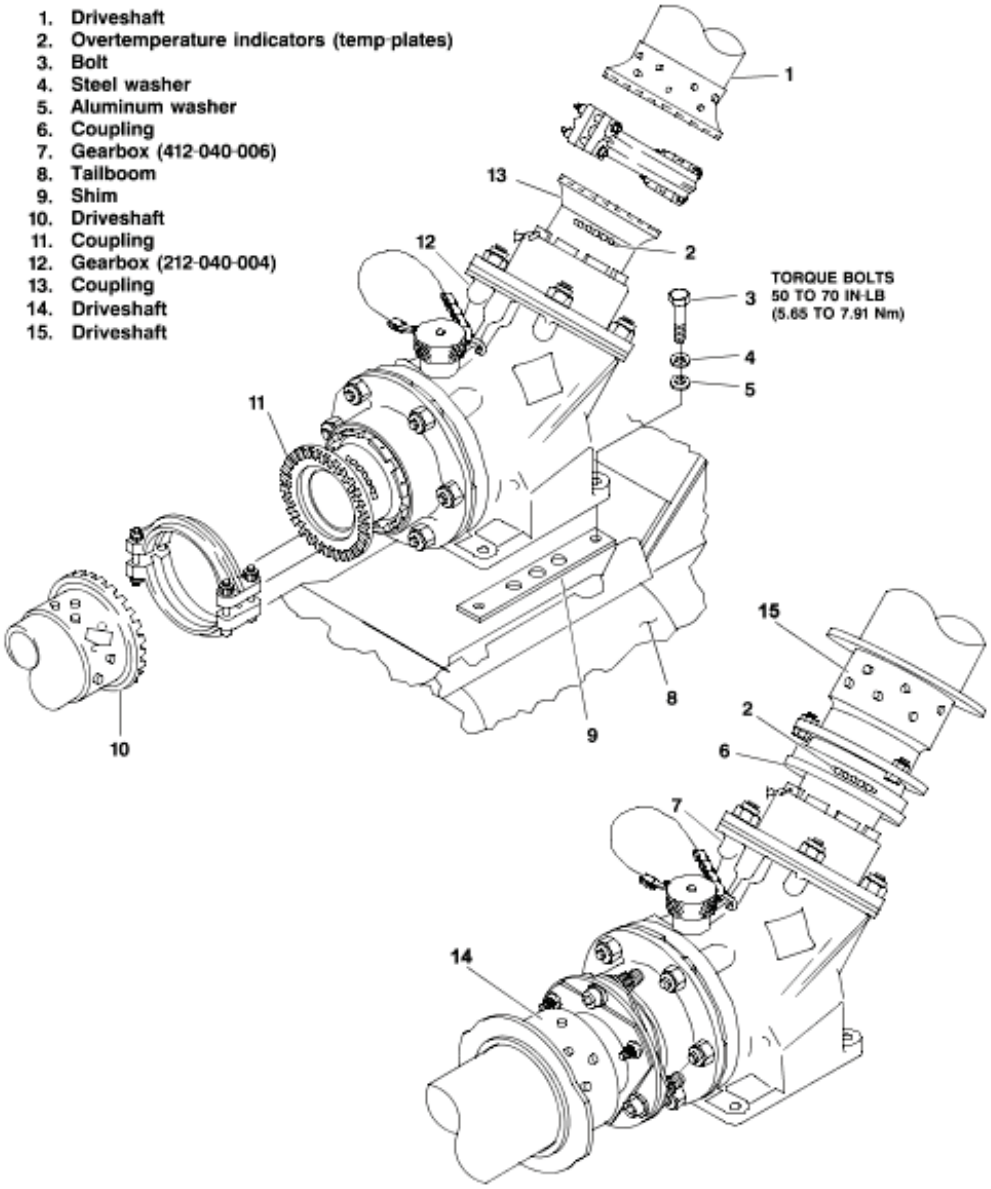
14. Retainer ring
15. Plate
16. Packing
17. Outer coupling
18. Inner coupling
19. Nut and seal assembly

NOTE

Applicable to helicopters S/N 36020 and subsequent and helicopters modified by 412-570-001-103.

ICN-412-A-652001-B-97499-00003-A-001-01

Figure 1. Intermediate Gearbox Flexible Couplings (Sheet 2 of 2)



ICN-412-A-651002-B-97499-00002-A-001-01

Figure 2. Intermediate Gearbox (Sheet 1 of 1)

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Tail Rotor Gearbox - Maintenance Procedures

Procedure

1. *For the maintenance procedures consisting of troubleshooting, inspection (installed and removed from helicopter and conditional inspection), removal, cleaning, repair and installation, refer to the below table:*

Troubleshooting	412-A-65-20-01-00A-422A-A / 01299
Inspection (Installed)	412-A-65-20-01-00A-280A-A / 01301
Inspection (Not Installed)	412-A-65-20-01-00A-280B-A / 01303
Inspection (Conditional)	412-A-65-20-01-00A-280C-A / 01304
Removal	412-A-65-20-01-00A-520A-A / 01302
Cleaning	412-A-65-20-01-00A-251A-A / 01300
Repair	412-A-65-20-01-00A-685A-A / 01305
Installation	412-A-65-20-01-00A-720A-A / 01306

2. *When any doubt exists as to the serviceability of a gearbox, perform serviceability checks defined in* 412-A-63-00-00-00A-028B-A / 01202*.*

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Tail Rotor Gearbox - Troubleshooting

Procedure

1. Refer to Table 1.

Table 1. Tail Rotor Gearbox - Troubleshooting

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
CHIP 42/90 BOX caution segment illuminated.	Metal chips have accumulated on chip detector.	Remove electric chip detector and inspect for metal chips. If chips are found, retain them in a clean container for subsequent investigation (412-A-65-40-01-00A-520B-A / 01310, 412-A-65-40-01-00A-280B-A / 01311, 412-A-65-40-01-00A-685B-A / 01312 and 412-A-65-40-01-00A-720B-A / 01313 and 412-A-63-00-00-00A-009A-A / 00987).
	Faulty wiring or faulty chip detector.	Check wiring to electric chip detector for continuity and grounding 412-A-98-00-00-21B-051A-A / 02581. Repair faulty wiring or replace electric chip detector as required. (412-A-96-03-01-13A-320A-A / 02059).
Oil leakage from tail rotor gearbox.	Faulty seals, packings, or wear sleeves.	Clean gearbox and recheck for oil leak, both static and dynamic (helicopter running). Measure and record rate of leakage. Maximum allowable static or dynamic leakage at input drive quill is two drops per minute. Total allowable leakage from all sources on gearbox is six drops per minute.
		Replace gearbox seals, packings, or wear sleeves if maximum allowable leakage rate is exceeded (BHT-412-CR&O).
	Cracked gearbox housing.	Inspect gearbox for cracks in housing. Replace gearbox if cracks are found.

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Tail Rotor Gearbox - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean gearbox with clean cloths damp with drycleaning solvent (C-304).



DO NOT FORCE SOLVENT AND/OR DIRT INTO FLEXIBLE COUPLING WHEN USING COMPRESSED AIR.

2. Dry with filtered, compressed air.

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Tail Rotor Gearbox - Inspection (Installed)

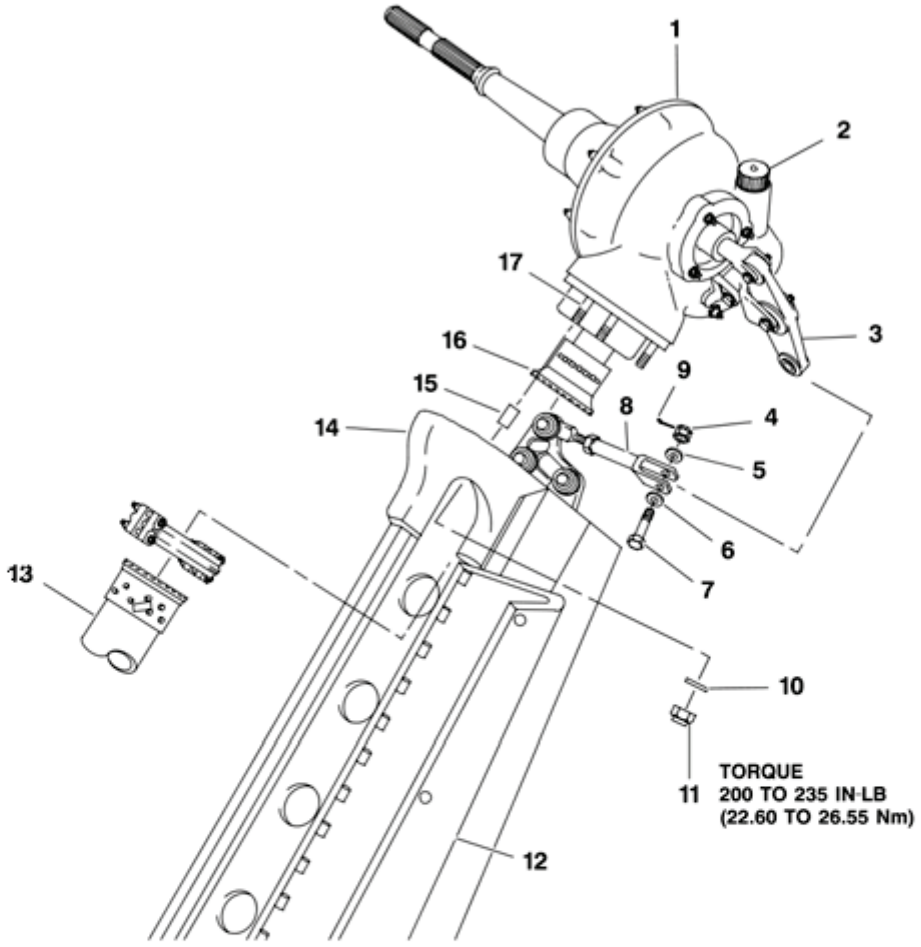
Procedure

1. Open vertical fin cover (12, Figure 1).
2. Shake tail rotor gearbox (1) and check for looseness on tail rotor gearbox support fitting (14).
3. Inspect for evidence of fretting corrosion at mating surface between gearbox (1) and support fitting (14) that could be caused by movement of gearbox on support. A gray residue is an indication of fretting corrosion. If residue is present, remove gearbox and inspect in accordance with BHT-412-CR&O and inspect support fitting in accordance with 412-A-53-50-00-00A-280A-A / 00913.

NOTE

Inspection Step 4. through Step 9. are not applicable if gearbox is removed at Step 3..

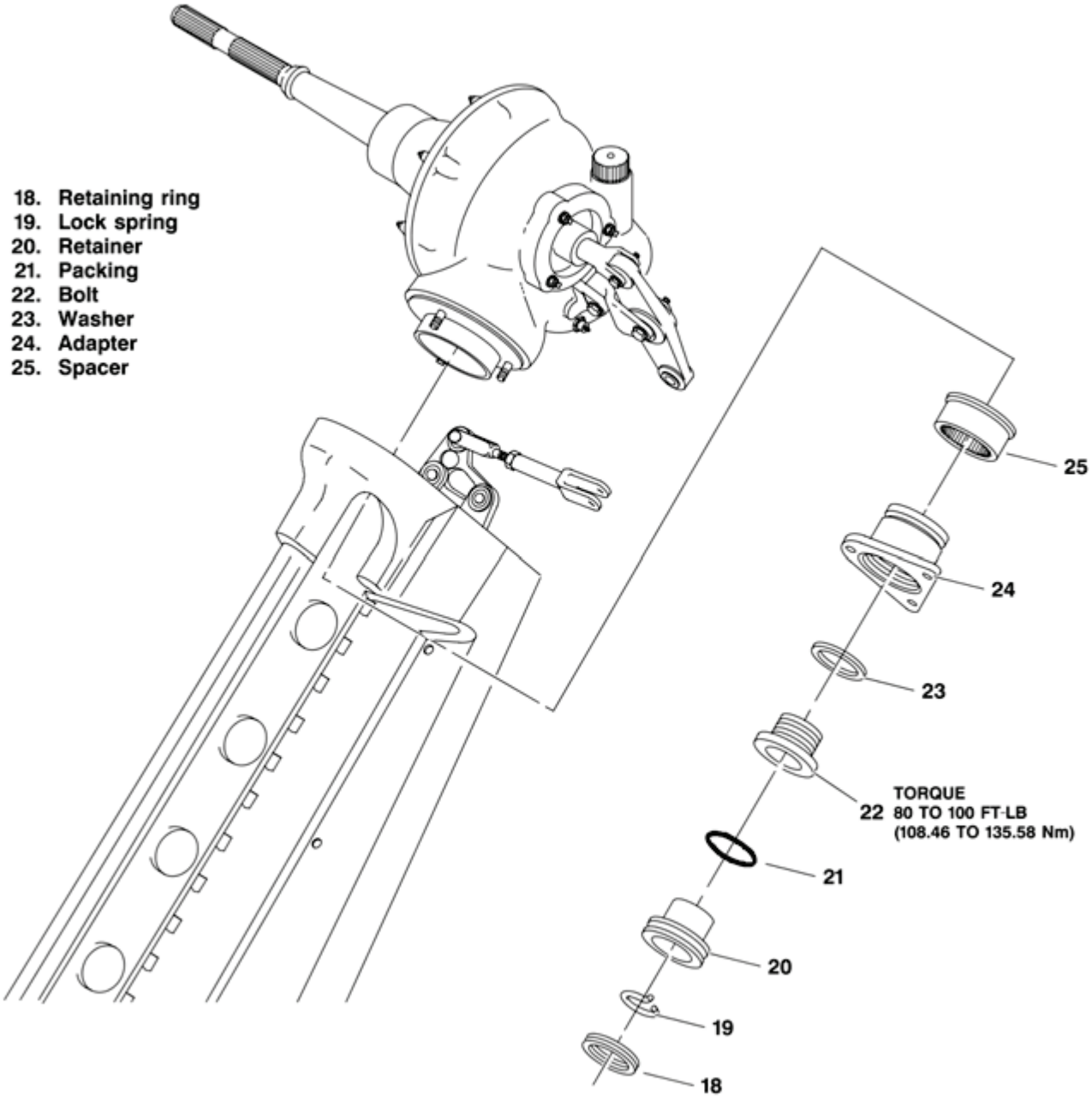
4. Inspect six nuts (11) for correct torque and thread engagement (412-A-65-20-01-00A-720A-A / 01306).
5. Inspect link assembly (8) and lever (3) for secure installation.
6. Inspect coupling (16) for grease leakage and visual overtemperature indicators (TEMP-PLATES) for discoloration and overheat condition. A change in color of TEMP-PLATES dots (from white to black) will indicate a possible overheat condition and/or component degradation. The cause of indicators discoloration shall be determined and corrected prior to continued operation (412-A-65-10-01-00A-280A-A / 01266).
7. Inspect oil filler cap assembly (2) for damage and for correct amount of aluminum wool (412-A-65-20-01-02A-280B-A / 01315).
8. Inspect oil sight glass for correct oil level. Inspect sight glass and indicator for staining and for cracks and crazing. Damage that could cause oil leakage or make oil level difficult to determine is not acceptable.
9. Inspect electrical chip detector for metal particles (412-A-65-40-01-00A-280B-A / 01311). Refer to 412-A-63-00-00-00A-280B-A / 01203 for serviceability check required if metal particles are found.



1. Tail rotor gearbox
2. Cap assembly
3. Lever
4. Nut
5. Washer
6. Washer
7. Bolt
8. Link assembly
9. Cotter pin
10. Thin steel washer
11. Nut
12. Vertical fin cover
13. Driveshaft
14. Support fitting
15. Spacer (NAS 43DD6-49) (2 reqd - removed prior to installation)
16. Coupling
17. Stud

ICN-412-A-652001-B-97499-00004-A-001-01

Figure 1. Tail Rotor Gearbox (Sheet 1 of 2)



ICN-412-A-652001-B-97499-00005-A-001-01

Figure 1. Tail Rotor Gearbox (Sheet 2 of 2)

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Tail Rotor Gearbox - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Drain Line	DB 3703-30	1	
Tail Rotor Gearbox Pusher	T101264-103	1	
SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: Holding Fixture	*412-240-008-101*	*1*	*(412Classic/SP to 412HP Upgrade)*

Procedure

1. When tail rotor gearbox is to be replaced, unless conditions prevent operation, perform a 10 minute ground run-up then drain operating oil using drain line (DB 3703-30). If run-up is not practical, remove tail rotor gearbox and flush with new lubricating oil of same type used in gearbox. Attach tag to tail rotor gearbox stating: "PRESERVED WITH OPERATING LUBRICANT".

2. Remove tail rotor assembly (412-A-64-00-00-00A-520A-A / 01229).

3. Remove the cotter pin (9, Figure 1), nut (4), and washer (5). Remove the bolt (7) and washer (6) and disconnect the link assembly (8) from the lever (3).

4. Disconnect the electrical wire and remove the chip detector (412-A-65-40-01-00A-520B-A / 01310).

5. Open the vertical fin cover (12) and remove the intermediate gearbox fairing.

6. Remove the driveshaft (13) (412-A-65-10-00-00A-520A-A / 01259 or 412-A-65-10-02-00A-520A-A / 01270).

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

7. *The tail rotor gearbox adapter (24) must be removed to permit removal of the gearbox. Proceed as follows:*

7.1. *Remove the retaining ring (18) and the lock spring (19).*

7.2. *Insert the 1/4 inch X 20 UNF bolt into the retainer (20) and remove the retainer with the packing (21). Discard the packing.*

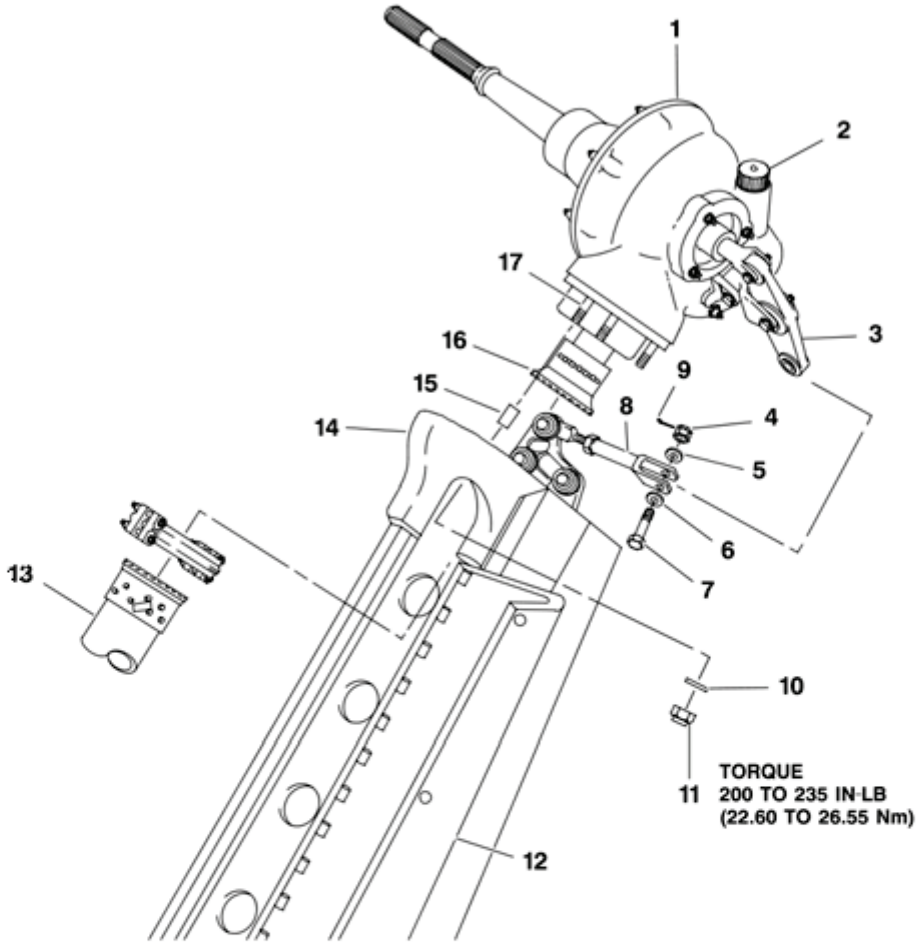
7.3. *While holding the adapter with the holding fixture (412-240-008-101), loosen the bolt (22). Remove the bolt, washer (23), and adapter (24) from the gearbox.*

8. Remove the nuts (11) with the thin steel washers (10).

9. Remove the gearbox (1) from the support fitting (14) using the tail rotor gearbox pusher (T101264-103) (Figure 2).

NOTE

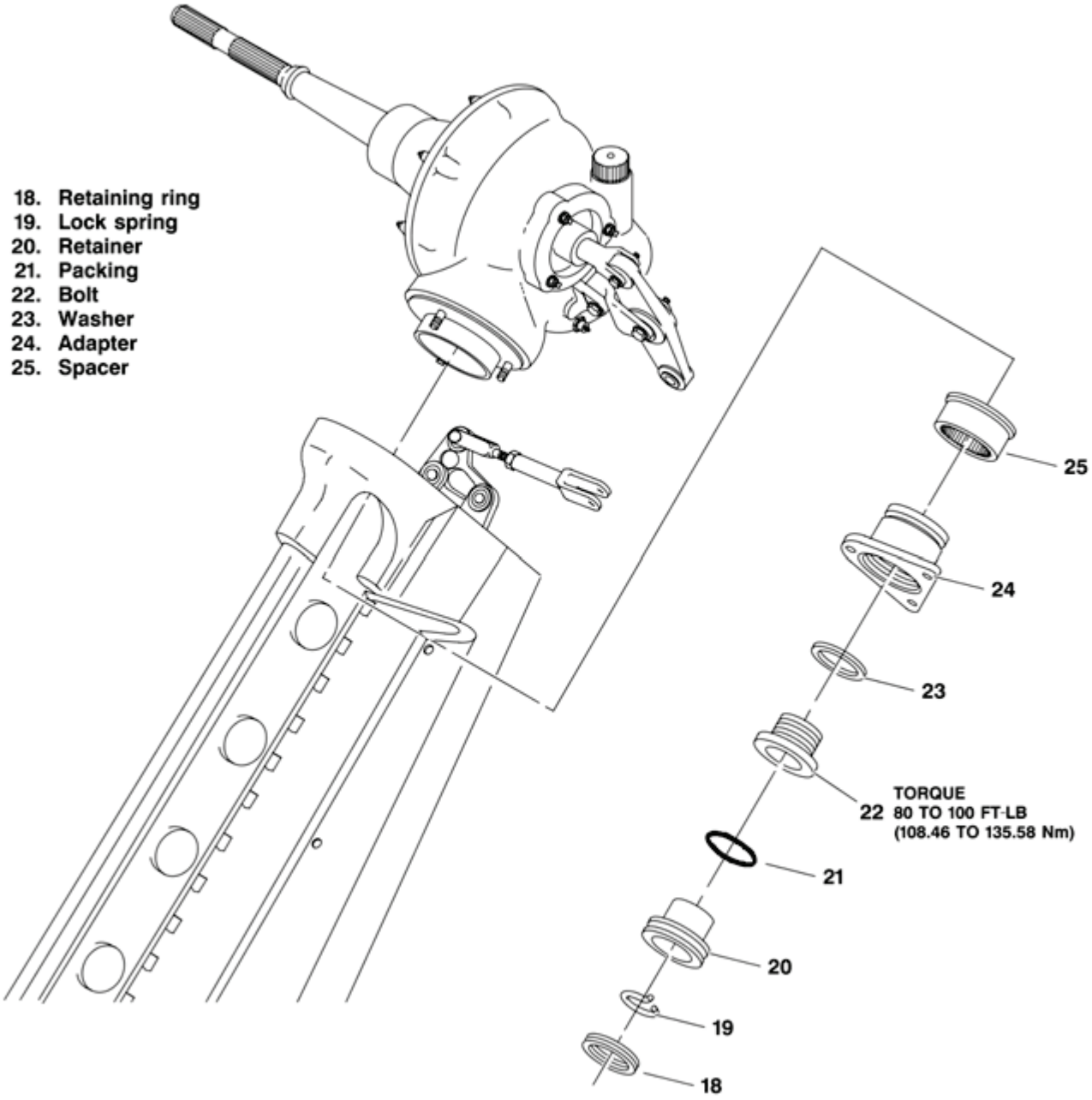
Install two spacers (15, Figure 1) and nut (11) with thin steel washer (10) on gearbox studs (17) to hold input quill in tail rotor gearbox (1) while gearbox is removed from helicopter.



1. Tail rotor gearbox
2. Cap assembly
3. Lever
4. Nut
5. Washer
6. Washer
7. Bolt
8. Link assembly
9. Cotter pin
10. Thin steel washer
11. Nut
12. Vertical fin cover
13. Driveshaft
14. Support fitting
15. Spacer (NAS 43DD6-49) (2 reqd - removed prior to installation)
16. Coupling
17. Stud

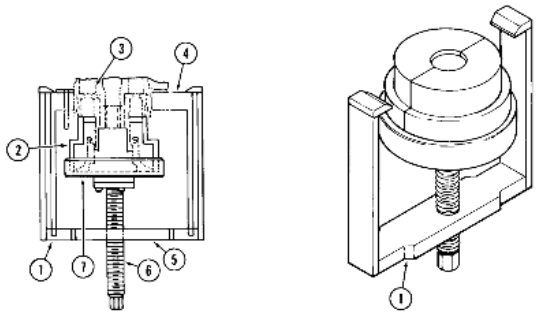
ICN-412-A-652001-B-97499-00004-A-001-01

Figure 1. Tail Rotor Gearbox (Sheet 1 of 2)



ICN-412-A-652001-B-97499-00005-A-001-01

Figure 1. Tail Rotor Gearbox (Sheet 2 of 2)



1. Tail rotor gearbox pusher (T101264-103)
2. Adapter assembly (T101264-109)
3. Tail rotor gearbox assembly
4. Tail rotor gearbox support
5. Clamp assembly (T101264-107)
6. Screw Assembly (T101264-105)
7. Adapter retainer (T101264-121)

NOTES:

1. If not previously accomplished, thread screw assembly (6) into clamp assembly (5). Allow sufficient clearance for positioning on support (4) and adapter assembly (2).
2. Position both halves of adapter assembly (2) on tail rotor gearbox (3) against input quill as shown.
3. Position tail rotor gearbox pusher (1) on gearbox and support with hooks of clamp assembly (5) centered securely over shoulders of support (4) and adapter retainer (7) over both halves of adapter assembly (2).
4. Tighten screw assembly (6) clockwise with a suitable wrench until tail rotor gearbox (3) separates from support (4). Remove tools and lift gearbox from support.

ICN-412-A-652001-B-97499-00006-A-001-01

Figure 2. Tail Rotor Gearbox Pusher (T101264-103) - Tool Application (Sheet 1 of 1)

Tail Rotor Gearbox - Inspection (Not Installed)

Procedure

1. Inspect the gearbox in accordance with the BHT-412-CR&O.

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Tail Rotor Gearbox - Inspection (Conditional)

Procedure

1. For conditional inspection of the tail rotor gearbox, refer to the BHT-412-CR&O.

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Tail Rotor Gearbox - Repair

Procedure

1. Polish out mechanical and corrosion damage within limits (BHT-412-CR&O).
2. Repair defective filler cap (412-A-65-20-01-02A-685B-A / 01316).
3. Replace defective sight glass (412-A-65-20-01-08A-520B-A / 01308 and 412-A-65-20-01-08A-720B-A / 01309).
4. Replace the chip detector and/or self-closing valve that failed to pass inspection (412-A-65-40-01-00A-520B-A / 01310, 412-A-65-40-01-00A-280B-A / 01311, 412-A-65-40-01-00A-685B-A / 01312 and 412-A-65-40-01-00A-720B-A / 01313).

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Tail Rotor Gearbox - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: Holding Fixture	*412-240-008-101*	*1*	*(412Classic/SP to 412HP Upgrade)*

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	
Sealant	C-251	AR	
Sealant	C-308	AR	
MEK	C-309	AR	

Procedure



SN: 33001-33213 or 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)):

PRIOR TO INSTALLATION OF TAIL ROTOR GEARBOX, ACCOMPLISH THE FOLLOWING:

ENSURE GEARBOX FLEXIBLE COUPLING IS PROPERLY LUBRICATED (BHT-412-CR&O).

ROTATE FLEXIBLE COUPLING AND CHECK FOR BINDING AND FOR PRESENCE OF BACKLASH BETWEEN PINION ON INPUT QUILL AND GEAR ON OUTPUT SHAFT.

DRAIN PRESERVATIVE OIL (IF ANY) FROM GEARBOX AND SERVICE WITH APPROVED OIL (412-A-12-10-04-00A-292A-A / 00137).

1. Check the mounting flange of the tail rotor gearbox (1, Figure 1) for sealant protruding from the three jackscrew holes. Trim off sealant flush to allow proper seating of the tail rotor gearbox flange on the support fitting (14).

2. Remove the nuts (11), spacers (15), and thin steel washers (10) from the studs (17) around the input drive quill.

3. Using MEK (C-309) and a non-metallic scraper, clean the mating surfaces of the tail rotor gearbox (1) and tailboom support fitting (14).

4. Coat all faying surfaces of the gearbox, studs, tailboom mount fitting, holes for studs in the mount, and washers with epoxy polyamide primer (C-204). Do not apply primer to the threads of the studs. Install the gearbox while primer is wet.



MAKE SURE A MINIMUM OF TWO THREAD PITCHES, INCLUDING CHAMFER, EXTEND THROUGH NUTS (11). ENSURE NUTS DO NOT BOTTOM ON GRIP PORTION OF STUDS. USE ADDITIONAL THIN STEEL WASHERS (10) IF REQUIRED.

NOTE

Perform special inspection after installation (Between 1 and 5 Hours of Flight After Each Installation) of tail rotor gearbox (412-A-05-47-00-32A-283A-A / 00033.1).

5. Position the tail rotor gearbox (1) on the support fitting (14) with the coupling (16) and the studs (17) through the holes in the support fitting.

6. Install the thin steel washers (10) and nuts (11). Torque evenly in a star pattern 200 to 235 inch-pounds (22.60 to 26.55 Nm). Repeat the torque pattern until all nuts retain torque initially applied to first the nut in the pattern. The torque value of the first nut will decrease as other nuts are torqued. Use a 0.005 inch (0.127 mm) feeler gauge to verify that no gap exists between the shim plate installed between the gearbox input quill and the gearbox.

7. Apply a fillet of sealant (C-251) around the periphery of the mating surface of the tail rotor gearbox (1) and support fitting (14).

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

8. *Install the adapter (24) to the tail rotor gearbox input pinion as follows:*



DO NOT ALLOW MEK (C-309) TO CONTACT LIP OF SEAL.

8.1. *Clean the inboard end of the adapter (24) and the mating face of the spacer (25) with a cotton swab dampened with MEK (C-309).*

8.2. *Install the adapter (24), washer (23), and bolt (22). Tighten the bolt to where a gap of approximately 0.10 inch (2.54 mm) remains between the inboard end of the adapter and the mating face of the spacer (25).*

8.3. *Apply a light coat of sealant (C-308) faying surfaces of the adapter (24) and the spacer (25) and fill the three slots in the inboard end of the adapter with sealant.*

8.4. *Within 5 minutes after sealant application, torque the bolt (22) 80 to 100 foot-pounds (108.46 to 135.58 Nm).*

8.5. *Wipe all sealant from the outside diameter of the adapter (24) and spacer (25). No sealant residue is permitted on the lip of the seal or the mating wear sleeve.*

8.6. *Install the retainer (20), with new packing (21), so the square extension engages the bolt (22).*

8.7. *Install the lock spring (19) in the groove of the retainer (20) so the tang extends through a hole in the retainer and engages a slot in the adapter (24).*

8.8. *Install the retaining ring (18).*

8.9. *Install the driveshaft (13) (412-A-65-10-02-00A-720A-A / 01275).*

9. Connect the link assembly (8) to the lever (3). Install the bolt (7) with the washer (6) under the bolt head and the washer (5) under the nut (4). Additional washers (5) may be used to align the nut for cotter pin installation. Install the cotter pin (9).

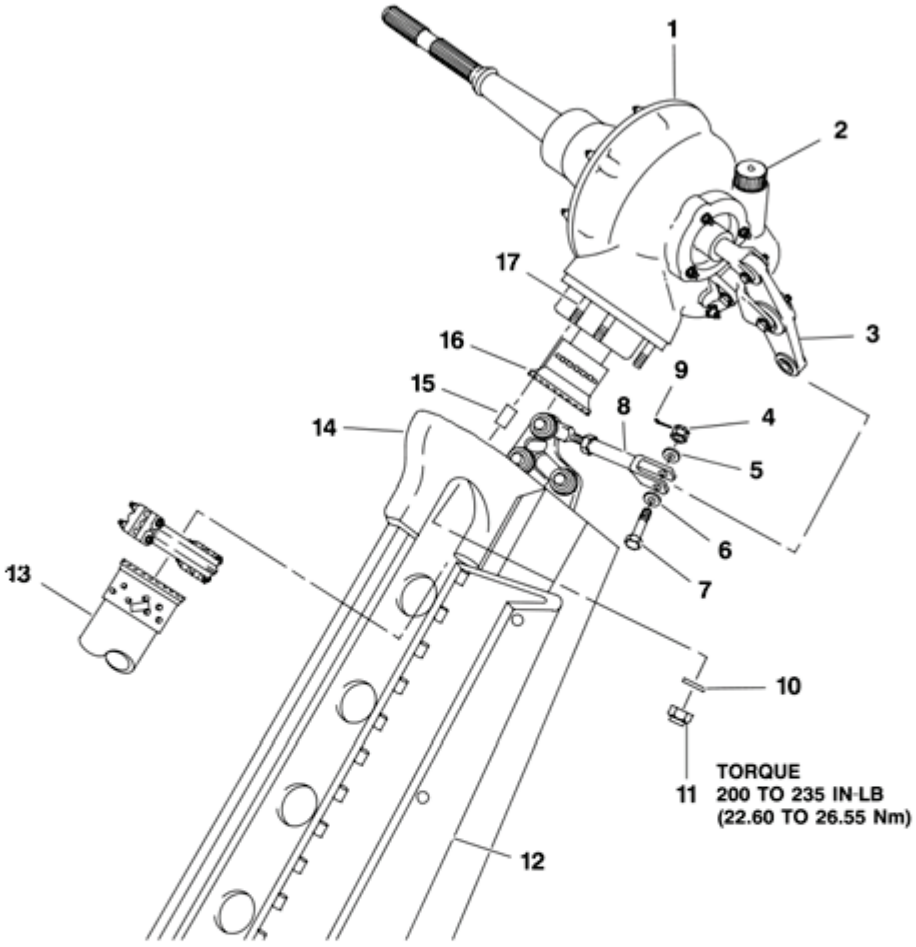
10. Install the chip detector (412-A-65-40-01-00A-720B-A / 01313).

11. Verify actual presence of oil in the sight gauge and check oil level to the full mark on the indicator.

12. Close the vertical fin cover (12) and install the intermediate gearbox fairing.

13. Install and rig the tail rotor (412-A-67-01-00-11A-276A-A / 01551).

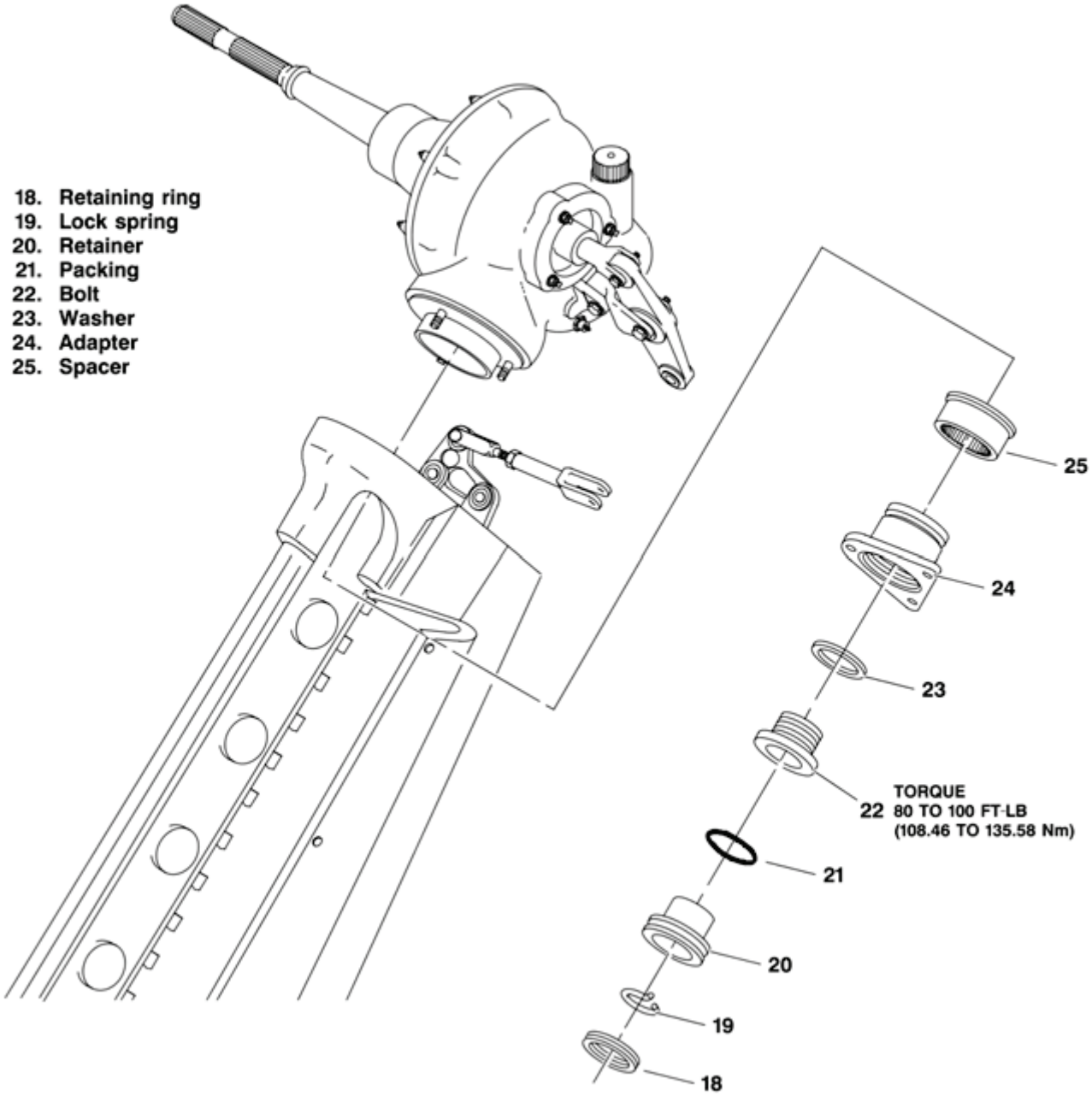
14. If the gearbox has not been previously qualified on the run-in test stand, accomplish the alternate run-in and test (412-A-65-20-01-00A-322A-A / 01307).



1. Tail rotor gearbox
2. Cap assembly
3. Lever
4. Nut
5. Washer
6. Washer
7. Bolt
8. Link assembly
9. Cotter pin
10. Thin steel washer
11. Nut
12. Vertical fin cover
13. Driveshaft
14. Support fitting
15. Spacer (NAS 43DD6-49) (2 reqd - removed prior to installation)
16. Coupling
17. Stud

ICN-412-A-652001-B-97499-00004-A-001-01

Figure 1. Tail Rotor Gearbox (Sheet 1 of 2)



ICN-412-A-652001-B-97499-00005-A-001-01

Figure 1. Tail Rotor Gearbox (Sheet 2 of 2)

Tail Rotor Gearbox - Alternate Run-In Test

Procedure

NOTE

It is desirable that tail rotor gearbox be test run and inspected after overhaul in accordance with publication furnished with run-in stand for qualification for flight. In the event a run-in stand is not available, the following function test may be accomplished as a minimum acceptable alternate test.

1. Install sufficient ballast in the cabin to obtain gross weight BHT-412-FM.

2. Start the engines and run at 60% RPM for 5 minutes.

3. Gradually increase RPM to 88% at the rate of 7% each 6 minutes.

4. Hold at 88% for 2 minutes with collective pitch applied to the maximum up position that can be maintained without becoming airborne or exceeding 100% torque.

5. Shut down the engines.

6. Remove, inspect, clean, and reinstall the chip detector.

7. Start the engines and gradually increase to 88% RPM. Run at 88% RPM for 6 minutes. Apply collective pitch to the maximum up position that can be maintained without becoming airborne or exceeding 100% torque. Gradually increase RPM to 100%, without becoming airborne or exceeding 100% torque at the rate of 3% each 6 minutes.

8. Run at 100% for 30 minutes with collective pitch in the maximum up position that can be maintained without becoming airborne or exceeding 100% torque.

9. Shut down the engines and drain oil from the gearbox. Check the gearbox electrical chip detector for an excessive amount of metal particles, which may indicate internal failure. Refer to 412-A-65-00-00-00A-009A-A / 01257 for required action if metal particles are found.

10. Inspect the tail rotor gearbox input quill coupling and surrounding area for grease leakage. If there is evidence of grease leakage, remove the coupling and replace defective parts (BHT-412-CR&O). Verify the servicing of the gearbox (412-A-12-10-04-00A-040A-A / 00136). Repeat the leak test run.

11. Check the visual overtemperature indicators (TEMP-PLATES) for discoloration and overheat condition. A change in color of TEMP-PLATES dots (from white to black) indicates a possible overheat condition and/or component degradation. The cause of dots discoloration shall be determined and corrected prior to continued operation (412-A-65-10-01-00A-280A-A / 01266).

12. Replace the visual overtemperature indicators if mutilated or discolored (BHT-412-CR&O).

13. Install the electrical chip detector (412-A-65-40-01-00A-720B-A / 01313) and service gearbox (412-A-12-10-04-00A-040A-A / 00136).

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Sight Glass - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Drain Line	DB 3703-30	1	

Procedure

1. Remove the electrical chip detector (6, Figure 1) (412-A-65-40-01-00A-520B-A / 01310) from the tail rotor gearbox (1).

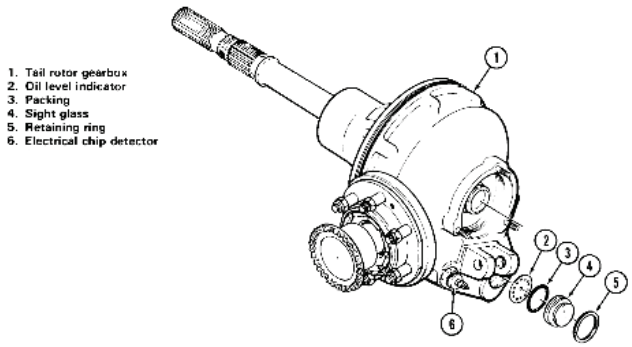
2. Insert one end of the drain line (DB 3703-30) into a container. Insert one end in the hole where the electrical chip detector (6) was installed. Oil will automatically drain.

3. Remove the retaining ring (5).

4. Remove the sight glass (4) and packing (3). Discard the packing.

5. Remove the oil level indicator (2).

6. Remove the drain line (DB 3703-30).



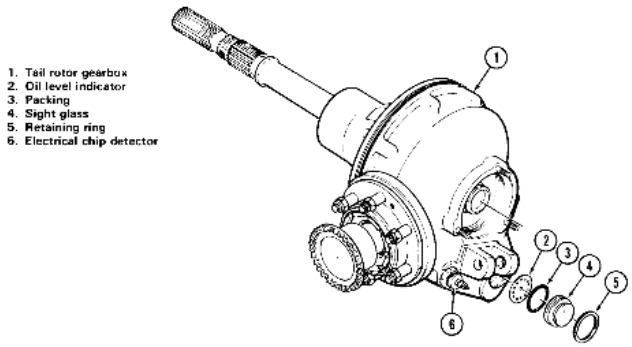
ICN-412-A-652001-B-97499-00007-A-001-01

Figure 1. Tail Rotor Gearbox Sight Glass (Sheet 1 of 1)

Sight Glass - Installation

Procedure

1. Install the oil level indicator (2, Figure 1).
2. Lubricate a new packing (3) with gearbox oil (412-A-12-01-00-00A-040A-A / 00141) and install the packing on the sight glass (4). Install the sight glass.
3. Install the retaining ring (5).
4. Install the electrical chip detector (6) (412-A-65-40-01-00A-720B-A / 01313).
5. Service the gearbox (412-A-12-10-04-00A-040A-A / 00136).



ICN-412-A-652001-B-97499-00007-A-001-01

Figure 1. Tail Rotor Gearbox Sight Glass (Sheet 1 of 1)

Electrical Chip Detector - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Drain Line	DB 3703-30	1	

Procedure

NOTE

The electrical chip detector element (5, Figure 1) may be removed for inspection without draining gearbox oil.

1. Move the nipple (8) back on the electrical wire (6) to expose the nut (7). Remove the nut and electrical wire.

2. Press in on the knurled body of the electrical chip detector element (5), turn counterclockwise to disengage the locking mechanism, and withdraw the element from the electrical chip detector self-closing valve (3).

NOTE

If the self-closing valve (3) is to be removed, drain gearbox oil into a container with the drain line (DB 3703-30) installed in the valve.

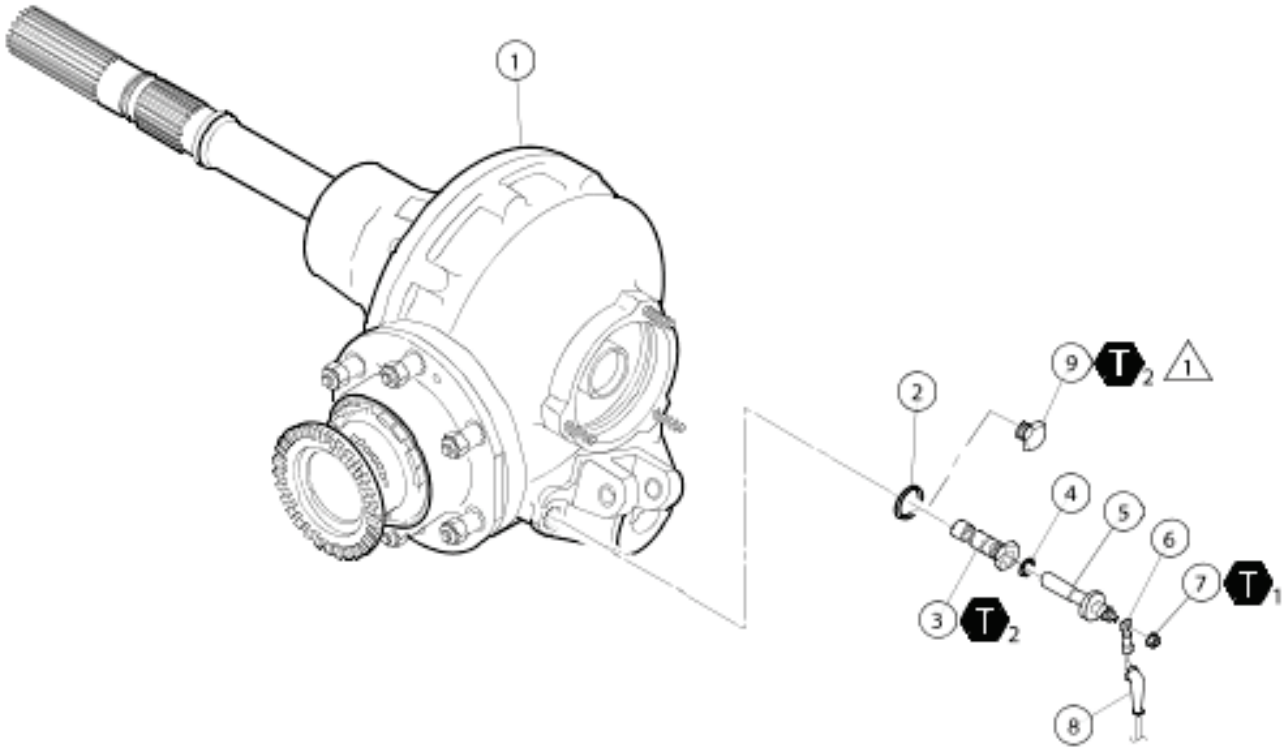
3. Cut and remove the lockwire from the electrical chip detector self-closing valve (3). Remove the valve. Remove the packing (2) from the valve.

NOTE

If the tail rotor gearbox will not be reinstalled in a timely manner, or shipped for service or repair, do the following steps.

4. Lubricate a new packing (2) with gearbox oil (412-A-12-01-00-00A-040A-A / 00141) and install on the plug (9).

5. Install the plug (9) into the gearbox. Torque the plug 120 to 150 inch-pounds (13.56 to 16.95 Nm).



1. Gearbox
2. Packing
3. Electrical chip detector self-closing valve
4. Packing
5. Electrical chip detector element
6. Electrical wire
7. Nut
8. Nipple
9. Plug

T1 4 IN-LBS (0.45 Nm)

T2 120 TO 150 IN-LBS (13.56 TO 16.95 Nm)

NOTE

1 For helicopters S/N 36694 and subsequent.

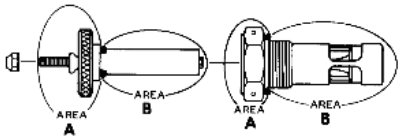
ICN-412-A-654001-B-97499-00002-A-001-01

Figure 1. Tail Rotor Gearbox Electrical Chip Detector (Sheet 1 of 1)

Electrical Chip Detector - Inspection

Procedure

1. Inspect the electrical chip detector for metal particles (412-A-63-00-00-00A-009A-A / 00987).
2. Inspect the electrical chip detector components for damage (Figure 1).
3. Perform an operational check on the electrical chip detector (412-A-96-00-00-00A-009A-A / 01870).



AREA	LIMITS
All	No cracks allowed.
A	Maximum depth of pitting is 0.030 inch (0.762 millimeter) with no more than 40 percent of any 1.0 inch (25.4 millimeters) square or 20 percent of total area of any pitted surface.
B	Maximum depth of pitting is 0.020 inch (0.508 millimeter) with no more than 40 percent of any 1.0 inch (25.5 millimeters) square or 20 percent of total area of any pitted surface. Thread damage is not permitted.

ICN-412-A-654001-B-97499-00003-A-001-01

Figure 1. Tail Rotor Gearbox Electrical Chip Detector - Damage Limits (Sheet 1 of 1)

Electrical Chip Detector - Repair

Procedure

1. Repair of the electrical chip detector is restricted to replacement of defective parts.

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Electrical Chip Detector - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

If the tail rotor gearbox is new or returned from service or repair, remove the plug (9, Figure 1) and discard the packing.

1. Lubricate a new packing (2) with gearbox oil (412-A-12-01-00-00A-040A-A / 00141). Install the packing on the electrical chip detector self-closing valve (3).

2. Install the electrical chip detector self-closing valve (3) on the gearbox (1). Torque the valve 120 to 150 inch-pounds (13.56 to 16.95 Nm). Secure the valve to the gearbox (1) with lockwire (C-405).

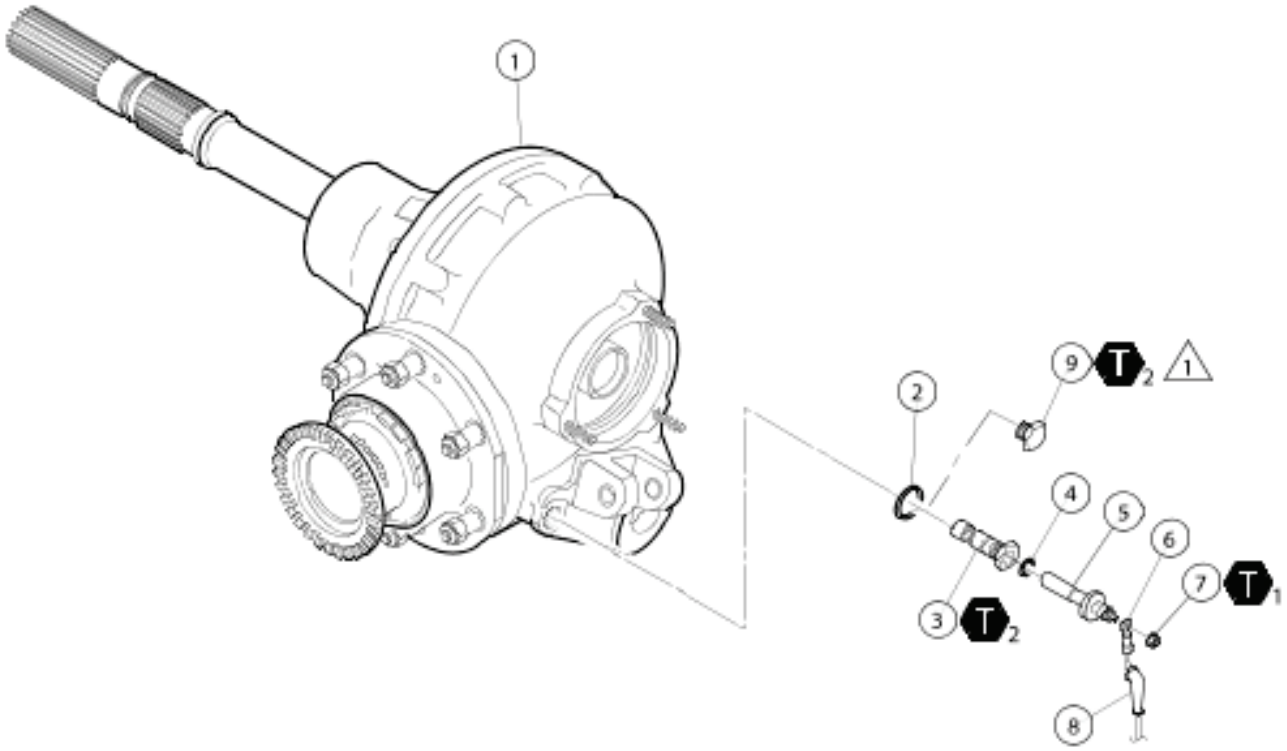
3. Lubricate a new packing (4) with gearbox oil (412-A-12-01-00-00A-040A-A / 00141). Install the packing on the electrical chip detector element (5). Insert the element into the electrical chip detector self-closing valve (3). Push down and twist the element until the locking detents engage.



DO NOT EXCEED 4.0 INCH-POUNDS (0.45 NM) TORQUE ON NUT (7).

4. Position electrical wire (6) on the electrical chip detector element (5) and install the nut (7). Torque the nut to a maximum of 4.0 inch-pounds (0.45 Nm). Position the nipple (8) over the nut.

5. Service the gearbox (412-A-12-10-04-00A-040A-A / 00136) if the electrical chip detector self-closing valve (3) was removed.



1. Gearbox
2. Packing
3. Electrical chip detector self-closing valve
4. Packing
5. Electrical chip detector element
6. Electrical wire
7. Nut
8. Nipple
9. Plug

T1 4 IN-LBS (0.45 Nm)

T2 120 TO 150 IN-LBS (13.56 TO 16.95 Nm)

NOTE

1 For helicopters S/N 36694 and subsequent.

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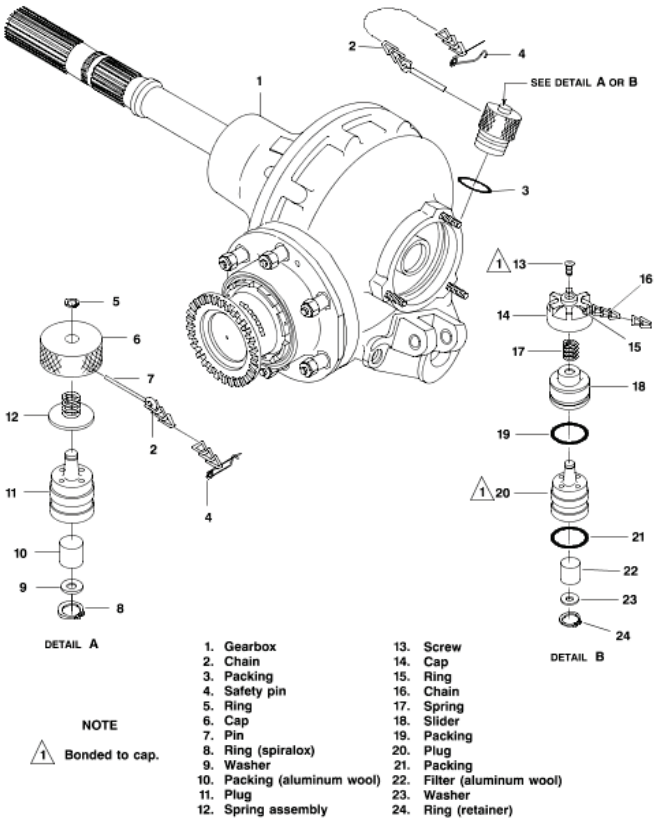
Figure 1. Tail Rotor Gearbox Electrical Chip Detector (Sheet 1 of 1)

Oil Filler Cap - Removal

Procedure

1. Push down on the cap assembly (6 or 14, Figure 1) and twist to disengage the pin (7, Detail A) from the gearbox (1). The new cap (14) does not have the pin (7). The pins to engage the gearbox adapter are part of the cap assembly.

2. Unsnap the safety pin (4) from the gearbox (1).



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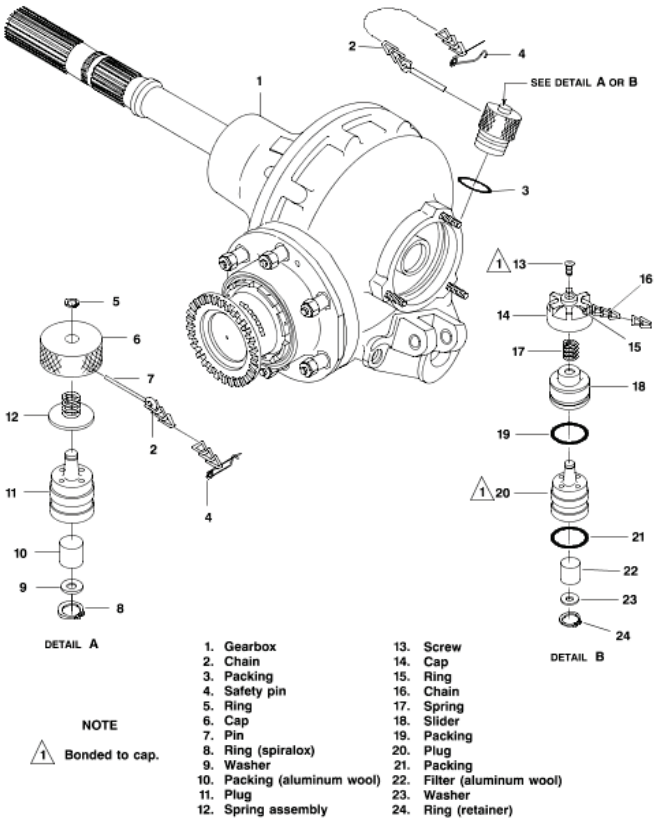
Figure 1. Tail Rotor Gearbox Oil Filler Cap (Sheet 1 of 1)

Oil Filler Cap - Inspection

Procedure

1. Inspect the filler cap for distortion of the pin (7, Figure 1) or other damage that would affect serviceability.

2. Invert the filler cap to expose the washer (9 or 23). Depress the washer approximately 0.06 inch (1.52 mm) and release. The washer should spring back against the ring (8 or 24). If spring back is not present, the aluminum wool filter (10 or 22) is dirty or an insufficient amount of aluminum wool is installed.



ICN-412-A-652001-B-97499-00008-A-001-01

Figure 1. Tail Rotor Gearbox Oil Filler Cap (Sheet 1 of 1)

Oil Filler Cap - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	
Aluminum wool	C-422	AR	

Procedure

1. Repair is restricted to the replacement of the aluminum wool filter (10 or 22, Figure 1) as outlined in Step 2. through Step 10..

NOTE

The screw (13) and plug (20, Detail B) are bonded to the cap (14). It is not necessary to remove either to replace the filter (22).

2. Remove the pin (7, Detail A).

3. Remove the ring (5), cap (6), and spring assembly (12) from the plug (11).

4. Remove the ring (8 or 24) and washer (9 or 23).

5. Remove the aluminum wool filter (10 or 22) and discard.

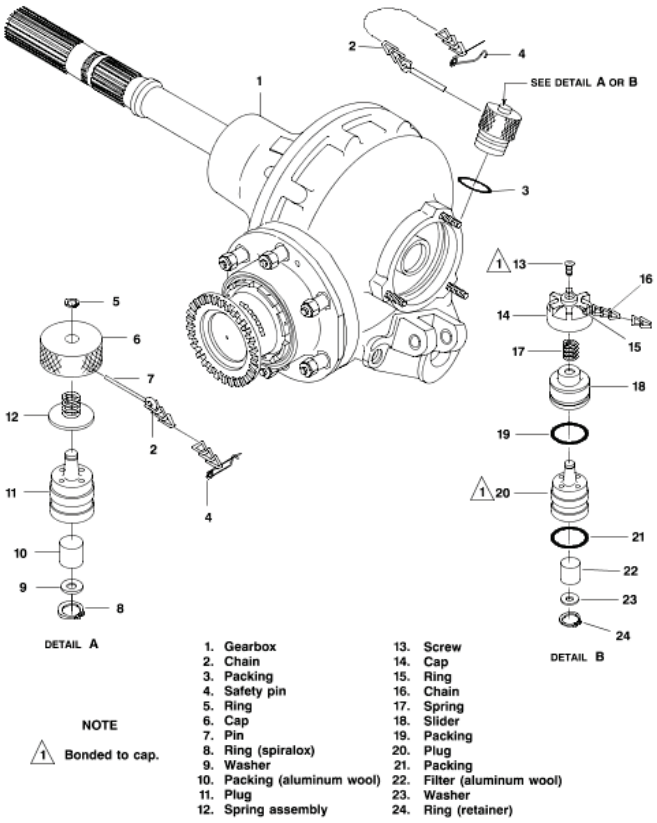
6. Clean parts with drycleaning solvent (C-304).

7. Fill the plug (11 or 20) with new aluminum wool (C-422) until the washer (9 or 23) spring back to the original position after being pressed in 0.06 inch (1.52 mm). Add or remove aluminum wool as required.

8. Install the washer (9 or 23) and ring (8 or 24).

9. Make sure the packing (3 or 21) is in place and install the spring assembly (12, Detail A), cap (6), and ring (5) on the plug (11).

10. Insert the pin (7, Detail A) through the cap (6) and bend the end of the pin to secure.



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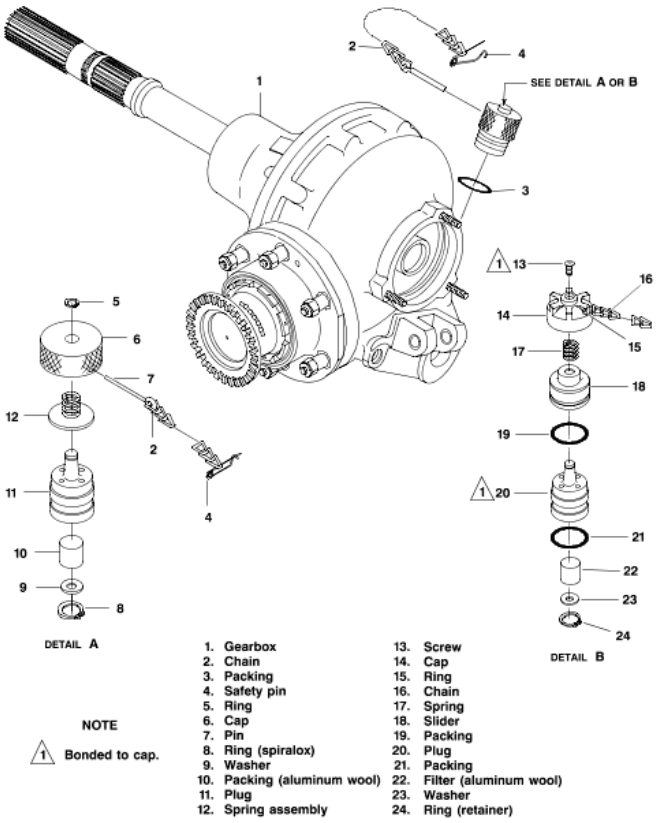
Figure 1. Tail Rotor Gearbox Oil Filler Cap (Sheet 1 of 1)

Oil Filler Cap - Installation

Procedure

1. Position the cap assembly on the gearbox (1, Figure 1). Push down on the cap (6 or 14) and twist clockwise to engage in the gearbox adapter.

2. Secure the chain (2 or 16) to the gearbox (1) with the safety pin (4).



ICN-412-A-652001-B-97499-00008-A-001-01

Figure 1. Tail Rotor Gearbox Oil Filler Cap (Sheet 1 of 1)

Flexible Coupling - Lubrication

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lubricant	C-015	AR	

Procedure

1. Remove the tail rotor driveshaft section between the tail rotor gearbox and the intermediate gearbox (412-A-65-10-00-00A-520A-A / 01259 or 412-A-65-10-02-00A-520A-A / 01270).

2. Remove the retaining ring (1, Figure 1) and at same time hold the plate (2) against the spring (3). Remove the plate and spring.

3. Hold the outer coupling (5) at the full inboard position and use clean, lint-free cloths to remove all old grease. Clean the splines thoroughly.

4. Visually inspect the splines of the inner coupling (4) and outer coupling (5) for unusual wear patterns, nicks, dents, and cracks. The maximum acceptable wear is 0.005 inch (0.127 mm) (BHT-412-CR&O).

5. Inspect the plate (2) for damage that could cause leakage of lubricant.

6. Inspect the seal (6) in the outer coupling (5) for protrusion or other damage that could cause leakage of lubricant. Replace the seal as required (BHT-412-CR&O).

7. Move the outer coupling (5) on the inner coupling (4) forward and aft with clockwise and counterclockwise preload and feel for roughness. If any roughness or resistance is felt, reinspect the coupling splines for damage.

8. Hold the outer coupling (5) at the full outboard position. Hand-pack lubricant (C-015) 0.12 inch (3.05 mm) over the top of the outer coupling spline teeth. A workaid may be fabricated to help obtain the proper depth of grease (412-A-63-00-00-00A-009A-A / 00987).

9. Hold the outer coupling (5) at the full outboard position. Install the spring (3), plate (2), and retaining ring (1).

10. Install the tail rotor driveshaft section between the intermediate gearbox and the tail rotor gearbox (412-A-65-10-00-00A-720A-A / 01263 or 412-A-65-10-02-00A-720A-A / 01275).

NOTE

A ground run and leak check must be performed following the installation of new, repaired, or relubricated flexible couplings.

11. After the flexible coupling is lubricated, perform the following test to make sure the coupling is not leaking.

11.1. Clean all lubricant from the exterior surfaces of the coupling and clamp set that secures the tail rotor driveshaft to the coupling. Also, clean all lubricant from the surrounding helicopter structure.

11.2. Operate the helicopter BHT-412-FM for 5 minutes at 100%.

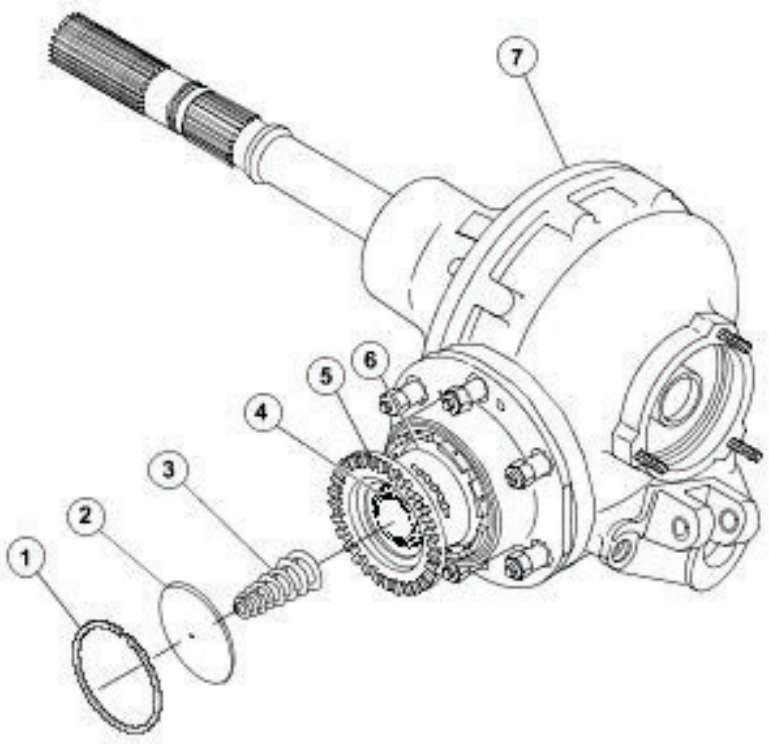
11.3. Shut down the helicopter and inspect the coupling and surrounding structure for evidence of grease leakage. If there is any evidence of grease leakage, wipe the surface clean and rerun the test to ensure leakage is not from excessive grease applied at build-up. If the test indicates the coupling is leaking, remove the coupling and replace the defective parts. Repeat the leak test run.

11.4. Inspect the input quill coupling (BHT-412-CR&O) and surrounding area for grease leakage. If there is evidence of grease leakage, remove the coupling and replace defective parts. Repeat the leak test run.

11.5. Check the visual overtemperature indicators (TEMP-PLATES) for discoloration and overheat condition. A change in color of TEMP-PLATES dots (from white to black) indicates a possible overheat condition and/or component degradation. The cause of indicator discoloration shall be determined and corrected prior to continued operation (412-A-65-10-01-00A-280A-A / 01266).

11.6. Replace the visual overtemperature indicators if mutilated or discolored (BHT-412-CR&O).

12. Make an entry in the helicopter log and in the flex couplings lubrication log of the date lubricated, date grease manufactured, and helicopter hours (412-A-12-01-00-00A-240B-A / 00143).



1. Retaining ring
2. Coupling seal plate assembly
3. Spring
4. Inner coupling
5. Outer coupling
6. Seal
7. Gearbox

ICN-412-A-652001-B-97499-00009-A-001-01

Figure 1. Tail Rotor Gearbox Flexible Coupling (Sheet 1 of 1)

CHAPTER 67
FLIGHT CONTROLS

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Introduction - Description	DMC-412-A-67-00-00-00A-028A-A / 01320	2023-05-31	SN: 36001-36086
Flight Controls - Troubleshooting	DMC-412-A-67-00-00-00A-422A-A / 01321	2023-05-31	SN: 36001-36086
Flight Control Rigging - Description	DMC-412-A-67-01-00-00A-042A-A / 01322	2023-05-31	ALL
Collective Controls - Rigging	DMC-412-A-67-01-00-01A-276A-A / 01323	2023-05-31	ALL
Cyclic Controls (Primary Method) - Rigging	DMC-412-A-67-01-00-03A-276A-A / 01324	2025-05-13	ALL
Cyclic Controls (Alternate Method) - Rigging	DMC-412-A-67-01-00-04A-276A-A / 01325	2025-05-13	ALL
Cyclic Position Sensors - Rigging	DMC-412-A-67-01-00-05A-276A-A / 01326	2023-05-31	ALL
Anti-Torque Controls - Rigging	DMC-412-A-67-01-00-06A-276A-A / 01327	2023-05-31	SN: 33001-33213 or 36001-36086
Aerodynamically Actuated Elevator - Rigging	DMC-412-A-67-01-00-07A-276A-A / 01328	2023-05-31	ALL
Collective Controls - Description	DMC-412-A-67-11-00-00A-028A-A / 01329	2023-05-31	ALL
Hydraulic Servo Cylinder Boot - Removal	DMC-412-A-67-11-00-01A-520A-A / 01330	2023-05-31	ALL
Hydraulic Servo Cylinder Boot - Inspection	DMC-412-A-67-11-00-01A-280A-A / 01331	2023-05-31	ALL
Hydraulic Servo Cylinder Boot - Installation	DMC-412-A-67-11-00-01A-720A-A / 01332	2023-05-31	ALL
Hydraulic Cylinder Extension Tube - Removal	DMC-412-A-67-11-00-02A-520A-A / 01333	2023-05-31	ALL
Hydraulic Cylinder Extension Tube - Inspection	DMC-412-A-67-11-00-02A-280A-A / 01334	2023-05-31	ALL
Hydraulic Cylinder Extension Tube - Installation	DMC-412-A-67-11-00-02A-720A-A / 01335	2023-05-31	ALL
Hydraulic Cylinder Extension Universal - Removal	DMC-412-A-67-11-00-03A-520A-A / 01336	2023-05-31	ALL
Hydraulic Cylinder Extension Universal - Inspection	DMC-412-A-67-11-00-03A-280A-A / 01337	2023-05-31	ALL
Hydraulic Cylinder Extension Universal - Installation	DMC-412-A-67-11-00-03A-720A-A / 01338	2023-05-31	ALL
Piston Rod Clevis - Removal	DMC-412-A-67-11-00-04A-520A-A / 01339	2023-05-31	ALL
Piston Rod Clevis - Inspection	DMC-412-A-67-11-00-04A-280A-A / 01340	2023-05-31	ALL
Piston Rod Clevis - Installation	DMC-412-A-67-11-00-04A-720A-A / 01341	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Servo Valve Balance Spring - Description	DMC-412-A-67-11-00-05A-042A-A / 01342	2023-05-31	ALL
Servo Valve Balance Spring - Removal	DMC-412-A-67-11-00-05A-520A-A / 01343	2023-05-31	ALL
Servo Valve Balance Spring - Inspection	DMC-412-A-67-11-00-05A-280A-A / 01344	2023-05-31	ALL
Servo Valve Balance Spring - Installation	DMC-412-A-67-11-00-05A-720A-A / 01345	2023-05-31	ALL
Collective Flight Control Cylinder - Removal	DMC-412-A-67-11-00-06A-520A-A / 01346	2023-05-31	ALL
Collective Flight Control Cylinder - Inspection	DMC-412-A-67-11-00-06A-280A-A / 01347	2023-05-31	ALL
Collective Flight Control Cylinder - Installation	DMC-412-A-67-11-00-06A-720A-A / 01348	2023-05-31	ALL
Left Cyclic and Collective Boost Support - Removal	DMC-412-A-67-11-00-07A-520A-A / 01349	2023-05-31	ALL
Left Cyclic and Collective Boost Support - Inspection	DMC-412-A-67-11-00-07A-280A-A / 01350	2023-05-31	ALL
Left Cyclic and Collective Boost Support - Installation	DMC-412-A-67-11-00-07A-720A-A / 01351	2023-05-31	ALL
Adjustable Rod End Tube - Removal	DMC-412-A-67-11-00-08A-520A-A / 01352	2023-05-31	ALL
Adjustable Rod End Tube - Inspection	DMC-412-A-67-11-00-08A-280A-A / 01353	2023-05-31	ALL
Adjustable Rod End Tube - Installation	DMC-412-A-67-11-00-08A-720A-A / 01354	2023-05-31	ALL
Pitch Control Bellcrank - Removal	DMC-412-A-67-11-00-09A-520A-A / 01355	2023-05-31	ALL
Pitch Control Bellcrank - Inspection	DMC-412-A-67-11-00-09A-280A-A / 01356	2023-05-31	ALL
Pitch Control Bellcrank - Installation	DMC-412-A-67-11-00-09A-720A-A / 01357	2023-05-31	ALL
Collective Force Gradient - Removal	DMC-412-A-67-11-00-10A-520A-A / 01358	2023-05-31	ALL
Collective Force Gradient - Inspection	DMC-412-A-67-11-00-10A-280A-A / 01359	2023-05-31	ALL
Collective Force Gradient - Repair	DMC-412-A-67-11-00-10A-685A-A / 01360	2023-05-31	ALL
Collective Force Gradient - Installation	DMC-412-A-67-11-00-10A-720A-A / 01361	2023-05-31	ALL
Collective Bellcrank - Removal	DMC-412-A-67-11-00-11A-520A-A / 01362	2023-05-31	ALL
Collective Bellcrank - Inspection	DMC-412-A-67-11-00-11A-280A-A / 01363	2023-05-31	ALL
Collective Bellcrank - Installation	DMC-412-A-67-11-00-11A-720A-A / 01364	2023-05-31	ALL
Collective Link - Removal	DMC-412-A-67-11-00-12A-520A-A / 01365	2023-05-31	SN: 36069-Subsequent
Collective Link - Inspection	DMC-412-A-67-11-00-12A-280A-A / 01366	2023-05-31	SN: 36069-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Collective Link - Installation	DMC-412-A-67-11-00-12A-720A-A / 01367	2023-05-31	SN: 36069-Subsequent
Collective Rotary Actuator - Removal	DMC-412-A-67-11-00-13A-520A-A / 01368	2023-05-31	4-Axis DAFCS: Installed
Collective Rotary Actuator - Inspection	DMC-412-A-67-11-00-13A-280A-A / 01369	2023-05-31	4-Axis DAFCS: Installed
Collective Rotary Actuator - Repair	DMC-412-A-67-11-00-13A-685A-A / 01370	2023-05-31	4-Axis DAFCS: Installed
Collective Rotary Actuator - Installation	DMC-412-A-67-11-00-13A-720A-A / 01371	2023-05-31	4-Axis DAFCS: Installed
Adjustable Collective Control Tube - Removal	DMC-412-A-67-11-00-14A-520A-A / 01372	2023-05-31	ALL
Adjustable Collective Control Tube - Inspection	DMC-412-A-67-11-00-14A-280A-A / 01373	2023-05-31	ALL
Adjustable Collective Control Tube - Installation	DMC-412-A-67-11-00-14A-720A-A / 01374	2023-05-31	ALL
Control Lever and Stop Assembly - Removal	DMC-412-A-67-11-00-15A-520A-A / 01375	2023-05-31	ALL
Control Lever and Stop Assembly - Inspection	DMC-412-A-67-11-00-15A-280A-A / 01376	2023-05-31	ALL
Control Lever and Stop Assembly - Installation	DMC-412-A-67-11-00-15A-720A-A / 01377	2023-05-31	ALL
Pilot Collective Boot - Description	DMC-412-A-67-11-00-16A-042A-A / 01378	2023-05-31	ALL
Pilot Collective Boot - Removal	DMC-412-A-67-11-00-16A-520A-A / 01379	2023-05-31	ALL
Pilot Collective Boot - Inspection	DMC-412-A-67-11-00-16A-280A-A / 01380	2023-05-31	ALL
Pilot Collective Boot - Installation	DMC-412-A-67-11-00-16A-720A-A / 01381	2023-05-31	ALL
Collective Jackshaft - Description	DMC-412-A-67-11-00-17A-042A-A / 01382	2023-05-31	ALL
Collective Jackshaft - Removal	DMC-412-A-67-11-00-17A-520A-A / 01383	2023-05-31	ALL
Collective Jackshaft - Inspection	DMC-412-A-67-11-00-17A-280A-A / 01384	2023-05-31	ALL
Collective Jackshaft - Repair	DMC-412-A-67-11-00-17A-685A-A / 01385	2023-05-31	ALL
Collective Jackshaft - Installation	DMC-412-A-67-11-00-17A-720A-A / 01386	2023-05-31	ALL
Collective Transducers - Removal	DMC-412-A-67-11-00-18A-520A-A / 01387	2023-05-31	SN: 33001-36086
Collective Transducers - Removal	DMC-412-A-67-11-00-18A-520B-A / 01388	2023-05-31	SN: 36087-36693
Collective Transducers - Removal	DMC-412-A-67-11-00-18A-520C-A / 01389	2023-05-31	SN: 36694-Subsequent
Collective Transducers - Inspection	DMC-412-A-67-11-00-18A-280A-A / 01390	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Collective Transducers - Repair	DMC-412-A-67-11-00-18A-685A-A / 01391	2023-05-31	ALL
Collective Transducers - Installation	DMC-412-A-67-11-00-18A-720A-A / 01392	2023-05-31	SN: 33001-36086
Collective Transducers - Installation	DMC-412-A-67-11-00-18A-720B-A / 01393	2023-05-31	SN: 36087-36693
Collective Transducers - Installation	DMC-412-A-67-11-00-18A-720C-A / 01394	2023-05-31	SN: 36694-Subsequent
Collective Transducers - Adjustment	DMC-412-A-67-11-00-18A-271A-A / 01395	2023-05-31	SN: 36037-Subsequent
Friction Clamp - Removal	DMC-412-A-67-11-00-19A-520A-A / 01396	2023-05-31	ALL
Friction Clamp - Inspection	DMC-412-A-67-11-00-19A-280A-A / 01397	2023-05-31	ALL
Friction Clamp - Repair	DMC-412-A-67-11-00-19A-685A-A / 01398	2023-05-31	ALL
Friction Clamp - Installation	DMC-412-A-67-11-00-19A-720A-A / 01399	2023-05-31	ALL
Cyclic Controls - Description	DMC-412-A-67-12-00-00A-028A-A / 01400	2023-05-31	ALL
Hydraulic Cylinder Extension Tube - Removal	DMC-412-A-67-12-00-01A-520A-A / 01401	2025-05-13	ALL
Hydraulic Cylinder Extension Tube - Inspection	DMC-412-A-67-12-00-01A-280A-A / 01402	2025-05-13	ALL
Hydraulic Cylinder Extension Tube - Installation	DMC-412-A-67-12-00-01A-720A-A / 01403	2025-05-13	ALL
Hydraulic Servo Cylinder Boot - Removal	DMC-412-A-67-12-00-02A-520A-A / 01405	2025-05-13	ALL
Hydraulic Servo Cylinder Boot - Inspection	DMC-412-A-67-12-00-02A-280A-A / 01406	2025-05-13	ALL
Hydraulic Servo Cylinder Boot - Installation	DMC-412-A-67-12-00-02A-720A-A / 01407	2025-05-13	ALL
Hydraulic Cylinder Extension Universal - Description	DMC-412-A-67-12-00-03A-042A-A / 01408	2023-05-31	ALL
Hydraulic Cylinder Extension Universal - Removal	DMC-412-A-67-12-00-03A-520A-A / 01409	2023-05-31	ALL
Hydraulic Cylinder Extension Universal - Inspection	DMC-412-A-67-12-00-03A-280A-A / 01410	2023-05-31	ALL
Hydraulic Cylinder Extension Universal - Bearing Replacement	DMC-412-A-67-12-00-03A-921A-A / 01411	2023-05-31	ALL
Hydraulic Cylinder Extension Universal - Installation	DMC-412-A-67-12-00-03A-720A-A / 01412	2023-05-31	ALL
Left Cyclic Piston Rod Clevis - Removal	DMC-412-A-67-12-00-04A-520A-A / 01413	2023-05-31	ALL
Left Cyclic Piston Rod Clevis - Inspection	DMC-412-A-67-12-00-04A-280A-A / 01414	2023-05-31	ALL
Left Cyclic Piston Rod Clevis - Repair	DMC-412-A-67-12-00-04A-685A-A / 01415	2023-05-31	ALL
Left Cyclic Piston Rod Clevis - Installation	DMC-412-A-67-12-00-04A-720A-A / 01416	2023-05-31	ALL

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Right Cyclic Piston Rod Clevis - Removal	DMC-412-A-67-12-00-05A-520A-A / 01417	2023-05-31	ALL
Right Cyclic Piston Rod Clevis - Inspection	DMC-412-A-67-12-00-05A-280A-A / 01418	2023-05-31	ALL
Right Cyclic Piston Rod Clevis - Installation	DMC-412-A-67-12-00-05A-720A-A / 01419	2023-05-31	ALL
Left Cyclic Flight Control Cylinder - Removal	DMC-412-A-67-12-00-06A-520A-A / 01420	2023-05-31	ALL
Left Cyclic Flight Control Cylinder - Inspection	DMC-412-A-67-12-00-06A-280A-A / 01421	2023-05-31	ALL
Left Cyclic Flight Control Cylinder - Installation	DMC-412-A-67-12-00-06A-720A-A / 01422	2023-05-31	ALL
Right Cyclic Flight Control Cylinder - Removal	DMC-412-A-67-12-00-07A-520A-A / 01423	2023-05-31	ALL
Right Cyclic Flight Control Cylinder - Inspection	DMC-412-A-67-12-00-07A-280A-A / 01424	2023-05-31	ALL
Right Cyclic Flight Control Cylinder - Installation	DMC-412-A-67-12-00-07A-720A-A / 01425	2023-05-31	ALL
Right Cyclic Boost Cylinder Support - Removal	DMC-412-A-67-12-00-08A-520A-A / 01426	2023-05-31	ALL
Right Cyclic Boost Cylinder Support - Inspection	DMC-412-A-67-12-00-08A-280A-A / 01427	2023-05-31	ALL
Right Cyclic Boost Cylinder Support - Installation	DMC-412-A-67-12-00-08A-720A-A / 01428	2023-05-31	ALL
Cyclic Control Tubes - Removal	DMC-412-A-67-12-00-09A-520A-A / 01429	2023-05-31	ALL
Cyclic Control Tubes - Inspection	DMC-412-A-67-12-00-09A-280A-A / 01430	2023-05-31	ALL
Cyclic Control Tubes - Repair	DMC-412-A-67-12-00-09A-685A-A / 01431	2023-05-31	ALL
Cyclic Control Tubes - Installation	DMC-412-A-67-12-00-09A-720A-A / 01432	2023-05-31	ALL
Mixing Lever Assembly - Removal	DMC-412-A-67-12-00-10A-520A-A / 01433	2023-05-31	ALL
Mixing Lever Assembly - Inspection	DMC-412-A-67-12-00-10A-280A-A / 01434	2023-05-31	ALL
Mixing Lever Assembly - Repair	DMC-412-A-67-12-00-10A-685A-A / 01435	2023-05-31	ALL
Mixing Lever Assembly - Installation	DMC-412-A-67-12-00-10A-720A-A / 01436	2023-05-31	ALL
Fore and Aft Proximity Switch - Removal	DMC-412-A-67-12-00-11A-520A-A / 01437	2023-05-31	ALL
Fore and Aft Proximity Switch - Inspection	DMC-412-A-67-12-00-11A-280A-A / 01438	2023-05-31	ALL
Fore and Aft Proximity Switch - Repair	DMC-412-A-67-12-00-11A-685A-A / 01439	2023-05-31	ALL
Fore and Aft Proximity Switch - Installation	DMC-412-A-67-12-00-11A-720A-A / 01440	2023-05-31	ALL
Lateral Proximity Switch - Removal	DMC-412-A-67-12-00-12A-520A-A / 01441	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Lateral Proximity Switch - Inspection	DMC-412-A-67-12-00-12A-280A-A / 01442	2023-05-31	ALL
Lateral Proximity Switch - Repair	DMC-412-A-67-12-00-12A-685A-A / 01443	2023-05-31	ALL
Lateral Proximity Switch - Installation	DMC-412-A-67-12-00-12A-720A-A / 01444	2023-05-31	ALL
Lateral Series Actuator - Removal	DMC-412-A-67-12-00-13A-520A-A / 01445	2023-05-31	ALL
Lateral Series Actuator - Inspection	DMC-412-A-67-12-00-13A-280A-A / 01446	2023-05-31	ALL
Lateral Series Actuator - Repair	DMC-412-A-67-12-00-13A-685A-A / 01447	2023-05-31	ALL
Lateral Series Actuator - Installation	DMC-412-A-67-12-00-13A-720A-A / 01448	2023-05-31	ALL
Longitudinal Series Actuator - Removal	DMC-412-A-67-12-00-14A-520A-A / 01449	2023-05-31	ALL
Longitudinal Series Actuator - Inspection	DMC-412-A-67-12-00-14A-280A-A / 01450	2023-05-31	ALL
Longitudinal Series Actuator - Repair	DMC-412-A-67-12-00-14A-685A-A / 01451	2023-05-31	ALL
Longitudinal Series Actuator - Installation	DMC-412-A-67-12-00-14A-720A-A / 01452	2023-05-31	ALL
Rotary Actuator (Lateral Cyclic) - Removal	DMC-412-A-67-12-00-15A-520A-A / 01453	2025-05-13	ALL
Rotary Actuator (Lateral Cyclic) - Inspection	DMC-412-A-67-12-00-15A-280A-A / 01454	2023-05-31	ALL
Rotary Actuator (Lateral Cyclic) - Installation	DMC-412-A-67-12-00-15A-720A-A / 01455	2025-05-13	ALL
Rotary Actuator (Fore and Aft Cyclic) - Removal	DMC-412-A-67-12-00-16A-520A-A / 01456	2025-05-13	ALL
Rotary Actuator (Fore and Aft Cyclic) - Inspection	DMC-412-A-67-12-00-16A-280A-A / 01457	2025-05-13	ALL
Rotary Actuator (Fore and Aft Cyclic) - Installation	DMC-412-A-67-12-00-16A-720A-A / 01458	2025-05-13	ALL
Force Gradient (Fore and Aft Cyclic) - Removal	DMC-412-A-67-12-00-17A-520A-A / 01459	2024-05-09	SN: 33001-33213 or 36001-36086
Force Gradient (Fore and Aft Cyclic) - Inspection	DMC-412-A-67-12-00-17A-280A-A / 01460	2024-05-09	SN: 33001-33213 or 36001-36086
Force Gradient (Fore and Aft Cyclic) - Repair	DMC-412-A-67-12-00-17A-685A-A / 01461	2024-05-09	SN: 33001-33213 or 36001-36086
Force Gradient (Fore and Aft Cyclic) - Installation	DMC-412-A-67-12-00-17A-720A-A / 01462	2024-05-09	SN: 33001-33213 or 36001-36086

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Force Gradient (Lateral Cyclic) - Removal	DMC-412-A-67-12-00-18A-520A-A / 01463	2024-05-09	SN: 33001-33213 or 36001-36086
Force Gradient (Lateral Cyclic) - Inspection	DMC-412-A-67-12-00-18A-280A-A / 01464	2024-05-09	SN: 33001-33213 or 36001-36086
Force Gradient (Lateral Cyclic) - Repair	DMC-412-A-67-12-00-18A-685A-A / 01465	2024-05-09	SN: 33001-33213 or 36001-36086
Force Gradient (Lateral Cyclic) - Installation	DMC-412-A-67-12-00-18A-720A-A / 01466	2024-05-09	SN: 33001-33213 or 36001-36086
Transducers (Fore and Aft Cyclic) - Removal	DMC-412-A-67-12-00-19A-520A-A / 01467	2023-05-31	ALL
Transducers (Fore and Aft Cyclic) - Inspection	DMC-412-A-67-12-00-19A-280A-A / 01468	2023-05-31	ALL
Transducers (Fore and Aft Cyclic) - Repair	DMC-412-A-67-12-00-19A-685A-A / 01469	2023-05-31	ALL
Transducers (Fore and Aft Cyclic) - Installation	DMC-412-A-67-12-00-19A-720A-A / 01470	2023-05-31	ALL
Jackshaft (Fore and Aft) Tube and Lever Assembly (Right Side) - Removal	DMC-412-A-67-12-00-20A-520A-A / 01471	2023-05-31	ALL
Jackshaft (Fore and Aft) Tube and Lever Assembly (Right Side) - Inspection	DMC-412-A-67-12-00-20A-280A-A / 01472	2023-05-31	ALL
Jackshaft (Fore and Aft) Tube and Lever Assembly (Right Side) - Installation	DMC-412-A-67-12-00-20A-720A-A / 01473	2023-05-31	ALL
Jackshaft (Fore and Aft) Tube and Lever Assembly (Left Side) - Removal	DMC-412-A-67-12-00-21A-520A-A / 01474	2023-05-31	ALL
Jackshaft (Fore and Aft) Tube and Lever Assembly (Left Side) - Inspection	DMC-412-A-67-12-00-21A-280A-A / 01475	2023-05-31	ALL
Jackshaft (Fore and Aft) Tube and Lever Assembly (Left Side) - Installation	DMC-412-A-67-12-00-21A-720A-A / 01476	2023-05-31	ALL
Jackshaft Housing - Removal	DMC-412-A-67-12-00-22A-520A-A / 01477	2023-05-31	ALL
Jackshaft Housing - Inspection	DMC-412-A-67-12-00-22A-280A-A / 01478	2023-05-31	ALL
Jackshaft Housing - Installation	DMC-412-A-67-12-00-22A-720A-A / 01479	2023-05-31	ALL
Arm Assembly - Removal	DMC-412-A-67-12-00-23A-520A-A / 01480	2023-05-31	ALL
Arm Assembly - Inspection	DMC-412-A-67-12-00-23A-280A-A / 01481	2023-05-31	ALL
Arm Assembly - Installation	DMC-412-A-67-12-00-23A-720A-A / 01482	2023-05-31	ALL
Lateral Transducers (Cyclic Controls) - Removal	DMC-412-A-67-12-00-24A-520A-A / 01483	2023-05-31	ALL
Lateral Transducers (Cyclic Controls) - Inspection	DMC-412-A-67-12-00-24A-280A-A / 01484	2023-05-31	ALL

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Lateral Transducers (Cyclic Controls) - Repair	DMC-412-A-67-12-00-24A-685A-A / 01485	2023-05-31	ALL
Lateral Transducers (Cyclic Controls) - Installation	DMC-412-A-67-12-00-24A-720A-A / 01486	2023-05-31	ALL
Cyclic Stick Boot - Removal	DMC-412-A-67-12-00-25A-520A-A / 01487	2023-05-31	ALL
Cyclic Stick Boot - Inspection	DMC-412-A-67-12-00-25A-280A-A / 01488	2023-05-31	ALL
Cyclic Stick Boot - Installation	DMC-412-A-67-12-00-25A-720A-A / 01489	2023-05-31	ALL
Pilot Cyclic Control Stick Assembly - Removal	DMC-412-A-67-12-00-26A-520A-A / 01490	2023-05-31	ALL
Pilot Cyclic Control Stick Assembly - Inspection	DMC-412-A-67-12-00-26A-280A-A / 01491	2023-05-31	ALL
Pilot Cyclic Control Stick Assembly - Installation	DMC-412-A-67-12-00-26A-720A-A / 01492	2023-05-31	ALL
Anti-Torque Controls - Description	DMC-412-A-67-20-00-00A-028A-A / 01493	2023-05-31	ALL
Pitch Change Control Bellcrank - Removal	DMC-412-A-67-20-00-01A-520A-A / 01494	2023-05-31	ALL
Pitch Change Control Bellcrank - Inspection	DMC-412-A-67-20-00-01A-280A-A / 01495	2023-05-31	ALL
Pitch Change Control Bellcrank - Repair	DMC-412-A-67-20-00-01A-685A-A / 01496	2023-05-31	ALL
Pitch Change Control Bellcrank - Installation	DMC-412-A-67-20-00-01A-720A-A / 01497	2023-05-31	ALL
Anti-Torque Control Tube Guides - Removal	DMC-412-A-67-20-00-02A-520A-A / 01498	2023-05-31	ALL
Anti-Torque Control Tube Guides - Inspection	DMC-412-A-67-20-00-02A-280A-A / 01499	2023-05-31	ALL
Anti-Torque Control Tube Guides - Installation	DMC-412-A-67-20-00-02A-720A-A / 01500	2023-05-31	ALL
Flight Control Tubes and Bellcranks - Description	DMC-412-A-67-20-00-03A-042A-A / 01501	2023-05-31	ALL
Flight Control Tubes and Bellcranks - Removal	DMC-412-A-67-20-00-03A-520A-A / 01502	2023-05-31	ALL
Flight Control Tubes and Bellcranks - Inspection	DMC-412-A-67-20-00-03A-280A-A / 01503	2023-05-31	ALL
Flight Control Tubes and Bellcranks - Installation	DMC-412-A-67-20-00-03A-720A-A / 01504	2023-05-31	ALL
Walking Beam Support Assembly - Removal	DMC-412-A-67-20-00-04A-520A-A / 01505	2023-05-31	ALL
Walking Beam Support Assembly - Inspection and Repair	DMC-412-A-67-20-00-04A-280A-A / 01506	2023-05-31	ALL
Walking Beam Support Assembly - Installation	DMC-412-A-67-20-00-04A-720A-A / 01507	2023-05-31	ALL
Control Cylinder - Removal	DMC-412-A-67-20-00-05A-520A-A / 01508	2023-05-31	ALL
Control Cylinder - Inspection	DMC-412-A-67-20-00-05A-280A-A / 01509	2023-05-31	ALL

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Control Cylinder - Repair	DMC-412-A-67-20-00-05A-685A-A / 01510	2023-05-31	ALL
Control Cylinder - Installation	DMC-412-A-67-20-00-05A-720A-A / 01511	2023-05-31	ALL
Anti-torque Series Actuator - Maintenance	DMC-412-A-67-20-00-06A-913A-A / 01512	2023-05-31	ALL
Anti-torque Series Actuator - Removal	DMC-412-A-67-20-00-06A-520A-A / 01513	2023-05-31	ALL
Anti-torque Series Actuator - Inspection	DMC-412-A-67-20-00-06A-280A-A / 01514	2023-05-31	ALL
Anti-torque Series Actuator - Installation	DMC-412-A-67-20-00-06A-720A-A / 01515	2023-05-31	ALL
Walking Beam and Friction Clamp - Removal	DMC-412-A-67-20-00-07A-520A-A / 01516	2023-05-31	ALL
Walking Beam and Friction Clamp - Inspection	DMC-412-A-67-20-00-07A-280A-A / 01517	2023-05-31	ALL
Walking Beam and Friction Clamp - Repair	DMC-412-A-67-20-00-07A-685A-A / 01518	2023-05-31	ALL
Walking Beam and Friction Clamp - Installation	DMC-412-A-67-20-00-07A-720A-A / 01519	2023-05-31	ALL
Magnetic Brake - Removal	DMC-412-A-67-20-00-08A-520A-A / 01520	2023-05-31	ALL
Magnetic Brake - Inspection	DMC-412-A-67-20-00-08A-280A-A / 01521	2023-05-31	ALL
Magnetic Brake - Repair	DMC-412-A-67-20-00-08A-685A-A / 01522	2023-05-31	ALL
Magnetic Brake - Installation	DMC-412-A-67-20-00-08A-720A-A / 01523	2023-05-31	ALL
Control Transducer - Removal	DMC-412-A-67-20-00-09A-520A-A / 01524	2023-05-31	ALL
Control Transducer - Inspection	DMC-412-A-67-20-00-09A-280A-A / 01525	2023-05-31	ALL
Control Transducer - Repair	DMC-412-A-67-20-00-09A-685A-A / 01526	2023-05-31	ALL
Control Transducer - Installation	DMC-412-A-67-20-00-09A-720A-A / 01527	2023-05-31	ALL
Force Gradient - Removal	DMC-412-A-67-20-00-10A-520A-A / 01528	2023-05-31	ALL
Force Gradient - Inspection	DMC-412-A-67-20-00-10A-280A-A / 01529	2023-05-31	ALL
Force Gradient - Repair	DMC-412-A-67-20-00-10A-685A-A / 01530	2023-05-31	ALL
Force Gradient - Installation	DMC-412-A-67-20-00-10A-720A-A / 01531	2023-05-31	ALL
Pedal Adjuster Assembly - Removal	DMC-412-A-67-20-00-11A-520A-A / 01532	2023-05-31	ALL
Pedal Adjuster Assembly - Inspection	DMC-412-A-67-20-00-11A-280A-A / 01533	2023-05-31	ALL
Pedal Adjuster Assembly - Installation	DMC-412-A-67-20-00-11A-720A-A / 01534	2023-05-31	ALL

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Pedals - Description	DMC-412-A-67-20-00-12A-042A-A / 01535	2023-05-31	ALL
Pedals - Removal	DMC-412-A-67-20-00-12A-520A-A / 01536	2023-05-31	ALL
Pedals - Inspection	DMC-412-A-67-20-00-12A-280A-A / 01537	2023-05-31	ALL
Pedals - Installation	DMC-412-A-67-20-00-12A-720A-A / 01538	2023-05-31	ALL
Aerodynamically Actuated Elevator - Description	DMC-412-A-67-02-00-01A-042A-A / 01539	2023-05-31	ALL
Aerodynamically Actuated Elevator - Removal	DMC-412-A-67-02-00-01A-520A-A / 01540	2023-05-31	ALL
Aerodynamically Actuated Elevator - Inspection	DMC-412-A-67-02-00-01A-280A-A / 01541	2023-05-31	ALL
Aerodynamically Actuated Elevator - Repair	DMC-412-A-67-02-00-01A-685A-A / 01542	2023-05-31	ALL
Aerodynamically Actuated Elevator - Installation	DMC-412-A-67-02-00-01A-720A-A / 01543	2023-05-31	ALL
Elevator Control Tube - Removal	DMC-412-A-67-02-00-02A-520A-A / 01544	2023-05-31	ALL
Elevator Control Tube - Inspection	DMC-412-A-67-02-00-02A-280A-A / 01545	2023-05-31	ALL
Elevator Control Tube - Installation	DMC-412-A-67-02-00-02A-720A-A / 01546	2023-05-31	ALL
Digital AFCS Flight Controls - Description	DMC-412-A-67-03-00-00A-042A-A / 01547	2023-05-31	SN: 36087-Subsequent
Flight Controls - Rigging	DMC-412-A-67-01-00-08A-276A-A / 01548	2023-05-31	SN: 36087-Subsequent
Collective Controls - Rigging	DMC-412-A-67-01-00-09A-276A-A / 01549	2023-05-31	SN: 36087-Subsequent
Cyclic Controls - Rigging	DMC-412-A-67-01-00-10A-276A-A / 01550	2023-05-31	SN: 36087-Subsequent
Anti-Torque Controls - Rigging	DMC-412-A-67-01-00-11A-276A-A / 01551	2023-05-31	SN: 36087-Subsequent
Cyclic Controls - Description	DMC-412-A-67-12-01-00A-042A-A / 01552	2023-05-31	SN: 36087-Subsequent
Rotary Actuator (Fore and Aft Cyclic) - Removal	DMC-412-A-67-12-01-01A-520A-A / 01553	2023-05-31	SN: 36087-Subsequent
Rotary Actuator (Fore and Aft Cyclic) - Inspection	DMC-412-A-67-12-01-01A-280A-A / 01554	2023-05-31	SN: 36087-Subsequent
Rotary Actuator (Fore and Aft Cyclic) - Installation	DMC-412-A-67-12-01-01A-720A-A / 01555	2023-05-31	SN: 36087-Subsequent
Rotary Actuator (Lateral Cyclic) - Removal	DMC-412-A-67-12-01-02A-520A-A / 01556	2023-05-31	SN: 36087-subsequent
Rotary Actuator (Lateral Cyclic) - Inspection	DMC-412-A-67-12-01-02A-280A-A / 01557	2023-05-31	SN: 36087-Subsequent
Rotary Actuator (Lateral Cyclic) - Installation	DMC-412-A-67-12-01-02A-720A-A / 01558	2023-05-31	SN: 36087-Subsequent
Force Gradient (Fore and Aft Cyclic) - Removal	DMC-412-A-67-12-01-03A-520A-A / 01559	2023-05-31	SN: 36087-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Force Gradient (Fore and Aft Cyclic) - Inspection	DMC-412-A-67-12-01-03A-280A-A / 01560	2023-05-31	SN: 36087-Subsequent
Force Gradient (Fore and Aft Cyclic) - Repair	DMC-412-A-67-12-01-03A-685A-A / 01561	2023-05-31	SN: 36087-Subsequent
Force Gradient (Fore and Aft Cyclic) - Installation	DMC-412-A-67-12-01-03A-720A-A / 01562	2023-05-31	SN: 36087-Subsequent
Force Gradient (Lateral Cyclic) - Removal	DMC-412-A-67-12-01-04A-520A-A / 01563	2023-05-31	SN: 36087-Subsequent
Force Gradient (Lateral Cyclic) - Inspection	DMC-412-A-67-12-01-04A-280A-A / 01564	2023-05-31	SN: 36087-Subsequent
Force Gradient (Lateral Cyclic) - Repair	DMC-412-A-67-12-01-04A-685A-A / 01565	2023-05-31	SN: 36087-Subsequent
Force Gradient (Lateral Cyclic) - Installation	DMC-412-A-67-12-01-04A-720A-A / 01566	2023-05-31	SN: 36087-Subsequent
Anti-Torque Controls - Description	DMC-412-A-67-12-02-00A-042A-A / 01567	2023-05-31	SN: 36087-Subsequent
Anti-Torque Transducers - Removal	DMC-412-A-67-12-02-01A-520A-A / 01568	2023-05-31	SN: 36087-Subsequent
Anti-Torque Transducers - Inspection	DMC-412-A-67-12-02-01A-280A-A / 01569	2023-05-31	SN: 36087-Subsequent
Anti-Torque Transducers - Repair	DMC-412-A-67-12-02-01A-685A-A / 01570	2023-05-31	SN: 36087-Subsequent
Anti-Torque Transducers - Installation	DMC-412-A-67-12-02-01A-720A-A / 01571	2023-05-31	SN: 36087-Subsequent
Anti-Torque Force Gradient - Description	DMC-412-A-67-12-02-02A-042A-A / 01572	2023-05-31	SN: 36087-Subsequent
Anti-Torque Force Gradient - Removal	DMC-412-A-67-12-02-02A-520A-A / 01573	2023-05-31	SN: 36087-Subsequent
Anti-Torque Force Gradient - Inspection	DMC-412-A-67-12-02-02A-280A-A / 01574	2023-05-31	SN: 36087-Subsequent
Anti-Torque Force Gradient - Repair	DMC-412-A-67-12-02-02A-685A-A / 01575	2023-05-31	SN: 36087-Subsequent
Anti-Torque Force Gradient - Installation	DMC-412-A-67-12-02-02A-720A-A / 01576	2023-05-31	SN: 36087-Subsequent
Anti-Torque Rotary Actuator - Removal	DMC-412-A-67-12-02-03A-520A-A / 01577	2023-05-31	SN: 36087-Subsequent
Anti-Torque Rotary Actuator - Inspection	DMC-412-A-67-12-02-03A-280A-A / 01578	2023-05-31	SN: 36087-Subsequent
Anti-Torque Rotary Actuator - Repair	DMC-412-A-67-12-02-03A-685A-A / 01579	2023-05-31	SN: 36087-Subsequent
Anti-Torque Rotary Actuator - Installation	DMC-412-A-67-12-02-03A-720A-A / 01580	2023-05-31	SN: 36087-Subsequent
Anti-Torque Series Actuators - Removal	DMC-412-A-67-12-02-04A-520A-A / 01581	2023-05-31	SN: 36087-Subsequent
Anti-Torque Series Actuators - Inspection	DMC-412-A-67-12-02-04A-280A-A / 01582	2023-05-31	SN: 36087-Subsequent
Anti-Torque Series Actuators - Installation	DMC-412-A-67-12-02-04A-720A-A / 01583	2023-05-31	SN: 36087-Subsequent

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Introduction - Description

The following paragraphs contains information and instructions for maintenance of both standard and automatic flight control systems. Refer to 412-A-67-03-00-00A-042A-A / 01547 for digital flight control system.

The standard flight control system includes a cyclic control stick for fore and aft and lateral controls, a collective pitch control stick for vertical control and anti-torque pedals for directional control. An aerodynamically actuated elevator increases stability about the longitudinal axis.

Electrically operated force trim systems consisting of electromechanical rotary actuators and force gradients (cyclic controls) and magnetic brake and force gradient (anti-torque controls) are connected in parallel to the cyclic and anti-torque controls, position the flight controls in response to a signal from either the pilot or AFCS system to establish a flight reference point, provide artificial force feel and stabilization of the cyclic and anti-torque controls.

The stick centering indicator system consists of two proximity switches and two cyclic centering (CYC CTR) caution lights (kit 412-704-115-101). The lights are located on the pilot and copilot instrument panel. System circuitry is activated whenever the rotor RPM warning light is illuminated. During ground operations below normal operating range, excessive cyclic inputs will illuminate the CYC CTR caution lights. Properly positioning cyclic stick will extinguish the lights.

The Automatic Flight Control System (AFCS) consists of the standard flight control system plus added AFCS components in the cyclic and directional control systems.

The AFCS cyclic control systems incorporate four electromechanical series actuators, two connected in series in the roll system and two connected in series in the pitch system. Four transducers, two in roll system and two in pitch system are used to convert mechanical movement to electrical signals for injection of pilot control commands to the AFCS.

AFCS anti-torque directional system incorporates a single electrical servo actuator and a transducer. The servo actuator is located forward of the directional hydraulic actuator. The transducer is connected to the bellcrank aft of the pedals.

Forward servo actuator in each cyclic system and directional series actuators are controlled by helipilot, HP-1 system. Aft actuators in each system are controlled by HP-2 system.

When the optional flight director kit is installed an additional transducer is added to the collective control system.

Applicable: SN: 36037-Subsequent:

Dual element transducers are installed. The second element is used when a Flight Data Recorder (FDR) is installed.

Refer to 412-A-98-00-00-10B-051A-A / 02570 for AFCS system wiring diagrams.

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Flight Controls - Troubleshooting

Procedure

1. Refer to 412-A-29-00-00-00A-422A-A / 00613 for flight control hydraulic systems troubleshooting.

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Flight Control Rigging - Description

The maintenance procedures paragraphs provide rigging instructions for the collective, cyclic, anti-torque (tail rotor), and aerodynamically actuated elevator control systems. For engine power lever controls (throttle), refer to 412-A-76-10-00-00A-276A-A / 01725.

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Collective Controls - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Push-Pull Dial Gauge (Spring Scale)	DPPH200	1	
Collective Lever Positioning Tool	T103287-101	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

1. Comply with the following general procedures when rigging collective controls:

1.1. When a rigging procedure requires a specific position to accomplish the following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ± 0.03 inch (0.76 mm).

1.3. All adjustable control tubes shall have a maximum of 1.0 inch (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, threads shall be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable end bearings and clevises after adjustment.

1.5. Torque control jam nuts to 80 to 100 inch-pounds (9.04 to 11.30 Nm) after adjustment unless other torque is specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. Loosen jam nuts, adjust end fittings, and torque jam nuts, if necessary. Remove and replace lockwire (C-405) as necessary.

2. Disconnect the following controls tubes:

2.1. Disconnect extension tube (2, Figure 1) from collective lever (1) by removing cotter pin, nut, four washers, and bolt.

2.2. Disconnect control tube (4) from collective hydraulic cylinder (3) by removing cotter pin, nut, two washers, bolt, and clip.

2.3. Disconnect control tube (7) from lever (8) by removing cotter pin, nut, two washers, and bolt.

NOTE

The following two substeps are only applicable to helicopters with 4-axis AFCS/ Flight Director kit installed.

2.4. Disconnect force gradient (1, Figure 2) from control lever (6).

2.5. Disconnect link (3) from rotary actuator (2).

3. If available, connect hydraulic power cart to helicopter. Do not apply hydraulic power at this time.

4. Position pilot collective stick (14, Figure 1) against the UP stop, hold in position, and tighten pilot collective friction adjuster (13).

5. Position top edge of bellcrank (6) 0.52 inch (13.21 mm) above top of support (5), as shown in Detail C.

6. Adjust control tube (7) to lever (8). Tighten jam nut on control tube clevis. Install control tube on lever using bolt, two washers, nut, and cotter pin.

7. Ensure pilot collective stick (14) is positioned as described in Step 4..

8. Move actuator valve lever (16) up to bottom collective hydraulic cylinder (3) in UP position. Set actuator valve lever 0.070 inch (1.778 mm) above center line neutral valve position (Detail B) and hold in position.

9. Adjust control tube (4) to fit actuator valve lever (16). Shorten rod end bearing three turns. Tighten jam nut against rod end bearing on control tube. Install control tube on actuator valve lever 412-A-67-11-00-06A-720A-A / 01348.

NOTE

If collective lever positioning tool (27) is available, skip Step 10. and perform Step 11..

10. Position collective lever (1) 4.67 to 4.70 inches (118.62 to 119.38 mm) above base of swashplate support (18) to center of pin (17) (Detail A).

11. Secure collective lever positioning tool (27) to pin (17) on collective lever (1) with screw (28), then pull up on collective lever to seat collective lever positioning tool base on plate assembly (29).

12. Apply hydraulic power from hydraulic cart to helicopter.



WHEN PERFORMING OPERATIONAL CHECK OF COLLECTIVE CONTROLS WITH HYDRAULIC POWER APPLIED, ENSURE CYCLIC STICK IS IN CENTER POSITION.

NOTE

If hydraulic power cart is not available, refer to note preceding Step 16..

13. With collective lever (1) set as per Step 10. or Step 11., position pilot collective stick (14) against DOWN stop. Tighten pilot collective friction adjuster (13).

14. Adjust extension tube (2) to fit collective lever (1). Torque jam nut 1500 to 1800 inch-pounds (169.47 to 203.37 Nm) on extension tube (2), and apply lockwire (C-405). Apply sealant (C-308) around nut, lockwire, and threads of clevis.



DO NOT MOVE PILOT COLLECTIVE STICK (14) WITH COLLECTIVE LEVER POSITIONING TOOL (27) INSTALLED.

15. Install extension tube (2) to collective lever (1) using bolt, washers, nut, and cotter pin (412-A-67-11-00-05A-520A-A / 01343).

15.1. If applicable, remove collective lever positioning tool (27).

NOTE

Alternate method of adjusting extension tube (2) without pressurizing hydraulic system is outlined in Step 16. through Step 20..

16. Position collective lever (1) and pilot collective stick (14), as outlined in Step 13..

17. Exert sufficient downward pressure on collective hydraulic cylinder (3) to hold actuator valve lever (16) at top of travel (Detail B).



DO NOT ALLOW COLLECTIVE CONTROLS BELOW HYDRAULIC CYLINDER (3) TO MOVE WHILE ACCOMPLISHING THE FOLLOWING STEP.

18. Adjust extension tube (2) to fit collective lever (1), then shorten tube by one complete turn of upper clevis.

19. Torque jam nut 1500 to 1800 inch-pounds (169.47 to 203.37 Nm) on extension tube (2) and secure with lockwire (C-405). Apply sealant (C-308) around nut, lockwire, and threads of clevis.

20. Install the extension tube (2) to the collective lever (1) using the bolt, washers, nut, and cotter pin (412-A-67-11-00-02A-720A-A / 01335).

Applicable: SN: 36069-Subsequent:

21. Adjust the stop (9), as shown in Detail F. Torque the jam nut 80 to 100 inch-pounds (9.04 to 11.30 Nm) and secure with lockwire (C-405).

NOTE

The following step is only applicable to helicopters with 4-axis AFCS/Flight Director kit installed.

22. Rig the collective trim system as follows:

22.1. Position the pilot collective stick (10, Figure 2) against its down stop.

22.2. Connect the force gradient (1) to the control lever (6).

22.3. Position the output arm of the rotary actuator (2) as shown in Detail A.

22.4. Position and hold the output arm of the rotary actuator (2) against its outboard left side stop.

22.5. Adjust the link (3) to fit between the bellcrank (4) and the rotary actuator (2) positioned per Step 22.4., then extend the rod end bearing 2 1/2 complete turns and secure to the output arm of the rotary actuator (2).

23. Adjust the minimum friction as follows:

NOTE

Step 23.1. and Step 23.2. are applicable to helicopters without 4-axis AFCS/Flight Director kit installed when a hydraulic power cart is not used.

23.1. Disconnect the control tube (10, Figure 1) from the collective jackshaft (12) by removing the cotter pin, nut, two washers, and bolt.

23.2. Attach a spring scale at the center of the upper throttle grip within 0.50 inch (12.7 mm). The correct measurement is 13 to 14 pounds (57.8 to 62.3 N) up-collective stick force. Proceed to Step 23.4..

NOTE

Measurement in Step 23.3. requires the use of a hydraulic power cart.

For helicopters with AFCS/4-axis Flight Director kit installed, measurement is taken with collective force trim OFF.

23.3. Attach a spring scale at center of upper throttle grip within 0.50 inch (12.7 mm). Correct measurement is 8 to 11 pounds (35.6 to 48.9 N) (helicopters without 4-axis Flight Director kit installed) or 4 to 4.5 pounds (17.8 to 20.0 N) (helicopters with 4-axis AFCS/Flight Director kit installed) up-collective stick force.

23.4. Tighten or loosen nut (20) to adjust minimum collective friction clamp (11), as shown in Detail D, to obtain correct sliding force.

23.5. After friction is adjusted within limits, connect control tube (10) if disconnected in Step 23.1. (412-A-67-11-00-17A-720A-A / 01386).

24. Move collective controls through full travel and check for binding and/or obstruction of control movement.

25. Turn off hydraulic power cart and disconnect from helicopter.

26. Ensure all collective control bolts are installed and cotter pins and lockwire are installed, where applicable.

27. Check throttle controls for binding or obstruction as follows:

27.1. Disconnect left and right adjustable throttle control tubes (25 and 26) from each bellcrank by removing nut, two washers, and bolt.

27.2. Check marked tooth of flex shaft is engaged with 10th tooth space of mating gear sectors (22 and 23) on both pilot and copilot collective elbow.

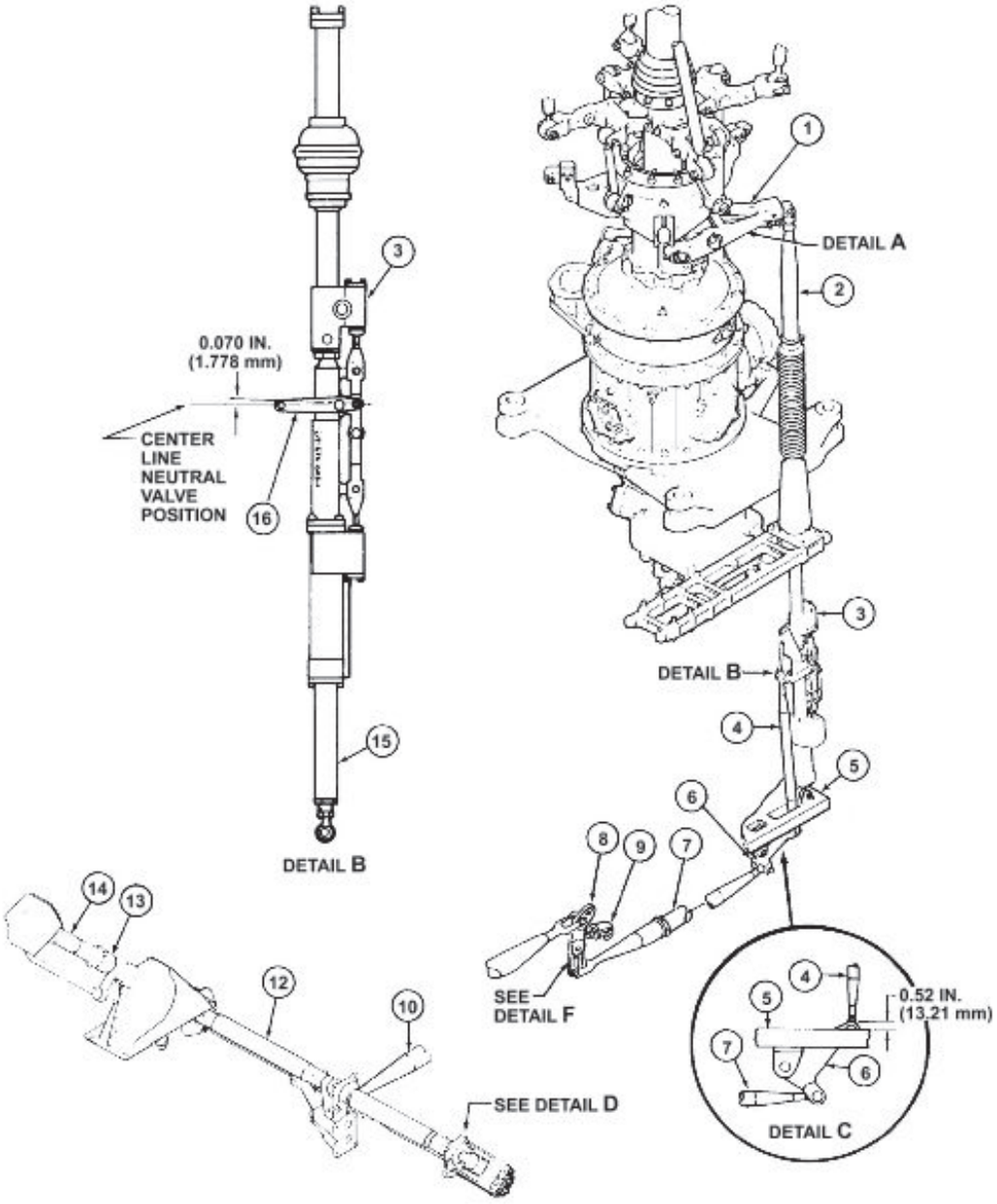
27.3. Roll both throttle grips on pilot (and copilot, if installed) collective sticks to full INCR position against stop.

NOTE

If binding or obstruction occurs in copilot throttle grips when rotated, adjust shims (24) between airframe structure and elbow support (21) to minimize lateral deflection of elbow assembly flex shafts.

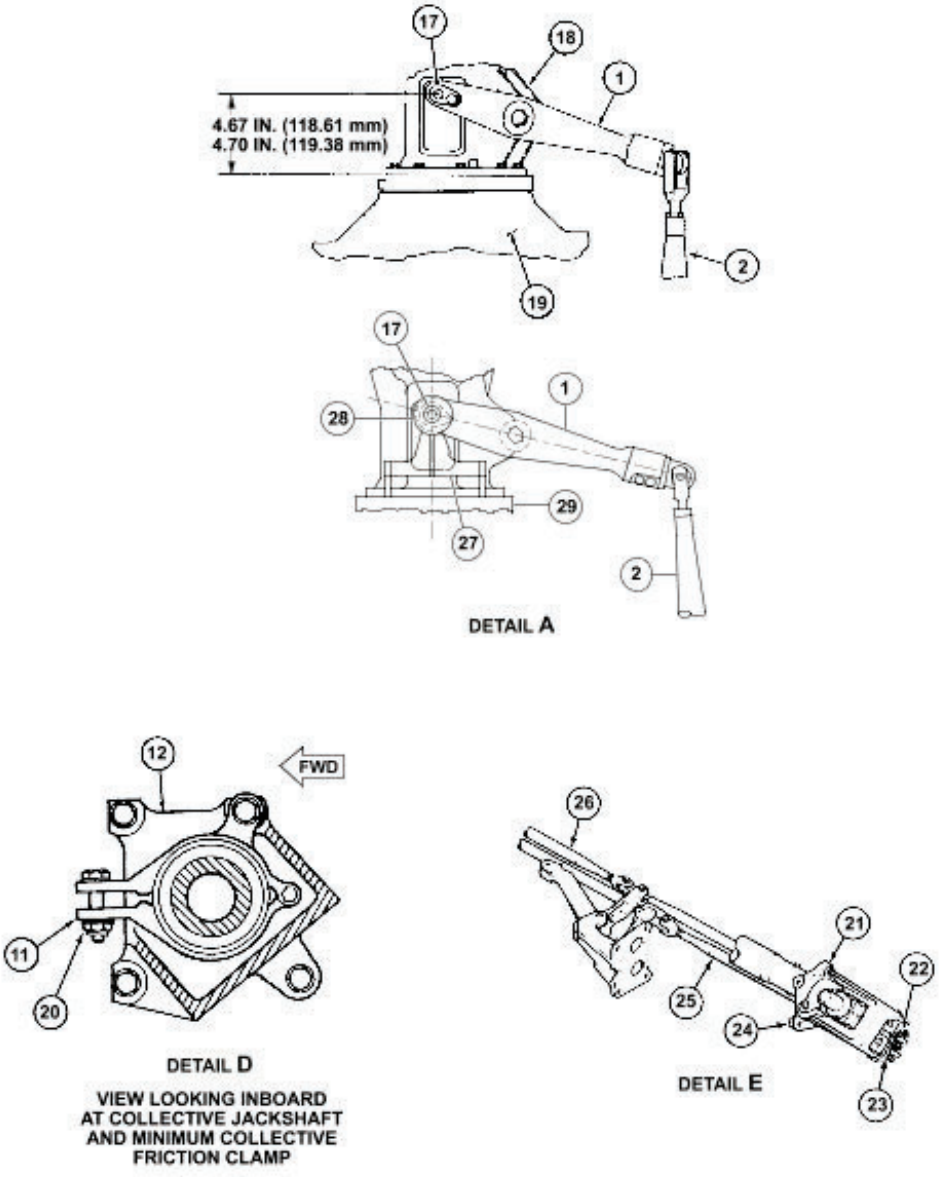
27.4. Adjust control tubes (25 and 26) to fit, torque jam nuts, and install.

27.5. Cycle both throttle grips through full travel and ensure each gear sector on both pilot and copilot elbow support is within no less than one tooth of rolling off its mating pinion.



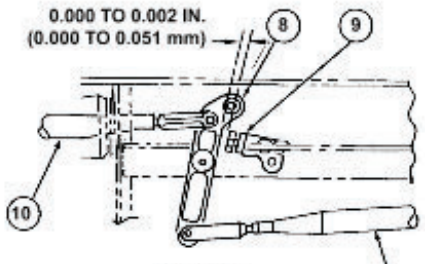
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Figure 1. Collective Rigging (Sheet 1 of 3)



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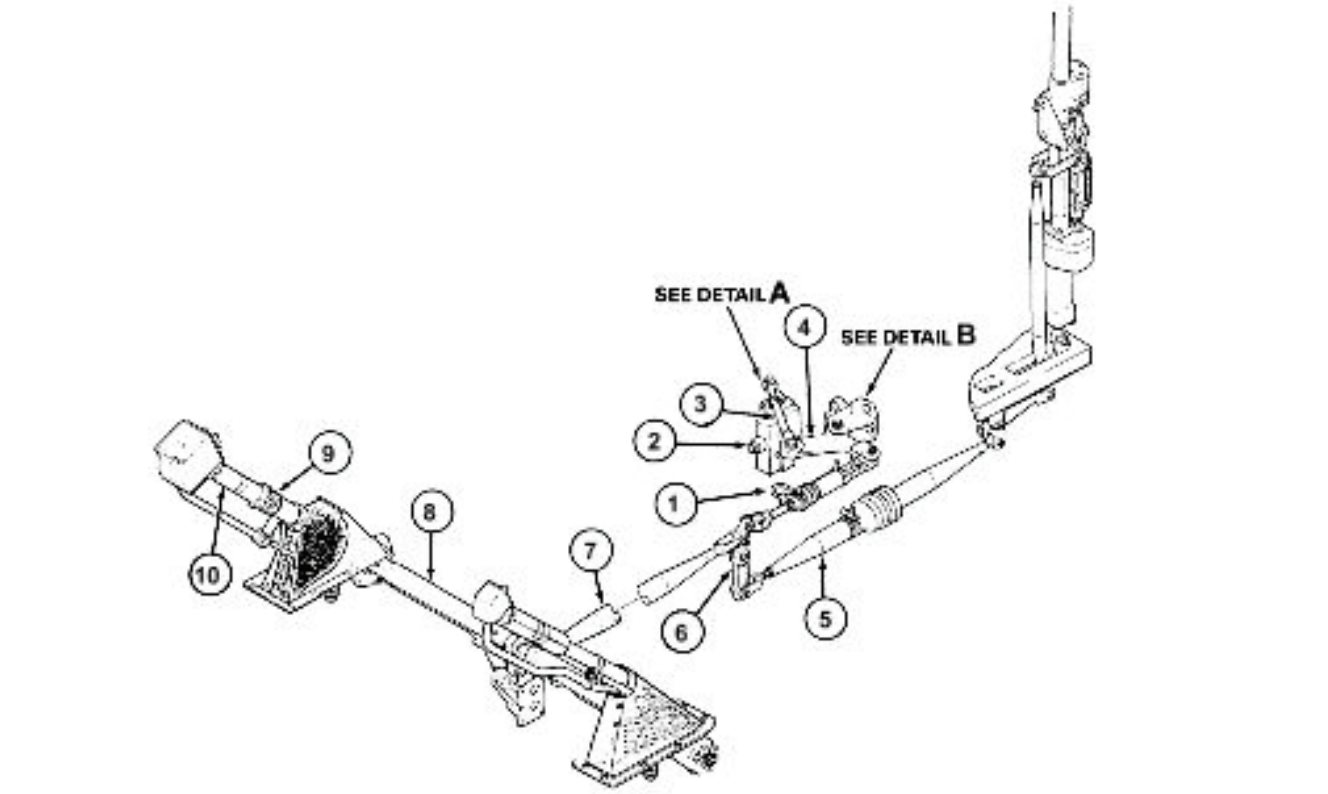
Figure 1. Collective Rigging (Sheet 2 of 3)

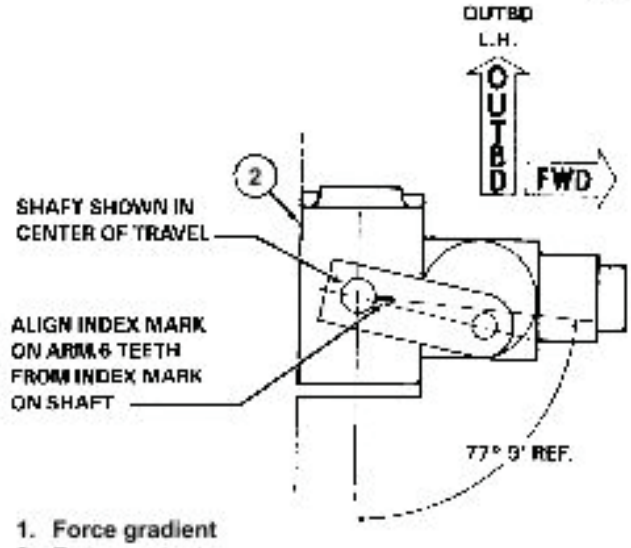


1. Collective lever
2. Extension tube
3. Collective hydraulic cylinder
4. Control tube
5. Support
6. Bellcrank
7. Control tube
8. Lever
9. Stop
10. Control tube
11. Minimum collective friction clamp
12. Collective jackshaft
13. Pilot collective friction adjuster
14. Pilot collective stick
15. Piston
16. Actuator valve lever
17. Pin
18. Swashplate support
19. Transmission
20. Nut
21. Elbow support
22. Gear sector
23. Gear sector
24. Shim
25. Control tube
26. Control tube
27. Collective lever positioning tool (T103287-101)
28. Screw
29. Plate assembly

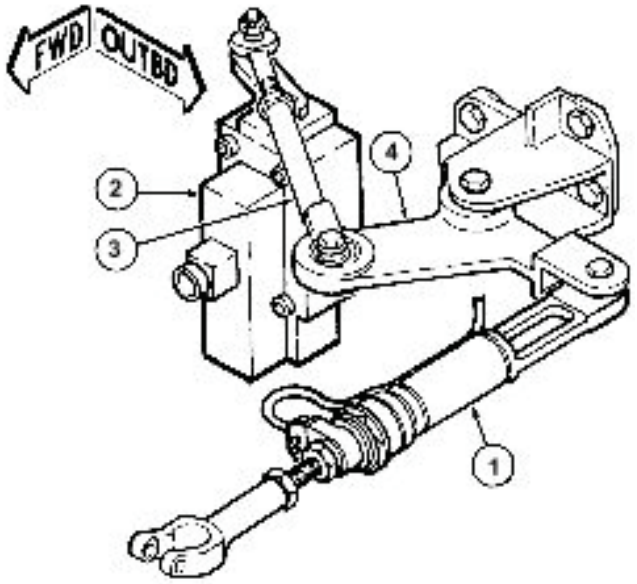
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Figure 1. Collective Rigging (Sheet 3 of 3)





DETAIL A
VIEW LOOKING DOWN



DETAIL B
VIEW LOOKING DOWN

1. Force gradient
2. Rotary actuator
3. Link
4. Bellcrank
5. Control tube
6. Control lever
7. Control tube
8. Collective jackshaft
9. Pilot collective friction adjuster
10. Pilot collective stick

ICN-412-A-670100-B-97499-00004-A-001-01

Figure 2. Collective Controls (4-Axis Configuration) (Sheet 1 of 1)

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Cyclic Controls (Primary Method) - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Inclinometer	Commercial	1	
Swashplate Rigging Tool	T101291-103	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

NOTE

Rigging procedures contained herein pertain to helicopters with dual controls installed in basic control systems.

1. Comply with the following general procedures when rigging cyclic controls (Figure 1).

1.1. When a rigging procedure requires a specific position to accomplish the following steps, manually hold controls in position or secure controls in position with friction adjustment.

1.2. Tolerance for rigging dimensions is ±0.03 inch (0.76 mm).

1.3. All adjustable control tubes shall have no more than 1.0 inch (25.40 mm) of exposed threads or less than 1.5 threads, after adjustment.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable rod bearings and clevises after adjustment.

1.5. Torque control jam nuts to 80 to 100 inch-pounds (9.04 to 11.30 Nm) after adjustment unless other torque is specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. Loosen jam nuts, adjust end fittings, and torque jam nuts, if necessary.

2. Connect hydraulic power cart to helicopter. Do not apply hydraulic power at this time.

NOTE

During rigging, care must be taken to prevent helicopter from moving. Keep helicopter as level as possible.

3. Retract stop bolts as far as possible (Figure 2 and Figure 3).

4. Disconnect the following control tubes and force gradients.

4.1. Disconnect extension tube (2, Figure 1) from swashplate (1) by removing cotter pin, nut, washers, and bolt.

4.2. Disconnect extension tube (3) from swashplate (1) by removing cotter pin, nut, washers, and bolt.

4.3. Disconnect control tube (8) from bellcrank (7) by removing two nuts, four washers, and two bolts, and from valve arm (4) by removing two nuts, four washers, and two bolts. Remove control tube and tag.

4.4. Disconnect control tube (9) from bellcrank (10) by removing two nuts, four washers, and two bolts, and from valve arm (13) by removing two nuts, four washers, and two bolts. Remove control tube and tag.

4.5. Disconnect fore and aft cyclic force gradient (17) and clevis (20) from jackshaft (16) by removing nut and washer.

4.6. Disconnect fore and aft cyclic transducers (34) from jackshaft (16) by removing nut, lock washer, two washers, and bolt.

4.7. Disconnect control tube (22) from jackshaft (16) by removing nut, three washers, and bolt, and remove control tube (24) from bellcrank (31) by removing nut, three washers, and bolt.

4.8. Disconnect lateral cyclic transducers (29) from bellcrank (31) by removing nut, two washers, two spacers, and bolt.

4.9. Disconnect lateral cyclic force gradient (26) from bellcrank (31) by removing cotter pin, nut, two washers, bushing and bolt.

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4.10. Disconnect lateral control tube (30) from pilot cyclic stick (33) by removing nut, two washers, and bolt. Remove control tube and tag.

5. Verify length of centered fore and aft cyclic series actuator (14, Figure 1) of 27.505 to 27.555 inches (698.627 to 699.897 mm).

6. Verify length of centered lateral cyclic series actuator (35) of 56.295 to 56.345 inches (1429.893 to 1431.163 mm).

7. Level helicopter (412-A-08-40-00-00A-169A-A / 00085).

8. Position pilot cyclic stick (33) with respect to deck using an inclinometer, bubble protractor, or equivalent as follows:

8.1. Place pilot cyclic stick (33) against extreme forward stop. Measure and record angle of stick relative to deck.

8.2. Place pilot cyclic stick (33) against extreme aft stop. Measure and record angle of stick relative to deck.

8.3. Place pilot cyclic stick (33) against extreme right lateral stop. Measure and record angle of stick relative to deck.

8.4. Place pilot cyclic stick (33) against extreme left lateral stop. Measure and record angle of stick relative to deck.

8.5. Position pilot cyclic stick (33) at mid-position within 0.5° utilizing 1/2 total stick deflections as measured in Step 8.1. and Step 8.2. (fore and aft cyclic) and Step 8.3. and Step 8.4. (lateral cyclic). Tighten friction adjustment knob (32) to maintain this position.

NOTE

SN: 33001-33061:

Helicopters have non-adjustable control tube (30) 204-001-016-013 installed. Control tube 204-001-016-013 replaced by adjustable control tube 412-001-307-101.

NOTE

SN: 33062-33213 or 36001-36999 or 37002-37999 or 38001-38999 or 39101-39999:

Helicopters use adjustable control tube 412-001-307-101.

9. Check and adjust (if necessary) control tube (30) between pilot cyclic stick (33) and bellcrank (31). Verify dimension A and B between bellcrank and bulkhead are equal within 0.030 inch (0.762 mm) (Detail C). Torque jam nut and install control tube.

10. Adjust rod end bearing (52, Detail E) on left cyclic cyclic hydraulic cylinder (5) as follows:

10.1. Remove cotter pin (41), nut (42), washer (43), and bolt (39) from support (40).

10.2. Push lower actuator body (48) full up and pull lower piston (49) full down (Detail E).

10.3. Remove lockwire and loosen jam nut (51) and adjust rod end bearing (52) to align bolt holes in rod end bearing and support (40).



IMPROPER TORQUE TECHNIQUE CAN DESTROY TAB ON LOCKING DEVICE (50). ROD END BEARING (52) MUST BE HELD SECURELY TO PREVENT DAMAGING LOCKING DEVICE.

10.4. Torque jam nut (51) 200 to 250 inch-pounds (22.60 to 28.25 Nm) and secure to locking device (50) with lockwire (C-405).

10.5. Inspect locking device (50) to ensure tab is in groove of rod end bearing (52) and that it was not damaged when jam nut (51) was tightened. Replace locking device if tab is twisted or damaged.

NOTE

Alignment procedures are applicable to all three cylinders.

11. Align left cyclic hydraulic cylinder (5) to support (40) as follows:

11.1. Make sure extension tube (2) is disconnected from swashplate (1).

11.2. Move lower piston rod up and clear of support (40). Lower piston rod end bearing (52) must be able to be moved freely down, up, and then down, and bolt (39) replaced without any pressure, forcing, binding, or bending of piston rod in fore, aft, left, or right direction. One person must guide extension tube (2) to prevent damage when conducting this step.

11.3. If cylinder meets requirements of Step 11.2., it is considered in alignment. If cylinder is aligned, proceed to Step 11.4.. If cylinder is misaligned, accomplish Step 11.5. through Step 11.13..

11.4. Install bolt (39), washer (43), nut (42), and cotter pin (41). Loosely install upper extension tube (2) to swashplate by installing washer and bolt.

11.5. Remove lockwire (47) from jam nut (46), bearing nut (45), and keyed washer (53).

11.6. Loosen jam nut (46) and bearing nut (45) so cylinder is loose.

11.7. Move rod end bearing (52) up approximately 0.5 inch (12.7 mm) above support (40). Rotate lower end of cylinder in a 3.0 inch (76.2 mm) circle.

11.8. Position rod end bearing (52) directly over lugs of support (40) and move piston down and install bolt (39).

11.9. Tighten bearing nut (45) by hand (approximately 70 inch-pounds (7.91 Nm).

11.10. Remove bolt (39) and move rod end bearing up and out of support (40) then move down. Rod end bearing (52) must be free of any side loading. Repeat Step 11.5. through Step 11.10. until free of side loading.

11.11. Install bolt (39). Torque bearing nut (45) to 200 inch-pounds (22.60 Nm).

11.12. If cylinder meets requirements of Step 11.2., install washer (43) and nut (42) and torque to 95 to 110 inch-pounds (10.73 to 12.41 Nm). Install cotter pin (41). Complete installation of upper extension tube (2) by installing two washers, cotter pin, and the nut.

11.13. Torque bearing nut (45) to 900 to 1100 inch-pounds. (101.68 to 124.28 Nm). Torque jam nut (46) to 900 to 1100 inch-pounds (101.68 to 124.28 Nm). Secure nuts (45 and 46) to keyed washer (53) with lockwire (C-405).

11.14. Connect control tube (8) to valve arm (4), by installing bolt, washers, and nut. Leaving tube disconnected from bellcrank (7).

NOTE

Repeat Step 10. and Step 11. on right cyclic hydraulic cylinder (12). With main rotor pitch links connected, all cyclic and collective adjustments or checks shall be accomplished using a hydraulic boost.

12. Turn on hydraulic power cart and apply hydraulic power to helicopter.



WHEN PERFORMING OPERATIONAL CHECK OF CYCLIC CONTROLS WITH HYDRAULIC SYSTEM PRESSURIZED, ENSURE COLLECTIVE STICK IS POSITIONED NEAR MID TRAVEL.

13. Adjust control tubes (2 and 3, Figure 2) as follows:

13.1. Adjust control tubes (2 and 3) to a length of 20.86 inches (529.84 mm).

13.2. Torque jam nuts to 80 to 100 inch-pounds (9.04 to 11.30 Nm) on control tubes (2 and 3).

13.3. *Connect each control tube (8 and 9, Figure 1) to each bellcrank (7 and 10) temporarily with bolts, washers, nuts, and cotter pins.*

14. Make sure that pilot cyclic stick (33) is perpendicular to deck within 0.5° (as determined in Step 8.) and is held in this position.

15. Disconnect four pitch link assemblies from pitch horns (412-A-62-30-00-02A-520A-A / 00961).

16. Set swashplate angle with respect to mast using swashplate rigging tool (T101291-103) as follows (Figure 4):

16.1. Ensure helicopter is level (412-A-08-40-00-00A-169A-A / 00085).

16.2. Position swashplate rigging tool (T101291-103) on rotating portion of swashplate with swashplate positioned fore and aft.

16.3. Set swashplate angle with respect to mast utilizing an inclinometer, or equivalent, located on upper surface of rigging tool as follows:

16.3.1. Fore and aft: set swashplate 0° 33' to 1.0° 3' down forward.

16.3.2. Lateral: set swashplate 1.0° 0' to 1.0° 30' down left.

NOTE

Provide a minimum of 1.5 exposed threads to left extension tube (2, Figure 1) upper clevis. If required at completion of Step 18., it is permissible to equally shorten control tubes (8 and 9) a maximum of five complete turns of adjustable rod end bearing. Shortening control tube assemblies (8 and 9) adjustable rod end bearing 1.5 turns is equal to approximately one thread of extension tube (2 and 3) upper clevis.

17. Hold swashplate in position, adjust length of right hydraulic cylinder extension tube (8, Figure 4) to fit swashplate. Adjust length of left hydraulic cylinder extension tube to fit swashplate.

18. Torque nuts (9) on both extension tubes (8) to 1500 to 1800 inch-pounds (169.47 to 203.37 Nm) and secure with lockwire (C-405). Apply sealant (C-308) to nut, lockwire, threads of clevis, and both extension tubes.

19. Connect clevis (10) to swashplate horn (1) by installing washer (3), bolt (2), washers (4 and 5), nut (6), and cotter pin (7).

20. Remove swashplate rigging tool (T101291-103) and inclinometer.

21. Connect four pitch links to pitch horns (412-A-62-30-00-02A-720A-A / 00965).

22. Set stop bolts on bellcranks (6 and 11, Figure 2) as follows:

22.1. Position pilot cyclic stick (33, Figure 1) perpendicular to deck within 0.5° and tighten friction adjustment knob (32).

NOTE

Aluminum strip work aids may be used in lieu of making marks on hydraulic cylinders as described in Step 22.2.. through Step 22.4.. Fabricate two strips of aluminum from aluminum sheet. Scribe index marks on aluminum strips to correspond to marks described in Step 22.2.. through Step 22.4.. Attach strips to cylinder (1 and 4, Figure 2) with tape or rubber cushion clamps.

22.2. Make an index mark on each cyclic hydraulic cylinder and a corresponding mark on any suitable point on airframe structure with a soft lead pencil as shown on Figure 2. Label index mark "N" (neutral, and record dimensions for each actuator.

22.3. Measure 1.61 in. (40.89 mm) above index mark “N” (neutral) on the hydraulic cylinder and mark with soft lead pencil on cylinder. Label index mark “F” (forward).

22.4. Measure 1.61 in. (40.89 mm) below index mark “N” neutral on the hydraulic cylinder and mark another mark with soft lead pencil on cylinder. Label index mark “A” (aft).

22.5. Loosen friction adjustment knob (32, Figure 1) on pilot cyclic (33) and position stick against extreme left lateral stop, then move stick toward forward left corner position until index mark “F” on left hydraulic cylinder (4, Figure 2) coincides with recorded dimensions for that actuator. Hold this position and adjust stop bolt (7) in bellcrank (6) to make contact with actuator support (8).

22.6. Position pilot cyclic against extreme left lateral stop, then move stick toward aft left corner position until index mark “A” on right hydraulic cylinder (1) coincides with recorded dimension for that actuator. Hold this position and adjust stop bolt (10) in stop (14) to make contact with bellcrank (11).

22.7. Position pilot cyclic against extreme right lateral stop then move stick toward forward right corner position until index mark “F” on right hydraulic cylinder (1) coincides with recorded dimensions for that actuator. Hold this position and adjust stop bolt (3, Figure 3) in bellcrank (2) to make contact with actuator support (4).

22.8. Position pilot cyclic against extreme right lateral stop. Move stick toward aft right corner position until index mark “A” on left hydraulic cylinder (4, Figure 2) coincides with recorded dimension for that actuator. Hold this position and adjust stop bolt (6, Figure 3) in stop (10) to make contact with bellcrank (7).

23. Turn off hydraulic power cart.

24. Adjust copilot control tubes (22 and 24, Figure 1) as follows:

24.1. Place pilot cyclic (33) and copilot cyclic (15) in extreme left position against stops and hold. Adjust control tube (24) to fit bellcrank (31). Torque jam nut on clevis on control tube (24). Install control tube (24) on bellcrank (31) by installing bolt, washer, nut and cotter pin.

24.2. Place pilot and copilot cyclics in extreme forward position against stops and hold. Adjust control tube (22) to fit jackshaft (16). Torque jam nut on clevis on control tube (22). Install control tube (22) on jackshaft (16) by installing bolt, washer, nut and cotter pin.

25. Turn on hydraulic power cart and apply hydraulic power to helicopter.



WHEN PERFORMING OPERATIONAL CHECK OF CYCLIC CONTROLS WITH HYDRAULIC SYSTEM PRESSURIZED, ENSURE COLLECTIVE STICK IS POSITIONED NEAR MID TRAVEL.

26. Move cyclic controls through full travel and check for binding and/or obstruction of control movement.

27. Turn off hydraulic power cart.

28. Perform fore and aft cyclic stick centering as follows:

28.1. Position output arm on fore and aft cyclic rotary actuator (Figure 1, Detail A).

28.2. Place pilot cyclic (33, Figure 1) against extreme forward stop.

28.3. Place output arm of the fore and aft rotary actuator (18) against aft stop.

29. Adjust clevis (20) attached to force gradient (17) in fore and aft cyclic control system to fit jackshaft (16). Extend clevis 3.5 to 4.0 turns and torque jam nut.

30. Install fore and aft cyclic force gradient (2, Figure 8) onto fore and aft cyclic rotary actuator (5) with nut (4) and washers (14 and 15). Safety the nut with a new cotter pin (3).

31. Install clevis (13) onto tube and lever assembly (1), with bolt (12) and washer (11). Tighten the bolt. Safety the bolt with lockwire (C-405)

NOTE

Check for approximately 0.9 inch (22.86 mm) cushion at stick grip when against forward cyclic stick stop.

32. Perform lateral cyclic stick centering as follows:

32.1. *Position output arm on lateral cyclic rotary actuator (Figure 5, Detail B).*

32.2. Check clearance between lateral cyclic force gradient (26, Figure 1) and structure at extreme positions of lateral cyclic rotary actuator (25) arm. If required, re-index arm relative to shaft by one serration for clearance.

32.3. Place pilot cyclic against right lateral stop.

32.4. Place output arm of lateral cyclic rotary actuator (25) to extreme left position against stop.

32.5. If necessary to extend clevis (28) to adjust clevis on lateral cyclic force gradient (26), remove control tube (24) from bellcrank (31) by removing cotter pin, nut, three washers, and bolt.

32.6. Adjust clevis (28) attached to lateral cyclic force gradient (26) to fit bellcrank (31) and temporarily install.

32.7. Place one end of a 6 inch (15.24 cm) (or similar measuring tool) scale against end cap of lateral cyclic force gradient (26) and measure to approximate center of clevis (28). Mark center with soft pencil and record dimension.

32.8. Place and hold pilot cyclic against left lateral stop. Measure and record total travel distance of clevis (28). Extend clevis one half recorded distance. Torque jam nut and install on bellcrank (1, Figure 9) with nut (10), washers (11 and 3), and bolt (2). Safety the nut with a new cotter pin (9).

32.9. Install control tube (24, Figure 1) to bellcrank (31) with nut, three washers (one under bolt head and two under nut) and bolt. Safety the nut with a new cotter pin.

33. Perform fore and aft cyclic transducer (34) adjustment as follows:

NOTE

SN: 33001-33213 or 36001-36086:

The following procedure applies to helicopters with transducer P/N 412-074-101-101 installed.

NOTE

SN: 36037-36086:

For helicopters with transducer P/N 412-074-101-103 installed, refer to 412-A-67-01-00-10A-276A-A / 01550.

33.1. With fore and aft cyclic transducer attached to structure, position pilot cyclic against extreme forward stop, then against extreme aft stop. Check if transducers would bottom out if connected to jackshaft (Figure 6).

33.2. If bottoming occurs, loosen jam nut, adjust rod end bearings on transducer until bottoming is eliminated, torque jam nut, then connect transducers to jackshaft. Overtravel of transducers should be approximately equal.

34. Perform lateral cyclic transducer (29, Figure 1) adjustment as follows:

NOTE

SN: 33001-33213 or 36001-36086:

The following procedure applies to helicopters with transducer P/N 412-074-101-101 installed.

NOTE

SN: 36037-36086:

For helicopters with transducer P/N 412-074-101-103 installed, refer to 412-A-67-01-00-10A-276A-A / 01550.

34.1. With transducer attached to structure (Figure 6), position pilot cyclic against extreme right lateral stop, then against extreme left lateral stop. Check if transducers bottom out if connected to bellcrank (Figure 6).

34.2. If bottoming occurs, loosen jam nut, adjust rod end bearings on transducers until bottoming is eliminated, torque jam nut, then connect to bellcrank. Overtravel of transducers should be approximately equal.

35. Turn on hydraulic power cart and apply hydraulic power to helicopter.



WHEN PERFORMING OPERATIONAL CHECK OF CYCLIC CONTROLS WITH HYDRAULIC SYSTEM PRESSURIZED, ENSURE COLLECTIVE STICK IS POSITIONED NEAR MID TRAVEL.

36. Perform cyclic minimum friction adjustment (Figure 7):

NOTE

Initial adjustment and check of fore and aft and lateral cyclic force shall be accomplished by utilizing a hydraulic power cart connected to helicopter using Step 36.1. through Step 36.7.. Final check and adjustment shall be accomplished during ground run of helicopter using Step 36.2. through Step 36.7..

36.1. Reduce cyclic friction to zero by loosening clamp. Rotate nut counterclockwise.

36.2. With force trim OFF, slowly increase stick friction while simultaneously pulling on cyclic stick with spring scale attached at midpoint of pilot cyclic grip until a fore and aft cyclic breakout force of 0.50 to 1.0 pound (2.22 to 4.44 N) is obtained.

36.3. Forces to be measured perpendicular to axis of cyclic stick with stick near center.

36.4. Check lateral cyclic breakout force of 0.50 to 1.00 pound (2.22 to 4.44 N).

36.5. If lateral cyclic stick force exceeds above requirement, adjust fore and aft cyclic friction breakout within above limits and recheck lateral cyclic breakout force (Step 36.2. through Step 36.4.).

36.6. Tighten clamp around sleeve with clamp resting on top surface of nut (Figure 7). Tighten screws to obtain equal gap between clamp halves.

NOTE

SN: 36087-Subsequent:

Step 36.7. through Step 36.9. are not applicable.

36.7. With force trim ON, check breakout force of fore and aft cyclic of 2.25 to 2.75 pounds (10.01 to 12.23 N) and lateral force of 0.50 to 2.00 pounds (2.22 to 8.90 N).

36.8. If fore and aft or lateral breakout forces exceed above requirements, repeat Step 36.1. through Step 36.6..

36.9. If fore and aft or lateral breakout forces still exceed above requirements, readjust either fore and aft cyclic force gradient or lateral cyclic force gradient spring preloads within respective limits to obtain required breakout force. Repeat Step 36.1. through Step 36.7. with hydraulic power applied.

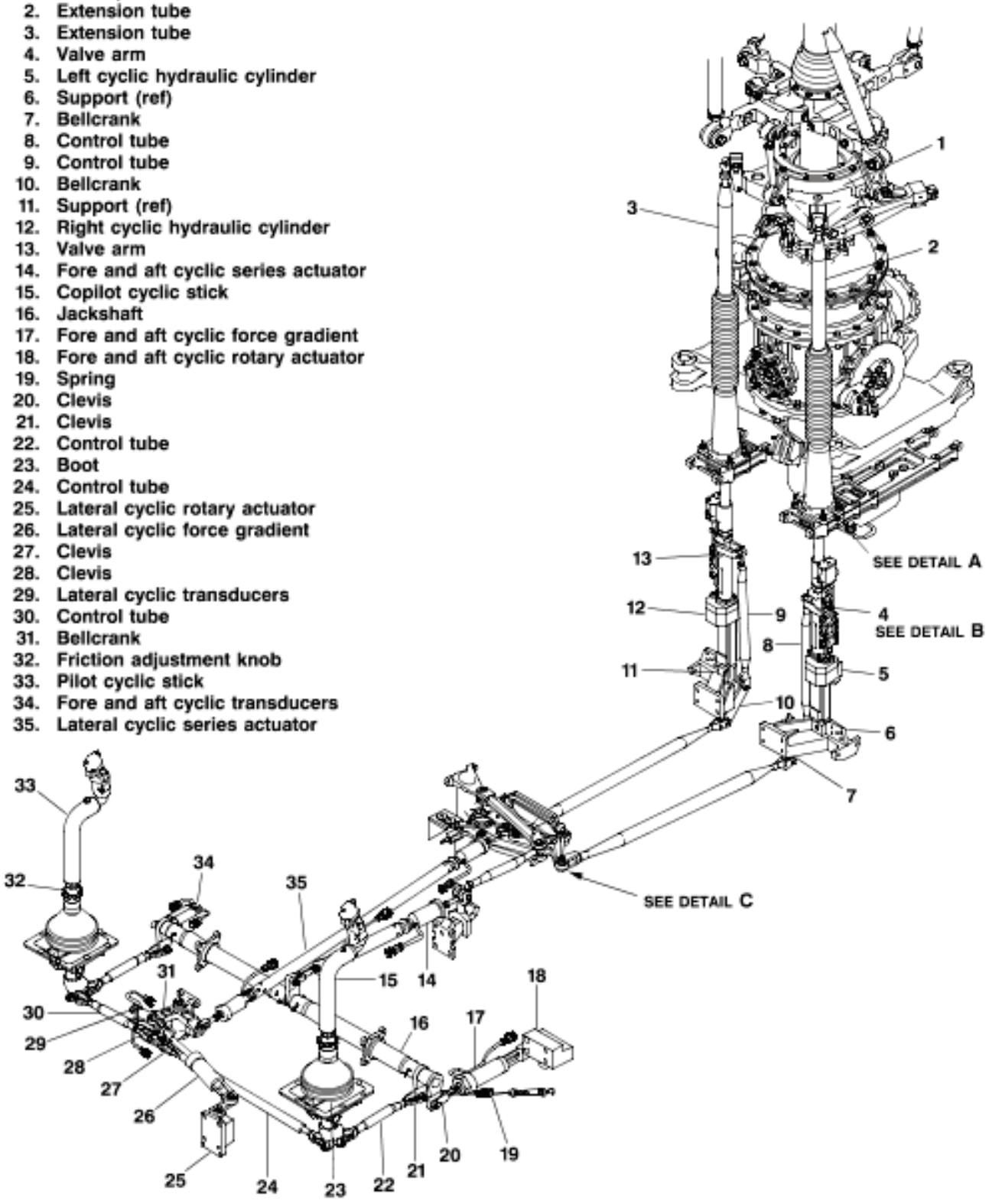
NOTE

For cyclic force gradient spring preload adjustments and cyclic control stick mass balance adjustment, refer to 412-A-67-12-00-16A-720A-A / 01458 and 412-A-67-12-00-17A-720A-A / 01462.

37. Remove hydraulic power.

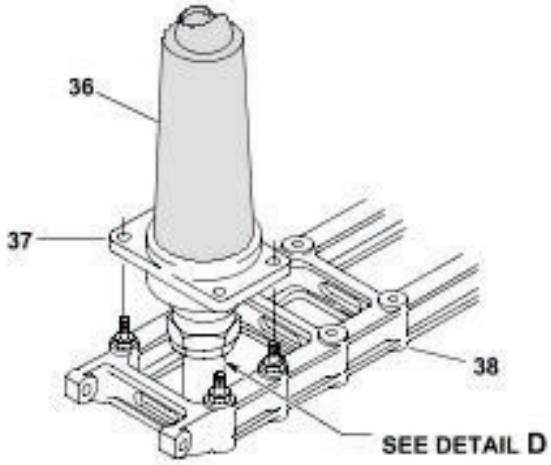
38. Accomplish rigging of cyclic controls; position sensors in (412-A-67-01-00-05A-276A-A / 01326).

1. Swashplate
2. Extension tube
3. Extension tube
4. Valve arm
5. Left cyclic hydraulic cylinder
6. Support (ref)
7. Bellcrank
8. Control tube
9. Control tube
10. Bellcrank
11. Support (ref)
12. Right cyclic hydraulic cylinder
13. Valve arm
14. Fore and aft cyclic series actuator
15. Copilot cyclic stick
16. Jackshaft
17. Fore and aft cyclic force gradient
18. Fore and aft cyclic rotary actuator
19. Spring
20. Clevis
21. Clevis
22. Control tube
23. Boot
24. Control tube
25. Lateral cyclic rotary actuator
26. Lateral cyclic force gradient
27. Clevis
28. Clevis
29. Lateral cyclic transducers
30. Control tube
31. Bellcrank
32. Friction adjustment knob
33. Pilot cyclic stick
34. Fore and aft cyclic transducers
35. Lateral cyclic series actuator

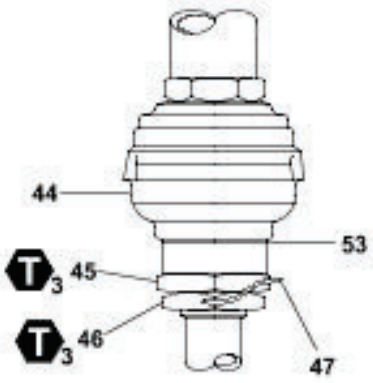


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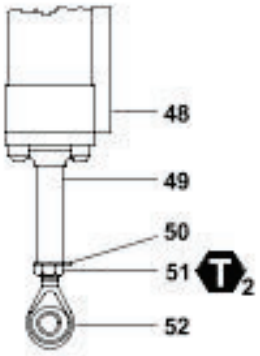
Figure 1. Cyclic Rigging (Sheet 1 of 3)



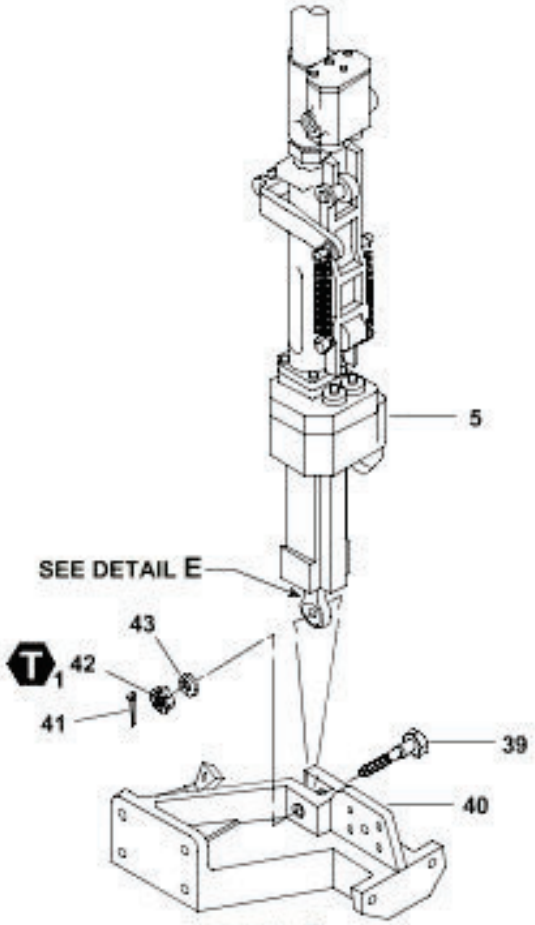
DETAIL A



DETAIL D



DETAIL E



DETAIL B
LEFT CYCLIC FLIGHT
CONTROL CYLINDER

36. Boot
37. Support
38. Support
39. Bolt
40. Support (ref)
41. Cotter pin
42. Nut
43. Washer
44. Support bracket
45. Nut
46. Nut
47. Lockwire
48. Lower actuator body
49. Lower piston
50. Locking device
51. Jam nut
52. Rod end bearing
53. Keyed washer

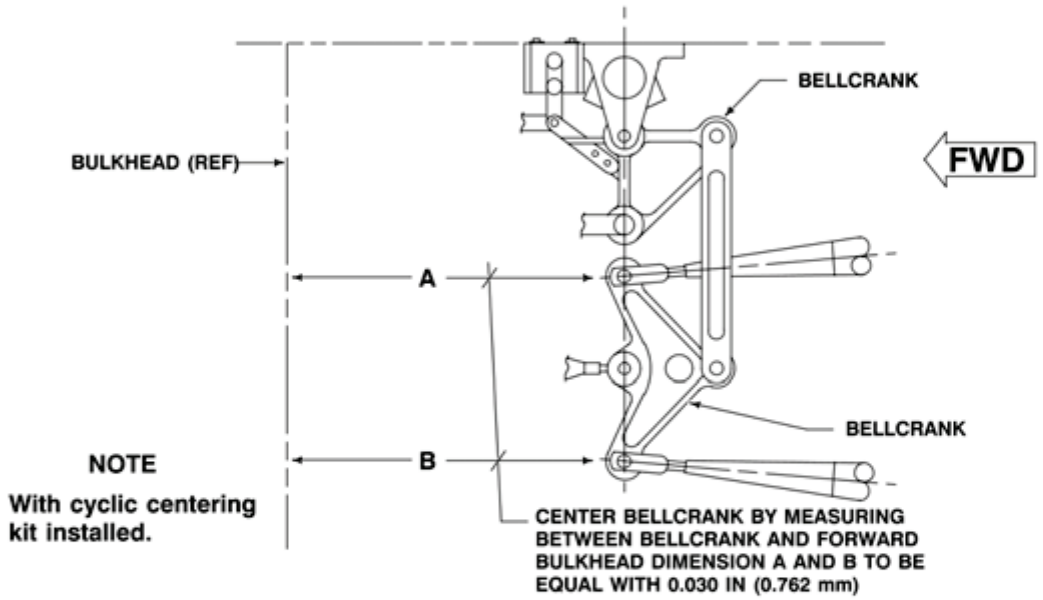
T1 95 TO 110 IN-LBS (10.73 TO 12.43 Nm)

T2 200 TO 250 IN-LBS (22.60 TO 28.25 Nm)

T3 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

ICN-412-A-670100-B-97499-00006-A-001-01

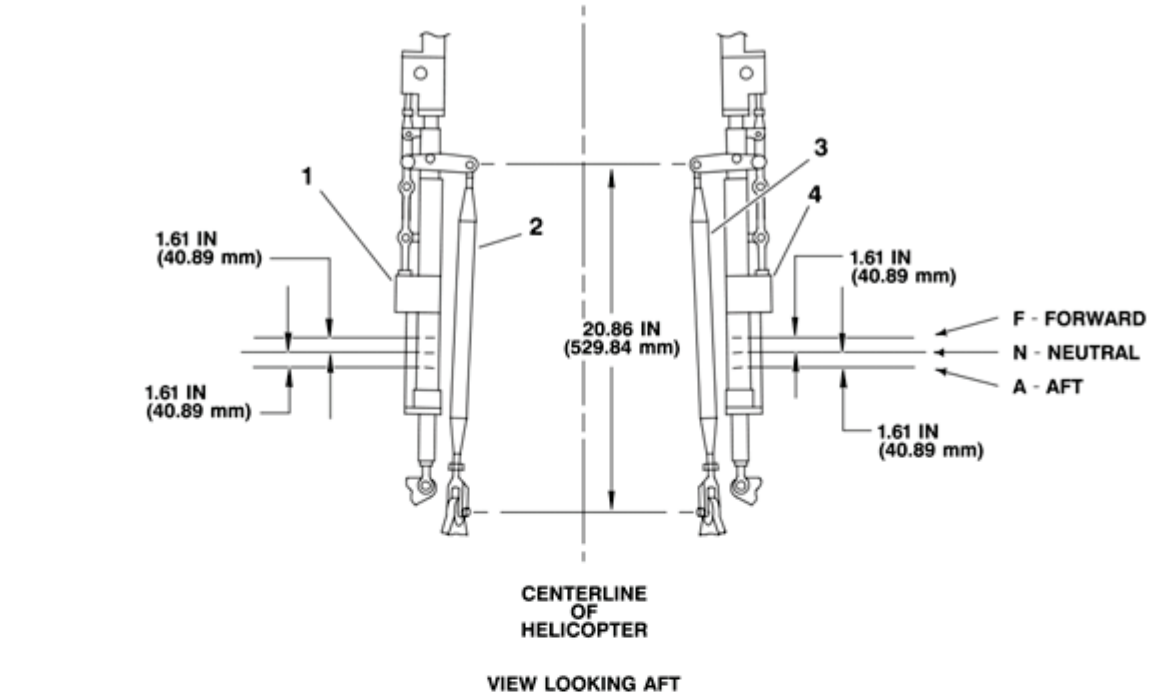
Figure 1. Cyclic Rigging (Sheet 2 of 3)

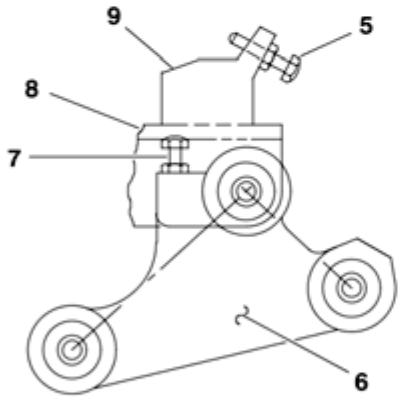


DETAIL C
VIEW LOOKING DOWN
MIXING LEVER

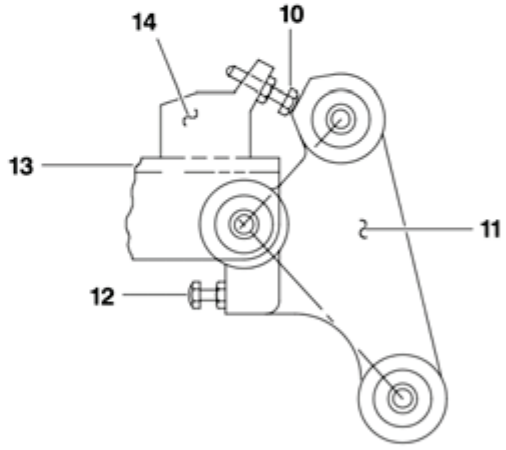
ICN-412-A-670100-B-97499-00007-A-001-01

Figure 1. Cyclic Rigging (Sheet 3 of 3)





POSITION OF BELLCRANK (6)
BELOW LEFT CYCLIC CYLINDER
(4) FOR EXTREME FORWARD
LEFT CORNER STICK POSITION

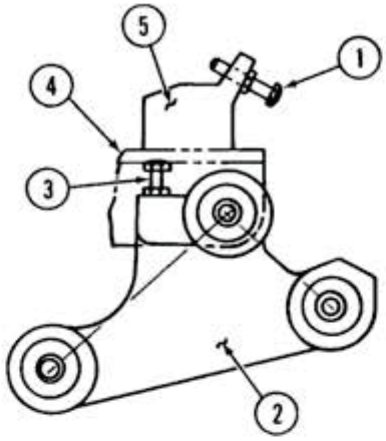


POSITION OF BELLCRANK (11)
BELOW RIGHT CYCLIC CYLINDER
(1) FOR EXTREME AFT LEFT
CORNER STICK POSITION

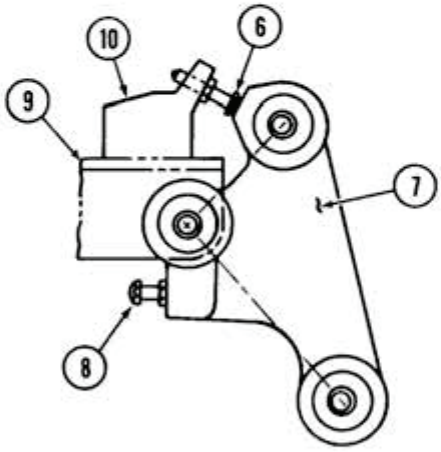
1. Right cyclic hydraulic cylinder
2. Control tube
3. Control tube
4. Left cyclic hydraulic cylinder
5. Stop bolt
6. Bellcrank
7. Stop bolt
8. Actuator support
9. Stop
10. Stop bolt
11. Bellcrank
12. Stop bolt
13. Actuator support
14. Stop

ICN-412-A-670100-B-97499-00008-A-001-01

Figure 2. Cyclic Rigging - Hydraulic Cylinders (Sheet 1 of 1)



POSITION OF BELLCRANK (2)
BELOW RIGHT CYCLIC CYLINDER
(1) FOR EXTREME FORWARD
RIGHT CORNER STICK POSITION

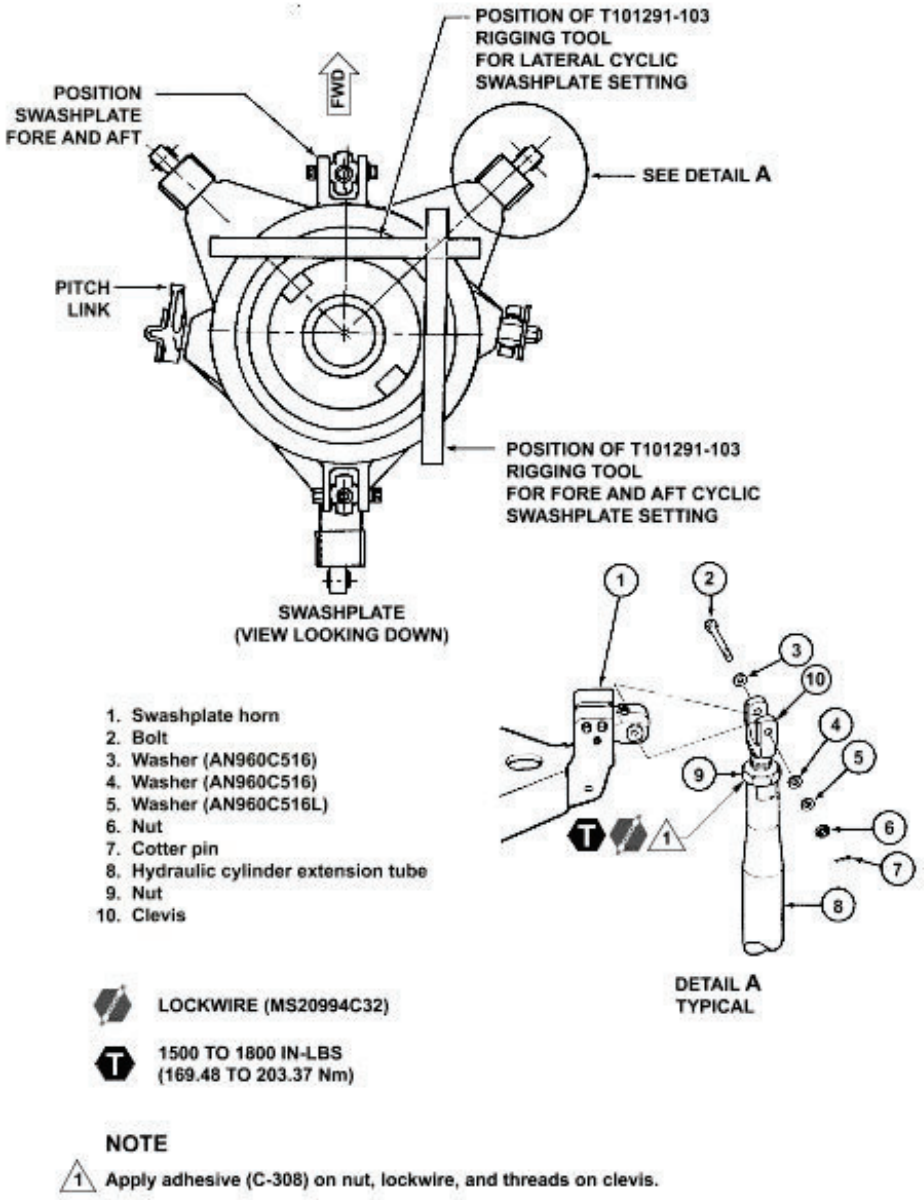


POSITION OF BELLCRANK (7)
BELOW LEFT CYCLIC CYLINDER
(4) FOR EXTREME AFT RIGHT
CORNER STICK POSITION

1. Stop bolt
2. Bellcrank
3. Stop bolt
4. Actuator support
5. Stop
6. Stop bolt
7. Bellcrank
8. Stop bolt
9. Actuator support
10. Stop

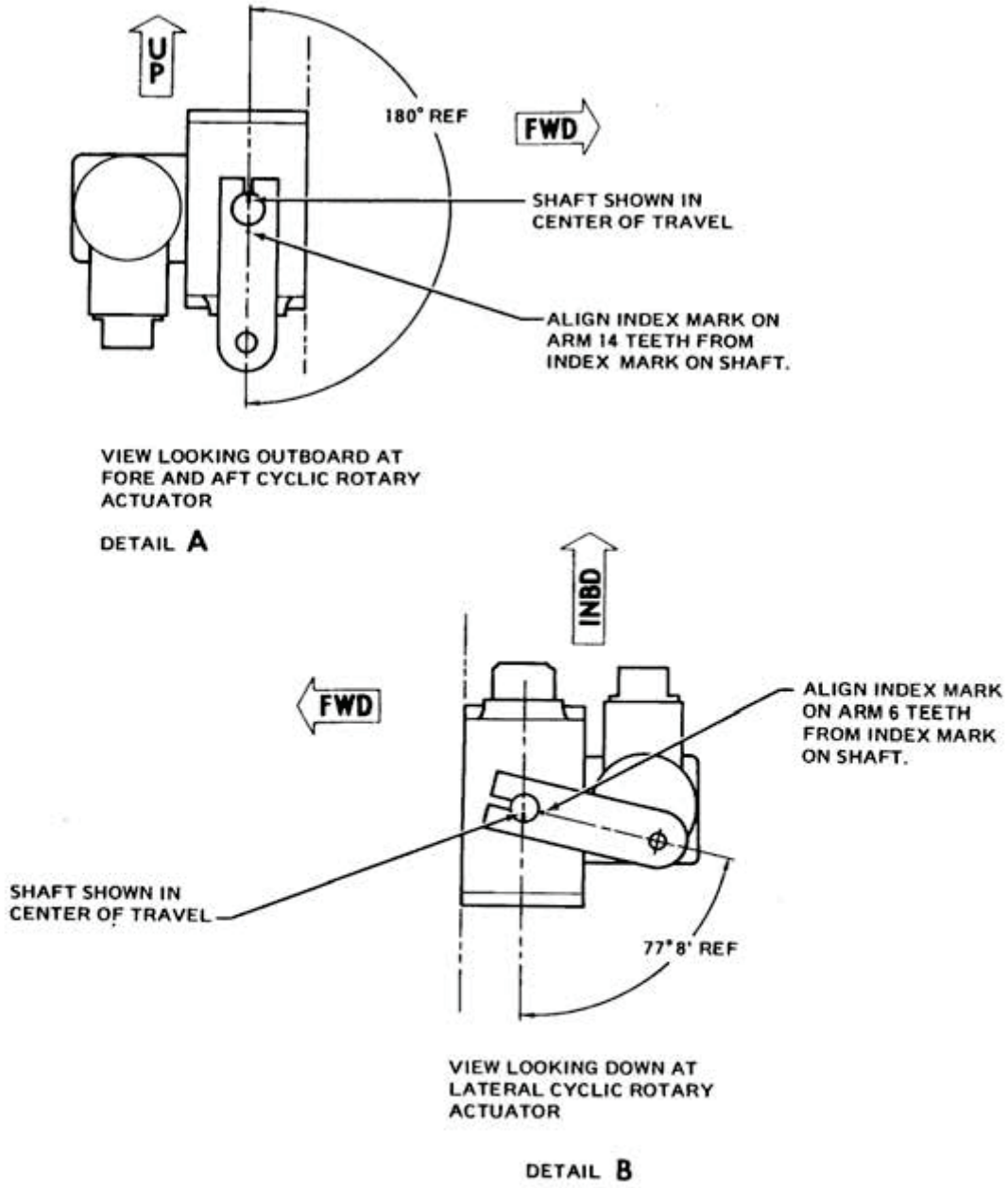
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Figure 3. Cyclic Rigging - Bellcrank Stops (Sheet 1 of 1)



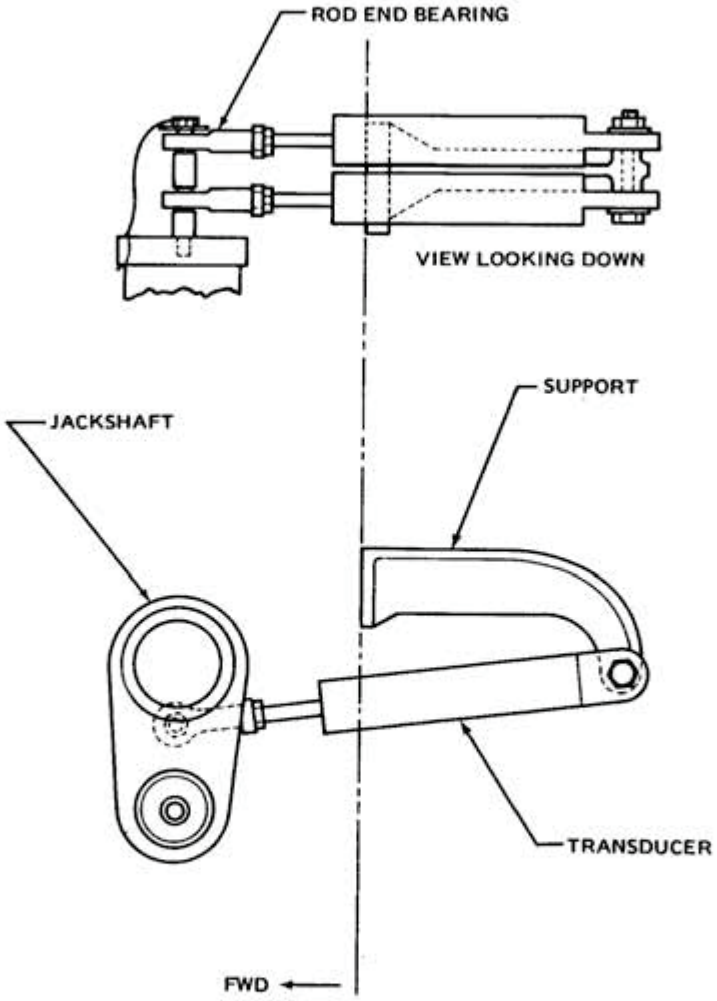
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Figure 4. Cyclic Rigging - Swashplate (Sheet 1 of 1)



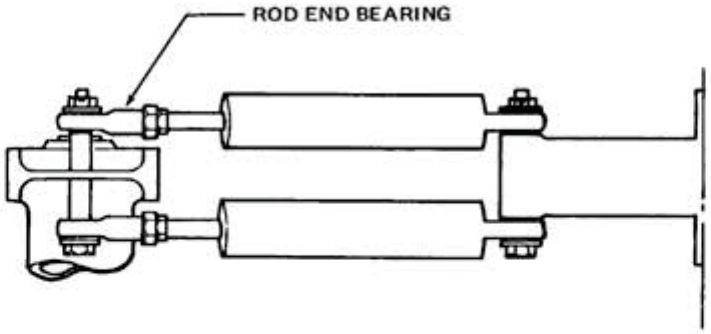
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Figure 5. Cyclic Rigging - Actuators (Sheet 1 of 1)

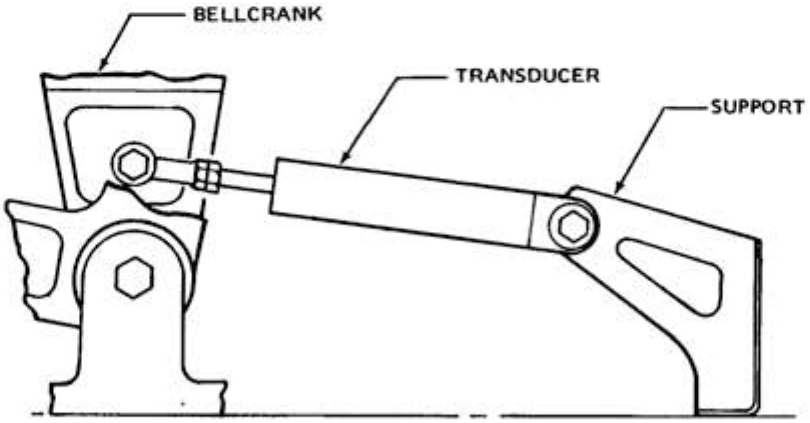


ICN-412-A-670100-B-97499-00012-A-001-01

Figure 6. Cyclic Rigging - Transducer (Sheet 1 of 2)



VIEW LOOKING FORWARD
LATERAL CYCLIC TRANSDUCER

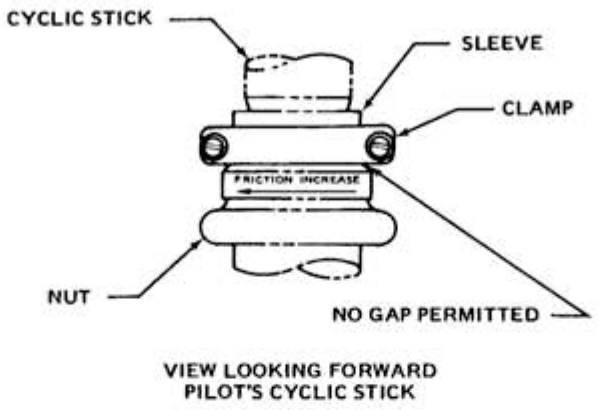


VIEW LOOKING DOWN LATERAL

CYCLIC TRANSDUCER

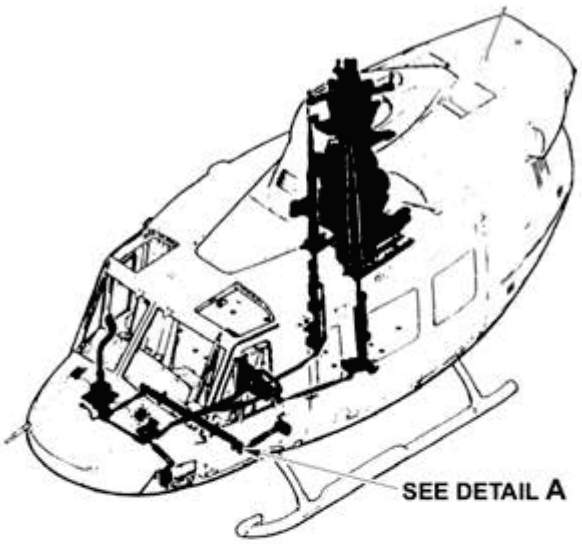
ICN-412-A-670100-B-97499-00013-A-001-01

Figure 6. Cyclic Rigging - Transducer (Sheet 2 of 2)



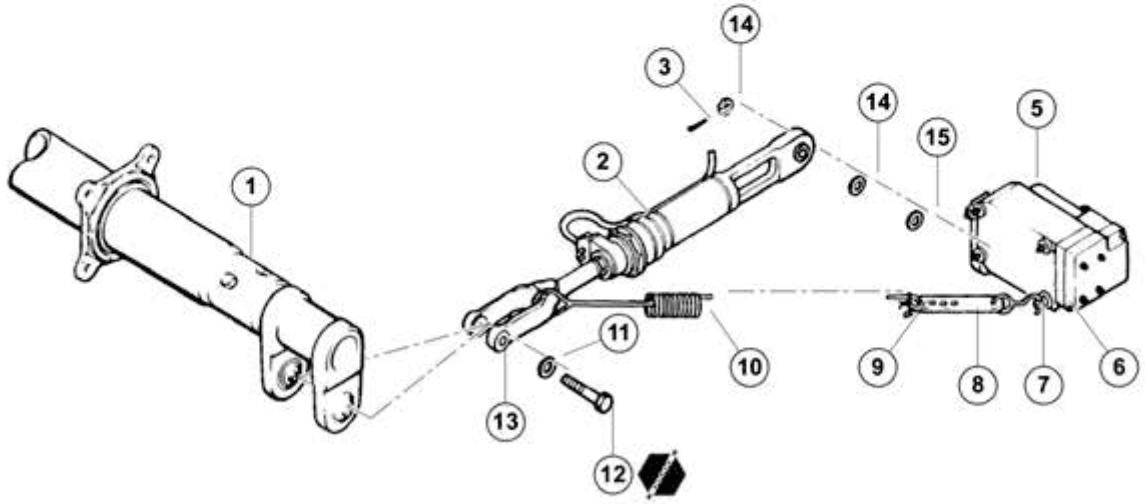
ICN-412-A-670100-B-97499-00014-A-001-01

Figure 7. Cyclic Friction Adjustment (Sheet 1 of 1)



1. Tube and lever assembly
2. Fore and aft cyclic controls force gradient
3. Cotter pin
4. Nut
5. Rotary actuator
6. Bolt (212-001-315-1)
7. Hook
8. Plate
9. Lockwire
10. Spring
11. Aluminum washer
12. Bolt
13. Clevis
14. Washer
15. Washer

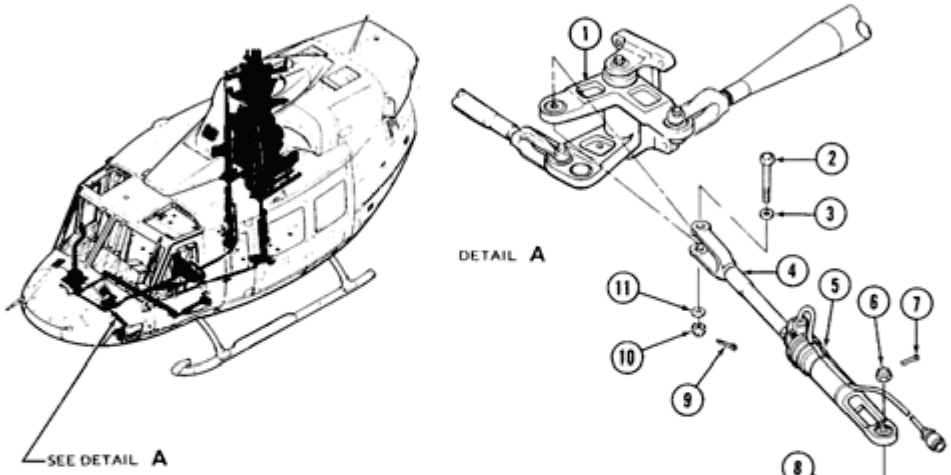
LOCKWIRE (C-405)



DETAIL A

ICN-412-A-671200-B-97499-00021-A-001-01

Figure 8. Fore and Aft Cyclic Force Gradient (Sheet 1 of 1)



1. Lateral cyclic control bellcrank
2. Bolt
3. Aluminum washer
4. Link
5. Lateral cyclic controls force gradient
6. Nut
7. Cotter pin
8. Lateral cyclic controls rotary actuator
9. Cotter pin
10. Nut
11. Aluminum washer

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Figure 9. Lateral Cyclic Force Gradient (Sheet 1 of 1)

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Cyclic Controls (Alternate Method) - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Cyclic Corner Stop Rigging Tool	T103281-101	1	
Cyclic Corner Stop Rigging Tool	T103281-102	1	
Cyclic Mixing Bellcrank Rigging Tool	T103286-101	1	
Collective Lever Rigging Tool	T103287-111	1	
Swashplate Rigging Tool	T103288-101	1	
Cyclic Stick Rigging Tool	T103347-101	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

NOTE

The procedures contained herein describe the alternate method of rigging cyclic control system using special rigging tools listed above.

1. Comply with the following general procedures when rigging cyclic controls (Figure 1).

1.1. When a rigging procedure requires a specified position to accomplish the following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ± 0.03 inch (0.76 mm).

1.3. All adjustable control tubes shall have no more than 1.0 inch (25.4 mm) of exposed threads, or less than 1.5 threads, after adjustment.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable rod bearings and clevises after adjustment.

1.5. Torque control jam nuts 80 to 100 inch-pounds (9.04 to 11.30 Nm) after adjustment unless other torque is specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. Loosen jam nuts, adjust end fittings, and torque jam nuts, if necessary.

2. Connect hydraulic power cart to helicopter. Do not apply hydraulic power at this time.

NOTE

During rigging of cyclic controls, take care to keep helicopter as level as possible.

3. Retract stop bolts as far as possible (Figure 2 and Figure 3).

4. Disconnect the following control tubes and force gradients.

4.1. Disconnect extension tube (2, Figure 1) from swashplate (1) by removing cotter pin, nut, washers, and bolt.

4.2. Disconnect extension tube (3) from swashplate (1) by removing cotter pin, nut, washers, and bolt.

4.3. Disconnect control tube (8) from bellcrank (7) by removing nut, two washers, and bolt, and from valve arm (4) remove nut, two washers, and bolt. Tag control tube and valve arm.

4.4. Disconnect control tube (9) from bellcrank (10) by removing nuts, two washers, and bolts, and from valve arm (13) by removing nuts, two washers, and bolts. Remove control tube and tag.

4.5. Disconnect fore and aft cyclic transducers (34) from jackshaft (16).

4.6. Disconnect control tube (22) from jackshaft (16) and control tube (24) from bellcrank (31).

4.7. Disconnect lateral cyclic transducers (29) from bellcrank (31).

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4.8. Disconnect control tube (30) from pilot cyclic stick (33).

5. Verify length of centered fore and aft cyclic series actuator (14) of 27.505 to 27.555 inches (698.627 to 699.897 mm).

6. Verify length of centered lateral cyclic series actuator (35) of 56.295 to 56.345 inches (1429.893 to 1431.163 mm).

7. Level helicopter (412-A-08-40-00-00A-169A-A / 00085).

8. Install cyclic stick rigging tool (T103347-101) (2, Figure 4). Align tool with bolt (3) at station 38.65.

9. Place pilot cyclic stick (1) in rigging tool (2), as shown in Figure 4. Position clamp (4) around pilot cyclic stick (1) and secure with bolt (6) and wing nut (5).

10. Install cyclic mixing bellcrank rigging tool (T103286-101) (15) with bolts (8).

11. Tighten screw (17) in slide (16) until slight friction is felt between slide and cyclic mixing bellcrank rigging tool (T103286-101) (15).

12. Position bushing (10) in support (12) over outboard bolt heads (11) of bellcrank (9).

13. Secure support (12) to slide (16) with screw (13) and wing nut (14).

14. With support (12) positioned per Step 12.., move slide (16) aft to align bellcrank (9) and tighten screw (17) to secure slide (16).

NOTE

SN: 33001-33061:

Helicopters have non-adjustable control tube (30) 204-001-016-013 installed. Control tube 204-001-016-013 replaced by adjustable control tube 412-001-307-101.

NOTE

SN: 33062-33213 or 36001-36999 or 37002-37999 or 38001-38999 or 39101-39999:

Helicopters use control tube 412-001-307-101.

15. Check and adjust (if necessary) control tube (30, Figure 1) located between pilots cyclic stick (33) and bellcrank (31). Torque jam nut 80 to 100 in.-lbs. (9.04 to 11.30 N-m) and install control tube (30).

16. Remove cyclic mixing bellcrank rigging tool (T103286-101).

17. Turn on hydraulic power cart and apply hydraulic power to helicopter.



WHEN PERFORMING OPERATIONAL CHECK OF CYCLIC CONTROLS WITH HYDRAULIC SYSTEM PRESSURIZED, ENSURE COLLECTIVE STICK IS POSITIONED NEAR MID TRAVEL.

18. Adjust control tubes (8 and 9, Figure 1) in accordance with Figure 2 as follows:

18.1. Connect control tubes (2 and 3) to valve arms with bolts, washers, nuts, and cotter pins, leaving tubes disconnected from bellcranks.

18.2. Adjust control tubes (2 and 3) to length of 20.86 in. (529.84 mm).

18.3. Torque jam nuts 80 to 100 in.-lbs. (9.04 to 11.30 N-m) on control tube.

18.4. Connect each control tube (8 and 9) to each bellcrank (7 and 10) with bolts, washers, nuts, and cotter pins.

19. Disconnect four pitch link assemblies from pitch horns. (412-A-62-20-00-02A-720A-A / 00952).

20. Adjust copilot control tubes (22 and 24, Figure 1) as follows:

20.1. Remove cyclic stick rigging tool (T103347-101).

20.2. Place pilot cyclic (33) and copilot cyclic (15) in extreme left position against stops and hold. Adjust control tube (24) to fit bellcrank (31). Torque jam nut on clevis (27) on control tube (24). Install control tube (24) on bellcrank (31).

20.3. Place pilot and copilot cyclics in extreme forward position against stops and hold. Adjust control tube (22) to fit jackshaft (16). Torque jam nut on clevis (21) on control tube (22). Install control tube (22) on jackshaft (16).

21. Disconnect hydraulic cylinder extension tube (25, Figure 4) from collective lever (24).

22. Reconnect cyclic stick rigging tool (T103347-101) and secure collective lever rigging tool (T103287-111) to pin (23) on collective lever (24) with screw (28), then pull up on lever to seat tool base on plate assembly (26).

23. Install swashplate rigging tool (T103288-101) halves (21 and 22) around collective sleeve with index pin (19) engaged in swashplate outer ring. Secure tool halves (21 and 22) with screws (18).

24. Lightly tighten screw (20) in aft tool half (21) to seat sleeve in forward rigging tool half (22), then rotate tool assembly clockwise to bottom index pin (19) in cutout of swashplate outer ring.

25. With swashplate rigging tool (T103288-101) positioned per Step 24., push tool assembly down until all three swashplate positioning lugs are in contact with outer swashplate bearing retainer. Hold tool and swashplate in this position and secure with screw (20).

NOTE

If necessary, rotate swashplate outer ring until bearing retainer ring fasteners clear tool swashplate positioning lugs.

26. Ensure collective lever rigging tool (T103287-111) is positioned per Step 22., and swashplate rigging tool (T103288-101) is positioned per Step 23. through Step 25. before proceeding.

27. Adjust length of extension tubes (2 and 3, Figure 1) to fit swashplate.

NOTE

Provide a minimum of 1.5 exposed threads to extension tubes (2 and 3) upper clevis. If required at completion of Step 27., it is permitted to shorten control tubes (8 and 9) equally a maximum of 5 complete turns of adjustable rod end bearing. Make sure extension tubes are disconnected from swashplate before adjusting control tubes. Shortening the control tubes adjustable rod end bearing 1.5 turns is equal t o approximately one thread of extension tube upper clevis.

28. Torque nuts on both extension tubes 1500 to 1800 inch-pounds (169.47 to 203.37 Nm) and secure with lockwire (C-405). Apply sealant (C-308) on nut, lockwire, threads of clevis, and both extension tubes.

29. Connect extension tubes (2 and 3) to swashplate (1) by installing bolts, washers, nut, and cotter pins.

30. Remove swashplate rigging tool (T103288-101).

31. Remove collective lever rigging tool (T103287-111) and reconnect hydraulic cylinder extension tube (25, Figure 4) to collective lever (24).

32. Install cyclic corner stop rigging tool (T103281-101) (34) on left series actuator and secure in place with screw (30). Attach rod clamp (31) to left series actuator support and secure with screw (35). Align slide (33) index mark N with pointer on rigging tool base (T103287-101) and secure with screw (32).

33. Install cyclic corner stop rigging tool (T103281-102) (34) on right series actuator and secure in place with screw (30). Attach rod clamp (31) to right series actuator support and secure with screw (35). Align slide (33) index mark N with pointer on cyclic corner stop rigging tool (T103281-102) and secure in position with screw (32).

34. Remove cyclic stick rigging tool (T103347-101).

35. Place pilot cyclic against extreme left lateral stop. Move stick forward toward forward left corner position until pointer on left hand rigging tool base (34) aligns with slide (33) index mark F.

36. Adjust corner stop bolt (7, Figure 2) in bellcrank (6) to make contact with actuator support (8).

37. Move cyclic stick from forward left corner position aft toward aft left corner position until pointer on right rigging tool base (34, Figure 4) aligns with slide (33) index mark A.

38. Adjust corner stop bolt (10, Figure 2) to make contact with bellcrank (11).

39. Place pilot cyclic against extreme right lateral stop. Move stick forward toward forward right corner position until pointer on right rigging tool base (34, Figure 4) aligns with slide (33) index mark F.

40. Adjust corner stop bolt (3, Figure 3) in bellcrank (2) to make contact with actuator support (4).

41. Move cyclic from forward right corner position aft towards aft right corner position until pointer on left rigging tool base (34, Figure 4) aligns with slide (33) index mark A.

42. Adjust corner stop bolt (6, Figure 3) to make contact with bellcrank (7).

43. Turn hydraulic power cart off and disconnect from helicopter.

44. Remove cyclic corner stop rigging tool (T103281-101) and cyclic corner stop rigging tool (T103281-102).

45. Connect four pitch links to pitch horns (412-A-62-30-00-02A-720A-A / 00965).

46. Perform fore and aft cyclic transducer (34, Figure 1) adjustments as follows:

NOTE

SN: 33001-33213 or 36001-36086:

Following procedure applies to helicopters with transducer P/N 412-074-101-101 installed.

NOTE

SN: 36037-36086:

For helicopters with transducer P/N 412-074-101-103 installed refer to 412-A-67-01-00-08A-276A-A / 01548.

46.1. Attach fore and aft cyclic transducers to structure (Figure 5).

46.2. Position pilot cyclic against extreme right lateral stop, then against extreme left lateral stop. Check if transducers would bottom out if connected to bellcrank.

46.3. If bottoming occurs, adjust rod end bearings on transducers until bottoming is eliminated and connect transducers to bellcrank. Overtravel to transducers should be approximately equal.

47. Perform lateral cyclic transducer (29, Figure 1) adjustment as follows:

NOTE

SN: 33001-33213 or 36001-36086:

Following procedure applies to helicopters with transducer P/N 412-074-101-101 installed.

NOTE

SN: 36037-36086:

For helicopters with transducer P/N 412-074-101-103 installed refer to 412-A-67-01-00-08A-276A-A / 01548.

47.1. Attach lateral cyclic transducer to structure (Figure 5).

47.2. Position pilot cyclic against extreme right lateral stop, then against extreme left lateral stop. Check if transducers would bottom out if connected to bellcrank.

47.3. If bottoming occurs, adjust rod end bearings on transducers until bottoming is eliminated and connect transducers to bellcrank. Overtravel of transducers should be approximately equal.

48. Turn on hydraulic power cart and apply hydraulic power to helicopter.



WHEN PERFORMING OPERATIONAL CHECK OF CYCLIC CONTROLS WITH HYDRAULIC SYSTEM PRESSURIZED, ENSURE COLLECTIVE STICK IS POSITIONED NEAR MID TRAVEL.

49. Perform cyclic minimum friction adjustment as follows:

NOTE

Initial adjustment and check of fore and aft and lateral cyclic stick force shall be accomplished by utilizing a hydraulic power cart connected to helicopter using Step 49.1.. through Step 49.7.. Final check and adjustment shall be accomplished during ground run of helicopter using Step 49.2.. through Step 49.7..

49.1. Reduce cyclic stick friction to zero by loosening clamp (Figure 6). Rotate nut counterclockwise.

49.2. With force trim OFF, slowly increase stick friction while simultaneously pulling on cyclic stick with spring scale attached at midpoint of grip until a fore and aft cyclic breakout force of 0.5 to 1.0 lbs. (2.224 to 4.448 N) is obtained.

49.3. Forces to be measured perpendicular to axis of stick with stick near center.

49.4. Check lateral cyclic breakout force of 0.5 to 1.0 lbs. (2.224 to 4.448 N).

49.5. If lateral stick force exceeds above requirement, readjust fore and aft friction breakout within above limits and recheck lateral breakout force.

49.6. Tighten clamp around sleeve with clamp resting on top surface of nut (Figure 6). Tighten screws to obtain equal gap between clamp halves.

NOTE

SN: 36087-Subsequent:

Step 49.7. through Step 49.9. are not applicable.

49.7. With the force trim on, check for a breakout force of the fore and aft cyclic stick force of 2.25 to 2.75 pounds (10.01 to 12.23 N) and lateral stick force of 1.5 to 2.0 pounds (6.67 to 8.90 N).

49.8. If the fore and aft or lateral breakout forces exceed the previous requirements, repeat Step 49.1. through Step 49.6..

49.9. If the fore and aft or lateral breakout forces still exceed the previous requirements, readjust either the fore and aft force gradient or lateral force gradient spring preloads within their respective limits to obtain the required breakout force. Repeat Step 49.1. through Step 49.7. with hydraulic power applied.

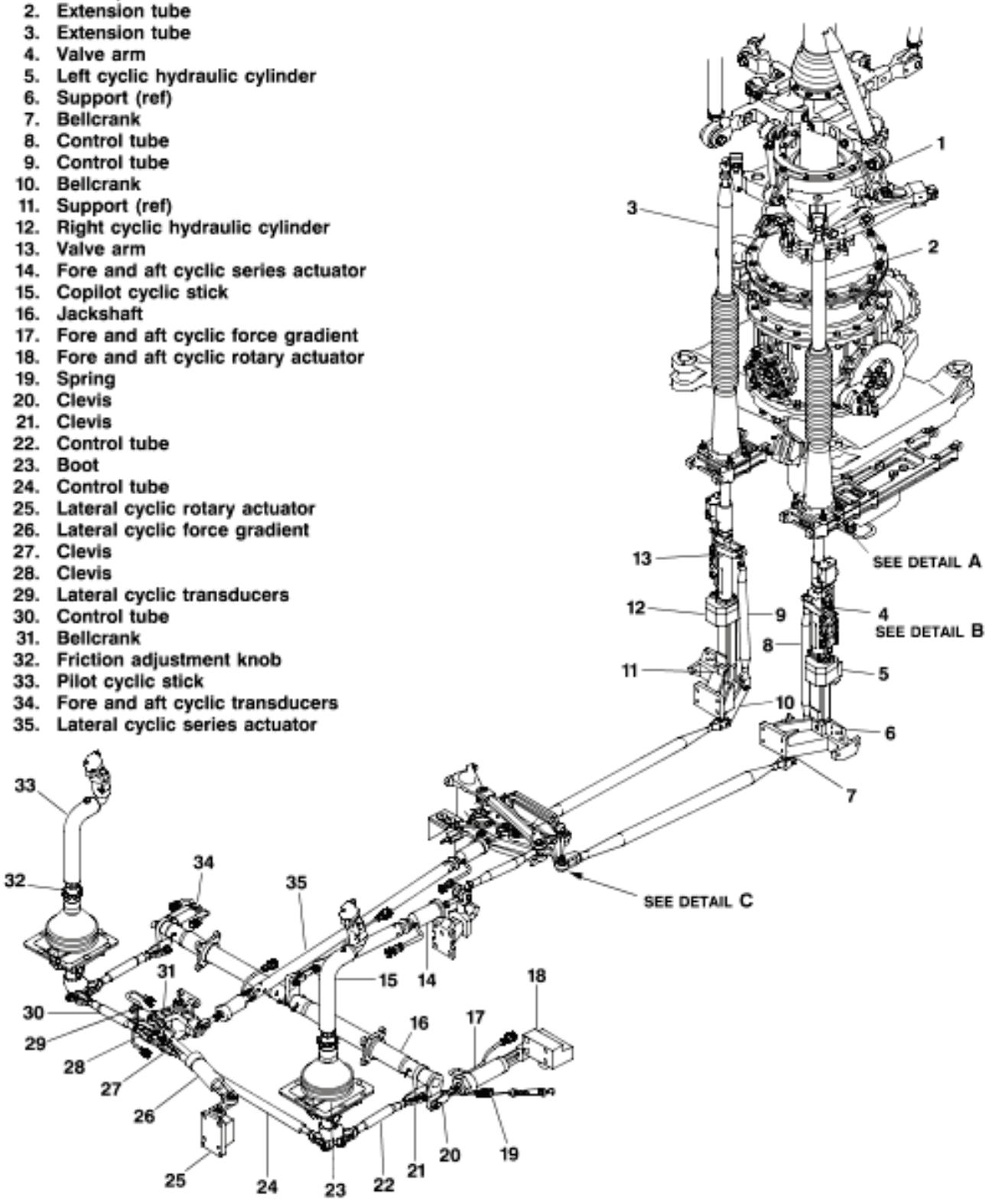
NOTE

For cyclic force gradient spring preload adjustments and cyclic control stick mass balance adjustment, refer to 412-A-67-12-00-16A-720A-A / 01458 and 412-A-67-12-00-17A-720A-A / 01462.

50. Remove the hydraulic power.

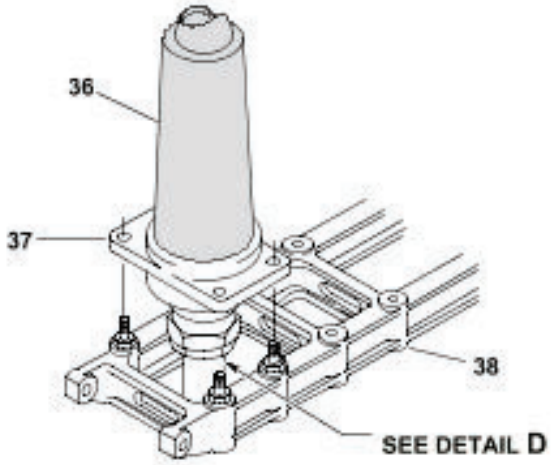
51. Accomplish rigging of cyclic controls position sensors (412-A-67-01-00-05A-276A-A / 01326).

1. Swashplate
2. Extension tube
3. Extension tube
4. Valve arm
5. Left cyclic hydraulic cylinder
6. Support (ref)
7. Bellcrank
8. Control tube
9. Control tube
10. Bellcrank
11. Support (ref)
12. Right cyclic hydraulic cylinder
13. Valve arm
14. Fore and aft cyclic series actuator
15. Copilot cyclic stick
16. Jackshaft
17. Fore and aft cyclic force gradient
18. Fore and aft cyclic rotary actuator
19. Spring
20. Clevis
21. Clevis
22. Control tube
23. Boot
24. Control tube
25. Lateral cyclic rotary actuator
26. Lateral cyclic force gradient
27. Clevis
28. Clevis
29. Lateral cyclic transducers
30. Control tube
31. Bellcrank
32. Friction adjustment knob
33. Pilot cyclic stick
34. Fore and aft cyclic transducers
35. Lateral cyclic series actuator

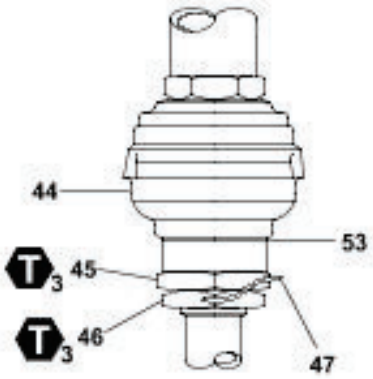


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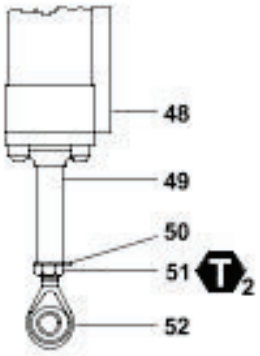
Figure 1. Cyclic Rigging (Sheet 1 of 3)



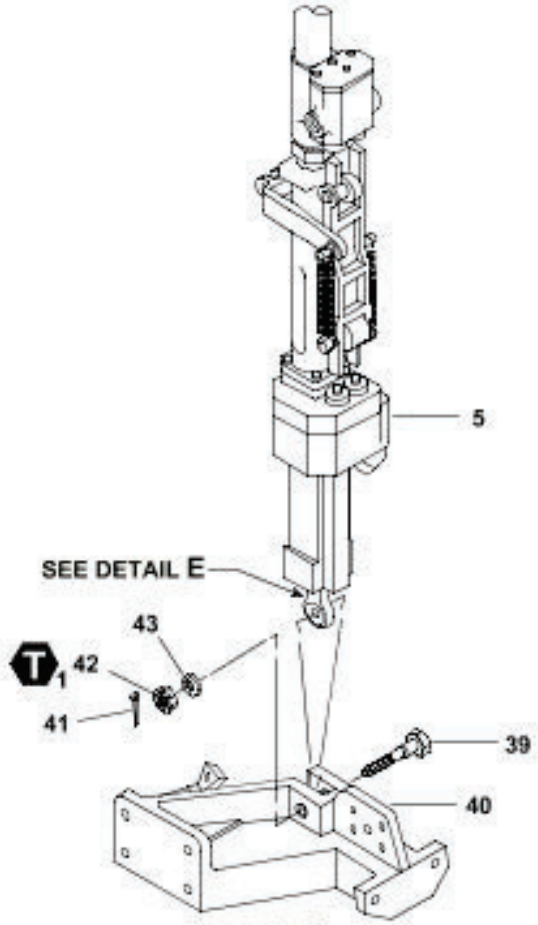
DETAIL A



DETAIL D



DETAIL E



DETAIL B
LEFT CYCLIC FLIGHT
CONTROL CYLINDER

36. Boot
37. Support
38. Support
39. Bolt
40. Support (ref)
41. Cotter pin
42. Nut
43. Washer
44. Support bracket
45. Nut
46. Nut
47. Lockwire
48. Lower actuator body
49. Lower piston
50. Locking device
51. Jam nut
52. Rod end bearing
53. Keyed washer

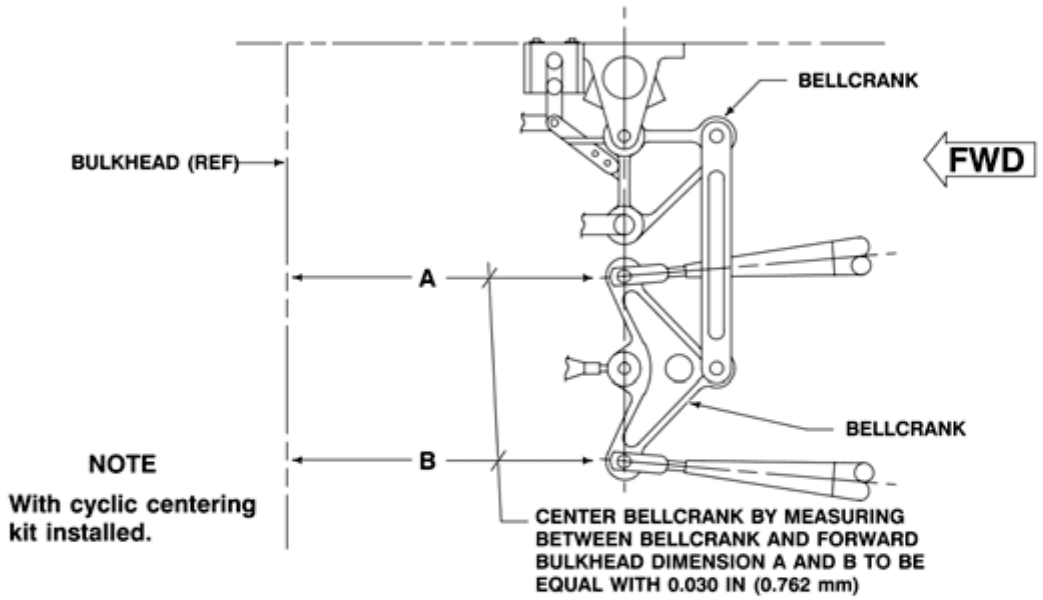
T1 95 TO 110 IN-LBS (10.73 TO 12.43 Nm)

T2 200 TO 250 IN-LBS (22.60 TO 28.25 Nm)

T3 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

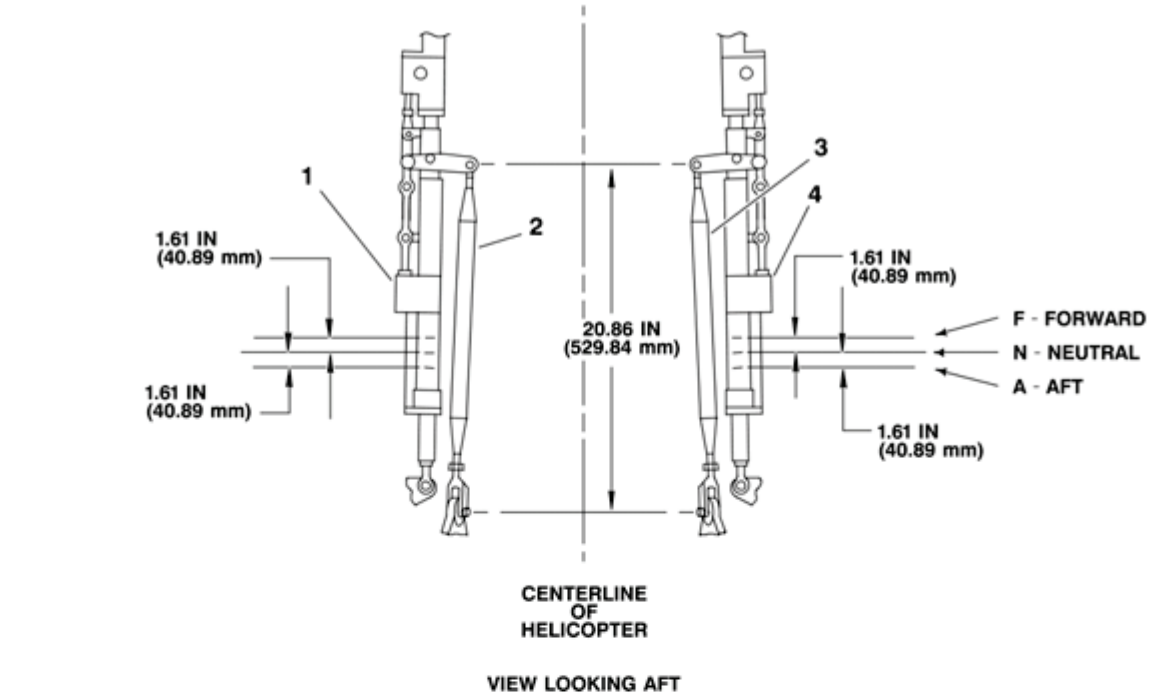
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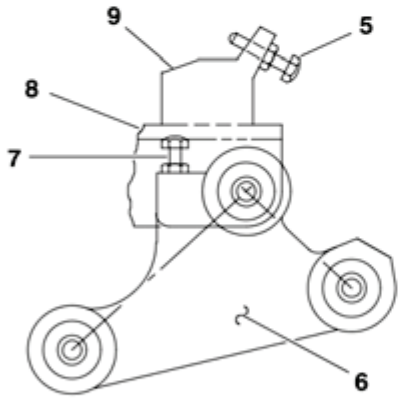
Figure 1. Cyclic Rigging (Sheet 2 of 3)



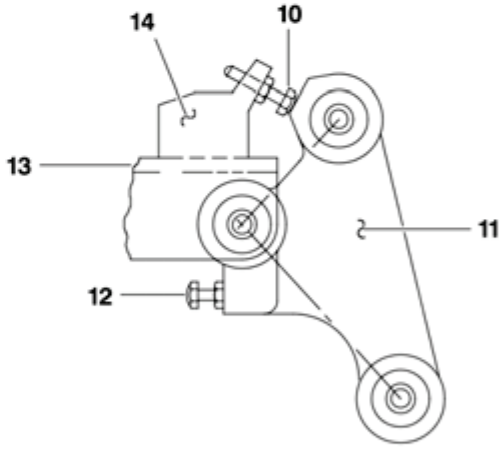
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Figure 1. Cyclic Rigging (Sheet 3 of 3)





POSITION OF BELLCRANK (6)
BELOW LEFT CYCLIC CYLINDER
(4) FOR EXTREME FORWARD
LEFT CORNER STICK POSITION

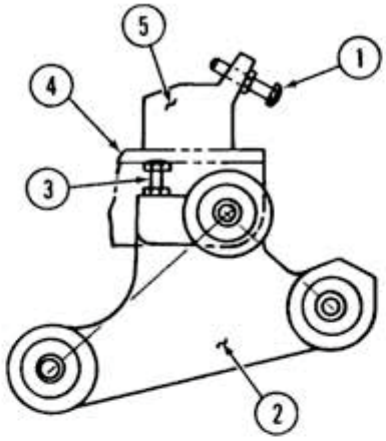


POSITION OF BELLCRANK (11)
BELOW RIGHT CYCLIC CYLINDER
(1) FOR EXTREME AFT LEFT
CORNER STICK POSITION

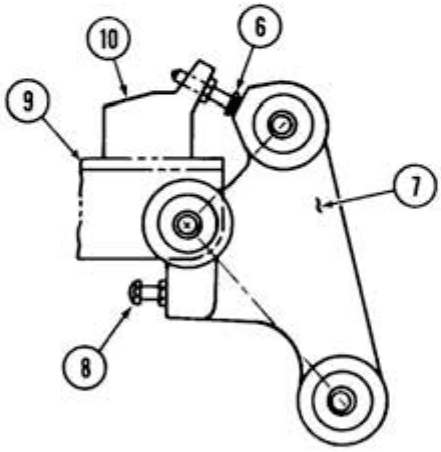
1. Right cyclic hydraulic cylinder
2. Control tube
3. Control tube
4. Left cyclic hydraulic cylinder
5. Stop bolt
6. Bellcrank
7. Stop bolt
8. Actuator support
9. Stop
10. Stop bolt
11. Bellcrank
12. Stop bolt
13. Actuator support
14. Stop

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Figure 2. Cyclic Rigging - Hydraulic Cylinders (Sheet 1 of 1)



POSITION OF BELLCRANK (2)
BELOW RIGHT CYCLIC CYLINDER
(1) FOR EXTREME FORWARD
RIGHT CORNER STICK POSITION

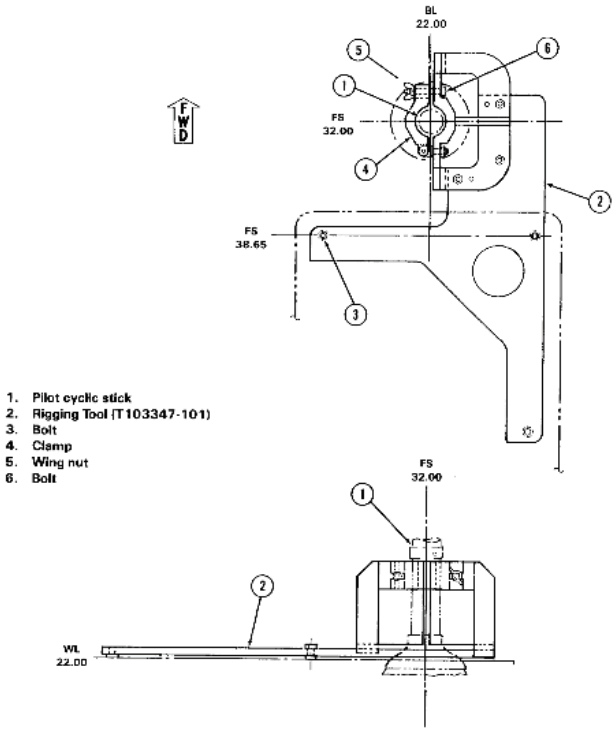


POSITION OF BELLCRANK (7)
BELOW LEFT CYCLIC CYLINDER
(4) FOR EXTREME AFT RIGHT
CORNER STICK POSITION

1. Stop bolt
2. Bellcrank
3. Stop bolt
4. Actuator support
5. Stop
6. Stop bolt
7. Bellcrank
8. Stop bolt
9. Actuator support
10. Stop

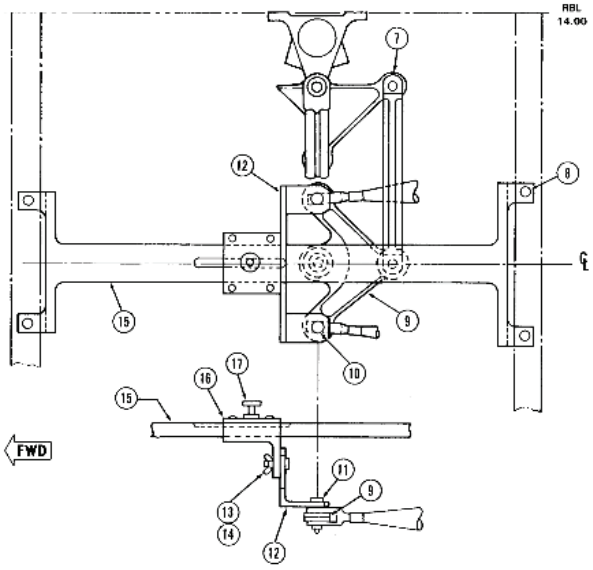
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Figure 3. Cyclic Rigging - Bellcrank Stops (Sheet 1 of 1)



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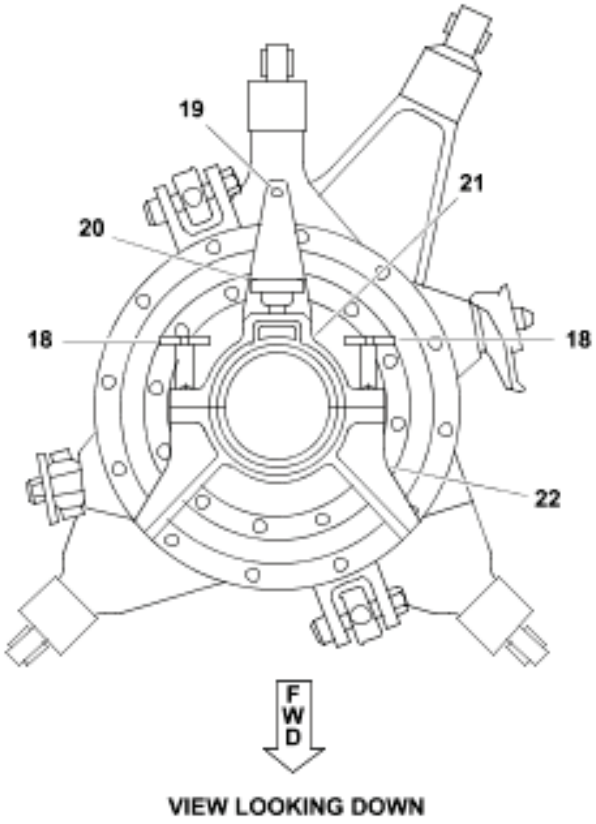
Figure 4. Cyclic Rigging (Alternate Method) (Sheet 1 of 4)



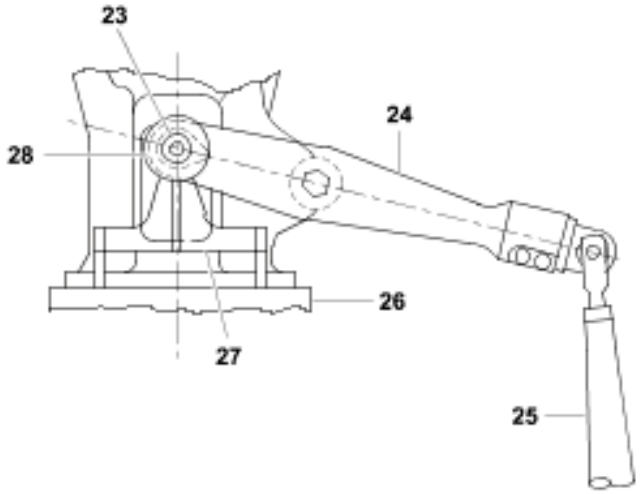
7. Bellcrank
8. Bolt
9. Bellcrank
10. Bushing
11. Bolt
12. Support
13. Screw
14. Wing nut
15. Rigging tool base (T103286-101)
16. Slide
17. Screw

ICN-412-A-670100-B-97499-00016-A-001-01

Figure 4. Cyclic Rigging (Alternate Method) (Sheet 2 of 4)

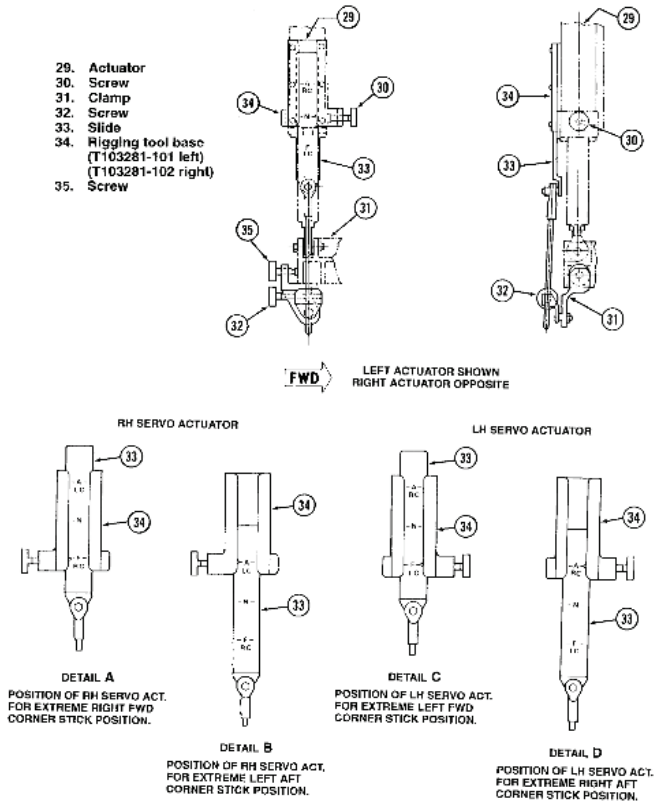


18. Screw
19. Index pin
20. Screw
21. Rigging tool half (T103288-101)
22. Rigging tool half (T103288-101)
23. Pin
24. Collective lever
25. Extension tube
26. Plate assembly
27. Rigging tool (T103287-111)
28. Screw



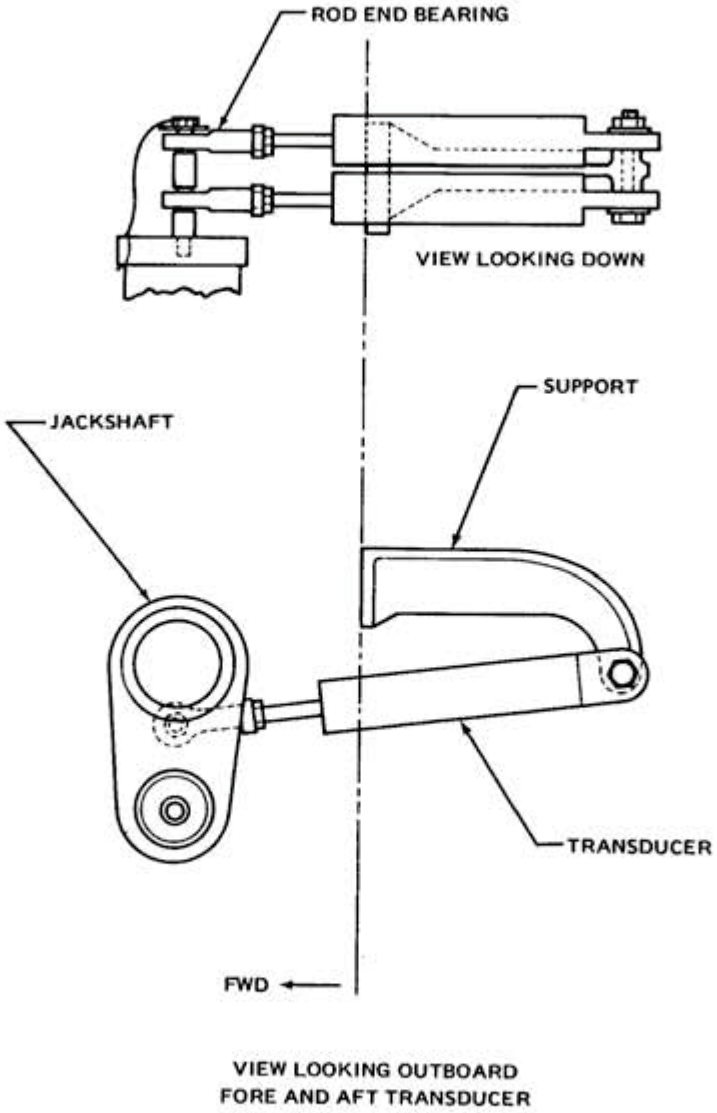
ICN-412-A-670100-B-97499-00017-A-001-01

Figure 4. Cyclic Rigging (Alternate Method) (Sheet 3 of 4)



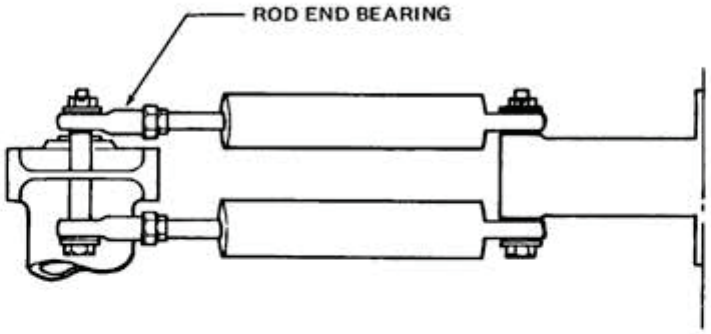
ICN-412-A-670100-B-97499-00018-A-001-01

Figure 4. Cyclic Rigging (Alternate Method) (Sheet 4 of 4)

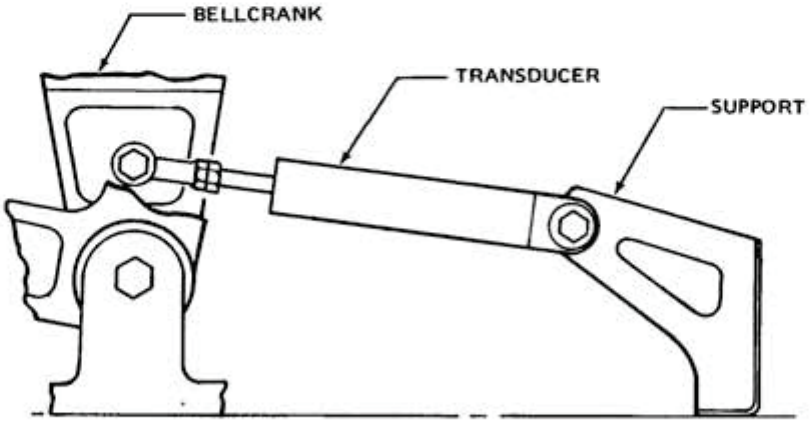


ICN-412-A-670100-B-97499-00012-A-001-01

Figure 5. Cyclic Rigging - Transducer (Sheet 1 of 2)



VIEW LOOKING FORWARD
LATERAL CYCLIC TRANSDUCER

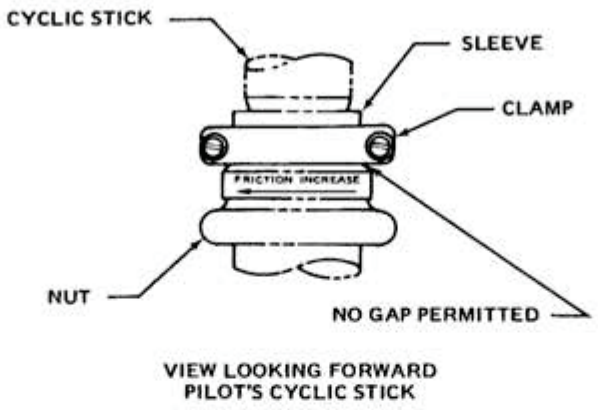


VIEW LOOKING DOWN LATERAL

CYCLIC TRANSDUCER

ICN-412-A-670100-B-97499-00013-A-001-01

Figure 5. Cyclic Rigging - Transducer (Sheet 2 of 2)



ICN-412-A-670100-B-97499-00014-A-001-01

Figure 6. Cyclic Friction Adjustment (Sheet 1 of 1)

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Cyclic Position Sensors - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Swashplate Rigging Tool	T101291-103	1	
Position Sensor Rigging Tool	T103356-101	1	
Inclinometer	Commercial	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Verify the cyclic rigging (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325) prior to adjustment of the cyclic controls position sensor and fore-and-aft cyclic sensor.

2. Prepare the helicopter for rigging as follows:

2.1. Place jacks under the helicopter as in the jacking procedure (412-A-07-10-00-00A-172A-A / 00080) and raise the jacks just high enough to prevent the helicopter from shifting when maintenance personnel are working around the swashplate and controls.

2.2. Verify that both actuator assemblies, longitudinal and lateral, are centered when electrical power is applied to the helicopter. Centering measurement will be taken from the center of the bolt hole on the clevis on one end, to the center of the rod end bearing on the other end of the actuator. The centered length of the longitudinal series actuator (14, Figure 1) is 27.505 to 27.555 inches (698.627 to 699.897 mm).

2.3. Set the swashplate to a level position as follows:

2.3.1. Apply hydraulic power to the helicopter.

2.3.2. Position the swashplate rigging tool (2, Figure 2) on the outer cap portion of the swashplate (1).

2.3.3. Position the swashplate approximately perpendicular with respect to the mast by moving the cyclic stick as required.

2.3.4. Place an inclinometer (propeller protractor) on the upper surface of the rigging tool (2).

2.3.5. Move the cyclic stick slightly to obtain the desired attitude of:

2.3.5.1. Fore and aft cyclic - position swashplate to 0° ±15' down forward with respect to the mast.

2.3.5.2. Lateral cyclic - position swashplate to 0° ±15' with respect to the mast.

3. Apply the cyclic stick friction to hold the position established in the preceding Step 2.3.5..

4. Rig the fore and aft cyclic sensor (Figure 3) as follows:

4.1. Position the clamp assembly (13) on the idler (5) to 3.05 inches (77.47 mm) dimension, as shown. Secure the clamp assembly to the idler with the yoke (12). Adjust the screws (1) to obtain an equal gap between both sides of the clamp assembly and yoke. Secure the screw with lockwire (C-405) (21).

4.2. Position and hold the tool (18) F/A (fore and aft) slot on the sensor target index attached to the clamp assembly (13).

4.3. Position the switch (15) attached to the sensor plate (10) in the opposite slot of the tool to align the switch with the target (14). Secure the sensor plate to the support assembly (9) with the screw (8).

4.4. With the tool (18) still positioned per Step 4.2., position and hold the switch (15) against the tool to set the air gap 0.064 to 0.065 inch (1.625 to 1.651 mm). Secure the switch to the sensor plate (10). Refer to Figure 4 for installation.

NOTE

The idler (5) may be disconnected and rotated 180° (around vertical axis) on the support assembly (6) to facilitate adjustment of the switch (15).

4.5. Use the tool (18) to recheck the switch (15) alignment and air gap. Readjust as required. Secure the retaining nuts with lockwire (C-405) (21) on either side of the sensor plate (10). Remove the tool.

5. Rig the lateral sensor (Figure 4) as follows:

5.1. Position and hold the tool (8) on the target (9) attached to the arm assembly (6) and bellcrank assembly (7).

5.2. Position the switch (5) attached to the sensor plate (3) in the opposite slot of the tool to align the switch with the target (9). Secure the plate to the support assembly (1) with the screws (2 and 4).

5.3. With the tool (8) still positioned per Step 5.1. and Step 5.2., position and hold the switch (5) against the tool to set the gap 0.078 to 0.080 inch (1.981 to 2.032 mm). Secure the switch to the sensor plate (3) with the retaining nuts (10 and 11).

NOTE

The support assembly (1) may be adjusted to facilitate the adjustment of the sensor.

5.4. Use the tool (8) to recheck the switch alignment and air gap. Adjust as required. Secure the retaining nut used to adjust the switch (5) with lockwire (C-405). Remove the tool and remove the swashplate rigging tool (2, Figure 2).

6. Electrical function and sensor rigging checkout.

6.1. Remove the electrical power and remove the center connector 8Z1P3 from the RPM limit warning detector. The RPM detector is located on the shelf in the nose compartment (412-A-96-03-03-00A-042A-A / 02091).

6.2. Install a jumper wire between pin F and pin G of the connector 8Z1P3.

6.3. Apply the electrical power and the Batt bus 1 and 2 on, and with hydraulic pressure applied, move the pilot cyclic laterally both directions. Note the CYC CTR caution lights illuminate after the stick moves approximately 10% either side of center.

6.4. Repeat Step 6.3. for the fore and aft cyclic.

6.5. From a centered position (CYC CTR caution lights extinguished), move the pilot cyclic forward until the caution light comes on. Hold the stick in this position and check for a longitudinal swashplate angle of 1° to 1° 30’ down forward with respect to the mast.

6.6. From a centered position (CYC CTR caution lights extinguished), move the pilot cyclic backward until the caution light comes on. Hold the stick in this position and check for longitudinal swashplate angle of 1° to 1° 30’ down aft with respect to the mast.

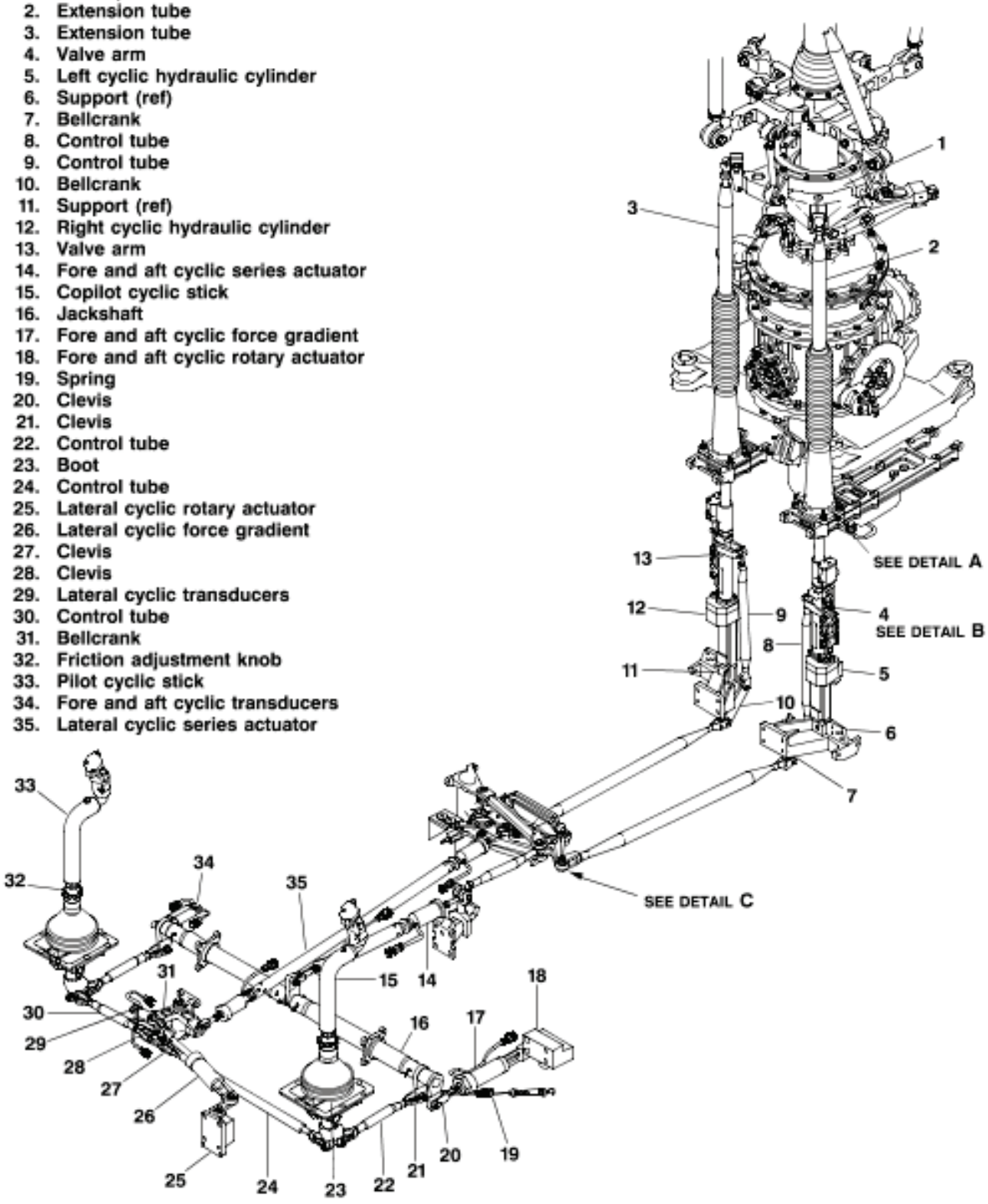
6.7. From a centered position, move the pilot cyclic to the left until the caution light comes on. Hold the stick in this position and check for lateral swashplate angle of 0° 30' to 1° down left with respect to mast.

6.8. From a centered position, move the pilot cyclic to the right until the caution light comes on. Hold the pilot cyclic in this position and check for lateral swashplate angle of 0° 30’ to 1° down right with respect to the mast.

6.9. If the swashplate angle does not fall within the limits of Step 6.5. through Step 6.8., rerig as required.

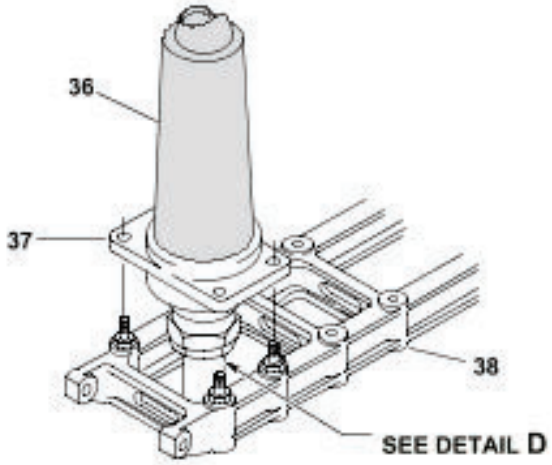
6.10. Remove the jumper and connect the connector 8Z1P3 to the RPM limit warning detector.

1. Swashplate
2. Extension tube
3. Extension tube
4. Valve arm
5. Left cyclic hydraulic cylinder
6. Support (ref)
7. Bellcrank
8. Control tube
9. Control tube
10. Bellcrank
11. Support (ref)
12. Right cyclic hydraulic cylinder
13. Valve arm
14. Fore and aft cyclic series actuator
15. Copilot cyclic stick
16. Jackshaft
17. Fore and aft cyclic force gradient
18. Fore and aft cyclic rotary actuator
19. Spring
20. Clevis
21. Clevis
22. Control tube
23. Boot
24. Control tube
25. Lateral cyclic rotary actuator
26. Lateral cyclic force gradient
27. Clevis
28. Clevis
29. Lateral cyclic transducers
30. Control tube
31. Bellcrank
32. Friction adjustment knob
33. Pilot cyclic stick
34. Fore and aft cyclic transducers
35. Lateral cyclic series actuator

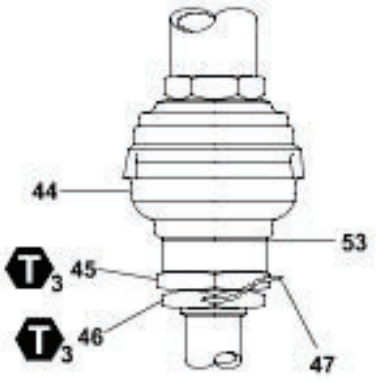


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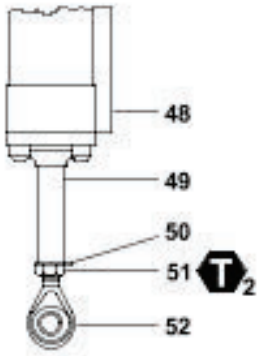
Figure 1. Cyclic Rigging (Sheet 1 of 3)



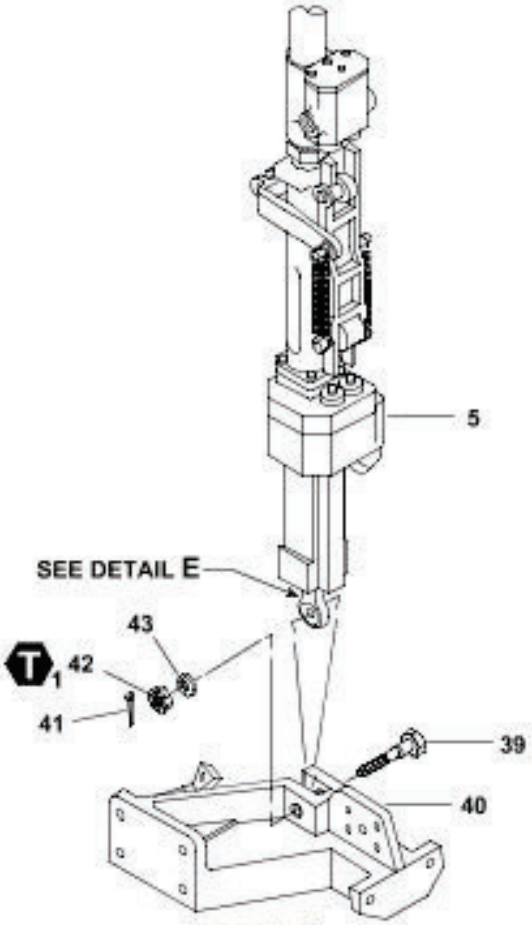
DETAIL A



DETAIL D



DETAIL E



DETAIL B
LEFT CYCLIC FLIGHT
CONTROL CYLINDER

36. Boot
37. Support
38. Support
39. Bolt
40. Support (ref)
41. Cotter pin
42. Nut
43. Washer
44. Support bracket
45. Nut
46. Nut
47. Lockwire
48. Lower actuator body
49. Lower piston
50. Locking device
51. Jam nut
52. Rod end bearing
53. Keyed washer

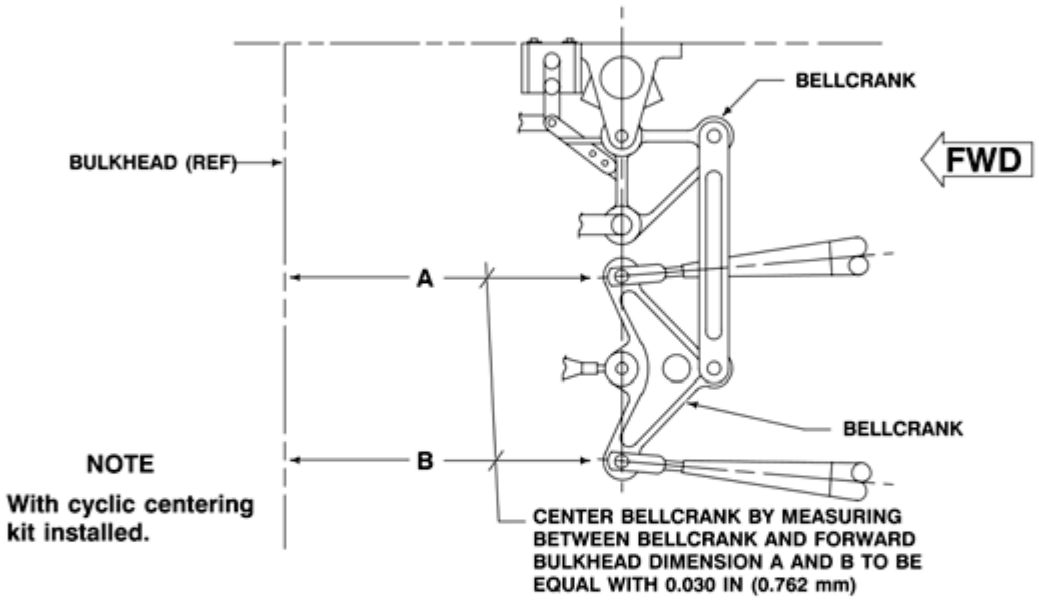
T1 95 TO 110 IN-LBS (10.73 TO 12.43 Nm)

T2 200 TO 250 IN-LBS (22.60 TO 28.25 Nm)

T3 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

ICN-412-A-670100-B-97499-00006-A-001-01

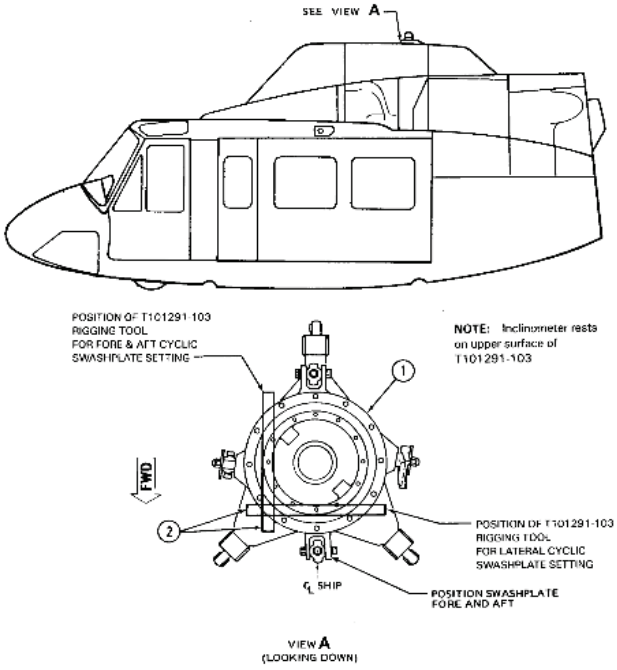
Figure 1. Cyclic Rigging (Sheet 2 of 3)



DETAIL C
VIEW LOOKING DOWN
MIXING LEVER

ICN-412-A-670100-B-97499-00007-A-001-01

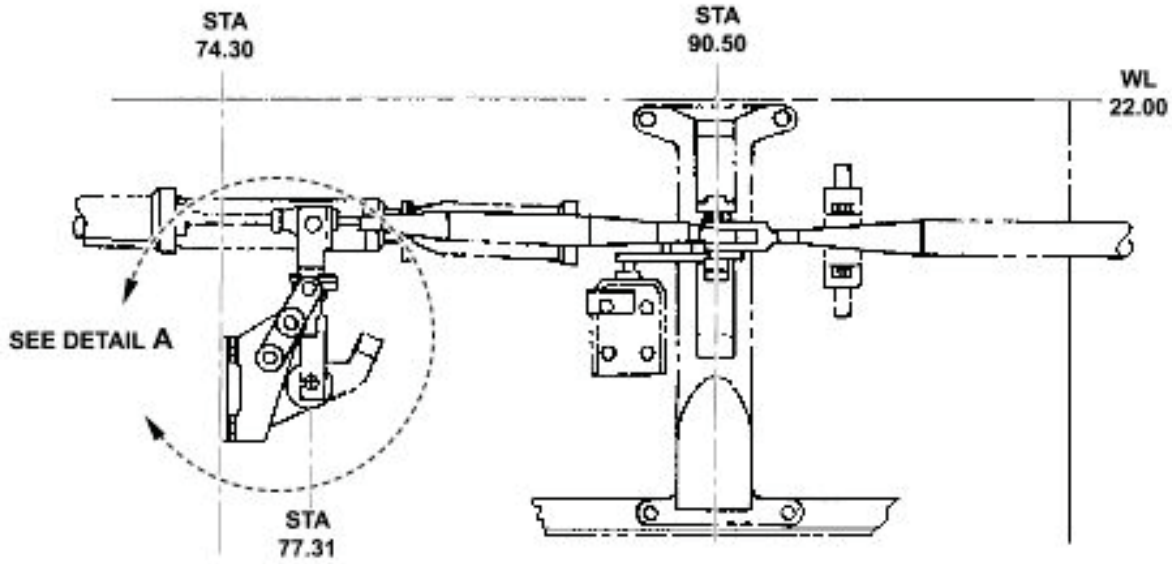
Figure 1. Cyclic Rigging (Sheet 3 of 3)



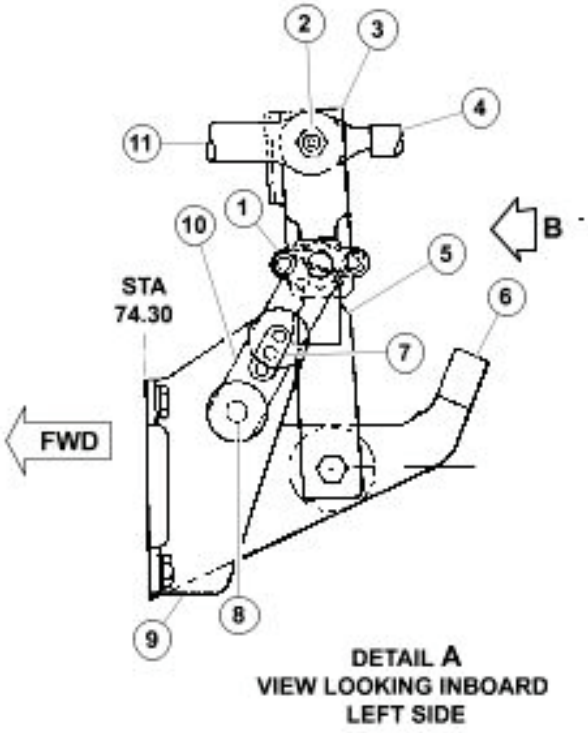
1. Swashplate and support assy
2. Rigging Tool

ICN-412-A-670100-B-97499-00019-A-001-01

Figure 2. Position Verification (Sheet 1 of 1)



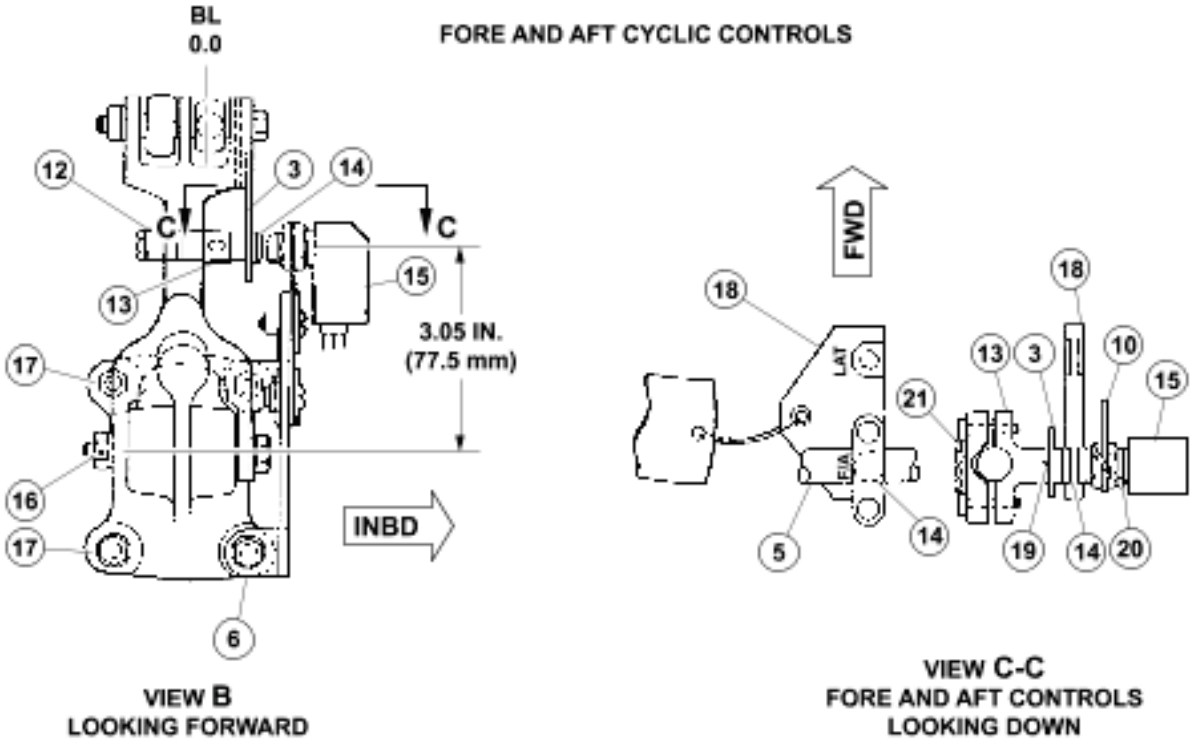
FORE AND AFT AND LATERAL CONTROLS
(VIEW LOOKING INBOARD LEFT SIDE)



DETAIL A
VIEW LOOKING INBOARD
LEFT SIDE

ICN-412-A-670100-B-97499-00020-A-001-01

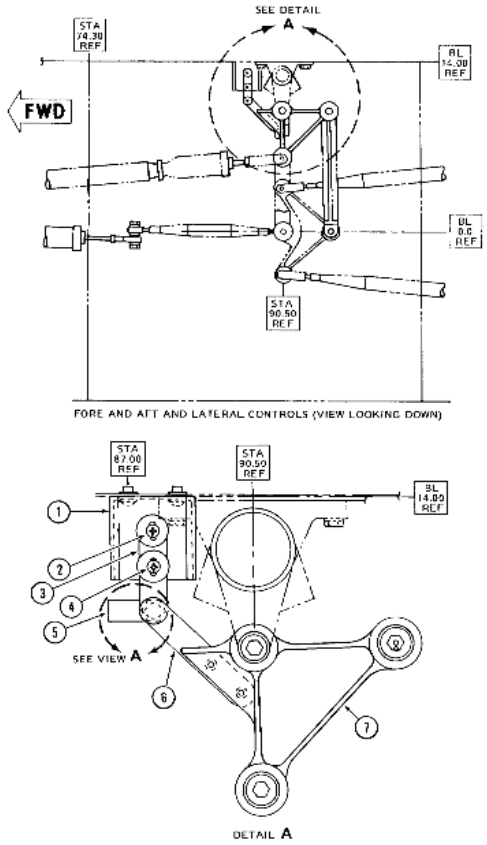
Figure 3. Fore and Aft Cyclic Sensor Adjustment (Sheet 1 of 2)



1. Screw
2. Bolt
3. Retainer
4. Tube assembly
5. Idler
6. Support assembly
7. Screw
8. Screw
9. Support assembly
10. Sensor plate
11. Actuator
12. Yoke
13. Clamp assembly
14. Target
15. Switch
16. Cotter pin
17. Bolt
18. Tool (T103356-101)
19. Cotter pin
20. Lockwire
21. Lockwire

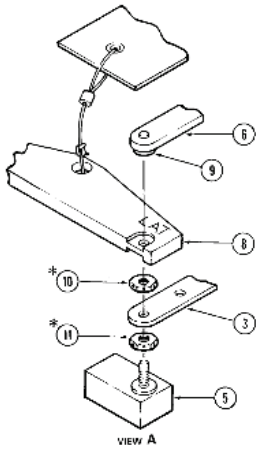
ICN-412-A-670100-B-97499-00021-A-001-01

Figure 3. Fore and Aft Cyclic Sensor Adjustment (Sheet 2 of 2)



ICN-412-A-670100-B-97499-00022-A-001-01

Figure 4. Lateral Cyclic Sensor Adjustment (Sheet 1 of 2)



1. Support assembly
2. Screw
3. Sensor plate
4. Screw
5. Switch
6. Arm assembly
7. Bellcrank assembly
8. Tool (T103356-101)
9. Target
10. Retaining nut
11. Retaining nut

* LOCKWIRE (MS20995C32) TOGETHER AFTER FINAL ADJUSTMENT

ICN-412-A-670100-B-97499-00023-A-001-01

Figure 4. Lateral Cyclic Sensor Adjustment (Sheet 2 of 2)

Anti-Torque Controls - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Tail Rotor Rigging Tool	T103206-101	1	
Spring Scale	Commercial	1	Range to Measure 5 to 6 pounds (22.2 to 26.7 N)

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-104	AR	

Procedure

NOTE

SN: 36087-Subsequent:

For Model 412EP anti-torque controls rigging, refer to 412-A-67-01-00-11A-276A-A / 01551.

NOTE

Rigging procedures contained herein pertain to helicopters with dual controls installed in basic controls systems.

1. Comply with the following general procedures when rigging anti-torque controls:

1.1. When a rigging procedure requires a specific position to accomplish the following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ±0.03 inch (0.76 mm).

1.3. All adjustable control tubes must have a maximum of 1.0 inch (2.54 mm) of exposed threads after adjustment. If control tubes have inspection holes, threads shall be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-104) to threads of all adjustable rod bearings and clevises after adjustment.

1.5. Torque control jam nuts 80 to 100 inch-pounds (9.04 to 11.30 Nm) after adjustment unless other torque is specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. Loosen jam nuts, adjust end fittings, and torque jam nuts if necessary.

2. Disconnect the following control tubes and force gradient.

2.1. Disconnect anti-torque force gradient (14, Figure 1) and transducer (20) from pilot pedal adjuster (21).

2.2. Disconnect control tube (16) from copilot pedal adjuster (17).

2.3. Disconnect control tube (7) from bellcrank (8).

2.4. Disconnect link (4) from lever (3).

3. Retract two stop bolts on bellcrank (10) as far as possible.

4. Verify the length of centered series actuator (11), 62.865 to 62.915 inches (1596.771 to 1598.041 mm).

5. Push both pilot and copilot left directional pedals against extreme forward stops. Adjust control tube (16) to fit, torque jam nut and install.

NOTE

Adjustment of pitch change link length not required for fixed length pitch link (Post TB 412-01-173).

6. Check two pitch links (2) for 6.180 ±0.010 inches (156.972 ±0.254 mm) between bolt hole centers (412-A-64-30-00-02A-520A-A / 01236).

7. Remove bolt (2, Figure 2) retaining pitch link to crosshead (1).

8. Install tail rotor rigging tool (T103206-101) (3) to crosshead and against tail rotor hub trunnion (5). Retain with bolt (2).

9. Push inboard on crosshead (1) to cause tool to make firm contact with tail rotor hub trunnion (5).

10. Adjust link (4, Figure 1) to obtain 0.58 inch (14.73 mm) clearance with bellcrank (6) (Detail A). Torque jam nut on link, and install link on bellcrank and lever (3).

NOTE

If hydraulic power cart is not available, omit Step 11. through Step 13. and proceed to the note preceding Step 14.. If hydraulic power is available, accomplish Step 11. through Step 13. and omit Step 14. through Step 17..

11. Connect hydraulic cart to helicopter.

12. Apply hydraulic power from hydraulic cart to helicopter.

13. Position pilot left anti-torque pedal forward against stop and retain in position. Adjust control tube (7) to fit bellcrank (8). Torque jam nut on tube. Install tube on bellcrank.

NOTE

Alternate method of rigging anti-torque controls without pressurizing the hydraulic system: If hydraulic power cart is not available, adjust control tube (7) as outlined in Step 14. through Step 17..

14. Hold pilot left pedal forward against stop.

15. Push down and hold anti-torque hydraulic cylinder (9) piston rod to bottom anti-torque hydraulic cylinder.

16. Position bellcrank (6), as shown in Detail A, and hold.

17. Adjust control tube (7) assembly to fit bellcrank (8), then shorten by 1/2 turn of clevis on control tube. Torque jam nut on clevis on bellcrank. Install control tube on bellcrank.

18. Remove tail rotor rigging tool (T103206-101) (3, Figure 2) and install pitch link (6) to crosshead (1).

19. Position pilot left pedal forward against stop. Adjust forward stop bolt in bellcrank (10, Figure 1, Detail D) 0.005 to 0.015 inch (0.127 to 0.381 mm).

20. Position pilot right pedal forward against stop. Adjust aft stop bolt in bellcrank (10) arm 0.005 to 0.015 inch (0.127 to 0.381 mm).

21. Turn off hydraulic power and disconnect cart from helicopter.

22. Flap tail rotor assembly to one extreme and check for clearance between blade and tail rotor yoke using a 0.010-inch (0.254-mm) feeler gauge (Figure 3). Perform clearance check on the same blade with tail rotor flapped to opposite extreme. Accomplish clearance check on opposite blade in the same manner.

23. If clearance is greater than 0.010 inch (0.254 mm), rigging has been performed correctly.

24. Rig anti-torque force gradient (14, Figure 1) and magnetic brake (15) as follows:

24.1. Make sure anti-torque magnetic brake arm (26) is on anti-torque magnetic brake (15) with index mark D aligned (Detail C).

24.2. Place pilot pedals in neutral position and hold.

24.3. Position anti-torque magnetic brake arm (26) on anti-torque magnetic brake (15) at center of travel.

24.4. Adjust anti-torque force gradient (14) to fit pilot pedal adjuster (21). Torque jam nut on anti-torque force gradient clevis.

24.5. Install anti-torque force gradient (14) on pilot pedal adjuster (21).

24.6. Move pilot pedals through full travel and check clearance between anti-torque force gradient (14) and structure. If there is interference, correct as described in Step 24.7..

24.7. Remove arm (26) from anti-torque magnetic brake (15). Move one serration and reinstall. Repeat clearance check described in Step 24.6..

25. Perform directional control motion transducer adjustment as follows:

NOTE

SN: 36087-Subsequent:

Refer to 412-A-67-01-00-11A-276A-A / 01551.

25.1. Attach transducer (3, Figure 4) in directional control system to support (4) only.

25.2. Place pilot pedal against one stop, then other stop. Check if transducer (3) bottoms out in either retracted or extended position if connected to bellcrank (1).

25.3. If bottoming occurs, adjust rod end bearing (2) on transducer (3) until bottoming is eliminated. Connect transducer to bellcrank (1). Overtravel of transducer should be approximately equal.

26. Adjust anti-torque minimum friction as follows:

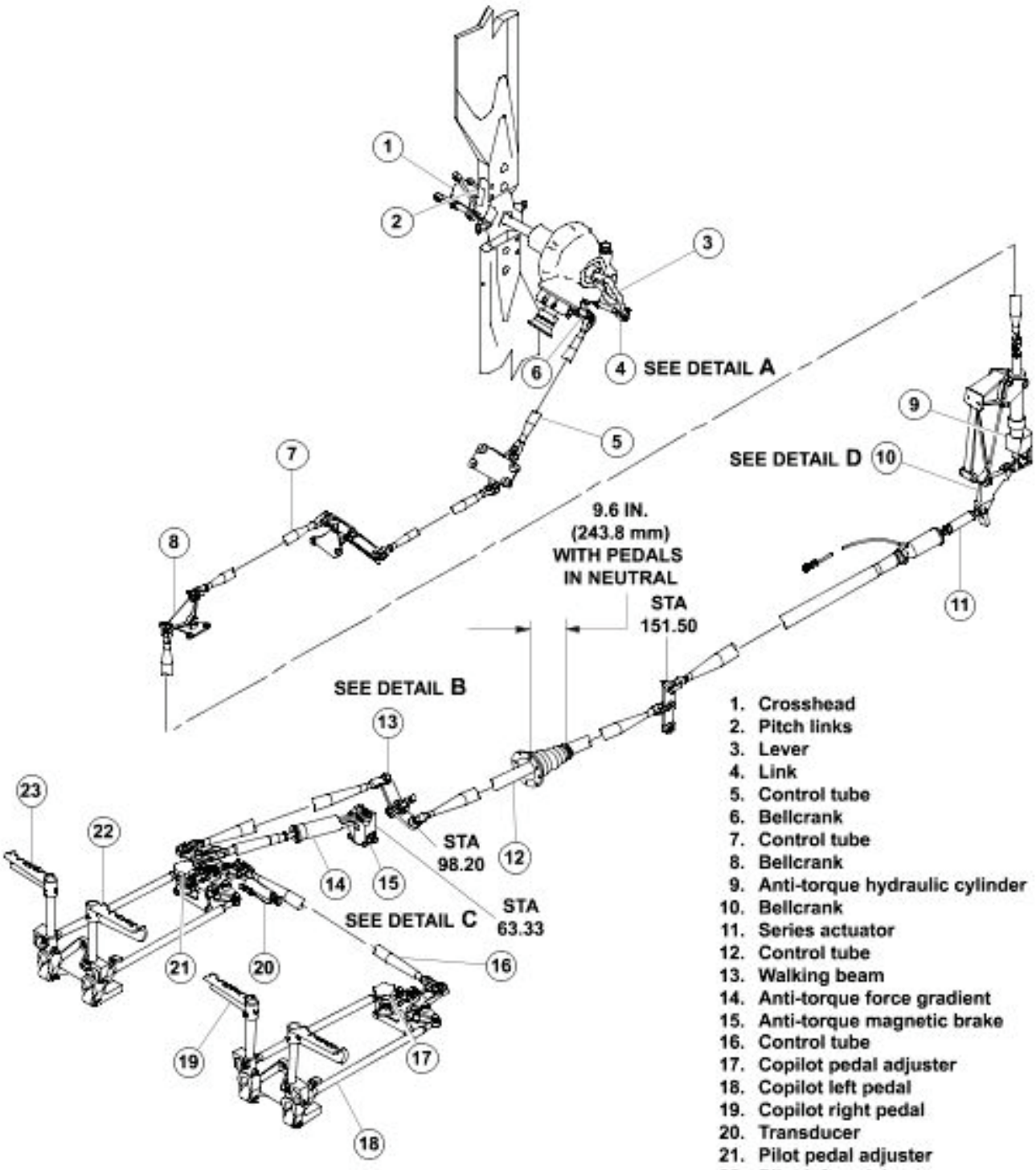
26.1. Attach spring scale to one pilot pedal.

26.2. With boost on, measure force required to move pedal through neutral position.

26.3. If required, adjust screw (25, Figure 1) on clamp (24) to obtain friction of 4.5 ±0.25 pounds (20.16 ±1.112 N).

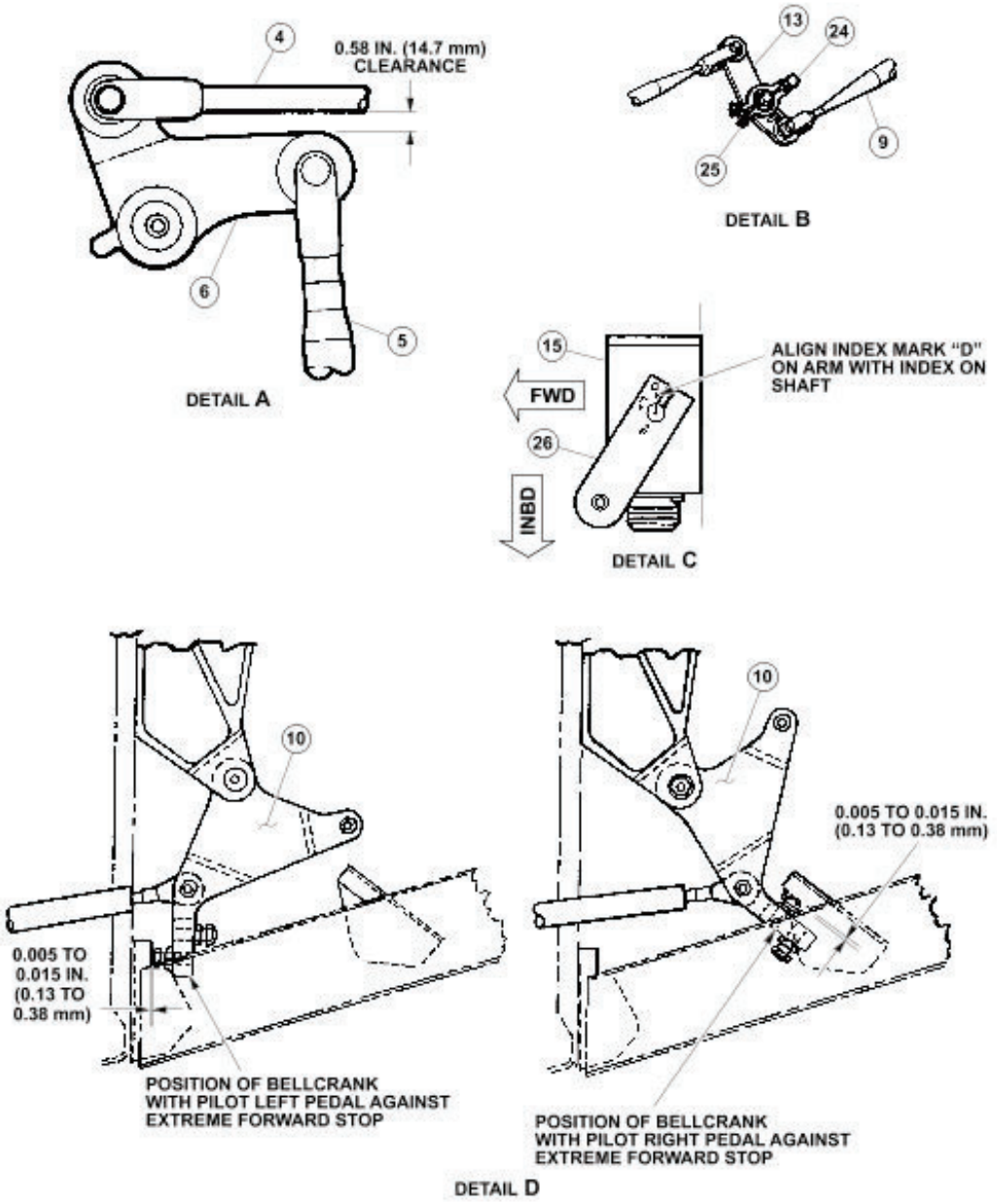
27. Move anti-torque controls through full travel and check for binding and/or obstruction of control movement.

28. Make sure all control bolts are installed and cotter pins and lockwire installed where applicable.



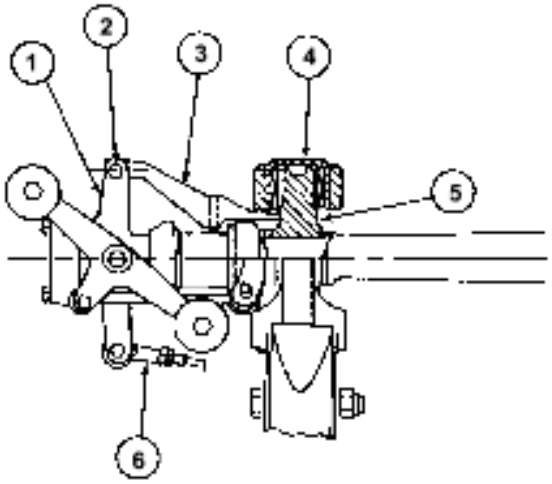
ICN-412-A-670100-B-97499-00024-A-001-01

Figure 1. Anti-Torque Rigging (Sheet 1 of 2)



ICN-412-A-670100-B-97499-00025-A-001-01

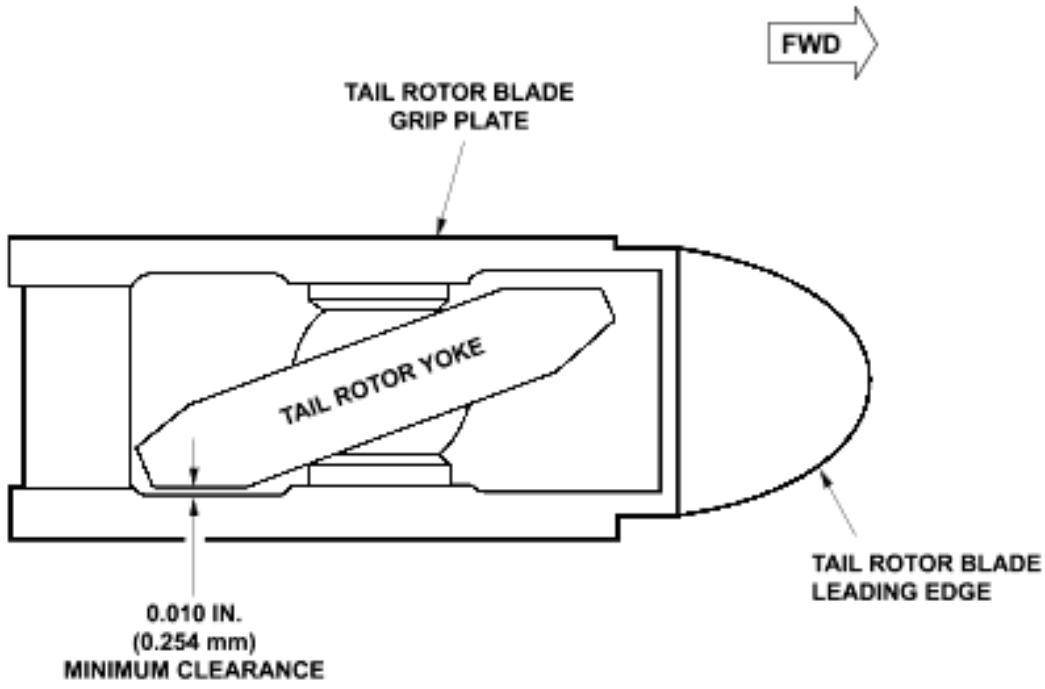
Figure 1. Anti-Torque Rigging (Sheet 2 of 2)



1. Crosshead
2. Bolt
3. Tail rotor rigging tool (T103206-101)
4. Tail rotor
5. Tail rotor hub trunnion
6. Pitch link

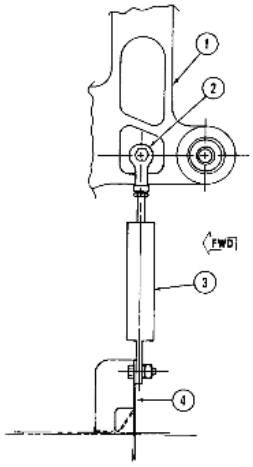
ICN-412-A-670100-B-97499-00026-A-001-01

Figure 2. Rigging Tool (T103206-101) (Sheet 1 of 1)



ICN-412-A-670100-B-97499-00027-A-001-01

Figure 3. Tail Rotor Blade to Yoke Clearance (Sheet 1 of 1)



VIEW LOOKING DOWN
DIRECTIONAL CONTROL TRANSDUCER

1. Bellcrank
2. Rod end bearing
3. Transducer
4. Support

ICN-412-A-670100-B-97499-00028-A-001-01

Figure 4. Control Motion Transducer Rigging (Sheet 1 of 1)

Aerodynamically Actuated Elevator - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Propeller Protractor	GS18217	1	Referred to as “protractor” in the following text.
Rigging Tool	T103296-101	1	
Clinometer	TB107	1	Referred to as “protractor” in the following text.
Feeler Gauge	Commercial	1	0.050 inch (1.270 mm) Shim Stock
Inclinometer	Commercial	1	Referred to as “protractor” in the following text.
Shot Bags	Commercial	1	or Equivalent, 20 to 30 pounds (9.1 to 13.6 Kg)

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Lockwire	C-405	AR	

Procedure

NOTE

Rigging may be accomplished with the helicopter leveled or not leveled. Caution must be taken not to move the helicopter regardless of leveling.

1. Remove the tailboom panel below the elevators (2 and 4, Figure 1) to get access to the control tube (5).

2. Remove the lockwire from the clevis jam nut (10) and locking washer (9). Loosen the clevis jam nut (10).

3. Remove the forward clevis (7) from the tube support (6) (412-A-67-02-00-02A-520A-A / 01544).

4. Install the rigging tool (14) on the shaft (15) as follows:

4.1. Separate the rigging tool (14) into two parts by removing the two tool screws (11) and tool washers (12).

4.2. Put the two parts of the rigging tool (14) on the shaft (15) with shouldered ends against the end cap (13). Attach the parts together with the two tool screws (11) and tool washers (12).

5. Hold the shaft (15) at wrenching flat. Tighten the clevis jam nut (10) against the rigging tool (14) until the shaft is extended to maximum length (control tube (5) internal stop is bottomed).

6. Find the correct elevator nose down position for the rigging as follows:



MAKE SURE THAT THE INSTRUMENT USED TO MEASURE DECK ANGLE, WHEN MOVED FROM THE LEFT SEAT RAIL TO THE LEFT ELEVATOR, IS PUT IN SAME FORE-AND-AFT DIRECTION AS WHEN THE DECK ANGLE WAS MEASURED.

6.1. Put the protractor (3) fore and aft on the copilot seat outboard seat rail.

NOTE

Removal of the copilot seat is not required.

6.2. Read and record the protractor (3) indication. This shows the helicopter nose up or down position.

6.3. Add/subtract above indication to/from 6° 4' (6.067°) and 6° 15' (6.25°). The result is the correct elevator nose down position for the rigging.

Example 1: Step 6.2. indicates that the helicopter is in a level position.

6 °04’(6.067 °) + 0 °0’(0.0 °) = 6 °04’(6.067 °)

6 °15’(6.25 °) + 0 °0’(0.0 °) = 6 °15’(6.25 °)

Therefore, correct elevator nose down position is **6 °04’ (6.067 °) to 6 °15’ (6.25 °)** for the rigging.

Example 2: Step 6.2. indicates that helicopter is in a 1 °50’(1.833 °) nose up position.

6 °04’(6.067 °) - 1 °50’(1.833 °) = 4 °14’(4.234 °)

6 °15’(6.25 °) - 1 °50’(1.833 °) = 4 °25’(4.417 °)

Therefore, correct elevator nose down position is **4 °14’ (4.234 °) to 4 °25’ (4.417 °)** for the rigging.

Example 3: Step 6.2. indicates that helicopter is in a 1 °50’ (1.833 °) nose down position.

6 °04’(6.067 °) + 1 °50’(1.833 °) = 7 °54’(7.9 °)

6 °15’(6.25 °) + 1 °50’(1.833 °) = 8 °05’(8.083 °)

Therefore, correct elevator nose down position is **7 °54’ (7.9 °) to 8 °05’ (8.083 °)** for the rigging.

7. Put the left elevator (4) in the correct nose down position, as done in the Step 6.2. as follows:



THE PROTRACTOR (3) SHALL BE POSITIONED ON THE LEFT ELEVATOR (4), AS SHOWN IN THE Figure 1, TO OBTAIN CORRECT RIGGING.

7.1. Put the protractor (3) on the left elevator (4).

7.2. Put the shot bags (1) near the leading edge of the right elevator (2) to temporarily hold the left elevator (4) in correct nose down position.

NOTE

It is permitted to remove the lockwire, loosen and adjust aft stop bolt (17), then reinstall the lockwire, if the correct position is not satisfactory.

7.3. Adjust the forward and aft stop bolts (17 and 18) to firmly hold the left elevator (4) in the correct nose down position.

7.4. Make sure that the protractor (3) is in the correct position and left elevator (4) is in the correct nose down position (Step 7.).

8. Adjust the length of the control tube (5) to align mounting holes of the forward clevis (7) and the tube support (6) as follows:

8.1. Hold the shaft (15) at wrenching flat. Move the forward clevis (7) until the clevis reaches the tube support (6).

8.2. Using the shaft (15) wrenching flat, move the shaft and forward clevis (7) until mounting holes in the forward clevis (7) and the tube support (6) are aligned.

9. Install the forward clevis (7) on the tube support (6) (412-A-67-02-00-02A-720A-A / 01546).

10. Examine for minimum thread engagement of the shaft (15) into the forward clevis (7) with the inspection hole in the forward clevis.

11. Make sure that the protractor (3) is in the correct position and left elevator (4) is in the correct nose down position (Step 7.).

12. Adjust the aft stop bolt (17) as follows:

12.1. Loosen the aft stop bolt (17).

12.2. Put the feeler gauge on aft surface of the elevator horn (8) in line with the aft stop bolt (17).

12.3. Tighten the aft stop bolt (17) until it contacts feeler gauge.

12.4. Attach the aft stop bolt (17) with the aft jam nut (16). Safety the jam nut (16) to stop support with the lockwire (C-405).

13. Make sure that the protractor (3) is in the correct position and left elevator (4) is in the correct nose down position (Step 7.).

14. Remove the rigging tool (14) as follows:

14.1. Hold the shaft (15) at wrenching flat. Loosen the clevis jam nut (10).

14.2. Separate the rigging tool (14) into two parts by removing the two tool screws (11) and tool washers (12).

15. Move the shaft no more than 45° to align keyway in the shaft (15) with nearest keyway in the forward clevis (7). Use the wrenching flat to move the shaft.

16. Put the locking washer (9) tab in the forward clevis (7) keyway. Tighten the clevis jam nut (10) against the locking washer. Safety the clevis jam nut to the locking washer with the lockwire (C-405).

17. Adjust the forward stop bolt (18) as follows:

17.1. Cut and remove the lockwire (C-405) and loosen the forward stop bolt (18).

17.2. Put a feeler gauge on forward surface of the elevator horn (8) in line with the forward stop bolt (18).

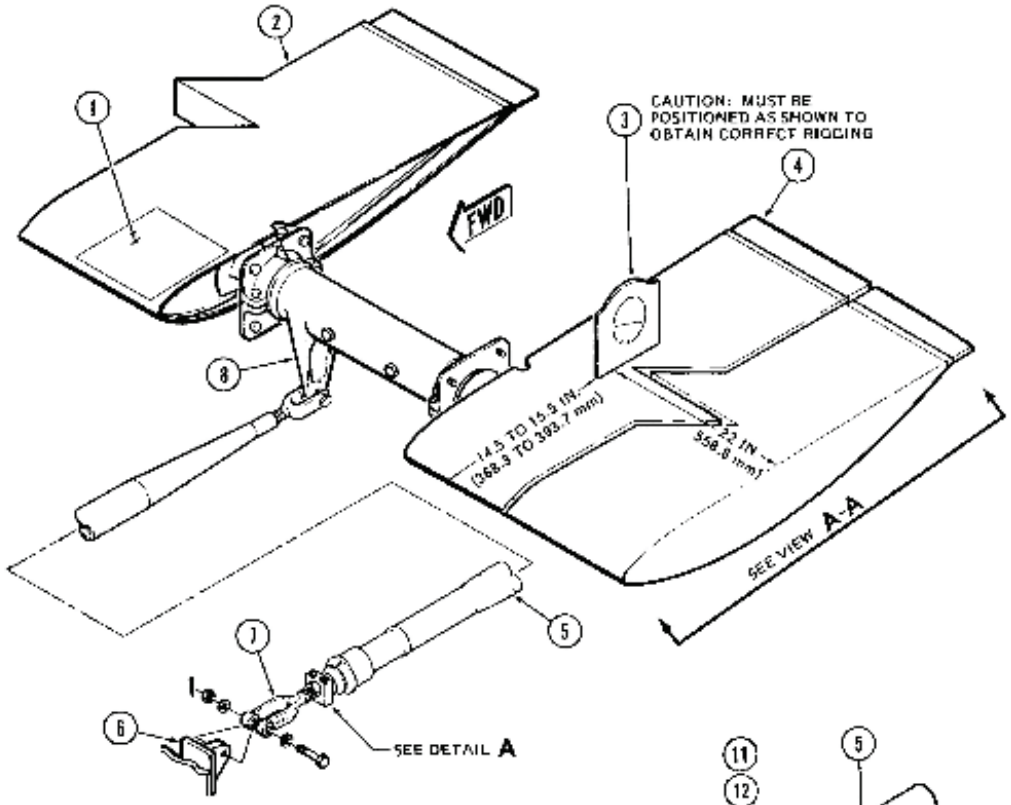
17.3. Tighten the forward stop bolt (18) until it contacts feeler gauge.

17.4. Attach the forward stop bolt (18) with the forward jam nut (19). Safety the forward jam nut (19) to stop support with the lockwire (C-405).

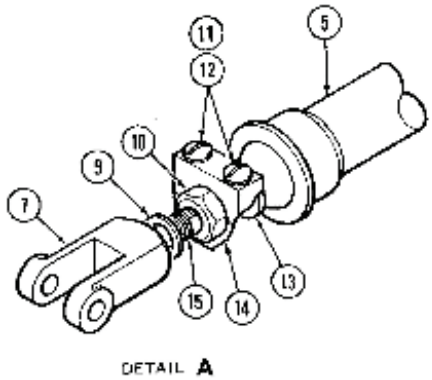
18. Remove the shot bags (1) and the protractor (3).

19. Apply the corrosion preventive compound (C-101) to threads of the shaft (15) and the forward clevis (7).

20. Install the tailboom panel below the right and left elevators (2 and 4).



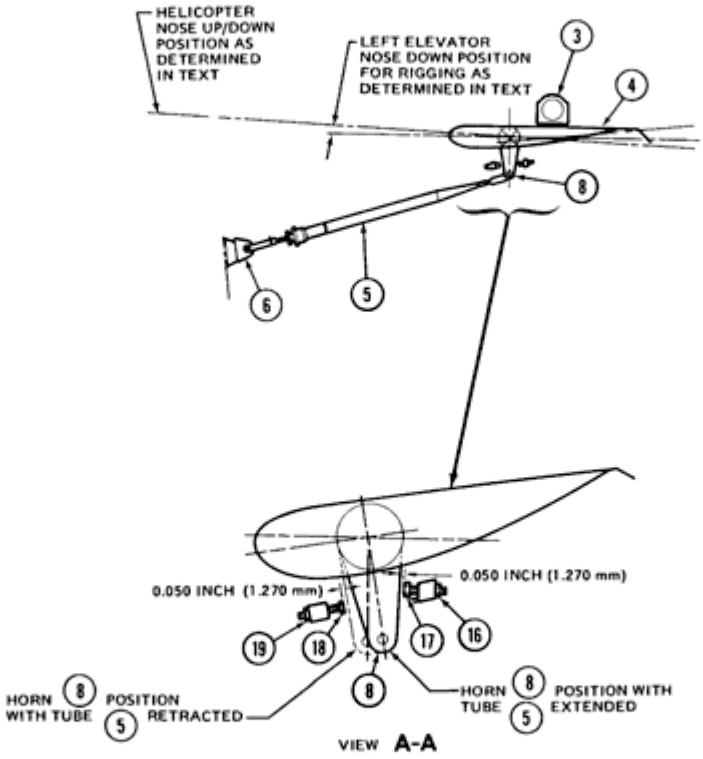
1. Shot bag — 20 to 30 lbs. (9.1 to 13.6 kg)
2. Right elevator
3. Protractor
4. Left elevator
5. Control tube
6. Tube support
7. Forward clevis
8. Elevator horn
9. Locking washer
10. Clevis jam nut
11. Tool screw
12. Tool washer
13. End cap
14. Rigging tool
15. Shaft
16. Aft jam nut
17. Aft stop bolt
18. Forward stop bolt
19. Forward jam nut



DETAIL A

ICN-412-A-670100-B-97499-00029-A-001-01

Figure 1. Elevator Rigging (Sheet 1 of 2)



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Figure 1. Elevator Rigging (Sheet 2 of 2)

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Collective Controls - Description

Collective controls (Figure 1) consists of linkage components from pilot collective stick (13) up to, but not including, collective lever (1).

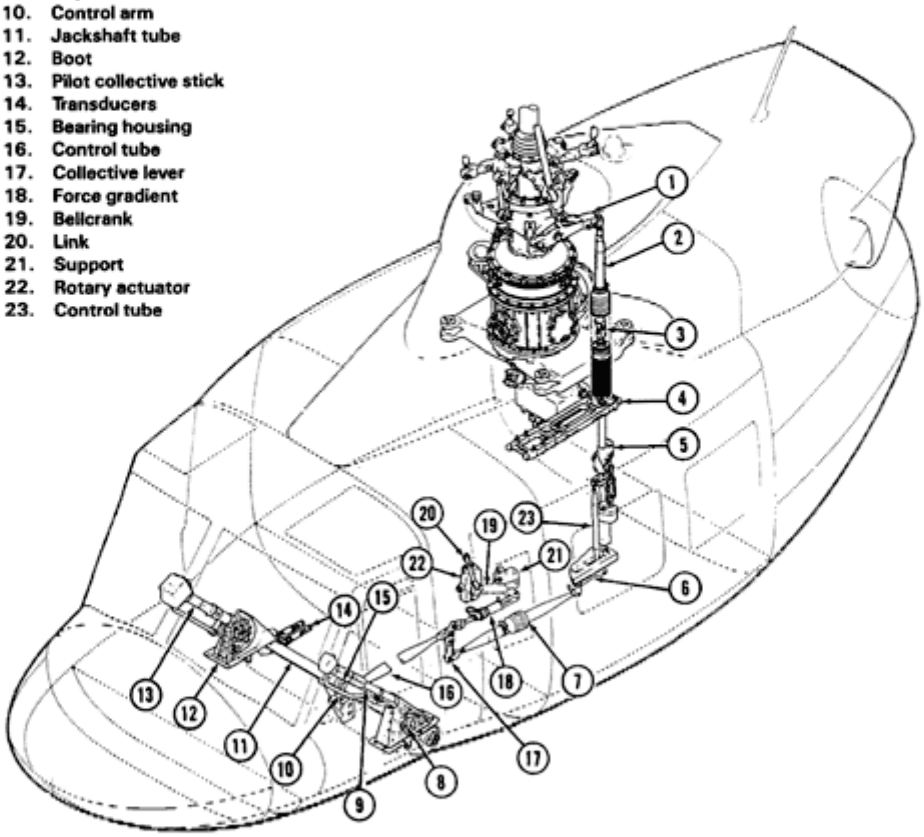
NOTE

Automatic Flight Control System (AFCS) is functional in collective control system only when a Flight Director Kit is installed in helicopter.

Cylinder (5) reduces effort required for control and reduces feedback forces from main rotor.

Operation of collective stick (13) causes hydraulic power assisted collective pitch change of main rotor blades.

1. Collective lever
2. Control tube
3. Universal
4. Support
5. Cylinder
6. Bellcrank
7. Adjustable control tube
8. Clamp
9. Copilot collective stick
10. Control arm
11. Jackshaft tube
12. Boot
13. Pilot collective stick
14. Transducers
15. Bearing housing
16. Control tube
17. Collective lever
18. Force gradient
19. Bellcrank
20. Link
21. Support
22. Rotary actuator
23. Control tube



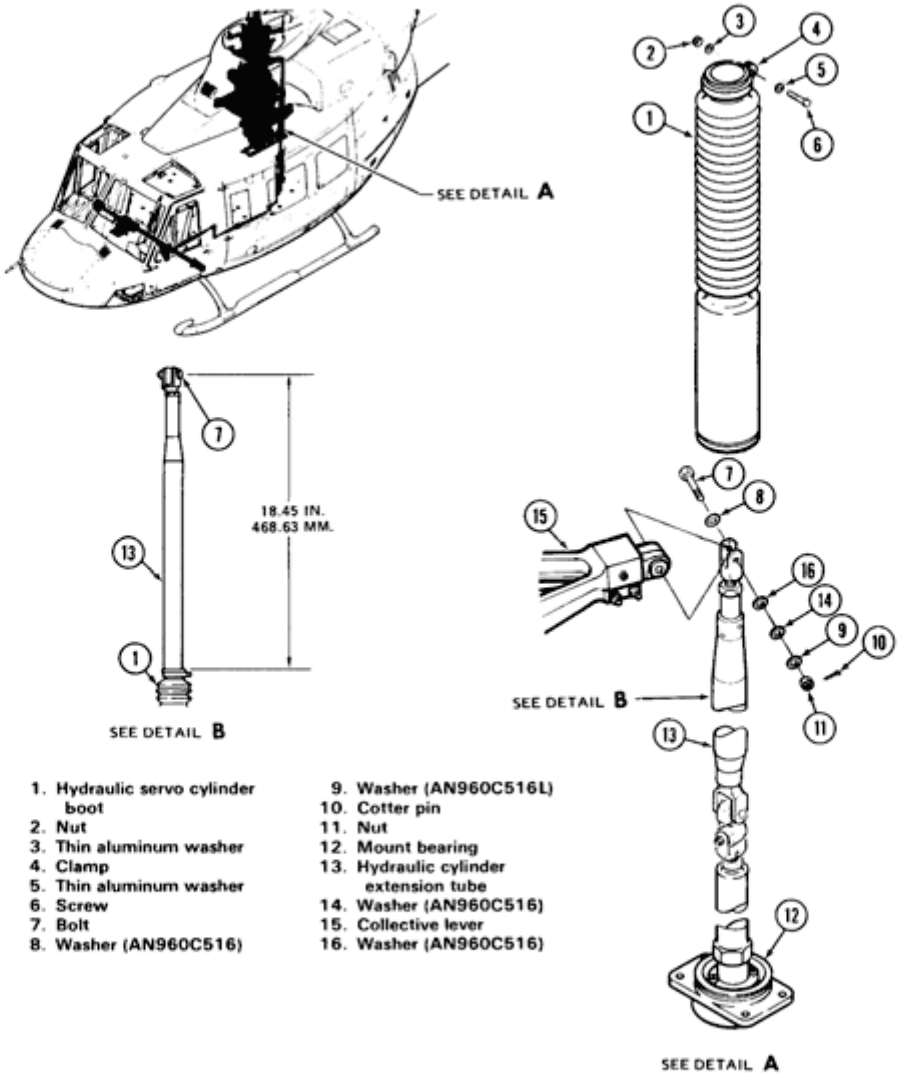
ICN-412-A-671100-B-97499-00001-A-001-01

Figure 1. Collective Controls (Sheet 1 of 1)

Hydraulic Servo Cylinder Boot - Removal

Procedure

1. Open forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).
2. Open aft pylon access doors (412-A-52-40-00-02A-520A-A / 00837).
3. Remove cotter pin (10, Figure 1), nut (11), washers (8, 9, 14 and 16) and bolt (7).
4. Remove nut (2), thin aluminum washers (3 and 5) and screw (6) and clamp (4).
5. Loosen the hydraulic servo cylinder boot (1) from retaining groove in support bracket at mount bearing (12) and remove boot (1).



1. Hydraulic servo cylinder boot
2. Nut
3. Thin aluminum washer
4. Clamp
5. Thin aluminum washer
6. Screw
7. Bolt
8. Washer (AN960C516)
9. Washer (AN960C516L)
10. Cotter pin
11. Nut
12. Mount bearing
13. Hydraulic cylinder extension tube
14. Washer (AN960C516)
15. Collective lever
16. Washer (AN960C516)

ICN-412-A-671100-B-97499-00002-A-001-01

Figure 1. Hydraulic Servo Cylinder Boot (Sheet 1 of 1)

Hydraulic Servo Cylinder Boot - Inspection

Procedure

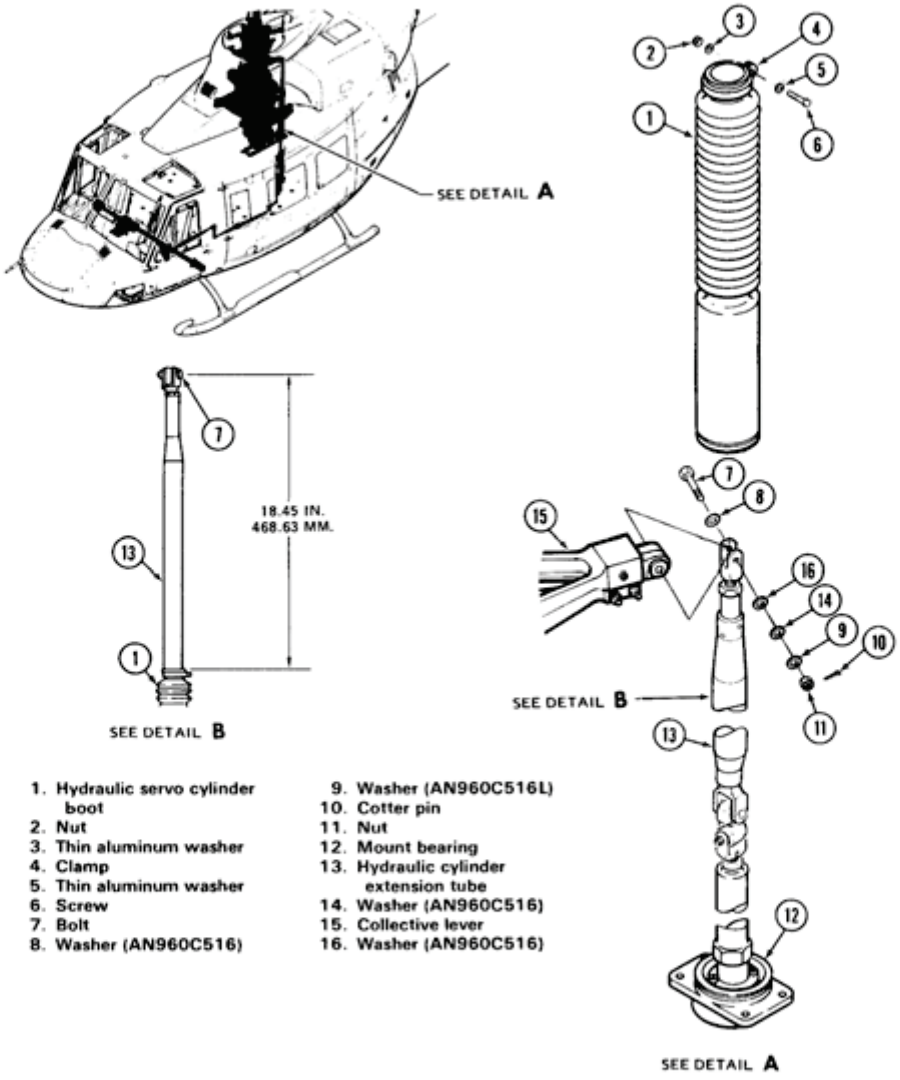
1. Examine the hydraulic servo cylinder boot (1) for tears and deterioration.

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Hydraulic Servo Cylinder Boot - Installation

Procedure

1. Slide hydraulic servo cylinder boot (1, Figure 1) over hydraulic cylinder extension tube (13). Secure bottom of boot (1) into lip in mount bearing (12).
2. Put the top of boot (1, detail B) on hydraulic cylinder extension tube (13) 18.45 in. (468.63 mm) down from centerline of bolt hole in clevis at top of tube (13).
3. Install the clamp (4), screw (6), thin aluminum washers (3 and 5) and nut (2).
4. Put the tube (13) onto the collective lever (15) and install bolt (7), washers (8, 9, 14, and 16), nut (11) and cotter pin (10).
5. Move collective controls through full travel and check for binding and/or obstruction of control movement.
6. Make sure that the collective control rigging is within limits (412-A-67-01-00-01A-276A-A / 01323).
7. Stow hold-open brace assembly and close aft pylon access door(s).
8. Install forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



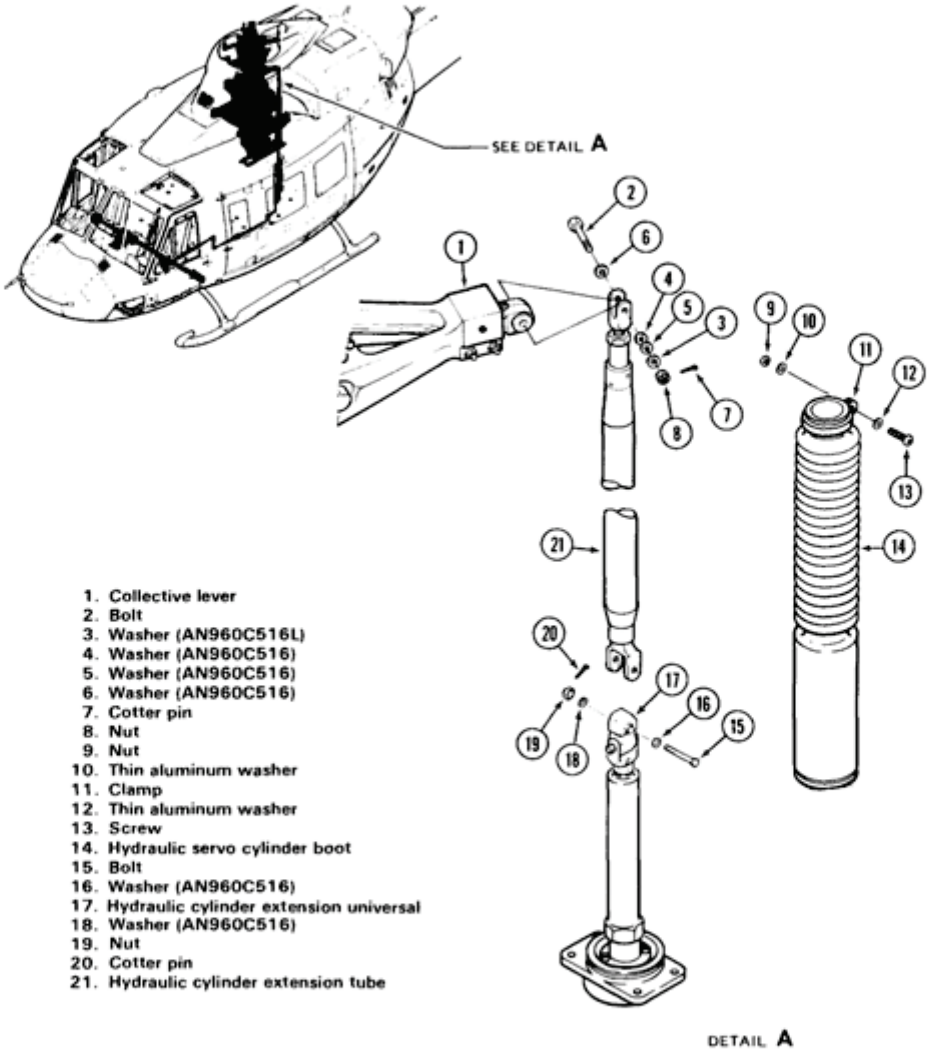
ICN-412-A-671100-B-97499-00002-A-001-01

Figure 1. Hydraulic Servo Cylinder Boot (Sheet 1 of 1)

Hydraulic Cylinder Extension Tube - Removal

Procedure

1. Open forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).
2. Open aft pylon access door(s) (412-A-52-40-00-02A-520A-A / 00837).
3. Remove screw (13, Figure 1) and loosen clamp (11). Slide hydraulic servo cylinder boot (14) up to expose hydraulic cylinder extension universal (17).
4. Remove cotter pin (20), nut (19), washers (16 and 18) and bolt (15).
5. Remove cotter pin (7), nut (8), washers (3, 5, 4 and 6) and bolt (2). Remove hydraulic cylinder extension tube (21) from helicopter.



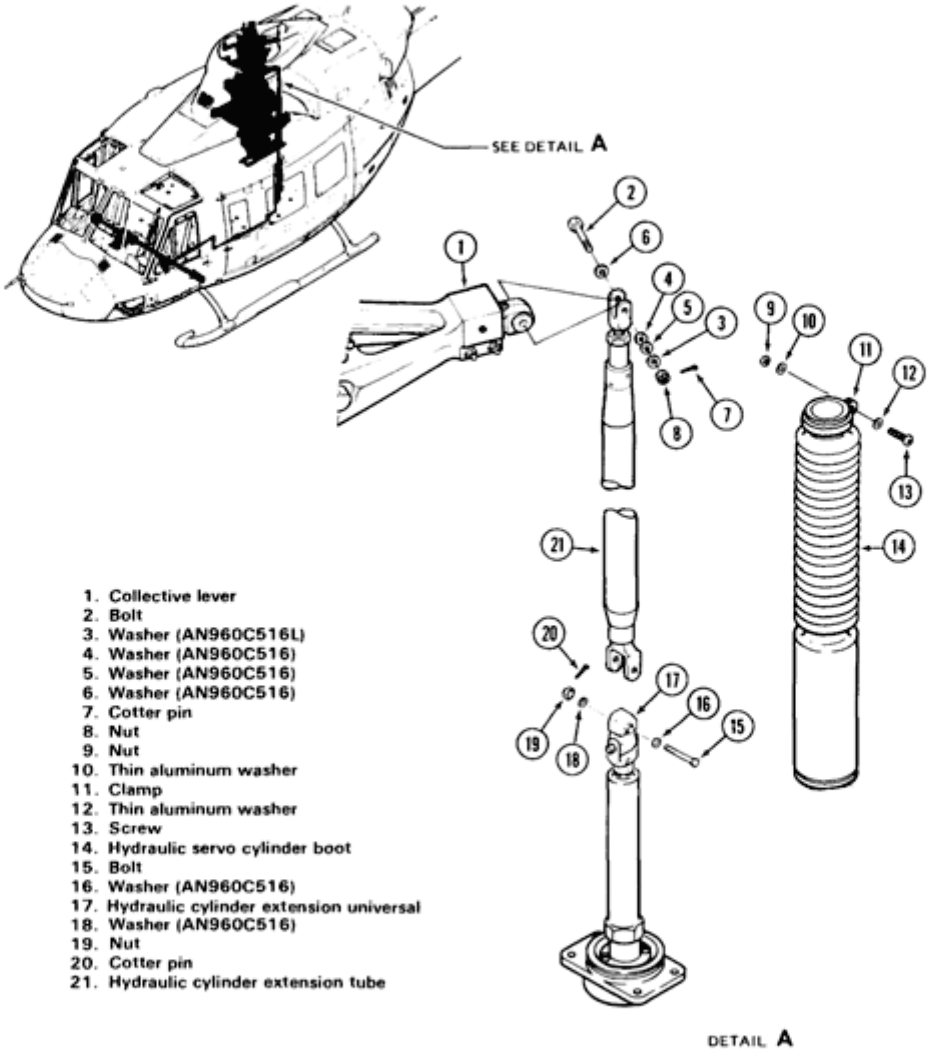
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Figure 1. Hydraulic Cylinder Extension Tube (Sheet 1 of 1)

Hydraulic Cylinder Extension Tube - Inspection

Procedure

1. Examine the tube (21, Figure 1) for serviceability in accordance with BHT-412-CR&O.



ICN-412-A-671100-B-97499-00003-A-001-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 1 of 1)

Hydraulic Cylinder Extension Tube - Installation

Procedure

1. Place hydraulic cylinder extension tube (21, Figure 1) into position on collective lever (1) and install bolt (2), washer (6), washers (4, 5 and 3) as required, nut (8), and cotter pin (7).

2. Place tube (21) onto hydraulic cylinder extension universal (17) and install bolt (15), washer (16), washer (18), nut (19), and cotter pin (20).

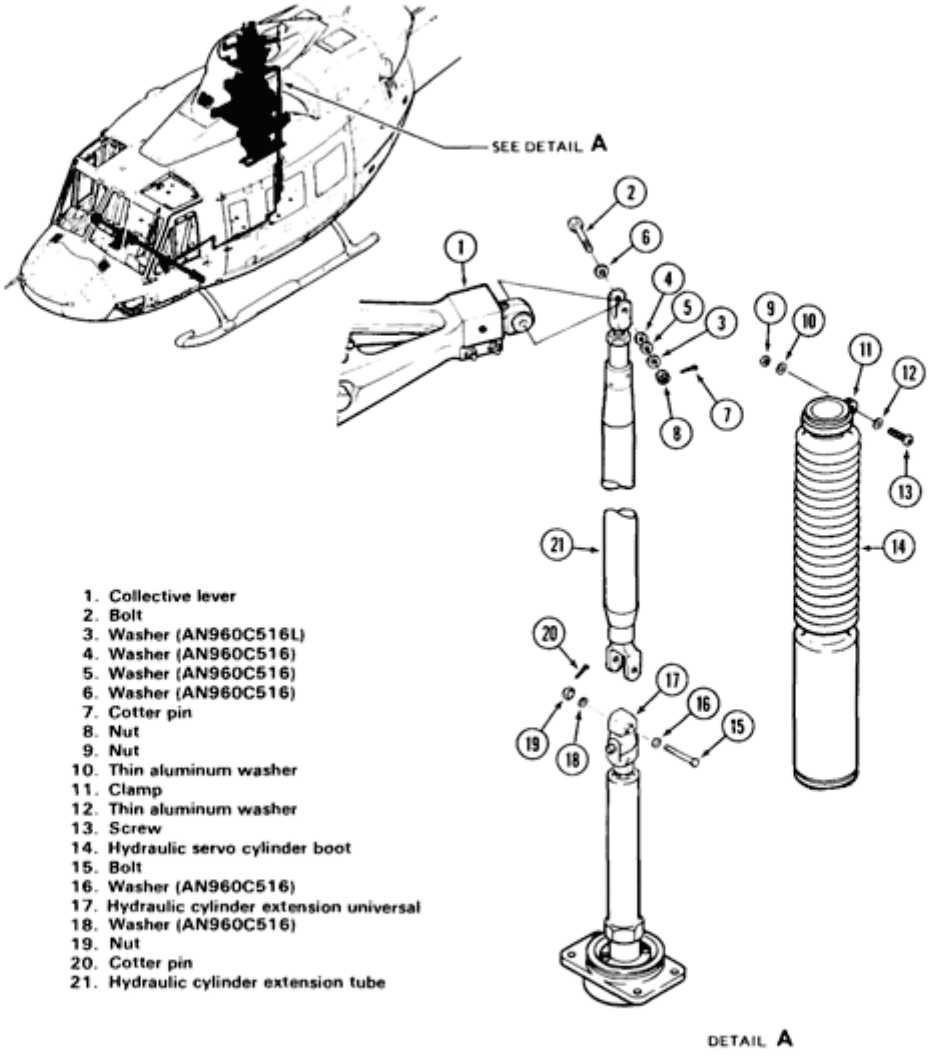
3. Position hydraulic servo cylinder boot (1, Figure 2) on hydraulic cylinder extension tube (13) to obtain dimension shown. Install clamp (11, Figure 1) with screw (13), washer (12), washer (10), and nut (9).

4. Move collective controls through full travel and check for interference and binding. Verify hydraulic servo cylinder boot (14) is correctly positioned and does not bind when controls are moved through full travel.

5. Verify collective rigging is within limits, (412-A-67-01-00-01A-276A-A / 01323).

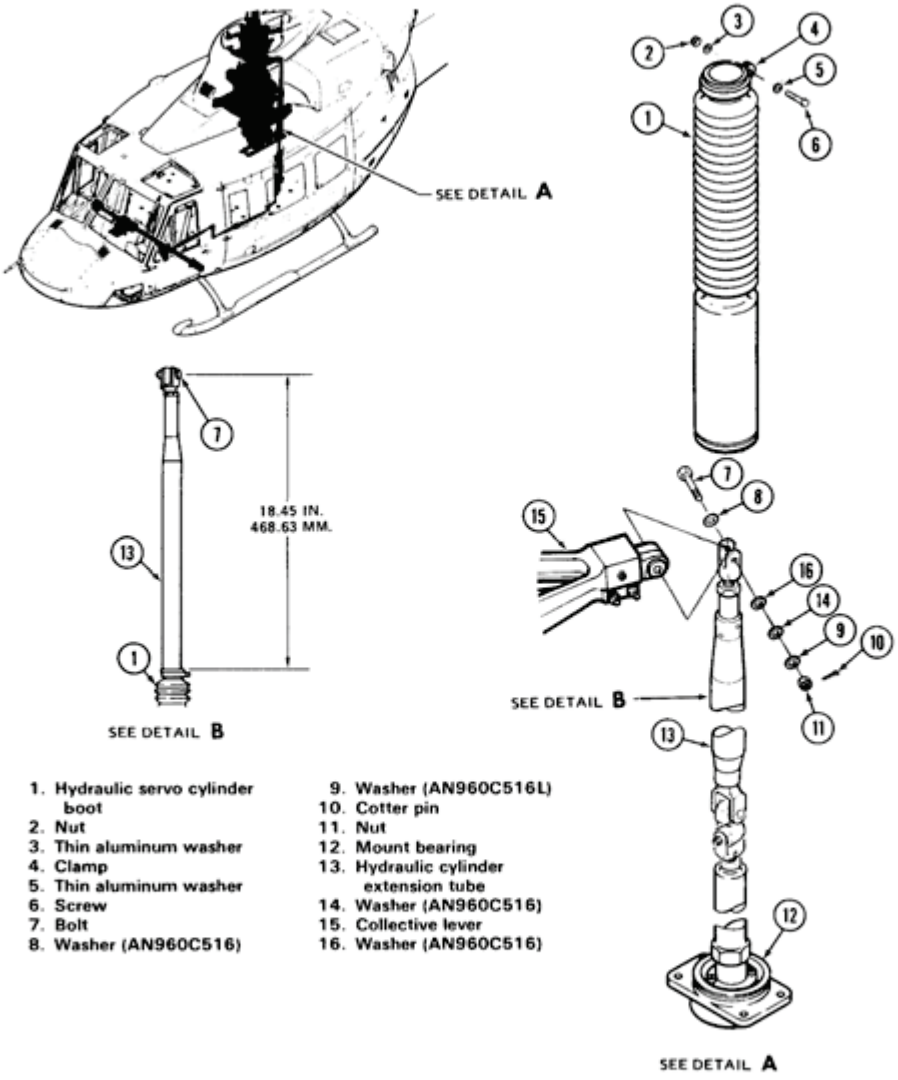
6. Close aft pylon access door(s) (412-A-52-40-00-02A-720A-A / 00840).

7. Close forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



ICN-412-A-671100-B-97499-00003-A-001-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 1 of 1)



1. Hydraulic servo cylinder boot
2. Nut
3. Thin aluminum washer
4. Clamp
5. Thin aluminum washer
6. Screw
7. Bolt
8. Washer (AN960C516)
9. Washer (AN960C516L)
10. Cotter pin
11. Nut
12. Mount bearing
13. Hydraulic cylinder extension tube
14. Washer (AN960C516)
15. Collective lever
16. Washer (AN960C516)

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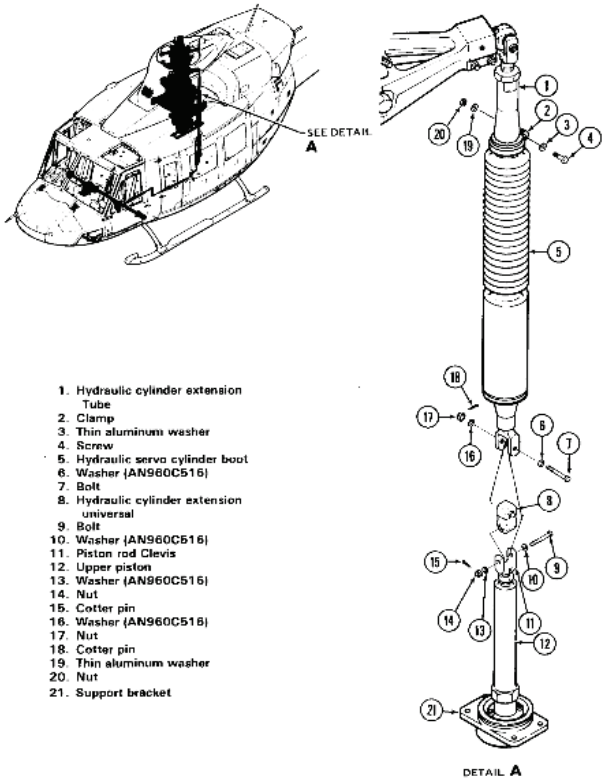
Figure 2. Hydraulic Servo Cylinder Boot (Sheet 1 of 1)

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Hydraulic Cylinder Extension Universal - Removal

Procedure

1. Open forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).
2. Open aft pylon fairing access door(s) (412-A-52-40-00-02A-720A-A / 00840).
3. Remove screw (4, Figure 1). Loosen bottom of hydraulic servo cylinder boot (5) from retaining groove in support bracket (21). Slide hydraulic servo cylinder boot upward to expose hydraulic cylinder extension universal (8).
4. Remove cotter pin (15), nut (14), washers (10 and 13) and bolt (9). Remove hydraulic cylinder extension universal (8) from piston rod clevis (11) and upper piston (12).
5. Remove cotter pin (18), nut (17), washers (16 and 6) bolt (7) and hydraulic cylinder extension universal (8).



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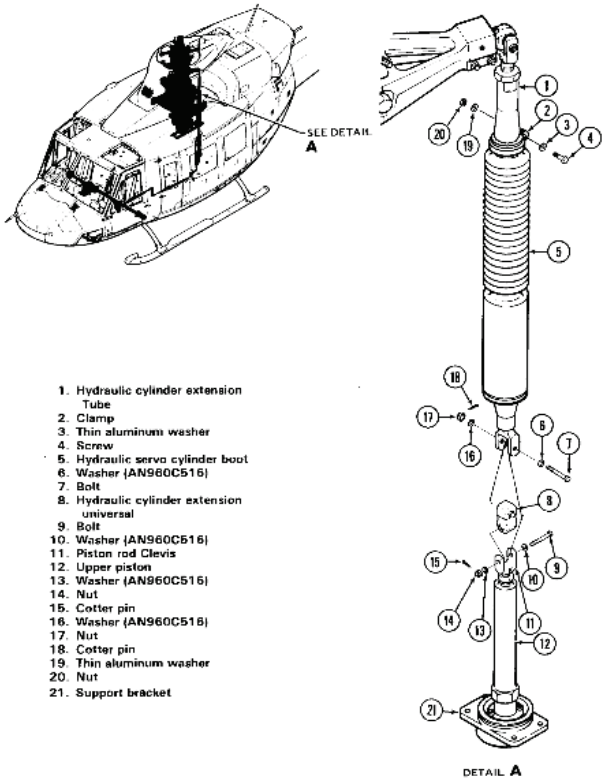
Figure 1. Hydraulic Cylinder Extension Universal (Sheet 1 of 1)

Hydraulic Cylinder Extension Universal - Inspection

Procedure

1. Inspect hydraulic cylinder extension universal (8, Figure 1) for mechanical and corrosion damage (BHT-412-CR&O).

2. Inspect bearings for secure installation in hydraulic cylinder extension universal and for wear. Replace as required (BHT-ALL-SPM, Standard Practices Manual).



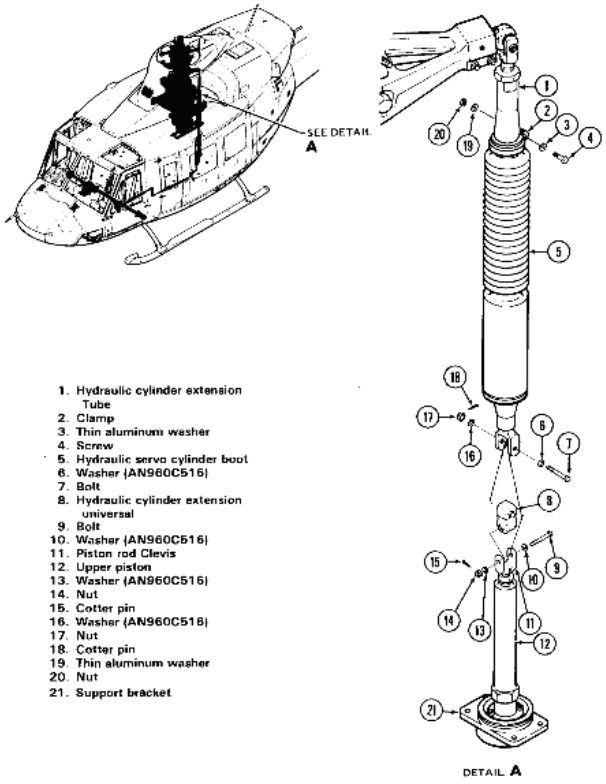
ICN-412-A-671100-B-97499-00004-A-001-01

Figure 1. Hydraulic Cylinder Extension Universal (Sheet 1 of 1)

Hydraulic Cylinder Extension Universal - Installation

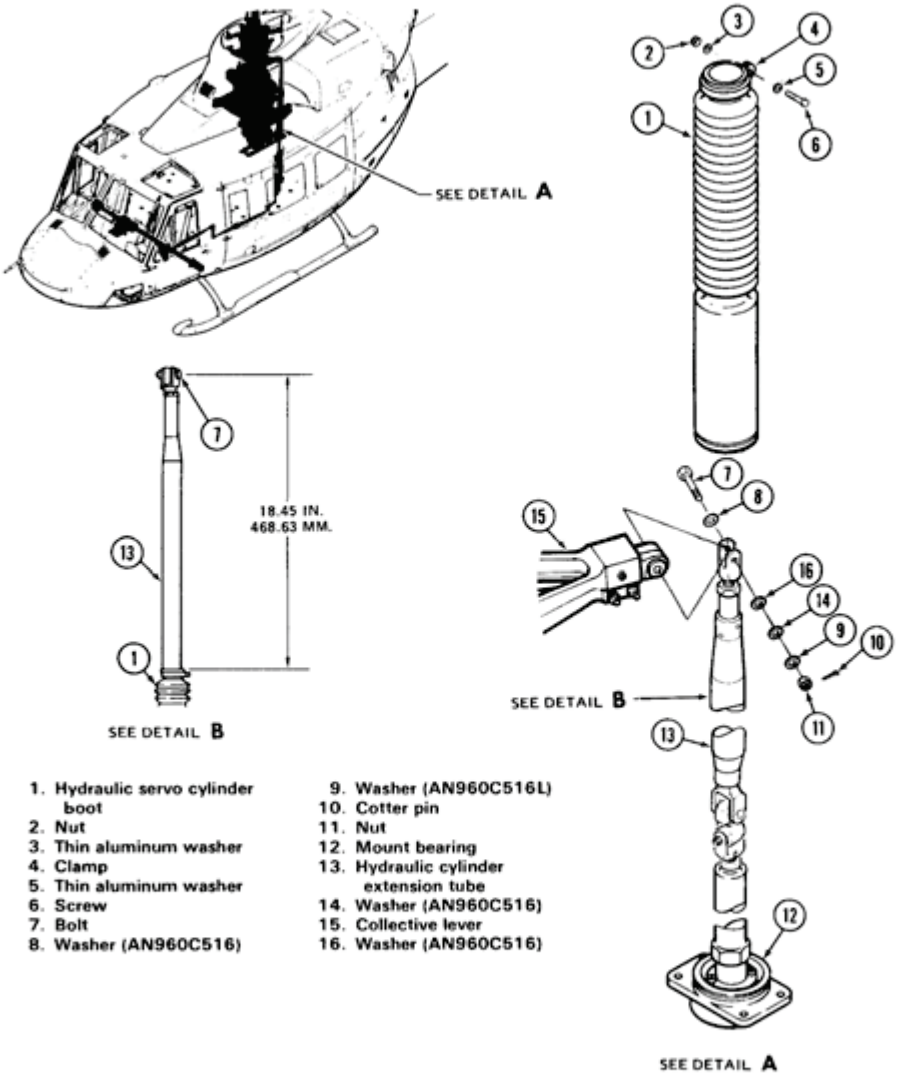
Procedure

1. Place hydraulic cylinder extension universal (8, Figure 1) in position in piston rod clevis (11) and install bolt (9), washer (10), washer (13), nut (14), and cotter pin (15).
2. Place clevis on end of hydraulic cylinder extension tube (1) on universal (8) and install bolt (7), washer (6), washer (16), nut (17), and cotter pin (18).
3. Slide hydraulic servo cylinder boot (5) down and secure lower end into retaining groove at support bracket (21).
4. Position top of hydraulic servo cylinder boot (1, Figure 2) 18.45 in. (468.63 mm) on hydraulic cylinder extension tube (13) down from centerline of bolt hole in clevis at top of tube (13).
5. Secure boot with clamp (2, Figure 1), screw (4), washer (3), washer (19), and nut (20).
6. Close aft pylon fairing access door(s) (412-A-52-40-00-02A-720A-A / 00840).
7. Close forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



ICN-412-A-671100-B-97499-00004-A-001-01

Figure 1. Hydraulic Cylinder Extension Universal (Sheet 1 of 1)



1. Hydraulic servo cylinder boot
2. Nut
3. Thin aluminum washer
4. Clamp
5. Thin aluminum washer
6. Screw
7. Bolt
8. Washer (AN960C516)
9. Washer (AN960C516L)
10. Cotter pin
11. Nut
12. Mount bearing
13. Hydraulic cylinder extension tube
14. Washer (AN960C516)
15. Collective lever
16. Washer (AN960C516)

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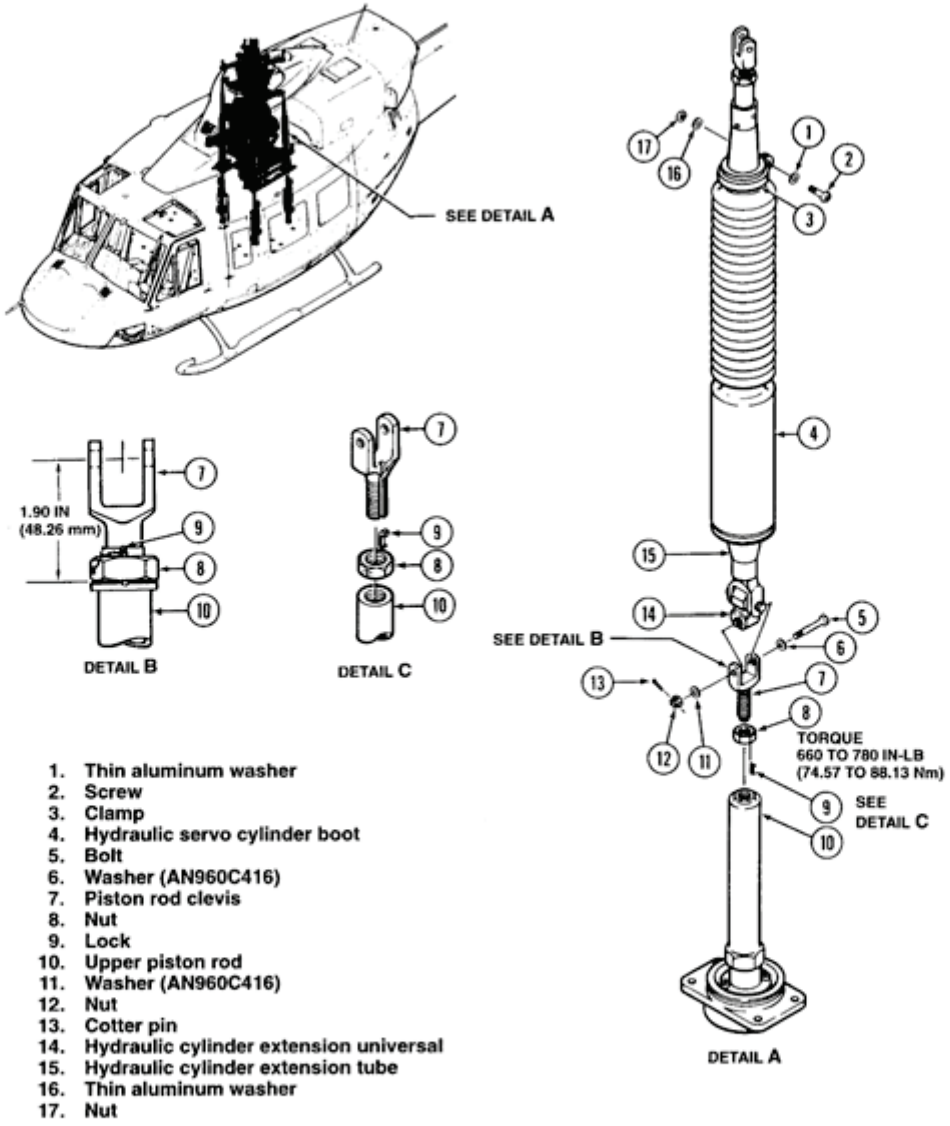
Figure 2. Hydraulic Servo Cylinder Boot (Sheet 1 of 1)

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Piston Rod Clevis - Removal

Procedure

1. Open forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).
2. Open aft pylon fairing access door(s) (412-A-52-40-00-02A-520A-A / 00837).
3. Remove nut (17, Figure 1), washers (1 and 16) and screw (2) loosen clamp (3) and slide boot (4) up to expose piston rod (10) and hydraulic cylinder extension tube (15).
4. Remove cotter pin (13), nut (12), washers (11 and 6) and bolt (5).
5. Remove lockwire, loosen nut (8) and remove piston rod clevis (7).
6. Remove nut (8) and lock (9) from piston rod clevis (7).



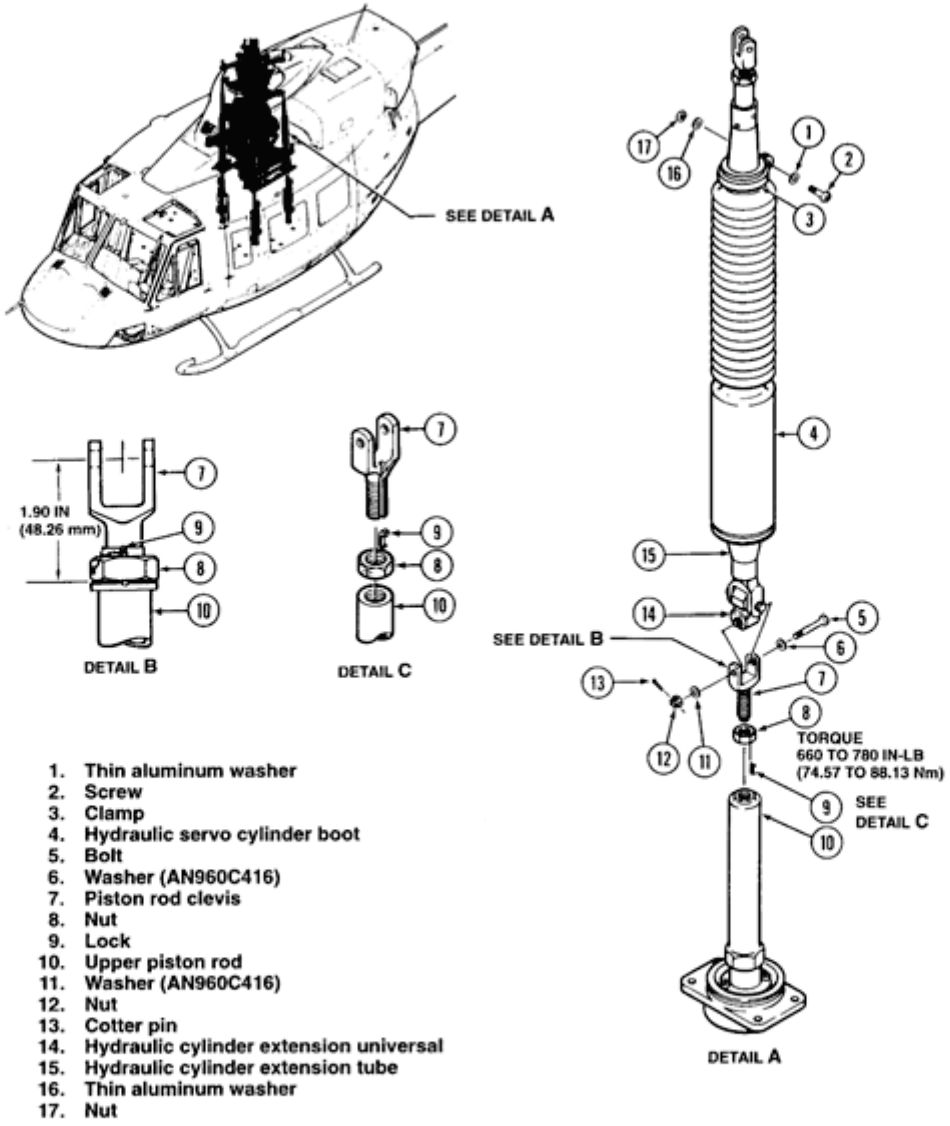
ICN-412-A-671100-B-97499-00005-A-001-01

Figure 1. Piston Rod Clevis (Sheet 1 of 1)

Piston Rod Clevis - Inspection

Procedure

1. Inspect piston rod clevis (7, Figure 1) for nicks, scratches, and damaged threads.
2. Inspect piston rod clevis (7) for elongated bolt holes.



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Figure 1. Piston Rod Clevis (Sheet 1 of 1)

Piston Rod Clevis - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-104	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

1. Coat threads of piston rod clevis (7, Figure 1) with corrosion preventive compound (C-104). Place lock (9) into nut (8, detail C). Install nut (8) and lock (9) on clevis (7) with hole in lock (9) up as illustrated.

2. Thread clevis (7) into piston rod (10) and adjust to obtain 1.90 in. (48.26 mm) from centerline of bolt hole to top of piston rod (detail B).

3. Torque nut (8) 660 to 780 in.-lbs. (74.57 to 88.13 N-m). Secure with lockwire (C-405).

4. Cover nut (8), lock (9), lockwire and threads on clevis (7) with sealant (C-308). Refer to figure Figure 3, Sheet 3, Detail G.



AVOID GETTING SEALANT ON PISTON ROD SURFACE OR SCRAPER AND ROD SEAL DAMAGE MAY OCCUR.

5. Place hydraulic cylinder extension universal (14, Figure 1) into clevis (7) and install bolt (5), washer (6), washer (11), nut (12), and cotter pin (13).

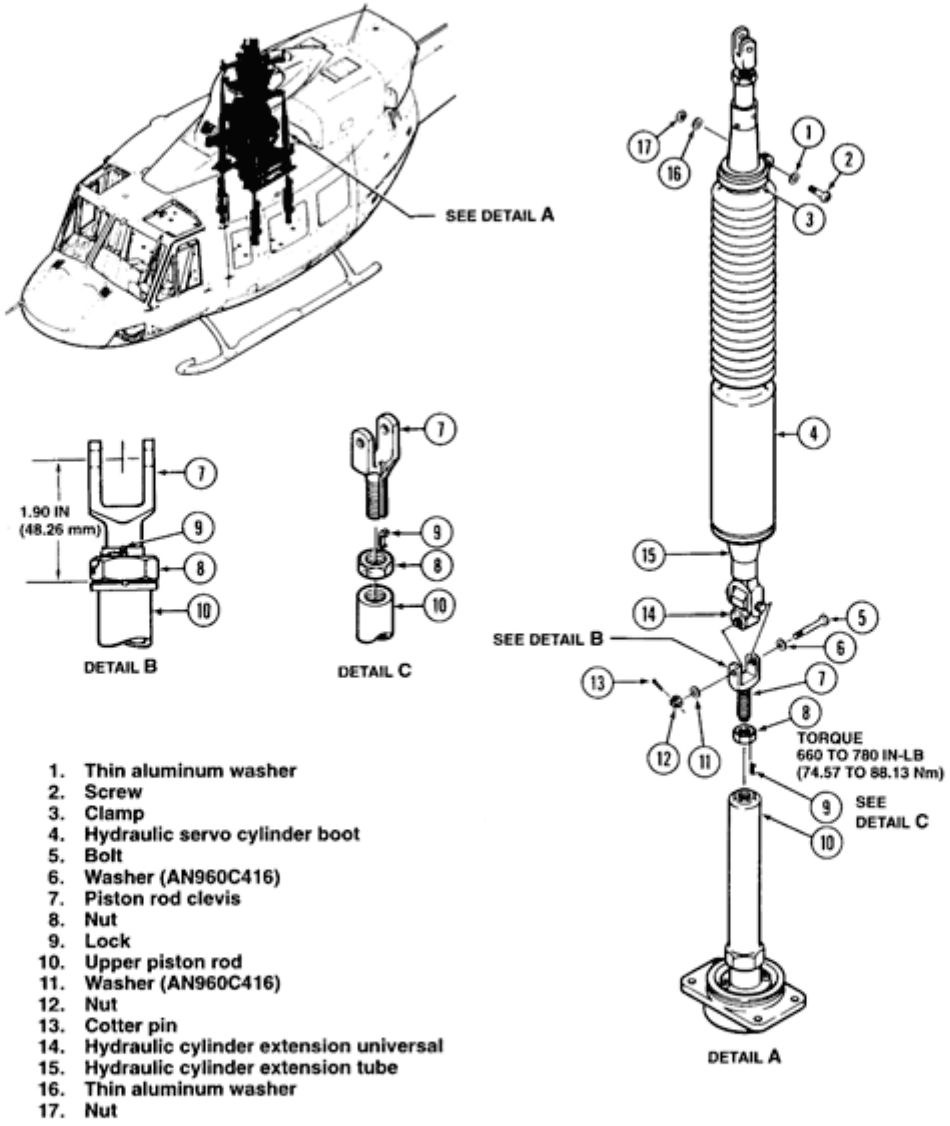
6. Slide hydraulic servo cylinder boot down over universal and position top of boot 18.45 in. (468.63 mm) from centerline of bolt hole in top of hydraulic cylinder extension tube (13, Figure 2, detail B). Secure clamp (3, Figure 1) with screw (2), thin aluminum washer (1), washer (16), and nut (17).

7. Move collective controls through full travel and check for interference and binding.

8. Verify collective and/or cyclic control rigging is within limits (412-A-67-01-00-01A-276A-A / 01323, 412-A-67-01-00-03A-276A-A / 01324, 412-A-67-01-00-04A-276A-A / 01325 and 412-A-67-01-00-05A-276A-A / 01326).

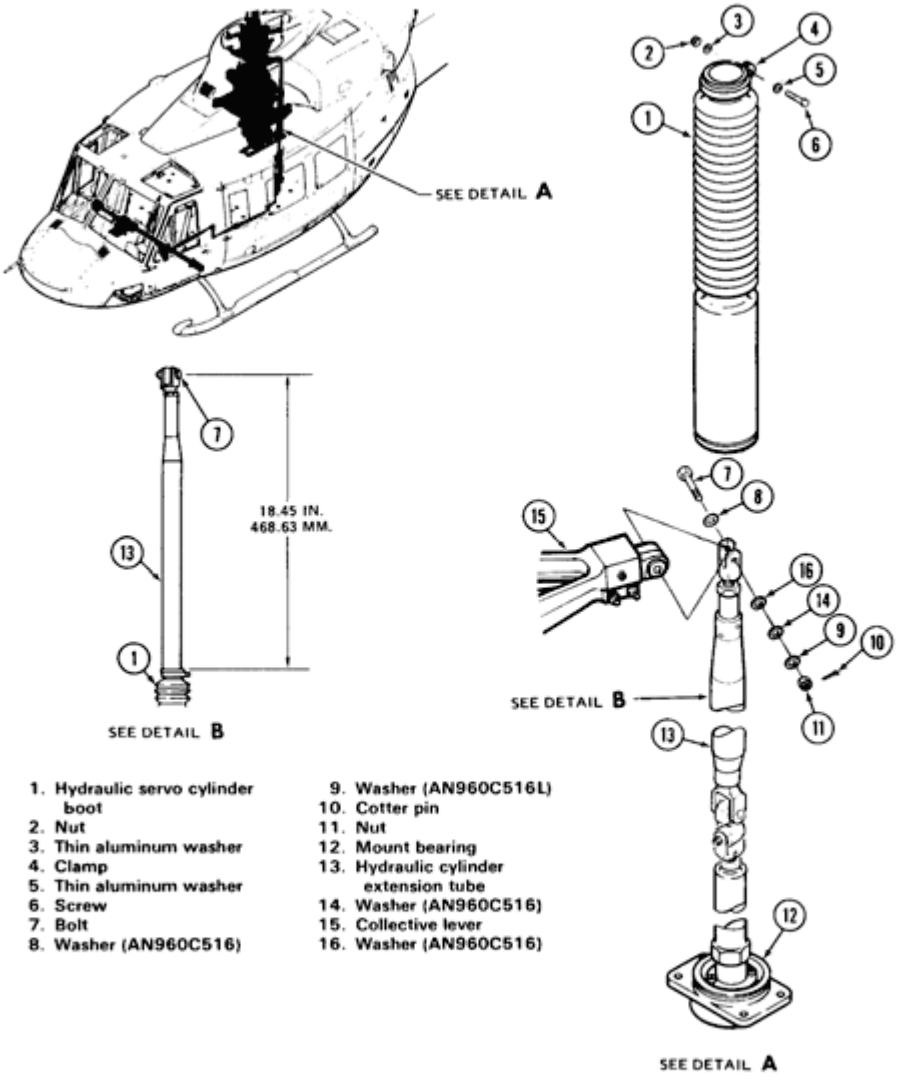
9. Close aft pylon fairing access door(s) (412-A-52-40-00-02A-720A-A / 00840).

10. Close forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



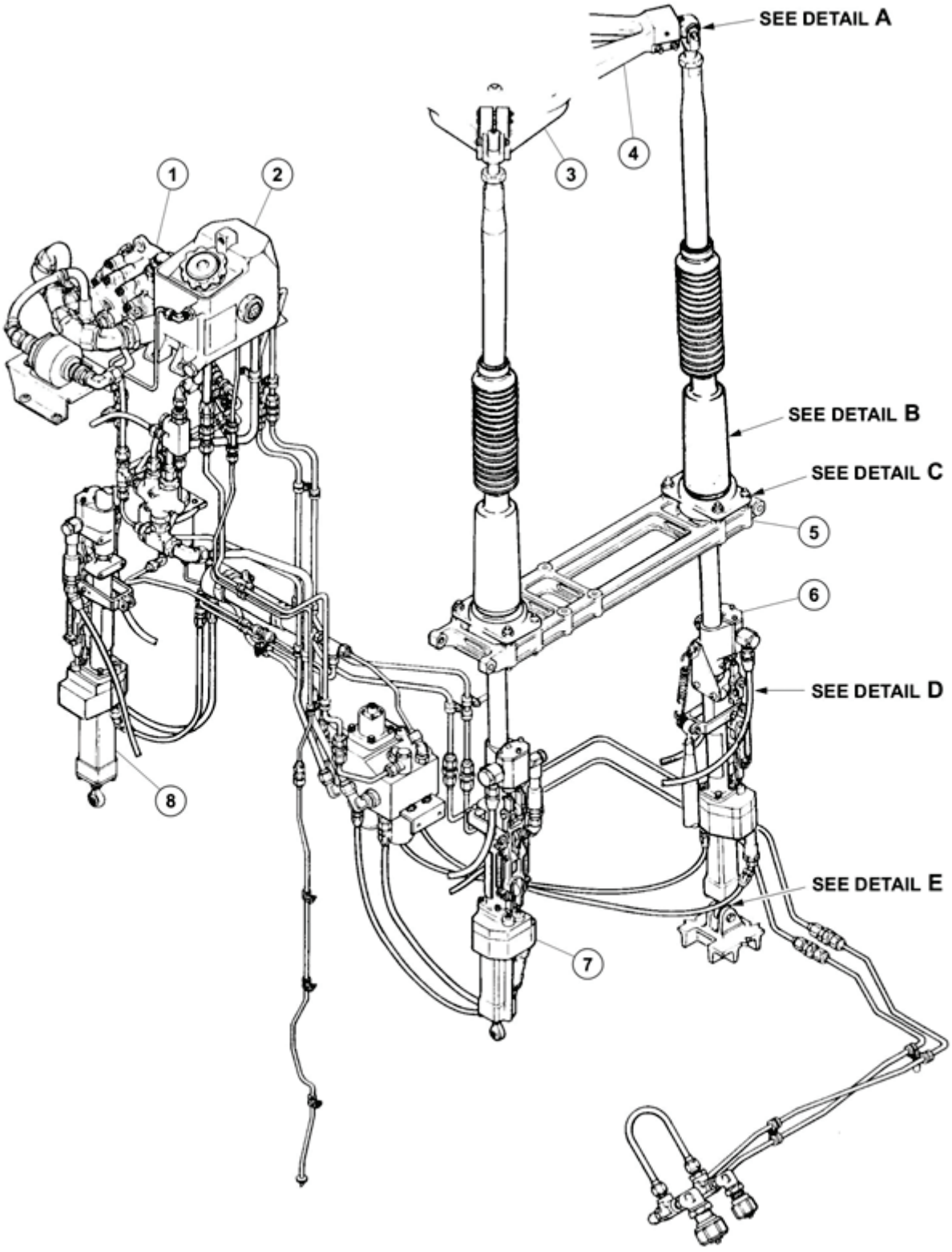
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Figure 1. Piston Rod Clevis (Sheet 1 of 1)



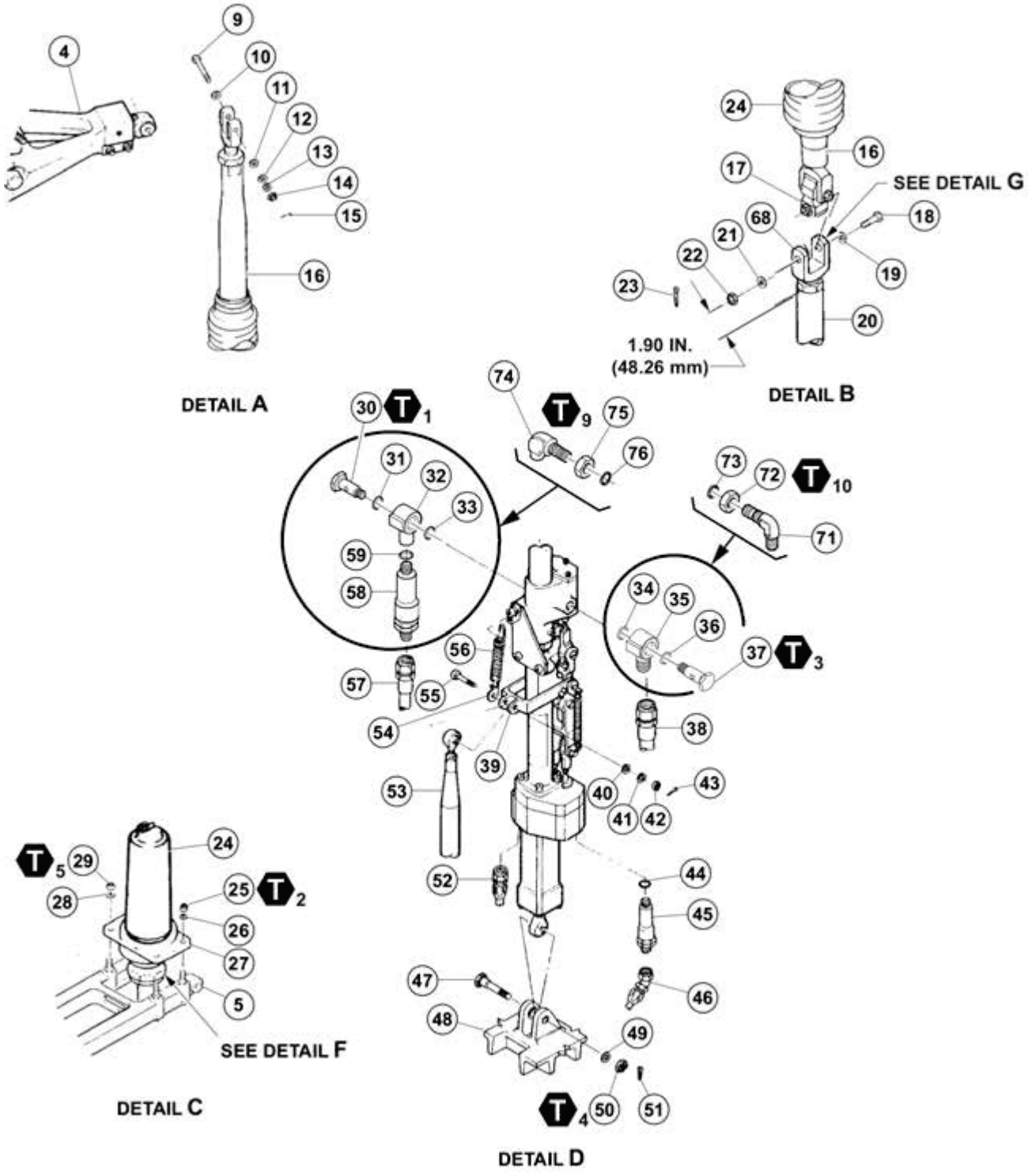
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Figure 2. Hydraulic Servo Cylinder Boot (Sheet 1 of 1)



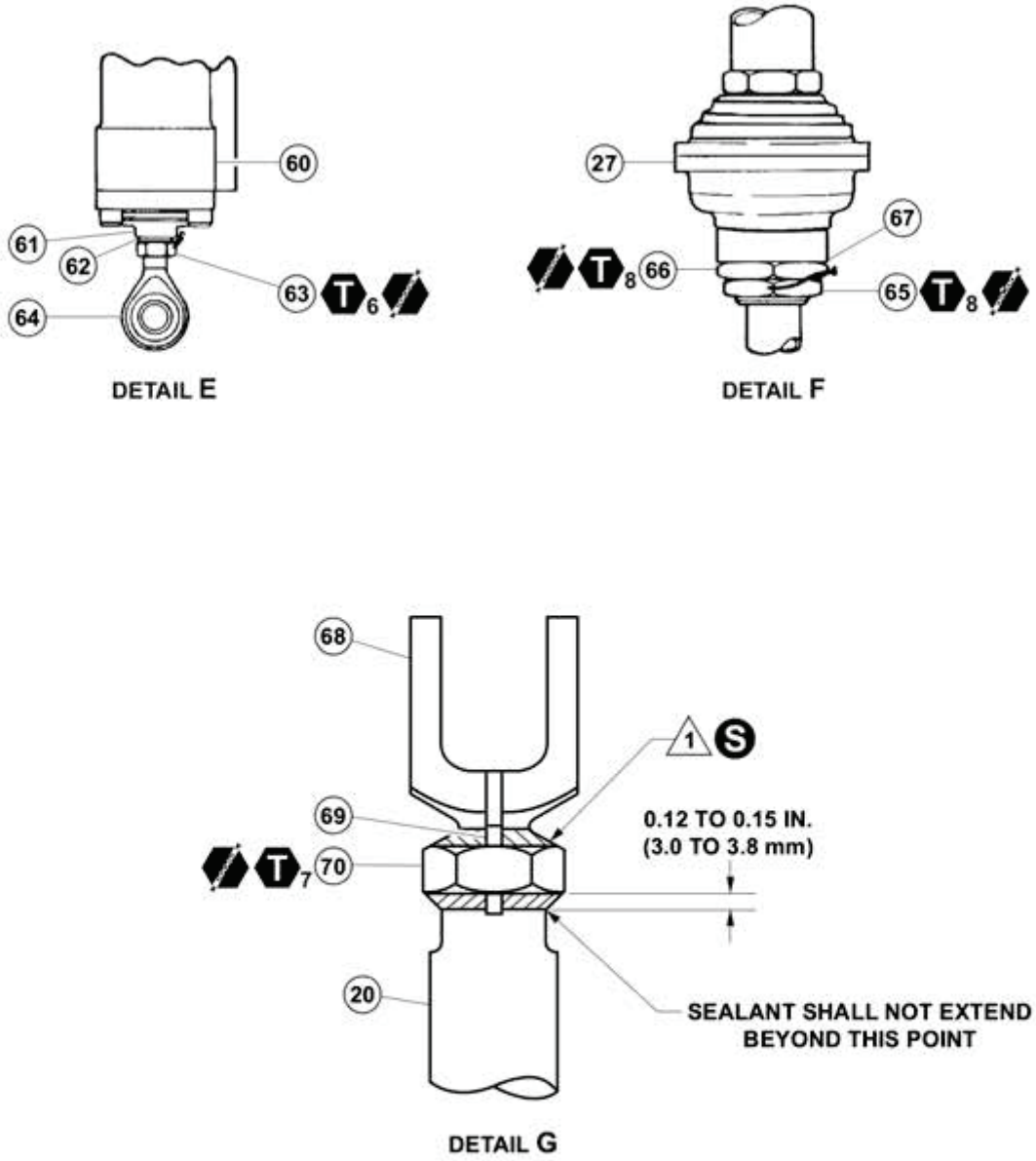
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Figure 3. Collective Flight Control Cylinder (Sheet 1 of 4)



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Figure 3. Collective Flight Control Cylinder (Sheet 2 of 4)



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Figure 3. Collective Flight Control Cylinder (Sheet 3 of 4)

1. Pump
2. Reservoir
3. Swashplate horn
4. Collective lever
5. Support
6. Collective flight control cylinder
7. Cyclic flight control cylinder
8. Cyclic flight control cylinder
9. Bolt
10. Washer (AN960C516)
11. Washer (AN960C516)
12. Washer (AN960C516)
13. Washer (AN960C516L)
14. Nut
15. Cotter pin
16. Extension tube
17. Universal
18. Bolt
19. Washer (AN960C516)
20. Upper piston
21. Washer (AN960C516)
22. Nut
23. Cotter pin
24. Boot
25. Nut
26. Aluminum washer
27. Support bracket
28. Aluminum washer
29. Nut
30. Bolt
31. Preformed packing
32. Elbow
33. Preformed packing
34. Preformed packing
35. Elbow
36. Preformed packing
37. Bolt
38. Hose
39. Lever
40. Washer
41. Washer
42. Nut
43. Cotter pin
44. Preformed packing
45. Bypass check valve
46. Hose
47. Bolt
48. Support
49. Aluminum washer
50. Nut
51. Cotter pin
52. Hose
53. Tube
54. Spring clip
55. Bolt
56. Spring
57. Hose
58. Bypass check valve
59. Preformed packing
60. Actuator
61. Lower piston
62. Locking device
63. Nut
64. Rod end bearing
65. Nut
66. Bearing nut
67. Keyed washer
68. Clevis
69. Lock
70. Jam nut
71. Elbow
72. Nut
73. Packing
74. Elbow
75. Nut
76. Packing

LOCKWIRE (C-405)

S SEALANT (C-308)

T1 40 TO 65 IN-LBS (4.51 TO 7.34 Nm)

T2 75 TO 95 IN-LBS (8.47 TO 10.73 Nm)

T3 75 TO 125 IN-LBS (8.47 TO 14.12 Nm)

T4 95 TO 110 IN-LBS (10.73 TO 12.42 Nm)

T5 120 TO 160 IN-LBS (13.55 TO 18.07 Nm)

T6 200 TO 250 IN-LBS (22.59 TO 28.54 Nm)

T7 660 TO 780 IN-LBS (74.56 TO 88.12 Nm)

T8 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

T9 135 TO 145 IN-LBS (15.25 TO 16.38 Nm)

T10 215 TO 245 IN-LBS (24.30 TO 27.68 Nm)

NOTE

1. Apply sealant (C-308) around jam nut (70), lock (69), and exposed threads of clevis (68).

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Figure 3. Collective Flight Control Cylinder (Sheet 4 of 4)

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Servo Valve Balance Spring - Description

The servo valve balance spring is mounted on collective flight control cylinder to apply an upward force to control lever.

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Servo Valve Balance Spring - Removal

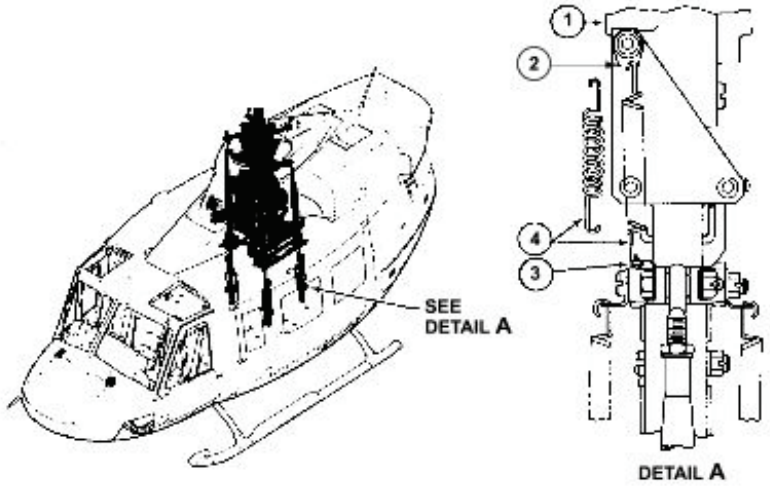
Procedure

NOTE

Gain access to collective hydraulic flight control cylinder (1, Figure 1) through large access in bottom of helicopter (412-A-52-40-00-02A-520A-A / 00837).

1. Disconnect top of spring (4) from spring clip (2).

2. Disconnect bottom of spring (4) from spring clip (3). Remove spring.



1. Collective hydraulic flight control cylinder
2. Spring clip
3. Spring clip
4. Spring

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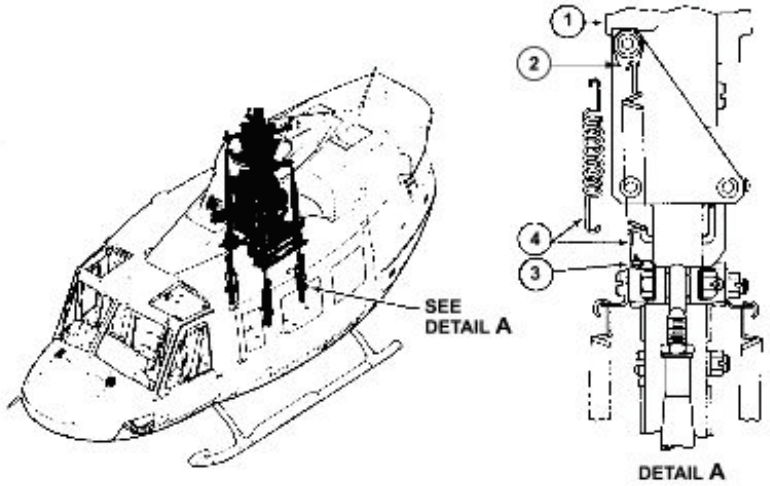
Figure 1. Servo Valve Balance Spring (Sheet 1 of 1)

Servo Valve Balance Spring - Inspection

Procedure

1. Examine the spring (4, Figure 1) for deformation and loss of tension. Free length should be approximately 3.5 inches (88.9 mm).

2. Use a spring scale to make sure that a force of 42.4 pounds (188.6 N) will stretch spring (4) 1.0 inch (25.4 mm).



1. Collective hydraulic flight control cylinder
2. Spring clip
3. Spring clip
4. Spring

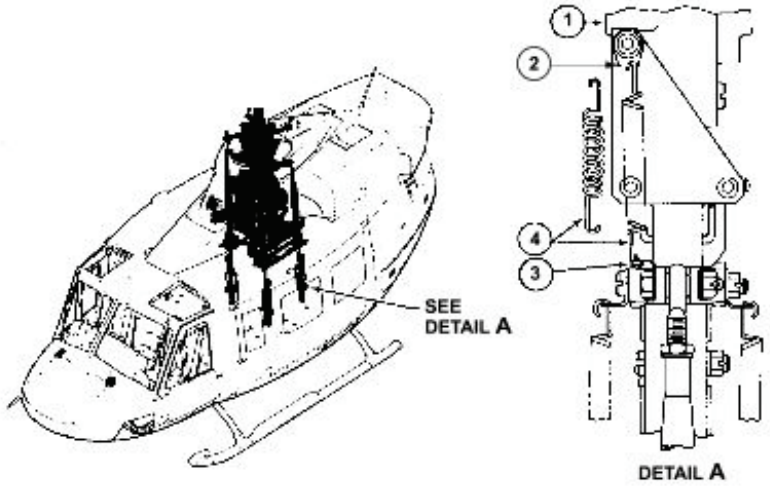
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Figure 1. Servo Valve Balance Spring (Sheet 1 of 1)

Servo Valve Balance Spring - Installation

Procedure

1. Connect the bottom of spring (4, Figure 1) to the spring clip (3).
2. Connect the top of spring (4) to the spring clip (2).



1. Collective hydraulic flight control cylinder
2. Spring clip
3. Spring clip
4. Spring

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Figure 1. Servo Valve Balance Spring (Sheet 1 of 1)

Collective Flight Control Cylinder - Removal

Procedure

1. Remove the access panel on the forward side of the pylon support inside the cabin (412-A-52-40-00-02A-520A-A / 00837).

2. Remove the aft pylon fairing assembly (412-A-71-10-00-01A-520A-A / 01605).

3. Refer to 412-A-29-00-00-00A-042A-A / 00612 before performing the maintenance on the hydraulic systems.

Applicable: SN: 33001-33213 or 36001-36300:

4. Disconnect the hoses (38, 46, 52, and 57, Figure 1, Detail D). Remove the bolts (30 and 37) and elbows (32 and 35). Cap or plug the open ports to prevent contamination.

Applicable: SN: 36301-Subsequent:

5. Disconnect the hoses (38, 46, 52, and 57, Figure 1, Detail D). Remove the elbows (71 and 74), nuts (72 and 75) and packings (73 and 76). Cap or plug open ports to prevent contamination.

6. Disconnect the spring (56) from the spring clip (54). Remove the cotter pin (43), nut (42), washers (40 and 41), and bolt (55), and disconnect the tube (53). Keep the bolt (55), spring clip (54), and spring (56).

7. Remove the cotter pin (51), nut (50), aluminum washers (49), and the bolt (47).

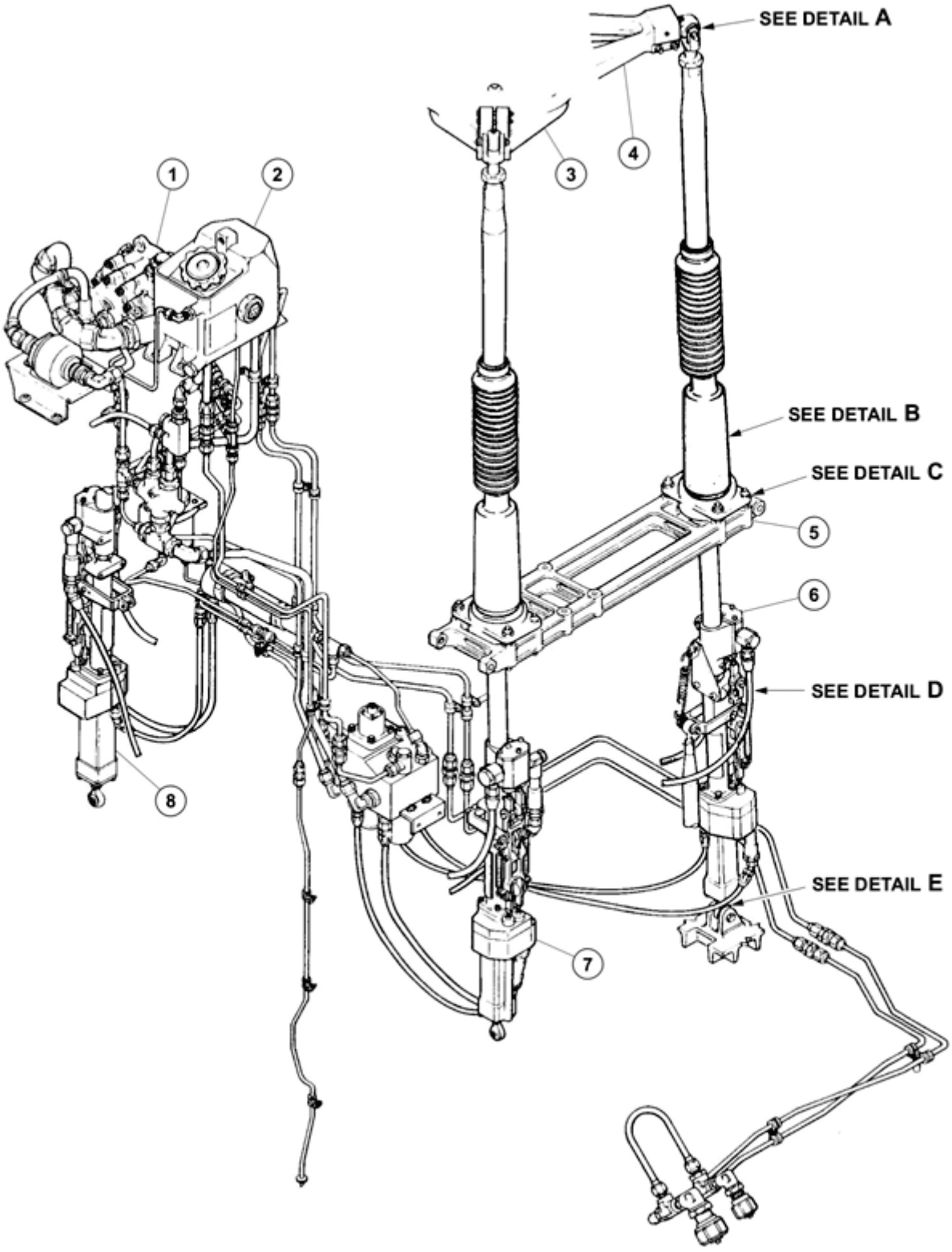
8. Remove the nuts (25 and 29) and aluminum washers (26 and 28) (Detail C).

9. Remove the cotter pin (15), nut (14), washers (10, 11, 12, and 13), and the bolt (9) (Detail A).

10. Carefully remove the collective flight control cylinder (6) and tube (16).

11. Adjust the boot (24) to allow the removal of the cotter pin (23), nut (22), washers (19 and 21), and bolt (18) (Detail B). Keep the bolt (18), universal (17), and tube (16).

12. Loosen the jam nut (70) and remove the clevis (68) from the upper piston (20).



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Figure 1. Collective Flight Control Cylinder (Sheet 1 of 4)